



IP Multicast

Volume 1

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Josh Loveless, CCIE No. 16638

Ray Blair, CCIE No. 7050

Arvind Durai

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Title Page

IP Multicast, Volume 1

Cisco IP Multicast Networking

Josh Loveless, CCIE No. 16638

Ray Blair, CCIE No. 7050

Arvind Durai, CCIE No. 7016

Cisco Press

800 East 96th Street

Indianapolis, Indiana 46240 USA

Copyright Page

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About the Authors

About the Authors

Josh Loveless, CCIE No. 16638, is a customer solutions architect for Cisco Systems. He has been with Cisco for four years, providing architecture and support services for Tier 1 service providers as well as for many of Cisco's largest enterprise customers, specializing in large-scale routing and switching designs. Currently, Josh is helping Cisco's customers in the defense and intelligence industries meet the challenges of an ever-changing technology landscape, designing secure automated networks with advanced capabilities, including IP multicast. Prior to joining Cisco, he spent 15 years working for large service providers and enterprises as both an engineer and an architect, as well as providing training and architecture services to some of Cisco's trusted partners. Josh maintains two CCIE certifications, Routing and Switching and Service Provider.

Ray Blair, CCIE No. 7050, is a distinguished systems engineer and has been with Cisco Systems since 1999. He uses his years of experience to align technology solutions with business needs, insuring customer success. Ray started his career in 1988, designing industrial monitoring and communication systems. Since that time, he has been involved with server/database administration and the design, implementation, and management of networks that included networking technologies from ATM to ZMODEM. He maintains three CCIE certifications in Routing and Switching, Security, and Service Provider (No. 7050), is also a Certified Information Systems Security Professional (CISSP), and a Certified Business Architect (No. 00298). Ray is coauthor of two Cisco Press books, *Cisco Secure Firewall Services Module* and *Tcl Scripting for Cisco IOS*. He speaks at many industry events and is a Cisco Live distinguished speaker.

Arvind Durai, CCIE No. 7016, is a director of solution integration for Cisco Advanced Services. His primary responsibility in the past 17 years has been in supporting major Cisco customers in the enterprise sector, including financial, retail, manufacturing, ecommerce, state government, utility (smart grid networks), and healthcare sectors. Some of his focuses have been on security, multicast, network virtualization, and data center, and he has authored several white papers and design guides on various technologies. He has been involved in multicast designs for several enterprise customers in different verticals. He is also one of the contributors in providing the framework for Advanced Services Multicast Audit tool that helps customers assess their operational multicast network to Industry best practices.

Arvind maintains two CCIE certifications: Routing and Switching and Security and also is a Certified Business Architect. He holds a Bachelor of Science degree in Electronics and Communication, a Master's degree in Electrical Engineering (MS), and a Master's degree in Business Administration (MBA). He is a coauthor of three Cisco Press books: *Cisco Secure Firewall Services Module*, *Virtual Routing in the Cloud*, and *Tcl Scripting for Cisco IOS*. He has coauthored IEEE WAN smart grid architecture and presented in many industry forums, such as IEEE and Cisco Live.

About the Technical Reviewers

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Nick Garner, CCIE No. 17871, is a solutions integration architect for Cisco Systems. He has been in Cisco Advanced Services, supporting customers in both transactional and subscription engagements, for 8 years. In his primary role, he has deployed and supported large-scale data center designs for prominent clients in the San Francisco Bay Area. His primary technical focus, outside of data center routing and switching designs, has been security and multicast. Prior to joining Cisco, Nick worked for a large national financial institution as a network security engineer. Nick maintains two CCIE certifications, Routing and Switching and Security.

Tianji Jiang, Ph.D., is a dual CCIE (No. 19987, R&S and Data Center) and has more than 20 years of experience in the networking field. For the past 15 years, he has worked at Cisco in various roles, spanning development, escalation, and advanced service delivery. Currently as a Cisco networking consultant, he has been a key member of many service provider projects, including planning, designing, implementing, and deploying large IP networks. Tianji graduated with a Doctorate degree from Georgia Tech, with his Ph.D. dissertation focusing on investigating multicast scalability and heterogeneity.

Dedications

Dedications

Josh Loveless: This book is dedicated to all those friends and family that have believed in me, especially to my family who sacrificed to help me make this book a reality and who continue to support me in my career every day.

Ray Blair: As with everything in my life, I thank my Lord and Savior for his faithful leading that has brought me to this place. This book is dedicated to my wife, Sonya, and my children, Sam, Riley, Sophie, and Regan. You guys mean the world to me!

Arvind Durai: This book is dedicated to my wife Monica, who is my best friend and advisor. I would also like to thank my son Akhhill for putting up with the long hours I spent working on this book.

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Arvind Durai: Thanks to my wife, Monica, for encouraging me to write my fourth book. You always kept my spirits high. Thank you!!!

Ray, this is my third book with you. We go a long way back, and working with you has always been a pleasure. Josh, I appreciate your dedication in this project more than you will ever know. Every success is a result of teamwork; you both have made the experience of writing this book a pleasure. Thank you!

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Command Syntax Conventions

Command Syntax Conventions

The conventions used to present command syntax in this book are the same conventions used in the IOS Command Reference. The Command Reference describes these conventions as follows:

- **Boldface** indicates commands and keywords that are entered literally as shown. In actual configuration examples and output (not general command syntax), boldface indicates commands that are manually input by the user (such as a show command).
- *Italics* indicate arguments for which you supply actual values.
- Vertical bars (|) separate alternative, mutually exclusive elements.
- Square brackets [] indicate optional elements.
- Braces { } indicate a required choice.
- Braces within brackets [{ }] indicate a required choice within an optional element.

Introduction

Introduction

IP Multicast, Volume I covers basic IP multicast principles and routing techniques specific to Cisco Systems routers and switches. Included is a pragmatic discussion of common features, deployment models, and field practices for IP multicast networks. The discussion culminates with commands and methodologies for implementing and troubleshooting Cisco IP multicast networks.

Who Should Read This Book?

IP Multicast, Volume I is intended for any professional supporting IP multicast networks. This book primarily targets the following groups, but network managers and administrators will also find value from the included case studies and feature explanations:

- IP network engineers and architects
- Network operations technicians
- Network consultants
- Security professionals
- Collaboration specialists and architects

How This Book Is Organized

This book is organized in seven chapters. It starts with an introduction to data communication in IP networks. Network access, Layer 2, and Layer 3 multicast is explained, along with protocol independent multicast (PIM). [Chapter 5](#) introduces multicast scoping and covers design considerations. The final chapters explain IPv6 multicast, and the operation and troubleshooting of IP multicast networks. The following are the chapters covered in this book:

■ [Chapter 1](#) , “[Introduction to IP Multicast](#) ”: This chapter introduces the reader to the basic concepts of multicast through the use of graphics and short explanations. It answers the question: *What is multicast and why is it needed?* We discuss the basic reasons for using multicast, give some practical examples of multicast applications, and describe the basics of IP multicasting and how it differs from unicasts and broadcasts.

■ [Chapter 2](#) , “[Network Access and Layer 2 Multicast](#) ”: This chapter introduces the concepts of multicast at the access layer, including layered encapsulation, IGMP group management, and the concept of switching multicast frames. It also discusses Layer 2 switching domains and IPv4 group address to MAC address mapping.

■ [Chapter 3](#) , “[IP Multicast at Layer 3](#) ”: This chapter introduces the concepts of multicast at Layer 3 of the OSI model. It covers the types of multicast hosts, and gives an introduction to the various modes of PIM.

■ [Chapter 4](#) , “[Protocol Independent Multicast](#) ”: The PIM discussion includes a description of basic forwarding trees, rendezvous point (RP) mechanics, and ways to propagate RP mapping information. The chapter concludes with a discussion of the different forwarding modes for multicast networks, ASM, SSM, and the PIM Bidir.

■ [Chapter 5](#) , “[IP Multicast Design Considerations and Implementation](#) ”: This chapter discusses the considerations required for a basic multicast network design. It focuses on properly

scoping a multicast enterprise network. It also discusses the concepts of forwarding replication as well as MFIB and MRIB, which allow the reader to make educated decisions about which implementation options to choose. The chapter also covers deployment IP multicast best practices with respect to security and resiliency.

- [Chapter 6](#) , “[IPv6 Multicast Networks](#) ”: This chapter introduces the concepts of IPv6 multicast. It provides the reader an overview of Layer 2 and 3 multicast for IPv6. It also explains RP deployment concepts unique to IPv6.

- [Chapter 7](#) , “[Operating and Troubleshooting IP Multicast Networks](#) ”: This chapter is a simple guide to operating and troubleshooting basic multicast networks essential for network administrators managing multicast networks.

Chapter 1. Introduction to IP Multicast

Chapter 1. Introduction to IP Multicast

There are three data communication methods in IP networks: unicast, broadcast, and multicast. To establish a baseline, it is important to understand the fundamental elements of unicast and broadcast before we explore the details of multicast communication methods.

Unicast communication at Layer 3 (the Network layer) of the Open Systems Interconnect (OSI) model is based on the IP address of the destination device. Routers learn about routes by static or dynamic methods and forward traffic by looking at the destination IP address. Layer 2 (the Data Link layer) uses another mechanism to establish communication between devices using media access control (MAC) addresses.

Take a look at [Figure 1-1](#) . The sender is sending a message to Receiver A, and this requires a combination of both L2 and L3 services. The sender learns the MAC address of the gateway using the Address Resolution Protocol (ARP) and sends IP traffic destined to any network other than the local subnet to the gateway/router. The router looks at the destination IP address and forwards the packet to the next router based on the information in the routing table (Layer 3). The destination router receives the packet and forwards it to the receiver (Receiver A, in this example). As you can see, the Sender never learns the MAC address of Receiver A because Receiver A is not on the same subnet.

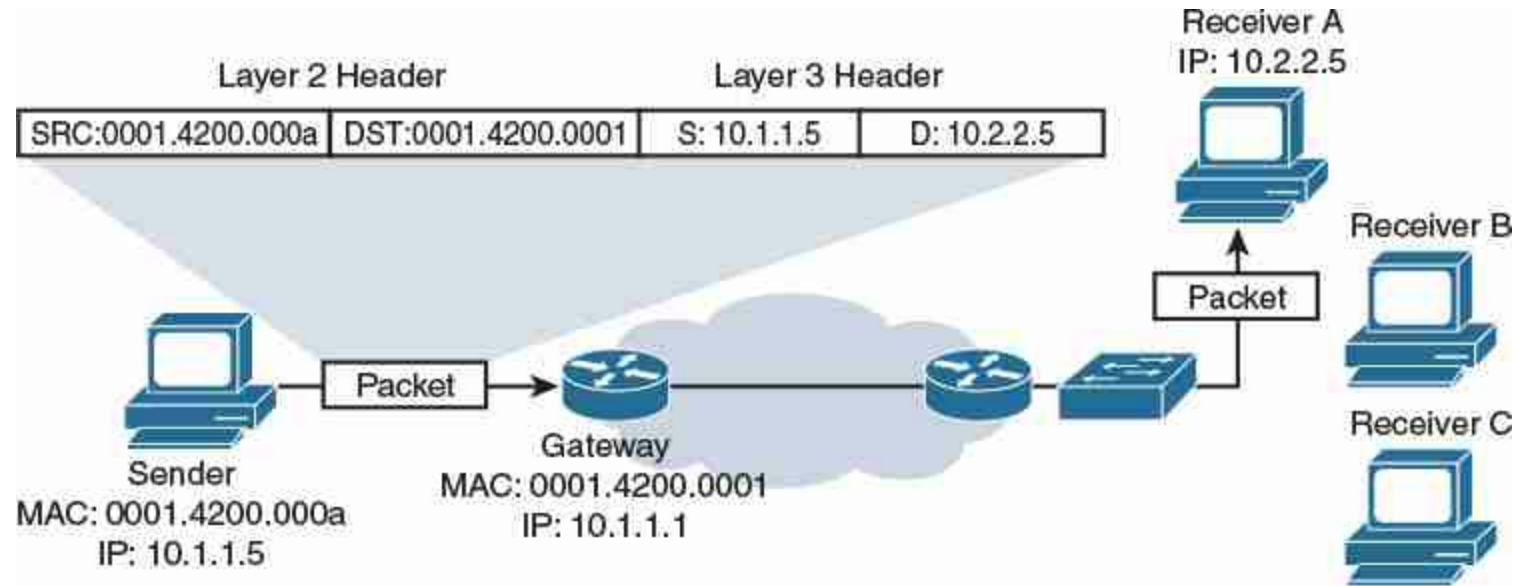


Figure 1-1 *Unicast Packet Forwarding Example*

Most nontechnical people are familiar with broadcasting. A *broadcast* , in this classic sense, is a signal that is propagated through electromagnetic waves in the air. The signal saturates the broadcast area, and retrieving the communication is a simple matter of tuning in to a specific frequency. In the networking world, a broadcast is a message sent to all devices on a subnet or a Layer 2 domain, and every device has an obligation to check the message to validate whether they are an intended recipient.

Note

An Ethernet broadcast is very different from an IP broadcast. Remember that IP packets (Layer 3) are encapsulated inside an Ethernet frame (Layer 2). There are source addresses and destination addresses at each layer, and each receives different treatment by networking devices. The destination

address of a Layer 3 broadcast in IP is all ones in the host portion of the address. Thus, all hosts would be 255.255.255.255. At Layer 2, an all-hosts broadcast is ffff.ffff.ffff, and switches must replicate these packets when forwarding in order to reach all intended recipients, regardless of physical segment (known as *flooding*). If a device was looking for a particular IP host but did not know the destination MAC address, it could send an IP unicast message encapsulated in an all-hosts Ethernet broadcast. Every machine on the Ethernet segment receives the packet, but only the intended IP host processes the packet fully and responds. In fact, this is exactly what an initial ARP request looks like.

We have traditionally used a router or Layer 3-capable switch to segment broadcast domains and protect devices from unwanted traffic. This means the router or Layer 3 switch must inspect Ethernet and IP headers and search for broadcast messages. If a packet is a broadcast packet, the router must make a forwarding decision on the destination address. If the address is a local broadcast (intended for only local hosts) the router should drop the packet because this is usually in the format of an all-hosts broadcast (255.255.255.255). If the address is a directed broadcast (a broadcast intended for a specific segment or subnet, such as, for example, 10.1.1.255) the router may forward the broadcast toward the targeted segment, or it may respond if the receiving interface is on the intended segment.

A *multicast message* is a similar, more efficient version of a broadcast, which is a replicated packet that is sent to a set of hosts. The primary exception is that with a multicast message, the intended recipients of the flow could be on any segment instead of on one specific segment. This process of receiving a single packet, replicating it, and sending copies to additional network interfaces is the key to understanding how multicast works, from the highest to the lowest level. Routers and switches must replicate the packet from the source interface and forward toward receivers. *Multicast* incorporates the concept of selective learning of data flow by multiple receivers in Layer 2 and 3 domains. Multicast data distribution methods optimize bandwidth usage and create an efficient method of data distribution.

What Problem Does Multicast Solve?

The purpose of multicast is to send information to a select group of devices and to avoid duplication of traffic streams, consequently saving precious network resources such as bandwidth utilization.

Consider the other two mechanisms to propagate information throughout a network—unicast and broadcast. Using the example of an Internet radio station transmitting information, unicast requires that each device on the network interested in listening to the audio stream has a unique session established. As shown in [Figure 1-2](#), unicast has one sender and three receivers. A *sender* is a device that generates information and advertises it to a group, and a *receiver* is a device that listens to the information.

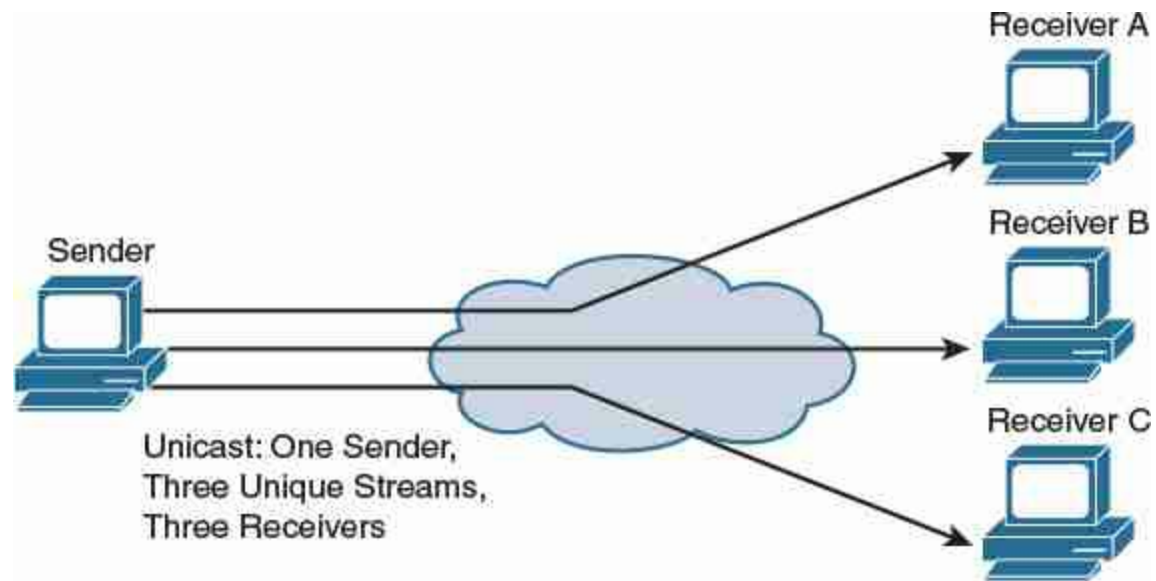


Figure 1-2 *Unicast Using Multiple Streams*

Because each stream is being replicated, the sender must replicate the information for each client and the network connection takes three times the bandwidth. This may not be a big problem for a low-bandwidth audio session with only three users, but consider scaling to tens of thousands or millions of sessions. Replicating information for every client creates a significant impact on network resources.

By using broadcasts, the radio station can minimize the number of sessions the sender has to be concerned with, which reduces bandwidth from the sender to the network; however, in this case, the radio station has a different challenge. Referring to [Figure 1-3](#), the problem of replicating numerous streams has been eliminated, and the bandwidth challenge has been mitigated, but now every device receives the radio station whether they are interested in listening to it or not. As the number of radio stations increases, every device on the network becomes more and more inundated with information it might not want. The receivers are now obligated to process the radio streams and to decide whether that information is relevant. Welcome to networking in the 1980s and 1990s, when broadcast storms were a common occurrence.

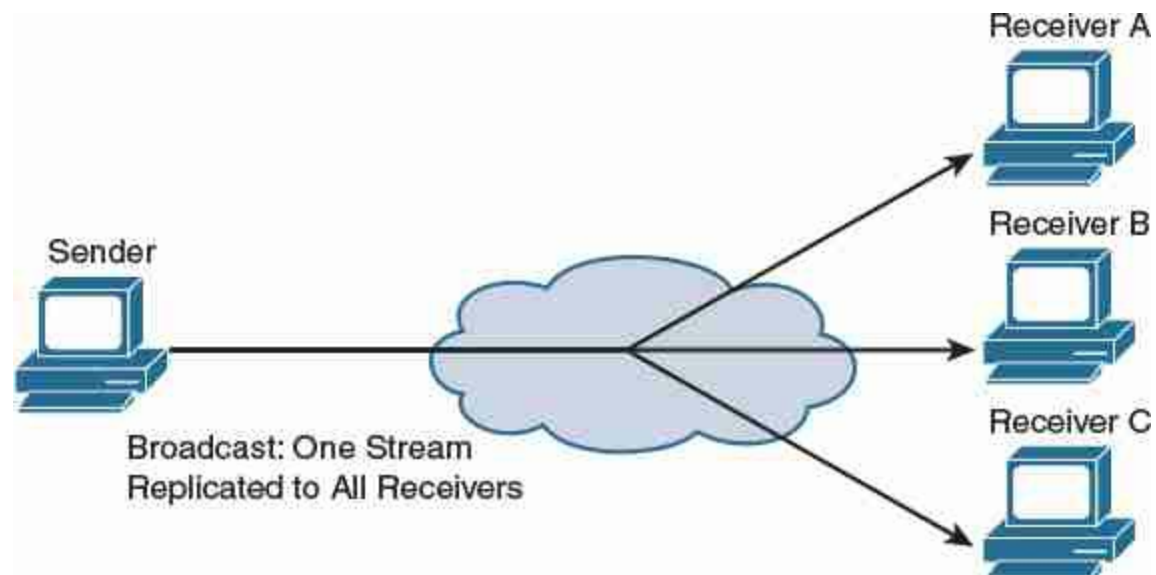


Figure 1-3 *Broadcast*

Multicast to the rescue! We have the best of both worlds with multicast, minimizing the number of sessions required by the sender and reducing the network load to a single traffic flow. We also obtain

the benefits of unicast by sending the radio station packets only to the devices interested in receiving them. Referring to [Figure 1-4](#) , with multicast, only a single stream is sent and replicated to the interested receivers.

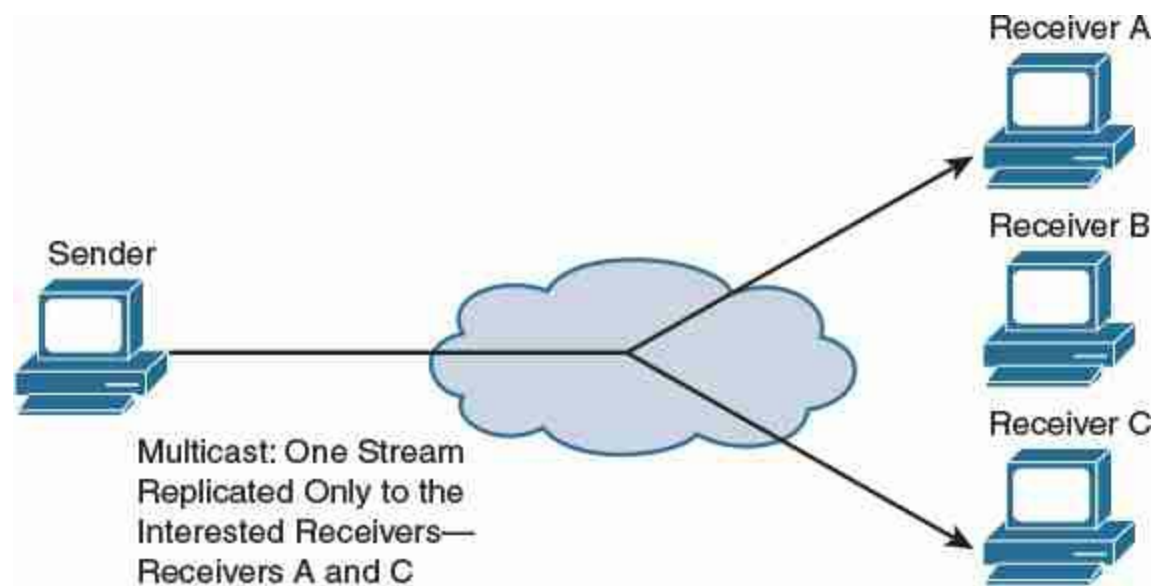


Figure 1-4 *Multicast*

IP multicast messages are transmitted from one-to-many, from many-to-many, and/or from many-to-one over an IP infrastructure that may span Layer 3 boundaries. The destination nodes (receivers) send join and leave messages that create an on-demand community of devices interested in receiving the multicast stream. Multicast optimally uses network resources by requiring the source to send a single packet stream, even though large numbers of receivers might need to receive the messages. The replication of messages takes place at the network node(s) or Layer 3 device(s) and diverges the stream to the receivers. The optimization of the message flow of multicast is leveraged by many applications. These applications are one of the major driving forces for the adoption of multicast in the network infrastructure. Some of the more common applications that rely on multicast are as follows:

- Stock exchanges
- Computer imaging applications
- Music-on-hold services
- Sensor updates
- Video distribution
- Corporate webcasts

Having a good understanding of the nuances of multicast enables you to effectively support those applications and services that rely on multicast as a transport mechanism. This chapter covers some of the applications and services that rely on multicast, examines the history of multicast, and delves into some of the standards.

Multicast Applications and Services

The network infrastructure is in place to support applications and services. These applications and services allow an entity—a government agency, bank, retail organization, hospital, emergency services, or any other business or institution—to fulfill its mission or business objective. Establishing an infrastructure that enables the effective use of these multicast applications and services helps to

ensure the success of that organization.

One-to-Many Multicast Applications

The most common form of multicast applications is *one-to-many*, as shown in [Figure 1-5](#).

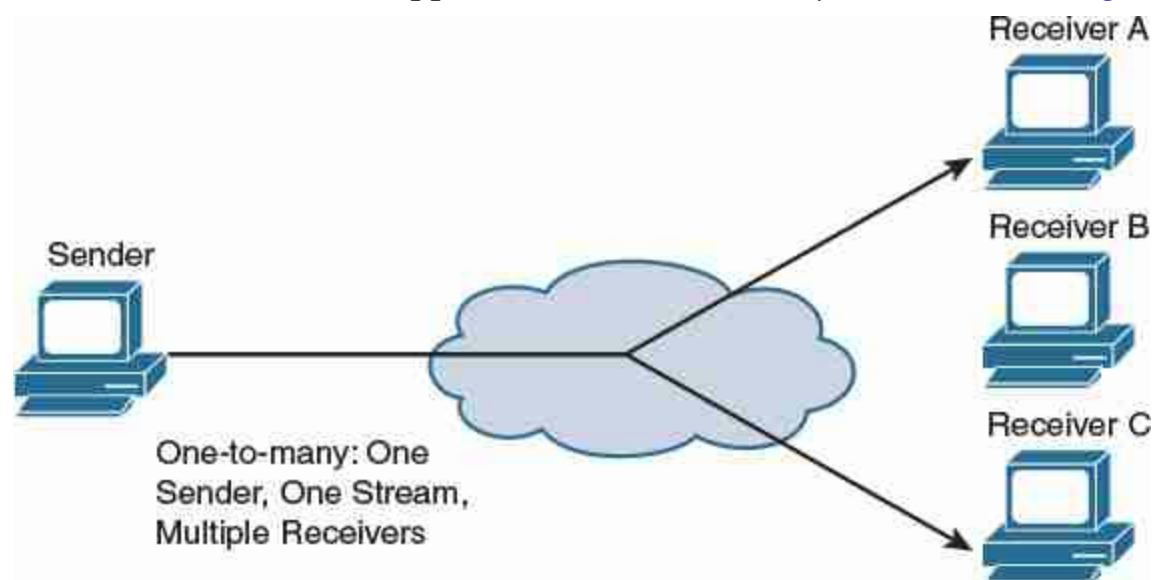


Figure 1-5 *Multicast Using One-to-Many*

As the name implies, there is one sender and multiple simultaneous receivers. Typical applications include video and audio broadcast, but there are many other applications as well, including the following:

- Television
- Radio
- Distance learning
- Presentation sharing and whiteboard applications
- Computer imaging software and application updates
- Data distribution and caching
- Informational updates:
 - Weather
 - News
 - Time—Network Time Protocol (NTP)
 - Sensors that collect environment information (water levels, temperatures, seismic readings, and so on).

Many-to-Many Multicast Applications

With *many-to-many* multicast applications, the senders also act as receivers. This permits all the devices in the group to simultaneously communicate with one another, as shown in [Figure 1-6](#).

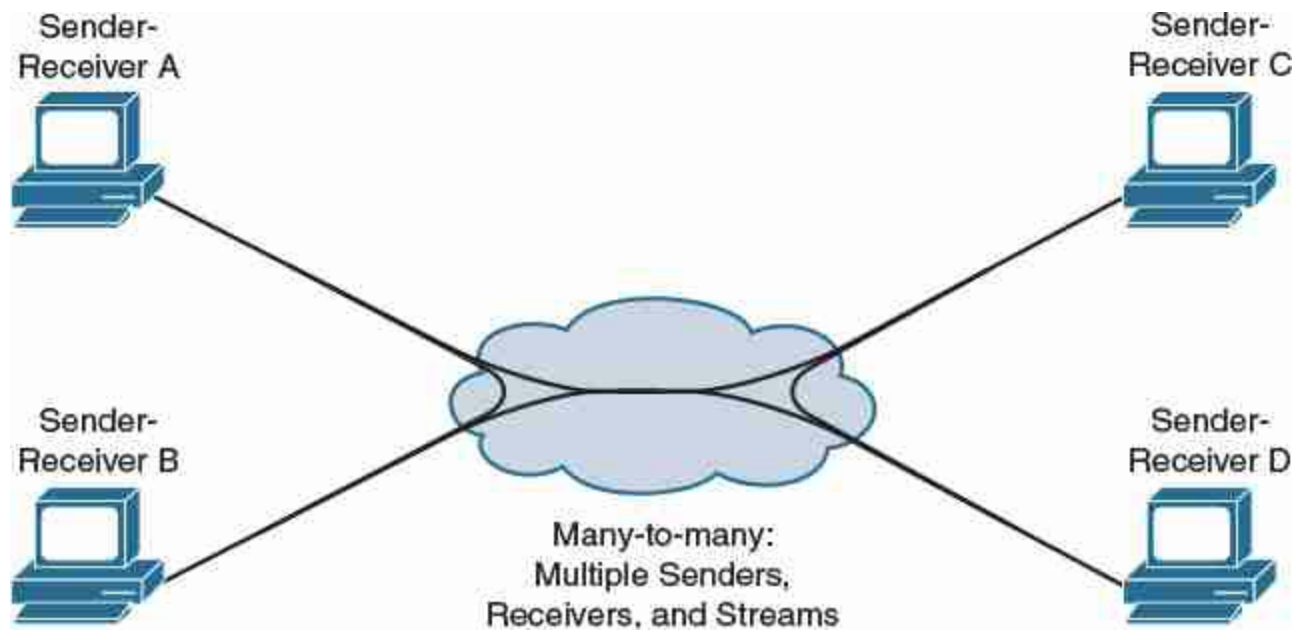


Figure 1-6 *Multicast Using Many-to-Many*

Many-to-many applications include the following:

- Audio and video communication
- Document sharing and whiteboard applications
- Data distribution, caching, and synchronization
- Group chat applications
- Financial applications
- Polling applications
- Multiplayer games.

Many-to-One Multicast Applications

Many-to-one applications have many senders but only one or few receivers, as shown in [Figure 1-7](#). This is not a common multicast offering, and it has a significant challenge in that the receiver might not be able to process the information when many devices are sending multicast flows. Scalability is a significant concern with this model. In some ways, this model is not an improvement over using unicast streams but instead allows configuration flexibility for the application. In fact, in many cases, the receiver will respond to senders via a unicast flow. See RFC 3170 for more details on many-to-one applications.

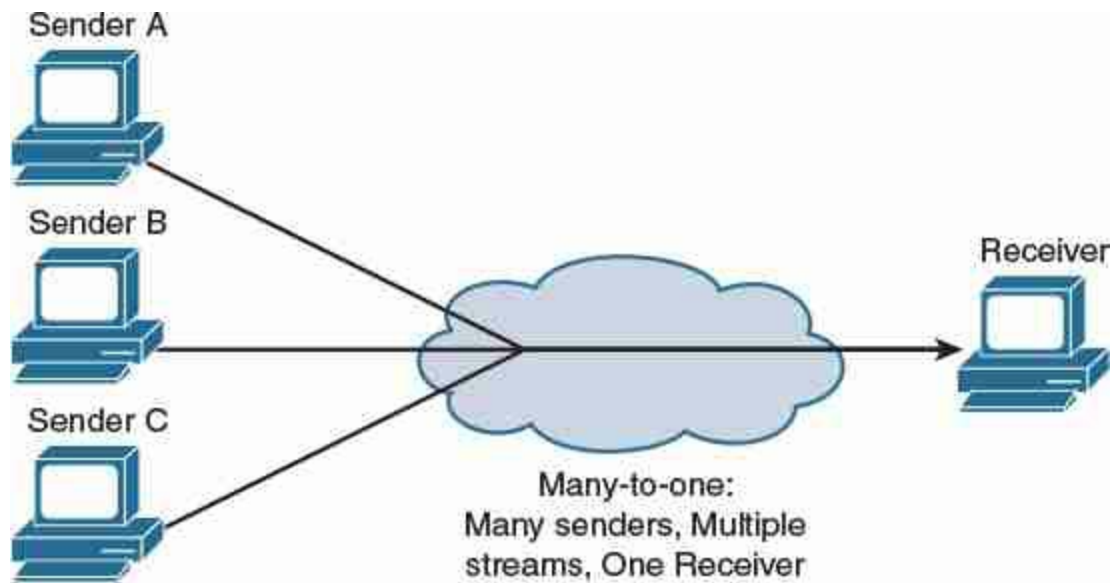


Figure 1-7 *Multicast Using Many-to-One*

Some of the applications for many-to-one include the following:

- Data collection
- Service discovery
- Polling

Multicast applications can consume a tremendous amount of bandwidth, as with high-definition video, or it can consume very little network bandwidth, as in time updates. All of these applications rely on the infrastructure for the successful operation of the previously mentioned applications and services.

Multicast Packet

As mentioned previously, multicast is a communication method for reaching many participants with a single message flow. In an Ethernet network running Internet Protocol (IP), the active components that make up the infrastructure, primarily routers and switches, are responsible for replicating the single packet flow into multiple packets or messages that are efficiently distributed to other devices interested in receiving those messages.

This warrants a brief review of the Open Systems Interconnect (OSI) model and an explanation of where multicast is applied to the different layers. The OSI model is comprised of the elements shown in [Table 1-1](#) .

Layer	Name	Comments
7	Application	
6	Presentation	
5	Session	
4	Transport	UDP
3	Network	IP routing and addressing
2	Data Link	MAC addressing
1	Physical	

Table 1-1 *Multicast Applied to the OSI Reference Model*

Multicast applications commonly use User Datagram Protocol (UDP) on IP. Consequently, the Transport layer is used, and this would not be possible without the Network layer running the IP protocol. The IP layer is also where routers primarily function. The Data Link layer is where Ethernet switches replicate multicast traffic on a subnet using multicast media access control (MAC) addresses.

You need to have a solid understanding of the OSI model to comprehend the dynamics of IP multicast technologies and how each layer interacts with one another.

Note

We refer to a *frame* as a Layer 2 message in that we are focusing on the source and/or destination MAC address. A *packet* is a Layer 3 message that includes a source and a destination IP address.

It is important to understand exactly how routers build a multicast routing information base (MRIB) and how they use the unicast routing information base (RIB) to ensure loop-free forwarding paths. Mathematically, a tree is the best way for a routing or switching device to guarantee loop free topologies. Multicast routers and switches are master tree builders, which will be covered in more detail throughout this book.

What Is a Source?

When speaking about multicast, there are always two types of hosts, a multicast source and a multicast receiver. A *multicast source* can be any device with an IP address in the network. To become a source, a host only needs to send a message to a multicast group IP address. (See [Figure 1-8](#).) The three senders in this diagram are all sending to the same multicast destination address of 239.1.1.1.

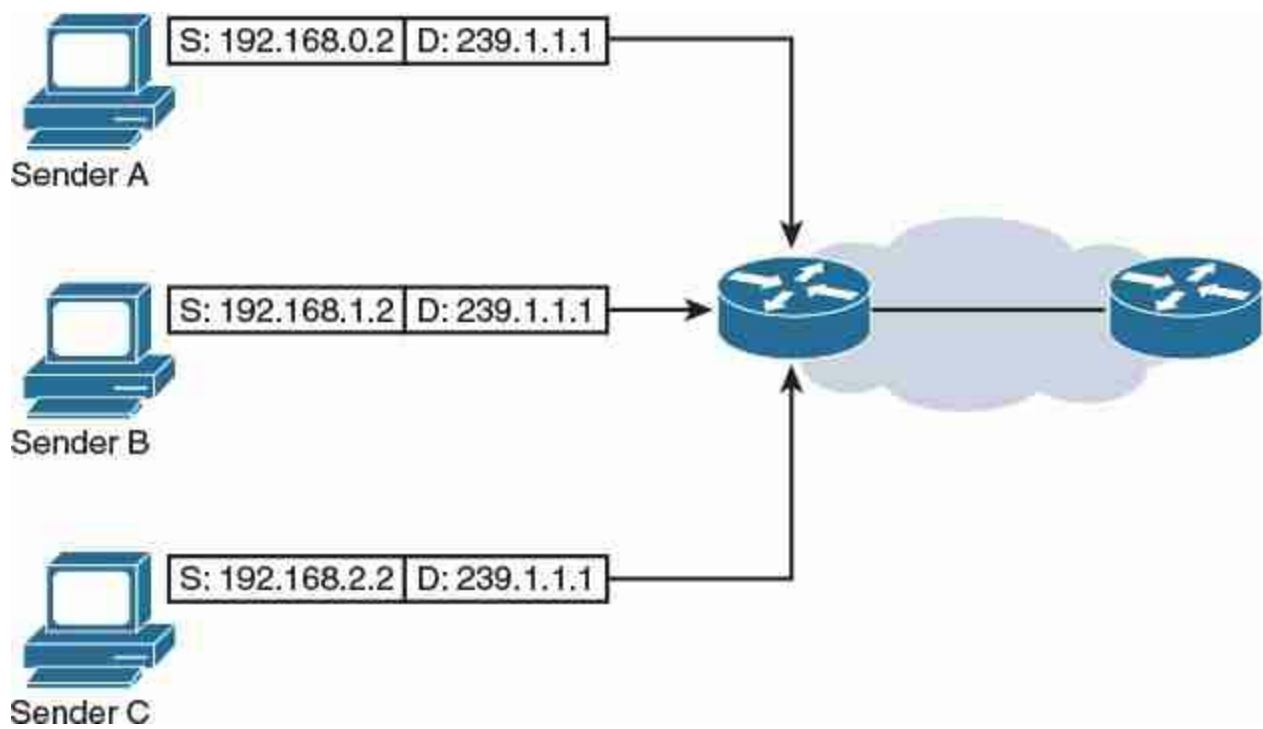


Figure 1-8 *Multicast Source*

A source does not need to indicate any intention to send to a group before sending a multicast message. Any IP enabled device, including a router or switch, can send packets toward a multicast group. When an IP multicast router processes a message destined to a multicast group, a new multicast forwarding entry is created. This is the Source, Group entry and is called *source comma group*, (S, G). The (S, G) entry for Sender A in [Figure 1-8](#) would be (192.168.0.2, 239.1.1.1).

The (S, G) entry refers to the source IP address and the destination group separated by a comma. The IP addresses of the senders in [Figure 1-8](#) are the source (S) addresses, whereas the destination IP address of the group is G. Notice that the three devices are all sending to the same group address. Could this be a problem? Keep that thought in the back of your mind as you continue reading.

What Is a Receiver?

A *receiver* is any multicast-enabled device that has expressed interest in a particular multicast group or a specific (S, G) pair. Unless the multicast is link-local (not passed on through the network by any router, such as, for example, the “all-routers” IPv4 group of 224.0.0.2), the IP devices must be configured by the administrator or by an application to subscribe to a specific multicast group. After it’s subscribed, a multicast receiver listens for packets that arrive with the multicast group destination address, like group 239.1.1.1 used in [Figure 1-8](#).

Group subscription is managed by the Internet Group Messaging Protocol (IGMP). When a receiver subscription for a specific group or set of groups is received or initiated by a router, the router adds what is called a *star comma G*, (*, G) entry to the MRIB. The (*, G) entry simply represents that the router has an interest in the group.

Note

Multicast forwarding information based (MFIB) and *multicast routing information base (MRIB)* are terms used to understand the multicast flow in Cisco routers.

The MRIB is responsible for routing information that is generated by multicast protocols. This information feeds into the MFIB responsible for forwarding multicast packets and collecting statistics on the multicast flow.

L3 Multicast Is Built on the TCP/IP Protocol Stack

IP multicast is built on the TCP/IP protocol stack. That means that the protocols necessary to transport multicast frames and packets are controlled by the Internet Engineering Task Force (IETF). IETF members develop and manage protocols through the RFC process, which means that IP multicast protocols are open standards.

Note

Multicast protocol IETF development applies to both IPv4 and IPv6 multicast technologies; however, as with other IP-based protocols, that does not mean that every manufacturer treats multicast the same. It also does not mean that every implementation of multicast protocols is perfectly standard-compliant.

Using the TCP/IP stack also means that IP multicast is subject to the Internet Assigned Numbers Authority (IANA). IANA controls and coordinates the assignment and use of IP addresses in public spaces. This includes multicast address assignment.

It's a Group Thing

A contrast and comparison between L3 unicasts, broadcasts, and multicasts will illustrate the uniqueness of multicast transmissions. The primary difference between a broadcast and a multicast is that multicast receivers can connect anywhere on any network segment or subnet, whereas subnets define broadcast boundaries, called *broadcast domains*. Routers and switches must therefore have a way of recognizing which segments and subnets connect interested multicast hosts. Senders and receivers manage this process through *group membership*.

A broadcast uses a specific destination IP address in the packet to reach each receiving host. Routers and switches immediately recognize broadcasts without any additional protocol overhead because the subnet defines the broadcast boundary. The following packet header breakdowns show the difference between a unicast IPv4 packet, a broadcast IPv4 packet, and a multicast IPv4 packet. [Figure 1-9](#) illustrates a basic unicast IP packet.

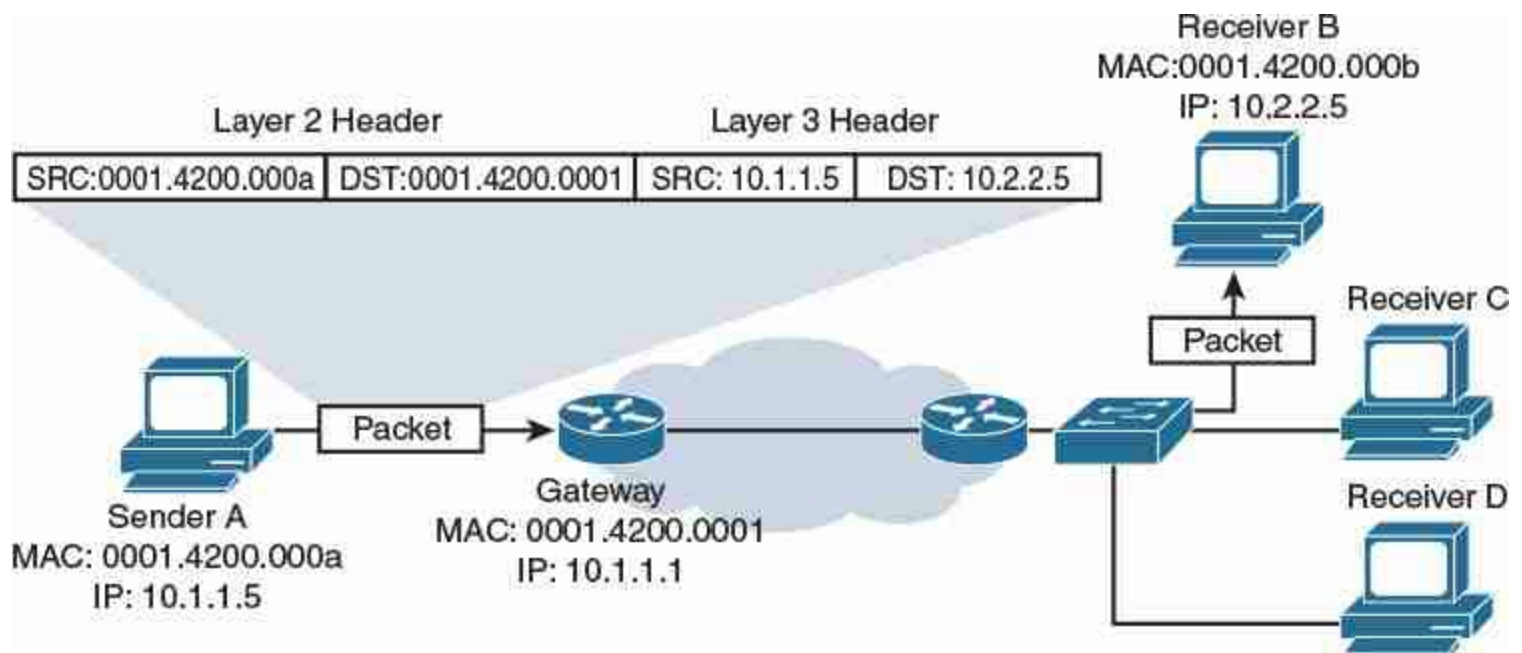


Figure 1-9 IPv4 Unicast Packet

Forwarding a unicast message is a simple task—follow the route to the IP destination. Broadcasts are equally simple to forward. Broadcast packets also follow the route to the destination, but they are

replicated on any switch ports that belong to the appropriate logical Ethernet segment (VLAN or subnet). There are two types of broadcasts, *all-hosts broadcasts* and *directed broadcasts*. An all-hosts broadcast is intended for every IP host on every subnet in which it is forwarded. A directed broadcast is intended for every IP host only within a given subnet/supernet or VLAN.

Note

Replication is the simple process of making a copy of a packet to forward through the network. Replication is required anytime a single packet has more than one intended recipient, as, for example, with broadcasts and multicasts. Replication is a basic function of any active networking device such as a switch or router.

If the broadcast includes all-hosts, then the local switch simply replicates the broadcast packet for each interface in the logical segment on which the packet was received. [Figure 1-10](#) depicts an all-hosts broadcast on a local segment. Routers, by their nature, do not forward all-hosts broadcasts received on an interface in a given subnet, consequently segmenting the broadcast domain from the rest of the network.

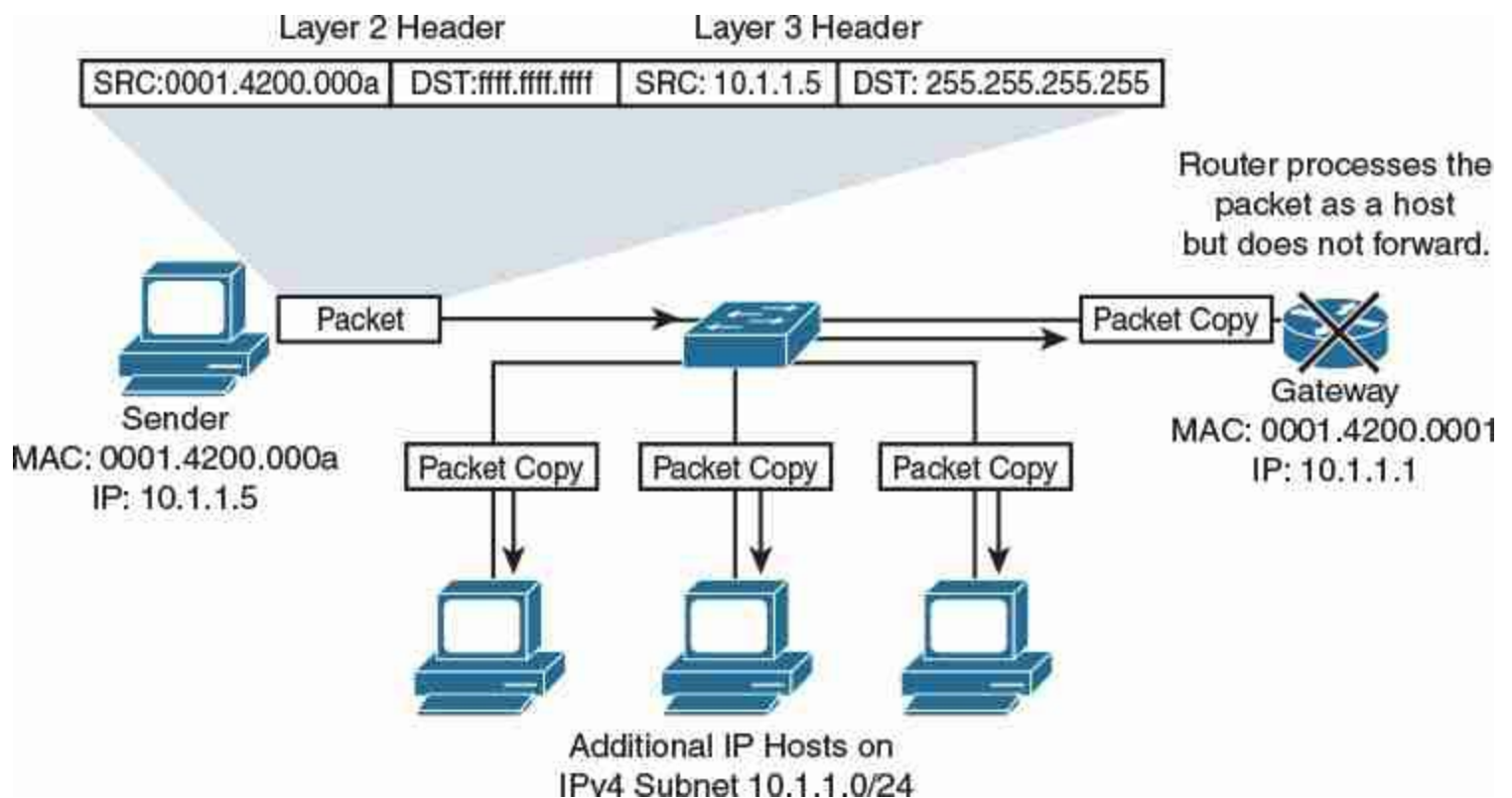


Figure 1-10 All-Hosts Broadcast Packet (Indicated by 255.255.255.255 as the Destination IPv4 Address and ffff.ffff.ffff as the Destination MAC Address)

If a broadcast is directed, it is sent through the network toward the intended subnet and replicated by any switch on that subnet, as shown in [Figure 1-11](#).

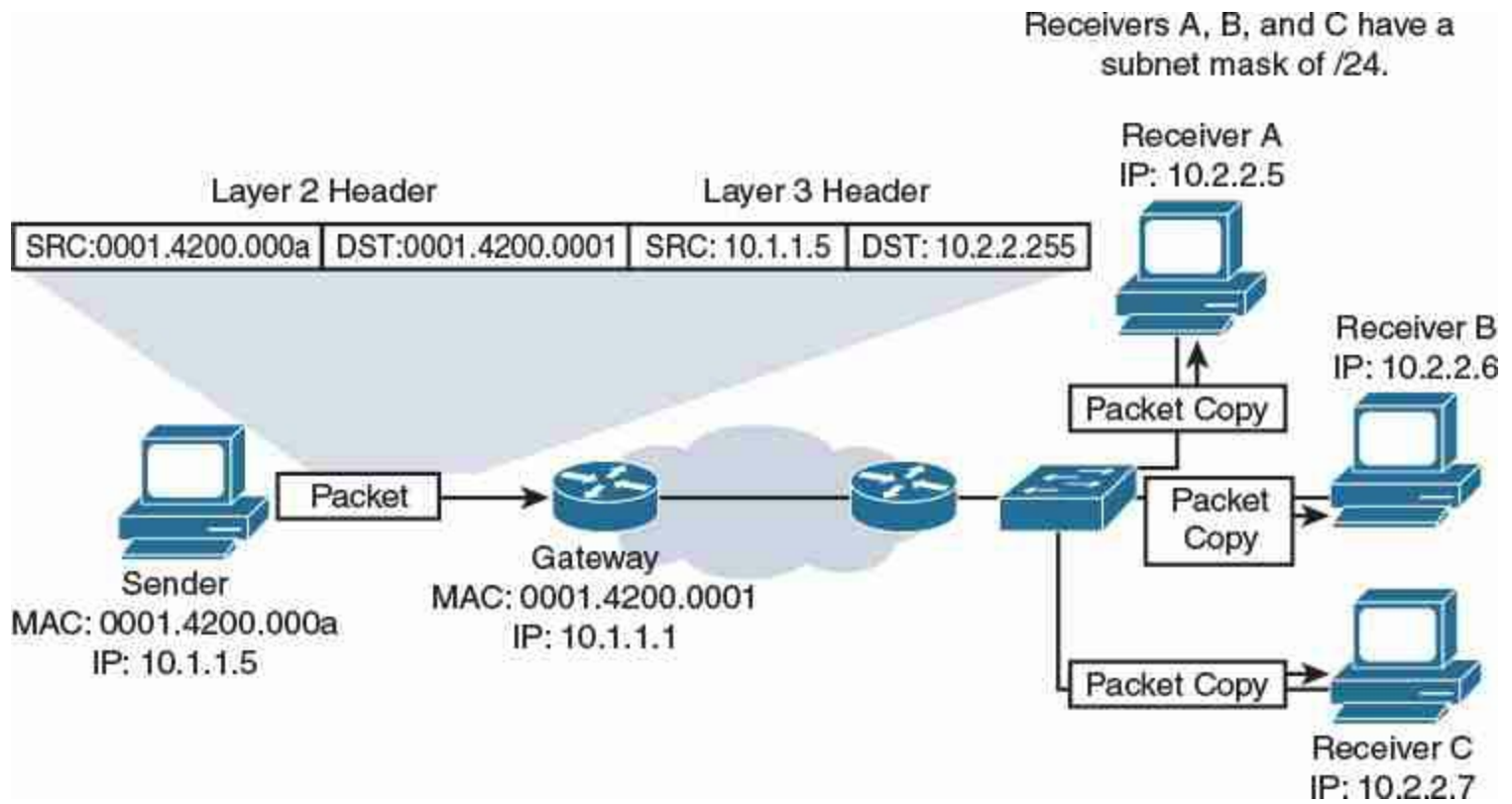


Figure 1-11 *Directed Broadcast Packet (Indicated by the 10.2.2.255 Destination IPv4 Address, or All Hosts on the 10.2.2.0 Subnet)*

The difference between a multicast and a broadcast with hosts on a single subnet is subtle. What if only a few of the hosts on the subnet need to receive the packets? Using a group address to which hosts can subscribe would ease the burden of sending packets to the select hosts on that segment, reduce replication overhead, and use the bandwidth only in the LAN where the hosts are located. [Figure 1-12](#) illustrates just such a scenario.

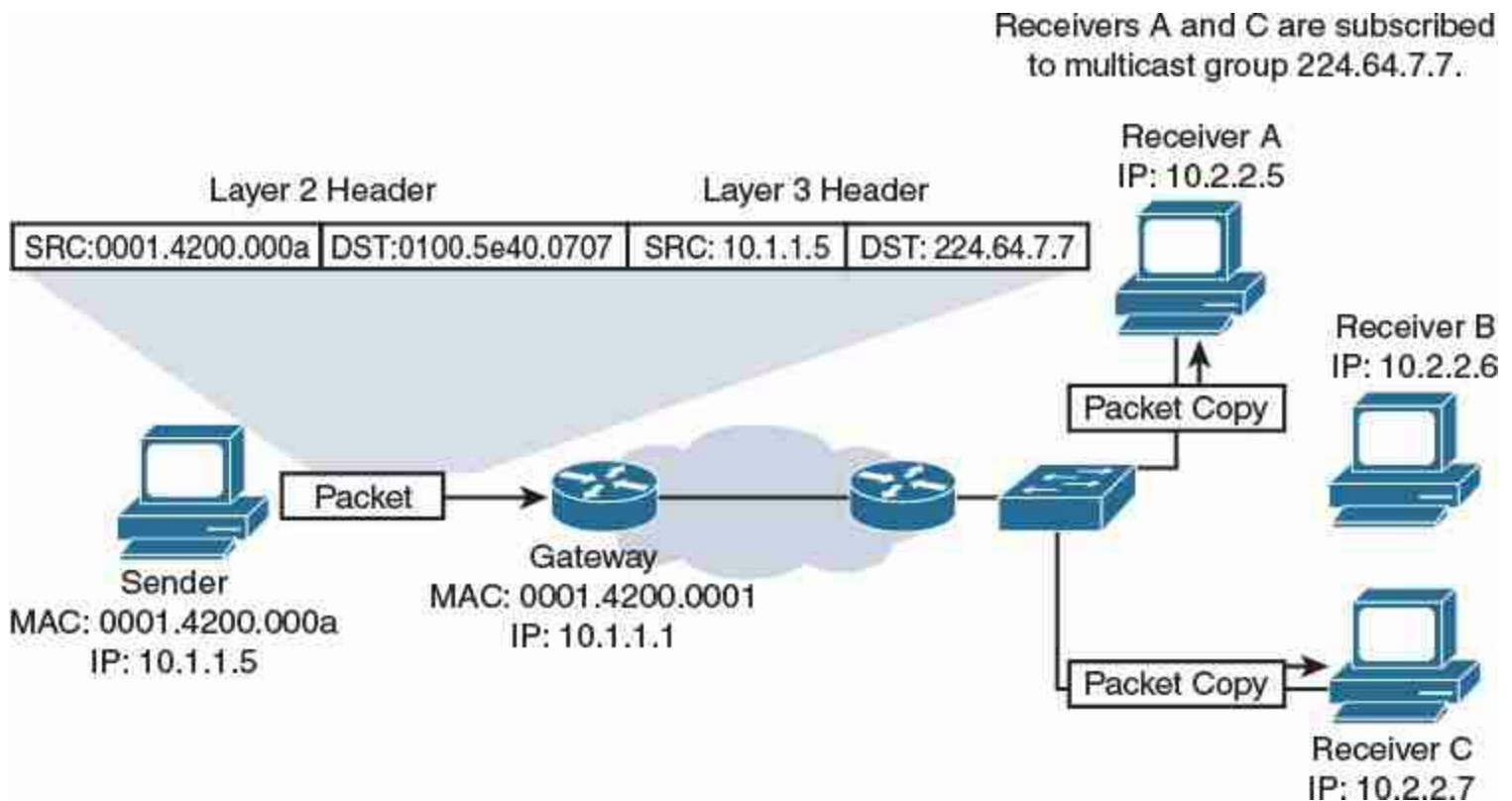


Figure 1-12 *Multicast Forwarding to Hosts on a Single Segment*

The major difference between a broadcast and multicast is that subscribed hosts of a multicast flow

may exist on multiple segments. How can routers and switches then forward to only those hosts without replicating the packet for every segment in a network or on the Internet? Multicast senders transmit IP packets using a specific destination IP address known as a *group address* using a specific multicast MAC address. You may have noticed the L2 destination address is not an address on the local subnet. This will be discussed further in [Chapter 2](#) , “[Network Access and Layer 2 Multicast](#) .”

IPv4 Layer 3 Multicast Addressing Defines Groups

The group address simply identifies a group of nodes interested in a flow. The group address combined with the source IP address identifies the multicast flow. Host receivers express interest in the flow by notifying upstream network devices of group membership. This expression of interest is known as *joining the group* .

IPv4 and IPv6 group addressing is controlled by a combination of IETF RFCs. The most important and current IPv4 RFC is 5771, which finalizes assignment of the multicast group address space and the individual group types within that space. The multicast group block address space is derived from IPv4 classful address space. [Table 1-2](#) details classful routing addressing assignment.

Class	Leading Bits	Start	End	Default Mask	CIDR Notation
A	0	0.0.0.0	127.255.255.255	255.0.0.0	/8
B	10	128.0.0.0	191.255.255.255	255.255.0.0	/16
C	110	192.0.0.0	223.255.255.255	255.255.255.0	/24
D	1110	224.0.0.0	239.255.255.255	Undefined	N/A
E	1111	240.0.0.0	255.255.255.255	Undefined	N/A

Table 1-2 *IPv4 Classful Addressing*

Class D addressing is reserved by RFC 5771 for multicast group assignments. RFC 5771 replaced or updated several RFCs, including RFC 988, which in 1986 originally proposed 1110 as the high-order bit reservation for IPv4 multicast group identification. RFC 988 supersedes RFC 966, the original IP multicast theory RFC. Both RFC 988 and 966 are important to the history of multicast, and a thorough understanding of the theory in each is a great starting point for anyone wishing to dive more deeply into the roots of multicast technology and terminology.

IPv4 Multicast Group Address Assignments

RFC 5771 further expands the Class D address for suggested IANA multicast group address assignment and deployment. [Table 1-3](#) illustrates this assignment.

Address Range	Size	Designation
224.0.0.0–224.0.0.255	/24	Local Network Control
224.0.1.0–224.0.1.255	/24	Internetwork Control
224.0.2.0–224.0.255.255	(65024)	AD-HOC I
224.1.0.0–224.1.255.255	/16	RESERVED
224.2.0.0–224.2.255.255	/16	SDP/SAP
224.3.0.0–224.4.255.255	2 total /16's	AD-HOC II
224.5.0.0–224.255.255.255	251 total /16's	RESERVED
225.0.0.0–231.255.255.255	7 /8's	RESERVED
232.0.0.0–232.255.255.255	/8	Source-Specific Multicast
233.0.0.0–233.251.255.255	(16515072)	GLOP
233.252.0.0–233.255.255.255	/14	AD-HOC III
234.0.0.0–238.255.255.255	5 total /8's	RESERVED
239.0.0.0–239.255.255.255	/8	Administratively Scoped

Table 1-3 *IPv4 Multicast Address Assignment*

The following is a brief explanation of each block type, and the proper usage:

- **Local Network Control block (224.0.0/24):** The local control block is used for specific protocol control traffic. Router interfaces listen to but do not forward local control multicasts; for example, OSPF “all routers” (224.0.0.5). Assignments in this block are publicly controlled by IANA. You can find a complete list of Local Network Control address assignments at the IANA website (www.iana.org).

Note

Routers listen for local control packets only if the control group feature is enabled on the node. For example, a router interface would only process RIPv2 control group packets (group 224.0.0.9) if RIPv2 is enabled on that interface.

- **Internetwork Control block (224.0.1/24):** The Internetwork Control block is for protocol control traffic that router interfaces *may* forward through the autonomous system number (ASN) or through the Internet. Examples include 224.0.1.1 Network Time Protocol (NTP), defined in RFC 4330, and 224.0.1.68 mdhcpdiscover, defined in RFC 2730. Internetwork Control group assignments are also publicly controlled by IANA.

- **AD-HOC blocks (I: 224.0.2.0–224.0.255.255, II: 224.3.0.0–224.4.255.255, and III: 233.252.0.0–233.255.255.255):** Traditionally assigned to applications that do not fit in either the Local or Internetwork Control blocks. Router interfaces *may* forward AD-HOC packets globally. Most applications using AD-HOC blocks require few group addresses (such as, for example, less than a /24 space). IANA controls any public AD-HOC Block assignments and future assignments will come from AD-HOC block III, if they are not more suited to Local Control or Internetwork Control. Public use of unassigned AD-HOC space is also permitted.

■ **SDP/SAP block (224.2.0.0/16):** The Session Description Protocol/Session Announcement Protocol (SDP/SAP) block is assigned to applications that receive addresses through the SAP as described in RFC 2974.

■ **Source-Specific Multicast block (232.0.0.0/8):** SSM addressing is defined by RFC 4607. SSM is a group model of IP Multicast in which multicast traffic is forwarded to receivers from only those multicast sources for which the receivers have explicitly expressed interest. SSM is mostly used in one-to-many applications. No official assignment from IANA is required to use the SSM block because the application is local to the host; however, according to IANA policy, the block is explicitly reserved for SSM applications and must not be used for any other purpose. SSM will be discussed further in subsequent sections of this and other chapters.

Note

The 232.0.0.0/8 block was originally assigned by IANA for use by the Versatile Message Transaction Protocol (VMTP).

■ **GLOP block (233.0.0.0/8):** These addresses are statically assigned with a global scope. Each GLOP static assignment corresponds to a domain with a public 16-bit autonomous system number (ASN), which is issued by IANA. The ASN is inserted in dotted-decimal into the middle two octets of the multicast group address (X.Y). An example GLOP assignment for an ASN of X.Y would be 233.X.Y.0/24. Domains using an assigned 32-bit ASN should apply for group assignments from the AD-HOC III Block. Another alternative is to use IPv6 multicast group addressing. Because the ASN is public, IANA does not need to assign the actual GLOP groups. The GLOP Block is intended for use by public content, network, and Internet service providers. IANA considers GLOP addressing to be experimental, and 233.252–255.0.0 is reserved.

■ **Administratively Scoped block (239.0.0.0/8):** Administratively Scoped addresses are intended for local use within a private domain as described by RFC 2365. These group addresses serve a similar function as RFC 1918 private IP address block (such as, for example, 10.0.0.0/8 or 172.16-31.0.0/16 blocks). Network architects can create an address schema using this block that best suits the needs of the private domain and can further split scoping into specific geographies, applications, or networks. Further information on best practices for scoping this block can be found in [Chapter 5](#), “[IP Multicast Design Considerations and Implementation](#) .”

Important Multicast Groups and Group Considerations

There are many multicast groups, each subdivided from a larger block of multicast group ranges. Each of the group block ranges has a specific application or scope. The scope of each block ranges from single segments to large enterprise multicast networks to the global Internet. It is important to understand the RFC and standards for multicast groups when designing multicast networks. Multicast group addresses play a very important part in “scoping” the multicast domains. [Chapter 5](#) explains this concept in more detail.

Note

IANA manages globally scoped address assignments as well as any protocol assignments for applications. Without control over these addresses, there would be little possibility of using them for protocol interchange or standards-driven communication.

IPv4 Local Network Control

The *Local Network Control block*, also known as the *Link-Local block*, consists of IANA assigned

addresses between 224.0.0.0 and 224.0.0.255. These multicast groups are intended for use within a single subnet or segment. They are not to be forwarded by routers, and therefore have a time-to-live (TTL) value of 1. Many well-known applications and communications protocols have reserved address assignments.

Application developers and network administrators should not use group addresses in this block for any purpose other than the IANA assigned application. [Table 1-4](#) lists some of the most relevant assignments from the link-local multicast address, taken directly from the IANA database. The table lists the reserved multicast addresses, the protocol function assigned, and relevant RFCs.

Address(es)	Description	References
224.0.0.0	Base address (reserved)	[RFC1112]
224.0.0.1	All systems on this subnet	[RFC1112]
224.0.0.2	All routers on this subnet	
224.0.0.4	DVMRP routers	[RFC1075]
224.0.0.5	OSPF IGP—All routers	[RFC2328]
224.0.0.6	OSPF IGP—Designated routers	[RFC2328]
224.0.0.9	RIPv2 routers	[RFC1723]
224.0.0.10	(E)IGRP routers	
224.0.0.11	Mobile-agents	
224.0.0.12	DHCP server/Relay agent	
224.0.0.13	All PIM routers	
224.0.0.14	RSVP-ENCAPSULATION	
224.0.0.18	VRRP	[RFC3768][RFC5798]
224.0.0.19–21	ISIS IP	
224.0.0.22	IGMP	
224.0.0.24	OSPF IGP-TE	[RFC4973]
224.0.0.25	router-to-switch	
224.0.0.27	AI MPP Hello	
224.0.0.101	cisco-nhap	
224.0.0.102	HSRP	
224.0.0.103	MDAP	
224.0.0.112	MMA device discovery	
224.0.0.251	mDNS	[RFC6762]
224.0.0.252	Link-local multicast name resolution	[RFC4795]

Table 1-4 IPv4 Link-Local Multicast Addresses

As the table illustrates, many important network functions rely on link-local multicasting. Routing protocols, including EIGRP, RIPv2, and OSPF use these protocols to send updates to neighbor routers. IGMP also uses a link-local multicast address to notify gateway routers of group subscription. The important point to remember is that the Layer 3 devices will not replicate or forward these packets. Layer 2 devices will forward link-local multicast frames only on ports within the same Layer 2 domain (VLAN or subnet).

IPv4 Inter-Network Control

The Inter-Network Control block is similar to the Local Network Control block, with the exception that multicast applications using this block may require that multicast packets be forwarded beyond the local segment. The block ranges from 224.0.1.0 to 224.0.1.255. [Table 1-5](#) provides a partial list of some of the more important Internetwork Control block assignments as made by IANA.

Address(es)	Description	References
224.0.1.0	VMTP managers group	[RFC1045]
224.0.1.1	NTP Network Time Protocol	[RFC1119][RFC5905]
224.0.1.9	MTP Multicast Transport Protocol	
224.0.1.10	IETF-1-LOW-AUDIO	
224.0.1.11	IETF-1-AUDIO	
224.0.1.12	IETF-1-VIDEO	
224.0.1.13	IETF-2-LOW-AUDIO	
224.0.1.14	IETF-2-AUDIO	
224.0.1.15	IETF-2-VIDEO	
224.0.1.16	MUSIC-SERVICE	
224.0.1.21	DVMRP on MOSPF	
224.0.1.24	microsoft-ds	
224.0.1.33	RSVP-encap-1	
224.0.1.34	RSVP-encap-2	
224.0.1.39	cisco-rp-announce	
224.0.1.40	cisco-rp-discovery	

224.0.1.41	gatekeeper	
224.0.1.49	dcap-servers	[RFC2114]
224.0.1.50	dcap-clients	[RFC2114]
224.0.1.51	mcntp-directory	
224.0.1.52	mbone-vcr-directory	
224.0.1.53	heartbeat	
224.0.1.60	hp-device-disc	
224.0.1.114	Mac Srv Loc	
224.0.1.115	Simple multicast	
224.0.1.116	SpectraLinkGW	
224.0.1.141	DHCP-SERVERS	
224.0.1.173	host-request	
224.0.1.174	host-announce	
224.0.1.185	ASAP	[RFC5352]
224.0.1.188– 224.0.1.255	Unassigned	

Table 1-5 *Inter-Network Multicast Address Addresses*

Many infrastructure protocols use assigned Inter-Network Control block group addresses for protocol communication. A good example of Inter-Network Control multicast is the Cisco Auto-RP protocol. Cisco Auto-RP uses multicast groups 224.0.1.39 and 224.0.1.40 to dynamically assign, advertise, and discover rendezvous points (RPs) in a PIM sparse mode network domain; 224.0.1.39 is used for Cisco Announce; and 224.0.1.40 is used for Cisco Discovery.

Using IP multicast for infrastructure communication simplifies the protocol design process. After a multicast address is assigned to a particular application or protocol, developers need only send packets to the assigned address to facilitate protocol intercommunication between many devices. Several of the existing assignments from the Inter-Network Control block may be for applications or protocols that will never be deployed on a large scale. Approximately one-third of the block space remains for future development.

The History of Multicast

Necessity is the mother of invention. Steve Deering was a student of Stanford University in the early 1980s, working on a distributed processing project. One of the underlying communication mechanisms allowed one device to send a message to many other devices as a single update. As the project grew, so did the requirement for compute resources. These resources were distributed throughout the campus, which required a mechanism to communicate to those devices over a router (L3) infrastructure. This could have been accomplished using multiple unicast messages, or by broadcasting the messages across the entire network. Neither of these methods were an acceptable solution in this case because they were too inefficient. The resolution required a unique single group

address, the ability for the routers to participate in sending the messages, and the capability to have the hosts join or leave the group at will—hence, the birth of multicast.

The MBone

The *multicast backbone (MBone) project* was started ten years after Dr. Deering's invention of multicast. The routers that comprised the Internet at that time did not have the capability to support native multicast; consequently, the MBone was a handful of Unix hosts connected over tunnels using Distance Vector Multicast Routing Protocol (DVMRP), running a daemon called *mrouted*. The MBone at that time was driven by higher-education institutions and was used to deliver content such as the Internet Engineering Task Force (IETF) meetings, concerts, and so on, all with a very limited scope of viewing.

Native Internet Multicast

Native Internet multicast would allow anyone with an Internet connection to view content. Could you imagine being able to watch any TV channel, listen to any radio station, participate in distance learning sessions, all via multicast? Unfortunately, the experiment of the MBone driven by academia in the 1990s has still not moved to a mainstream service offered by Internet Service Providers (ISPs), because many service providers do not support the transport of multicast traffic across their network. The adoption of multicast by ISPs has been slowed by security concerns, the complexity of implementation, and the capability to easily share multicast routing information.

This has not diminished the need for multicast within private networks. As we reviewed earlier, there are many applications that benefit from an infrastructure that transports multicast traffic. We can still take advantage of the transport services of the Internet by tunneling multicast traffic over it, even if it is not supported natively.

IPv6 Multicast

The rapid growth of the Internet is causing the depletion of IPv4 address space. Consequently, IPv6 is taking hold to provide for the expansion of the Internet and to permit our ability to access any device on the planet.

IPv4 addresses use 32 bits for a numerical value to distinguish individual devices, whereas IPv6 uses 128 bits. This increase offers tremendous scalability. The implementation of IPv6 has another interesting characteristic in that it no longer supports network broadcasts. The two methods of communication with IPv6 are either unicast or multicast. Because multicast was considered during the creation of the protocol, there are inherent capabilities that improve the operation. Besides the greater amount of address space, there are other features in IPv6 that make the multicast designs simpler. You will learn much more about the functionality of IPv6 and multicast in [Chapter 6](#), “[IPv6 Multicast Networks](#).”

Multicast Development and Standardization

Just as with many other networking technologies, the effort being made to improve multicast has continued. There have been many enhancements to address the shortfalls and feature enhancements that were not foreseen during the creation of the protocol.

Could you imagine if every developer decided to write code based on how they thought it should work? Fortunately, standardization groups collaborate on how to solve technical challenges and create documentation on how those solutions are to be addressed to achieve compatibility and interoperability. There are two major standards bodies that help to drive common implementation

methodologies, the Internet Engineering Task Force (IETF) and the Institute of Electrical and Electronic Engineers (IEEE).

Note

The charter of the Internet Engineering Task Force (IETF) is to make the Internet work better by producing high-quality and relevant technical documents that influence the way people design, use, and manage the Internet. The IETF generates documents such as requests for comment (RFC) and best current practices (BCP).

The Institute of Electrical and Electronics Engineers (IEEE) is the largest technical professional society and is an association dedicated to advancing innovation and technological excellence for the benefit of humanity. Besides developing standards for Ethernet, the IETF develops standards for many other areas.

Summary

The three data communication methods in IP networks are unicast, broadcast, and multicast. Each of these methods has advantages and disadvantages depending on the application. Multicast offers an efficient communication mechanism for sending messages to multiple recipients in separate locations. It is also capable of supporting many-to-many and many-to-one communication.

Multicast applications commonly use User Datagram Protocol (UDP) on IP. Messages are sent by a source (called the sender) and will send messages (termed as a stream) even if there is not another device on the network interested in receiving that information. Receivers, on the other hand, must subscribe to a particular multicast stream in order to inform the network to forward those messages.

IP addressing for multicast is also unique. There are many public and private addresses that have been allocated for different applications and services. It is imperative to understand what addresses you plan to use prior to building out a multicast network.

Multicast was developed in the early 1980s from a research project at Stanford University. Improvements continue as new applications and services are developed and to address security concerns.

Chapter 2. Network Access and Layer 2 Multicast

Chapter 2. Network Access and Layer 2 Multicast

[Chapter 1](#) , “[Introduction to IP Multicast](#) ,” examined the differences between unicast, broadcast, and multicast messages. This chapter takes an in-depth look at IP multicast messages at Layer 2 and how they are transported in a Layer 2 domain. This chapter covers the basic elements of multicast functionality in Layer 2 domains as well as design considerations for multicast deployments.

Layered Encapsulation

Before reviewing multicast in Layer 2, we must discuss fundamental packet-forwarding concepts to establish a baseline of the process. Encapsulation is an important component of the OSI model for data communication and is absolutely essential in IP networks. *Encapsulation* is the method by which information is added at each layer of the OSI reference model, used for processing and forwarding purposes. Think of it like an onion, with many layers. This information is added in the form of headers and footers. At each layer of the OSI reference model, the data is processed, encapsulated, and sent to the next layer. Take a look at [Figure 2-1](#) to understand how this works.

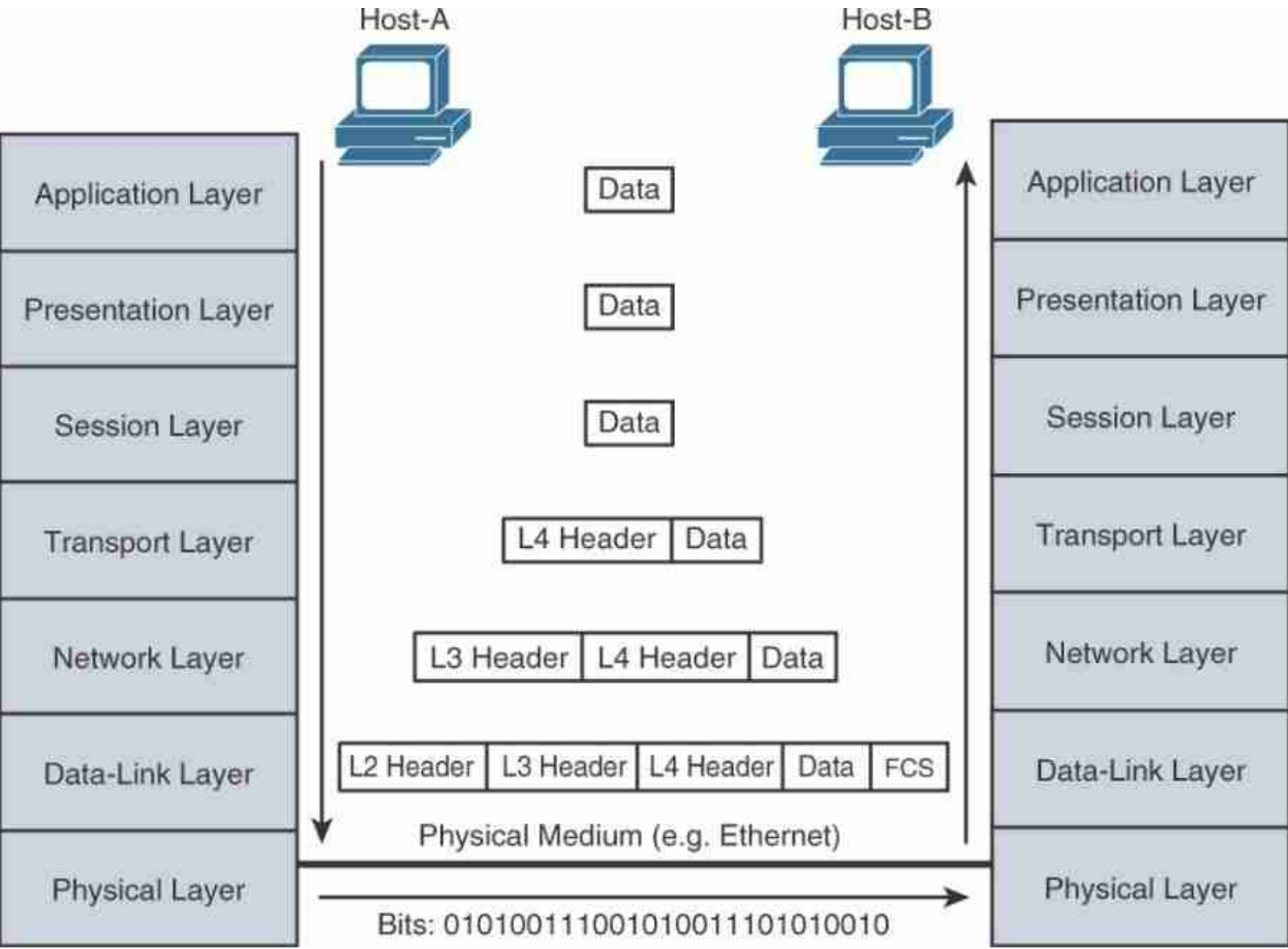


Figure 2-1 OSI Model and Data Encapsulation

Data from the application, presentation, and session layers (Layers 1, 2, and 3) is encapsulated at

Layer 4 with transport protocol information. This information is encapsulated in TCP and/or UDP with specific port numbers. For example, TCP port 80 is typically web traffic. This allows an operating system to forward data to an appropriate application or subroutine. Layer 3 adds logical forwarding details (source and destination IP addresses) so that networking devices can determine the best path toward a destination. Finally, Layer 2 adds hardware forwarding information (source and destination MAC addresses). This allows data to be passed physically to the appropriate machine or the next hop in a network. A data transmission at Layer 4 is called a *segment*, a packet at Layer 3, and a frame at Layer 2.

At each step through a network, routers and switches use these encapsulation headers and footers to make decisions about how to treat data transmissions. The diagrams in [Chapter 1](#) show much of this processing in action. In that chapter, you learned the uniqueness of IP multicast packets compared to unicast and broadcast packets. From a Layer 2 perspective, this is only part of the story.

To better explain what happens to a packet traversing the network, we will walk you through the Layer 2 and Layer 3 transmission process. [Figure 2-2](#) illustrates the first step in this process. Before the sender has the ability to encapsulate any data, it must first understand where the destination is located. The sender performs a simple check to verify if the receiving device is on the same subnet. This is accomplished by checking the destination IP address against the local subnet. In this example, the receiver is not on the same subnet. The sender must use the configured route to the destination. In most cases, this is the default-gateway.

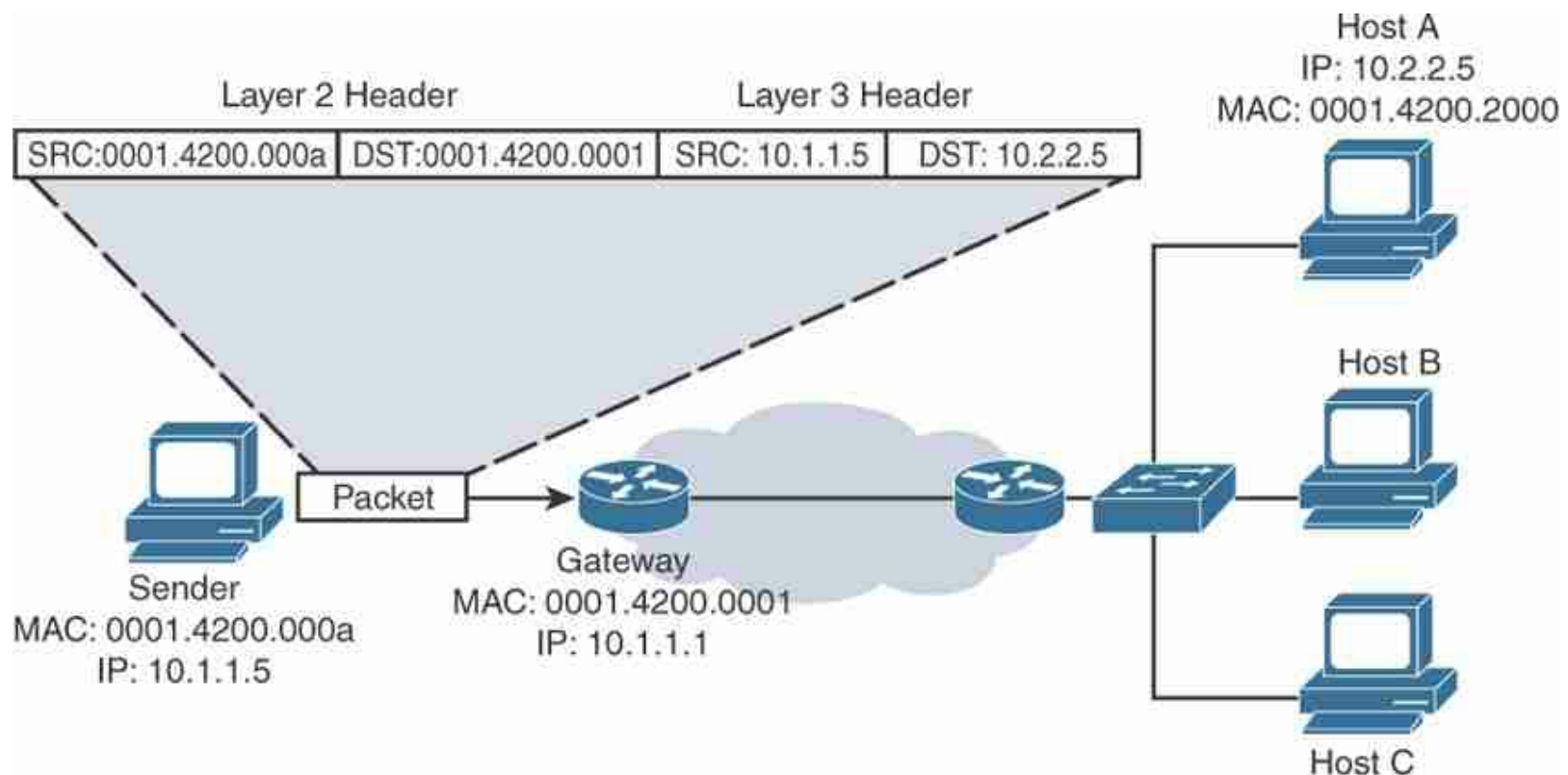


Figure 2-2 *Layer 2 and Layer 3 Transport Process on the Local Segment*

Before the sender can communicate with the default gateway, it must know the media access control (MAC) address of that device. Because the destination is on a different segment, the sender will need to discover the MAC address of the default gateway (IP address 10.1.1.1) using an Address Resolution Protocol (ARP) request. The default gateway responds to the ARP request with its MAC address. Finally, the sender has enough information to encapsulate the data with the destination IP address of Host A and the MAC addresses of the default gateway, as shown in [Figure 2-2](#).

The default gateway or router has Layer 3 IP routing information that determines where Host A is physically connected. This information determines the appropriate outgoing interface to which the message should be sent. The router should already know the MAC address of the neighbor router if there is an established routing protocol adjacency. If not, the same ARP request process is conducted. With this information, the router can now forward the message. Understand that both Layer 2 addresses (SA and DA) change at each logical hop in the network, but the Layer 3 addresses never change and are used to perform route lookups.

When the packet is forwarded to the final router, that router must do a lookup and determine the MAC address of the destination IP. This problem exists in part because of the historical implications of Ethernet. Ethernet is a physical medium that is attached to a logical bus network. In a traditional bus network, many devices can be connected to a single wire. If the gateway router does not have an entry from a previous communication, it will send out an ARP request and finally encapsulate with the destination MAC address of the host, as shown in [Figure 2-3](#).

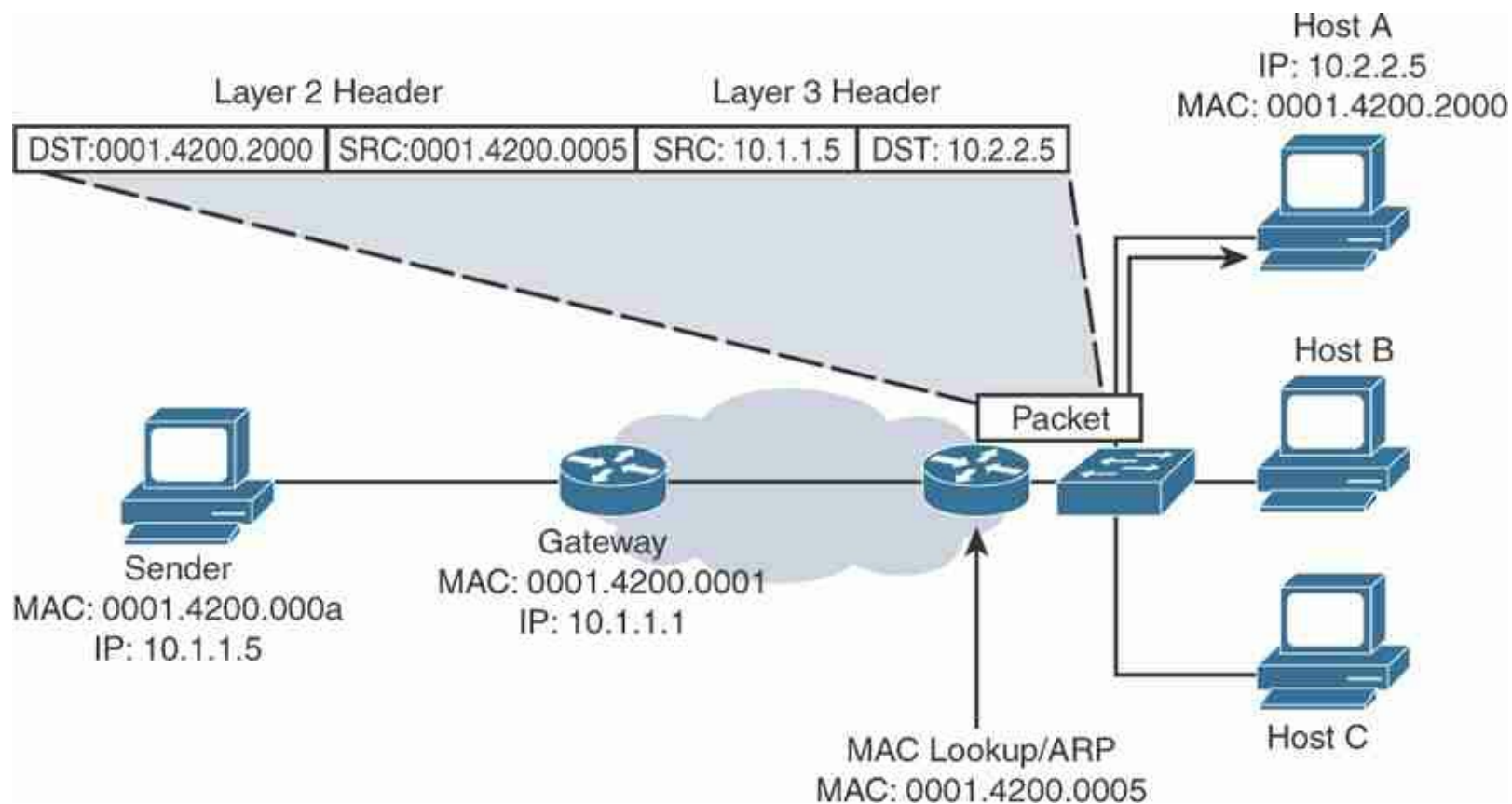


Figure 2-3 *Layer 2 and Layer 3 Transport Process on the Destination Router*

After the final router properly encapsulates the message, it is the responsibility of the switch to send the packet to the appropriate host, and only to that host. This is one of the primary functions of a traditional Layer 2 switch—to discover the location of devices connected to it. It does this by cataloging the source MAC addresses in frames received from connected devices. In this way, the switch builds a table of all known MAC addresses and keeps Ethernet networks efficient by making intelligent Layer 2 forwarding decisions.

This process is easy to understand for the unicast packet shown. Items to consider while you read this chapter include the following:

- What happens if the packet is a multicast packet and many hosts connected to a switch are subscribed to the destination multicast group?

- Can a switch still make efficient forwarding decisions if there are multiple ports that require a copy of the packet (meaning there are multiple endpoints on multiple segments that need a copy of the frame)?
- If the MAC address in a frame is not the physical address of the host, will it process the packet, assuming it is not the intended recipient?
- How do you identify multicast groups at Layer 2?

MAC Address Mapping

A traditional Ethernet switch (Layer 2 device) works with Ethernet frames, and a traditional router (Layer 3 device) looks at packets to make decisions on how messages will be handled. As discussed in [Chapter 1](#), when a device sends a broadcast frame, the destination address is all ones, and a unicast message is the destination MAC address. What happens when it is a multicast message? To optimize network resources, an Ethernet switch also needs to understand multicast. This is where the magic happens. The sending device must convert the destination IP multicast address into a special MAC address as follows:

- The high-order 25 bits is the official reserved multicast MAC address range from 0100.5E00.0000 to 0100.5E7F.FFFF (request for Comment 1112). These bits are part of the organizational unit identifiers (OUI).
- The lower-order 23 bits of the destination IP multicast address are mapped to the lower-order 23 bits of the MAC address.
- The high-order 4 bits for the destination IP multicast address are set to 1110 binary (0b1110). This represents the Class D address range from 224.0.0.0 (0b11100000) to 239.255.255.255 (0b11101111).
- Of the 48 bits used to represent the multicast MAC address, the high-order 25 bits are reserved as part of the OUI, and the last 23 bits of the multicast IP address are used as the low-order bits, as shown in [Figure 2-4](#).

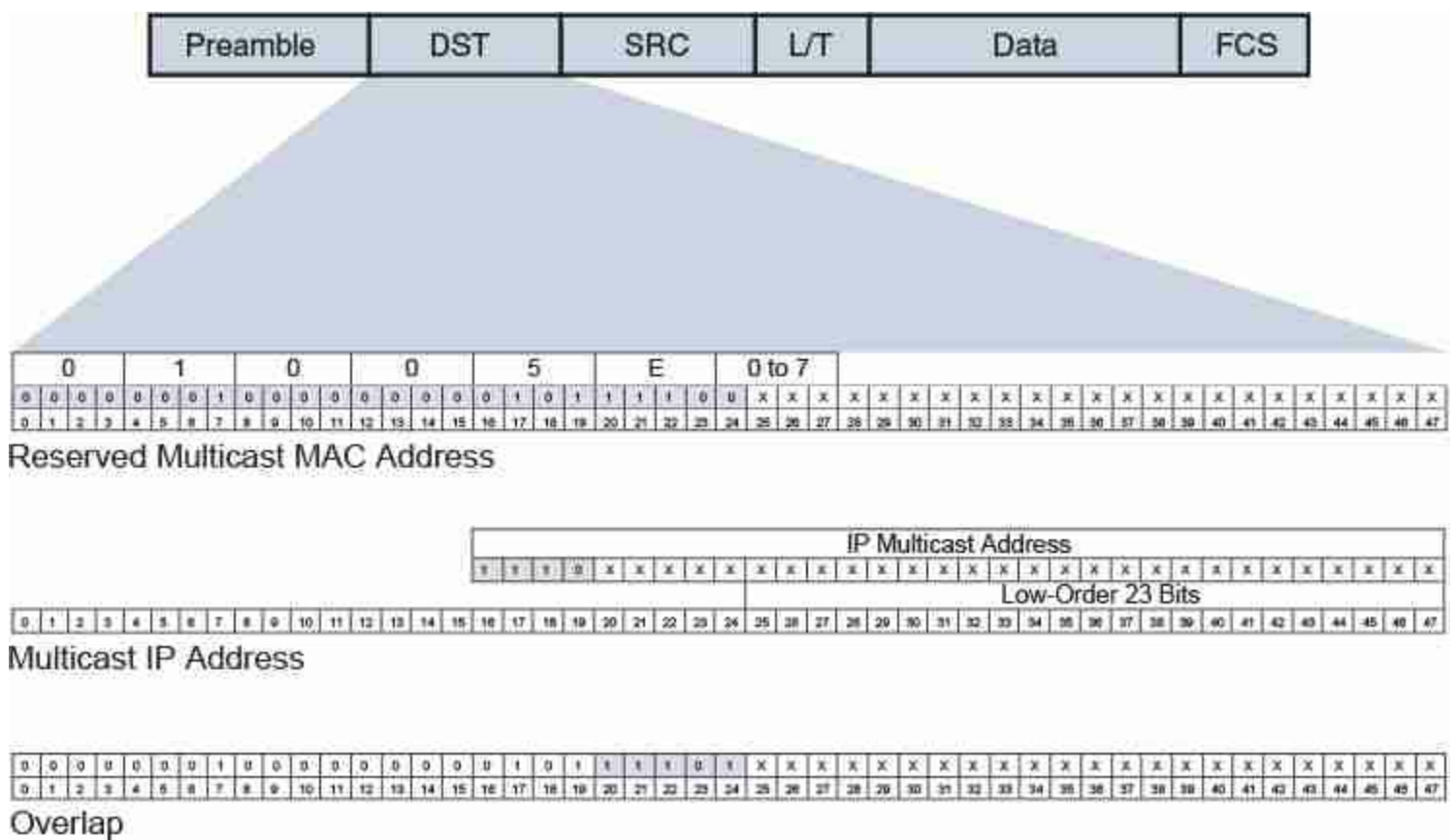


Figure 2-4 *Layer 2 Multicast Address Format*

A switch can use this calculated multicast MAC address to distinguish a frame as a multicast and make efficient forwarding decisions. End hosts can listen for frames with a specific multicast MAC, allowing them to process only those multicast streams to which they have subscribed. There's a small wrinkle in this process, however.

Did you notice a slight challenge with the number of IP addresses and MAC addresses? Five bits of the IP address are overwritten by the OUI MAC address. This causes a 32-to-1 IP multicast address-to-multicast MAC address ambiguity ($2^5 = 32$).

This means that a host subscribing to a multicast stream could potentially receive multiple multicast streams that it did not subscribe to, and the host will have to discard the unwanted information. A host subscribing to the multicast stream of 224.64.7.7 would map to MAC address of 0x0100.5E40.0707, and so would 225.64.7.7 and 224.192.7.7. It all boils down to 1s and 0s. [Figure 2-5](#) shows the ambiguity. The "X" in the binary row represents the bits that are overwritten and shows how 32 multicast IP addresses map to a single multicast MAC address.

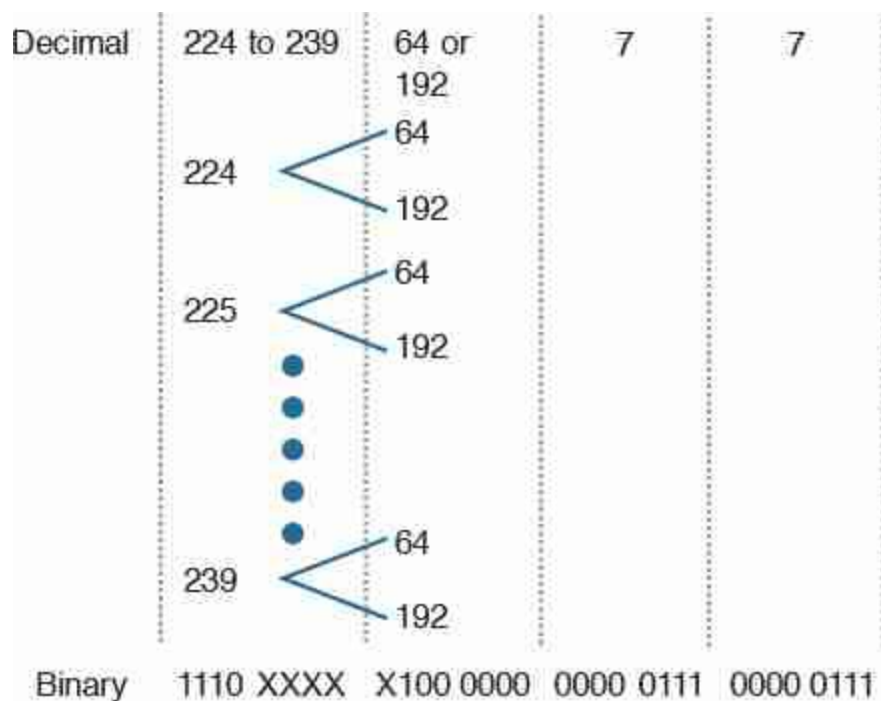


Figure 2-5 *Layer 2 Multicast MAC Address Overlap*

Switching Multicast Frames

Layer 2 switches send frames to a physical or logical interface based on the destination MAC address. Multicast MAC addresses are a different animal than unicast MAC addresses, because a unicast MAC address should be unique and have only a single destination interface. Multicast MAC frames may have several destination interfaces, depending upon which devices have requested content from the associated IP multicast stream.

Before the Layer 2 switch can forward multicast frames, it must know the destination interfaces on which those messages should be sent. The list of destination interfaces includes only those interfaces connected to a device subscribed to the specific multicast flow. The destination can be added as static entries that bind a port to a multicast group, or the switch can use a dynamic way of learning and updating the ports that need to receive the flow.

There are several ways in which a Layer 2 switch can dynamically learn where the destinations are located. The switch may use Cisco Group Management Protocol (CGMP) or Internet Group Management Protocol (IGMP) snooping for IPv4 multicast. These methods will be discussed later in this chapter.

If a Layer 2 switch does not have a mechanism to learn about where to send multicast messages, it treats all multicast frames as broadcast, which is to say it floods the packet on every port or VLAN port! As you can imagine, this is a very bad thing. Many networks have melted down due to large multicast streams. For example, when sending computer operating system image files, a tremendous amount of data is sent to every device in the broadcast domain, every computer, router, printer, and so on. The unfortunate side effect of these messages is that network performance may be affected in locations on the network that do not need the multicast stream. How could this happen if these are broadcast messages and will not go beyond the local network? These messages will not go beyond any local Layer 3 devices, but local Layer 3 devices must process each one of the broadcast messages. While the Layer 3 device is inundated processing these messages, it may not have the available cycles to process other more important messages, such as routing updates or spanning-tree messages. As you can imagine, or may have already experienced, this can impact or “melt-down” the

entire network.

Group Subscription

You have seen that in order for IP multicast forwarding to work on the local segment and beyond, switches and gateway routers need to be aware of multicast hosts interested in a specific group and where those hosts are located. Without this information, the only forwarding option is to flood multicast datagrams throughout the entire network domain. This would destroy the efficiency gains of using IP multicast.

Host group membership is a dynamic process. When a host joins a multicast group, there is no requirement to continue forwarding group packets to the segment indefinitely, nor is group membership indefinite. The only way to manage alerting the network to a multicast host location is to have multicast host group members advertise interest or membership to the network. [Figure 2-6](#) depicts a high-level example of this requirement, known as a *join*.

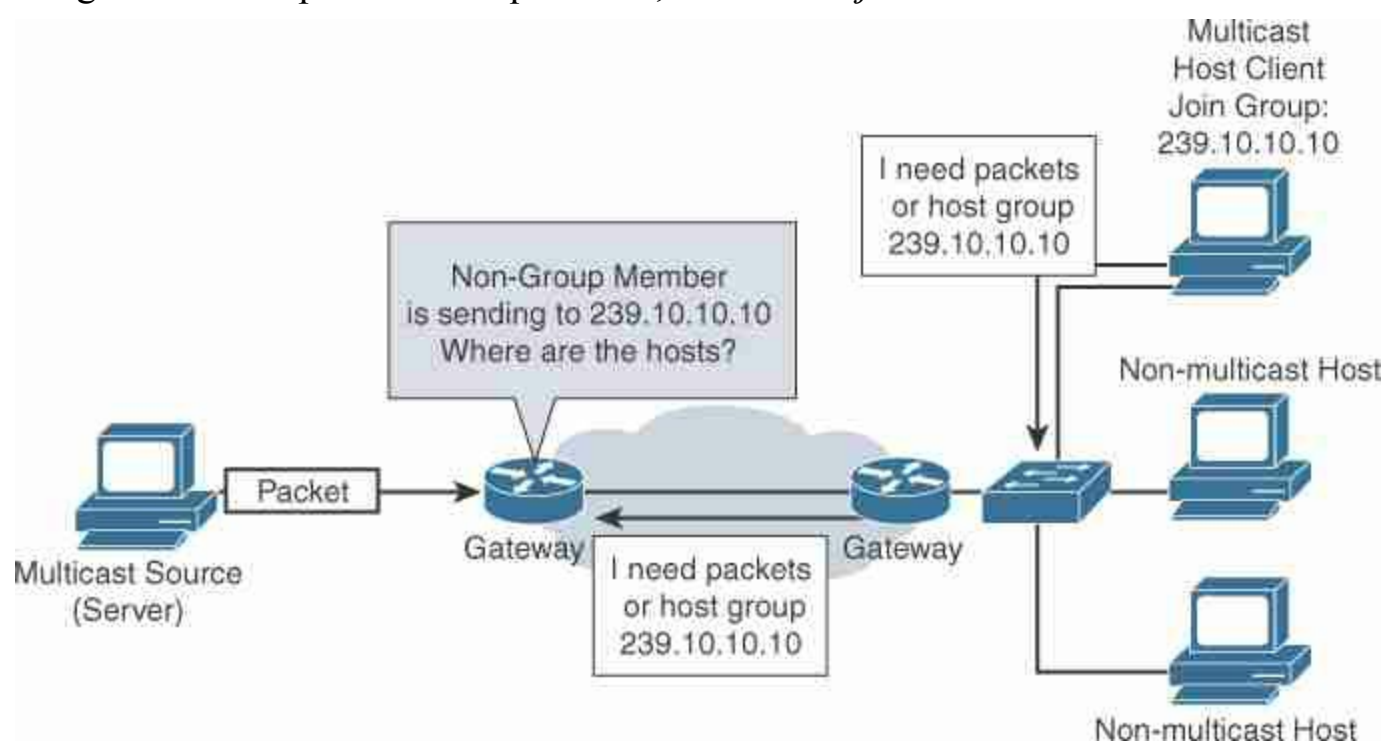


Figure 2-6 *Host Joins a Multicast Group*

A Layer 3 gateway provides access to the larger network for hosts on a given subnet. The gateway is the network demarcation between Layers 2 and 3 and is the most appropriate device to manage host group membership for the larger network. Hosts forward group management information, like joins, to the network. The gateway receives these management messages and adds host segment interfaces to the local multicast table (multicast forwarding information base [FIB]). After the FIB is updated, the gateway router communicates group interest using protocol independent multicast (PIM) to the larger network domain.

It is important to note that without multicast-aware Layer 2 protocols, all hosts on a given Layer 2 segment will receive multicast packets for any groups joined by a host on that segment. For this reason, it is also logical that hosts and routers have the capability to dynamically leave a group or to prune a group from a particular segment. [Figure 2-7](#) describes a high-level example of this process in action, known as a *leave*.

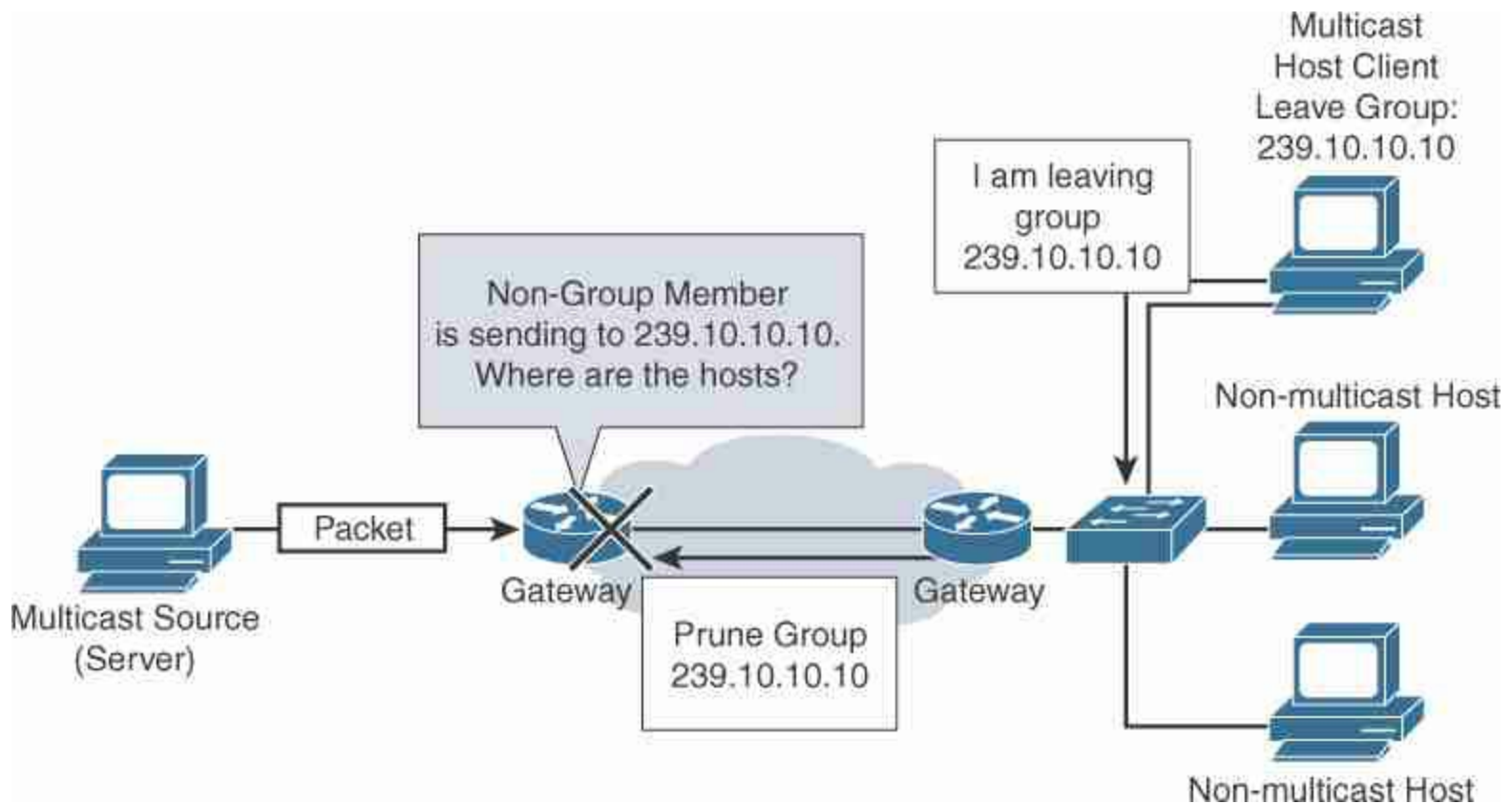


Figure 2-7 *Host Leaves a Multicast Group*

Administrators can configure the gateway router to statically process joins for specific groups using router interfaces. This alleviates the need to have a dynamic join/leave process; however, having a dynamic process simplifies the operational aspects for the administrator. In the next section, we show you the dynamic process needed to get this intelligence to the Layer 2 networks.

IGMP on the Gateway Router

Internet Group Management Protocol, or IGMP, is the protocol used to manage group subscription for IPv4 multicast. On the gateway router, called the *querier*, IGMP is used to track multicast group subscriptions on each segment. The router sends query messages to discover the hosts that are members of a group. The hosts send membership report messages to inform the router that they are interested in receiving or leaving a particular multicast stream, and they also send report messages in response to a router query message.

Note

When protocol independent multicast (PIM) is enabled on an interface of the router, IGMP (version 2) is also enabled.

IGMP Versions

The selection of which IGMP version(s) to run on your network is dependent on the operating systems and behavior of the multicast application(s) in use. Generally speaking, the capability of the operating system determines the IGMP version(s) you are running on your network. There are three versions of IGMP, version 1, 2, and 3. Each of these has unique characteristics. As of this writing, the default IGMP version enabled on most Cisco devices is version 2.

IGMPv1

The original specification for IGMP was documented in RFC 988 back in 1986. That RFC, along with RFC 1054, was made obsolete by RFC 1112, which is known as IGMPv1 today. IGMPv1 offers

a basic query-and-response mechanism to determine which multicast streams should be sent to a particular network segment.

IGMPv1 works largely like the explanation given in [Figure 2-7](#) , with two major exceptions, a primary issue with using version one. IGMPv1 has no mechanism for a host to signal that it wants to leave a group. When a host using IGMPv1 leaves a group, the router will continue to send the multicast stream until the group times out. As you can imagine, this can create a large amount of multicast traffic on a subnet if a host joins groups very quickly. This will occur if the host is “channel-surfing” using IPTV, for example.

In order to determine the membership of a group, the querier (router) sends a message to every host on the subnet. The functionality of the querier is to maintain a list of hosts in the subnet interested in multicast flows. Yes, even those that were never interested in receiving any multicast streams. This is accomplished by sending the query to the “all-hosts” multicast address of 224.0.0.1. When a single host responds to the query, all others suppress sending a report message.

IGMPv1 also does not have the capability of electing a querier. If there are multiple queriers (routers) on the subnet, a designated router (DR) is elected using PIM to avoid sending duplicate multicast packets. The elected querier is the router with the highest IP address. IGMPv1 is rarely used in modern networks and the default for Cisco devices has been set to v2 because of these limitations.

IGMPv2

As with every invention, we make improvements as we find shortcomings. IGMPv2, as defined in RFC 2236, made improvements over IGMPv1. One of the most significant changes was the addition of the leave process. A host using IGMPv2 can send a leave-group message to the querier indicating that it is no longer interested in receiving a particular multicast stream. This eliminates a significant amount of unneeded multicast traffic by not having to wait for the group to timeout; the trade-off is that routers need to track membership to efficiently prune when required.

IGMPv2 added the capability of group-queries. This feature allows the querier to send a message to the host(s) belonging to a specific multicast group. Every host on the subnet is no longer subjected to receiving a multicast message.

The querier election process offers the capability to determine the querier without having to use PIM. In addition, the querier and the DR function are decoupled. This process requires that each device send a general query message to all hosts 224.0.0.1. If there are multiple routers on a subnet, the DR is the device with the highest IP address and the querier is the device with the lowest IP address.

IGMPv2 also added the *Maximum Response Time* field, which is used to tune the query-response process to optimize leave latency.

Food for thought: Is a multicast message sent to all-host 224.0.0.1 a broadcast?

[Figure 2-8](#) shows the format for IGMPv1 and IGMPv2 messages.

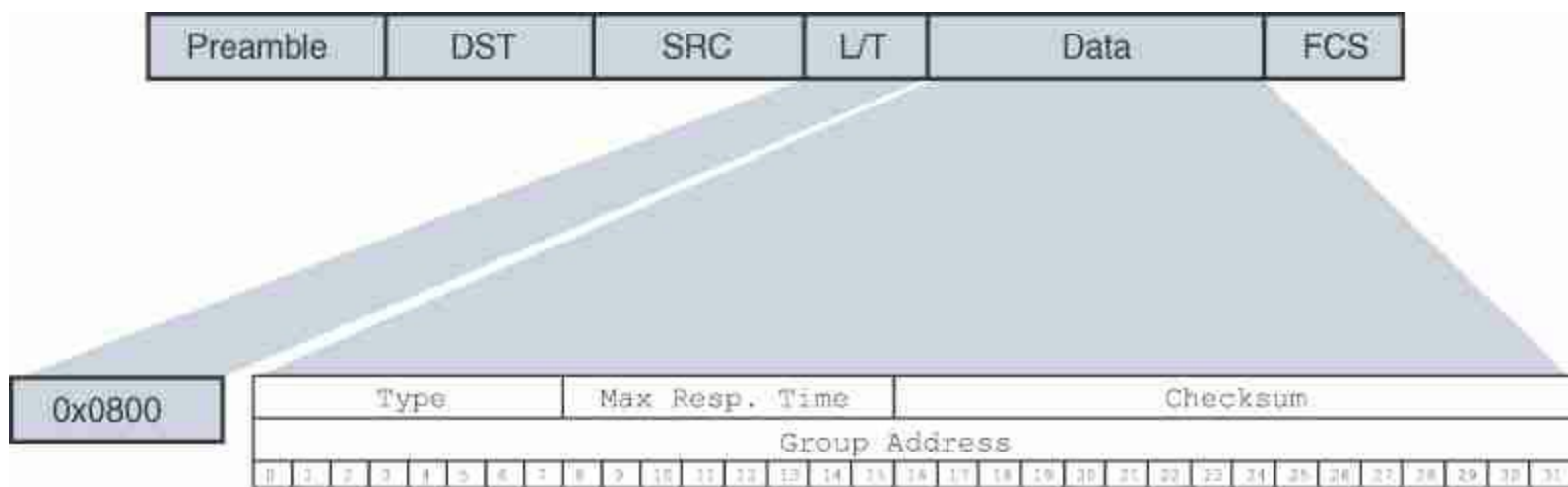


Figure 2-8 *IGMPv1 and IGMPv2 Message Format*

IGMP message types for IGMPv1 and IGMPv2 are as follows:

- 0x11—Membership query
- General query message used to determine group membership of any group
- Group-specific query used to verify if any hosts are part of a specific group
- 0x12—IGMPv1 membership report
- 0x16—IGMPv2 membership report
- 0x17—Leave-group message

The maximum response time (MRT) is calculated in one-tenth of a second increments and is used only with membership query messages. This parameter allows routers to manage the time between the moment the last host leaves a group and the moment the routing protocol is notified. When a host receives an IGMP query packet, it kicks off a timer that begins with a random value that is less than the MRT. If no other host responds with a membership report before this random timer expires, the host will then reply with a report. This decreases the number of total IGMP reports needed to maintain the group state as well as preserves local bandwidth, because the host suppresses its own reports unless absolutely necessary. IGMPv1 does not use MRT; instead, it has a timer that is always set to 10 seconds. Of course, this means the MRT cannot be less than the *query-interval*, making the maximum configurable MRT 25 seconds (1 byte MRT field; $1/10s * 255 = 25$ seconds).

The checksum is a value calculated using information within the message used to detect errors.

[Example 2-1](#) shows a packet capture of an IGMPv2 membership query. Items of interest include the source and destination MAC address. The source of this request is the router (192.168.12.1) and the destination is the multicast MAC address for 224.0.0.1, which includes all devices on the subnet. Referring to the packet capture in [Example 2-1](#), you see the IGMP type is 0x11, the maximum response time is 0x64 (hex for 10 seconds, the default for IGMPv2), the checksum, and the group address of 0.0.0.0, which indicates that it is a general query message. Also, pay particular attention to the time to live (TTL) field. This message has the TTL set to 1, which means that it will not be sent to multiple subnets. If you are troubleshooting multicast problems, you should always make sure the multicast sender has a TTL value greater than or equal to the diameter of your network.

Example 2-1 *IGMPv2 Membership Query Packet Capture*

[Click here to view code image](#)

Ethernet Packet: 60 bytes

Dest Addr: 0100.5E00.0001, Source Addr: 0022.5561.2501

Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x6, TOS: 0xC0 (Prec=Internet Contrl)

Length: 32, ID: 0x03E6, Flags-Offset: 0x0000

TTL: 1, Protocol: 2 (IGMP), Checksum: 0x7387 (OK)

Source: 192.168.12.1, Dest: 224.0.0.1

Options: Length = 4

Router Alert Option: 94 0000

IGMP VersionType: 0x11, Max Resp: 0x64, Checksum: 0xEE9B (OK)

Version 2 Membership Query

Group Address: 0.0.0.0

Remember that IGMP is a LAN-based protocol, used to manage hosts. Managing hosts is often considered a chatty process. Several configurable timers, including the MRT, within the IGMP implementation can be adjusted to modify protocol message timing and processing. Look at the IGMP interface configuration timers that are listed in the **show ip igmp interface x/x** command output in [Example 2-2](#) .

Example 2-2 show ip igmp interface Command Output

[Click here to view code image](#)

```
Router#show ip igmp interface e1/0
```

```
Loopback0 is up, line protocol is up
```

```
Internet address is 192.168.2.2/32
```

```
IGMP is enabled on interface
```

```
Current IGMP host version is 2
```

```
Current IGMP router version is 2
```

```
IGMP query interval is 60 seconds
```

```
IGMP configured query interval is 60 seconds
```

```
IGMP querier timeout is 120 seconds
```

```
IGMP configured querier timeout is 120 seconds
```

```
IGMP max query response time is 10 seconds
```

```
Last member query count is 2
```

```
Last member query response interval is 1000 ms
```

```
Inbound IGMP access group is not set
```

```
IGMP activity: 3 joins, 0 leaves
```

```
Multicast routing is enabled on interface
```

```
Multicast TTL threshold is 0
```

```
Multicast designated router (DR) is 192.168.2.2 (this system)
```

```
IGMP querying router is 192.168.2.2 (this system)
```

```
Multicast groups joined by this system (number of users):
```

```
224.0.1.40(1) 224.0.1.39(1) 239.1.1.1(1)
```

The respective timers in this output are all using implementation default values. In generic multicast deployments, these timers are not tweaked and are kept “default.” Administrators may

tweak them based on specific application requirements (not commonly seen). It is beneficial to understand the functionality of these timers:

- **ip igmp query-interval [*interval in secs*]**: Hosts on a segment will send a report of their group membership in response to queries received from the IGMP querier. The query interval defines the amount of time the router will store the IGMP state if it does not receive a report for the particular group. This hold period is three times the query interval time.

- **ip igmp query-max-response-time [*time-in-seconds*]**: When a host receives a query from the IGMP querier, it starts the countdown of the maximum response time before sending a report to the router. This feature helps reduce the chatter between hosts and the first hop router. The max-response time cannot be less than the query interval value.

- **ip igmp query-timeout [*timeout*]**: This timer is used for the querier election process described earlier, especially when multiple routers are in the LAN segment. A router that loses the election will assume querier malfunction based on the expiry of this timer. When the timer is expired, the router restarts the querier election process.

- **ip igmp last-member-query-count [*number*]**: This timer tracks the time the router must wait after the receipt of the leave message before removing the group state from local state tables. The timer is overwritten if a router is configured with the command **ip igmp immediate-leave group-list [*list*]** . With the **ip igmp immediate-leave group** command, the router treats these groups as having a single host member. After the reception of a leave message, the router immediately removes the multicast group.

IGMPv3

The addition of IGMPv3 (RFCs 3376 and 4604) brought with it significant changes over IGMPv1 and v2. Although there are vast improvements, backward compatibility between all three versions still exists. To understand why, examine [Figure 2-9](#) , which shows the IGMPv3 header format. New header elements of importance include a Number of Sources field, a Source Address(es) field, and a change from a Max Response Time field to a Max Response Code field.

Preamble	DST	SRC	LT	Data	FCS
----------	-----	-----	----	------	-----

Type (0x11)				Max Resp. Code				Checksum																											
Group Address																																			
RESV		S	QRV		QQIC										Number of Sources [N]																				
Source Address [1]																																			
Source Address [2]																																			
Source Address [3]																																			
.																																			
.																																			
.																																			
Source Address [N]																																			
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31				

Figure 2-9 IGMPv3 Message Format

As the header shows, the most significant addition to IGMPv3 is the capability to support specific source filtering. Why is this a big deal? With IGMPv1 and v2, you could not specify the host from which you wanted to receive a multicast stream; consequently, multiple sources could be sending to the same multicast IP address and port number, and the host would now have a conflict with which stream to receive. Source filtering allows the host to signal membership with either an include or an exclude group list. This way, the host can specify which device(s) it is interested in receiving a stream from, or it can indicate which devices that it is not interested in receiving a stream from. This adds an additional security component that can be tapped at the application level. IGMPv3 is used at Layer 2 for source-specific multicast (SSM). SSM is covered in [Chapter 3](#).

In addition to this change, the MRT was updated once again in IGMPv3; in fact, it was changed in RFC 3376 to a maximum response code (MRC). Similar to the MRT field in IGMPv2, the max response code field indicates the maximum time allowed before a report for a group must be received. The maximum response code (MRC) can still incorporate the MRT, which is represented in units of one-tenth of a second. There are 8 bits in the MRC field, and the value of those bits indicates how the MRC is to be read. If the MRC is less than 128, the Max Response Time is equal to the Max Response Code value. If the MRC is greater than or equal to 128, the MRC has a floating point value to reflect much longer periods of time. This makes the total maximum timer configurable up to 55 minutes.

The response time was modified in IGMPv3 to better accommodate different types of network connectivity. Using a smaller timer allows the network administrator to more accurately tune the leave latency of hosts. Using a larger timer can accommodate network types where the burstiness of

group management traffic is less desirable, e.g. low bandwidth wireless networks.

[Example 2-3](#) shows a packet capture of a membership report from an IGMPv3 host with the IP address of 192.168.7.14 with a group membership request to receive a multicast stream from 224.64.7.7 from the source of 192.168.8.10.

Example 2-3 *IGMPv3 Membership Report Packet Capture*

[Click here to view code image](#)

Ethernet II, Src: (80:ee:73:07:7b:61), Dst: (01:00:5e:00:00:16)

Type: IP (0x0800)

Internet Protocol Version 4, Src: 192.168.7.14, Dst: 224.0.0.22

Version: 4

Header length: 24 bytes

Differentiated Services Field: 0xc0 (DSCP 0x30: Class Selector 6; ECN: 0x00:

Not-ECT (Not ECN-Capable Transport))

Total Length: 52

Identification: 0x0000 (0)

Flags: 0x02 (Don't Fragment)

Fragment offset: 0

Time to live: 1

Protocol: IGMP (2)

Header checksum: 0x3c37 [validation disabled]

Source: 192.168.7.14

Destination: 224.0.0.22

Options: (4 bytes), Router Alert

Internet Group Management Protocol

[IGMP Version: 3]

Type: Membership Report (0x22)

Header checksum: 0x4a06 [correct]

Num Group Records: 2

Group Record : 224.64.7.7 Mode Is Include

Record Type: Mode Is Include (1)

Aux Data Len: 0

Num Src: 1

Multicast Address: 224.64.7.7

Source Address: 192.168.8.10

Group Record : 224.0.0.251 Mode Is Exclude

Record Type: Mode Is Exclude (2)

Aux Data Len: 0

Num Src: 0

Multicast Address: 224.0.0.251

Notice the destination IP address of the IPv4 packet; it is being sent to 224.0.0.22. This is the IP address to which all hosts send their membership report.

Configuring IGMP on a Router

A router by default is not configured to support multicast traffic. There are a few basic

configuration tasks that need to be performed on Cisco IOS routers in order to enable multicast on the network.

Step 1. Enable multicast routing on the router in global configuration mode.

```
ip multicast-routing
```

Step 2. Configure PIM routing on the associated interface(s) in the interface configuration mode. Take into consideration that multicast traffic may traverse many different paths and it is always a good idea to enable it on every interface that may need to participate. This might save you some troubleshooting time in the future. When you configure an interface for PIM, it is automatically configured to use IGMPv2 on most current operating systems.

```
ip pim sparse-mode
```

On Cisco switches, IGMP version 2 is enabled by default.

Step 3. If necessary, change the IGMP version supported on an interface.

```
ip igmp version [1,2,3]
```

Mixed Groups: Interoperability Between IGMPv1, v2, and v3

It all boils down to the least common denominator. When there is a mix of hosts on a subnet, the only method for them to communicate is with the lowest IGMP version number used by a host. This is due to the fact that higher versions understand lower versions for backward compatibility.

The other situation you may encounter is having a mix of clients and several routers on a subnet. There is no mechanism for routers configured with a lower IGMP version to detect a router running a higher version IGMP. This requires manual intervention, and someone must configure the routers to use the same version.

Layer 2 Group Management

As mentioned earlier, Layer 2 devices treat multicast messages as broadcasts when no group management mechanism is present. This not only causes an increase of traffic on a particular subnet, but these messages are sent to every device within that subnet (flooding). These devices may treat the processing of multicast messages differently, depending on the behavior of the operating system and associated hardware. Multicast messages may be processed in hardware, software, or the combination of both. Consequently, multicast messages, or too many multicast messages, may have an adverse effect on a device. It is better to handle multicast messages in the network and only send messages to the devices that are interested in receiving them.

Two protocols were created to help manage this behavior on LAN segments, Cisco Group Management Protocol (CGMP) and Router-port Group Management Protocol (RGMP). While both of these protocols still exist in networks today, they are generally not used in favor of IGMP snooping, which is discussed in the next section. For this reason, we only provide a brief introduction to these protocols.

Cisco Group Management Protocol

In order to address the Layer 2 challenges with multicast as a broadcast message, Cisco first developed a proprietary solution called CGMP. At the time of development, the capabilities of Layer 2 switches did not offer the Layer 3 inspection or snooping of messages as they do today. CGMP was used between the connected router and switch. The router would send IGMP information to the switch using CGMP, regarding which clients have registered. The receiving switch could then determine

which interfaces to send the outgoing multicast messages.

CGMP behaves in the following manner: When a host is interested in receiving a stream from a particular Group Destination Address (GDA), it sends an IGMP report message. This message is received by the router and the router in turn sends a CGMP Subnetwork Access Protocol (SNAP) frame to the destination MAC address of 0x0100.0CDD.DDD with the following information:

- Version: 1 or 2
- Message field: Join or Leave
- Count: Number of address pairs
- MAC address of the IGMP client
- Group multicast address/Group destination address (GDA)
- Unicast source address (USA): MAC address of the host sending the join message

The attached switch, configured to support CGMP, receives the frame from the router. The switch looks at the USA and performs a lookup in the content addressable memory (CAM) table to determine the interface of the requesting host. Now that the interface of the requesting host has been determined, the switch places a static entry for the GDA and links the host address in the CAM table if this is the first request for the GDA. If there was already an entry for the GDA, the switch just adds the USA to the GDA table.

Note

Due to conflicts with other protocols such as HSRPv1, CGMP may have certain features disabled. Always check current configuration guides before enabling CGMP. Using IGMP snooping is an easy way to avoid such conflicts.

The CGMP Leave Process

The leave process is dependent on the IGMP version of the host. IGMPv1 does not provide a mechanism to inform the network when it no longer wants to participate in receiving the stream. When an IGMPv1 host leaves, the only way the network realizes that the host is no longer participating in the multicast stream is through an IGMP query message. You can imagine that having a host join a stream, then join another stream, and so on, can cause a significant impact on the amount of traffic on the network. Consider someone watching IPTV and quickly changing channels. To address this problem, the router periodically sends IGMP query messages to determine if devices are still interested in receiving the stream. If a router does not receive a response after sending three IGMP query messages, it informs the switch via CGMP, which then removes the entries for the GDA.

IGMPv2 added the functionality of a *leave message*; this provides the ability for a host to gracefully leave a session that it is no longer interested in receiving. When a host sends an IGMP leave message, the router sends a query message and starts a query-response message timer. This process is done to determine if there are hosts on that specific network that are still interested in receiving the multicast stream. If the router does not receive a response, it will send a CGMP message to the switching informing it to remove the entries for the GDA.

Router-Port Group Management Protocol

Along the same lines as CGMP, another protocol, RGMP, was developed to address the multicast communication of routers over a switched network. When several routers are connected to the same Layer 2 switched network, multicast messages are forwarded to all protocol independent multicast

(PIM) routers, even those that are not interested in receiving the multicast streams.

RGMP is configured on the switch in conjunction with IGMP snooping (IGMP snooping is discussed at length in the next section of this chapter). The routers that are connected to the switch are configured with PIM sparse-mode and RGMP. A router configured with RGMP sends a RGMP hello message to the attached switch. The switch creates an entry indicating the receiving interface is a RGMP router and will not forward multicast traffic to that interface unless it receives a join message from the router. A router interested in receiving a specific multicast stream sends a RGMP join message to the switch with the GDA. The switch in turn creates an entry for the GDA and links the router interface in the CAM table.

We have covered two of the four RGMP message types, “hello” and “join.” The other messages are “bye” and “leave.” The “bye” RGMP message tells the switch to place the interface in a normal forwarding state. Finally, the “leave” message is sent from the router when it no longer is interested in receiving a specific multicast stream.

Snooping

According to the Merriam-Webster dictionary, snooping is “to look or pry especially in a sneaking or meddlesome manner.” When we use this term referring to multicast, it means the same thing, with the exception of the meddlesome manner. When a device monitors the conversation or messages sent between devices on the network, we can gain a great deal of information which can be used in turn to tune network behavior to be much more efficient. Over the last several years, Cisco has made great improvements in the intelligence of the components that make up a switch. Switches can now perform Layer 3 services, capture analytics, rewrite information, and so on, all at line rate. This increased intelligence has now provided the capability of a Layer 2 switch to look at more than just the destination MAC address; it offers the capability to look deep into the message and make decisions based on Open Systems Interconnect (OSI) Layer 2 to L7 information.

IGMP Snooping

IGMP snooping is one of those features that does exactly what it says. A network component, generally a Layer 2 switch, monitors frames from devices, and, in this case, it listens specifically for IGMP messages. During the snooping process, the switch listens for IGMP messages from both routers and hosts. After discovering a device and determining which GDA that particular device is interested in, the switch creates an entry in the CAM table that maps the GDA to the interface.

Switches learn about routers using several mechanisms:

- IGMP query messages
- PIMv1 and/or PIMv2 hellos

Examine [Figure 2-10](#) because it will be used to explain IGMP snooping.

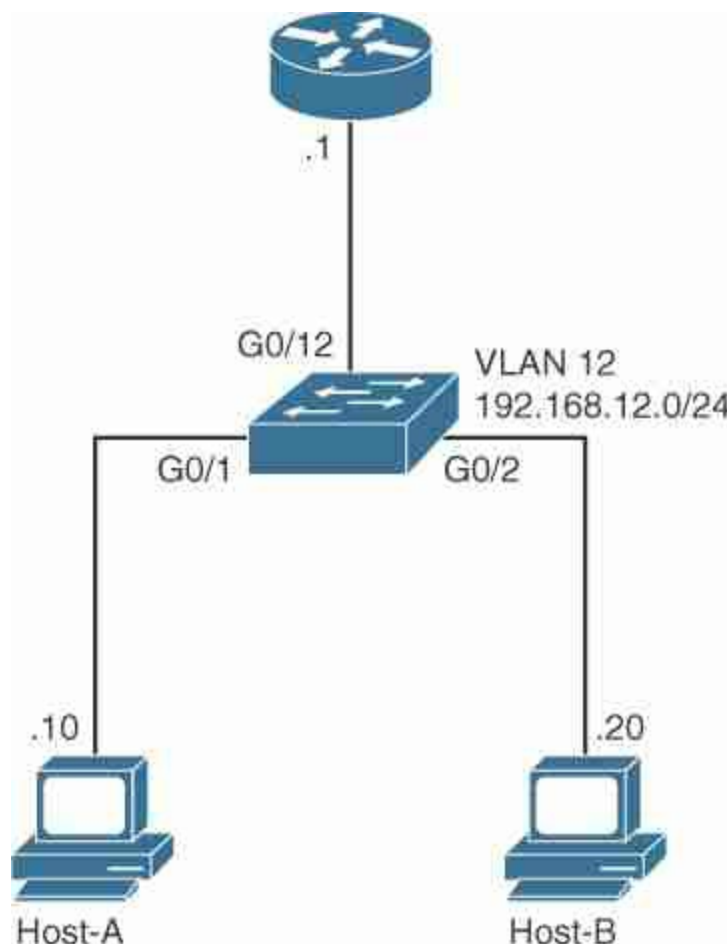


Figure 2-10 IGMP Snooping

[Example 2-4](#) shows a packet capture of an IGMP query report generated from a router.

Example 2-4 IGMP Query Packet Capture

[Click here to view code image](#)

Ethernet Packet: 60 bytes

Dest Addr: 0100.5E00.0001, Source Addr: 0C85.2545.9541

Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x6, TOS: 0xC0 (Prec=Internet Contrl)

Length: 32, ID: 0x1FC5, Flags-Offset: 0x0000

TTL: 1, Protocol: 2 (IGMP), Checksum: 0x57A8 (OK)

Source: 192.168.12.1, Dest: 224.0.0.1

Options: Length = 4

Router Alert Option: 94 0000

IGMP VersionType: 0x11, Max Resp: 0x64, Checksum: 0xEE9B (OK)

Version 2 Membership Query

Group Address: 0.0.0.0

When the switch determines there is a router attached, it places an entry in the IGMP snooping table that specifies the interface, as [Example 2-5](#) demonstrates.

Example 2-5 IGMP Snooping Table

[Click here to view code image](#)

Switch#show ip igmp snooping mrouter

Vlan ports

12 Gi0/12(dynamic)

In this case, the switch has learned that a router is attached to interface g0/12. This interface is known as the *mrouter port* or *multicast router port* . The mrouter port is essentially a port that the switch has discerned is connected to a multicast enabled router that can process IGMP and PIM messages on behalf of connected hosts. An IGMP-enabled VLAN or segment should always have an mrouter port associated with it. We can also see the effect using the **debug ip igmp snooping router** command, which gives us greater insight into the process, as [Example 2-6](#) demonstrates.

Example 2-6 debug ip igmp snooping router Output

[Click here to view code image](#)

```
Switch#debug ip igmp snooping router
router debugging is on
01:49:07: IGMPSEN: router: Received non igmp pak on Vlan 12, port Gi0/12
01:49:07: IGMPSEN: router: PIMV2 Hello packet received in 12
01:49:07: IGMPSEN: router: Is not a router port on Vlan 12, port Gi0/12
01:49:07: IGMPSEN: router: Created router port on Vlan 12, port Gi0/12
01:49:07: IGMPSEN: router: Learning port: Gi0/12 as rport on Vlan 12
```

As you see from the output in [Example 2-6](#) , the switch received a PIMv2 hello packet from interface Gi0/12 and changed the state of the port to a router port.

When a host connected to the switch wants to join a multicast stream, it sends an IGMP membership report. In [Example 2-7](#) , the host connected to port Gi0/2 is interested in receiving data from 224.64.7.7. Using the **debug ip igmp snooping group** command, we can monitor the activity.

Example 2-7 debug ip igmp snooping group Output

[Click here to view code image](#)

```
Switch#debug ip igmp snooping group
router debugging is on
01:58:47: IGMPSEN: Received IGMPv2 Report for group 224.64.7.7 received on Vlan 12,
port Gi0/2
01:58:47: IGMPSEN: group: Adding client ip 192.168.12.20, port_id Gi0/2, on vlan 12
```

From the output in [Example 2-7](#) , we can ascertain that the host connected to Gi0/2 is attempting to connect to the multicast group 224.64.7.7.

Using the **show ip igmp snooping groups** command, we can also see the entry in the switch, as demonstrated in [Example 2-8](#) .

Example 2-8 show ip igmp snooping groups Output

[Click here to view code image](#)

```
Switch#show ip igmp snooping groups
```

Vlan	Group	Type	Version	Port List
12	224.0.1.40	igmp	v2	Gi0/12
12	224.64.7.7	igmp	v2	Gi0/2, Gi0/12

The output in [Example 2-8](#) specifies the VLAN, multicast group, IGMP version, and ports associated with each group.

The packet capture in [Example 2-9](#) shows the membership report generated from the host with the MAC address of 0x000F.F7B1.67E0. Notice how the destination MAC and destination IP are those of the multicast group the host is interested in receiving. The IGMP snooped mrouter port entry ensures this IGMP membership report is forwarded to the multicast router for processing, if necessary. See the next section on maintaining group membership.

Example 2-9 IGMP Membership Report Packet Capture

[Click here to view code image](#)

Ethernet Packet: 60 bytes
Dest Addr: 0100.5E40.0707, Source Addr: 000F.F7B1.67E0
Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x5, TOS: 0xC0 (Prec=Internet Contrl)
Length: 28, ID: 0x0000, Flags-Offset: 0x0000
TTL: 1, Protocol: 2 (IGMP), Checksum: 0x051D (OK)
Source: 192.168.12.20, Dest: 224.64.7.7

IGMP VersionType: 0x16, Max Resp: 0x00, Checksum: 0x02B8 (OK)

Version 2 Membership Report
Group Address: 224.64.7.7

The output in [Example 2-10](#) shows several hosts connecting to the multicast group.

Example 2-10 show ip igmp snooping groups Output

[Click here to view code image](#)

Switch#show ip igmp snooping groups

Vlan	Group	Type	Version	Port List

12	224.0.1.40	igmp	v2	Gi0/15
12	224.64.7.7	igmp	v2	Gi0/1, Gi0/2, Gi0/4, Gi0/15

Maintaining Group Membership

As hosts are added to or removed from the multicast group, the switch manages the interaction. The switch does not notify the router of any additions or removals to the group, with the exception of the last host. If there is only one host and it leaves the multicast group, the switch immediately sends a group leave message to the upstream router. One of the interesting aspects of this message is that the switch spoofs the IP address of the last client. Look carefully at the output in [Example 2-11](#) .

Example 2-11 IGMP Leave Capture Output

[Click here to view code image](#)

Ethernet Packet: 60 bytes
Dest Addr: 0100.5E00.0002, Source Addr: 0013.19C6.A60F
Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x6, TOS: 0xC0 (Prec=Internet Contrl)

Length: 32, ID: 0x0000, Flags-Offset: 0x0000
TTL: 1, Protocol: 2 (IGMP), Checksum: 0x7745 (OK)
Source: 192.168.12.40, Dest: 224.0.0.2

Options: Length = 4

Router Alert Option: 94 0000

IGMP VersionType: 0x17, Max Resp: 0x00, Checksum: 0x01B8 (OK)

Version 2 Leave Group

Group Address: 224.64.7.7

The benefit of this behavior is that when the last device leaves the multicast group, the router does not have to wait for a timeout. Notice also that the MAC address of the source in the packet in [Example 2-11](#) is the MAC address of the switch as depicted in the **show interface Gi0/12** output in [Example 2-12](#) . This is the mrouter interface for this segment.

Example 2-12 show interface *Output*

[Click here to view code image](#)

```
Switch#show interface Gi0/12
```

```
GigabitEthernet0/12 is up, line protocol is up
```

```
Hardware is Gigabit Ethernet, address is 0013.19C6.A60F (bia 0013.19C6.A60F)
```

```
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
```

Configuring IP IGMP Snooping

The configuration could not be easier for newer Catalyst or Nexus product switches. The command is as follows:

[Click here to view code image](#)

```
C2970(config)#ip igmp snooping
```

Here is the best part—it is on by default and obviously not necessary to type the previous command. You can confirm that IGMP snooping is functional and verify the specific operating parameters with the **show ip igmp snooping** command, as demonstrated in [Example 2-13](#) .

Example 2-13 Confirming IGMP Snooping Functionality and Parameters

[Click here to view code image](#)

```
Switch#show ip igmp snooping
```

```
Global IGMP Snooping configuration:
```

```
-----  
IGMP snooping           : Enabled  
IGMPv3 snooping (minimal) : Enabled  
Report suppression      : Enabled  
TCN solicit query       : Disabled  
TCN flood query count    : 2  
Robustness variable      : 2  
Last member query count  : 2  
Last member query interval : 1000
```

The output was truncated for brevity, but the IGMP snooping information will be displayed per VLAN.

The Process of Packet Replication in a Switch

Nearly all the protocols required to accomplish multicast forwarding are open standards, ratified by working groups like the IETF. However, the actual process of forwarding packets through a network device is not an open standard. The same is true of unicast packets. The way each manufacturer, or in some cases product lines, implement forwarding is what differentiates each platform from the next.

At the heart of IP multicast forwarding is the packet replication process. Packet replication is the process of making physical copies of a particular packet and sending a copy out any destination interfaces in the derived forwarding path.

What makes the process of replication so different from platform to platform is *where* the replication occurs. Each Cisco networking platform handles this process a little differently. Many routers use centralized processing to perform replication. Other more advanced routers and switches with distributed processing require specialized application-specific integrated circuits (ASICs) to perform packet replication and forwarding from one line-card to another. At the beginning of the Internet, the key objective of packet handling was to simply forward packets.

Packet forwarding at wire speed required the specialization of ASIC processing. As features and applications embedded on routers and switches grew (including critical components in modern networks like QoS, MPLS, multicast, SNMP, flow reporting, and so on) so did the need for packet replication and treatment at wire speed. Without ASICs, router processors would be overwhelmed by packet handling requirements. Cisco Systems spent over 25 years developing custom ASICs, many specifically for packet replication needs. With that understanding, the router manufacturer must make a choice about where and on which ASIC(s) to replicate packets. This is especially true in distributed routing and switching platforms. A distributed platform uses ASICs throughout the device to push forwarding decisions as close to the interface as possible.

Some distributed platforms may forward incoming multicast packets to the central processing card. That card may take special action on the packet, replicate the packet, or forward to other line-cards for replication. If replication occurs on the central processor, then the replication model used is centralized replication and acts as a traditional bus. In a centralized replication model, resource pressure occurs on the central processor and centralized resources like memory. Depending on the size of the multicast deployment or the number of packets requiring replication can result in serious performance problems for control plane traffic.

Other distributed platforms may use the ASICs associated with the inbound interface or line card to perform replication. This is known as *ingress replication*. In this case, the incoming interface line card replicates the multicast packet and sends copies via a fabric toward the exit interfaces in the path. Ingress replication distributes resource pressure across multiple processors and may still require occasional forwarding by the central processor depending on enabled features.

The ASICs associated with the exit interfaces can also perform replication. This, of course, is *egress replication*. In some instances, replicating only at the egress interface could mean an obvious loss in efficiency; however, exit line cards in many models terminate encapsulated multicast packets destined for certain domains. This means that the egress line card can be an ideal point of replication because that particular card might have many interfaces with subscribed receivers downstream.

Note

Downstream is the direction flowing away from the sender, toward the receiver. Fabric-facing ASICs on these cards will take on the role of replication.

Platforms may implement a combination of these replication methods, centralized, ingress, and egress. This is known as distributed replication. An incoming interface ASIC may perform one level of replication and send one packet copy to each line card with exit interfaces in the forwarding tree. The egress line cards can then create additional copies, one for each interface on that card in the path. This method further distributes resource pressure across as many ASICs as possible. [Figure 2-11](#) represents a basic distributed replication model using replication on both the incoming line card and the outgoing line card. (This is a common model used in Cisco devices, for example the Catalyst 6500 switch line.)

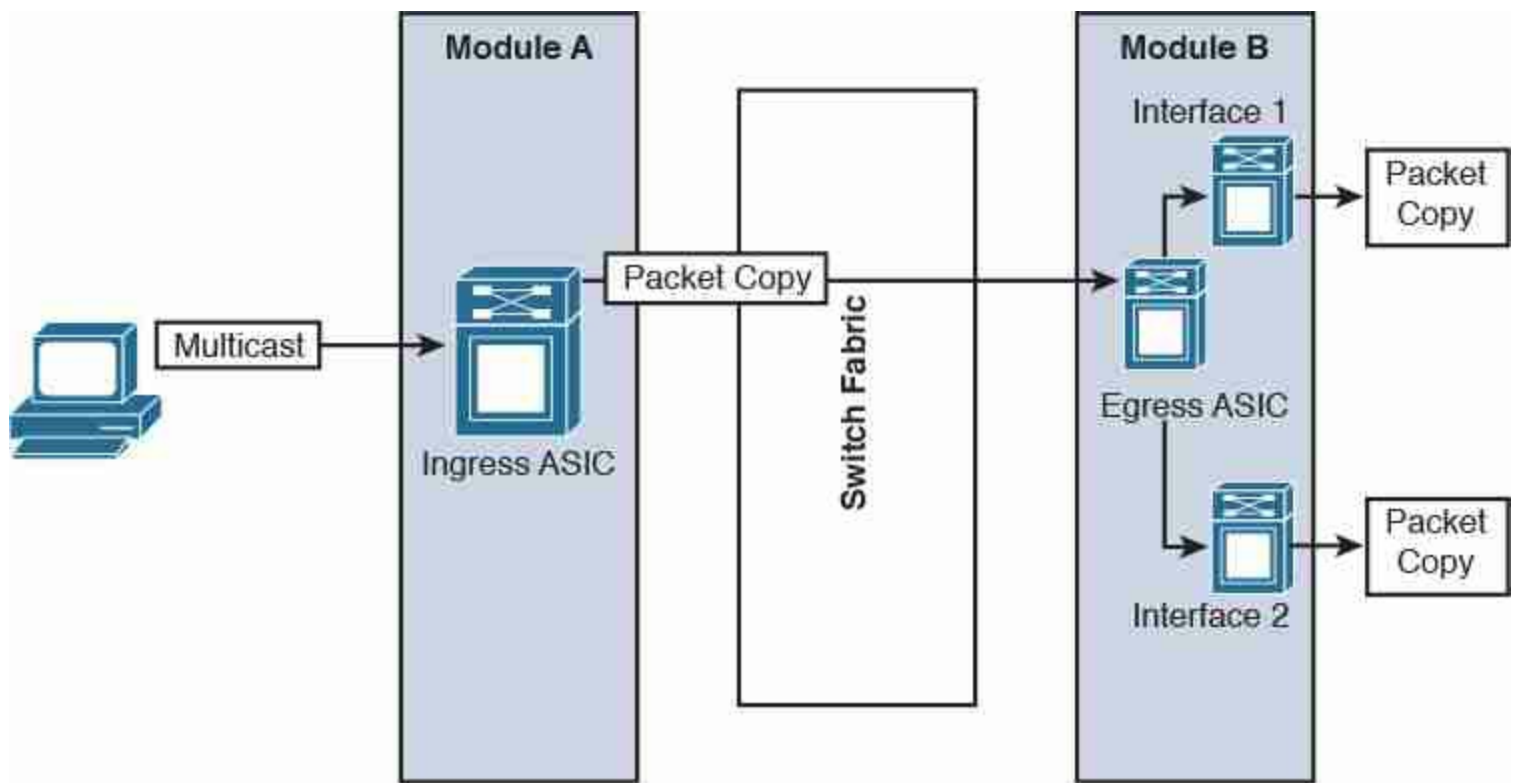


Figure 2-11 *Packet Replication*

The important thing to remember is that each manufacturer and each platform handles replication differently. To be truly competitive, each manufacturer must perform replication in a safe and efficient manner. That means the platform must forward as quickly as possible, while preventing loops and protecting precious control-plane resources. Any architect or engineer seeking to deploy IP multicast technologies should pay special attention to the replication process of each platform in the network path, as well as any specialized hardware and software feature enhancements.

Protecting Layer 2

IGMP snooping is a mechanism that we configure on a switch to minimize the impact of multicast traffic being directed to devices that are not interested in receiving it. This feature helps protect not only the infrastructure resources, but the devices that are attached to the network. Another feature that is well worth mentioning and will help to ensure the successful operation of your network is *storm control*.

Storm Control

Data storms in networks can be generated in several different ways, including an intentional denial of service (DoS) attack, a defective network interface card (NIC), a poorly programmed NIC driver,

and so on. In order to prevent broadcast, multicast, or even unicast traffic from overwhelming a switch by an inordinate amount of traffic, the storm control feature offers the capability to set thresholds for these types of traffic on a per-port basis.

Configuration options are on a port basis and offer the capability to specify traffic based on the percentage of bandwidth, bits per second (BPS) or packets per second (PPS). If the threshold is reached, you can either send a Simple Network Management Protocol (SNMP) trap message or shut down the port by placing it in an error-disable state. The configuration parameters are as follows:

[Click here to view code image](#)

```
storm-control broadcast level <0.00 - 100.00> / bps / pps
storm-control multicast level <0.00 - 100.00> / bps / pps
storm-control unicast level <0.00 - 100.00> / bps / pps
storm-control action trap
storm-control action shutdown
```

In the following example, the switch will be configured to send a SNMP message when the broadcast level exceeds 50 percent:

[Click here to view code image](#)

```
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#storm-control broadcast level 50
Switch(config-if)#storm-control action trap
```

The following is the SNMP message generated when the broadcast level has been exceeded:

[Click here to view code image](#)

```
%STORM_CONTROL-3-FILTERED: A Broadcast storm detected on Gi0/2. A packet filter
action has been applied on the interface.
```

You also have the ability to place the port in an error-disable state using the following command:

[Click here to view code image](#)

```
Switch(config-if)#storm-control action shutdown
```

The following output depicts the messages shown in the event of a port shutdown:

[Click here to view code image](#)

```
%PM-4-ERR_DISABLE: storm-control error detected on Gi0/2, putting Gi0/2 in
err-disable state
%STORM_CONTROL-3-SHUTDOWN: A packet storm was detected on Gi0/2. The interface has
been disabled.
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state
to down
%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down
```

We mentioned DoS attacks earlier in this section. When configuring the **storm-control action shutdown** command, you may have to manually enable the ports in the event the port is disabled. Using the **errdisable recovery** commands helps to mitigate that problem:

[Click here to view code image](#)

```
Switch(config)#errdisable recovery cause storm-control
```

Switch(config)#errdisable recovery interval 30

The following output shows the logging message after recovery:

[Click here to view code image](#)

```
2d07h: %PM-4-ERR_RECOVER: Attempting to recover from storm-control err-disable
state on Gi0/2
2d07h: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
2d07h: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed
state to up
```

Note

Use caution when setting storm levels because you may inadvertently create your own DoS attack and deny legitimate traffic.

Summary

The process of communication between devices on an IP network requires the handling or encapsulation of data at each layer of the OSI model. Packets are composed of MAC addresses, IP addresses, port numbers, and other necessary information. Multicast at Layer 2 has unique requirements regarding MAC addresses and the way IP addresses are mapped to them. In the mapping process, 5 bits of the IP address are overwritten by the OUI MAC address, which causes a 32-to-1 IP multicast address-to-multicast MAC address ambiguity. Client devices on the network use Internet Group Management Protocol (IGMP) to signal the intent to receive multicast streams and, in most cases, use IGMP to send leave messages. Modern switches have the capability to “snoop” or listen to IGMP messages and build appropriate forwarding tables. Timely delivery of messages is the most important role of the network and protecting those resources are critical to that function. Storm control can be used to aid in protecting network elements by limiting the types of traffic. Understanding the intricacies of how Layer 2 devices deliver multicast messages internally will help you in building an infrastructure to support your business initiatives.

References

Request for Comment (RFC) 1054: Host Extensions for IP Multicasting

RFC 1112: Internet Group Management Protocol, Version 1

RFC 2236: Internet Group Management Protocol, Version 2

RFC 3376: Internet Group Management Protocol, Version 3

RFC 4604: Internet Group Management Protocol, Version 3 and Multicast Listener Discovery Protocol Version 2 (MLDv2) for Source-Specific Multicast

RFC 2362: Protocol Independent Multicast-Sparse Mode (PIM-SM)

Chapter 3. IP Multicast at Layer 3

Chapter 3. IP Multicast at Layer 3

As discussed in [Chapter 1](#), many unique applications leverage multicast for efficient packet delivery. In most cases, these applications use the network merely as transport for application data. Multicast senders and receivers are typically IP host machines that run these applications.

Because of the abstraction of the logical layer from the physical in the OSI model, almost any computing device can be a multicast host. Potential multicast hosts include set-top cable boxes, smartphones, tablets, laptops, personal computers, servers, routers, and so on. With this kind of ubiquity, an engineer might assume that a fully converged IP unicast network is simply multicast ready and that hosts may begin participation in multicast by simple virtue of the host's IP connectivity. Unfortunately, that assumption is incorrect. It is true that most IP devices have some minor inherent multicast capability built in, such as support for specific link local applications. However, most operating systems do not come preconfigured to participate in specific administrative multicast groups; that function is dictated by the level of multicast participation (described by IETF RFC 1112) and by any multicast-enabled application(s) running on the host device.

It is important to understand multicast networking from the perspective of the IP host. It is equally important to understand which groups may apply to the operating system specifically and which may apply to specific applications. IP hosts that are fully multicast-aware must interact with the network to manage the flow of multicast packets, maintaining the highest level of efficiency in packet flow. After this interaction, network devices can overlay an IP multicast forwarding topology on top of the established IP unicast network, similar to other technologies like MPLS or IPsec VPNs.

In this chapter, we explain the functionality of clients and servers and what constitutes membership to a group. You learn about the different types of multicast trees and how they are built using protocol independent multicast (PIM). Finally, we explain the relationship between routing and forwarding and what this looks like from the perspective of a Layer 3 device. The elements in this chapter are critical to understanding how multicast works. Be sure you have a good understanding of what is explained in this chapter before you move on.

Multicast Hosts

Most major multicast applications function in a specific client/server model. Many engineers make the mistake of assuming that a server is always a network multicast sender, and that clients are the receivers. Remember, multicast application flow models (one-to-many, many-to-one, many-to-many) vary widely depending on the needs of the application.

In a many-to-many implementation, it is not uncommon for servers to both send and receive multicast messages. Clients in a many-to-one model may be the multicast packet senders, updating a single server acting as a receiver. Another one-to-many application may support the traditional server as sender, and clients as receivers (host model). More intricate still are multicast applications that include these functions only within the networking infrastructure, applications that have no traditional IP hosts listening to or sending to the infrastructure multicast groups. These applications often enable network operations or network transport of other packets. This warrants an exploration of how clients, servers, and network devices support host-based multicast applications, as well as which established IP multicast groups (or group blocks) are used to identify these applications.

Networked Groups: Client/Server

An IP multicast group is also known as a *host group*. The original idea behind a host group was that one IP host, perhaps a server, would act as a source, sending IP datagrams toward hosts acting as receivers. The host group is the identifier for those hosts. It is accepted that there are no restrictions on the hosts that can participate in a group or on where those hosts are located. All hosts in the host group are equal, and any host may send packets toward the group, regardless of group membership. A host may also be interested in, or send packets to, more than one group at any given time. To manage this process, hosts need to have a dynamic mechanism to join or leave a group.

[Chapter 2](#) introduced the basic concept of multicast routing, switching, and forwarding. Routers that act as gateways must have a way of knowing if a host on a given segment wishes to receive multicast packets. Gateway routers need that information to build forwarding trees with other routers in the forwarding path between sources and receivers. Without this membership information, routers could not build a multicast routing or forwarding tree. Consequently, a join or leave by an IP host in the group is the mechanism used to notify the network of membership to a specific group.

Note

It is important to note that the IGMP RFC language specifically calls these membership reports *join/leave messages*. However, to avoid confusion with PIM join/prune messages, many Cisco documents simply refer to the IGMP join/leave as a *membership report*. Both are accurate; this text uses the RFC terminology.

Hosts must manage group membership dynamically. A host may join a group at any time and may also leave at any time, or it can also simply leave through natural attrition with timers. Hosts can participate in many groups at once. Additionally, host group numbers can change, if necessary. Some host groups are permanent, meaning the group is assigned a “well-known address,” whereas other host groups can be dynamically assigned through dynamic group assignment protocols or applications. IPv4 hosts that need to join and leave host groups do so by sending updates to routers through Internet Group Messaging Protocol (IGMP), and IPv6 hosts use Multicast Listener Discovery (MLD). RFC 1112 specifies three levels of IP multicast host interest: levels zero through two. [Table 3-1](#) shows the differences between each host level.

Level 0	Level 1	Level 2
No multicast support.	Multicast source only.	Multicast source or receiver.
Does not send or process received IP multicast packets.	Can only send IP multicast packets.	Can send or receive multicast packets.
No IGMP/MLD.	No IGMP/MLD.	Employs IGMP/MLD to manage group membership.
Packets with Class D addresses in the header destination should be quietly dropped.	Packets with Class D addresses in the header destination should be quietly dropped.	Packets with Class D addresses in the header destination should be processed. If the host has expressed interest in the group the host should process the packet, otherwise the packet can be quietly discarded.

Table 3-1 *Multicast Host Support Levels*

Each of the various host operating systems can accommodate any of these three levels of multicast participation. Host and server operating systems can also use Dynamic Host Configuration Protocol (DHCP) and Multicast Address Dynamic Client Allocation Protocol (MADCAP) services to dynamically assign IP addresses and IP multicast group assignments for certain applications. To learn more about host multicast configuration, consult the operating system user manual for the device in question.

Network Hosts

In certain cases, network devices (routers and switches) can function as either an IP multicast receiver or sender as though they were a networked host system. For the most part, the applications that require network device participation in the group are related to network infrastructure. In fact, most of the IP multicast communication between network hosts is for protocol intercommunication, and much of this traffic only travels between network devices connected to a common local segment. The Local Network Control block of addresses facilitates this communication; these addresses are universally assigned and managed by Internet Assigned Numbers Authority (IANA). Routing protocols, such as OSPF and EIGRP, are common examples of this type of IP multicast traffic.

Note

Many protocols require special configuration and packet handling on non-broadcast media. Layer 2 non-broadcast domains, like Frame Relay, do not forward broadcast or multicast packets to all destination ports by default. Additional considerations for many protocols, including Enhanced IGRP (EIGRP), are often necessary within an OSI Layer 2 domain.

There are other important infrastructure protocols that use multicast communications and require packets to flow beyond local segments. Some of these protocols may not need routing beyond a local router or switch, from one local network to another. A fully functional IP multicast network is requisite for those protocols that require communication to travel beyond a particular networking device. Example infrastructure protocols that need full network forwarding are multicast virtual private network (mVPN), heartbeat and cluster applications, and virtual extensible local-area

network (VXLAN).

Note

The original VXLAN standard uses IP multicast for infrastructure communications—broadcast, unicast, and multicast messages (BUM). More recent standards can also use unicast protocols as well.

Some infrastructure protocols have specific IANA assigned host groups from the Internetwork Control, Ad-Hoc, or other multicast blocks. Other protocols require domain-specific administrative assignment and management from the Administratively Scoped IPv4 block or private IPv6 ranges. For troubleshooting purposes, network administrators may simply want to join a network device to a particular group. The network device then participates to the extent of its required capabilities in any multicast groups that have been assigned to it. This means that each IP multicast packet must be decapsulated, read, and sent to the device processor for additional processing, if the device is a member of the destination host group.

Multicast Routing: An Introduction to Protocol Independent Multicast and Multicast Trees

A routing protocol is used to facilitate communication between devices on different segments of a network. Providing a method of communication might be the primary function, but there are other purposes for a routing protocol, including loop prevention, path selection, failure detection, redundancy, and so on. Originally, multicast required a unique routing method that was completely autonomous to the unicast routing protocol. Distance Vector Multicast Routing Protocol (DVMRP) and Multicast Open Shortest Path First (MOSPF) are two examples of these protocols. Specific multicast routing protocols are no longer required because you can take advantage of the underlying unicast routing protocol. Consequently, DVMRP and MOSPF have fallen out of favor.

A unicast routing protocol is forward-looking, and the routing information or routes stored in the routing database provide information on the destination. The opposite is true for multicast routing. The objective is to send multicast messages away from the source toward all branches or segments of the network interested in receiving those messages. Messages must always flow away from the source, and never back on a segment from where the transmission originated. This means that rather than tracking only destinations, multicast routers must also track the location of sources, the inverse of unicast routing. Even though multicast uses the exact inverse logic of unicast routing protocols, you can leverage the information obtained by those protocols for multicast forwarding. Why? Because a multicast source is an IP host, a possible unicast destination, whose location is tracked by the unicast routing table, like any other IP host in the network.

This allows a network to avoid using another full routing protocol, minimizing the memory space that needs to be consumed. All of the functionality that is built into the unicast routing protocol, including loop prevention, path selection, failure detection and so on, can be used for multicast. Modern IP multicast routing uses *protocol independent multicast (PIM)*, which leverages unicast routing information, to forward messages to receivers. The process of determining where the multicast messages are to be sent is referred to as *building the tree*.

Seeing the Forest Through the Trees

Anyone who engages in the study of networking inevitably encounters the concept of network trees. For example, Spanning Tree Protocol is a well-known tool for switches to build Layer 2 forwarding trees that are devoid of network loops. Trees are important because they allow routers and switches

to calculate forwarding tables that follow a prescribed and specific path for frames and packets without making mistakes or causing network loops. In fact, the design of complex forwarding chips known as application-specific integrated circuits (ASICs), the “special ingredient” to hardware forwarding, depends on complex tree mathematics and discrete algebraic logic gates.

Trees are also convenient in that they can be visually represented. Visual trees can make complex forwarding decisions easier to understand by human beings. But what is a network tree really and why are they important? This section introduces the uninitiated to basic tree theory and, in particular, to those trees used in multicast forwarding. Understanding this information is a fundamental requirement for understanding how multicast protocols create loop-free forwarding topologies.

What Is a Network Tree?

Trees and tree building are an important part of mathematics and computer science. Mathematically, a network tree is essentially a graph, or *directed graph (digraph)*, that represents a hierarchical logical structure. Like real trees, graphical trees have roots, branches, and leaves. Leaves always diverge away from the root of the tree. [Figure 3-1](#) depicts a network that is translated into a graph with a root, branches, and a leaf. The graph is directional because traffic is forwarded from R1 and toward Host B, the end of the network tree being the last-hop router (LHR), or R7.

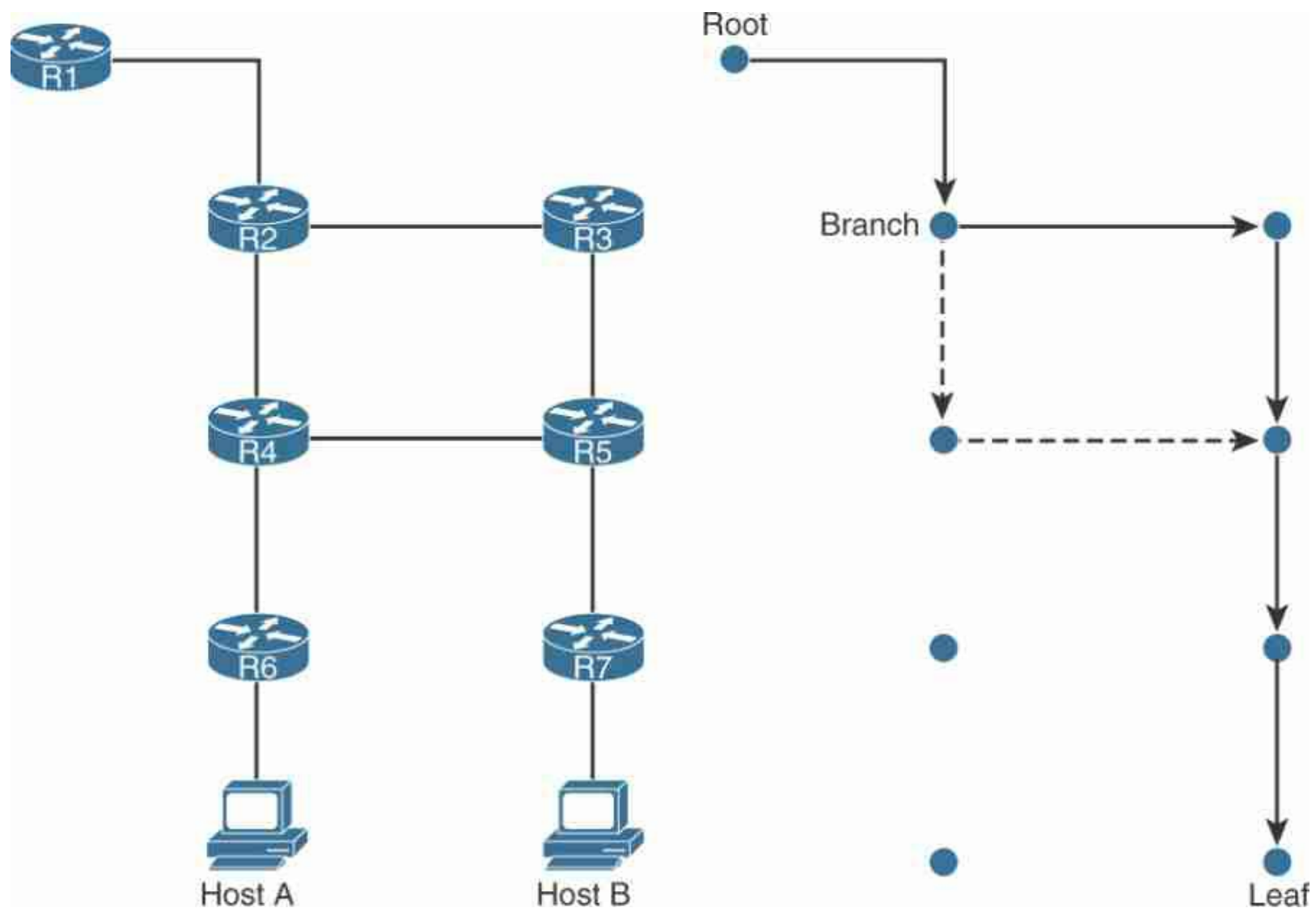


Figure 3-1 Network Depicted as a Digraph

The root of a tree is the beginning of the graph. A branch is anywhere the tree diverges into two or more separate paths and a leaf is any place in which the tree ends. Each of these components can be

represented mathematically, through mechanisms such as algebraic Boolean equations.

Network devices share information with each other in order to build trees. A network tree allows routers and switches to act together in creating forwarding paths. The network can build the forwarding path based on any desirable outcome (such as loop avoidance, efficiency, shortest path management, predictive failover, and others). Although it is true that each network device can make separate decisions, the hope is that the network will collectively build trees that map ideal paths through the network.

One example of a network tree are the shortest path trees built by Open Shortest Path First (OSPF) routers. OSPF routers build a database of possible paths from each router interface (root) to each possible IP destination (leaf). There may be many paths to a destination. The router compares and then ranks each path and creates a list of shortest (often meaning fastest) paths. Messages are forwarded toward the destination using the best path. The next-best paths are kept in memory so that the router may failover to a secondary path if the primary path becomes unusable.

Multicast routers make extensive use of network trees. The primary purpose of a multicast tree is to build an efficient loop-free forwarding path from the source of the multicast stream toward all the receivers. Looking at the entire network is a unidirectional way to look at the end-to-end path and not just next-hop in the forwarding path, which is the local view on any given network router.

The root of the tree from the perspective of the network is the router closest to the source (server) of the packet stream. A leaf is any router with an attached IP receiver (client) for the given stream. A branch in a multicast tree is any router that must perform replication to connect the root to the leaves that are not in the same segment.

Looking at each individual router, the tree is built from the interface(s) closest to the source toward the list of interfaces that are in the path, including leaves and branches. Understanding multicast trees requires an understanding of how routers represent each component of the tree, how they build a loop-free topology, and how multicast messages are forwarded to the destination.

Concepts of PIM Group States

Spanning Tree Protocol is a well-known tool for switches to build Layer 2 forwarding trees that are devoid of network loops. Multicast trees serve the same purpose. These trees are built by the PIM protocol. Command line routers cannot represent the tree visually, just like STP on command line switches. Instead, routers create and display a state table, which represents the state of each multicast tree maintained by the router. You can display the IP multicast state table in most of the Cisco platforms using the **show ip mroute** command, as demonstrated in [Figure 3-2](#).


```
r2-GRP#show ip mroute 239.1.1.1
```

```
IP Multicast Routing Table
```

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
L - Local, P - Pruned, R - RP-bit set, F - Register flag,
T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source-Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

Flag states—Platform
and code-dependent
flags

```
(* , 239.1.1.1), 5w0d/stopped, RP 192.168.2.2, flags: SJCL
```

```
Incoming interface: Null, RPF nbr 0.0.0.0
```

```
Outgoing interface list:
```

```
Loopback0, Forward/Sparse, 5w0d/00:02:40
```

Multicast group state
entries

```
(10.1.4.1, 239.1.1.1), 00:00:24/00:02:35, flags: L
```

```
Incoming interface: Ethernet1/0, RPF nbr 10.1.2.2
```

```
Outgoing interface list:
```

```
Loopback0, Forward/Sparse, 00:00:24/00:02:40
```

```
(10.1.3.2, 239.1.1.1), 00:00:24/00:02:35, flags: L
```

```
Incoming interface: Ethernet1/0, RPF nbr 10.1.2.2
```

```
Outgoing interface list:
```

```
Loopback0, Forward/Sparse, 00:00:24/00:02:40
```

Figure 3-2 *Displaying the IP Multicast State Table with **show ip mroute***

The output of the command is divided into two parts, network flags and multicast group state along with interface flags. The first part of the output provides generic platform and multicast state flags. These flags change based on the platform consideration and provide the information of multicast flow in the multicast group state section.

The multicast group-specific information in the multicast group state is (*,G) and (S,G). This provides tree information for each maintained multicast flow in the form of a “state table.”

The RP is the rendezvous point for the multicast group, which is discussed in greater detail later in the section “[Two Types of Trees](#) .”

Flags provide information such as the type of multicast stream and the control-plane information in use. These flags change slightly based on platform, and it is important to refer to the definitions in the first part of the output to clearly understand this information.

Note

Routers use reverse path forwarding (RPF) checks to keep the multicast topologies loop-free. At its most basic, an RPF check is a comparison of incoming packets to the vector established in the unicast routing table for the source of the packet. It is the inverse of a unicast forwarding lookup,

being more concerned with where the packet came from rather than where it is going. In this example, the RPF check is in the direction of the RP. RPF checks are discussed in more detail in the subsection “[Reverse Path Forwarding](#) .”

Outgoing interface list (OIL) and incoming interface list (IIL) are also provided in this output. The OIL lists the outgoing interfaces local to the node for the multicast state entry. The IIL in the output shows the incoming interface list for the multicast state for this node. These two lists are created by routers after the RPF check has been performed.

Generally speaking, **show ip mroute** is the first command that an administrator executes to understand the state of multicast flow on the Layer 3 device.

The (*,G) State Entry

A (*,G) entry in an mroute table represents a router’s relationship to the leaves of a tree. Remember that IGMPv1 and IGMPv2 hosts use the (*,G) to signal intended membership to upstream routers. The router adds the (*,G) to the mroute table. [Figure 3-3](#) shows an example network in which a host is connected to Router 7 (R7) and is sending an IGMP join request for multicast group 239.1.1.1. The sender is connected to R3 and has an IP address of 192.168.20.1.

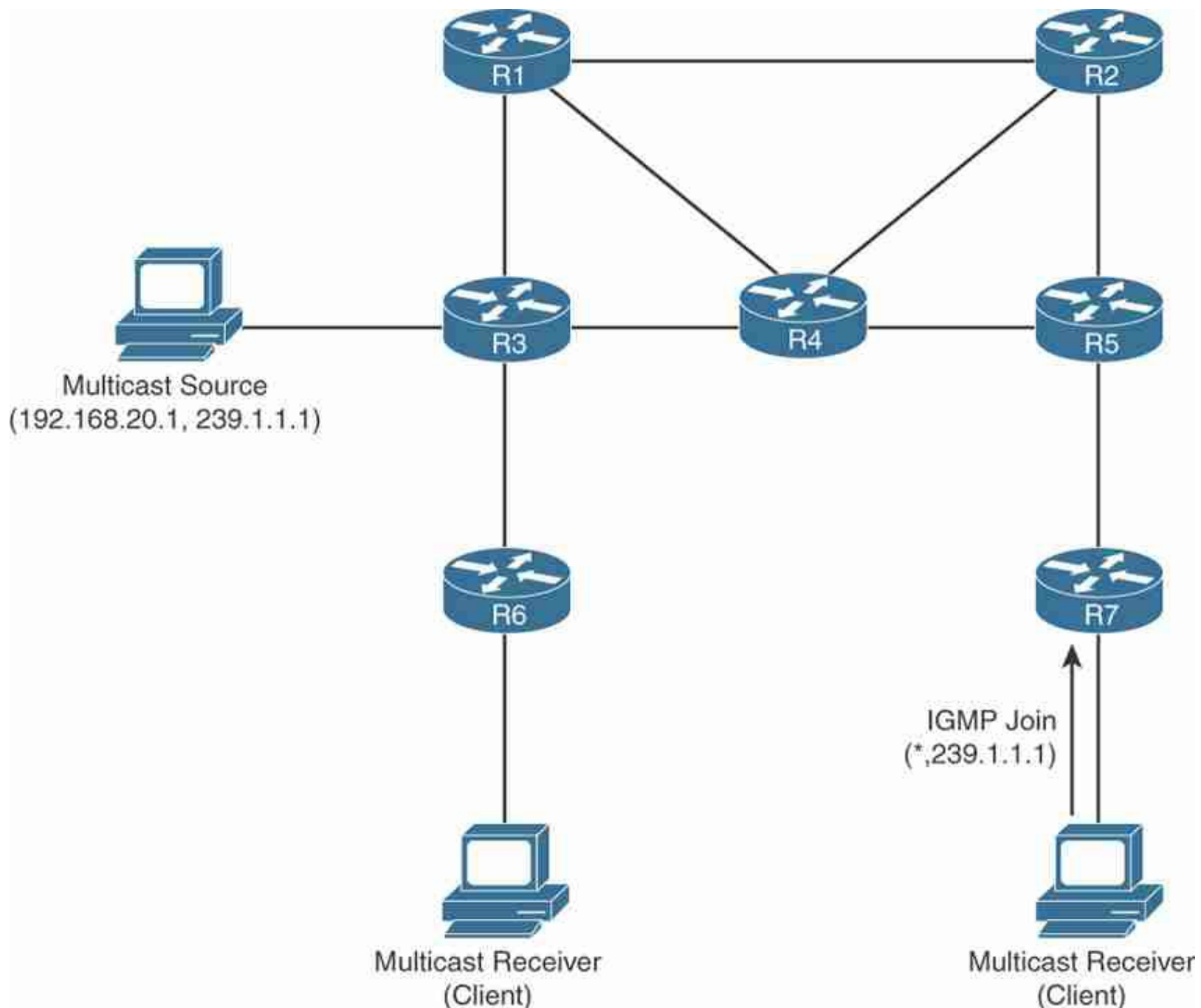


Figure 3-3 *Example Multicast Topology*

Examine the output from the **show ip mroute** command on R7 in [Example 3-1](#) .

Example 3-1 *show ip mroute Command Output*

[Click here to view code image](#)

R7#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, C - Connected, L - Local, P - Pruned

R - RP-bit set, F - Register flag, T - SPT-bit set, J - Join SPT

M - MSDP created entry, X - Proxy Join Timer Running

A - Advertised via MSDP

Outgoing interface flags: H - Hardware switched

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.1.1.1), 10:50:13/00:01:56, RP 192.168.1.1, flags: SJCF

Incoming interface: Null, RPF neighbor 0.0.0.0

Outgoing interface list: FastEthernet1/0, Forward/Sparse, 01:35:17/00:02:22

The table output shows that R7 has learned of group 239.1.1.1 and the router calculating the tree knows that the host members are downstream from interface *FastEthernet1/1* . In this case, the outgoing interface represents the path toward the host receivers. Although this router has only one leaf-facing interface, it is possible to have many downstream interfaces, consequently creating tree branches. All of the leaf-facing interfaces are added to an outgoing interface list (OIL). Using PIM messaging, the router forwards this (*,G) entry to routers upstream. Each PIM router in the path adds the (*,G), and the interface that the update was received on to the OIL for that multicast group.

The (S,G) State Entry

The (S,G) entry in the mroute table represents the router's relationship to the source of the multicast stream. In order to build a tree with an (S,G) the router needs to receive and identify packets from the source, or receive some type of notification either from the network or from hosts using an (S,G) join via IGMP. After a source for a group is known by the router it adds the (S,G) to the table using the **show ip mroute** command, as demonstrated in [Example 3-2](#) .

Example 3-2 (S,G) Entry Added to the mroute Table

[Click here to view code image](#)

R7#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, C - Connected, L - Local, P - Pruned

R - RP-bit set, F - Register flag, T - SPT-bit set, J - Join SPT

M - MSDP created entry, X - Proxy Join Timer Running

A - Advertised via MSDP

Outgoing interface flags: H - Hardware switched

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 239.1.1.1), 10:50:13/00:01:56, RP 192.168.1.1, flags: SJCF

Incoming interface: FastEthernet0/0, RPF nbr 192.168.2.2

Outgoing interface list: FastEthernet1/0, Forward/Sparse, 01:35:17/00:02:22

(192.168.20.1, 239.1.1.1), 01:55:30/00:03:26, flags: CFT

Incoming interface: FastEthernet0/0, RPF nbr 192.168.2.2

Outgoing interface list: FastEthernet1/0, Forward/Sparse, 01:55:30/00:03:12

In this output (using the previous example topology), the router has received packets destined to group 239.1.1.1 from host 192.168.20.1. The router updates the table, which now shows that the (*,G), entry (*, 239.1.1.1) has an incoming interface of **FastEthernet0/0** with Reverse Path Forwarding (RPF) Neighbor (nbr) 192.168.2.2. This indicates the router has determined that 192.168.20.1 is upstream on interface **FastEthernet0/0** , which is not in the path of the leaf, using a RPF check. The (S,G) entry (192.168.20.1, 239.1.1.1) also has an incoming interface and an OIL.

Routers can forward packets using either the (*,G) or the (S,G) entries, as long as the paths are determined to be loop-free. In fact, each type of entry represents a different type of tree; the incoming and outgoing interfaces being the path of the tree from the root, to branches, and finally to the leaves.

These two different types of trees are discussed in the next section.

Reverse Path Forwarding

Under normal unicast routing rules, packet-forwarding decisions are based on the destination address of the packet arriving at a router. The unicast routing table consists of learned networks and the associated exit interfaces for each destination network. Thus, the unicast routing table is used for destination lookups by routers so that each packet is forwarded in the most appropriate direction. The primary responsibility of each unicast routing protocol is to find and populate the routing table with the best path toward the destination, while preventing any packets from looping back toward the packet sender (source). The router's primary role is to select paths from among many protocols, again ensuring a loop-free topology.

Using IP multicast, the router forwards the packet based on a lookup of where the source is, the exact reverse of the unicast forwarding methodology. The router's multicast forwarding state runs more logically by organizing tables based on the reverse path, from the receiver back to the root of the distribution tree. Consequently, the multicast network is a true overlay of the unicast network and even uses the unicast routing table to ensure a loop-free multicast forwarding tree. This process is known as *reverse path forwarding (RPF)*. For (*,G) entries, the root of the distribution tree is the RP. The RP is at the center of the PIM sparse-mode state logic and it maintains the path toward all sources and receivers in each sparse-mode multicast domain. You will learn more about RP functionality in the PIM section. For the (S,G) entries, the root for the distribution tree is the router that connects to the multicast source. In short, incoming multicast packets will not be accepted/forwarded unless they are received on an interface that is the outgoing interface for the unicast route to the source of the packet. [Figure 3-4](#) and [Figure 3-5](#) represent this process occurring on R4.

[Figure 3-4](#) shows a multicast packet coming inbound to R4 on interface E0/0. The source IP of the multicast sender is 192.168.20.1. The router performs an RPF lookup on that source to check that E0/0 is indeed in the direction of the route for network 192.168.20.1. A **show ip route** on R4 shows that the IP unicast network 192.168.20.0/24 is in fact located via E0/0. The **show ip rpf** output for source 192.168.20.1 also shows that the router has confirmed the reverse path to be E0/0 for network 192.168.20.1. With this confirmation, the router can now confidently forward the multicast packet, knowing that a network loop is unlikely.

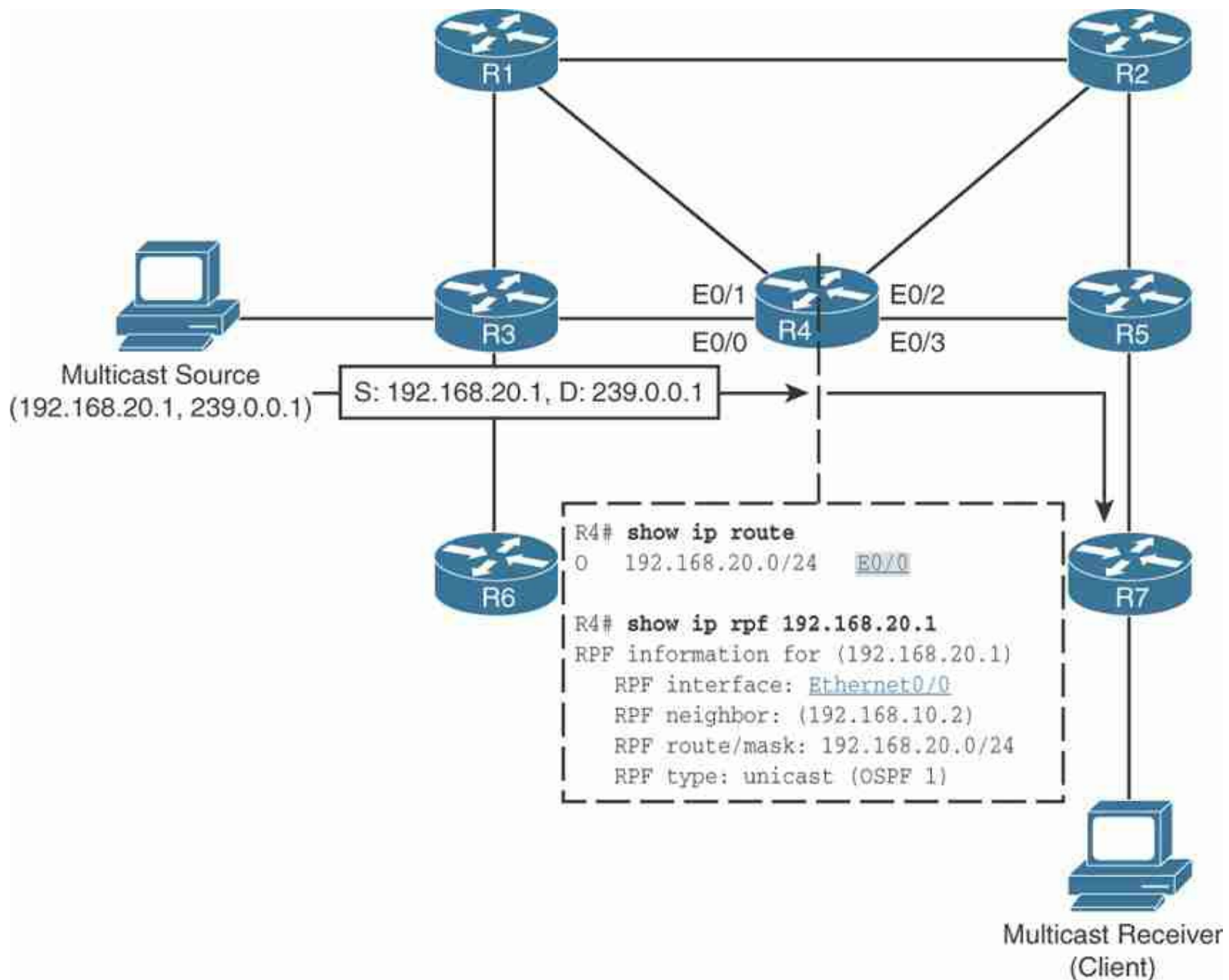


Figure 3-4 *Multicast Packet RPF Success*

[Figure 3-5](#) shows this same process, except that for some unknown reason the source multicast packet is received on interface E0/2. Using the same lookup information, the router can easily determine that the reverse unicast path for network 192.168.20.0 is not in fact located out E0/2. This indicates that there is likely a forwarding loop in the multicast topology. This can occur for many reasons, especially in the event of misconfiguration. The router drops this packet to prevent further propagation of the multicast packet loop. In this case, the PIM tree has served its purpose and prevented the router from creating or continuing a network loop.

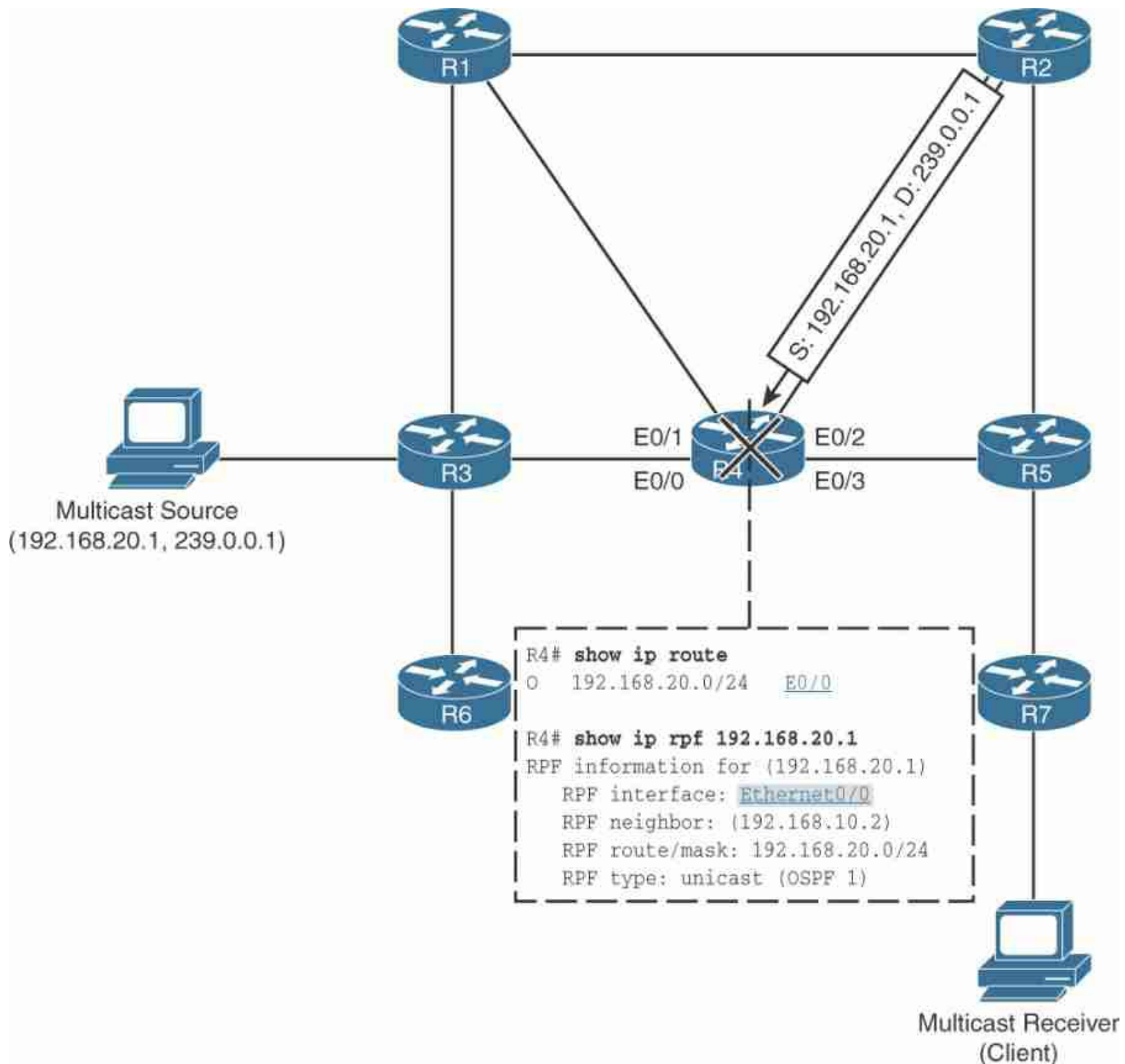


Figure 3-5 *Multicast Packet RPF Failure*

Note

This section is meant only to introduce you to the concept of RPF checking. For a more in-depth discussion of RPF checking and the associated mechanics, see [Chapter 5](#).

Two Types of Trees

The two types of trees used in multicast networks to control the distribution of traffic are *source trees* and *shared trees*. A source tree originates from the device sending the multicast messages. A shared tree originates from a device in the network called a rendezvous point (RP). The RP manages information about the sender or source of the multicast messages, but is not the real source of the tree. When a receiver is interested in getting multicast messages from a stream that uses a RP, its gateway

router contacts the RP, and, in turn, the RP matches the receiver with the real source of the tree. When the connection is made between the receiver and the source, the RP will no longer be in the path; more information on that in the following sections.

Source Trees (Shortest Path Trees)

The most common type of multicast tree is the source tree. A source tree is one in which the network routers build a tree for each (S,G) state in the network. From the perspective of the network, the root of the source tree is the router closest to the source. The tree extends from the root router in a direct path toward all of the group members. This creates a tree in which the path might look similar to the one shown in [Figure 3-6](#).

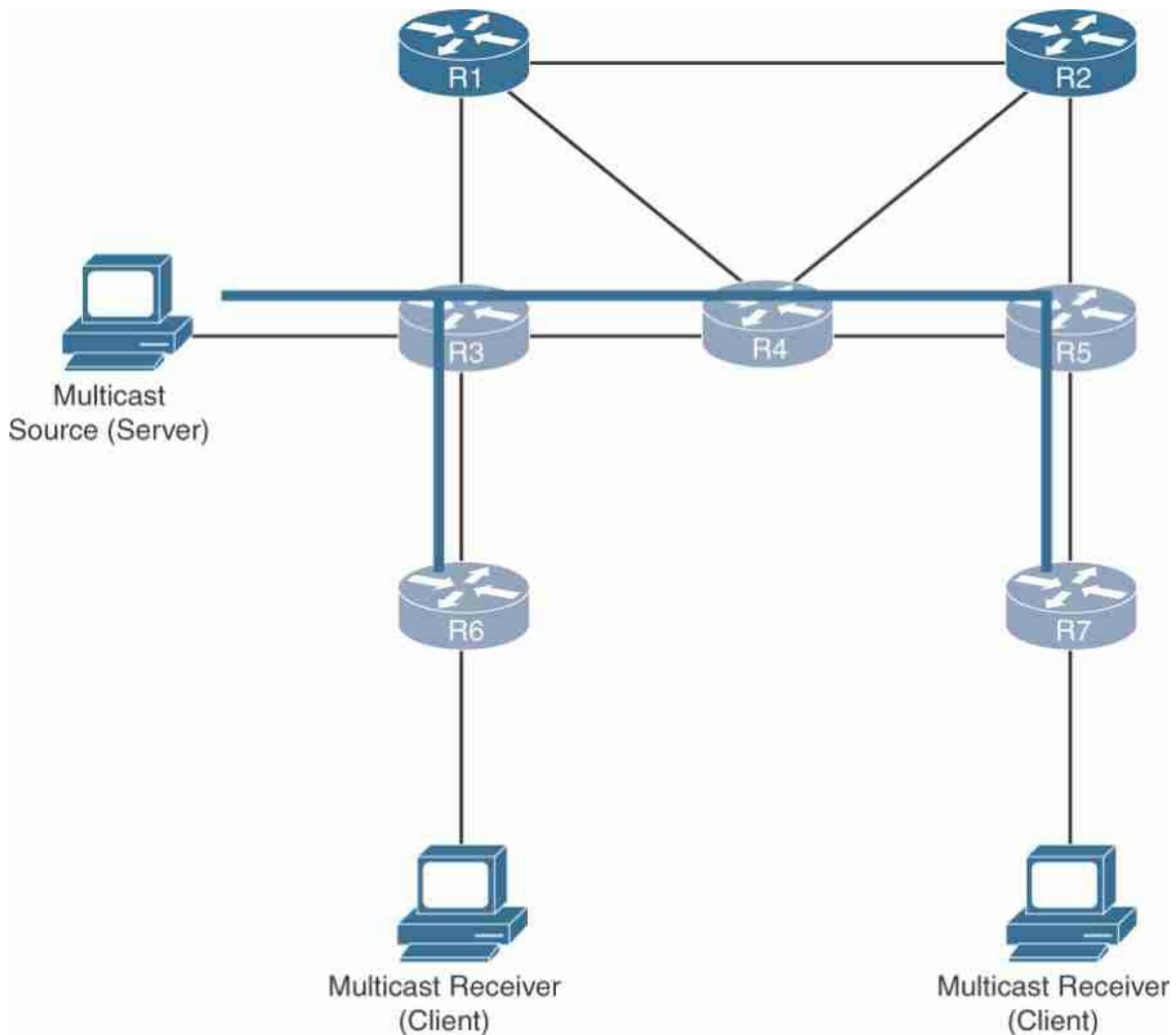


Figure 3-6 *Source Tree Path*

Network trees can sometimes be difficult to understand. This is true regardless of what protocols are building the tree. As shown in [Figure 3-6](#), the tree overlays the entire network. This is simple

enough; however, each router in the network must calculate the tree independently based on the information it has received, whether by dynamic protocol updates (PIM), or by configuration. From each router's perspective, the tree is smaller but has a similar structure built on a root, leaves, and replication points, or branches. In essence, the mroute table entries represent the smaller trees local to each router.

An incoming interface (the one closest to the source) and an OIL connect the source packets to the group members. Using reverse path forwarding, the router checks for loops and creates the OIL from the best (sometimes called shortest) paths from the unicast routing information base (RIB). For this reason, a source tree is also called the *shortest path tree*. The path is calculated and added to the mroute table, which is also called the *multicast routing information base (MRIB)*.

The tree can only be calculated if there is a valid, sending source; otherwise, it has no root. Any (S,G) installed in a router's MRIB essentially represents a source tree. If all the routers in the network have correctly calculated the direction of the source and all subscribed hosts, following the path should be as simple as following the packet path from each incoming interface through each OIL in the network.

A source tree will, of course, branch any time a router must forward group multicast packets for an OIL with more than one interface. After the local tree (the router's mroute table) is populated, it performs any necessary packet replication through the network tree. In the example network, the source tree is replicated only at R3, as depicted earlier in [Figure 3-6](#).

Note

Trees are usually drawn to only include Layer 3 (IP) path and replication. The tree does not generally include replication or flooding that may occur at Layer 2 of the OSI model, because an mroute entry is not required for this type of same-segment forwarding.

The network continues to build trees for each group and each source. Depending on the location of the packet sources and the group members, the shortest path may be different for each (S,G) pair. A separate and distinct tree is maintained for each unique (S,G) entry. [Figure 3-7](#) shows the same example network with two source trees, one for each (S,G) pair, for groups 239.1.1.1 and 239.2.2.2 respectively.

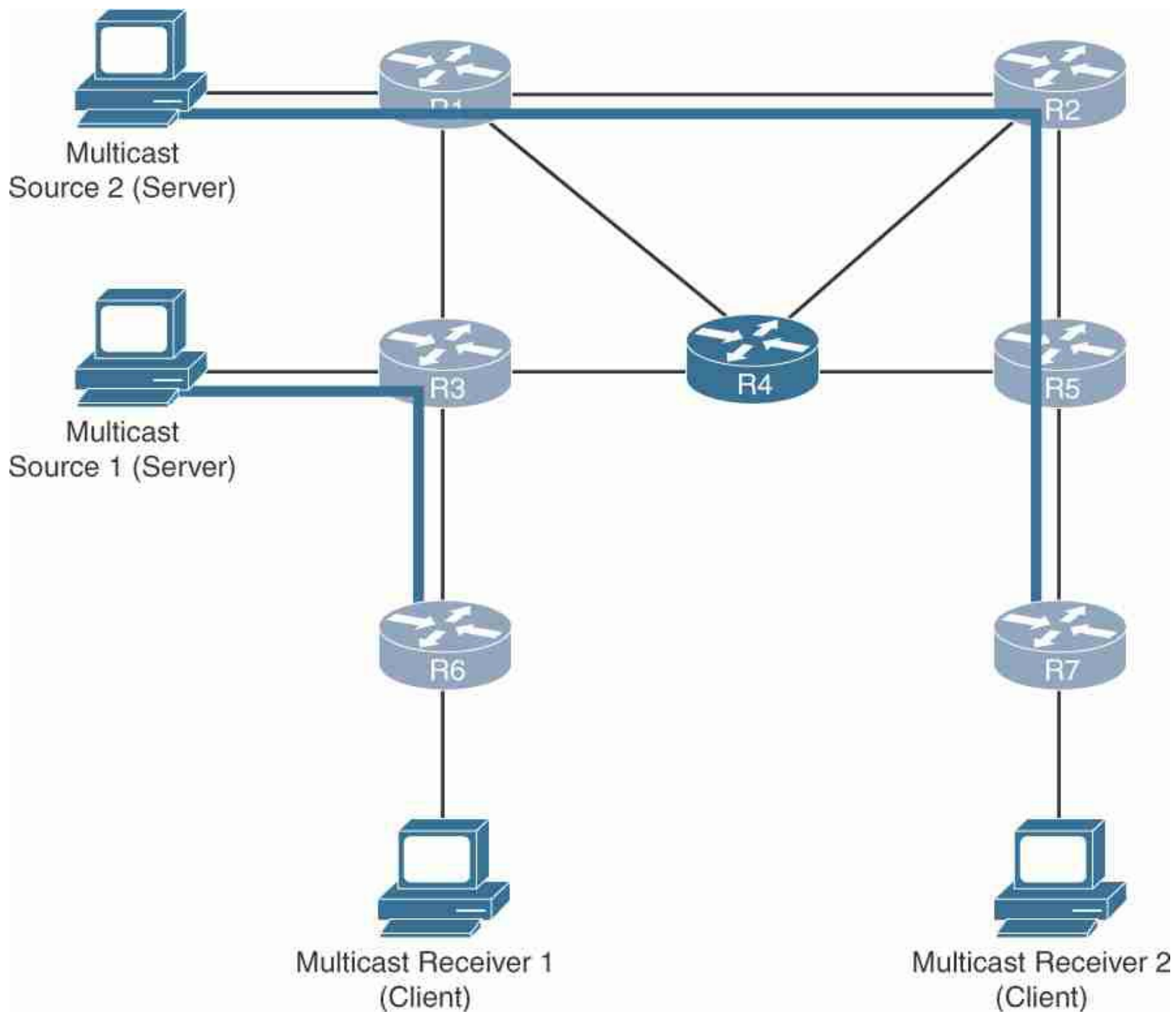


Figure 3-7 *Distinct Source Trees*

Shortest path trees are an efficient way to use bandwidth and links in an IP multicast network; however, depending on how the tree is constructed (or, in other words, which protocol is building the tree), the administrative and control-plane overhead can become overly burdensome. For this reason, the multicast engineers at the IETF introduced another type of network tree: the shared tree.

Shared Trees

A shared tree uses a different philosophy for distributing multicast packets. One problem with a source tree model is that each router in the path must share and maintain state information. Having a large number of groups, which consequently results in having a large number of trees, puts additional strain on control-plane resources.

The shared tree, on the other hand, uses a shared point in the network from which to build the distribution tree (this is used in PIM sparse mode). This location is called a rendezvous point or RP and is therefore the root of the tree. The $(*,G)$ state information is shared upstream from the IGMP

member routers to the RP and the routers in the path build the tree from the RP. Each router still uses reverse path forwarding (RPF) checks to keep the topology loop-free, but the RPF check is in the direction of the RP, not the source. [Figure 3-8](#) depicts a shared tree built from the root RP to the routers with host group members.

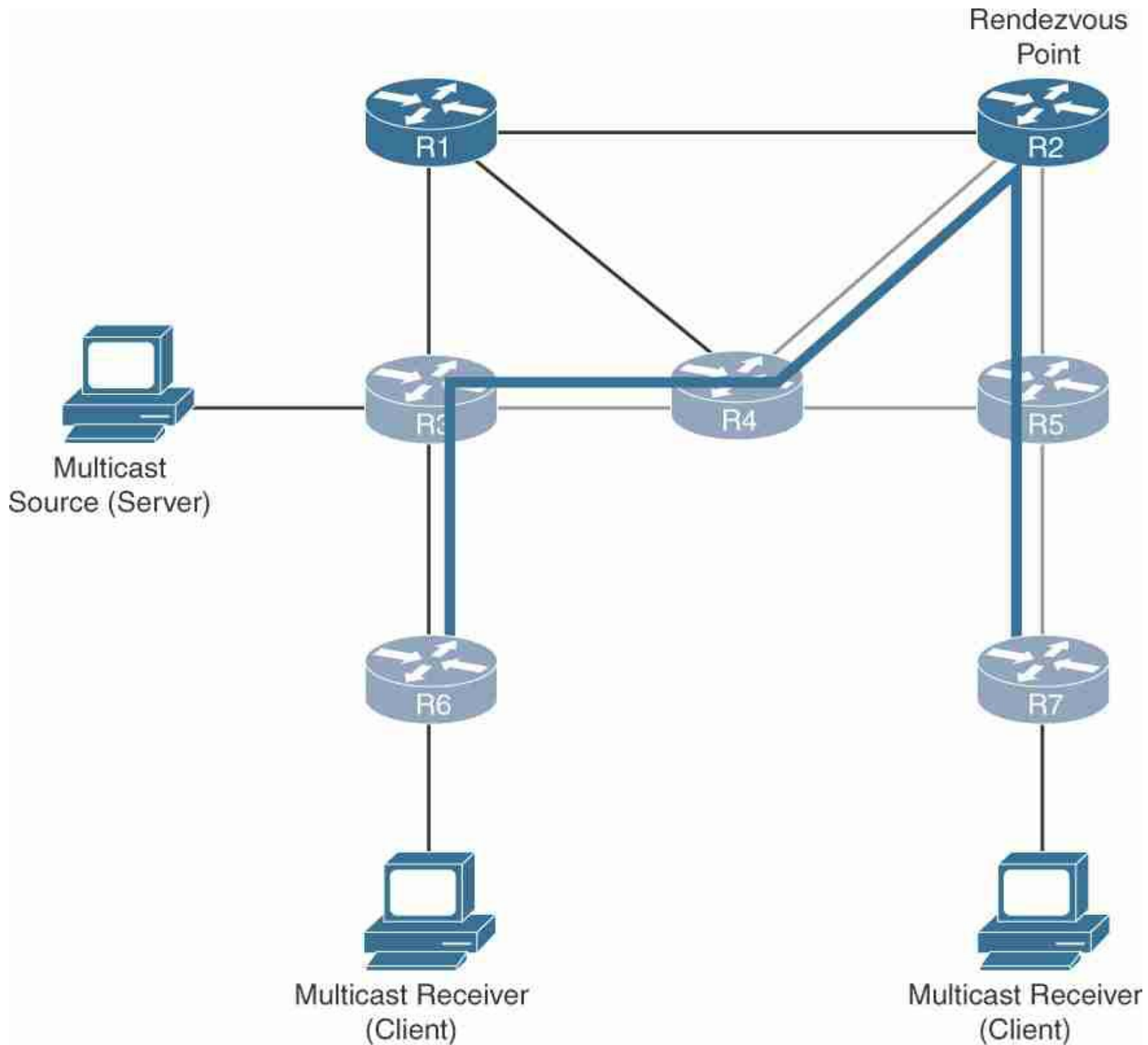


Figure 3-8 *Shared Tree*

Notice the RP placement has no relation to the IP multicast source. The RP may or may not be in the direct path (shortest path) between the group source and the group client(s). In fact, it can be placed outside of the path entirely. For this reason, a shared tree is also sometimes called a *rendezvous point tree (RPT)*.

The advantage to using this type of tree is that all non-RP routers can conserve control-plane resources during stream establishment or during host maintenance. Each (*,G) is a single tree, regardless of the location of the source and only one tree is required for the group, even if there are

many sources. This means that each non-leaf and non-RP router need only maintain the (*,G) for the entire group. In addition, the network path is pre-determined and predictable, which leads to consistent network operations.

Branches on a Tree

Branches are built when a router needs to send information to multiple destinations that require the use of multiple outgoing interfaces, such as, for example, more than one interface entry in the OIL. Remember, we are building a tree and must eliminate any loops that may cause duplicate information to be sent to a receiver.

Examining [Figure 3-9](#), we now see two receivers requesting information from the same multicast stream. Based on the shortest path learned from the routing protocol, R3 becomes a branch in the tree. In the case of the receiver connected to R6, the decision for R3 only has one option, which is to send the multicast information to R6. With the receiver connected to R7, R3 has the option of sending it to R1 or to R4 and needs to decide which neighbor will receive the stream. In this example, the best path to the client connected to R7 is via the R4-R5-R7 path.

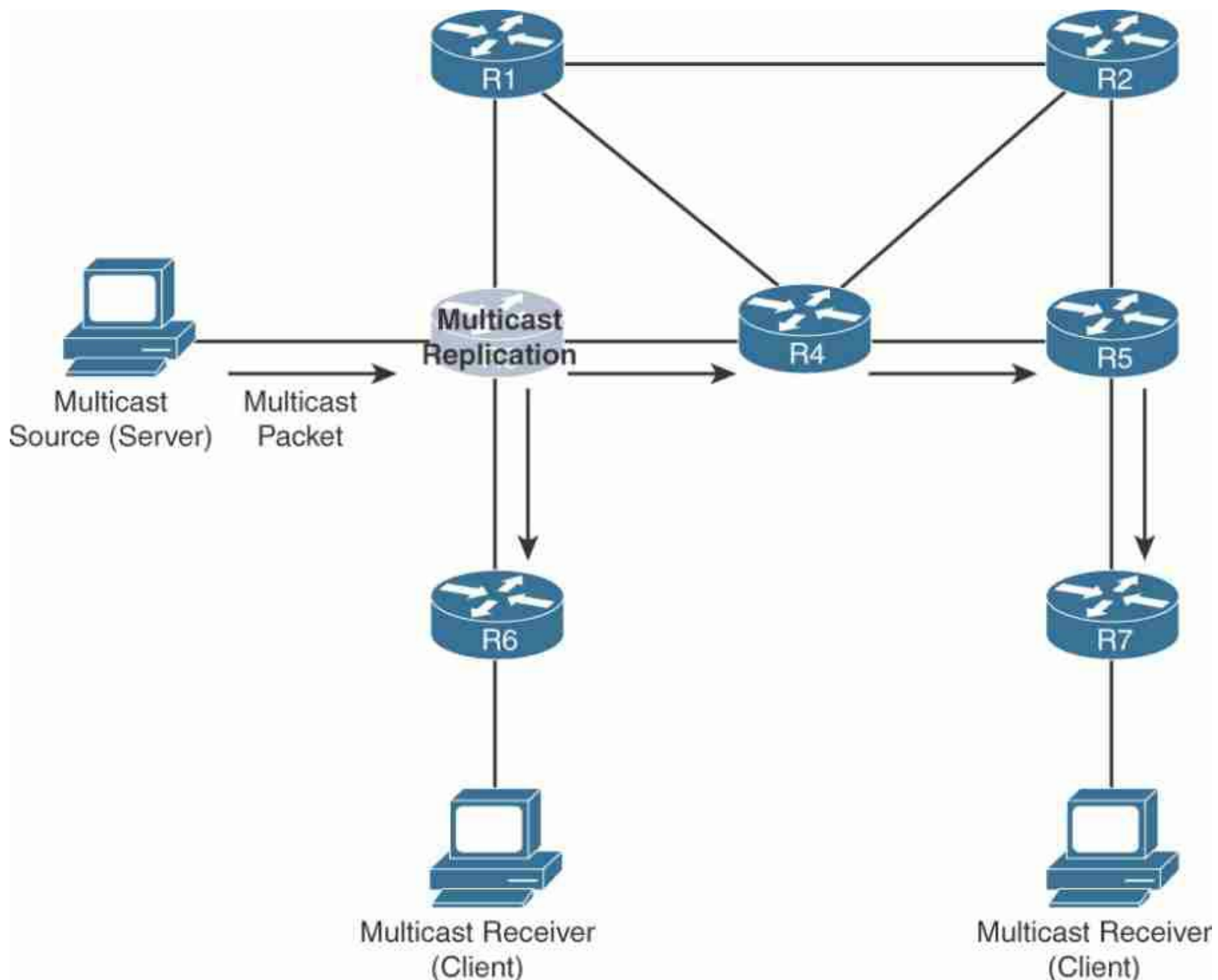


Figure 3-9 *Branches*

PIM Neighbors

The primary purpose of PIM is to share multicast source/receiver information and to build loop-free trees. With most unicast routing protocols, routers need to build neighbor relationships in order to share information. PIM has the same neighbor requirement as these unicast protocols. The PIM process begins when two or more routers establish a PIM neighbor adjacency on a given segment. To begin the adjacency process, any interfaces participating in multicast routing must be configured for PIM communications. In Cisco IOS, using the **ip pim** interface command with the selected PIM mode enables PIM communications on a specific interface.

Routers discover PIM neighbors by sending PIM hello messages to the link-local multicast address 224.0.0.13 out each PIM-enabled interface. The hello messages are sent every 30 seconds to refresh neighbor adjacencies. Hello messages also have a neighbor hold-time value, typically 3.5 times the hello interval, or 105 seconds. If this hold time expires without a refresh from the neighbor, the router assumes a PIM neighbor failure on that link and tears down the associated PIM adjacency. The router continues to send hello messages on any PIM-enabled links every 30 seconds to ensure adjacency occurs with any connected routers or if the failed router returns to service.

To improve security, an MD5 hash can be used to authenticate PIM hello messages with neighbors. You may also choose to add a PIM neighbor filter using an access control list (ACL). This provides you with the ability to control which devices you will receive PIM messages from.

In [Example 3-3](#), the output from R4 shows that it has learned about four PIM-enabled neighbor routers on interfaces Ethernet0/0 through Ethernet0/4. Each corresponding neighbor router also has PIM enabled on the interface connected to R4. Pay very close attention to the flags indicated by **Mode:** in the output.

Example 3-3 PIM Neighbors Table

[Click here to view code image](#)

R4#show ip pim neighbor

PIM Neighbor Table

Mode: B - Bidir Capable, DR - Designated Router, N - Default DR Priority,

P - Proxy Capable, S - State Refresh Capable, G - GenID Capable,

L - DR Load-balancing Capable

Neighbor Address	Interface	Uptime/Expires	Ver	DR Prio/Mode
192.168.54.5	Ethernet0/0	06:25:54/00:01:33	v2	1 / DR S P G
192.168.42.2	Ethernet0/1	06:27:10/00:01:20	v2	1 / S P G
192.168.41.1	Ethernet0/2	06:27:10/00:01:28	v2	1 / S P G
192.168.43.3	Ethernet0/3	06:27:10/00:01:34	v2	1 / S P G

Designated Routers

A single network interface and IP address may connect to a multipoint or broadcast network, such as Ethernet. If the network is PIM-enabled, many neighbor adjacencies can occur, one for each PIM-enabled IP neighbor. If every PIM router on the segment were to attempt to manage the tree state, confusion and excess traffic could occur. Similar to OSPF, IETF versions of PIM use a designated router (DR) on each segment. Much like an OSPF DR, the PIM DR manages state information and PIM updates for all the routers on the segment.

The DR selection process occurs when all neighbors on a segment have replied to PIM hello messages. Each router on a segment selects one router to function as the DR. The DR router is chosen using the PIM DR priority parameter, where the highest priority router is selected as the DR. The DR priority is configurable and sent in the PIM hello message. If the DR priority value is not supplied by all routers, or the priorities match, the highest IP address is used to elect the DR, which is the IP address of the interface sending the PIM message. By default, all interfaces have a PIM DR priority of 1. All routers on the segment should select the same DR router, consequently avoiding conflicts.

When the DR receives an IGMP membership report message from a receiver for a new group or source, the DR creates a tree to connect the receiver to the source by sending a PIM join message out the interface toward the rendezvous point, when using any-source multicast (ASM) or bidirectional PIM (BiDir), and to the source when using source-specific multicast (SSM). When the DR determines that the last host has left a group or source, it sends a PIM prune message to remove the path from the distribution tree.

To maintain the PIM state, the last-hop DR (the DR for the router pair closest to the group receivers using IGMP) sends join-prune messages once per minute. State creation applies to both (*, G) and (S, G) states as follows:

- **State creation of (*,G) tree:** IGMP-enabled router receives an IGMP (*, G) report. The last-hop DR sends a PIM join message toward the RP with the (*,G) state.

- **State creation of (S,G) tree at source:** Router receives multicast packets from the source. The DR sends a PIM register message to the RP. In the case of the (S,G) entry, this is the DR closest to the source or the first-hop DR.

- **State creation of (S,G) tree at receivers:** IGMP-enabled router receives an IGMP (S,G) report. The last-hop DR sends a PIM join message toward the source.

[Figure 3-10](#) shows the efficiency of using designated routers on a LAN segment in a PIM sparse-mode (PIM-SM) design. In the diagram, the connecting interface on SW1 has a default PIM DR priority of 1, whereas SW2's interface has a configured priority of 100.

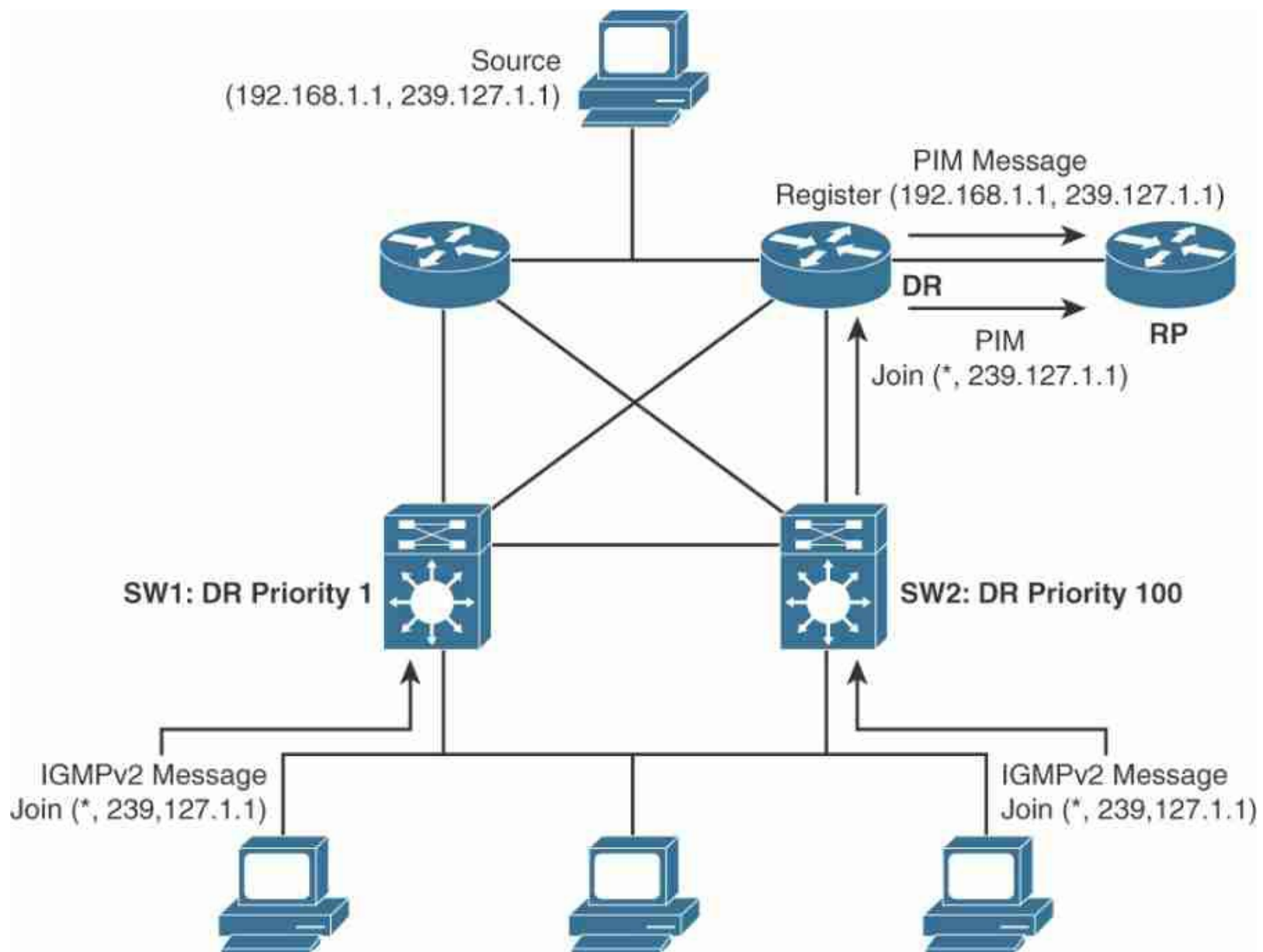


Figure 3-10 *PIM DR Election*

Note

Both SW1 and SW2 are operating in L3/routed mode.

SW2 is selected as DR, and only one PIM register message is sent to the rendezvous point (RP).

The configured DR priority and DR-elected router address for each interface can be shown by issuing the command **show ip pim interfaces** in IOS/XE or **show pim interface** in IOX-XR. For NX-OS, use the **show ip pim neighbors** command instead. The output in [Example 3-4](#) is from an IOS router. Notice the **DR Prior** field and the **DR** address fields in the output.

Example 3-4 *Displaying the DR Priority and DR-Elected Router Address for Each Interface*

[Click here to view code image](#)

CR1#show ip pim interface

Address	Interface	Ver/	Nbr	Query	DR	DR
	Mode	Count	Intvl	Prior		
192.168.63.3	Ethernet0/0	v2/D	1	30	1	192.168.63.6
192.168.43.3	Ethernet0/1	v2/S	1	30	1	192.168.43.4
192.168.31.3	Ethernet0/2	v2/S	1	30	1	192.168.31.3

PIM Messages: Join, Leave, Prune, Graft, and Assert

Routers build trees using multicast state information shared between all the IP multicast routers in the network. Different types of state messages help the router build the tree appropriately. These messages draw their nomenclature from tree-related words: join, leave, prune, and graft.

The most current version of PIM for IPv4 is PIMv2, whereas PIM6 is the current version for IPv6. When PIM is enabled globally on Cisco routers, the protocol defaults to PIMv2 and PIM6. PIM uses negotiated neighbor adjacencies similar to OSPF to build a network of routers for multicast information sharing.

There are several types of PIM messages, but all contain the same PIM message format or header, as shown in [Figure 3-11](#) , the key fields for which are described in the list that follows:

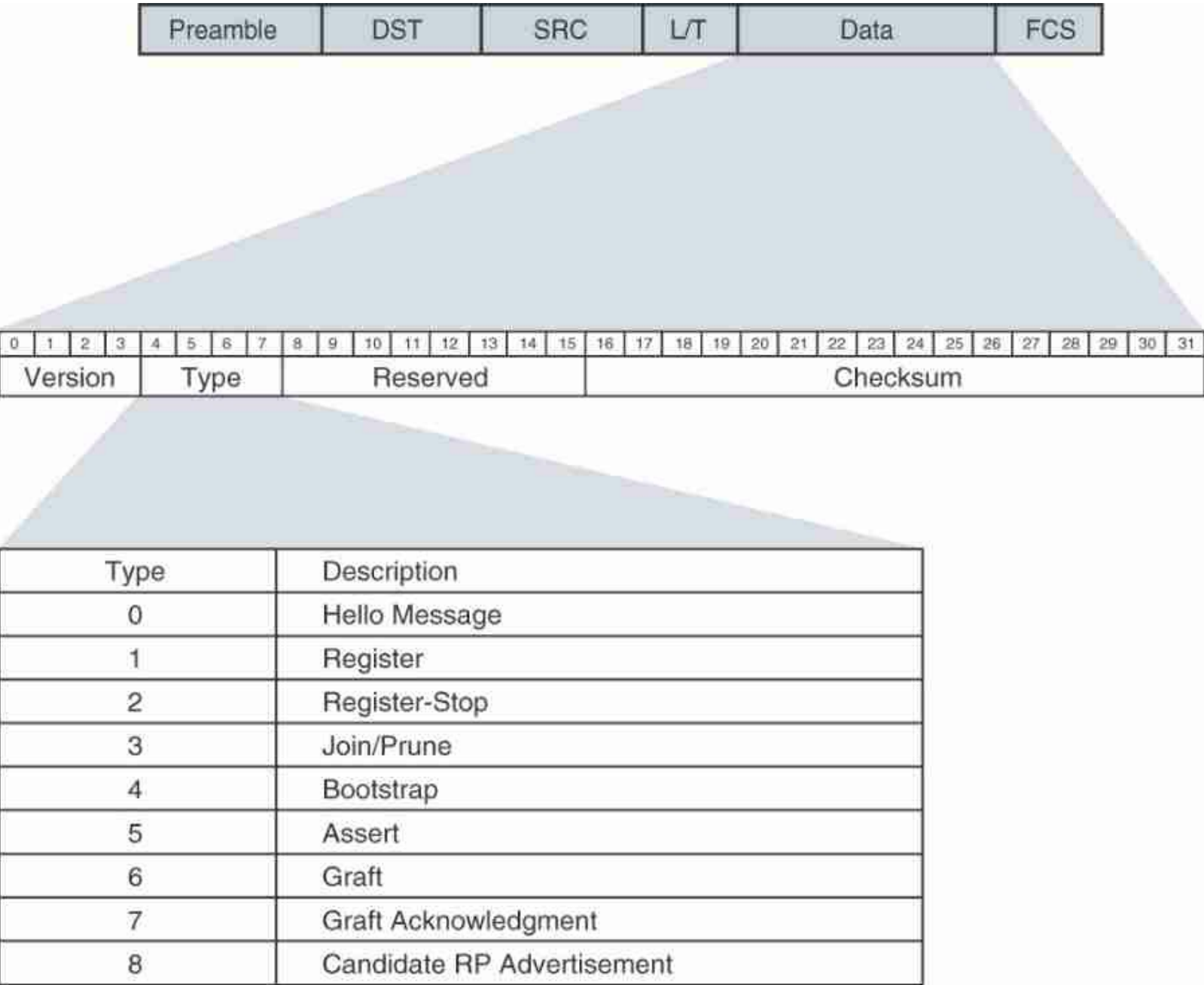


Figure 3-11 *PIM Message Format*

- Version is the PIM version used
- Type:

1. Hello message sent to all PIM neighbors and may contain a hold-time value and/or the LAN prune delay time. The hold-time signifies how long the neighbor will keep the neighbor relationship without receiving an update. The LAN prune delay is used on multi-access networks for propagation delay.
2. A register message is sent by a DR or a PIM multicast border router (PMBR) to the RP, indicating that a source is sending traffic to a particular multicast group.
3. The register-stop message is a unicast message sent by the RP to the DR after the SPT is created.
4. Join/prune messages are sent by routers to the upstream sources and RPs. As the name indicates, join messages are sent to join a stream and prune messages to leave a stream.
5. Bootstrap messages contain several entries for the election of the bootstrap router (BSR).
6. Assert messages are used to elect the PIM forwarder.
7. Graft messages are used in a PIM-DM environment to indicate the desire to receive a multicast stream.
8. A graft acknowledge or graft-ack is sent to the graft originator upon receipt of an (S,G) entry.
9. A candidate RP advertisement is used to send advertisements to the BSR.
10. Checksum is a one's complement sum of the entire message.

Routers and L3 switches running PIM send periodic hello messages on PIM-enabled interfaces to discover neighbors. These messages are sourced with the IP address of the outgoing interface and are sent to all PIM routers with the multicast destination address of 224.0.0.13. They contain information such as the hello period time, hello hold-time, capabilities, IP address, and so on. The packet capture in [Example 3-5](#) demonstrates a hello message. Pay special attention to the highlighted areas, the destination multicast addresses at Layers 2 and 3, the PIM version (v2), the DR priority, and the PIM message type (Hello).

Example 3-5 *Hello Message Packet Capture*

[Click here to view code image](#)

Ethernet Packet: 72 bytes

Dest Addr: 0100.5E00.000D, Source Addr: AABB.CC00.0430

Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x5, TOS: 0xC0 (Prec=Internet Contrl)

Length: 58, ID: 0x04CB, Flags-Offset: 0x0000

TTL: 1, Protocol: 103 (PIM), Checksum: 0xE818 (OK)

Source: 192.168.43.4, Dest: 224.0.0.13

PIM Ver:2, Type:Hello, Reserved: 0, Checksum: 0x23E1 (OK)

Type: 1, Length: 2, Holdtime: 105

Type: 20, Length: 4, GEN_ID: -389360719

Type: 19, Length: 4, DR Priority: 1

Type: 21, Length: 4, State Refresh: 16777216

The packet capture in [Example 3-6](#) shows an example of a join/prune message for the group 239.1.1.1. The output also indicates the source of the multicast stream (192.168.8.8) and the number of sources joined to the group. Although the destination address is the multicast address of

224.0.0.13, the PIM header contains the destination IP address, which is 192.168.43.3, and indicates a type of join/prune (see highlighted sections). This message was sent from 192.168.43.4 to 192.168.43.3.

Example 3-6 *Join/Prune Message Packet Capture*

[Click here to view code image](#)

Ethernet Packet: 68 bytes

Dest Addr: 0100.5E00.000D, Source Addr: AABB.CC00.0430

Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x5, TOS: 0xC0 (Prec=Internet Contrl)

Length: 54, ID: 0x0508, Flags-Offset: 0x0000

TTL: 1, Protocol: 103 (PIM), Checksum: 0xE7DF (OK)

Source: 192.168.43.4, Dest: 224.0.0.13

PIM Ver:2, Type:Join/Prune, Reserved: 0, Checksum: 0x308C (OK)

Addr Family: IP, Enc Type: 0, Uni Address: 192.168.43.3

Reserved: 0, Num Groups: 1, HoldTime: 210

Addr Family: IP, Enc Type: 0, Reserved: 0, Mask Len: 32

Group Address:239.1.1.1

Num Joined Sources: 1, Num Pruned Sources: 0

Joined/Pruned Srcs:

Addr Family: IP, Enc Type: 0, Reserved: 0

S: 1, W: 0, R:0, Mask Len: 32

Source Address:192.168.8.8

Join

When a host wishes to join an IP multicast stream, it does so with a join message. A join for IGMPv2 includes the (*,G) entry, and the router that receives the initial join prepares to join the larger network tree. Using a PIM join/prune message, the router shares the join with other upstream neighbors, joining the router to the larger network tree. This process requires the addition of a rendezvous point (RP) that will be discussed shortly.

Leave and Prune

When a host no longer wishes to receive the stream, it can issue a leave message to the upstream router. If the upstream router has no other downstream hosts with a desire to receive the stream, it will remove the associated (*,G) from the MRIB and send a PIM prune message to neighboring routers. A prune may also be sent if IGMP has not updated the upstream router's join timer, or if it is a non-leaf router that is not a part of the network path. As the name implies, a prune essentially cuts the router from any paths in the large network tree.

Graft

Because multicast networks are dynamic, a previously pruned router may need to rejoin the stream. In the case of dense-mode operation, the pruned router sends a graft message to upstream neighbors, informing the network that downstream host(s) once again desires the stream. The graft message is sent during the three-minute prune expiration time; otherwise, the router will send another join message.

Assert

Finally, what happens to a tree when there are two or more routers upstream from a single host or server? Remember, the purpose of multicast is to create more efficient forwarding over loop-free topologies. Having two routers sending replicated packets for a single stream while also managing upstream protocol communication would defeat this purpose. In a situation where multiple routers have the same distance and metric values, one of the routers must have a way to assert control over that part of the tree. The router with the highest IP address is the winner and the assert message tells upstream routers which router should forward the stream. Any router that loses the assert will suppress multicast tree building for each asserted group until such time as the assert expires, which may be caused by a failure in the asserting router.

The way routers use joins, leaves, prunes, grafts, and asserts (if at all) depends entirely on the type of multicast routing protocol being used in the network. This subsection serves only to introduce the terms as they relate to a network tree. For detailed information on how these messages are used to build complex topologies, refer to [Chapter 7](#), “[Operating and Troubleshooting IP Multicast Networks](#).”

PIM Modes

PIM uses unicast routing information to forward multicast messages. One of the key elements is the “P” in PIM. “Independent” indicates that any routing protocol can be used to build multicast trees and forward to the appropriate destination. Forwarding of the multicast messages can operate in the following modes: dense, sparse, and sparse-dense.

PIM Dense-Mode

In dense-mode (DM) operation, multicast messages are flooded throughout the entire DM network. When a downstream router receives the multicast packet but does not have a downstream receiver interested in the multicast stream, it responds with a prune message. Consequently, multicast messages are flooded and then pruned approximately every three minutes. This can be a serious issue for large networks or networks with slower connection speeds as in a wide-area network (WAN). The amount of traffic generated may cause network outages.

This does not mean that DM is bad. It means that you need to use the proper tool for the job. DM is extremely easy to implement because there is no need to have a RP and consequently no (*,G) entries. Where DM makes the most sense is in a small network with high-speed connections. The use of the PIM-DM state refresh feature keeps pruned state from timing out, allowing even greater scalability.

The IETF implementation of PIM dense-mode (PIM-DM) is often referred to as a push model for tree building. The name is derived from the fact that all dense-mode nodes assume that all other PIM neighbors have an interest in each source tree. The PIM-DM router floods (pushes) (S,G) path updates for each PIM neighbor, regardless of expressed interest. PIM-DM routers may have to process and maintain unnecessary PIM updates, making PIM dense-mode very inefficient for most multicast implementations. The use of dense mode assumes a very limited number of sources and a network in which each subnet is likely to have at least one multicast receiver for each (S,G) tree in the MRIB (multicast routing information base).

In the push model, joins, prunes, and grafts are very important to maintaining a tree. After the (S,G) tree is flooded throughout the network, any dense-mode routers without downstream listeners will prune themselves from the tree. The router places the removed path in a pruned state in the MRIB, and the path is not added to the mFIB (multicast forwarding information base).

PIM-DM routers that maintain the (S,G) state information must continue to flood multicast traffic (forwarding) down tree branches that have not been pruned. During the convergence of the PIM-DM network, unwanted traffic may reach routers that do not require the multicast stream, resulting in these routers sending the traffic to the bit bucket.

The pruning PIM-DM routers send PIM prune messages back upstream toward neighbors, and those neighbors then prune that particular tree branch, preventing traffic from flooding on that branch. This flood prune process repeats every three minutes. Excessive traffic flooding and PIM management processing make PIM dense-mode very inefficient and undesirable in larger networks.

Tip

In dense mode, the Layer 3 device assumes that all routers in the Layer 3 domain configured with PIM dense-mode needs the multicast stream, consequently forwarding to all Layer 3 devices. If there are no directly connected members, the router will send a prune message to the router connected to the source.

When a new receiver on a previously pruned branch of the tree joins a multicast group, the PIM DM device detects the new receiver and immediately sends a graft message up the distribution tree toward the source. When the upstream PIM-DM device receives the graft message, it immediately places the interface on which the graft was received into the forwarding state so that the multicast traffic begins flowing to the receiver.

All of this constant communication to maintain the source trees can be burdensome on the network. For this reason, PIM dense-mode is not optimal and suitable for large-scale deployment. The next generation Cisco network devices do not have the support for PIM dense-mode. (Nexus platform does not support PIM dense mode.)

PIM Sparse-Mode

Protocol independent multicast sparse-mode (PIM-SM) is a protocol that is used to efficiently route multicast packets in the network. In multicast terminology, PIM sparse-mode is an implementation of any-source multicast (ASM). It interacts with the router's IP routing table to determine the correct path to the source of the multicast packets (RPF). It also interacts with IGMP to recognize networks that have members of the multicast group. As the name implies, it does not depend on any particular unicast routing protocol. Based on the RFC 4601, the basic mechanism of PIM-SM can be summarized as follows:

A router receives explicit join/prune messages from those neighboring routers that have downstream group members. The router then forwards data packets addressed to a multicast group (G) only onto those interfaces on which explicit joins have been received.

A designated router (DR) sends periodic join/prune messages toward a group-specific rendezvous point (RP) for each group for which it has active members. The RP acts as the gatekeeper and meeting place for sources and receivers. In a PIM-SM network, sources must send their traffic to the RP. For this to work, the router must know where the RP is located, meaning it must know the IP address of the RP for a specific multicast group. Every group for which the router is receiving joins must have a group-to-RP mapping. To show this mapping in IOS, use the command **show ip pim rp-mapping**, as shown in [Example 3-7](#).

Example 3-7 PIM RP to Group Mappings

[Click here to view code image](#)

```
R7#show ip pim rp mapping
PIM Group-to-RP Mappings
```

```
Group(s) 224.0.0.0/4
```

```
RP 192.168.0.2 (?), v2v1
```

```
Info source: 192.168.0.2 (?), elected via Auto-RP
```

```
Uptime: 1d15h, expires: 00:02:53
```

After the traffic is sent to the RP, it is then forwarded to receivers down a shared distribution tree. By default, when the last-hop router in the tree (the router closest to the receiver) learns about the source, it verifies the best reachability path to the source from unicast routing table. If a path exists, then it will send a join message directly to the source, creating a source-based distribution tree from the source to the receiver. This source tree does not include the RP unless the RP is located within the shortest path between the source and receiver.

Each router along the path toward the RP builds a wildcard (any-source) state for the group and sends PIM join messages on towards the RP. When a data source first sends to a group, its DR (designated router, also the FHR) unicasts register messages to the RP with the source's data packets encapsulated within. The RP sends a register stop message towards the FHR, the FHR then sends the multicast data packets towards the RP. The RP forwards the source's multicast data packets down the RP-centered distribution tree toward group members (receivers for the multicast group). The last-hop router verifies the existence of a more optimal path to the source. This depends on the interior gateway protocol (IG) and congruent PIM relationships along the path. If such a path exists, a (S,G) join is sent, and the flow moves to the alternate path as defined by the IGP. Finally, the last-hop router sends a prune to the RP.

[Figure 3-12](#) and the list that follows review a step-by-step approach of a tree build-up using the multicast flow in the figure.

.....> S,G: Shortest Path Tree
 —> *,G: Shared Tree

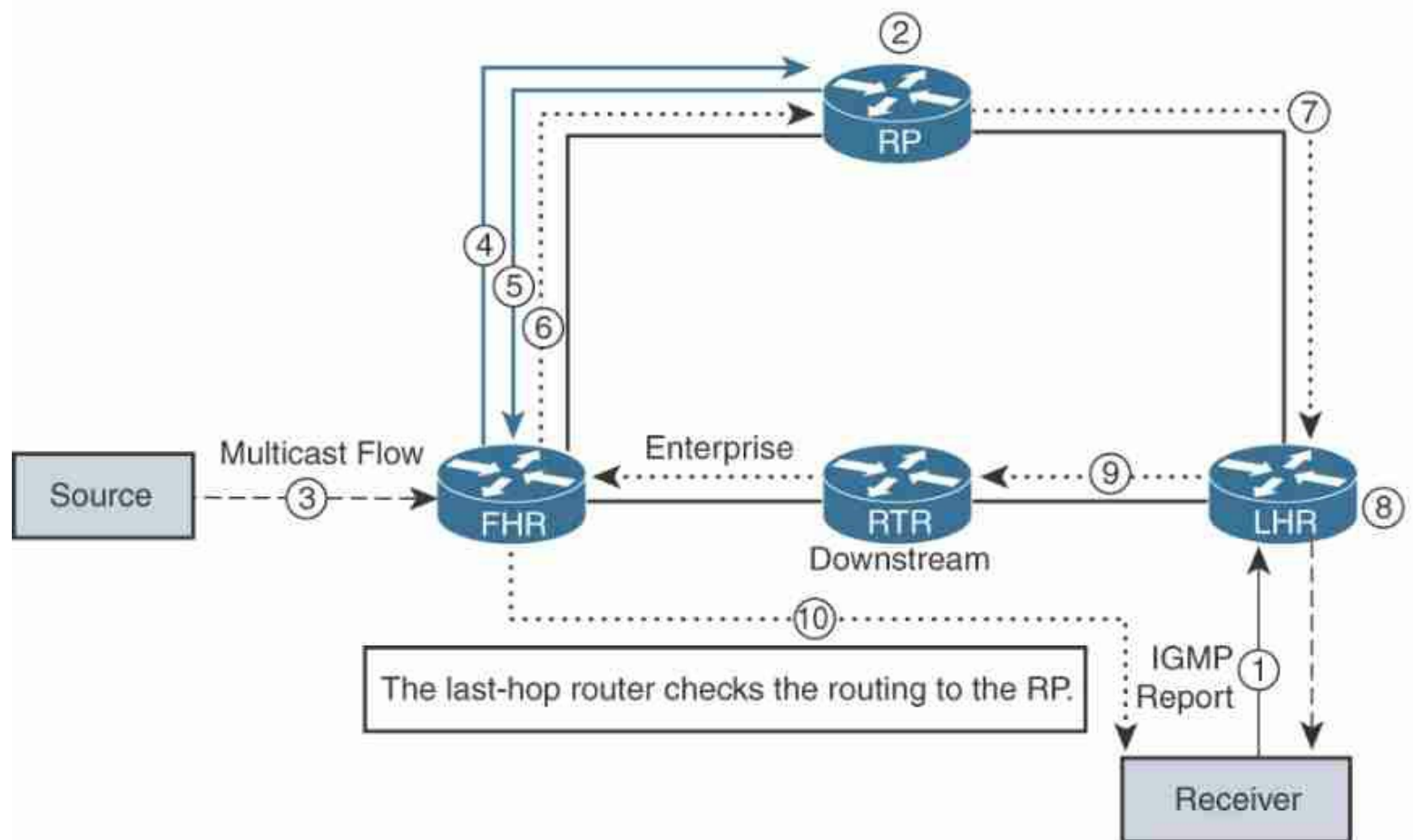


Figure 3-12 *PIM Sparse-Mode Tree Build-Up Process*

Step 1. Receiver sends an IGMP request to the last-hop router (LHR). The LHR reviews its configuration for RP information for the requested group and sends a request in PIM control protocol towards the RP. The RP is a gatekeeper that holds all information on active multicast receivers and sources.

Note

PIM sparse-mode and this process is based on RFC 4601.

The communication between the LHR and the RP is represented as a shared tree and is referred to as a (*, G).

Step 2. The (*, G) update is used to provide the RP information of an active receiver for the multicast group.

Step 3. The source sends the multicast stream to the first-hop router (FHR).

Step 4. The FHR, configured with PIM sparse-mode and relevant RP information, encapsulates the multicast stream in a unicast packet called the register message and sends it to the RP.

Step 5. In this example, the RP already has an interested receiver that wants to join the multicast group using the (*, G) entry. Consequently, the RP sends an (S,G) PIM join towards the FHR. It also sends a register stop message directly to the FHR.

Step 6. After receiving the register stop message, the FHR sends the multicast stream to the RP as an (S,G) flow.

Step 7. Based on the interested receiver, the outgoing interface list is populated, and the (S,G) flow is sent to the LHR via the PIM-enabled interfaces aligned to the unicast routing table. From the LHR, the multicast packet flows to the receiver.

Step 8. At the LHR, the router checks its unicast routing table to reach the source.

Step 9. If the check verifies an alternate path is more optimal based on the unicast routing protocol, the (S,G) join is sent to the source from the LHR.

Step 10. The source sends the multicast flow to the LHR via the (S,G) state and is created in the multicast routing table. When the data flow has migrated to the shortest path tree (S,G), the multicast LHR sends an RP prune message towards the RP.

PIM Sparse-Dense Mode

Sparse-dense mode was developed to address the issue of distributing RP information in some dynamic RP mapping environments. For example, when using the Cisco dynamic RP mapping protocol, Auto-RP, the address of the RP is distributed using dense mode, and additional multicast streams would use the RP as a shared tree.

One of the challenges with sparse-dense mode occurs during RP failure. If the network is not configured appropriately, all the multicast traffic reverts to DM, consequently causing undue stress on the network. If the RP fails, any new sessions requiring the interaction of the RP will also fail. Using sparse-dense mode (PIM-SD) is one method to ensure delivery of multicast messages for new connections. In the event of a RP failure, the multicast mode of operation reverts to or falls back to the least common denominator, which is dense mode (PIM-DM) flooding of multicast messages.

The behavior in a PIM-DM configuration is to flood multicast messages. Routers in the multicast network without downstream clients interested in receiving the multicast traffic send a prune message. This becomes a constant flood and prune, which may overwhelm a network with many senders and/or limited bandwidth. Caution should always be used any time PIM-DM could be initiated.

Tip

Because of this additional burden of maintaining dense-mode source trees during misconfiguration or router failures, the authors do not recommend using sparse-dense mode. Additional tools and features have been developed for Cisco routers to avoid dense-mode.

For this reason, it is desirable to have high availability of RP systems. There are many ways to achieve this, depending on the PIM mode in use. For information about dynamic RP mapping and propagation, as well as RP redundancy, please refer to [Chapters 4](#) and [5](#).

Multicast Flow at the Leaf

The overall multicast flow is a combination of the Layer 2 and 3 environments. IGMP is the most common mechanism to control multicast messages at Layer 2, and the PIM control plane is used to manage multicast traffic to flow from the source to the receiver at Layer 3. Let's examine the flow of the multicast tree establishment for a sender and receiver multicast flow at the last-hop router (LHR).

To understand this, let's look at a slightly different example network, shown in [Figure 3-13](#). There are three routers that makeup the network, R1, R2, and R3. R3 is connected to VLAN 8, and this is

where the source of the multicast stream is located. R2 is connected to VLAN 7, and this is where the client is located.

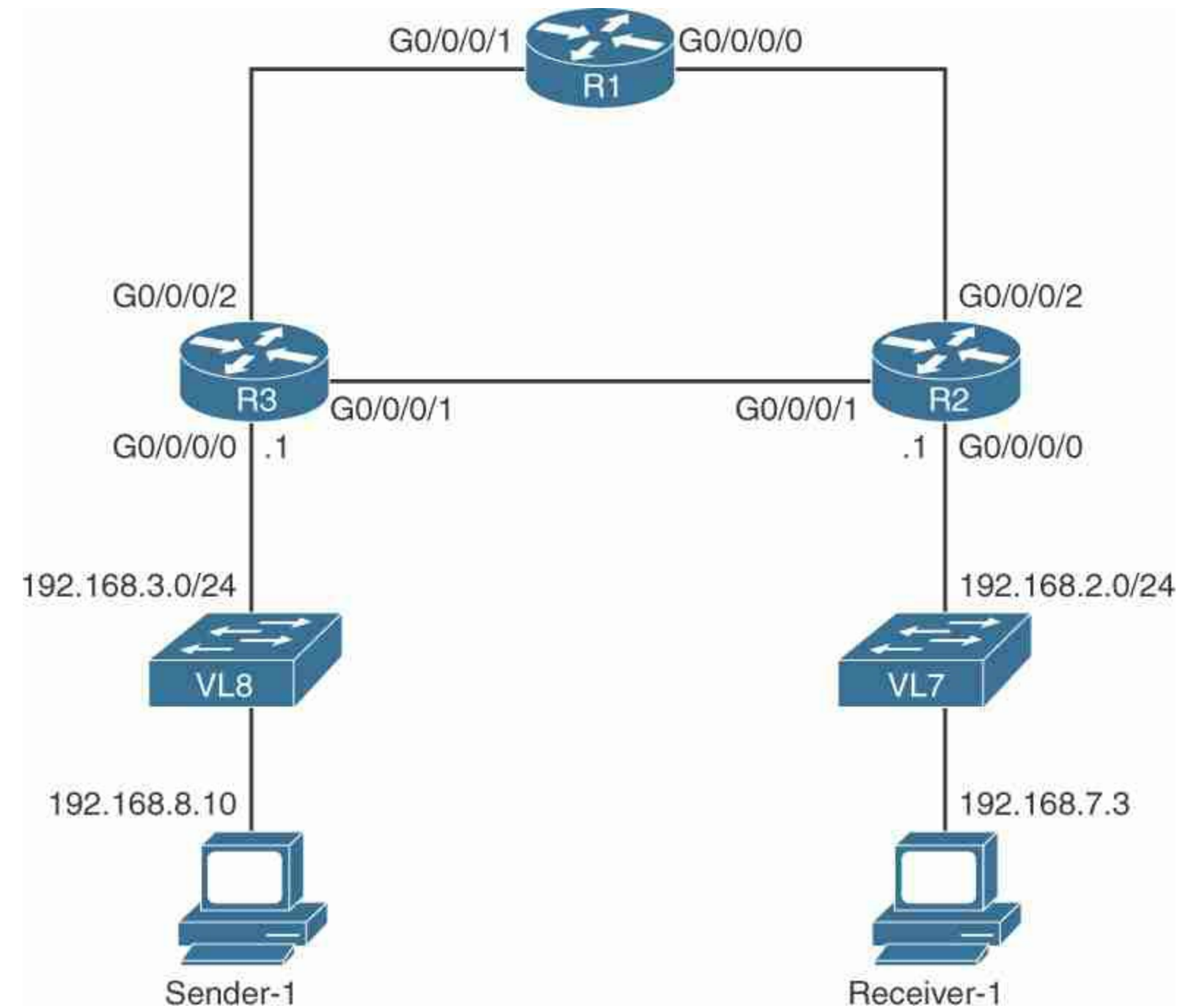


Figure 3-13 *IGMPv2 Example*

In this example, the source Sender-1 is sending a multicast stream to 224.64.7.7. Because the network is configured using PIM dense-mode, the multicast group is flooded and pruned throughout the network. Using the **show ip mroute 224.64.7.7 verbose** command demonstrated in [Example 3-8](#), you can see that the 'P' flag has been set, indicating that the multicast route has been pruned.

Example 3-8 *Verifying a Multicast Route Has Been Pruned*

[Click here to view code image](#)

```
R2#show ip mroute 224.64.7.7 verbose
```

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
V - RD & Vector, v - Vector

Outgoing interface flags: H - Hardware switched, A - Assert winner

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(*, 224.64.7.7), 00:12:38/stopped, RP 0.0.0.0, flags: D

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list:

GigabitEthernet0/0/2, Forward/Dense, 00:12:38/stopped

GigabitEthernet0/0/1, Forward/Dense, 00:12:38/stopped

(192.168.8.10, 224.64.7.7), 00:09:37/00:01:18, flags: PT

Incoming interface: GigabitEthernet0/0/1, RPF nbr 192.168.32.3

Outgoing interface list:

GigabitEthernet0/0/2, Prune/Dense, 00:00:35/00:02:24, A

When a host is interested in receiving a particular multicast stream, it sends an unsolicited membership report to the multicast group it is interested in joining. In our example, it is Receiver-1 (192.168.7.3).

Note

In this example, we have turned off IGMP snooping on the switch for VLAN 7 to better explain the effect of IGMP on the router.

The high-level steps involved for the host to begin receiving the multicast stream are as follows:

Step 1. The host (Receiver-1) sends a multicast IGMP join report to a group address, 224.64.7.7.

Step 2. The router receives the report from Receiver-1, and, using the **debug ip igmp** command on R2, we can see the logging message as shown here:

[Click here to view code image](#)

IGMP(0): Received v2 Report on GigabitEthernet0/0/0 from 192.168.7.3 for 224.64.7.7

Step 3. The (*,G) entry (*, 224.64.7.7) is added to the *multicast routing table (MRT)* —not to be confused with the *maximum response timer*. The entry matches packets from any source as long as the packet is received from an interface that matches the reverse path forwarding (RPF) check. In dense mode, the route is 0.0.0.0. The output is shown using the **debug ip mrouting** command on R2:

MRT(0): Update (*,224.64.7.7), RPF /0.0.0.0

Step 4. The router now updates the multicast routing information base (MRIB) table with the outgoing interface, this is where Receiver-1 is located and the outbound interface list (OIL) is added. The following output is generated using the **debug ip mrrib route** command on R2.

[Click here to view code image](#)

MRIB: Table = Default, Trans: (192.168.8.10, 224.64.7.7) midb update: ADD, IDB = GigabitEthernet0/0/0, next hop = 0.0.0.0

Step 5. The multicast forwarding information base (MFIB) for IPv4 is now updated with the appropriate source address (S,G) learned from PIM dense-mode. This can be viewed using the **debug ip mfib ps** (Cisco IOS) command:

[Click here to view code image](#)

```
MFIBv4(0x0): (*,224.64.7.7) MRIB update[1]: C K (Modified: )
MFIBv4(0x0): (*,224.64.7.7) GigabitEthernet0/0/0 MRIB update: RF F NS
(Modified: RF NS)
MFIBv4(0x0): (192.168.8.10,224.64.7.7) MRIB update[1]: K DDE
(Modified: )
MFIBv4(0x0): (192.168.8.10,224.64.7.7) GigabitEthernet0/0/0 MRIB
update: RF F NS (Modified: RF NS)
```

Step 6. Finally, the router performs an RPF check to validate where the sender (Sender-1) is located. In this example, the interface is GigabitEthernet0/0/1. To view the following output, use the **debug ip mrib route** command on R2.

[Click here to view code image](#)

```
MRIB: Table = Default, Trans: (192.168.8.10, 224.64.7.7) mroute update:
MOD FLAGS, rpf = GigabitEthernet0/0/1, flag = C0
MRIB: Table = Default, Trans: (192.168.8.10, 224.64.7.7) igmp update
local interest, rpf = GigabitEthernet0/0/1
MRIB: Table = Default, Trans: (*, 224.64.7.7) mroute update: MOD FLAGS,
rpf = Null, flag = 84
MRIB: Table = Default, Trans modify entry: (*,224.64.7.7/32) flags:
set C , success
MRIB: Table = Default, Trans: (*, 224.64.7.7) igmp update local
interest, rpf = Null
```

Now that the multicast stream is being received by Receiver-1, we can verify the IGMP group membership on R2 using the **show ip igmp groups** command, as demonstrated in [Example 3-9](#) .

Example 3-9 *Verifying IGMP Group Membership*

[Click here to view code image](#)

```
ASR1K-2#show ip igmp groups
IGMP Connected Group Membership
Group Address   Interface          Uptime    Expires    Last Reporter  Group
Accounted
224.64.7.7      GigabitEthernet0/0/0  00:05:47  00:02:44  192.168.7.3
```

We can also validate the active state for group 224.64.7.7, because the “P” flag has now been removed. This is shown using the **show ip mroute** command, as demonstrated in [Example 3-10](#) .

Example 3-10 *Validating Active State for a Multicast Group*

[Click here to view code image](#)

```
ASR1K-2#show ip mroute 224.64.7.7
IP Multicast Routing Table
Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
```

L - Local, P - Pruned, R - RP-bit set, F - Register flag,
T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
V - RD & Vector, v - Vector

Outgoing interface flags: H - Hardware switched, A - Assert winner

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 224.64.7.7), 00:05:19/stopped, RP 0.0.0.0, flags: C

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list:

GigabitEthernet0/0/2, Forward/Dense, 00:05:19/stopped

GigabitEthernet0/0/1, Forward/Dense, 00:05:19/stopped

GigabitEthernet0/0/0, Forward/Dense, 00:05:19/stopped

(192.168.8.10, 224.64.7.7), 00:05:19/00:01:34, flags: T

Incoming interface: GigabitEthernet0/0/1, RPF nbr 192.168.32.3

Outgoing interface list:

GigabitEthernet0/0/0, Forward/Dense, 00:05:19/stopped

GigabitEthernet0/0/2, Prune/Dense, 00:00:39/00:02:20, A

Note

It is important to understand the flags used in the state output given by the **show ip mroute** command. Here is an explanation of these flags as used in IOS:

Common flags:

■ **D** : PIM state is being processed as a dense-mode flow.

■ **S** : PIM state is being processed as a sparse-mode flow.

■ **B** : PIM state is being processed as a bidirectional flow.

■ **SSM** : PIM state is being processed as a source-specific mode flow.

■ **T** : SPT bit set, meaning the router has moved the flow to an (SPT).

■ **C** : Connected, when a group member is connected to the router.

■ **L** : Local, when a router itself is considered a member of the group.

■ **P** : Pruned, a multicast group state has been pruned by PIM but has not yet aged from the mroute table.

■ **R** : RP-bit is set, indicating that the (S,G) flow is currently pointed toward the RP.

■ **F** : Register flag appears when the source is being registered to the RP (appears on the FHR).

■ **J** : The router has processed a PIM join for the SPT state.

Leaving an IGMP Group

When using IGMPv1, there is no such thing as a leave process. When a host is no longer interested in receiving a multicast stream, it just stops processing the multicast information, but that doesn't stop

traffic from being sent! The only way a router can determine if there are valid hosts interested in the multicast stream is by sending query messages. Query messages are generally sent every 60 seconds and typically require a timeout interval of three query messages. This means that multicast traffic is being sent for an additional three minutes to an uninterested host. Yes, that is a terrible waste of bandwidth!

[Example 3-11](#) shows a packet capture of a membership query message sent from the router to the devices on the subnet.

Example 3-11 *IGMP Membership Query Message Packet Capture*

[Click here to view code image](#)

Internet Protocol Version 4, Src: 192.168.7.1, Dst: 224.0.0.1

Version: 4

Header length: 20 bytes

Differentiated Services Field: 0xc0 (DSCP 0x30: Class Selector 6; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))

Total Length: 28

Identification: 0x2116 (8470)

Flags: 0x00

Fragment offset: 0

Time to live: 1

Protocol: IGMP (2)

Header checksum: 0xf05f [validation disabled]

Source: 192.168.7.1

Destination: 224.0.0.1

Internet Group Management Protocol

[IGMP Version: 1]

Type: Membership Query (0x11)

Header checksum: 0xeeff [correct]

Multicast Address: 0.0.0.0 (0.0.0.0)

IGMPv2 and v3 are significantly more efficient at leaving a group. An IGMPv2 or v3 host can send a leave message to all routers on the subnet, indicating that it no longer wants to receive a multicast stream. Upon receipt of the leave message, the router stops forwarding the multicast traffic. [Example 3-12](#) shows a packet capture of a host leave message.

Example 3-12 *IGMP Host Leave Message Packet Capture*

[Click here to view code image](#)

Internet Protocol Version 4, Src: 192.168.7.3 Dst: 224.0.0.2

Version: 4

Header length: 24 bytes

Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))

Total Length: 32

Identification: 0x6d72 (28018)

Flags: 0x00

Fragment offset: 0

Time to live: 1

Protocol: IGMP (2)

Header checksum: 0x0fb8 [validation disabled]

Source: 192.168.7.3

Destination: 224.0.0.2

Options: (4 bytes), Router Alert

Internet Group Management Protocol

[IGMP Version: 2]

Type: Leave Group (0x17)

Max Response Time: 0.0 sec (0x00)

Header checksum: 0x01b8 [correct]

Multicast Address: 224.64.7.7

The Rendezvous Point and Shared Tree Dynamics

Let's take a much closer look at sparse-mode operations and the significance of the rendezvous point. In sparse mode, the RP of a network is always a router. The RP router is not a packet source but simply a location from which to root and calculate a loop-free topology. The RP can also offload the processing of network joins and leaves during the initial setup of a multicast stream, saving network resources by concentrating them in a single location. In the example network shown in [Figure 3-14](#), the clients notify last-hop routers (LHR) (R6 and R7) via IGMP of the intention to join a group. The LHR routers forward PIM join messages toward the RP, establishing the (*,G) state and the LHR as leaves in the shared tree.

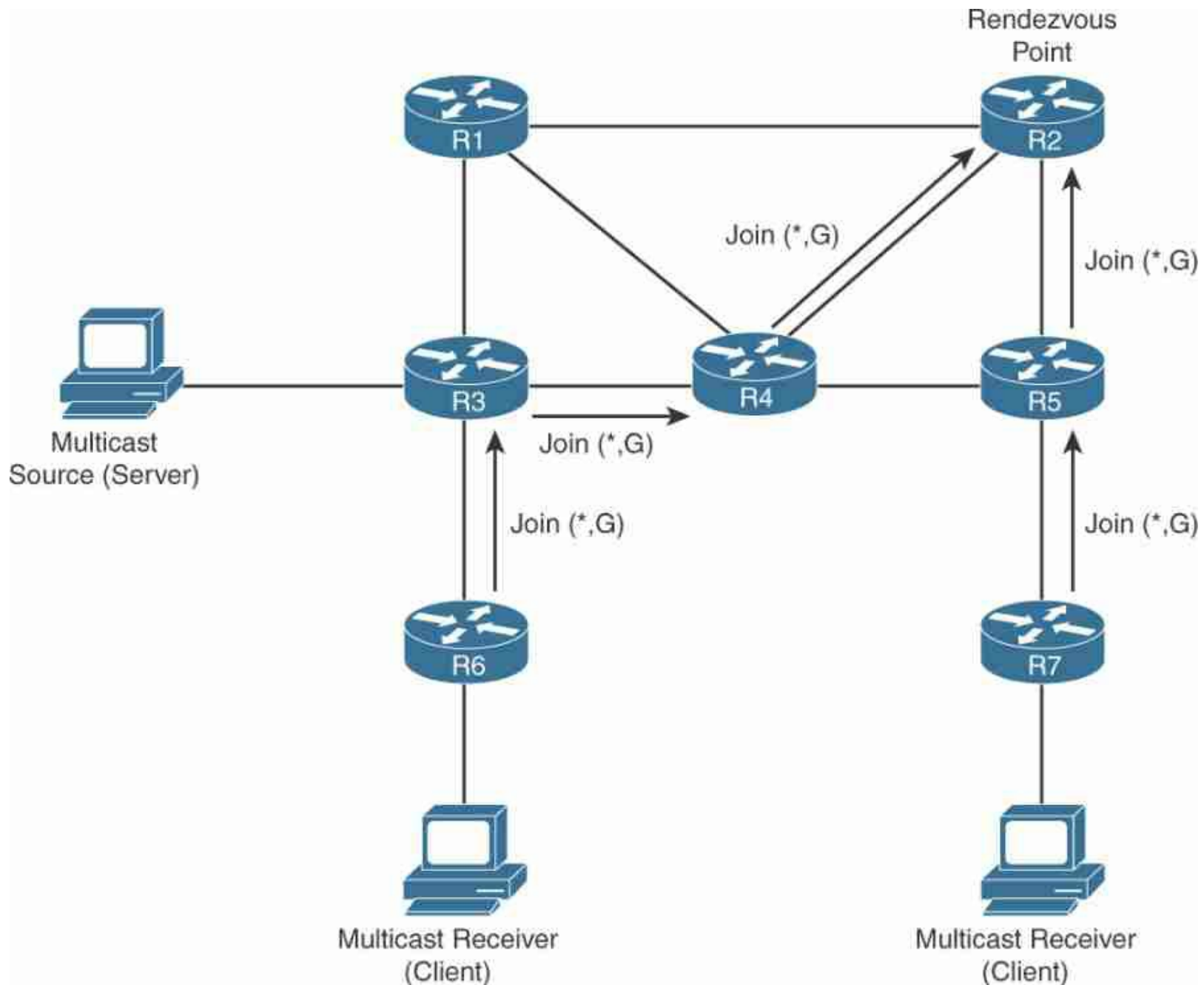


Figure 3-14 *Forwarding Joins Toward the RP*

Each router in the path between the RP and the LHR (which hosts receivers) follows the same procedure, recording the incoming interface of the join message. The router RPF checks each join against the location of the RP and adds the joined interface to the OIL. The incoming interface, of course, will be the multicast-enabled interface with the shortest path to the RP. The join messages stop at the RP for the shared tree, or if the receiving router already has an (S,G) entry for the group in the (*,G) join message. Using the **show ip mroute** command, you can view the conditions of both the (S,G) and (*,G) entries, as demonstrated in [Example 3-13](#).

Example 3-13 *Displaying the (S,G) and (*,G) Entry Condition*

[Click here to view code image](#)

```
R4#show ip mroute 239.2.2.2
```

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
 L - Local, P - Pruned, R - RP-bit set, F - Register flag,
 T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.2.2.2), 00:00:16/stopped, RP 172.16.1.1, flags: SPF

Incoming interface: Ethernet1/0, RPF nbr 10.1.3.1

Outgoing interface list: Null

(10.11.11.11, 239.2.2.2), 00:00:16/00:02:43, flags: FT

Incoming interface: Loopback0, RPF nbr 0.0.0.0, Registering

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:00:16/00:03:13

At this point in the process, a shared tree has been constructed from the RP to the gateway routers (LHRs) in the path. For the example network in [Figure 3-14](#) , the shared tree flows from the root at the RP to R7. R3 would already have an (S,G) and would therefore share the SPT with R6, and after it receives packets from a multicast source destined to the group, it begins replicating and forwarding packets to R6. Remember, the shared tree is represented by the (*,G) entries of the routers in the path and at the RP.

When a source begins sending packets to the multicast flow, the DR interface on the FHR (R3, or the router closest to the source) needs to notify the RP that a stream needs forwarding and a source tree needs to be built. It does this by sending a PIM source register message to the RP. The RP needs this information in order to join the source tree at the source. The RP is now joined to the SPT, and packets can be forwarded to the RP, the root of the shared tree for further replication and forwarding. [Figure 3-15](#) depicts this process.

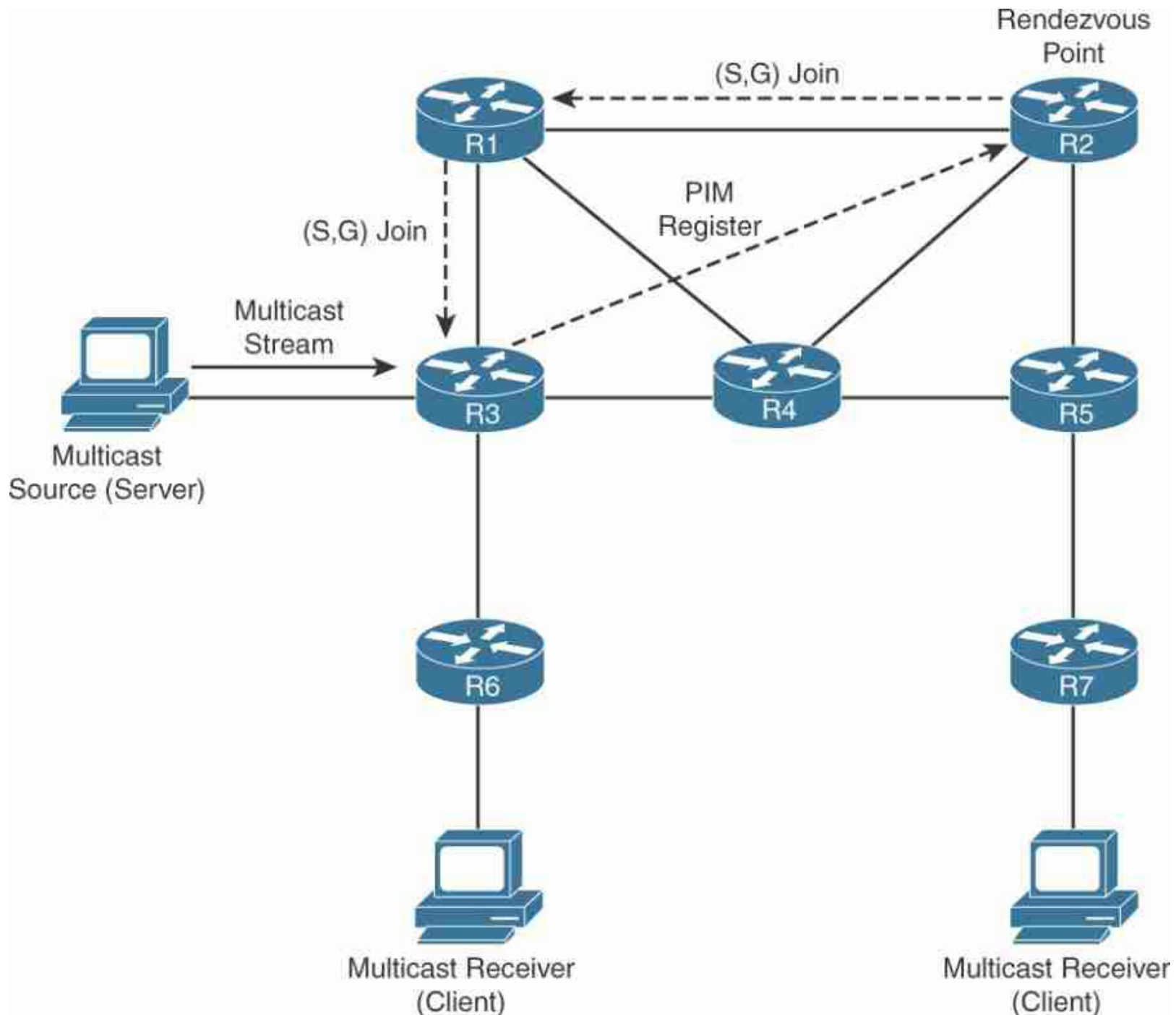


Figure 3-15 *Source Registration Process*

There is a small gap in this explanation: How then are packets forwarded from the source to the shared tree before the RP has joined the SPT? If the multicast source (server) still sends the first multicast packets toward its gateway router (in this case R3, the FHR) and R3 forwards the packets toward the RP, a serious problem could occur. For example, if the packet is forwarded through R1, but R1 did not get a PIM message yet and will therefore not have a (*,G) or (S,G) entry in its mroute table, it would drop the packet! Sending the multicast packet toward the RP through normal multicast forwarding will not work until the RP is joined to the SPT.

To resolve this problem, the router closest to the source must encapsulate the multicast packet inside of a unicast packet with the IP address of the rendezvous point as the IP destination. In other words, the multicast stream must be tunneled from the router at the source to the RP.

When configuring multicast on a router, you may have noticed the following message:

%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

This appears when the router learns about the RP and creates the tunnel interface. Upon further inspection using the **show interfaces tunnel 0** command, you see the destination of the tunnel is the RP 192.168.0.2 and sourced from a local interface, as demonstrated in [Example 3-14](#) .

Example 3-14 show interfaces tunnel 0 Command Output

[Click here to view code image](#)

```
R3#show interfaces tunnel 0
```

```
Tunnel0 is up, line protocol is up
```

```
Hardware is Tunnel
```

```
Description: Pim Register Tunnel (Encap) for RP 192.168.0.2
```

```
Interface is unnumbered. Using address of Ethernet0/2 (192.168.31.3)
```

```
MTU 17912 bytes, BW 100 Kbit/sec, DLY 50000 usec,
```

```
reliability 255/255, txload 1/255, rxload 1/255
```

```
Encapsulation TUNNEL, loopback not set
```

```
Keepalive not set
```

```
Tunnel source 192.168.31.3 (Ethernet0/2), destination 192.168.0.2
```

```
Tunnel Subblocks:
```

```
src-track:
```

```
Tunnel0 source tracking subblock associated with Ethernet0/2
```

```
Set of tunnels with source Ethernet0/2, 1 member (includes iterators),  
on interface <OK>
```

```
Tunnel protocol/transport PIM/IPv4
```

```
Tunnel TOS/Traffic Class 0xC0, Tunnel TTL 255
```

```
Tunnel transport MTU 1472 bytes
```

```
Tunnel is transmit only
```

```
Tunnel transmit bandwidth 8000 (kbps)
```

```
Tunnel receive bandwidth 8000 (kbps)
```

```
Last input never, output 00:17:06, output hang never
```

```
Last clearing of "show interface" counters 00:21:34
```

```
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
```

```
Queueing strategy: fifo
```

```
Output queue: 0/0 (size/max)
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
```

```
5 minute output rate 0 bits/sec, 0 packets/sec
```

```
0 packets input, 0 bytes, 0 no buffer
```

```
Received 0 broadcasts (0 IP multicasts)
```

```
0 runts, 0 giants, 0 throttles
```

```
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
```

```
4 packets output, 1216 bytes, 0 underruns
```

```
0 output errors, 0 collisions, 0 interface resets
```

```
0 unknown protocol drops
```

```
0 output buffer failures, 0 output buffers swapped out
```

Another item of interest is the number of output packets. This indicates multicast packets that were tunneled to the RP.

The tunnel is known as the *Encap Tunnel* . To make the encapsulation process more efficient, the

outer unicast header that is added to the multicast is actually the PIMv2 register message that is sent to the router. In this way, the multicast and the PIM register arrive at the RP at the same time. The RP strips the PIM register encapsulation from the multicast packets and forwards the packets to any receivers using the shared tree state(s). [Figure 3-16](#) depicts the tunneling, replication, and subsequent forwarding of the multicast stream with packet capture output from the links between R1, the RP, and R5.

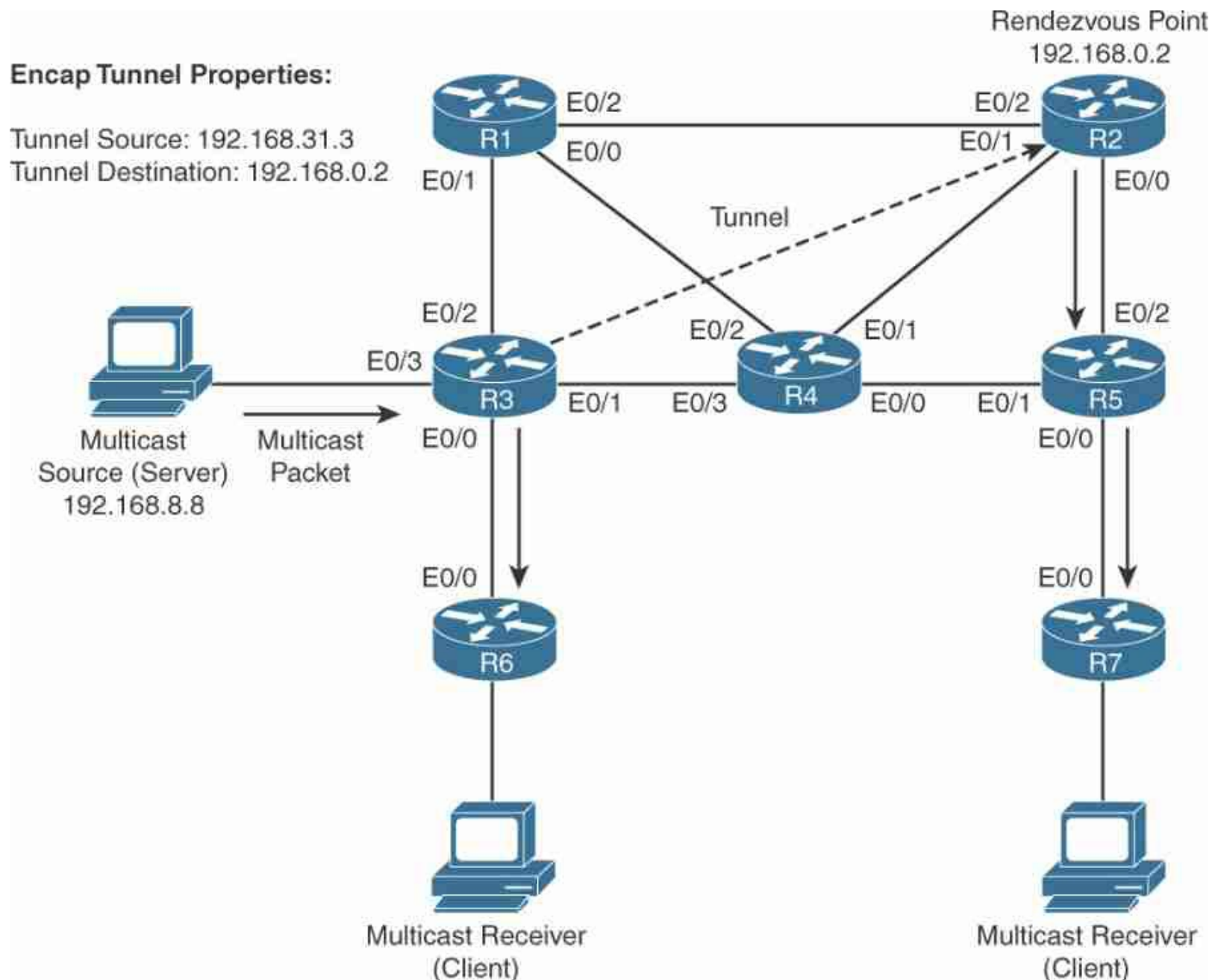


Figure 3-16 *Shared Tree Forwarding*

This process of encapsulating packets in the tunnel is only temporary. When there is a source tree (S,G) established between the FHR and the RP, the RP actually begins to receive two copies of each multicast packet, one from the now native source tree, and one inside the unicast encapsulation. This is obviously not the ideal state for the traffic to be in. When this condition occurs, the RP sends a register stop message via PIM to the FHR encapsulating the flow. This tells the FHR, in this case R3, to use only the native source tree to the RP to forward packets. The register stop message conditions are as follows:

- When the RP has receivers in its multicast entry table (*,G) for the new multicast group, the RP

sends an (S,G) join message towards the FHR and then sends a register stop message directly to the FHR.

■ When the RP has no receivers, it discards the multicast group register message and sends a register stop message towards the FHR. Each FHR maintains a register suppression timer that is initiated by the register stop message. Upon expiration of this timer, the FHR again sends a register packet to the RP to verify whether there are any active receivers for that group, provided the multicast source is also still active.

The register stop message with packet capture is shown in [Figure 3-17](#) . This tells the FHR, in this case R3, to use the native source tree to the RP to forward packets.

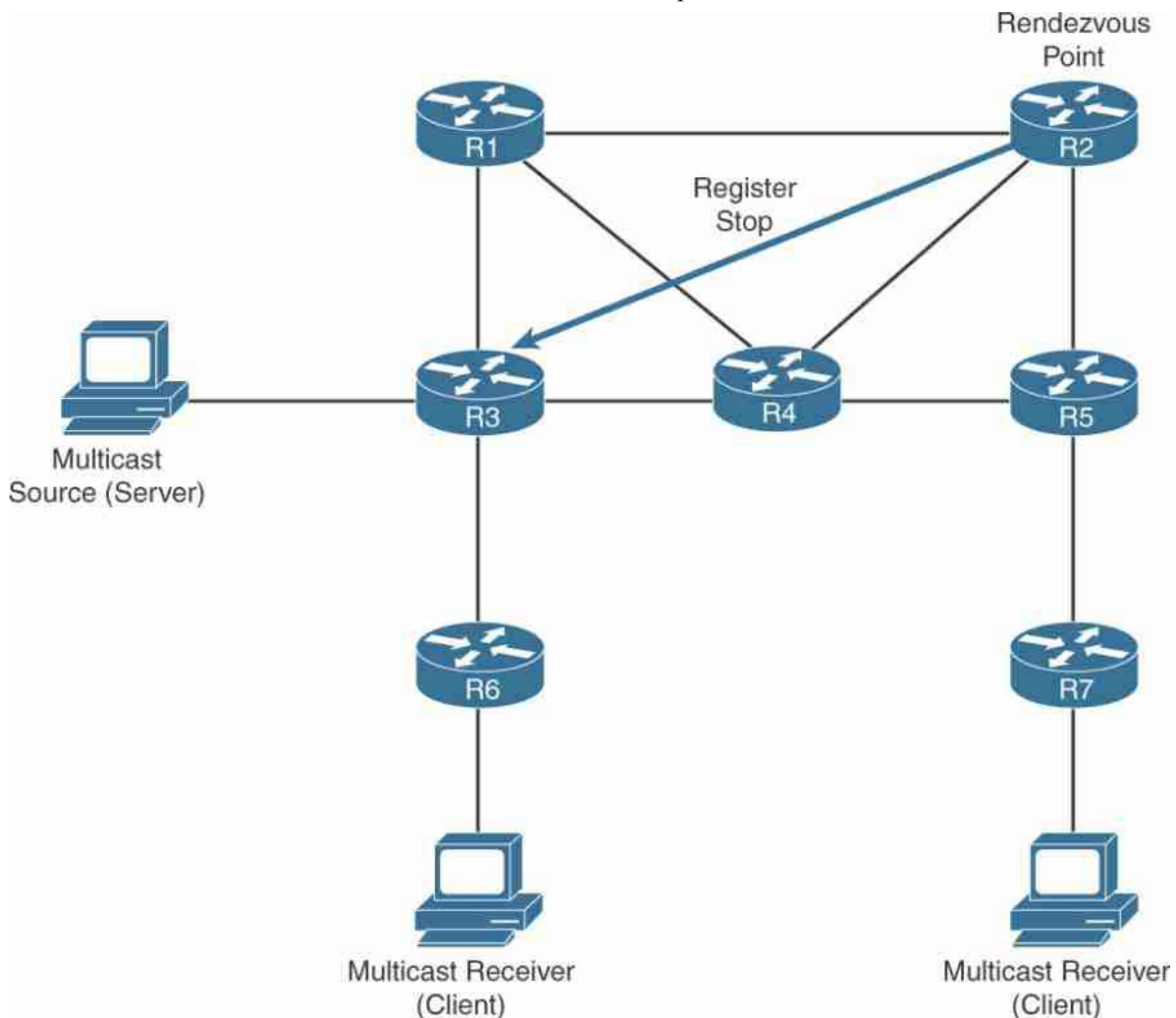


Figure 3-17 *Register Stop*

It may also be of value to examine the encapsulation and register stop process described in [Figures 3-16](#) and [3-17](#) through a sniffer trace of the packets traveling to and from the RP. [Table 3-2](#) shows a packet capture on the link between R1 and R2, whereas [Table 3-3](#) shows the multicast packets being

passed down the shared tree on the link between R2 and R5. In this capture, the IP address of the source is 192.168.8.8, the RP is address 192.168.0.2, and the tunnel source interface on R3 is 192.168.31.3, as shown in [Figure 3-16](#).

Sequence	Source	Destination	Protocol	Length	Info
1	192.168.31.3	192.168.0.2	PIMv2	582	Register + Source IP: 192.168.8.8 Destination IP: 239.1.1.1
2	192.168.21.2	224.0.0.13	PIMv2	68	Join/prune
3	192.168.31.3	192.168.0.2	PIMv2	582	Register + Source IP: 192.168.8.8 Destination IP: 239.1.1.1
4	192.168.8.8	239.1.1.1	UDP	554	Source port: 2048 Destination port: 20000
5	192.168.0.2	192.168.31.3	PIMv2	60	Register stop

Table 3-2 *Packet Capture Between R1 and R2*

Sequence	Source	Destination	Protocol	Length	Info
1	192.168.8.8	239.1.1.1	UDP	554	Source port: 2048 Destination port: 20000
3	192.168.8.8	239.1.1.1	UDP	554	Source port: 2048 Destination port: 20000
4	192.168.8.8	239.1.1.1	UDP	554	Source port: 2048 Destination port: 20000

Table 3-3 *Packet Capture Between R2 and R5*

The packet trace in these tables shows the flow, explained here by packet sequence number:

1. R3 packages the first packet (Seq. 1 on both traces) in the flow into the PIM register message and forwards it to the RP through the tunnel. The RP unpacks the inner multicast packet and forwards it down the shared tree to R5.
2. After the first packet is received, the RP sends a PIM join toward the FHR (R3). Because this packet is a PIM join, it is sent by the PIM interface closest to R1 with IP address 192.168.21.2 to the all-routers local multicast group 224.0.0.13.
3. The encapsulation process continues until the RP receives a native multicast packet from the source to that group, and the RP continues to forward the multicast packets toward R5.
4. The RP receives the native multicast packet via the newly formed source tree from the FHR to the RP. This packet is also forwarded down the shared tree. Any future copies received in the Encap Tunnel are squelched.

5. The RP sends a register-stop message to the FHR, indicating that the (S,G) to the RP is functional. The FHR no longer uses the encap tunnel for this multicast group.

From a Shared Tree to a Source Tree

When is a shared tree used and when is a source tree used? As shown in the previous section, multicast flows in a PIM sparse-mode network actually use both types of trees. The initial flow follows the source tree toward the RP and a shared tree from the RP to the receiver. Even though this forwarding arrangement completes the flow, that's not the end of the process. Sending all traffic through the RP is probably not optimal, especially if the RP is not in the preferred data path from the source to the receivers, as is the case in the example network shown in [Figure 3-16](#).

Don't forget the larger purpose of multicast: to gain more efficiency in the way packets are forwarded from the source to receivers. The shared-tree forwarding process is designed to create an efficient loop-free forwarding tree that conserves router memory and other control-plane resources across the network. To achieve the larger purpose of multicast, however, the network needs to further optimize the path(s) used for forwarding. To do this, routers only send packets through the shared tree until the network can properly calculate a source tree from the FHR to all receivers, eventually bypassing the RP altogether. The ideal state for all multicast flows in a sparse-mode network is an SPT that takes the most direct path(s) between sources and receivers.

Note

If a shared tree was always used as the distribution point for all traffic, this could create a significant amount of traffic through the RP and will likely introduce suboptimal traffic flows, depending on the placement of the RP in the network. It is the default behavior of Cisco routers to honor the forwarding paradigm of moving packets from the shared tree to the source tree as described above. When this occurs, the SPT bit is marked in PIM messages for this group, and the T flag appears in the router's state entry, as shown earlier in the flag explanation for [Example 3-9](#). However, there may be instances where this is less desirable and is therefore a configurable parameter. This parameter is known as the *spt-threshold*, and it is calculated in terms of bandwidth consumed (kbps) by the flow. The default setting on all Cisco routers is a threshold of 0 kbps, meaning that the router will always transition traffic to an SPT. If there is a reason that warrants delaying or preventing a stream from switching to an SPT, such as, for example, conserving memory resources on routers in the SPT path, then the **ip pim spt-threshold {kbps | infinity} [group-list access-list]** command in IOS can be used. The **infinity** command option can be used to permanently prevent STP switchover from occurring. When this command option is used, forwarding is completed only through the shared tree (*,G). In modern networks, memory for SPT calculations is rarely an issue, but it does exist for those circumstances that warrant it.

Let's look closer at the tree transition using our network from the diagram in [Figure 3-18](#), which includes IP addressing. In the following list we detail the flow of communication, as described earlier, step-by-step, including router debug output. In this example, a multicast source (multicast server) for group 239.1.1.1 connected to R3 is using the IP address of 192.168.8.8. A receiver for 239.1.1.1 is connected to R7 and is using the IP address of 192.168.10.10.

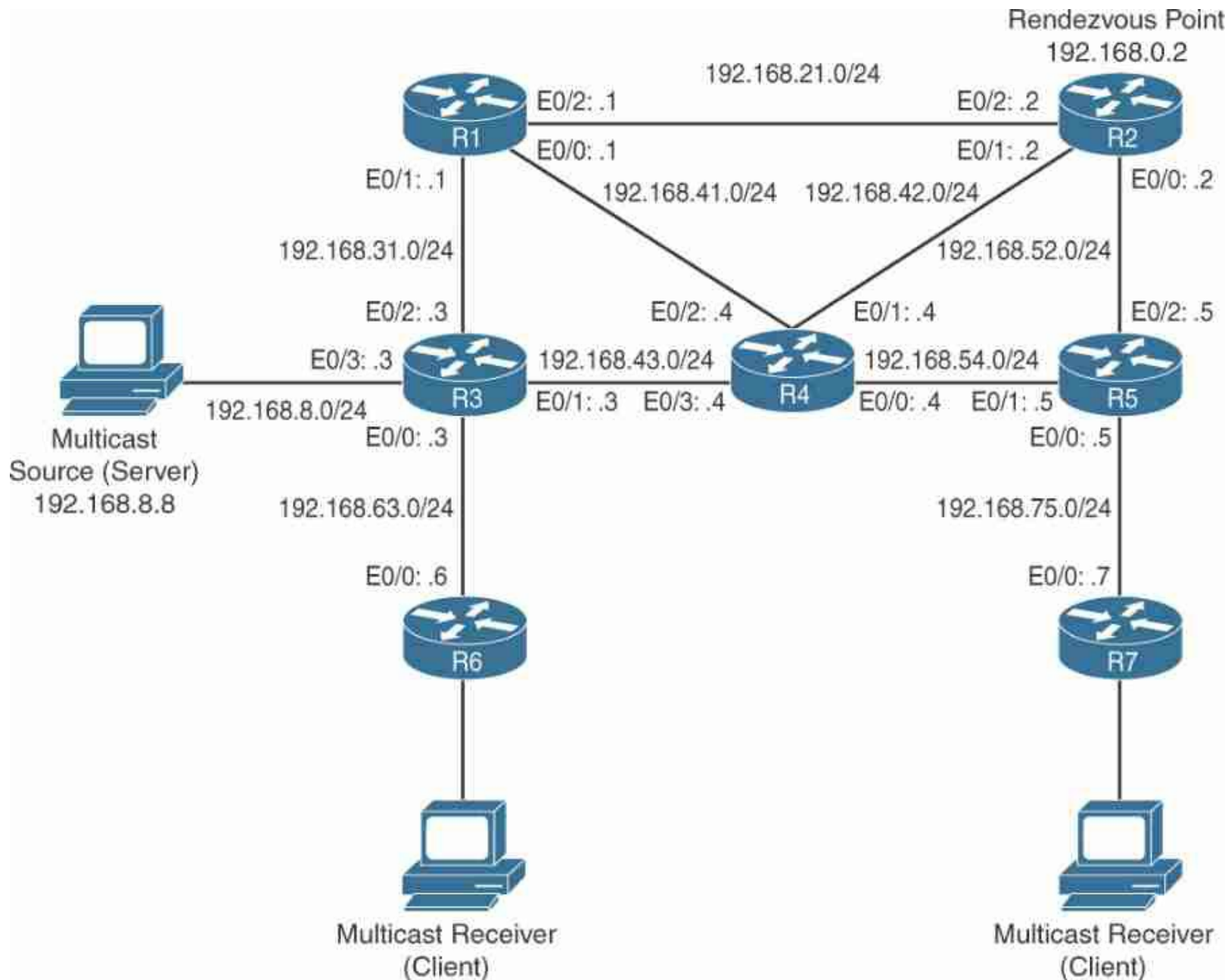


Figure 3-18 *Example Network with IP Addressing*

1. Source sends to FHR with no receivers: When the source begins to send multicast information, R3 registers the entry and informs the RP of the stream. This is seen on the RP (R2) using the **debug ip pim** command:

Note

R3 is using the IP address of 192.168.31.3.

[Click here to view code image](#)

```
PIM(0): Received v2 Register on Ethernet0/2 from 192.168.31.3 for 192.168.8.8,
group 239.1.1.1
PIM(0): Check RP 192.168.0.2 into the (*, 239.1.1.1) entry
PIM(0): Adding register decap tunnel (Tunnel1) as accepting interface of
(*, 239.1.1.1).
PIM(0): Adding register decap tunnel (Tunnel1) as accepting interface of
(192.168.8.8, 239.1.1.1).
```

Another item of interest is that the RP responds back to R3 from two interfaces because there is an

equal cost path.

At this stage in building the tree, the only routers that have any awareness of the stream are R3, the router directly upstream from the source, and the RP.

The source is sending a multicast stream, but it does not have any receivers, as verified in the output from the **show ip mroute 239.1.1.1** command demonstrated here:

[Click here to view code image](#)

```
R3#show ip mroute 239.1.1.1
```

... partial output shown for brevity ...

```
(* , 239.1.1.1), 00:00:14/stopped, RP 192.168.0.2, flags: SPF
```

```
Incoming interface: Ethernet0/2, RPF nbr 192.168.31.1
```

```
Outgoing interface list: Null
```

```
(192.168.8.8, 239.1.1.1), 00:00:14/00:02:45, flags: PFT
```

```
Incoming interface: Ethernet0/3, RPF nbr 0.0.0.0
```

```
Outgoing interface list: Null
```

The OIL is Null, which means that nothing is being sent.

Note

Recall the register stop conditions: If there are no established receivers for a group state, the RP will eventually start sending register stop messages to the FHR. This is the state shown in the **show ip mroute 239.1.1.1** output, and this is why the OIL is Null. Sending the register stop prevents unnecessary forwarding of any packets when there is no shared tree established for the flow until such time as there are both known receivers and an active source. The RP discontinues the register stop messages and allows the registration process to begin anew after it receives a (*,G) PIM join messages for that group.

From the **show ip mroute 239.1.1.1** output, you see the shared tree (*, 239.1.1.1) and the source tree (192.168.8.8, 239.1.1.1).

R7 currently does not have any knowledge of 239.1.1.1, as seen using the **show ip mroute 239.1.1.1** command demonstrated here:

```
R7#show ip mroute 239.1.1.1
```

```
Group 239.1.1.1 not found
```

When the host connected to R7 desires to join the multicast stream sourced of 239.1.1.1, a series of events takes place, as follows (note that the multicast source is already active and the FHR starts the registry process):

2. Receivers join at the LHR: R7 receives the IGMP join message (membership report) from the attached client and adds it to the IGMP group table, as seen from the **show ip igmp groups** command:

[Click here to view code image](#)

```
R7#show ip igmp groups 239.1.1.1
```

```
IGMP Connected Group Membership
```

Group Address	Interface	Uptime	Expires	Last Reporter
---------------	-----------	--------	---------	---------------

Group Accounted				
-----------------	--	--	--	--

239.1.1.1	Ethernet0/1	00:00:02	00:02:57	192.168.10.10
-----------	-------------	----------	----------	---------------

3. LHR establishes the shared tree toward the RP: R7 establishes the (*,G) and sends a join message to the upstream neighbor (R5), as seen from the following debug messages using the command **debug ip pim** :

[Click here to view code image](#)

```
PIM(0): Check RP 192.168.0.2 into the (*, 239.1.1.1) entry
PIM(0): Building Triggered (*,G) Join / (S,G,RP-bit) Prune message for
239.1.1.1
PIM(0): Building Triggered (*,G) Join / (S,G,RP-bit) Prune message for
239.1.1.1
PIM(0): Upstream mode for (*, 239.1.1.1) changed from 0 to 1
PIM(0): Insert (*,239.1.1.1) join in nbr 192.168.75.5's queue
PIM(0): Building Join/Prune packet for nbr 192.168.75.5
PIM(0): Adding v2 (192.168.0.2/32, 239.1.1.1), WC-bit, RPT-bit, S-bit Join
PIM(0): Send v2 join/prune to 192.168.75.5 (Ethernet0/0)
PIM(0): Insert (192.168.8.8,239.1.1.1) join in nbr 192.168.75.5's queue
PIM(0): Building Join/Prune packet for nbr 192.168.75.5
PIM(0): Adding v2 (192.168.8.8/32, 239.1.1.1), S-bit Join
PIM(0): Send v2 join/prune to 192.168.75.5 (Ethernet0/0)
PIM(0): Received RP-Reachable on Ethernet0/0 from 192.168.0.2
PIM(0): Received RP-Reachable on Ethernet0/0 from 192.168.0.2 for group
239.1.1.1
PIM(0): Update RP expiration timer (270 sec) for 239.1.1.1
PIM(0): Insert (192.168.8.8,239.1.1.1) join in nbr 192.168.75.5's queue
PIM(0): Building Join/Prune packet for nbr 192.168.75.5
PIM(0): Adding v2 (192.168.8.8/32, 239.1.1.1), S-bit Join
PIM(0): Send v2 join/prune to 192.168.75.5 (Ethernet0/0)
PIM(0): Building Periodic (*,G) Join / (S,G,RP-bit) Prune message for
239.1.1.1
PIM(0): Insert (*,239.1.1.1) join in nbr 192.168.75.5's queue
PIM(0): Building Join/Prune packet for nbr 192.168.75.5
PIM(0): Adding v2 (192.168.0.2/32, 239.1.1.1), WC-bit, RPT-bit, S-bit Join
PIM(0): Send v2 join/prune to 192.168.75.5 (Ethernet0/0)
```

4. RP builds the (*,G) state and receiver entry: Upon receiving the join message generated from R7, the RP (R2) builds the (*, G) state, as shown in the following **debug ip pim** message:

[Click here to view code image](#)

```
PIM(0): Received v2 Join/Prune on Ethernet0/0 from 192.168.52.5, to us
PIM(0): Join-list: (*, 239.1.1.1), RPT-bit set, WC-bit set, S-bit set
PIM(0): Add Ethernet0/0/192.168.52.5 to (*, 239.1.1.1), Forward state, by PIM
*G Join
PIM(0): Add Ethernet0/0/192.168.52.5 to (192.168.8.8, 239.1.1.1), Forward
state, by PIM *G Join
```

5. FHR registers the active source: The RP already received a register packet from the FHR because it has the (*,G) state with receiver information. The RP sends a (S,G) join message to

224.0.0.13 towards the FHR and then sends a register stop message directly to the FHR. The output from the RP is shown using the following **debug ip pim** message:

[Click here to view code image](#)

```
*May 11 21:36:48.311: PIM(0): Received v2 Register on Ethernet0/2 from
192.168.43.3
*May 11 21:36:48.311:    for 192.168.8.8, group 239.1.1.1
*May 11 21:36:48.311: PIM(0): Adding register decap tunnel (Tunnel1) as
accepting interface of (192.168.8.8, 239.1.1.1).
*May 11 21:36:48.311: PIM(0): Insert (192.168.8.8,239.1.1.1) join in nbr
192.168.42.4's queue
*May 11 21:36:48.311: PIM(0): Building Join/Prune packet for nbr 192.168.42.4
*May 11 21:36:48.311: PIM(0): Adding v2 (192.168.8.8/32, 239.1.1.1), S-bit
Join
*May 11 21:36:48.311: PIM(0): Send v2 join/prune to 192.168.42.4 (Ethernet0/1)
*May 11 21:36:48.310: PIM(0): Check RP 192.168.0.2 into the (*, 239.1.1.1)
entry
*May 11 21:36:48.310: PIM(0): Building Triggered (*,G) Join / (S,G,RP-bit)
Prune message for 239.1.1.1
*May 11 21:36:48.310: PIM(0): Adding register encap tunnel (Tunnel0) as for-
warding interface of (192.168.8.8, 239.1.1.1).
*May 11 21:36:48.313: PIM(0): Received v2 Join/Prune on Ethernet0/1 from
192.168.43.4, to us
*May 11 21:36:48.313: PIM(0): Join-list: (192.168.8.8/32, 239.1.1.1), S-bit
set
*May 11 21:36:48.313: PIM(0): Add Ethernet0/1/192.168.43.4 to (192.168.8.8,
239.1.1.1), Forward state, by PIM SG Join
```

At the FHR, the output of the **debug ip pim** message at this step is as follows:

[Click here to view code image](#)

```
*May 11 21:45:11.152: PIM(0): Check RP 192.168.0.2 into the (*, 239.1.1.1)
entry
*May 11 21:45:11.152: PIM(0): Building Triggered (*,G) Join / (S,G,RP-bit)
Prune message for 239.1.1.1
*May 11 21:45:11.152: PIM(0): Adding register encap tunnel (Tunnel0) as
forwarding interface of (192.168.8.8, 239.1.1.1).
*May 11 21:45:11.155: PIM(0): Received v2 Join/Prune on Ethernet0/1 from
192.168.43.4, to us
*May 11 21:45:11.155: PIM(0): Join-list: (192.168.8.8/32, 239.1.1.1), S-bit
set
*May 11 21:45:11.155: PIM(0): Add Ethernet0/1/192.168.43.4 to (192.168.8.8,
239.1.1.1), Forward state, by PIM SG Join
R3#
*May 11 21:45:15.133: PIM(0): Received v2 Register-Stop on Ethernet0/1 from
192.168.0.2
*May 11 21:45:15.133: PIM(0):    for source 192.168.8.8, group 239.1.1.1
```

*May 11 21:45:15.133: PIM(0): Removing register encap tunnel (Tunnel0) as forwarding interface of (192.168.8.8, 239.1.1.1).

*May 11 21:45:15.133: PIM(0): Clear Registering flag to 192.168.0.2 for (192.168.8.8/32, 239.1.1.1)

R3 receives the register stop and the (S,G) join from the RP. Now the FHR (R3) sends the multicast traffic towards the RP in a unicast tunnel. The flow of the multicast stream now progresses from the RP down to R7 and ultimately towards the receiver. This is the initial tree build up in PIM sparse-mode. For details on the SPT switch, refer to the earlier section, “[The Rendezvous Point and Shared Tree Dynamics](#) .”

6. The multicast flow moves from the shared tree to a source tree: R3 received the prune message from the RP and added the interface Ethernet0/1 to the OIL, as shown from the following **debug ip pim** message, creating a source tree from the FHR to the LHR:

[Click here to view code image](#)

PIM(0): Received v2 Join/Prune on Ethernet0/1 from 192.168.43.4, to us
PIM(0): Join-list: (192.168.8.8/32, 239.1.1.1), S-bit set
PIM(0): Add Ethernet0/1/192.168.43.4 to (192.168.8.8, 239.1.1.1), Forward state, by PIM SG Join

R5 sees the source tree formed from the FHR (R3) and uses the RPF in the unicast routing table to find a better metric. When this occurs, R5 sends a PIM prune toward the RP to remove itself from the shared tree, as shown from the **debug ip pim** output from R5 below.

[Click here to view code image](#)

PIM(0): Received v2 Join/Prune on Ethernet0/0 from 192.168.75.7, to us
PIM(0): Join-list: (*, 239.1.1.1), RPT-bit set, WC-bit set, S-bit set
PIM(0): Add Ethernet0/0/192.168.75.7 to (*, 239.1.1.1), Forward state, by PIM *G Join
PIM(0): Building Triggered (*,G) Join / (S,G,RP-bit) Prune message for 239.1.1.1
PIM(0): Upstream mode for (*, 239.1.1.1) changed from 0 to 1
PIM(0): Insert (*,239.1.1.1) join in nbr 192.168.52.2's queue
PIM(0): Insert (192.168.8.8,239.1.1.1) sgr prune in nbr 192.168.52.2's queue
PIM(0): Update Ethernet0/0/192.168.75.7 to (192.168.8.8, 239.1.1.1), Forward state, by PIM *G Join
PIM(0): Building Join/Prune packet for nbr 192.168.52.2
PIM(0): Adding v2 (192.168.0.2/32, 239.1.1.1), WC-bit, RPT-bit, S-bit Join
PIM(0): Adding v2 (192.168.8.8/32, 239.1.1.1), RPT-bit, S-bit Prune
PIM(0): Send v2 join/prune to 192.168.52.2 (Ethernet0/2)

[Figure 3-19](#) shows the traffic flow for the new source tree.

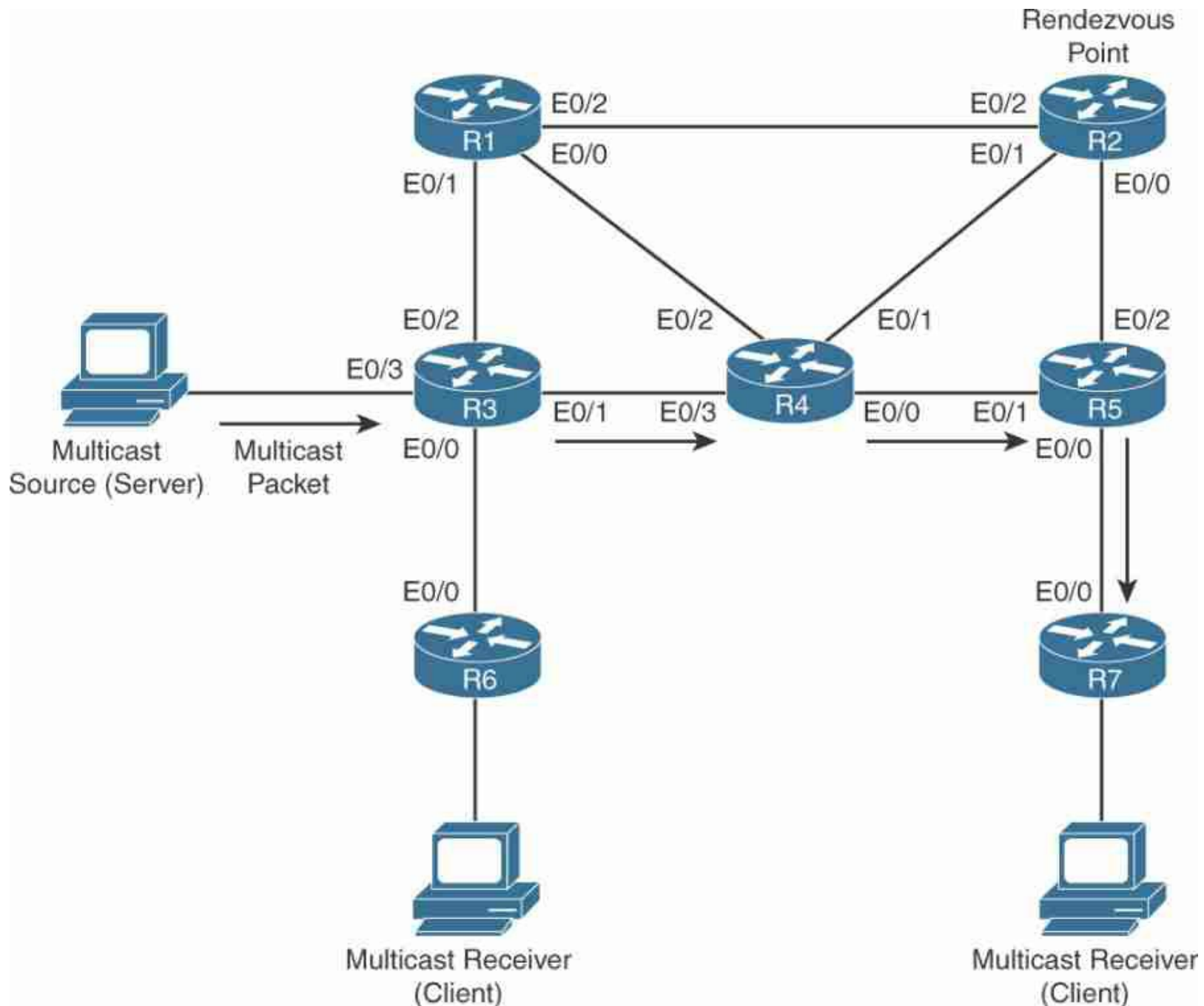


Figure 3-19 Source Tree Forwarding

7. **The new source tree is fully activated** : The source tree is now active, as seen using the **show ip mroute 239.1.1.1** command on R3:

[Click here to view code image](#)

```
R3#show ip mroute 239.1.1.1
```

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
 L - Local, P - Pruned, R - RP-bit set, F - Register flag,
 T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
 X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
 U - URD, I - Received Source Specific Host Report,
 Z - Multicast Tunnel, z - MDT-data group sender,
 Y - Joined MDT-data group, y - Sending to MDT-data group,
 G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route

Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:00:22/stopped, RP 192.168.0.2, flags: SPF

Incoming interface: Ethernet0/1, RPF nbr 192.168.43.4

Outgoing interface list: Null

(192.168.8.8, 239.1.1.1), 00:00:22/00:03:07, flags: FT

Incoming interface: Ethernet0/3, RPF nbr 0.0.0.0

Outgoing interface list:

Ethernet0/1, Forward/Sparse, 00:00:22/00:03:07

The shared tree has not been removed. When another receiver connected to a different router is interested in the multicast stream, the shared tree and the same process of moving from a shared tree to a source tree will occur again, like déjà vu all over again.

Note

While there is much going on under the hood in this section, the process of switching between the shared tree and the source tree is relatively quick (less than a second in a robust network). This is exactly what we want! If a sparse-mode network has trees stuck in the shared tree state, eating up RP router resources, this indicates that a group has no source(s), or that something has likely gone wrong in the PIM process. In fact, this is a common symptom that occurs in a network in which PIM communication has broken down. For information on how to find and fix such failures, please see [Chapter 7](#) , which covers PIM troubleshooting.

Building the Multicast Routing Information Base

The unicast routing information base (RIB) has been populated by unicast routing protocols. It is not necessary to build another multicast routing table with identical or additional route information that consumes additional memory. As discussed earlier with RPF checks, rather than forward packets to the source, multicast routers forward packets away from the source.

When a router receives a multicast packet, it performs the following actions.

1. Upon receipt of a multicast packet, the router checks the receiving interface to determine whether it is in the path to the source (sending IP address) by performing a lookup in the RIB. This is known as the *RPF check* .

2. If the interface is in the path to the source, the packet is forwarded.

3. If the receiving interface is not in the path to the source, the RPF fails and the packet is dropped.

Multicast takes a unique perspective when it comes to forwarding traffic in the network. Rather than looking for the destination address of where to send the information as with traditional routing, multicast routing does just the opposite. Because multicast routing is the opposite of unicast or traditional routing, you can take advantage of the RIB.

The RIB is a collection of all routing information contained on a router and is a culmination of connected routes, static routes, and routes learned from dynamic routing protocols. It also holds

adjacency information for neighboring routers. The RIB is used to populate the forwarding information base (FIB) with the best path to a particular destination. The FIB table is used to forward packets to the appropriate destination.

Multicast Routing Information Base and Multicast Forwarding Information Base

In the earlier section “[Two Types of Trees](#),” we addressed a couple of items that need some further clarification. These are the multicast routing information base (MRIB) and multicast forwarding information base (MFIB) and the way they interact with the other functions of the router.

The MRIB table is a collection of multicast entries, including source, group, group mask, and flags, and may also contain a list of associated interfaces. The entries in the MRIB are created/updated by the control plane when traffic is received from a connected source, a switchover occurs, or there is some other data-driven event. The MRIB table (for non-local multicast groups) can be viewed on older IOS platforms simply by using the **show ip mroute** command as it has been discussed. Newer versions of IOS, including IOS XE, can display the MRIB table using the **show ip mrrib route** command, as demonstrated in [Example 3-15](#).

Example 3-15 *Displaying the MRIB Table*

[Click here to view code image](#)

```
ASR1K-2#show ip mrrib route
```

IP Multicast Routing Information Base

Entry flags: L - Domain-Local Source, E - External Source to the Domain,

C - Directly-Connected Check, S - Signal, IA - Inherit Accept, D - Drop

ET - Data Rate Exceeds Threshold, K - Keepalive, DDE - Data Driven Event

Interface flags: F - Forward, A - Accept, IC - Internal Copy,

NS - Negate Signal, DP - Don't Preserve, SP - Signal Present,

II - Internal Interest, ID - Internal Disinterest, LI - Local Interest,

LD - Local Disinterest, MD - mCAC Denied

(* ,224.64.7.7) RPF nbr: 0.0.0.0 Flags: C

GigabitEthernet0/0/2 Flags: F NS

GigabitEthernet0/0/1 Flags: F NS

GigabitEthernet0/0/0 Flags: F NS

(192.168.8.10,224.64.7.7) RPF nbr: 192.168.32.3 Flags:

GigabitEthernet0/0/1 Flags: A

GigabitEthernet0/0/0 Flags: F NS

The MFIB is the multicast forwarding information derived from the MRIB and is presented as a unique table, which displays how the router intends to forward multicast messages. Information such as source, group, mask, counts, and rates of received, dropped, and forwarded packets are contained in this table. [Example 3-16](#) shows an abbreviated output from the **show ip mfib** command.

Example 3-16 *MFIB State Table*

[Click here to view code image](#)

```
ASR1K-2#show ip mfib
```

Entry Flags: C - Directly Connected, S - Signal, IA - Inherit A flag,

ET - Data Rate Exceeds Threshold, K - Keepalive

DDE - Data Driven Event, HW - Hardware Installed

I/O Item Flags: IC - Internal Copy, NP - Not platform switched,
NS - Negate Signaling, SP - Signal Present,
A - Accept, F - Forward, RA - MRIB Accept, RF - MRIB Forward,
MA - MFIB Accept

Forwarding Counts: Pkt Count/Pkts per second/Avg Pkt Size/Kbits per second

Other counts: Total/RPF failed/Other drops

I/O Item Counts: FS Pkt Count/PS Pkt Count

Default

(*224.64.7.7) Flags: C HW

SW Forwarding: 0/0/0/0, Other: 0/0/0

HW Forwarding: 0/0/0/0, Other: 0/0/0

GigabitEthernet0/0/0 Flags: F NS

Pkts: 0/0

GigabitEthernet0/0/1 Flags: F NS

Pkts: 0/0

GigabitEthernet0/0/2 Flags: F NS

Pkts: 0/0

(192.168.8.10,224.64.7.7) Flags: HW

SW Forwarding: 0/0/0/0, Other: 15/2/13

HW Forwarding: 685830/156/1369/1673, Other: 1111/1111/0

GigabitEthernet0/0/1 Flags: A

GigabitEthernet0/0/0 Flags: F NS

Pkts: 0/0

[Figure 3-20](#) depicts the interaction between the various components involved in the derivation of multicast forwarding information.

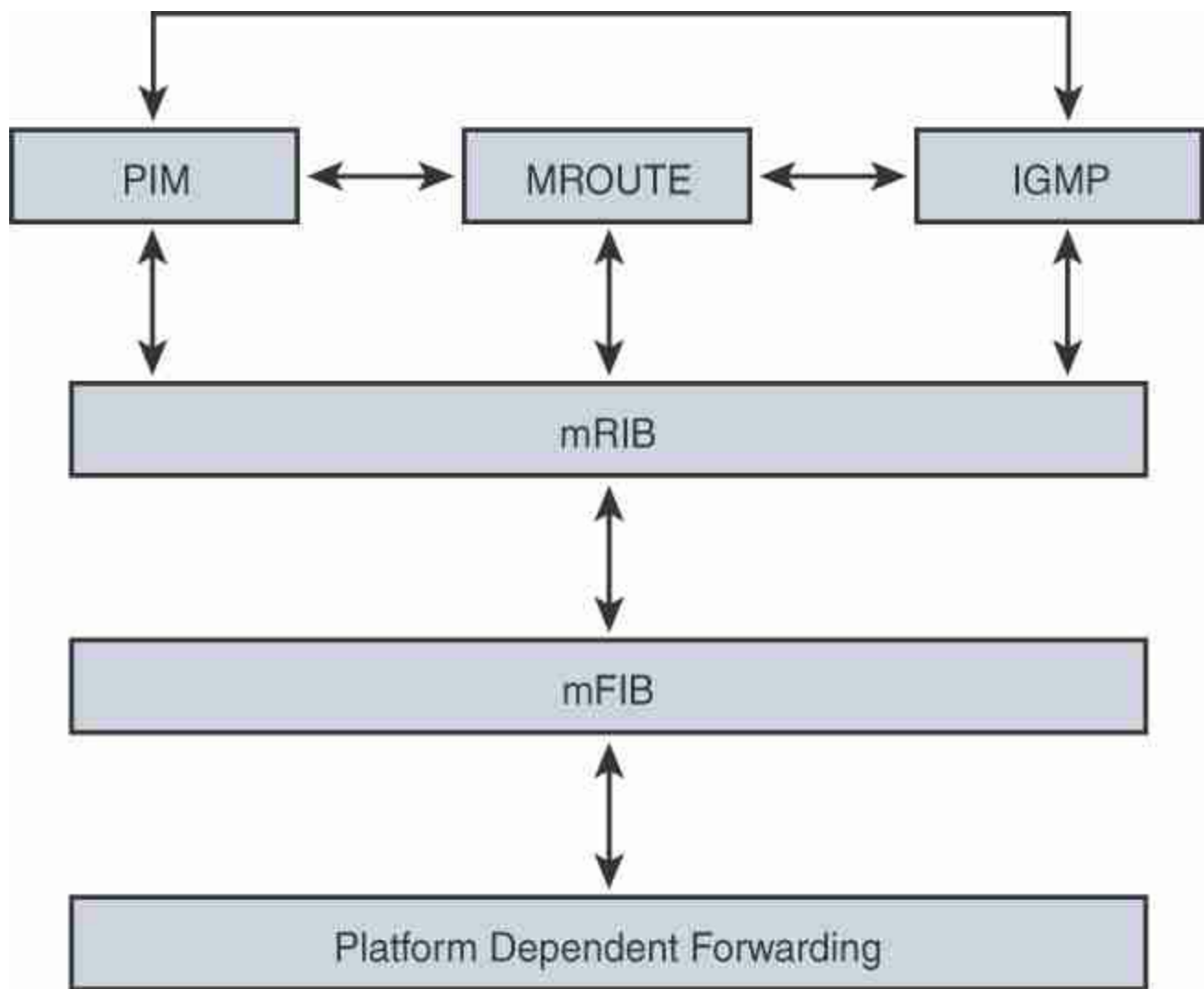


Figure 3-20 *MFIB and MRIB*

PIM-BiDir

RFC 5015 defines PIM-BiDir, which is a variant of protocol independent multicast (PIM). The traffic in PIM-BiDir is rooted at the rendezvous point (RP) for the group and the IP address of the RP acts as the key to having all routers establish a loop-free tree topology. Explicit join messages signal membership to the bidirectional group. Traffic from sources is unconditionally sent up the shared tree toward the RP and passed down the tree toward the receivers on each branch of the tree. This unconditional forwarding of multicast traffic towards the RP using a prebuilt tree is one of the major differences between PIM-BiDir and sparse-mode. The state is based on the prebuilt tree towards the RP in the PIM domain. Therefore, the PIM-BiDir mode only supports a (*,G) group state. (S,G) state is not seen in the mroute tables when using PIM-BiDir. Eliminating (S,G) state allows the PIM domain to scale to an extensive number of sources. The pre-built tree used in PIM-BiDir enables it to be used for many-to-many applications within individual PIM domains. Multicast groups in bidirectional mode can scale to an arbitrary number of sources without incurring overhead due to the number of sources.

PIM-BiDir is derived from the mechanisms of PIM sparse-mode (PIM-SM) and shares many shortest path tree (SPT) operations. BiDir also uses unconditional forwarding of source traffic toward the RP upstream on the shared tree, through the prebuilt state information; however, there is no registration process for sources as in PIM-SM. This means the RP is used as a root point for the tree, but not used as a tree transitioning platform, taking the tree management load off the RP altogether.

[Figure 3-21](#) shows the simple forwarding of multicast traffic using a PIM-BiDir tree.

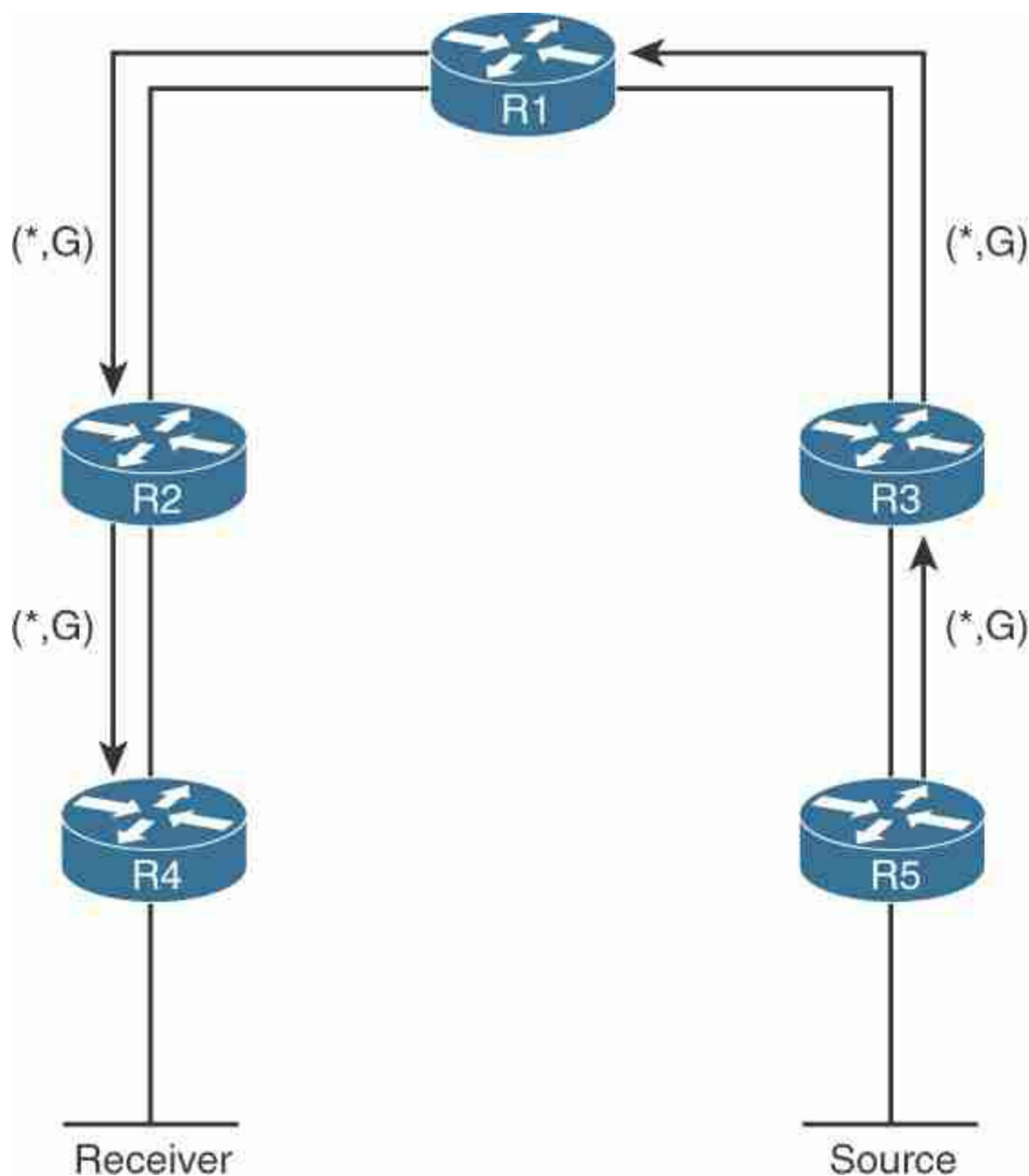


Figure 3-21 *PIM-BiDir $(*,G)$*

As you'll notice in [Figure 3-21](#), the multicast stream forwarding to the downstream receiver is similar to PIM sparse-mode. The only difference between PIM-BiDir and PIM sparse-mode is the upstream communication and multicast stream flow to the RP. The reason the upstream flow is different is because PIM sparse-mode requires an RPF check to pass through to avoid looping of multicast data stream. The packet forwarding rules have been enhanced in PIM-BiDir over PIM-SM, allowing the traffic to be forwarded directly to the RP. The multicast looping protection in PIM-BiDir is provided by a designated forwarder (DF), which establishes a loop-free SPT rooted at the RP. With PIM-BiDir, the RP address does not need to belong to the physical box; instead, it can be simply a routable address. This is covered in more detail in [Chapter 4](#). The concept of the RPF interface is important to understand with non-receiver segments, because the RPF interface is used in the $(*,G)$ to point towards the RP.

In a PIM-BiDir network, every segment (including point-to-point) elects a designated forwarder (DF). The DF is elected based on the RP for a group. In a scenario of multiple RPs for different groups, you will have multiple DFs. The DF router is responsible for creating a preexisting state that

is used in forwarding multicast packets towards the RP.

On every network segment and point-to-point link, all PIM routers participate in a procedure called *DF election*. The procedure selects one router as the DF for every RP in bidirectional multicast groups. This router is responsible for forwarding multicast packets received on that network “upstream” to the RP. The DF election process is based on unicast metrics and uses highest IP address in the case of a tie with the unicast metric. The DF forwards only one copy of the multicast flow upstream. It should be noted that for every RP assigned to a separate multicast group, a new DF is elected. A DF in PIM-BiDir takes the role of DR in PIM sparse-mode for first-hop router.

The conditions under which an update to the election process that creates state changes are:

- Unicast routing table changes to reach the RP
- Local interface changes the RPF
- A new PIM router is seen in the topology
- Neighbor failure that prompts a reelection

[Figure 3-22](#) and the list that follows review the two-step process of building state for PIM-BiDir.

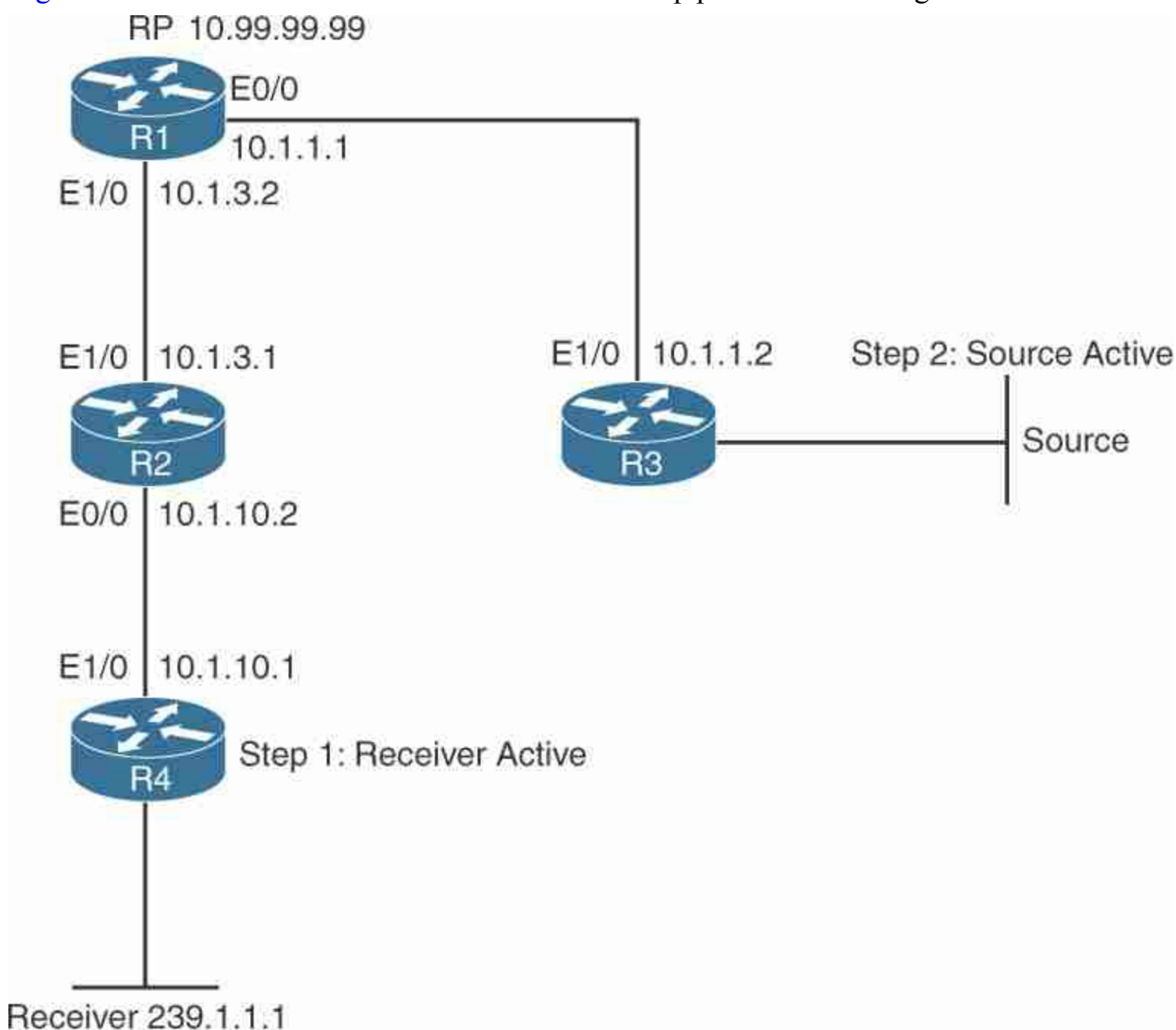


Figure 3-22 PIM-BiDir Walkthrough 1

Step 1. Receiver joins the R4, which can be seen using the commands **debug ip pim** and **show ip mroute** .

[Click here to view code image](#)

7: IGMP(0): MRT Add/Update Loopback0 for (*,239.1.1.1) by 4

Nov 21 23:29:50.457: IGMP(0): MRT Add/Update Loopback0 for (,239.1.1.1) by 4

*Nov 21 23:29:50.457: IGMP(0): Send v2 Report for 239.1.1.1 on Loopback0

*Nov 21 23:29:50.457: IGMP(0): Received v2 Report on Loopback0 from 10.11.11.11 for 239.1.1.1

*Nov 21 23:29:50.457: IGMP(0): Received Group record for group 239.1.1.1, mode 2 from 10.11.11.11 for 0 sources

*Nov 21 23:29:50.457: IGMP(0): Updating EXCLUDE group timer for 239.1.1.1

Nov 21 23:29:50.457: IGMP(0): MRT Add/Update Loopback0 for (,239.1.1.1) by 0

Nov 21 23:29:50.457: IGMP(0): MRT Add/Update Loopback0 for (,224.0.1.40) by 4

*Nov 21 23:29:50.457: IGMP(0): Send v2 Report for 224.0.1.40 on Loopback0

*Nov 21 23:29:50.457: IGMP(0): Received v2 Report on Loopback0 from 10.11.11.11 for 224.0.1.40

*Nov 21 23:29:50.457: IGMP(0): Received Group record for group 224.0.1.40, mode 2 from 10.11.11.11 for 0 sources

R4#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:06:11/00:02:19, RP 10.99.99.99, flags: BCL

Bidir-Upstream: Ethernet1/0, RPF nbr 10.1.3.1

Outgoing interface list:

Loopback0, Forward/Sparse, 00:06:11/00:02:19

Ethernet1/0, Bidir-Upstream/Sparse, 00:06:11/stopped

The status at R2 is shown using the **show ip mroute** command.

[Click here to view code image](#)

R2#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,

C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP

Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:10:38/00:03:23, RP 10.99.99.99, flags: B

Bidir-Upstream: Ethernet1/0, RPF nbr 0.0.0.0

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:10:38/00:03:23

Ethernet1/0, Bidir-Upstream/Sparse, 00:10:38/stopped

As you can see from this output, the state is built using the DF. The receiver sends the notification upstream without registry (due to the prebuilt DF information) towards the RP. All the downstream routers—in this topology, R2 and R4—have the state information for 239.1.1.1 learned via PIM-BiDir. Also notice that the upstream interface Ethernet1/0 uses PIM-BiDir in the state table. The downstream interface is Ethernet0/0 and uses PIM sparse-mode forwarding, as shown using the **show ip pim interface df** command. The RP configuration needs to be done on each router. In [Chapter 4](#) , you learn about various RP configurations for each PIM mode.

[Click here to view code image](#)

R2# show ip pim interface df

* implies this system is the DF

Interface	RP	DF Winner	Metric
Uptime			
Ethernet0/0	10.99.99.99	*10.1.3.1	11
00:17:13			
Ethernet1/0	10.99.99.99	0.0.0.0	11
00:00:00			

Step 2. The source joins at R3 and the state at R3 is shown using the **show ip mroute** command:

[Click here to view code image](#)

R3#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP

Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* ,224.0.0.0/4), 00:19:20/-, RP 10.99.99.99, flags: B

Bidir-Upstream: Ethernet1/0, RPF nbr: 10.1.1.1

Incoming interface list:

Ethernet0/0, Accepting/Sparse

Loopback0, Accepting/Sparse

Ethernet1/0, Accepting/Sparse

(* , 224.0.1.40), 00:20:15/00:02:53, RP 10.99.99.99, flags: BCL

Bidir-Upstream: Ethernet1/0, RPF nbr 0.0.0.0

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:19:19/00:02:53

Loopback0, Forward/Sparse, 00:19:19/00:02:53

Ethernet1/0, Bidir-Upstream/Sparse, 00:19:20/stopped

R3 will unconditionally forward multicast traffic toward the RP upstream using the shared tree. There is no registering process for sources as in PIM-SM, the traffic is forwarded based on (*,G) attributes between the source and receiver(s).

PIM-SSM

Source-specific multicast (SSM) is another flavor of PIM sparse-mode defined in RFC 3569. The multicast stream is transmitted using an IP source and multicast address of a group. These two attributes are the primary components necessary for the transportation of the multicast messages. This protocol provides a host application with complete identification of a channel. Because this includes a combination of the source IP address and the address of the group, there can be only one multicast stream (without duplicate IP addresses in the network). Using SSM, you can have one source and many multicast applications. The receiver must subscribe to the channel using IGMPv3. IGMPv3 provides the designated router with the source IP address and the multicast group address. This set of addresses in IGMPv3 identifies the channel. The designated router connected to the receiver segment already has the IP address of the source, and thereby the streams can directly be received from the source. This mode of PIM does not require the use of an RP because the receiver is able to receive the group from a specified source. In SSM and IGMPv3, there are no shared trees; consequently, the router does not need to maintain (*,G) state. The multicast state only builds using SPTs (shortest path trees) towards our sources.

Source-specific multicast (SSM) provides the capability to identify a set of multicast hosts not only by group address but also by source. An SSM group, called a *channel*, is identified as an (S,G) where *S* is the source address and *G* is the group address. An example for a channel is as follows:

Channel A (S,G) = (10.0.0.1, 232.1.1.1)

Channel B (S,G) = (10.0.0.2, 232.1.1.1)

Even though Channel A and B are represented using the same multicast group, 232.1.1.1, the receiver can access a separate flow for Channel A because the identification of multicast stream is based on the channel (the unicast address of the source and the multicast group address).

[Chapter 1](#) covered the SSM addressing scheme. For this IANA has reserved the IPv4 address range of 232.0.0.0/8, and it is recommended to allocate SSM multicast groups using that range. PIM-SSM requires that the successful establishment of an (S,G) forwarding path from the source *S* to any receiver(s) depends on hop-by-hop forwarding of the explicit join request from the receiver toward the source. The receivers send an explicit join to the source because they have the source IP address in their join message with the multicast address of the group. PIM-SSM leverages the unicast routing topology to maintain the loop-free behavior. Traffic for one (S, G) channel consists of datagrams with an IP unicast source address *S* and the multicast group address *G* as the IP destination address. Systems receive this traffic by becoming members of the (S, G) channel. The receivers can receive traffic only from designated (S, G) channels to which they are subscribed, which is in contrast to ASM, where receivers need not know the IP addresses of sources from which they receive their traffic.

IGMPv3 is a requirement for both the host and the next-hop device (router). If this support is not available, then the option for host-to-network device communication at Layer 2 can be IGMP v3lite and URL Rendezvous Directory (URD). These are two Cisco-developed transition solutions that were designed to enable the immediate deployment of SSM services without the need to wait for the availability of full IGMPv3 support on host operating systems. (Today, most host operating systems

include support for IGMPv3.) URD is a solution for content providers and content aggregators that allows them to deploy receiver applications that are not yet SSM enabled (through support for IGMPv3). IGMPv3, IGMPv3lite, and URD interoperate with each other, and can be used as transitional solutions toward full IGMPv3 support for hosts.

IGMP and INCLUDE mode membership reports are used for channel subscription signaling with IGMPv3. The INCLUDE can be statically created with IGMPv3lite or URD. The INCLUDE and EXCLUDE mode of membership adds additional security not available in other PIM modes.

Note

The INCLUDE and EXCLUDE mode are additional filters that add extra security parameters to the PIM control plane. In the INCLUDE mode, reception of packets is allowed for only those multicast addresses specified in the source-list parameter. In the EXCLUDE mode, reception of packets is allowed for *all* multicast addresses except those specified in the source-list parameter. The source list is a list of IP source addresses from which multicast reception is allowed and is configured in the form of an access control list (ACL).

The following examples explain the SSM process, beginning with [Figure 3-23](#) :

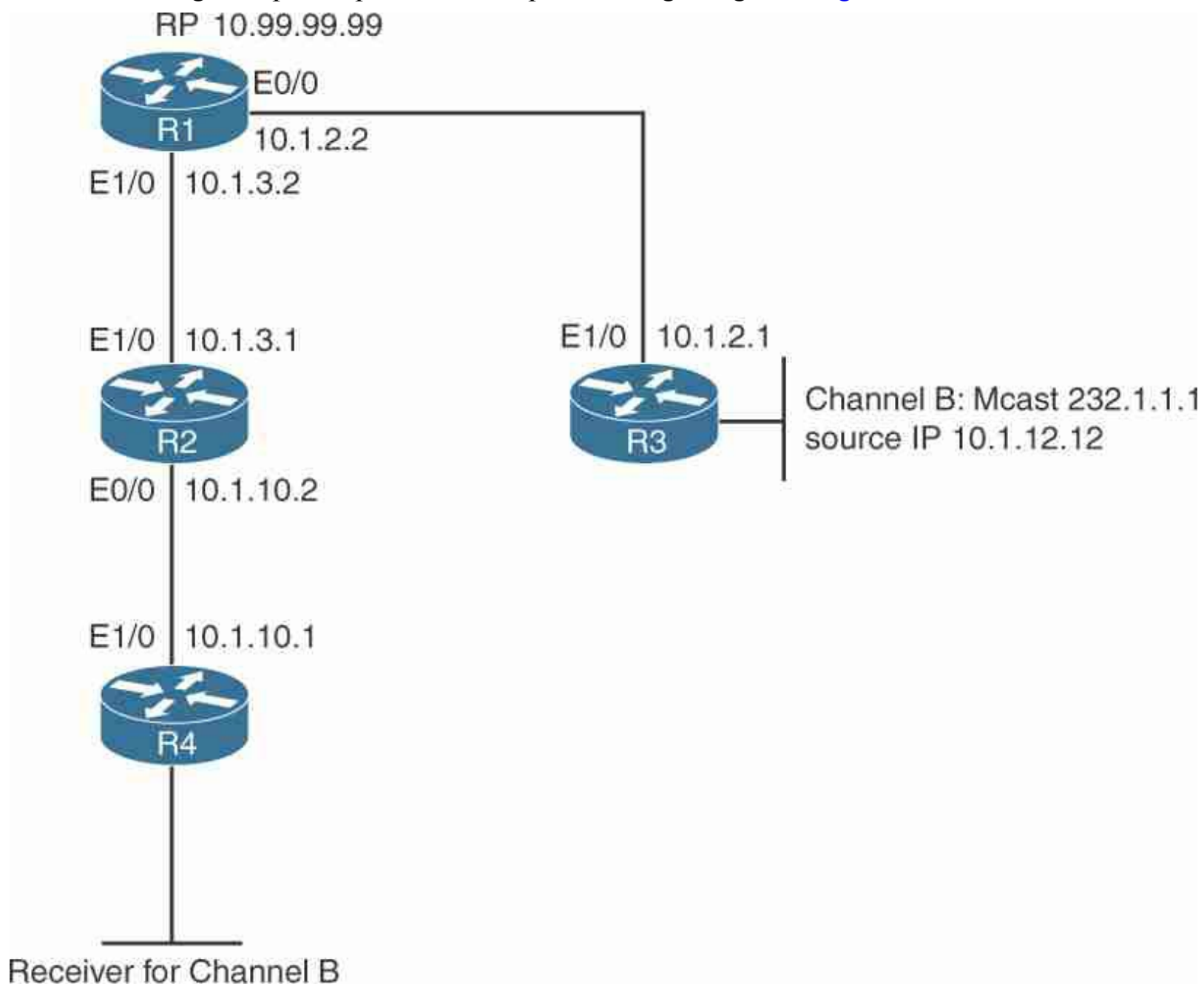


Figure 3-23 PIM SSM Single Channel A

Step 1. The receiver joins R4 using an IGMPv3 join message. The source IP is located off of R3. Unicast routing is used to build state. Using the **debug ip igmp** and **show ip mroute** commands, you can see the state information can on R1:

[Click here to view code image](#)

```
*Nov 21 23:01:24.899: IGMP(0): Updating expiration time on
(10.1.12.12,232.1.1.1) to 180 secs
*Nov 21 23:01:24.899: IGMP(0): Send v3 init Query on Loopback0
*Nov 21 23:01:24.899: IGMP(0): Set report delay to 0.7 seconds to
respond to General Query on Loopback0
*Nov 21 23:01:24.899: IGMP(0): Set report delay to 0.2 seconds to
respond to General Query on Loopback0
*Nov 21 23:01:25.132: IGMP(0): Building v3 Report on Loopback0
*Nov 21 23:01:25.132: IGMP(0): Add Group Record for 232.1.1.1, type 1
*Nov 21 23:01:25.132: IGMP(0): Add Source Record 10.1.12.12
*Nov 21 23:01:25.132: IGMP(0): Send v3 Report with 1 group records on
Loopback0
*Nov 21 23:01:25.132: IGMP(0): Received v3 Report for 1 group on
Loopback0 from 10.11.11.11
*Nov 21 23:01:25.132: IGMP(0): Received Group record for group
232.1.1.1, mode 1 from 10.11.11.11 for 1 sources
*Nov 21 23:01:25.132: IGMP(0): MRT Add/Update Loopback0 for
(10.1.12.12,232.1.1.1) by 0
*Nov 21 23:01:25.132: IGMP(0): Updating expiration time on
(10.1.12.12,232.1.1.1) to 180 secs
```

R4#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP
Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires
Interface state: Interface, Next-Hop or VCD, State/Mode
(*, 239.1.1.4), 00:30:49/00:02:17, RP 0.0.0.0, flags: SJC
Incoming interface: Null, RPF nbr 0.0.0.0
Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:30:49/00:02:17

(10.1.12.12, 232.1.1.1), 00:30:50/00:02:29, flags: sLTI

Incoming interface: Ethernet1/0, RPF nbr 10.1.3.1

Outgoing interface list:

Loopback0, Forward/Sparse, 00:01:33/00:02:29

(*, 224.0.1.40), 00:30:50/00:02:12, RP 0.0.0.0, flags: DCL

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:27:43/00:02:12

R1#sh ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP
Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(10.1.12.12, 232.1.1.1), 00:05:41/00:02:43, flags: sT

Incoming interface: Ethernet0/0, RPF nbr 10.1.1.2

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:05:41/00:02:43

(*, 224.0.1.40), 00:06:41/00:02:18, RP 0.0.0.0, flags: DPL

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list: Null

On R4, the IGMPv3 join is seen and the mroute table is updated. The “s” flag denotes the SSM

state. R1 shows the incoming interface as Ethernet0/0 and the outgoing interface as Ethernet1/0 for 232.1.1.1. The incoming interface is based on unicast table for 10.1.12.12, and the outgoing is based on receiver reachability via the unicast routing table. When an additional subscription for the group is processed by the network from a separate source (10.2.12.12) connected to R3, the SSM network adjusts, adding the second channel. [Figure 3-24](#) illustrates this dual-channel instantiation.

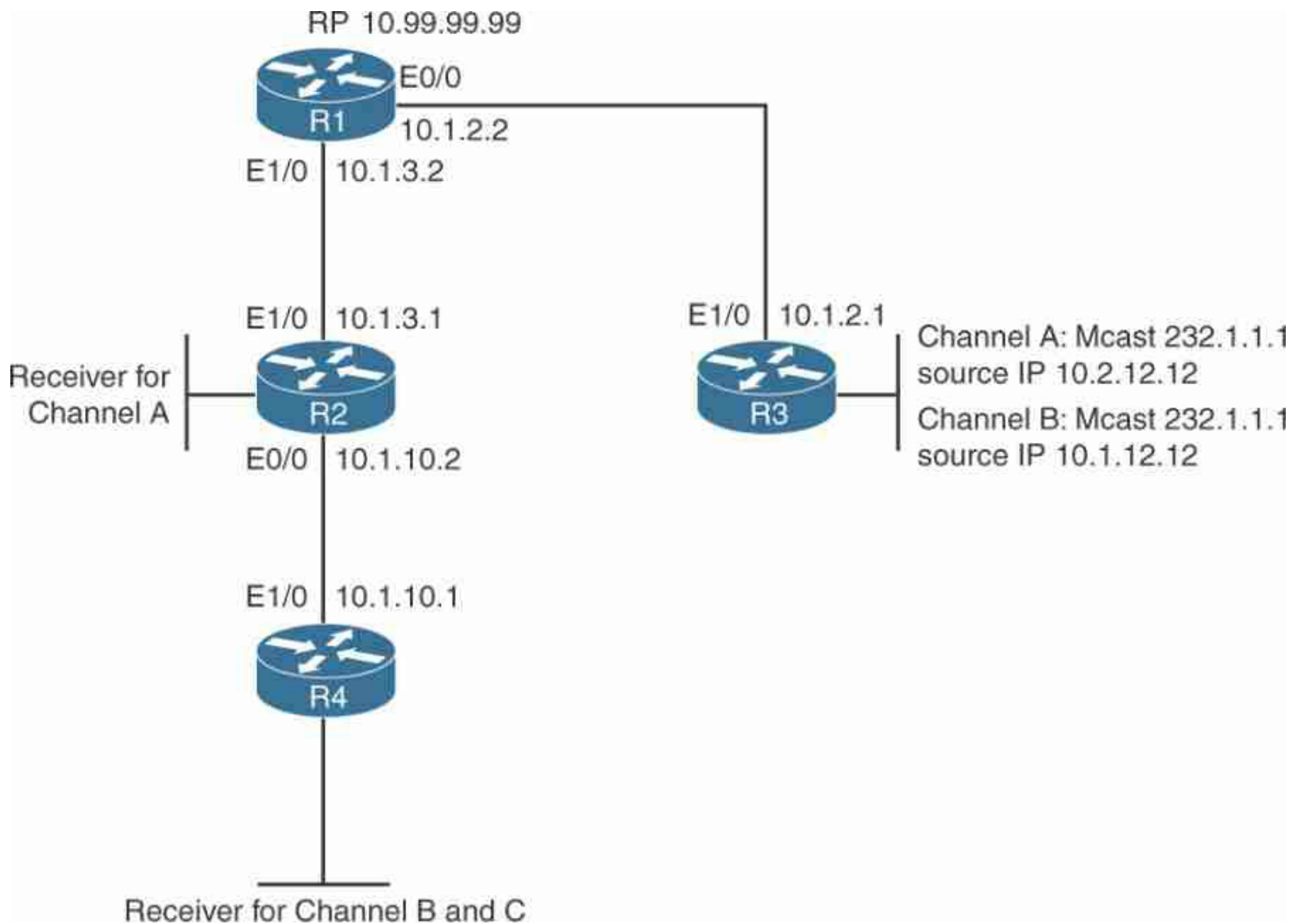


Figure 3-24 *PIM SSM Dual Channel*

Step 2. The **show ip mroute** command shows the dual channel for 232.1.1.1 with different source IP addresses.

[Click here to view code image](#)

R2#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP
Advertisement,

U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,
p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

...

(10.1.12.12, 232.1.1.1), 00:16:31/00:02:42, flags: sT

Incoming interface: Ethernet1/0, RPF nbr 10.1.10.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:16:31/00:02:42

(10.2.12.12, 232.1.1.1), 00:17:30/00:02:39, flags: sLTI

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list:

Loopback0, Forward/Sparse, 00:00:20/00:02:39

....

R1 has no state without the source IP address available in the routing table for 10.2.12.12, 232.1.1.1). This is shown using the **show ip mroute** command:

[Click here to view code image](#)

R1# show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP

Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner, p -
PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(10.1.12.12, 232.1.1.1), 00:19:04/00:03:09, flags: sT

Incoming interface: Ethernet0/0, RPF nbr 10.1.1.2

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:19:04/00:03:09

(*, 224.0.1.40), 00:20:04/00:02:01, RP 0.0.0.0, flags: DPL

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list: Null

Step 3. The source is active and available as seen in the unicast RIB. The state table on R1 changes with the state of (10.2.12.12, 232.1.1.1) seen in the routing table. These are viewed using the **debug ip pim** and **show ip mroute** commands:

[Click here to view code image](#)

*Nov 21 22:55:59.582: PIM(0): Insert (10.2.12.12,232.1.1.1) join in nbr 10.1.1.2's queue

*Nov 21 22:55:59.582: PIM(0): Building Join/Prune packet for nbr 10.1.1.2

*Nov 21 22:55:59.582: PIM(0): Adding v2 (10.2.12.12/32, 232.1.1.1), S-bit Join

*Nov 21 22:55:59.582: PIM(0): Send v2 join/prune to 10.1.1.2 (Ethernet0/0)

R1#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(10.2.12.12, 232.1.1.1), 00:03:38/00:02:33, flags: sT

Incoming interface: Ethernet0/0, RPF nbr 10.1.1.2

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:03:38/00:02:33

(10.1.12.12, 232.1.1.1), 00:25:29/00:02:38, flags: sT

Incoming interface: Ethernet0/0, RPF nbr 10.1.1.2

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:25:29/00:02:38

(*, 224.0.1.40), 00:26:28/00:02:31, RP 0.0.0.0, flags: DPL

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list: Null

For the same multicast group, two channels are shown in the output.

Step 4. The unicast routing table changes on R2, a new link is added, and the connection between R1 and R3 is brought down. Prior to the change, the mroute state on R2 is shown using the **show ip mroute** command:

[Click here to view code image](#)

R2#show ip mroute

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP

Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

..

(10.1.12.12, 232.1.1.1), 00:37:33/00:03:19, flags: sT

Incoming interface: Ethernet1/0, RPF nbr 10.1.10.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:09:15/00:03:19

..

The mroute state snapshot for (10.1.12.12, 232.1.1.1) shows the incoming interface of Ehternet1/0, which connects R1 to R3. [Figure 3-25](#) shows the topology change.

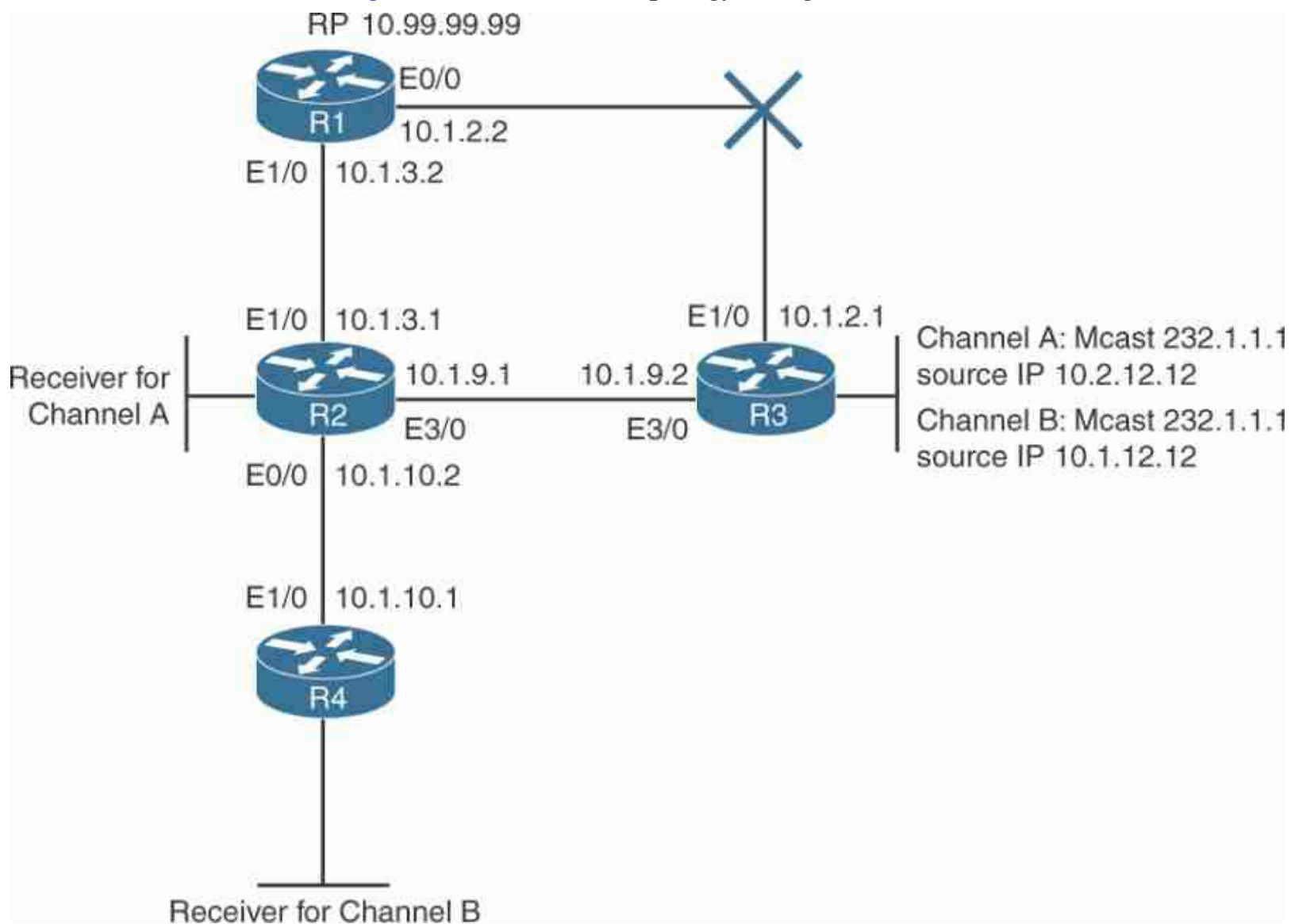


Figure 3-25 PIM SSM Dual Channel After Topology Change

At R2, the topology change can be seen using the **show ip mroute 232.1.1.1** command:

[Click here to view code image](#)

R2#show ip mroute 232.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(10.1.12.12, 232.1.1.1), 00:42:23/00:03:28, flags: sT

Incoming interface: Ethernet3/0, RPF nbr 10.1.9.2

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:00:01/00:03:28

(10.2.12.12, 232.1.1.1), 00:43:23/00:02:55, flags: sLTI

Incoming interface: Ethernet3/0, RPF nbr 10.1.9.2

Outgoing interface list:

Loopback0, Forward/Sparse, 00:26:12/00:02:55

The incoming interface is Ethernet3/0.

IGMPv3 subscribes the receiver to the specific source IP address in the join message. This greatly simplifies the tree build-up process, using an explicit source tree from the source to the receiver in SSM, and it leverages existing unicast routing information for multicast forwarding. This completely eliminates the need for an RP in the network while ensuring a loop-free forwarding topology.

Summary

Multicast networks consist of senders, receivers, and the network infrastructure. The communication of multicast messages uses the concept of a *tree*. The sender is the *root* of the tree and the receiver is a *leaf*, with the network being the *branches* or *infrastructure*. The responsibility of the infrastructure is to eliminate any duplication of messages, consequently removing any loops. This process is known as *building the tree*. Trees are built using one of two methods: a shared-tree shown as an (*,G) or a source-tree shown as an (S,G). A shared-tree is rooted at the rendezvous point (RP), and a source-tree is rooted at the source (S,G) or sender.

Layer 3 networking devices discover other directly attached networking devices and establish a “neighbor” relationship. Neighbor information is exchanged, which aids in understanding the capability of each device and in minimizing multicast joins from the same segment.

Protocol independent multicast (PIM) uses the underlying unicast routing protocol to forward multicast messages. Because multicast routing is the opposite of unicast routing, in that it sends messages away from the source, a new routing table for multicast specific routes is not required. Multicast routing uses the concept of reverse path forwarding (RPF).

Forwarding of multicast messages can operate in the following modes: dense, sparse, and sparse-dense. Dense mode uses a *push model*, in which multicast messages are sent to every PIM-enabled device in the network. By contrast, sparse mode uses a *pull model*, in which receivers request multicast messages. Finally, sparse-dense mode uses dense-mode flooding to propagate RP information and sparse-mode for sending messages. Be aware that a failure of the RP in a sparse-dense mode environment can cause dense-mode flooding.

Internet Group Management Protocol (IGMP) is the most common mechanism to control multicast messages at Layer 2. There are currently three versions of IGMP. IGMPv3 is the latest, and it does not require the use of an RP because it has awareness of the multicast source.

Any-source multicast (ASM) uses both a shared-tree and a source-tree. Typically, multicast messages start from a shared-tree using a RP but switch to a source-tree to provide better network efficiency. Bidirectional PIM (BiDir) is generally used to many-to-many communication and it supports only shared-trees. Source-specific multicast (SSM) requires receivers to be IGMPv3-aware, because this method uses only source-trees.

Chapter 4. Protocol Independent Multicast

Chapter 4. Protocol Independent Multicast

[Chapter 3](#) , “[IP Multicast at Layer 3](#) ,” addressed the different protocol independent multicast (PIM) modes and how those modes affected the growing or building of multicast trees. You saw the significant role that the rendezvous point (RP) played in establishing PIM sparse-mode and BiDir trees. In addition, you learned that source-specific multicast (SSM) does not require the use of an RP. This chapter covers the RP design mechanisms and configuration nuances using various Cisco platforms.

RP Overview

PIM-SM uses shared trees. With shared trees, multicast distribution is rooted at the RP. The process of building a shared tree requires that the network components register active sources as well as receivers for a given multicast group to the RP. To register to the RP, routers must encapsulate data in PIM control messages and send it by unicast to the RP. The source’s upstream designated router (DR) encapsulates and sends the register packet. Remember, the DR is a router on the source’s local network. A single DR is elected from all PIM routers on a subnet so that unnecessary control messages are not sent.

It should be noted that any number of routers could be configured as an RP for a given multicast group range. When the RP is configured, this information must be available to downstream Layer 3 devices on the network so that a shared tree for a specific group can be established. In simple sparse-mode deployments, all multicast groups are mapped to a single RP. [Figure 4-1](#) provides an example of the RP concept using a shared tree.

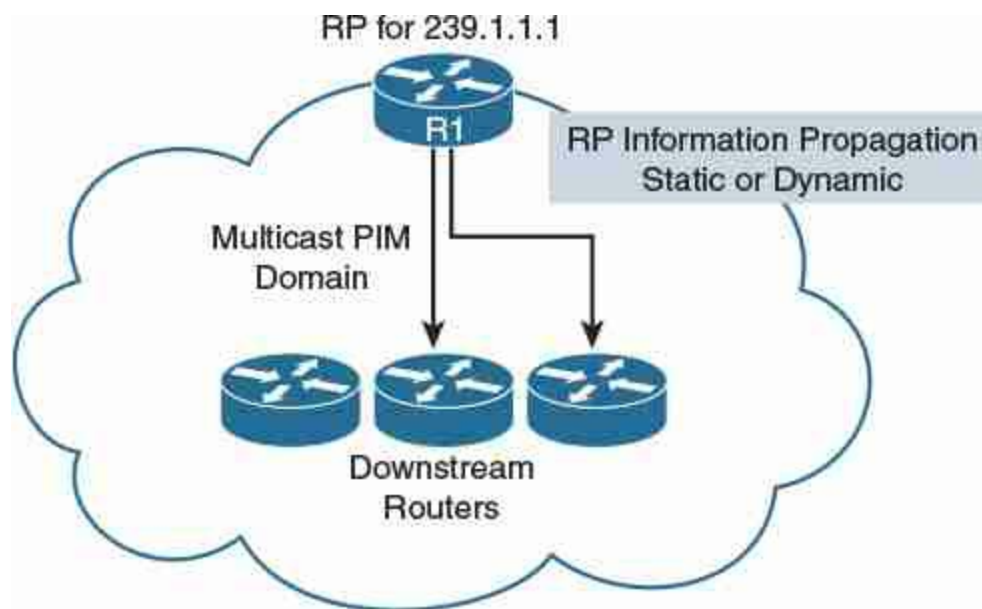


Figure 4-1 *Rendezvous Point in a Shared Tree*

In this example, router R1 is defined as the RP for multicast group 239.1.1.1. The other routers in the diagram are called the *downstream routers* . Each of the downstream routers needs to know about the RP. The RP information contains details about the IP address the RP is using (the location of the RP) and the group addresses for which the RP will require registration. It is imperative that the router acting as RP and the downstream routers all agree whom the RP is for a given group.

The downstream routers use this information to create mappings of groups to a given RP address.

This mapping is not necessarily one-to-one; there might be many RPs for different groups, or there might be only one RP for all groups. The number and placement of RPs is an administrative decision that is usually based on network scale and other best practices.

The RP-to-group mapping is a critical step in building sparse mode trees. If there is no group mapping, DRs will not be able to register group subscriptions, active sources cannot be tracked, and routers downstream will not know the direction in which to RPF check shared tree entries. In short, the shared tree will not form across the network, from the RP to the receivers, and the multicast path will fail.

Examining the following **show** command output from an IOS router can give us additional insight into how routers use this information:

[Click here to view code image](#)

```
R1#show ip pim rp 239.127.1.1
```

```
Group: 239.127.1.1, RP: 192.168.0.2, v2, v1, uptime 05:44:08, expires 00:02:53
```

Using the **show ip pim rp address** command, you see that the configured RP for multicast group 239.127.1.1 is the router with an interface using 192.168.0.2, but this only tells you what RP address the router is aware of for group 238.127.1.1. Using the **show ip pim rp mapping** command shows all the mappings learned by the router, even those that have been learned dynamically but are not installed. Adding the **in-use** option shows only those mappings that are installed in the mRIB and how the mapping was learned. [Example 4-1](#) shows that all multicast groups are mapped to RP 192.168.0.2, and the mapping was learned dynamically through Auto-RP, a Cisco proprietary RP information distribution protocol.

Example 4-1 *show ip pim rp mapping in-use Command Output*

[Click here to view code image](#)

```
R1#show ip pim rp mapping in-use
```

```
PIM Group-to-RP Mappings
```

```
Group(s) 224.0.0.0/4
```

```
RP 192.168.0.2 (?), v2v1
```

```
Info source: 192.168.0.2 (?), elected via Auto-RP
```

```
Uptime: 05:44:19, expires: 00:02:42
```

```
Dynamic (Auto-RP or BSR) RPs in cache that are in use:
```

```
Group(s): 224.0.0.0/4, RP: 192.168.0.2, expires: 00:00:58
```

Adding a group address to the **show ip pim rp mapping** command provides the mapping information for that specific group; in the case of [Example 4-2](#), the mapping is using static ACL 10.

Example 4-2 *show ip pim rp mapping group-address Command Output*

[Click here to view code image](#)

```
R1#show ip pim rp mapping 239.127.1.1
```

```
PIM Group-to-RP Mappings
```

```
Acl: 10, Static-Override
```

```
RP: 192.168.0.2 (?)
```

This output clearly shows that group-to-RP mappings can be learned by the router through various methods. This brings us to the concept of propagating the RP information to the downstream devices.

Any inconsistencies in RP group mappings by downstream routers also cause trees to fail because downstream routers will not have the appropriate RPF or other information from which to build a share tree.

The downstream devices can learn this RP information in one of two ways: static mapping through manual configuration, or dynamic sharing and mapping through protocols. A network administrator has two protocol options for dynamic distribution: Auto-RP (Cisco proprietary) or the IETF standard bootstrap router (BSR). These RP propagation methods are considered at length in later sections of this chapter.

Similar to other IP protocols, reliability and availability are very important in any network, especially if forwarding information is being propagated dynamically. Networks often address availability, frequently called *high availability (HA)*, through redundancy of critical systems. IP multicast networks are similar in function and RPs can be configured with HA in mind.

[Figure 4-2](#) shows a network with HA added to the RP configuration. RP redundancy provides high availability to RP resources using shared trees. High availability in networking terms can usually be classified as active/active or active/standby. The same is true for RP redundancy configurations.

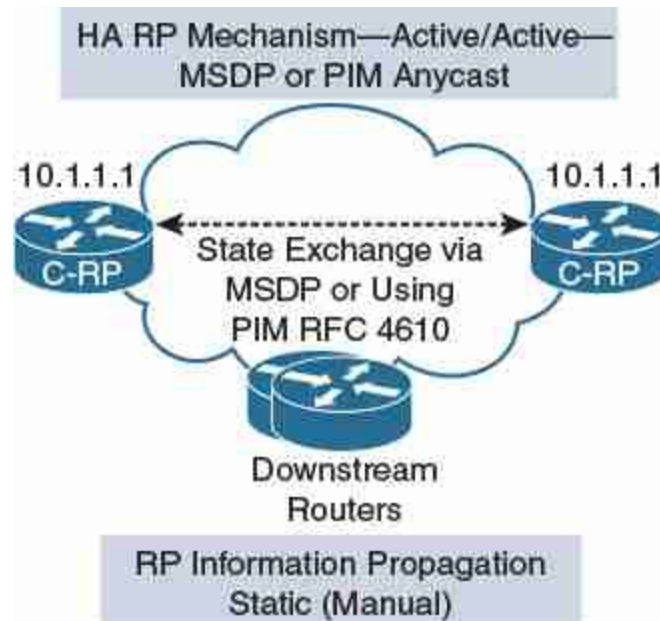


Figure 4-2 *RP High Availability*

IP Multicast Domains

In order to create effective multicast designs, we need to understand the meaning and use of a multicast domain. Just like the unicast routing protocols OSPF and EIGRP, PIM routers have the capability to dynamically share information about multicast trees. For most networks, there is only one routing protocol that is used internally, the interior gateway protocol, or IGP.

When the IGP network is properly designed, most routers in the network will have the same routes in their individual routing information base (RIB). The routes may be summarized into larger entries on some routers, even as far as having only one summary (default) route on stub routers. Often, this process is controlled by the use of regions (or areas, in the case of IS-IS or OSPF). The point is that when it's configured, the IGP dynamically provides the necessary routing information to all of the routers that need the information, and those routers interacting with each other are part of a domain.

Of course, the deployed IGP has a natural, physical boundary. When the interface of an IGP router is not configured to send/receive IGP information, that interface bounds the IGP. This serves the

purpose of preventing internal routing information from leaking to routers that should not or do not need internal information.

To share routes outside the network, administrators generally use an External Gateway Protocol (EGP). If a network needs to connect to the global Internet, it will generally use Border Gateway Protocol (BGP) as the EGP. The EGP only shares essential IGP routes with external routers, so that only internal networks meant for public access are reachable by external devices. The routing process of the IGP is kept separate and secure from external influence. For most networks, the natural boundary of the internal routing domain lies between the IGP and the EGP of the network.

We often call the internal network, as it appears to the outside, an *autonomous system (AS)*. In fact, BGP relies heavily on this concept because each public BGP network must use an assigned AS number (ASN) to participate in the global Internet. As with IP address and multicast IP group assignments, IANA assigns these ASNs to participating networks. To more deeply understand the division of networks into domains and autonomous systems, please refer to IETF RFCs 1136 and 1930.

Multicast networks also must have boundaries; however, the boundaries may be drastically different than those of the underlying unicast network. Why? It is important to remember that IP multicast networks are an overlay on the IGP network, and the network routers can only have one PIM process.

The unicast network could use multiple routing processes or even protocols through the use of redistribution. Sophisticated link-state protocols, like OSPF and IS-IS, provide built-in mechanisms for tighter controls on route sharing. PIM routers can share tree state information with any PIM neighbors and have no similar structures.

Under the definition of a domain, all PIM routers with the capability to share trees could be considered a part of the domain. Although convenient for the purposes of multicast forwarding, it may not be secure or desirable to have all multicast sources and groups available to all routers within that domain. It also may not be entirely efficient, depending on the location of sources, receivers, and RPs. PIM networks need tighter, more configurable boundaries and a different definition for a domain.

Network architects can define and create PIM domains in multiple ways. If the multicast overlay is very simple, then the domain may also be very simple. The domain could span an entire IGP network, with universal PIM neighbor relationships between all IGP routers. When required, a single RP could be used for all group mappings. In this type of domain, all multicast groups and sources would be available to all systems within the domain.

If the network requirements are more stringent, it is likely that a defined tiered domain model will be needed. Many network architects use group access requirements as a way to define a multicast domain. Let's look at a theoretical example, in this case, Multicaster's Bank Corporation.

[Figure 4-3](#) shows the Multicaster's global network. It is very likely, especially in a financial institution, that applications will require intense security between various portions of the network. For example, there might be a financial trading application that must update several local servers in real time. The security and accuracy of such information is of paramount concern. Thus, the network architect may want to define a single domain around that application.

Multicaster's Bank Corp.

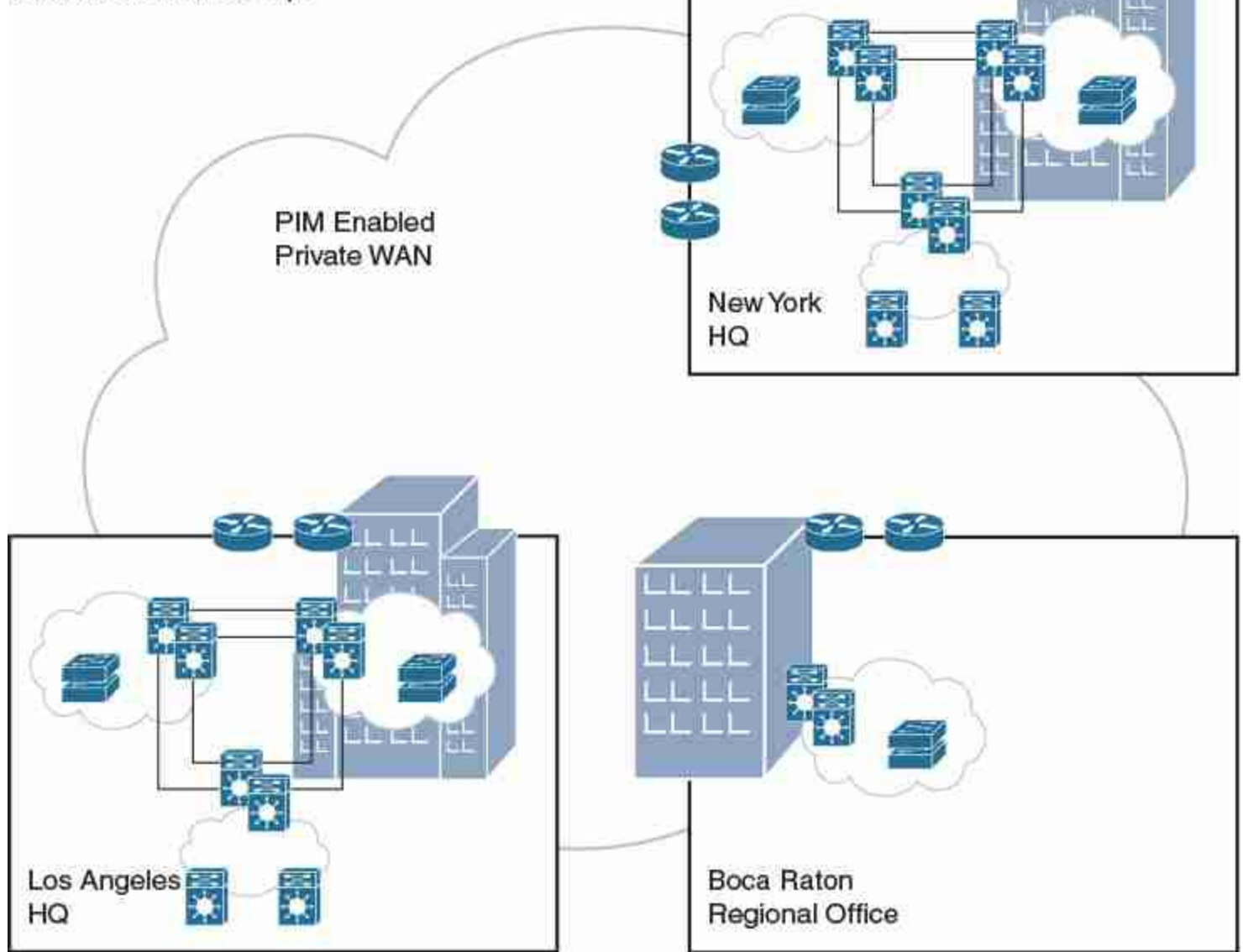


Figure 4-3 *Multicaster's Bank Corp*

In the case of Multicaster's there are two data centers at each of the two major trading locations, L.A. and N.Y. Each data center must process only the local trading information and should not share multicast data flows between them. A domain with secure boundaries must then be configured for each data center.

Multicaster's Bank Corp. system administrators also prefer to use multicast groups for local administrative tasks, or for local services like music on hold for local IP telephony systems. It would not be very secure or prudent to allow the local administrator's multicast flows to interrupt or possibly corrupt the flows updating the trade floor application in the L.A. data center.

One effective way to divide these two types of flows into separate domains is to simply use sparse mode on all router links but to use a unique RP for each domain. With different RP structures, the routers in each domain will build trees only within that domain. [Figure 4-4](#) takes a closer look at the L.A. location using such a model.

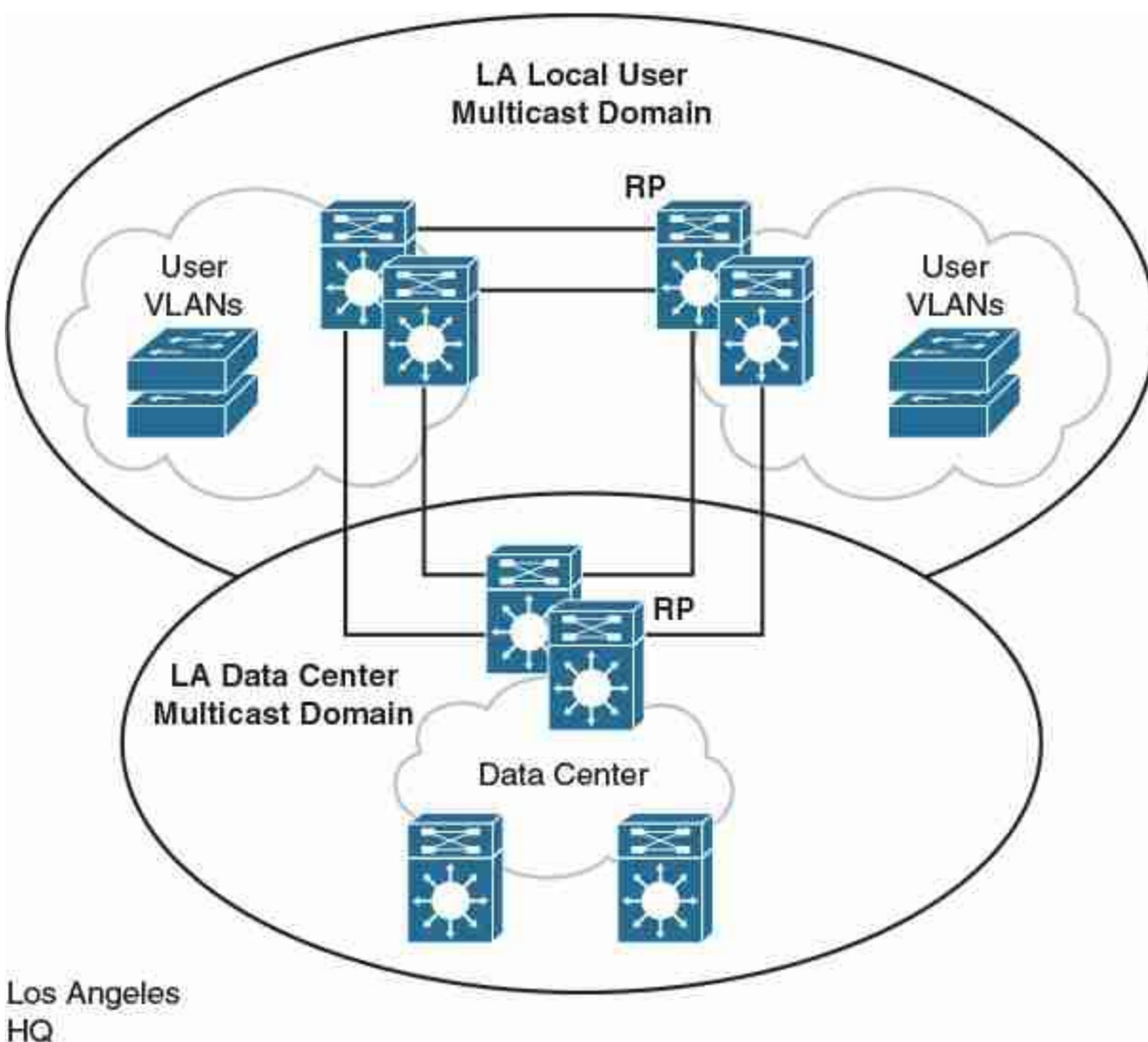


Figure 4-4 *Multicaster's L.A. Multicast Domains*

As discussed earlier, an RP is quantified and mapped for specific multicast groups. This multicast group presence with the mapped RP is called a *multicast scope range*, and that scope becomes the domain, the boundaries being any routers in the PIM network not configured with the RP information for a given group. In enterprise designs, you can have multiple scopes specific to a group. These scope ranges are aligned with RP and PIM domains. Scope ranges can superimpose over each other or be a subset of one another. [Figure 4-5](#) illustrates a scope range.

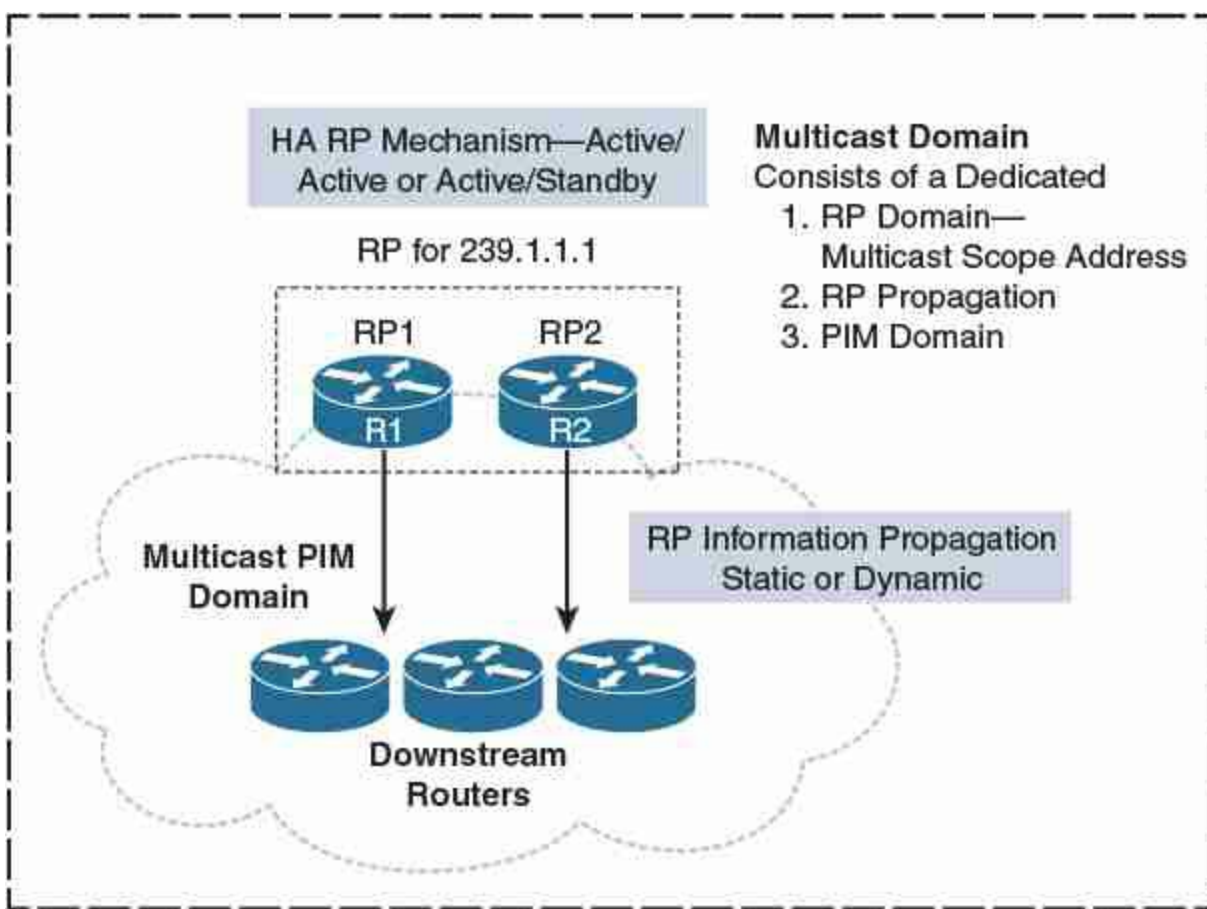


Figure 4-5 *Scope Range*

The scope of the domain is dependent on the capability of the downstream routers to properly map group addresses to the appropriate RP. In order to build shared trees and subsequent source trees, each router in the domain must map group addresses to the same RP for a given group in that domain. As mentioned previously, if this group mapping is not consistent, or present, tree formation cannot be completed and flows will fail.

The configuration items and criteria reviewed here, especially RP propagation, availability, and scoping, are important to consider while learning dynamic RP protocols. Scoped multicast domains play a very important role in multicast design. Multicast domains and the relationships between them will be discussed in greater detail in [Chapter 5](#).

Basic PIM Configuration

The PIM process begins when two or more routers establish a PIM neighbor adjacency on a given segment. To begin the adjacency process, any interfaces participating in multicast routing must be configured for PIM communications. To configure an IOS router interface for PIM, use the following interface level command:

[Click here to view code image](#)

```
ip pim { dense-mode [ proxy-register {list access-list | route-map map-name} ]
| passive | sparse-mode | sparse-dense-mode }
```

In IOS-XE and in NX-OS, the **ip pim** command is entered at the interface configuration mode. For IOS-XR, the command is entered as **router pim**, which enables PIM for the default context and enters the PIM configuration submenu. The **passive** command option prevents the sending of PIM messages from this interface and instead duplicates and forwards any IGMP reports received.

Note

Don't forget the importance of the underlying unicast network. Because of the heavy reliance on unicast, it is wise and recommended to simply include every interface in the domain in the PIM topology. That means configuring PIM on every IP interface within the domain. As the domain, or the network grows, you can simply make the PIM configuration a network standard. There might be cases where this does not make sense; see [Chapter 5](#) for more details.

Note

The **dense-mode proxy register** command option allows a dense-mode router to use a designated router to join a sparse-mode domain. The DR elected can generate and send a (*,G) register messages from the (S,G) dense-mode trees in the routers mRIB. This allows for full multicast domain participation where legacy routers cannot support sparse mode. The administrator can also assign a route map or access list to limit the groups allowed by the proxy.

Because dense-mode PIM is no longer a recommended mode for multicast networks, it is likely that most networks will be running in sparse-mode. In a sparse-mode deployment, it is not enough to have neighbor adjacencies. We also need an RP and a way to associate the group to RP mapping. As previously mentioned, RPs and group mappings can be learned statically (through manual configuration) at each router or dynamically (by using either the Auto-RP or BSR protocols for propagation).

Static RP

Enabling PIM sparse-mode using a static RP is perhaps the simplest way to configure a multicast domain. A static RP is defined through manual configuration on the RP itself and at every downstream router. The configuration, with multicast scoping (that is, multicast RP configuration with an access-list for multicast group definition), has to be defined at every downstream router in exactly the same way; otherwise, the domain will be incomplete.

Using a static RP might be a good choice for certain multicast configurations, including Anycast RP with MSDP for RP redundancy. This can also be an attractive option if the network is small. This RP mechanism does not provide any redundancy (unless coupled with Anycast RP). Anycast RP is discussed more fully in subsequent sections of this chapter.

Use the commands in [Table 4-1](#) , shown by platform, to configure a static RP.

Operating System	Command
IOS/XE	<code>ip pim [vrf <i>vrf-name</i>] rp-address <i>rp-address</i> [<i>access-list</i>] [<i>override</i>] [<i>bidir</i>]</code>
IOS XR	<code>rp-address <i>ip-address</i> [<i>group-access-list</i>] [<i>override</i>] [<i>bidir</i>]</code>
NX-OS	<code>ip pim rp-address <i>rp-address</i> [<i>group-list prefix</i> <i>override</i> <i>route-map policy-name</i>] [<i>bidir</i>]</code>

Table 4-1 *Commands to Configure a Static RP*

A static RP can coexist with dynamic RP mechanisms such as Auto-RP. By default, dynamically learned RP mappings take precedence over manually configured RPs. Thus, if a router receives Auto-RP information for a multicast group that has been manually configured, the Auto-RP information will be used unless the **override** keyword is specified. This scenario occurs only if there are multiple RPs available for mapping to the same group(s).

Let's look at a basic configuration example using only one static RP for all multicast group addresses. [Figure 4-6](#) shows the network diagram for this configuration.

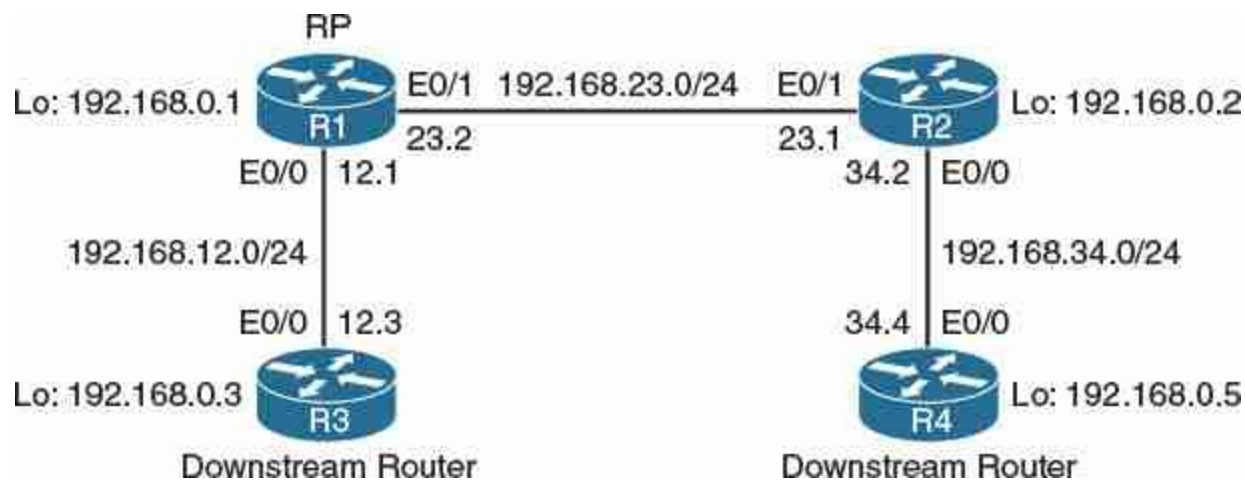


Figure 4-6 Static RP. The Icons Represent Layer 3 Functionality, Including IOS, IOS-XR, and NxOS

The following are step-by-step configuration instructions to enable PIM sparse-mode for IOS, IOS-XR and NX-OS using static RP.

The steps to configure static RP with IOS and enabling PIM sparse-mode are as follows:

Step 1. Enable IP multicast routing.

```
ip multicast-routing
```

Step 2. Enable interfaces in the Layer 3 domain, including the loopback with the **ip pim sparse-mode** command:

[Click here to view code image](#)

```
interface Loopback0
ip address 192.168.0.1 255.255.255.255
ip pim sparse-mode
!
interface Ethernet0/0
ip address 192.168.12.1 255.255.255.0
ip pim sparse-mode
```

Step 3. Add the static RP configuration:

[Click here to view code image](#)

```
R3(config)#ip pim rp-address 192.168.0.1 ?
<1-99>      Access-list reference for group
<1300-1999> Access-list reference for group (expanded range)
WORD        IP Named Standard Access list
override    Overrides dynamically learnt RP mappings
```

Configuring IOS-XR is significantly different from configuring IOS. The PIM configuration is accomplished through separate **multicast-routing** and **router pim** configuration modes. The following basic steps explain how to configure PIM using a static RP with PIM sparse-mode in IOS XR:

Step 1. Enable multicast-routing and enable Layer 3 interfaces:

[Click here to view code image](#)

```
multicast-routing
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
interface all enable
```

Step 2. Enable interfaces in the Layer 3 domain, including the loopback with the **ip pim sparse-mode** command:

[Click here to view code image](#)

```
router pim
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
```

Step 3. Add the static RP configuration:

[Click here to view code image](#)

```
RP/0/0/CPU0:R1(config-pim)#router pim
RP/0/0/CPU0:R1(config-pim)#address-family ipv4
RP/0/0/CPU0:R1(config-pim-default-ipv4)#rp-address 192.168.0.1?
WORD      Access list of groups that should map to given RP
bidir     Specify keyword bidir to configure a bidir RP
override  Static RP config overrides auto-rp and BSR
<cr>
```

The static RP configuration in NX-OS is similar to IOS and IOS-XE configurations and is as follows:

Step 1. Enable the **feature pim** .

```
feature pim
```

Step 2. Enable interfaces in the Layer 3 domain, including the loopback with **ip pim sparse-mode** :

[Click here to view code image](#)

```
interface Ethernet2/1
no switchport
mac-address 0001.4200.0001
ip address 192.168.23.1/24
ip router ospf 1 area 0.0.0.0
ip pim sparse-mode
```


no shutdown

Step 3. Add the static RP configuration:

[Click here to view code image](#)

```
nexusr1(config)# ip pim rp-address 192.168.0.1 ?
<CR>
bidir      Group range is treated in PIM bidirectional mode
group-list  Group range for static RP
override    RP address will override the dynamically learnt RPs
prefix-list Prefix List policy for static RP
route-map   Route Map policy for static RP
```

Note

Why is it important to configure the main network loopback interface with sparse-mode PIM as shown in the preceding examples? After all, the loopback interface is unlikely to have any PIM neighbors. This is a recommended practice for any multicast overlay. The reason for this recommendation is that the router can then fully participate in the multicast domain, even if errors are occurring on leaf facing interfaces. It also allows the loopback interfaces to be used as a RP addresses or mapping agents in dynamic RP propagation, making them more reliable. See [Chapter 5](#) , “[IP Multicast Design Considerations and Implementation](#) ,” for more information on this and other best practices for multicast networks.

PIM Dense Mode

To configure dense mode on older Cisco IOS routers and switches, use the following commands:

[Click here to view code image](#)

```
C6K-720(config)#ip multicast-routing [vrf vrf-name] [distributed]
C6K-720(config)#interface {type [number|slot/port[/port]]}
C6K-720(config-if)#ip pim dense-mode [proxy-register {list access-list |
route-map map-name}]
```

[Figure 4-7](#) depicts a small campus network with a very limited multicast deployment for minor host updates. The underlying configuration example enables PIM dense-mode multicast and utilizes IGMP snooping for improved Layer 2 efficiency. IGMP snooping should be enabled by default.

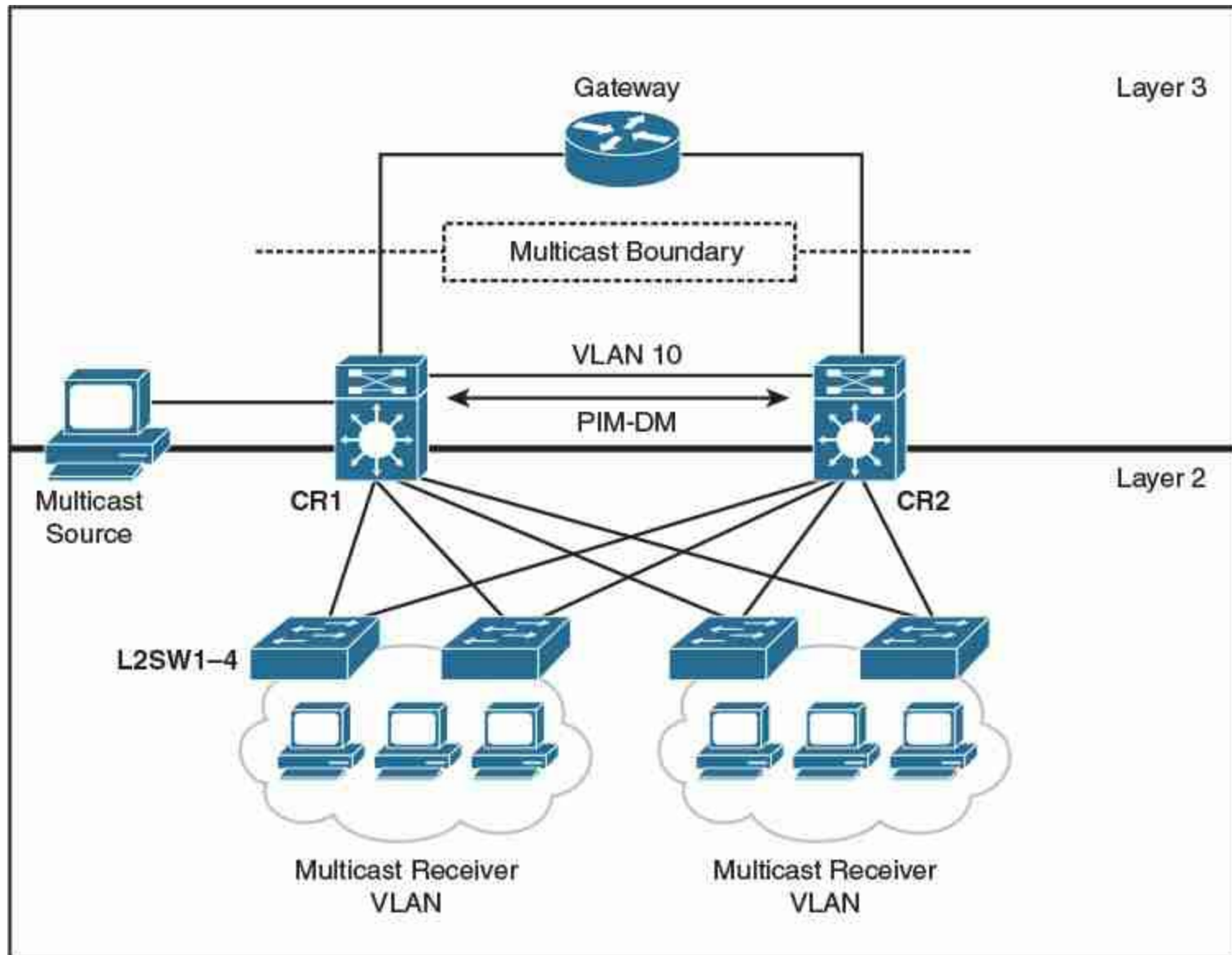


Figure 4-7 *Small Dense-Mode Deployment*

Note

As discussed previously, there are very few reasons to ever deploy a PIM-DM network. Because of this, many Cisco networking operating systems will not support dense-mode configuration or certain dense-mode features. At presstime, Cisco IOS-XR and NX-OS do not support any PIM dense-mode-deployment or configurations. The following sample configuration is only provided as supplemental for existing dense-mode-compatible systems.

Using the network topology in [Figure 4-7](#) , the IOS configuration commands in [Example 4-3](#) demonstrate how to configure dense-mode multicast.

Example 4-3 *Configuring Dense-Mode Multicast in IOS*

[Click here to view code image](#)

```
CR1(config)#ip multicast routing
```

```
CR1(config)#interface vlan 10
```

```
CR1(config-if)#ip pim dense-mode
```

```
CR2(config)#ip multicast routing
```

```
CR2(config)#interface vlan 10
```

```
CR2(config-if)#ip pim dense-mode
```

Dynamic RP Information Propagation

There are two ways to dynamically propagate RP information to routers within a domain:

- Auto-RP (Cisco proprietary)
- Bootstrap router (BSR)

Both solutions are acceptable because they provide a similar service and play a key role in a multicast design.

You may be asking yourself, “If static configurations work well, why have a dynamic protocol at all?” As discussed earlier, one of the most important concepts in group mapping is that all routers within a domain agree on the RP for a given group. If a network is very large, has many overlapping domains, many multicast applications, many rendezvous points, or all of the above, a consistent group mapping through static commands may become extremely difficult to manage. In fact, it is for this reason that these two dynamic protocols provide not only dynamic propagation, but also methods of ensuring RP mapping accuracy and consistency throughout a domain at all downstream routers. Remember, the RP itself does not make mapping decisions for downstream routers. Each router must learn of the RP individually and use the provided information to determine the best RP to map a group to. There are some similarities between Auto-RP and BSR that provide this consistency to downstream routers.

The control for this process is accomplished through the concept of centralized mapping. This means that some routers in the network are configured as RP routers and advertise themselves as such to other routers in the network. Centralized mapping routers receive the information about the RPs available within the network and establish group to RP mapping parameters, or compile available RP sets. When RP information is distributed from the centralized mapping routers, downstream routers need only listen to these advertisements and use the advertised information to create local RP mapping entries. This also serves the added purpose of limiting the number of protocol messages required throughout the domain.

Auto-RP and BSR both perform these mapping and advertising functions differently. But in the end, they provide the same essential functions to the network.

Auto RP

Auto-RP provides HA (active/standby) to the RP service. The propagation of RP information to the downstream routers is done via Auto-RP messages. The downstream routers do not require an explicit RP configuration. Rendezvous points using Auto-RP announce their availability to the mapping agents via the 224.0.1.39 multicast group. The RP mapping agent listens to the announced packets from the RPs, then sends RP-to-group mappings in a discovery message to 224.0.1.40. Downstream routers listen for mapping advertisements on group 224.0.1.40 and install the RP mappings as advertised from the mapping agent. It is acceptable to use the same interface address RP as a candidate and mapping agent. In larger systems to provide greater scalability, it would more efficient to use different interfaces, or to separate the candidate and agent functions to different routers. [Figure 4-8](#) shows the Auto-RP mechanism.

HARP Mechanism—Active/ Standby

Mapping Agent

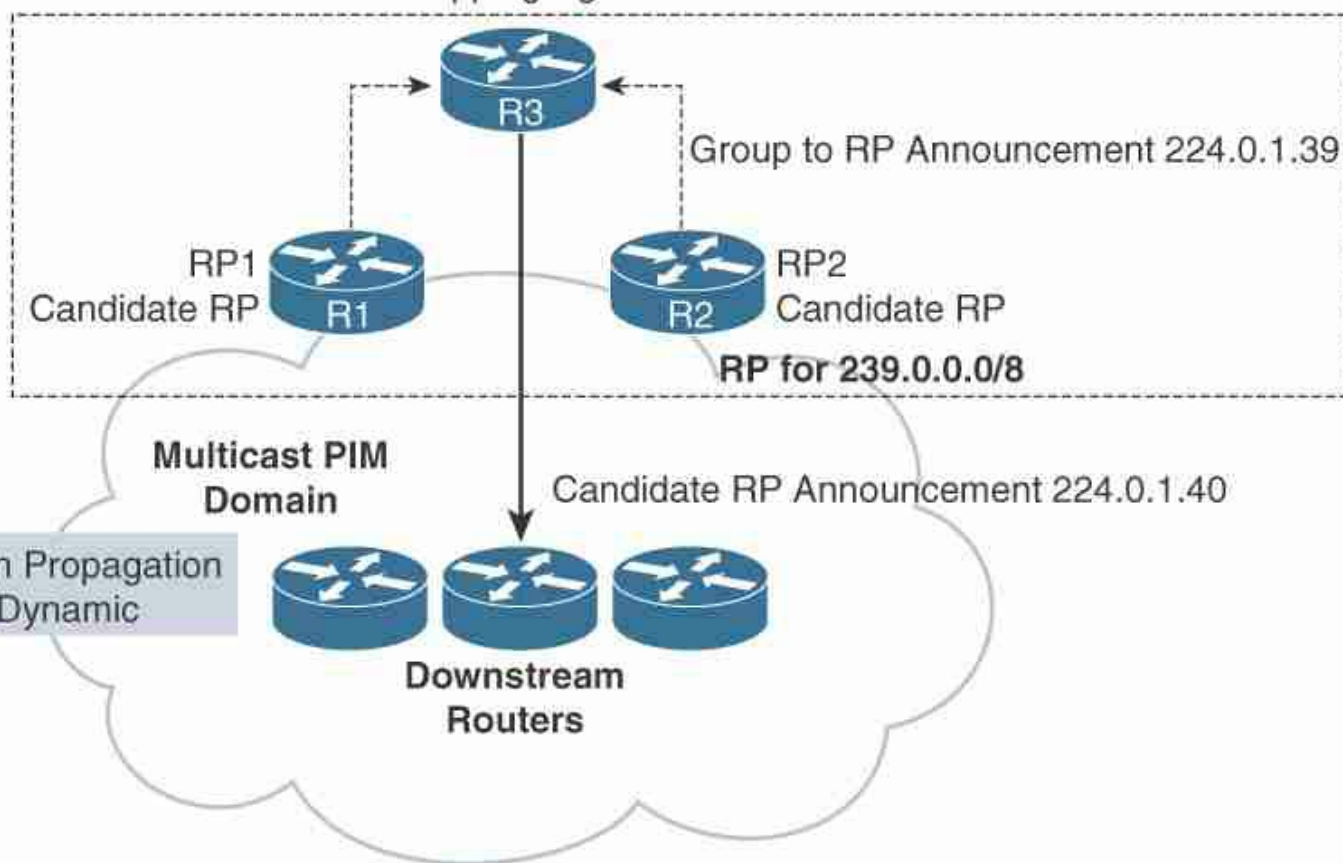


Figure 4-8 *Auto-RP Overview*

The two multicast groups for Auto-RP information are advertised via dense mode in the sparse-dense mode of interface operation. This flooding of message allows automatic propagation to the downstream routers. As mentioned earlier, some operating systems do not support dense mode. How can RP information be propagated in a sparse-mode-only environment? You can use “listen” configuration commands in global configuration mode to cause IP multicast traffic for the two Auto-RP groups of 224.0.1.39 and 224.0.1.40 to be protocol independent multicast (PIM) dense-mode flooded across interfaces operating in PIM sparse-mode.

The Auto-RP mapping agent uses the following logic to build the RP-cache:

- When there is a tie between two candidate RPs, the RP with the highest IP address wins the tie.
- Two candidate RPs contest where one group is a subset of another, but the RPs are different. Both will be sent in the discovery RP-cache.

Auto-RP is best-suited in a multicast scoped environment. The Auto-RP message has an inbuilt time to live (TTL) and various other boundary features that make it best-suited in scoped multicast environments. A scoped multicast domain has its own RP with a group address assigned, which makes it a separate PIM domain.

[Table 4-2](#) outlines some items to remember using Auto-RP:

Features	Support
Scoping design support	Yes
HA	Active/standby
Automatic downstream propagation	Yes

Table 4-2 *Auto-RP Feature Considerations*

[Table 4-3](#) outlines the IOS/XE, IOS XR, and NX-OS mapping agent commands.

Operating System	Command
IOS/XE	<code>ip pim [vrf vrf-name] send-rp-discovery [interface-type interface-number] scope ttl-value [interval seconds]</code>
IOS XR	<code>auto-rp mapping-agent type interface-path-id scope ttl-value [interval seconds]</code>
NX-OS	<code>ip pim auto-rp mapping-agent if-type if-number [scope ttl]</code>

Table 4-3 *Mapping Agent Commands for IOS/XE, IOS XR, and NX-OS*

[Table 4-4](#) outlines the IOS/XE, IOS XR, and NX-OS Candidate RP commands.

Operating System	Command
IOS/XE	<code>ip pim [vrf vrf-name] send-rp-announce [interface-type interface-number ip-address] scope ttl-value [group-list access-list] [interval seconds] [bidir]</code>
IOS XR	<code>auto-rp candidate-rp type interface-path-id scope ttl-value [group-list access-list-name] [interval seconds] [bidir]</code>
NX-OS	<code>ip pim auto-rp rp-candidate if-type if-number [group-list prefix] { [scope ttl] [interval interval] [bidir] }</code>

Table 4-4 *Candidate RP Commands for IOS/XE, IOS XR, and NX-OS*

[Table 4-5](#) outlines the IOS/XE, IOS XR, and NX-OS Auto-RP Listener commands.

Operating System	Command
IOS/XE	<code>ip pim autorp listener</code>
IOS XR	Default is to listen; to disable: <code>router(config)# router pim</code> <code>router(config-pim-default-ipv4)# auto-rp listen disable</code>
NX-OS	<code>ip pim auto-rp { listen forward } forward [listen]</code>

Table 4-5 *Auto-RP Listener Commands for IOS/XE, IOS XR, and NX-OS*

Note

With NX-OS, use the **listen** keyword to process the Auto-RP message and use the **forward**

keyword to send the Auto-RP message to next-downstream routers.

[Figures 4-9](#) through [4-11](#) illustrate a typical deployment example on how to configure Auto-RP in IOS, IOS-XR and NX-OS. In this example, R1 and R2 are the candidate RPs and mapping agents, a common deployment practice.

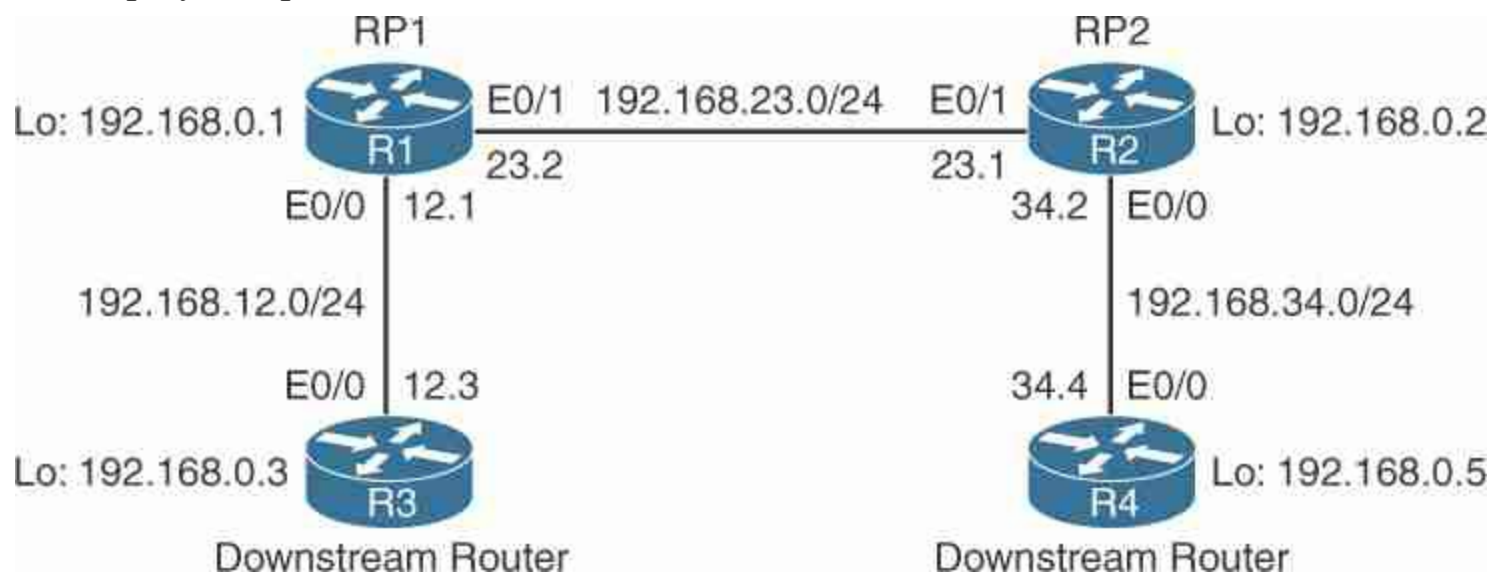


Figure 4-9 *IOS RP Configuration Topology*

Sample Configuration: Auto-RP for IOS

[Example 4-4](#) shows the RP configuration for Router R1 and R2.

Example 4-4 *Configuring RP for R1 and R2 in IOS*

RP: R1	RP: R2
<pre> ip multicast-routing ip cef ! interface Loopback0 ip address 192.168.0.1 255.255.255.255 ip pim sparse-mode ! interface Ethernet0/0 ip address 192.168.12.1 255.255.255.0 ip pim sparse-mode ! interface Ethernet0/1 ip address 192.168.23.2 255.255.255.0 ip pim sparse-mode ! ! router ospf 1 network 0.0.0.0 255.255.255.255 area 0 ! ip pim autorp listener ip pim send-rp-announce Loopback0 scope 255 group-list 1 interval 15 ip pim send-rp-discovery Loopback0 scope 255 interval 15 ! ! ! access-list 1 permit 239.0.0.0 0.255.255.255 </pre>	<pre> ip multicast-routing ip cef ! interface Loopback0 ip address 192.168.0.2 255.255.255.255 ip pim sparse-mode ! interface Ethernet0/0 ip address 192.168.34.2 255.255.255.0 ip pim sparse-mode ! interface Ethernet0/1 ip address 192.168.23.1 255.255.255.0 ip pim sparse-mode ! ! router ospf 1 network 0.0.0.0 255.255.255.255 area 0 ! ip pim autorp listener ip pim send-rp-announce Loopback0 scope 255 group-list 1 interval 15 ip pim send-rp-discovery Loopback0 scope 255 interval 15 ! ! ! access-list 1 permit 239.0.0.0 0.255.255.255 </pre>

[Example 4-5](#) shows the downstream router configuration:

Example 4-5 *Configuring the Downstream Router in IOS*

[Click here to view code image](#)

R3: Downstream router

```

ip multicast-routing
ip cef
!
interface Loopback0
 ip address 192.168.0.3 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet0/0

```

```

ip address 192.168.12.3 255.255.255.0
ip pim sparse-mode
!
!
!
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
!
ip pim autorp listener
!

```

Sample Configuration: Auto-RP for IOS-XR

[Figure 4-10](#) shows the topology for the following sample configuration of Auto-RP for IOS-XR.

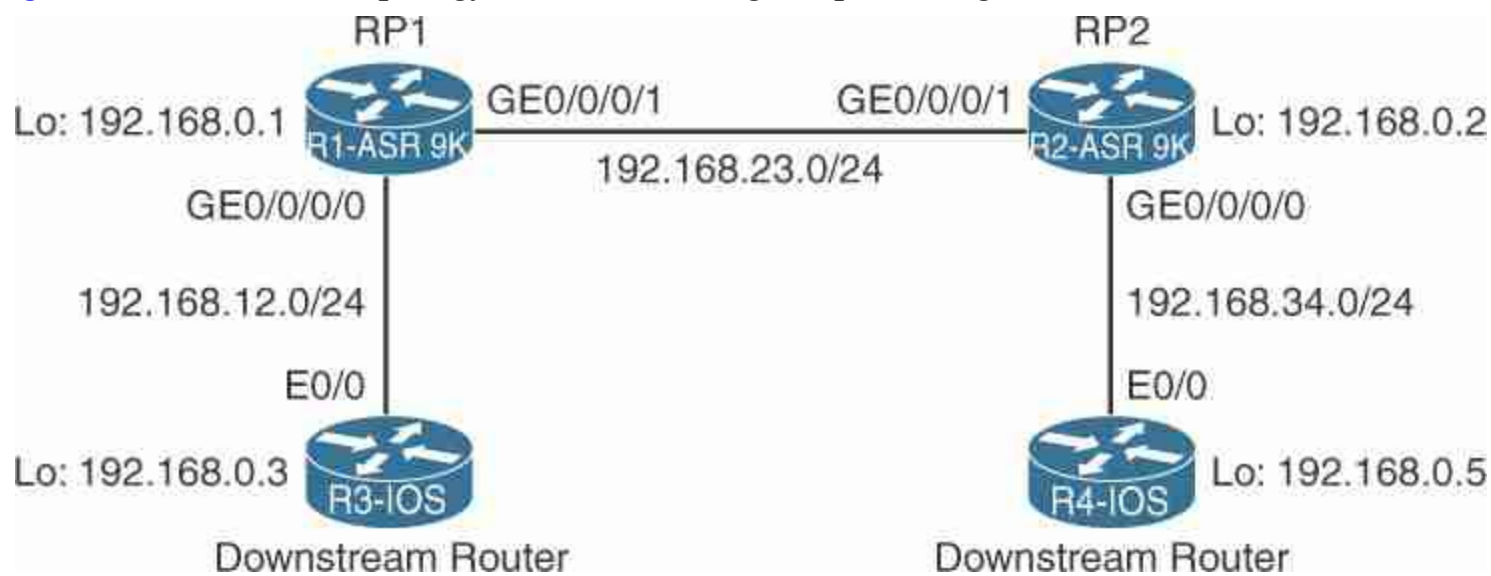


Figure 4-10 *IOS-XR Auto RP Configuration Topology*

[Example 4-6](#) shows the RP configuration for Router R1 and R2.

Example 4-6 *Configuring RP for R1 and R2 in IOS-XR*

RP: R1	RP: R2
<pre> hostname R1 ipv4 access-list 1 10 permit ipv4 239.0.0.0 0.255.255.255 any ! interface Loopback0 ipv4 address 192.168.0.1 255.255.255.255 ! interface GigabitEthernet0/0/0/0 ipv4 address 192.168.12.1 255.255.255.0 ! interface GigabitEthernet0/0/0/1 ipv4 address 192.168.23.1 255.255.255.0 ! router ospf 1 area 0 interface Loopback0 ! interface GigabitEthernet0/0/0/0 ! interface GigabitEthernet0/0/0/1 ! ! ! ! </pre>	<pre> hostname R2 ipv4 access-list 1 10 permit ipv4 239.0.0.0 0.255.255.255 any ! interface Loopback0 ipv4 address 192.168.0.2 255.255.255.255 ! interface GigabitEthernet0/0/0/0 ipv4 address 192.168.34.2 255.255.255.0 ! interface GigabitEthernet0/0/0/1 ipv4 address 192.168.23.2 255.255.255.0 ! router ospf 1 area 0 interface Loopback0 ! interface GigabitEthernet0/0/0/0 ! interface GigabitEthernet0/0/0/1 ! ! ! ! </pre>

```

multicast-routing
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
interface all enable
!
!
router pim
address-family ipv4
auto-rp mapping-agent Loopback0 scope
255 interval 60
auto-rp candidate-rp Loopback0 scope
255 group-list 1 interval 60
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!

```

```

multicast-routing
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
interface all enable
!
!
router pim
address-family ipv4
auto-rp mapping-agent Loopback0 scope
255 interval 60
auto-rp candidate-rp Loopback0 scope
255 group-list 1 interval 60
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!

```

[Example 4-7](#) shows the downstream router configuration.

Example 4-7 *Configuring the Downstream Router in IOS*

[Click here to view code image](#)

R3 : Downstream router

```

hostname R3
!
ip multicast-routing
ip cef
!
interface Loopback0
ip address 192.168.0.4 255.255.255.255
ip pim sparse-mode
!
interface Ethernet0/0
ip address 192.168.34.4 255.255.255.0
ip pim sparse-mode
!
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
!

```

ip pim autorp listener

Sample Configuration: Auto-RP for NX-OS

[Figure 4-11](#) shows the topology for the following sample configuration of Auto-RP for NX-OS.

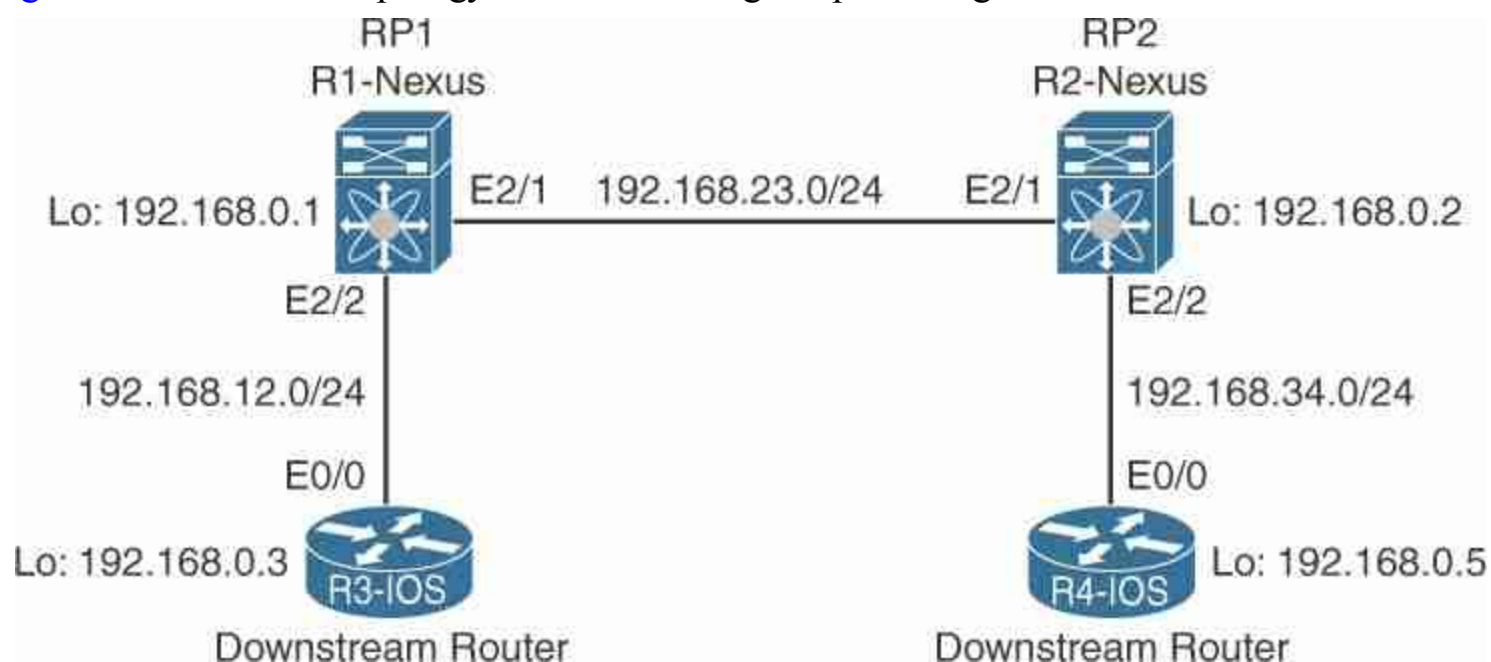


Figure 4-11 *NX-OS Auto-RP Configuration Topology*

Sample configuration—Auto-RP for NX-OS:

[Example 4-8](#) shows the RP configuration for Router R1 and R2.

Example 4-8 *Configuring RP for R1 and R2 in NX-OS*

RP: R1	RP: R2
<pre> hostname nexusr1 feature ospf feature pim ip pim auto-rp rp-candidate loopback0 group-list 239.0.0.0/8 ip pim auto-rp mapping-agent loopback0 ip pim auto-rp listen forward interface Ethernet2/1 no switchport ip address 192.168.23.1/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface Ethernet2/2 no switchport ip address 192.168.12.1/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface loopback0 ip address 192.168.0.1/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode router ospf 1 </pre>	<pre> hostname nexusr2 feature ospf feature pim ip pim auto-rp rp-candidate loopback0 group-list 239.0.0.0/8 ip pim auto-rp mapping-agent loopback0 ip pim auto-rp listen forward interface Ethernet2/1 no switchport ip address 192.168.23.2/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface Ethernet2/2 no switchport ip address 192.168.34.2/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface loopback0 ip address 192.168.0.2/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode router ospf 1 </pre>

[Example 4-9](#) shows the downstream router configuration.

Example 4-9 *Configuring the Downstream Router in IOS*

[Click here to view code image](#)

R3 : Downstream router

```

hostname R3
!
ip multicast-routing
ip cef
!
interface Loopback0
ip address 192.168.0.4 255.255.255.255
ip pim sparse-mode

```



```

!
interface Ethernet0/0
ip address 192.168.34.4 255.255.255.0
ip pim sparse-mode
!
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
!
ip pim autorp listener
!

```

BSR

BSR is an open standards protocol defined by the IETF. BSR is another RP high-availability mechanism that provides active/standby functionality and automatic downstream RP information propagation. The candidate RP in the BSR domain advertises its availability by sending RP cache information to all the downstream routers in the PIM domain. The function of the BSR is to collect and broadcast the RP set to all routers in the domain. The RP cache information is sent to all downstream routers using PIMv2 message group (224.0.0.13) hop-by-hop to all routers in the PIM domain. As [Figure 4-12](#) shows, the candidate RP sends its availability as RP only to the BSR router.

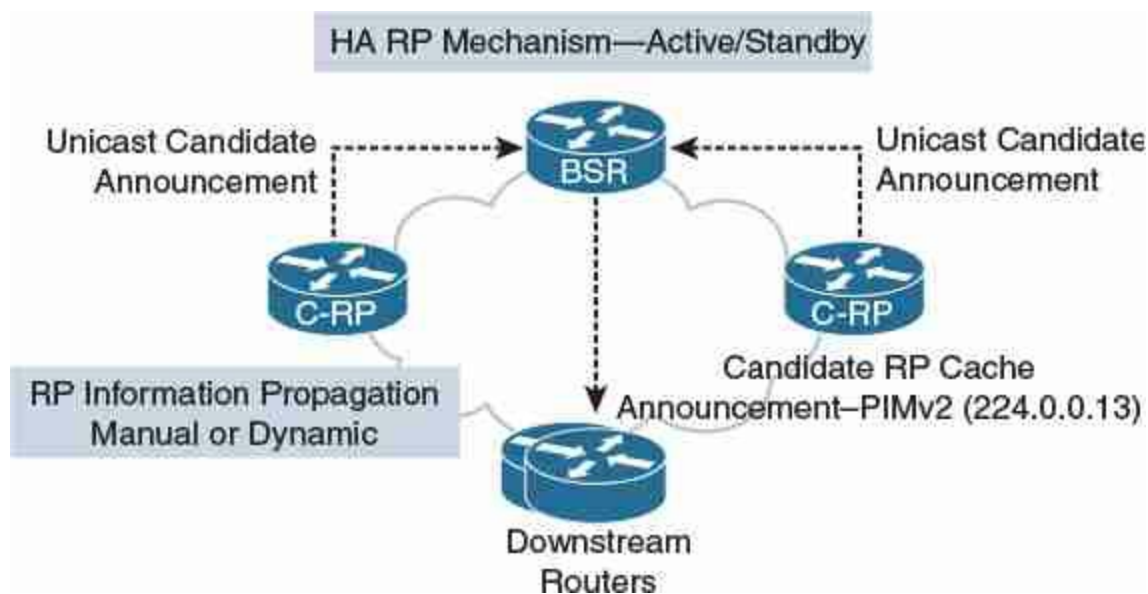


Figure 4-12 *BSR Overview*

The bootstrap message sent by the BSR includes information about all the candidate-RPs. Each router uses a common algorithm to select the same RP address for a given multicast group. The selection criterion for the RP is based on the common algorithm and is the responsibility of each downstream router. This is one of the major differences between BSR and Auto-RP. Downstream routers use the following criteria to select the RP in BSR:

- **Multicast group range:** Reviews the group that is attached to the RP for the selection.
- **RP priority:** Configured field on the candidate RP, the higher priority wins the RP selection process for a specific group.
- **Hash Mask:** Hash-mask-length bits of the group IP address are used to calculate the hash value. It uses a selection procedure for the whole multicast address space advertised in the RP multicast

group. This multicast group is partitioned among different RPs. Every RP gets approximately $2^{[32-\text{hash-mask-length}]}$ groups assigned, provided that there are enough RPs to evenly distribute the load. This does not offer active/active within a group range. It does however, provide address that use RP-1 within the multicast range based on the hashed value and addresses that use RP-2. The BSR mechanism is a nonproprietary method of defining RPs that can be used with third-party routers. Similar to Auto-RP, there is no configuration necessary on downstream router for BSR (except on candidate-BSRs and candidate-RPs).

BSR is a standards-based protocol available with PIMv2. Unlike Auto-RP, BSR does not use any dense-mode groups to flood candidate RP and RP mapping information. BSR uses PIM messages on a hop-by-hop basis with all messages flooded using the all-PIM routers group with TTL of 1. This keeps those multicast advertisements to a local subnet.

[Table 4-6](#) outlines the IOS/XE, IOS XR, and NX-OS commands for configuring the BSR.

Operating System	Command
IOS/XE	<code>ip pim [vrf vrf-name] bsr-candidate interface-type interface-number [hash-mask-length [priority]]</code>
IOS XR	<code>bsr candidate-bsr ip-address [hash-mask-len length] [priority value]</code>
NX-OS	<code>ip pim [bsr] bsr-candidate if-type if-number [hash-len hash-len] [priority priority]</code>

Table 4-6 BSR Configuration Commands for IOS/XE, IOS XR, and NX-OS

[Table 4-7](#) outlines the IOS/XE, IOS XR, and NX-OS commands for configuring the BSR candidate RP.

Operating System	Command
IOS/XE	<code>ip pim [vrf vrf-name] rp-candidate interface-type interface-number [bidir] [group-list access-list] [interval seconds] [priority value]</code>
IOS XR	<code>bsr candidate-rp ip-address [group-list access-list] [interval seconds] [priority value]</code>
NX-OS	<code>ip pim [bsr] rp-candidate if-type if-number [group-list prefix] [priority priority] [interval interval] [bidir]</code>

Table 4-7 Candidate RP Configuration Commands for IOS/XE, IOS XR, and NX-OS

[Figures 4-13](#) through [4-15](#) illustrate a typical deployment example on how to configure BSR in IOS, IOS-XR and NX-OS. In this example, R1 and R2 are the candidate RPs and BSR routers, a common deployment practice.

Sample Configuration: BSR in IOS

[Figure 4-13](#) shows the topology for the following sample configuration of BSR in IOS.

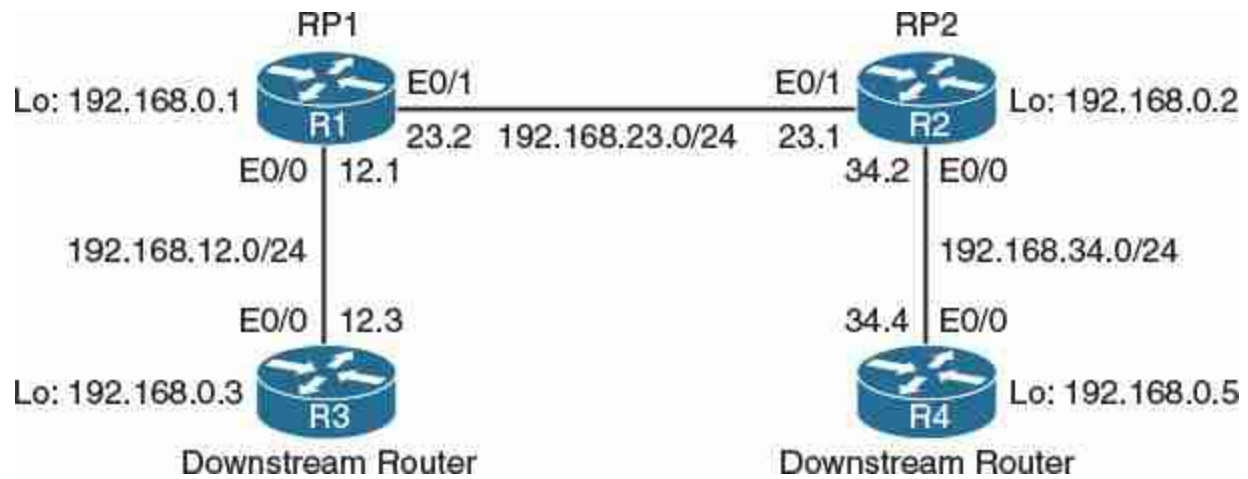


Figure 4-13 IOS BSR Configuration Topology

[Example 4-10](#) shows the BSR RP configuration for Router R1 and R2.

Example 4-10 Configuring BSR RP for R1 and R2 in IOS

RP: R1	RP: R2
<pre> ip multicast-routing ip cef ! interface Loopback0 ip address 192.168.0.1 255.255.255.255 ip pim sparse-mode ! interface Ethernet0/0 ip address 192.168.12.1 255.255.255.0 ip pim sparse-mode ! interface Ethernet0/1 ip address 192.168.23.2 255.255.255.0 ip pim sparse-mode ! ! router ospf 1 network 0.0.0.0 255.255.255.255 area 0 ! ip pim bsr-candidate Loopback0 0 ip pim rp-candidate Loopback0 interval 10 priority 10 </pre>	<pre> ip multicast-routing ip cef ! interface Loopback0 ip address 192.168.0.2 255.255.255.255 ip pim sparse-mode ! interface Ethernet0/0 ip address 192.168.34.2 255.255.255.0 ip pim sparse-mode ! interface Ethernet0/1 ip address 192.168.23.1 255.255.255.0 ip pim sparse-mode ! ! router ospf 1 network 0.0.0.0 255.255.255.255 area 0 ! ip pim bsr-candidate Loopback0 0 ip pim rp-candidate Loopback0 interval 10 priority 5 </pre>

Note

Remember, because BSR uses the local PIM multicast group (224.0.0.13) no additional configuration is required for downstream routers to process BSR updates. The downstream router will receive the BSR mapping advertisements, process the update, and update any group mappings as necessary. Multicast group state entries will, of course, use the RP(s) in the mapping as processed.

Sample Configuration: BSR in IOS-XR

[Figure 4-14](#) shows the topology for the following sample configuration of BSR in IOS-XR.

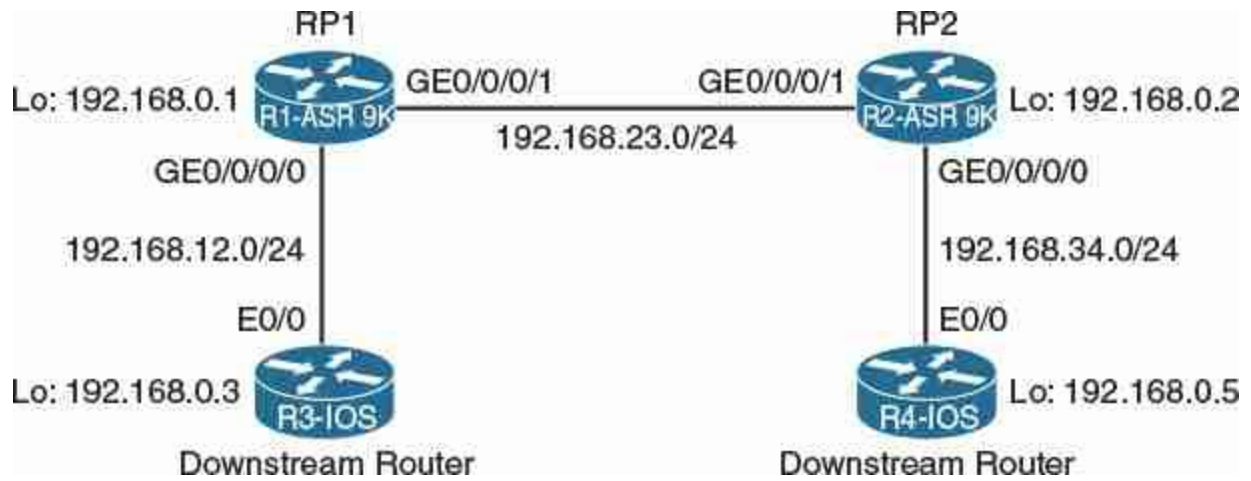


Figure 4-14 *IOS-XR BSR Configuration Topology*

[Example 4-11](#) shows the BSR RP configuration for Router R1 and R2.

Example 4-11 *Configuring BSR RP for R1 and R2 in IOS-XR*

RP: R1	RP: R2
<pre> hostname R1 ipv4 access-list 1 10 permit ipv4 239.0.0.0 0.255.255.255 any ! interface Loopback0 ipv4 address 192.168.0.1 255.255.255.255 ! interface GigabitEthernet0/0/0/0 ipv4 address 192.168.12.1 255.255.255.0 ! interface GigabitEthernet0/0/0/1 ipv4 address 192.168.23.1 255.255.255.0 ! router ospf 1 area 0 interface Loopback0 ! interface GigabitEthernet0/0/0/0 ! interface GigabitEthernet0/0/0/1 ! ! ! ! ! </pre>	<pre> hostname R2 ipv4 access-list 1 10 permit ipv4 239.0.0.0 0.255.255.255 any ! interface Loopback0 ipv4 address 192.168.0.2 255.255.255.255 ! interface GigabitEthernet0/0/0/0 ipv4 address 192.168.34.2 255.255.255.0 ! interface GigabitEthernet0/0/0/1 ipv4 address 192.168.23.2 255.255.255.0 ! router ospf 1 area 0 interface Loopback0 ! interface GigabitEthernet0/0/0/0 ! interface GigabitEthernet0/0/0/1 ! ! ! ! ! </pre>


```

multicast-routing
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
interface all enable
!
!
router pim
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
bsr candidate-bsr 192.168.0.1 hash-
mask-len 30 priority 10
bsr candidate-rp 192.168.0.1 group-
list 1 priority 10 interval 60

```

```

multicast-routing
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
interface all enable
!
!
router pim
address-family ipv4
interface Loopback0
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
bsr candidate-bsr 192.168.0.2 hash-
mask-len 30 priority 5
bsr candidate-rp 192.168.0.2 group-
list 1 priority 5 interval 60

```

Sample Configuration: BSR in NX-OS

[Figure 4-15](#) shows the topology for the following sample configuration of BSR in NX-OS.

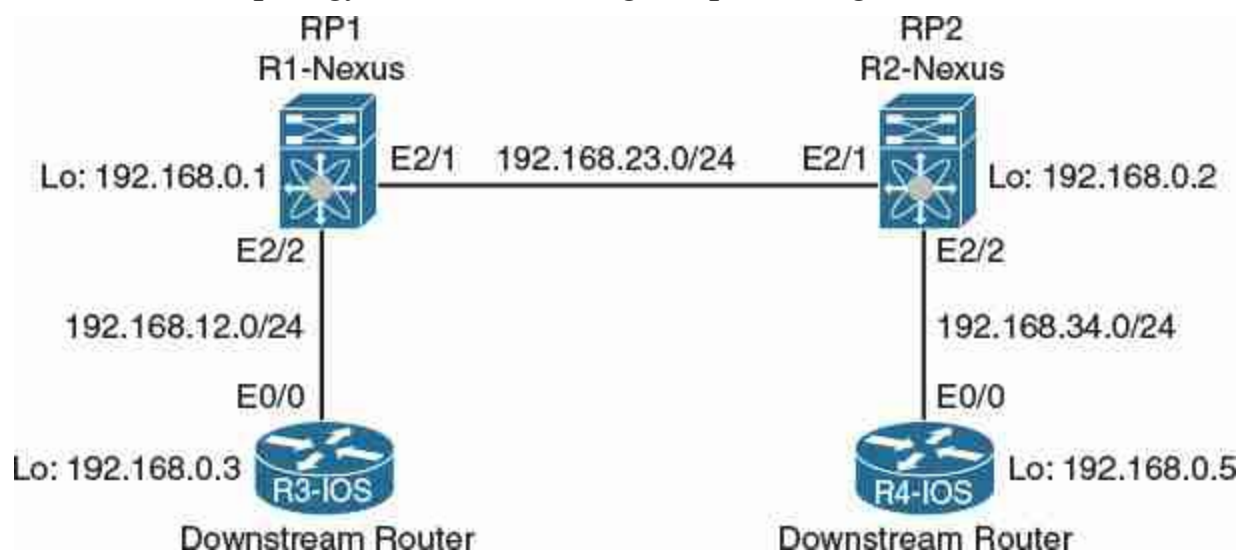


Figure 4-15 NX-OS BSR Configuration Topology

[Example 4-12](#) shows the BSR RP configuration for Router R1 and R2.

Example 4-12 Configuring BSR RP for R1 and R2 in NX-OS

RP: R1	RP: R2
<pre> feature pim hostname nexusr1 ip pim bsr-candidate loopback0 priority 10 ip pim bsr rp-candidate loopback0 group-list 239.0.0.0/8 ip pim bsr forward listen interface Ethernet2/1 no switchport mac-address 0001.0001.0001 ip address 192.168.23.1/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown </pre>	<pre> feature pim hostname nexusr2 ip pim bsr-candidate loopback0 priority 5 ip pim bsr rp-candidate loopback0 group-list 239.0.0.0/8 ip pim bsr forward listen interface Ethernet2/1 no switchport mac-address 0002.0002.0002 ip address 192.168.23.2/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown </pre>
<pre> interface Ethernet2/2 no switchport mac-address 5254.0012.345f ip address 192.168.12.1/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface loopback0 ip address 192.168.0.1/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode </pre>	<pre> interface Ethernet2/2 no switchport mac-address 5254.0012.345f ip address 192.168.34.2/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface loopback0 ip address 192.168.0.2/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode </pre>

Note

It is important to note that because of the differences between mapping process used by BSR and Auto-RP the two protocols should never be used simultaneously in the same network or PIM domain.

Anycast RP

Anycast RP uses a load-balancing technique that achieves active/active RP distribution for a multicast group. [Figure 4-16](#) illustrates the mechanics of active/active RP distribution using Anycast techniques.

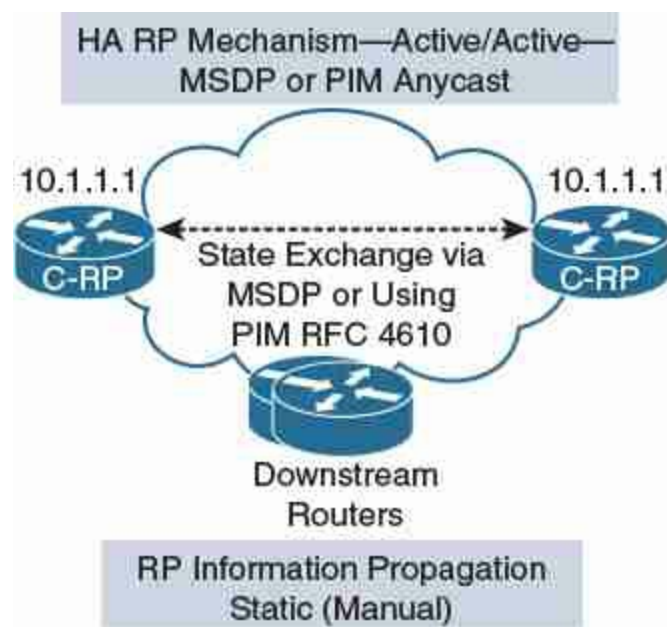


Figure 4-16 *Anycast RP Overview*

There are two variants of Anycast RP you should be aware of.

- RFC 3446 based on Multicast Source Discovery Protocol (MSDP) to maintain the active state between the RPs.
- RFC 4610 based on protocol independent multicast (PIM) to maintain the active state in the RPs.

Anycast allows two or more RPs to participate simultaneously within the multicast PIM domain. Based on one of the methods described in these RFCs, the state of the tree is maintained between two or more RPs. Using the protocols specified, the RPs also monitor and share active sources so that the state information is always current. Thus, the administrative domain can have more than one active RP.

What is very different about Anycast RP from other active/active network mechanisms is that Anycast requires both RPs to use the same IP address. This is the mechanism that allows receiver and source routers to register groups with the RP, regardless of the location, or which physical router is used. The registration messages are simply sent to the physical RP with the closest unicast routing metric. The IP address used for the RP for Anycast RP *must* be the same on each router.

Multicast Source Discovery Protocol

Multicast forwarding domains are designed to be secure and separate, just like their unicast counterparts. However, even private unicast routing domains can be, and most often are designed to connect to external domains or the Internet. What about multicast domains? Multicast domains are more fluid by definition and many domains may exist in a single IGP autonomous system. It stands to reason that some multicast applications will need external reach and therefore a way to bridge multicast domains and autonomous systems must also exist.

The way routers and networks build a multicast tree can make cross-domain forwarding a conundrum. In a sparse-mode network we must have both an active source and an active receiver to build the shared and source trees required to forward traffic. Routers and RPs use the unicast routing table to perform loop avoiding RPF checks against the sources in the network. If the source is outside the unicast domain, how do we learn about it? How can we trust that the source is legitimate when it is outside our control? How do we limit the sources to only those needed without flooding the entire network like the Internet with every known source?

There must be a protocol that can track and share active sources between domains. That protocol is Multicast Source Discovery Protocol (MSDP). MSDP tracks active sources in a network and shares those sources only with configured neighbors. It allows routers to extend multicast forwarding beyond internal boundaries, yet maintain a secure multicast posture.

Like BGP, MSDP uses a preconfigured peering relationship to exchange information. MSDP also uses TCP to securely establish the peering sessions, using port 639 for transport. Each MSDP peer must be explicitly configured to reach any other peer for which session establishment is required. Let's examine how this relationship is formed and what it does.

Referring to [Figure 4-17](#), R1 is a receiver for multicast group S1. RP1 and RP2 have the same IP address; however, the Anycast PIM relationship needs to be built via a unique IP address that is assigned to the local RP device. The following provides a step-by-step flow using Anycast, as illustrated in [Figure 4-17](#):

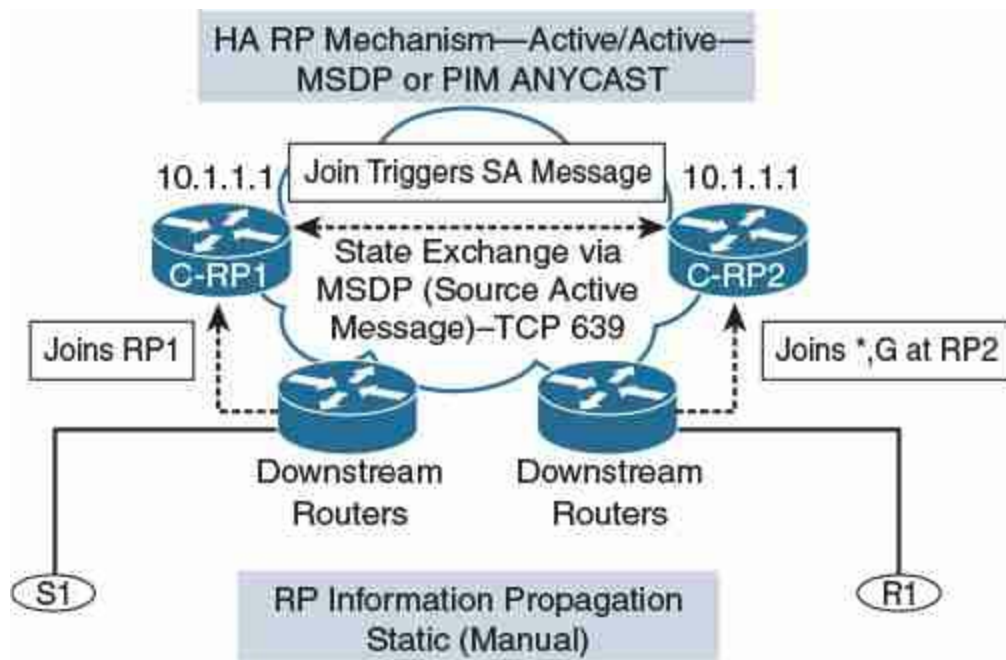


Figure 4-17 *MSDP Overview*

1. S1 sends a multicast packet to the first hop-designated router. The designated router sends a PIM register message to RP1.

2. RP1 and RP2 have an established MSDP relationship. RP1 sends a source active (SA) message to RP2. The SA message is sent via MSDP between RP1 and RP2. The SA message contains the following fields:

- The source address of the data source.
- The group address the data source is sending to.
- The IP address of the RP.

3. RP2 has the state information (*,G) for receiver R1. After RP2 receives the SA message, the shortest path tree is built for the S, G flow.

Anycast with MSDP is supported on all Cisco Layer 3 platforms: NX-OS, IOS-XR, and IOS.

PIM Anycast RP

PIM Anycast RP is a variant of Anycast deployments that use PIM to build the active/active multicast state distribution between the RPs. This process is illustrated in [Figure 4-18](#).

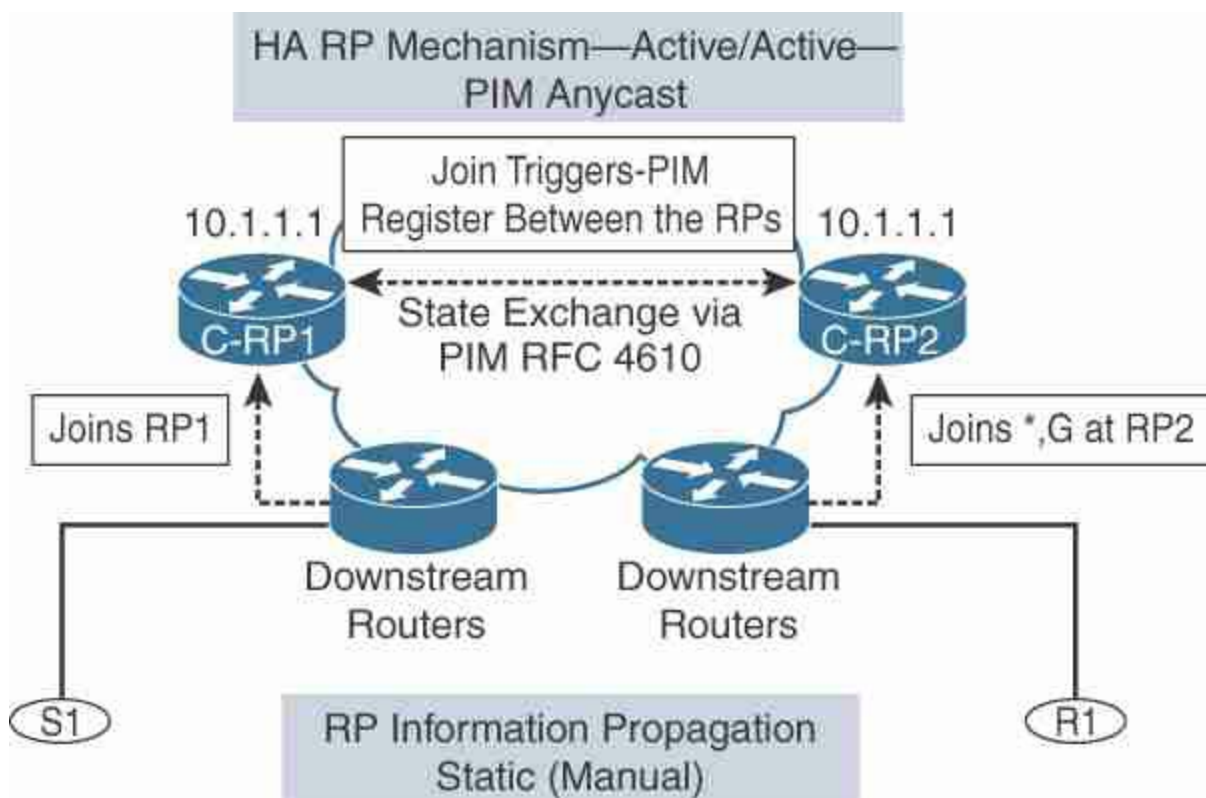


Figure 4-18 *PIM Anycast RP Overview*

R1 is a receiver for multicast group S1. RP1 and RP2 have the same IP address; however, the Anycast PIM relationship needs to be built via a unique IP address that is assigned to the local RP device. The following provides a step-by-step flow using Anycast as illustrated in the [Figure 4-18](#).

1. S1 sends a multicast packet to the first-hop designated router. The designated router sends a PIM register message to RP1.

2. RP1 and RP2 are configured with the same IP address. Because the register message did not come from one of the RPs in the Anycast RP set, RP1 then sends a copy of the register message to RP2. In this case, this register message will use RP1's own IP address as the source address for the PIM Register message. RP2 receives the register message from RP1 and checks the state table. R1 is in RP2's multicast state table, RP2 decapsulates the register packet from RP1 and sends the flow down the shared-tree to receiver R1. It should be noted if RP2 does not have a receiver in its multicast state table the register stop is sent in similar manner as the PIM register process.

RP2 sends a register stop back to RP1. This is the state maintenance mechanism between the RPs.

3. RP1 joins the multicast PIM state for S1 by triggering a (S1,G) join message toward S1 and (S1,G) state is created. After this, RP2 also joins to the source tree by creating a (S1,G) join towards S1.

Anycast with PIM using IPv4 is only supported with NX-OS. Anycast deployment in IPv6 environment supports PIM Anycast feature in all IOS, IOS-XE and Nexus platforms. The Nexus platform supports both Anycast PIM and MSDP modes.

Sample Configuration: Anycast RP with MSDP on IOS

The following configurations provide examples for configuring Anycast with IOS (with MSDP), IOS-XR (with MSDP), and NX-OS (Anycast PIM 4610). In this example, R1 and R2 are the candidate RPs using loopback 1 as the Anycast RP address. The loopback 0 interfaces of R1 and R2 are used to establish the MSDP relationship and also for PIM. The downstream routers are statically

configured using the RP address of 10.100.1.1. The unicast routing protocol controls the receivers and/or sources joining one of the active/active RP. [Figure 4-19](#) shows us this example network diagram.

Note

The recovery from a failure in an Anycast environment when compared to Auto-RP or BSR is significantly faster. This is because as soon as unicast routing protocol converges, the multicast control plan RP reachability will take place simultaneously.

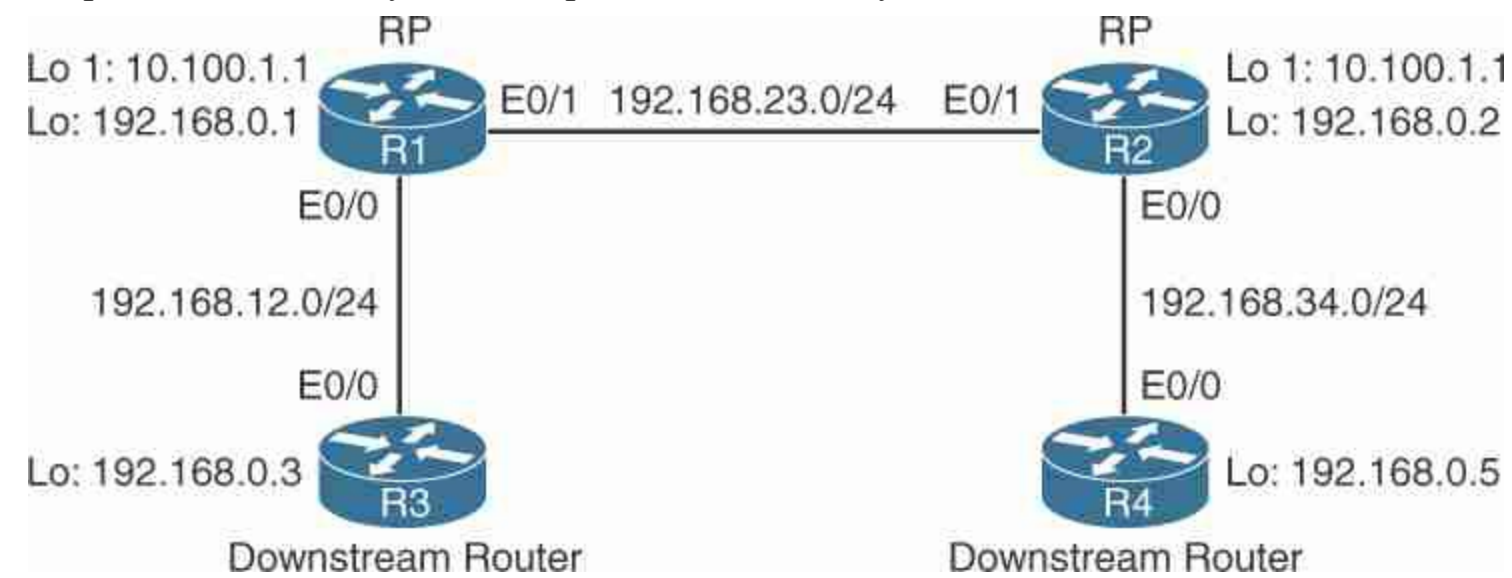


Figure 4-19 PIM Anycast RP Configuration

[Example 4-13](#) shows the PIM Anycast RP configuration for Router R1 and R2.

Example 4-13 Configuring PIM Anycast RP for R1 and R2 in IOS

RP: R1	RP: R2
<pre>hostname R1 ! ip multicast-routing ip cef ! ! interface Loopback0 ip address 192.168.0.1 255.255.255.255 ip pim sparse-mode ! interface Loopback1 ip address 10.100.1.1 255.255.255.255 ip pim sparse-mode ! interface Ethernet0/0 ip address 192.168.12.1 255.255.255.0 ip pim sparse-mode !</pre>	<pre>hostname R2 ! ip multicast-routing ip cef ! ! interface Loopback0 ip address 192.168.0.2 255.255.255.255 ip pim sparse-mode ! interface Loopback1 ip address 10.100.1.1 255.255.255.255 ip pim sparse-mode ! interface Ethernet0/0 ip address 192.168.34.2 255.255.255.0 ip pim sparse-mode !</pre>

```

interface Ethernet0/1
 ip address 192.168.23.2 255.255.255.0
 ip pim sparse-mode
!
router ospf 1
 router-id 192.168.0.1
 network 0.0.0.0 255.255.255.255 area 0
!
ip pim rp-address 10.100.1.1 GROUPS
ip msdp peer 192.168.0.2 connect-source
  Loopback0
ip msdp cache-sa-state
ip msdp originator-id Loopback0
!
!
ip access-list standard GROUPS
 permit 239.0.0.0 0.255.255.255

```

```

interface Ethernet0/1
 ip address 192.168.23.3 255.255.255.0
 ip pim sparse-mode
!
router ospf 1
 router-id 192.168.0.2
 network 0.0.0.0 255.255.255.255 area 0
!
ip pim rp-address 10.100.1.1 GROUPS
ip msdp peer 192.168.0.1 connect-source
  Loopback0
ip msdp cache-sa-state
ip msdp originator-id Loopback0
!
!
ip access-list standard GROUPS
 permit 239.0.0.0 0.255.255.255

```

Loopback 1 is the Anycast RP address, which is the same on both the RPs.

Note

Make sure you have the router-id configured for unique local node. The same loopback 1 is used in all the anycast examples.

[Example 4-14](#) shows the downstream router configuration.

Example 4-14 *Configuring the Downstream Router in IOS*

[Click here to view code image](#)

R4 : Downstream router

```

ip multicast-routing
ip cef
!
interface Loopback0
 ip address 192.168.0.4 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet0/0
 ip address 192.168.34.4 255.255.255.0
 ip pim sparse-mode
router ospf 1
 network 0.0.0.0 255.255.255.255 area 0
!
ip pim rp-address 10.100.1.1 GROUPS
!
ip access-list standard GROUPS
 permit 239.0.0.0 0.255.255.255

```


Sample Configuration: Anycast with MSDP on IOS-XR

If we change the RP routers in our network diagram to IOS-XR routers, in this case ASR 9Ks, we need to adjust our configurations accordingly. The diagram in [Figure 4-20](#) depicts this change to the network. Example configurations for XR follow the diagram.

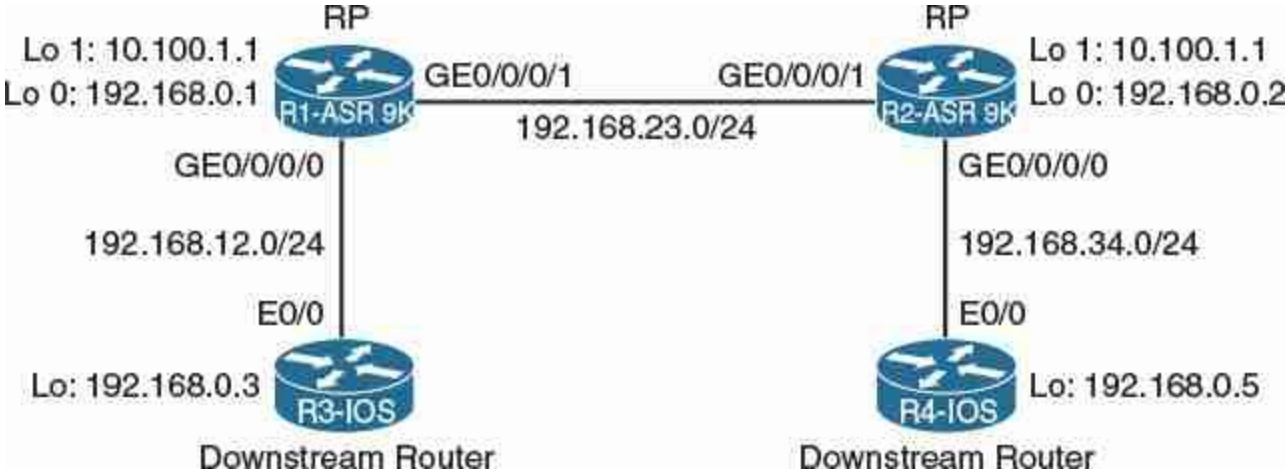


Figure 4-20 PIM Anycast RP Configuration

[Example 4-15](#) shows the PIM Anycast RP configuration for Router R1 and R2.

Example 4-15 *Configuring PIM Anycast RP for R1 and R2 in IOS-XR*

RP: R1	RP: R2
<pre>hostname R1 ipv4 access-list 1 10 permit ipv4 239.0.0.0 0.255.255.255 any ! interface Loopback0 ipv4 address 192.168.0.1 255.255.255.255 ! interface Loopback1 ipv4 address 10.100.1.1 255.255.255.255 ! interface GigabitEthernet0/0/0/0 ipv4 address 192.168.12.1 255.255.255.0 ! interface GigabitEthernet0/0/0/1 ipv4 address 192.168.23.1 255.255.255.0</pre>	<pre>hostname R2 ipv4 access-list 1 10 permit ipv4 239.0.0.0 0.255.255.255 any ! interface Loopback0 ipv4 address 192.168.0.2 255.255.255.255 ! interface Loopback1 ipv4 address 10.100.1.1 255.255.255.255 ! interface GigabitEthernet0/0/0/0 ipv4 address 192.168.34.2 255.255.255.0 ! interface GigabitEthernet0/0/0/1 ipv4 address 192.168.23.2 255.255.255.0</pre>

```
!
router ospf 1
  router-id 192.168.0.1
  address-family ipv4
    area 0
      interface Loopback0
      !
      interface Loopback1
      !
      interface GigabitEthernet0/0/0/0
      !
      interface GigabitEthernet0/0/0/1
      !
      !
  !
multicast-routing
  address-family ipv4
    interface Loopback0
    !
    interface Loopback1
    !
    interface GigabitEthernet0/0/0/0
```

```
!
router ospf 1
  router-id 192.168.0.2
  address-family ipv4
    area 0
      interface Loopback0
      !
      interface Loopback1
      !
      interface GigabitEthernet0/0/0/0
      !
      interface GigabitEthernet0/0/0/1
      !
      !
  !
multicast-routing
  address-family ipv4
    interface Loopback0
    !
    interface Loopback1
    !
    interface GigabitEthernet0/0/0/0
```

```

!
interface GigabitEthernet0/0/0/1
!
interface all enable
!
!
router msdp
connect-source Loopback0
peer 192.168.0.2
!
!
router pim
address-family ipv4
rp-address 10.100.1.1 1
interface Loopback0
!
interface Loopback1
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
!
!
end

```

```

!
interface GigabitEthernet0/0/0/1
!
interface all enable
!
!
router msdp
connect-source Loopback0
peer 192.168.0.1
!
!
router pim
address-family ipv4
rp-address 10.100.1.1 1
interface Loopback0
!
interface Loopback1
!
interface GigabitEthernet0/0/0/0
!
interface GigabitEthernet0/0/0/1
!
!
!
end

```

[Example 4-16](#) shows the downstream router configuration.

Example 4-16 *Configuring the Downstream Router in IOS*

[Click here to view code image](#)

R4 : Downstream router

```

hostname R4
!
ip multicast-routing
ip cef
no ipv6 cef
!
!
interface Loopback0
ip address 192.168.0.4 255.255.255.255
ip pim sparse-mode
!
interface Ethernet0/0
ip address 192.168.34.4 255.255.255.0
ip pim sparse-mode

```

```

!
router ospf 1
 network 0.0.0.0 255.255.255.255 area 0
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
ip pim rp-address 10.100.1.1 1
!
!
access-list 1 permit 239.0.0.0 0.255.255.255

```

Sample Configuration: Anycast on NX-OS

Let's take one more look at the same network, but here we replace the RP routers with Cisco Nexus switches using NX-OS. [Figure 4-21](#) reflects the changed topology for the accompanying configuration examples.

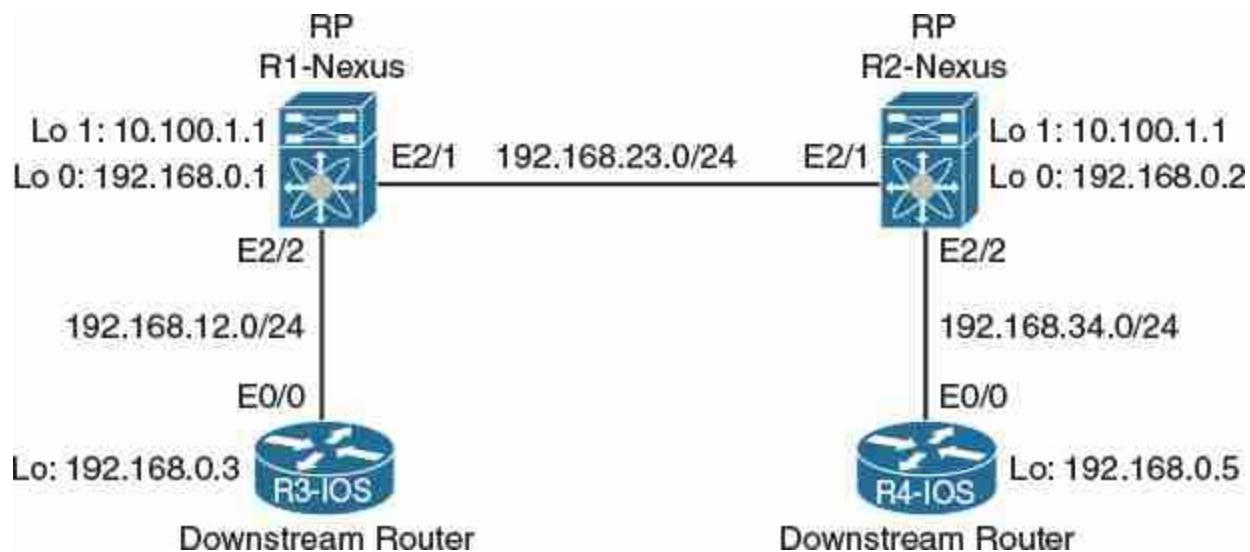


Figure 4-21 *PIM Anycast RP Configuration*

[Example 4-17](#) shows the PIM Anycast RP configuration for Router R1 and R2.

Example 4-17 *Configuring PIM Anycast RP for R1 and R2 in NX-OS*

RP: R1	RP: R2
<pre> hostname nexusr1 feature ospf feature pim ip pim rp-address 10.100.1.1 group-list 239.0.0.0/8 ip pim anycast-rp 192.168.0.1 10.100.1.1 ip pim anycast-rp 192.168.0.2 10.100.1.1 interface Ethernet2/1 no switchport mac-address 0001.0001.0001 ip address 192.168.23.1/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown </pre>	<pre> hostname nexusr2 feature ospf feature pim ip pim rp-address 10.100.1.1 group-list 239.0.0.0/8 ip pim anycast-rp 192.168.0.1 10.100.1.1 ip pim anycast-rp 192.168.0.2 10.100.1.1 interface Ethernet2/1 no switchport mac-address 0002.0002.0002 ip address 192.168.23.2/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown </pre>
<pre> interface Ethernet2/2 no switchport mac-address 5254.0012.345f ip address 192.168.12.1/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface loopback0 ip address 192.168.0.1/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode interface loopback1 ip address 10.100.1.1/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode router ospf 1 router-id 192.168.0.1 </pre>	<pre> interface Ethernet2/2 no switchport mac-address 5254.0012.345f ip address 192.168.34.2/24 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode no shutdown interface loopback0 ip address 192.168.0.2/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode interface loopback1 ip address 10.100.1.1/32 ip router ospf 1 area 0.0.0.0 ip pim sparse-mode router ospf 1 router-id 192.168.0.2 </pre>

[Example 4-18](#) shows the downstream router configuration.

Example 4-18 *Configuring the Downstream Router in IOS*

[Click here to view code image](#)

R4 : Downstream router

```
hostname R4
!
no ip domain lookup
ip multicast-routing
ip cef
no ipv6 cef
!
!
interface Loopback0
 ip address 192.168.0.4 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet0/0
 ip address 192.168.34.4 255.255.255.0
 ip pim sparse-mode
!
router ospf 1
 network 0.0.0.0 255.255.255.255 area 0
!
 ip forward-protocol nd
!
no ip http server
no ip http secure-server
ip pim rp-address 10.100.1.1 1
!
!
access-list 1 permit 239.0.0.0 0.255.255.255
```

Note

Use caution when configuring Anycast RP. The configuration is quite simple, but the selection of the RP address on the Anycast loopback can create problems if not designed properly. Although not a multicast-specific issue, remember that many protocols use loopback addresses as a router identification, including BGP, MPLS, OSPF, and IS-IS. In each of these protocols, if the router-ID is not explicitly configured, the highest IP address on a loopback interface will be used. If the Anycast loopback IP address happens to be the highest, you could end up with multiple routers having the same router-id! This causes peers and neighbors to crash and routing anomalies to occur. It is always best practice to explicitly configure router-IDs in all protocols that use them. It is also best practice to never use the Anycast loopback for any other purpose other than as an RP, such as, for example, a configured tunnel destination/source or any other non-Anycast mechanism.

Phantom RP

The concept of a phantom RP is used with BiDir PIM designs. BiDir PIM designs work with regular RP distribution methods that we described in the earlier sections. Unlike PIM ASM, with BiDir the RP does not handle any control plane load and RP information. A good BiDir RP design does not need to provide a physical RP, but rather use a phantom RP. As the name states, the phantom RP uses a virtual IP address as an RP that is advertised as the RP address but is not locally present on any router.

Figure 4-22 illustrates a BiDir domain.

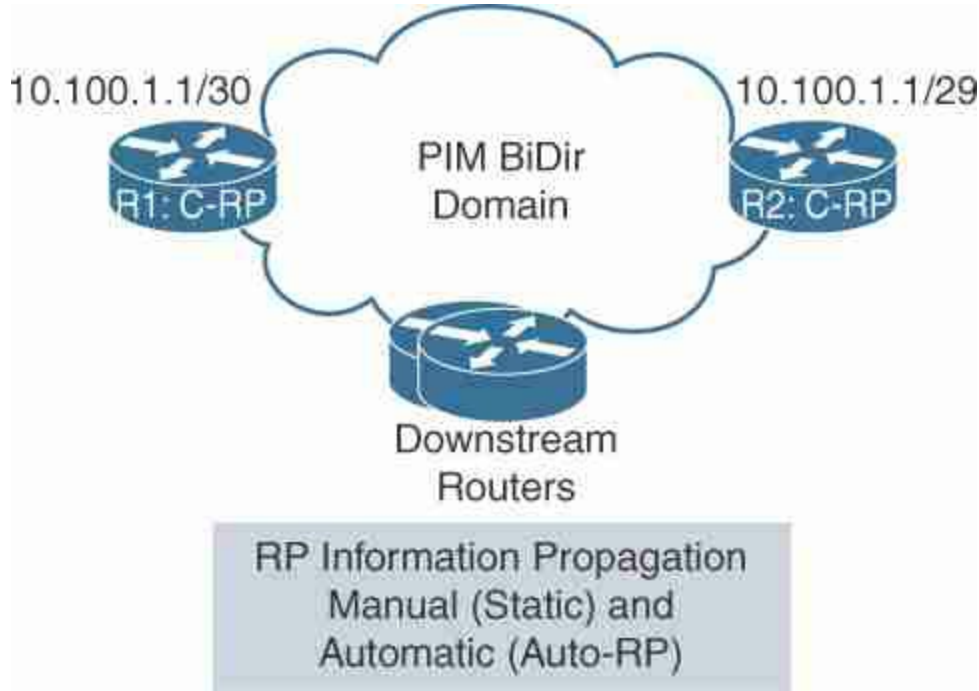


Figure 4-22 PIM BiDir Domain

Sample Configuration—Phantom RP on IOS

The following configuration provides a configuration example of a phantom RP using IOS. R1 and R2 are two routers that have RP information in the PIM BiDir domain.

Example 4-19 shows the Phantom RP configuration for Router R1 and R2.

Example 4-19 Configuring Phantom RP for R1 and R2 in IOS

RP: R1	RP: R2
<pre>interface Loopback0 ip address 192.168.0.1 255.255.255.255 ! interface Loopback1 ip address 10.100.1.1 255.255.255.252 ip pim sparse-mode ip ospf network point-to-point ! ip pim rp-address 10.100.1.2 bidir ! router ospf 1 router-id 192.168.0.1</pre>	<pre>interface Loopback0 ip address 192.168.0.2 255.255.255.255 ! interface Loopback1 ip address 10.100.1.1 255.255.255.248 ip pim sparse-mode ip ospf network point-to-point ! ip pim rp-address 10.100.1.2 bidir ! router ospf 1 router-id 192.168.0.2</pre>

In this example, the RP (10.100.1.2) does not physically exist on any device. The subnet to which the RP belongs, exists with a different mask. The different mask creates a virtual identity for 10.100.1.2. The command **ip ospf network point-to-point** makes the route of 10.100.1.1 appear with its configured mask, because by default OSPF advertises all loopbacks as /32. The downstream routers will see the 10.100.1.2 using the **show ip route** command, as shown in [Example 4-20](#).

Example 4-20 *Route to 10.100.1.2 on R4*

[Click here to view code image](#)

```
R4#show ip route
```

```
Routing entry for 192.168.3.0/24, 2 known subnets
```

```
Variably subnetted with 2 masks
```

```
 O   10.100.1.0/30 [110/21] via 192.168.34.1, 00:10:03, Ethernet0/0
```

```
 O   10.100.1.0/29 [110/11] via 192.168.34.1, 00:10:03, Ethernet0/0
```

The preferred route(s) is the one with the longest prefix.

```
R4#show ip route 10.100.1.2
```

```
Routing entry for 10.100.1.0/30
```

```
Known via "ospf 1", distance 110, metric 21, type intra area
```

```
Last update from 192.168.0.2 on Ethernet0/0, 00:10:19 ago
```

```
Routing Descriptor Blocks:
```

```
.....
```

The Phantom RP configuration example uses a static RP assignment. Phantom RP with PIM BiDir is compatible with dynamic RP propagation using Auto-RP protocol. Phantom RP propagation using BSR is not a configuration option at this time; because a virtual address is used that can exist on both R1 and R2, redundancy is built into the RP configuration. Any failure of the R1 router or the /30 route configured for the R1 loopback interface, would cause the network to reconverge on the larger /29 subnet route configured for the R2 loopback interface. Because the RP address is virtual, the mappings would use R2 as the next path to reach the RP until such time as the R1 loopback is reintroduced to the network.

PIM SSM Configuration

As you read earlier in [Chapter 3](#), SSM is the only configuration that does not require a RP. Remember that PIM SSM uses IGMPv3 source-specific joins to subscribe to the appropriate channel, and the network uses only source trees to forward each flow. Consequently, configuring SSM is very easy in an established IP multicast network using sparse or sparse-dense mode. Enabling PIM SSM and defining a range is the only required step and is only required on leaf routers. The SSM source trees within the existing network form a new SSM multicast domain that coexists but does not interfere with the existing sparse domain(s).

If the network is using only source-specific applications and is not PIM-configured, PIM-SSM is very simple to enable. Use the following steps and configuration commands to configure an IP network for PIM-SSM from scratch.

Step 1. Enable PIM-SSM and define a range for SSM forwarding. Remember the default group block for SSM is 232.0.0.0/8. If you do not configure a range, the default is assumed.

[Click here to view code image](#)

IOS/XE: ip pim ssm[default | range access-list]

IOS-XR: ip pim ssm range {ip-prefix | none | route-map policy-name}

NX-OS: ssm [allow-override | disable | range access-list]

Step 2. Enable Interfaces for PIM sparse-mode/ (There is no “ssm mode” configuration option, and no dense-mode operations are required for SSM; thus, sparse-mode configuration is sufficient.)

IOS/XE: ip pim sparse-mode

IOS-XR: router pim

NX-OS: ip pim sparse-mode

Note

IOS-XR supports only sparse mode and it is assumed on any interface configured under the **router pim** configuration.

Step 3. Gateway router interfaces, host interfaces, and L3 switch interfaces should be configured with **igmp version 3** .

IOS/XE: ip igmp version 3

IOS-XR: router igmp

NX-OS: ip igmp version 3

Note

For IOS-XR, IGMP version 3 is the default. Use the **version** command under **igmp config** to change versions if necessary.

Note

Make sure that all Layer 2 switches at the leaf edges of the network support IGMPv3 snooping. Refer to [Chapter 2](#) for additional information on IGMP versions and Layer 2 configurations.

[Example 4-21](#) (for IOS) configures PIM SSM as the only PIM mode available for applications using the default PIM group addressing. Remember that no RP is required for SSM.

Example 4-21 *Sample SSM Configuration*

[Click here to view code image](#)

RouterX

hostname rX

!

no ip domain lookup

ip multicast-routing

ip cef

no ipv6 cef

!

!

interface Loopback0

ip address 192.168.0.X 255.255.255.255

ip pim sparse-mode

ip igmp version 3

!

interface Ethernet0/0

```

ip address 192.168.X.X 255.255.255.0
ip pim sparse-mode
ip igmp version 3
!
interface Ethernet0/1
ip address 192.168.X.X 255.255.255.0
ip pim sparse-mode
ip igmp version 3
!
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
ip pim ssm default
!
ip igmp version 3

```

Summary

A rendezvous point (RP) is the service in the network where both multicast senders and receivers go to “connect” with their associated counterpart in a shared tree environment. The process of building a shared tree requires that the network components register active sources, as well as receivers for a given multicast group to the RP.

IP multicast domains are generally a subset of the entire network infrastructure. This offers an administrator the ability to have better control over multicast messaging. You can consider a group of PIM routers with the ability to share the same tree as part of a domain or scope. These scope ranges are aligned with RP and PIM domains. A scope range can superimpose over each other or be a subset of one another.

The first step in building a PIM domain for sparse-mode or BiDir is to establish a PIM neighbor adjacency on a given subnet. The second step is to agree on an RP. RPs and group mappings can be learned statically using manual configuration at each router or dynamically by using either Auto-RP or BSR.

There are two ways to dynamically propagate RP information to routers within a domain, Auto-RP and bootstrap router (BSR). Auto-RP is a Cisco proprietary solution. Rendezvous points using Auto-RP announce their availability to the mapping agents via the 224.0.1.39 multicast group. The RP mapping agent listens to the announced packets from the RPs, then sends RP-to-group mappings in a discovery message to 224.0.1.40. BSR is another RP high-availability mechanism that provides active/standby functionality of the RP and automatic downstream RP information propagation. The function of the BSR is to broadcast the RP set to all routers in the domain. The RP cache information is sent to all downstream routers using PIMv2 message group (224.0.0.13) hop-by-hop to all routers in the PIM domain.

Maintaining the integrity of the multicast infrastructure can be a concern in high-availability (HA) environments. Anycast RP uses a load-balancing technique that achieves active/active RP distribution for a multicast group. There are two variants of Anycast RP, one based on Multicast Source Discovery Protocol (MSDP) (RFC 3446) and one based on protocol independent multicast (PIM) (RFC 4610) to maintain the active state in the RPs.

A phantom RP is used with BiDir PIM designs for HA. BiDir PIM designs work with regular RP distribution methods, but unlike PIM ASM, BiDir does not handle any control plane load and RP information. The phantom RP uses a virtual IP address that is advertised as an RP address but not locally present on any router.

Both PIM dense-mode and SSM do not require the use of an RP. Dense mode is not a recommended solution for most implementations, as some networking operating systems will not support dense mode operation. SSM uses IGMPv3 source specific joins to subscribe to the appropriate channel and the network uses only source trees to forward each flow. Consequently, configuring SSM is very easy in an established IP multicast network using sparse.

Chapter 5. IP Multicast Design Considerations and Implementation

Chapter 5. IP Multicast Design Considerations and Implementation

In this chapter, we build on the material that has been introduced so far. We look at multicast group scoping and how multicast networks can be bounded to control the flow of information to provide security. We explain organizational and global group assignments and how to include address organization and schemas. Group scoping with hybrid designs and RP placement are examined, to include MSDP mesh groups and scoped multicast domains. We delve into traffic engineering, how the elements of Layer 3 devices make forwarding decisions, and how to manipulate those elements for path selection. IP multicast best practices and security are also covered concerning the network as a whole and the components that make up that network. Finally, we combine the elements we discuss in the chapter in a practical case study to solidify what you learned.

Multicast Group Scoping

With the depletion of public IPv4 addresses and the inability to obtain additional numbers from the Internet Assigned Numbers Authority (IANA), public IPv4 addresses are at a premium. Many technologies exist to make address management easier, including RFC 1918 and RFC 6598 private IP addresses and network address translation (NAT). These technologies impact the way we manage IP addresses internally. In addition, many routing protocols simply work better when address spaces can be summarized at particular boundaries. Thus, many organizations rely on an IP address schema to manage internal address assignments.

If you have ever administered an IPv4 or IPv6 network, you know that IP schemas are a very important part of network design and operation. An IP schema is essentially a map of how IP addresses are assigned and managed within the network. For example, the schema may prescribe very specific IP subnets for the network infrastructure while also making available other subnets for DHCP address assignments for end points and hosts. This is especially relevant for enterprises that may have limited access to public address space.

Many schemas use particular IP address octets to imply a specific meaning within the organization. For example, a network architect may assign the private network 172.16.0.0/16 to cover all network infrastructure address assignments. Administrators may break this block down further to provide additional control and meaning; for example, the routers in a given location may be assigned addresses from the 172.16.10.0/24 subnet, derived from the 172.16.0.0/16 supernet.

IPv4 multicast addresses are also a limited commodity. Organizations that roll-out multicast applications should create a detailed address schema. This schema helps control address assignment and assists in network operations. If the same IPv4 unicast schema principles are applied to the IPv4 multicast address schema, operations and design engineers can quickly identify applications and application properties derived through the assigned address.

Scoping is not just about which addresses to assign. Just like the underlying unicast network, multicast networks must be bounded in order to securely control the flow of information. In many cases, the boundary of the unicast autonomous system (AS) may coincide with the boundary of the multicast network, but this is not always the case. The scope of any IPv4 multicast domain should, at minimum, coincide with the scope of the unicast domain on which it is being overlain. Multiple multicast domains can overlay on a single unicast network, which can mean that multiple multicast scopes may be employed in the same unicast domain.

Some multicast boundaries occur naturally as part of the process of configuring the network. One obvious boundary is the one that exists between ASs. For unicast routing, the AS boundary is between the interior gateway protocol (IGP) and the exterior gateway protocol (EGP, most likely this will be BGP). Although route sharing may be configured between them, the external networks do not speak directly to IGP routers using the IGP protocol interface. For this reason, BGP routing information is often excluded from the processes of many internal overlay protocols, like Multiprotocol Label Switching (MPLS).

Multicast domains can use BGP routes for multicast RPF checks if they are using multicast BGP (MBGP), reviewed in this chapter. It is rare that all the necessary remote domain routes, like those of an internal multicast rendezvous point, are shared through native unicast BGP. It is assumed that these networks are internal only to the domain and therefore should be excluded by policy. It is also possible that the network may use different paths for external multicast and unicast flows. This can result in an incongruent network that causes RPF failures in the multicast path. Thus, for most multicast networks, unicast BGP still creates a natural boundary, in particular when it comes to RPF checking for loop-free paths. Properly scoping the multicast domain makes it significantly easier to summarize and secure the domain at the domain edge.

Organizational and Global Group Assignment Considerations

The public IPv4 multicast address blocks detailed in [Chapter 1](#) are assigned by IANA and are not open for use by an organization for internal independent applications. As with publicly assigned unicast addresses, nothing prevents deployment of any public address internal to a network, but this could potentially cause serious conflicts with external-facing routers that have Internet routes. The same logic applies to IP multicast addresses. When an organization uses multicast privately, they should select addresses from the IPv4 administratively scoped address block.

Both of these blocks provide a tremendous number of possible addresses. For example, the administratively scoped IPv4 block is a /8, providing the application architect with the ability to select from 16,777,216 possible host group addresses for a given application. Very few, if any, networks will ever need this many addresses. Still, the selection of a group should be rigorously controlled by some entity within the organization. Otherwise, group assignment conflicts can occur, and the groups themselves will have little meaning. It is best to create a group address schema to permanently address this within an organization.

The considerations necessary to create a multicast address schema are similar to those needed for a unicast schema. For example, summarization is just as important for multicast as it is for unicast. Even though each group address is routed as a single address (a /32, or having a mask of 255.255.255.255), it is best to further subdivide the administrative blocks by orderable bit boundaries that can take advantage of masking. Each contiguous sub-block of addresses can then represent a particular type of application, giving the address both meaning and additional scope. This makes security, routing, and other policies easier to implement.

There are several methods of determining the best addressing schema for an organization and several questions the architect must answer. These questions include:

- What is the structure of the organization and how will each line of business use multicast applications?
- What is the scale of the multicast deployment, including both planned and unplanned growth?
- What organizational security policy exists for multicast deployments?

- What is the geographical layout of the multicast-enabled network and what is the geographical scope of each application?
- Where are the hosts and where are the sources in the geography?
- What address ranges may overlap with Layer 2 MAC addresses?
- What plans does the organization have for the use of source-specific multicast?
- What is the ability of hosts, servers, and applications to support various address types?
- What are the resource utilization parameters for multicast applications?

The answers to each of these questions may affect how the architect subdivides the group address space. For example, a security policy may dictate that some multicast applications only exist in a local data center, whereas other applications may have an organization-wide boundary. Another example could include dividing groups by the amount of resource consumption or by the business criticality of each application. Important or high-bandwidth groups can receive different treatment than other groups.

If the blocks are properly subdivided, then creating policy boundaries is a cinch. If not, each group will need individualized policy statements. The important element to remember is that no one schema is best for all organizations.

IPv4 Considerations

An IPv4 group address is 32 bits in length and, when written in dotted decimal, is split into 4 octets. The first octet of the private Administrative scope range is set at 239.0.0.0/8. [Table 5-1](#) shows a simple schema created to separate a small service provider network into meaningful groups. The division begins with geography, followed by application priority, fulfilling some of the design concepts previously mentioned.

Octet 1	Octet 2	Octet 3	Octet 4
Fixed	Geography	Priority	Application Number

Table 5-1 *Group Scoping by Octet*

Some organizations may have very large topologies that require additional complexity. One way to achieve this is to break down the schema further within each octet. [Table 5-2](#) breaks down the geography octet into eight regions with up to eight Point of Presence (PoP) locations per region, and the priority octet into eight priorities, with eight resource consumption models (high bandwidth versus low bandwidth).

Geography		Priority	
Region: First Four Bits	PoP: Remaining Four Bits	Priority: First Four Bits	Consumption: Remaining Four Bits
0001: Core	0001: All	0000: Infrastructure	0000: Infrastructure
0010: Ops	0010: PoP 1	0001: Data	0001: Data
0011: Data Center	0011: PoP 2	0010: High	0010: High
0100: North East	0100: PoP 3	0011: Med High	0011: Med High
0101: South East	0101: PoP 4	0100: Medium	0100: Medium
0110: North West	0110: PoP 5	0101: Med Low	0101: Med Low
0111 South West	0111: PoP 6	0110 Low	0110 Low
1001: Central	1001: PoP 7	0111: Default	0111: Default
1010–1111: Growth	1010–1111: Growth	1000–1111: Growth	1000–1111: Growth

Table 5-2 *Group Scoping by Geography and Priority*

Using the schema from [Table 5-2](#) , if the provider needed a group assignment for a core application or protocol that spanned all PoPs and has a priority/consumption of Infrastructure, then 239.17.0.X would suffice. The provider would also use 239.84.34.X (239.[0101][0100].[0010][0010].X) as an assignment for a high-bandwidth, high-priority application with a scope of PoP 3 in the South East region. The advantage of such a schema is that routers and firewalls can employ wildcard masks to manage policy statements in the network architecture.

Note

Routers use wildcard masks with IP access control lists (ACLs) to specify what should be matched for further action, depending on how an ACL is applied. Interface subnet masks read from left to right; for example, IP address 172.18.100.129 with a 255.255.255.224 mask. This provides an IP device the delimiting bits between subnet and host. Wildcard masks for IP ACLs reverse this structure; for example, mask 0.0.0.31 is the reverse of 255.255.255.224 (replacing ones with 0s). When the value of the mask is broken down into binary (0s and 1s), the results determine which address bits are to be considered in processing the traffic. A 0 in the mask means that the corresponding address bits must match exactly with the address for comparison. A 1 in the mask means the corresponding address bit is variable. Thus, an ACL statement with subnet 10.0.0.0 and mask 0.0.0.255 means that any address that begins with 10.0.0. will match the statement, because the last octet can be variable.

If the provider wanted to place boundaries on all groups within the Central region, a simple ACL using a network/mask entry of 239.121.0.0 0.15.255.255 could accomplish the task. Similarly, a network/mask entry of 239.0.4.0 0.255.251.255 matches any application deemed high resource consumptive. This schema also has the advantage of allowing for growth or additional scope constraints that may arise in the future.

This schema also has potentially serious drawbacks. Wildcard mask overlap might occur if certain subblocks of groups are needed to match a single ACL statement. Layer 2 MAC address overlap could become a serious issue as well. Additionally, the region, PoP, priority, and consumption model

are not readily apparent in the address, and breakdown of the bits might be necessary to identify the application's scope. A simpler schema may do more for human interaction but be more difficult to draw boundaries around. ACL based boundaries are applicable to the multicast data plane. Efficient control should be considered in the design for multicast control plane isolation. This will be covered in detail in this chapter.

The point is, any group schema should address the needs of the organization; there is no one-size-fits-all approach. If the multicast design overlays an existing multicast network, it may not be possible to change the schema without disruption; however, the value of such a workable schema is immeasurable in a large multicast deployment. Keep in mind: If only a few multicast applications are on the network, there is no need to make a large and complex schema like the one shown in [Table 5-2](#). Instead, create a table and schema that has meaning and value for your specific multicast implementation.

Another IPv4 subblock consideration arises when using source-specific multicast (SSM). Remember that SSM can use both the 232/8 block for global and enterprise use as well as the 239.232/16 block private-only use. Administrators should never assign group space from the 232/8 block unless it is for SSM traffic. Many Layer 3 devices are preprogrammed to act on this block as SSM and will look to build SSM PIM trees accordingly.

It is also prudent when using SSM to subdivide the public and private SSM blocks further to give them scope and meaning (as with the preceding example schema). Using the 239.232/16 block for internal-only applications may provide fewer options for additional scope assignment, but it will still make bounding the groups easier. [Table 5-3](#) shows a possible subdivision of the 239.232/16 private SSM subblock using the third octet to identify geographic scope.

Octet 1	Octet 2	Octet 3	Octet 4
Fixed: 239	Fixed: 232	Geographic Scope	Application Number

Table 5-3 *Group Scoping by Octet Applied*

In addition to creating an addressing schema that makes sense for your organization, all administrators should follow several basic rules. Some of these rules are certainly flexible, in that they can easily and thoughtlessly be broken. Care should be taken to design a schema that meets these rules. Doing so streamlines configurations, makes troubleshooting easier, and ensures that specific router features do not interfere with proper multicast operations:

- Follow IANA's addressing guidelines, especially using 239/8 addresses for internal applications. RFC 2365 describes the use of administratively scoped IP multicast addresses. This address range should be used for all internal applications. Again, this block is similar in concept to the use of RFC 1918 addresses for unicast.
- Avoid using any group address with the x.0.0.x or x.128.0.x prefixes.

This rule should be somewhat obvious because the 224.0.0.X range encompasses link-local applications. Using an address in this range could interfere with critical network control traffic that uses multicast, such as, for example, EIGRP or OSPF. Let these addresses remain reserved as per the intention of IANA. Routers and switches, including IGMP snooping functions, will be unable to distinguish between the addresses. Furthermore, consider the 32:1 overlap of IP multicast addresses to Ethernet MAC addresses. This means you should avoid any multicast address in the [224-

239].0.0.x and [224-239].128.0.x ranges. As an example, notice that the schema in [Table 5-3](#) eliminates this problem by requiring the first elements to begin with bits 0001 and not 0000.

- Always use the 232/8 block for SSM applications, including inter-domain one-to-many applications. RFC 4608 describes the use of the 232/8 address range for PIM-SSM interdomain applications.

- Petition IANA for a publically recognized address from the 224 address range for any public-facing applications, but *only* if the application is truly public. Content providers that need to ensure against an address collision with any other provider or customer on a global scale should consider this block.

Using Group Scoping for Hybrid Designs and RP Placement

We reviewed the different RP design modes in [Chapter 4](#) . The key considerations for RP redundancy for any source multicast design are as follows:

1. **High-Availability mode** : Active/Active or Active/Standby options:

- We are aware that Anycast fits the bill for Active/Active mode.

- Active/Standby mode is supported by Auto-RP and BSR.

2. **Scoping requirement** : RP domains and multicast address scheme to scope regions for multicast:

- Scoping requirements need to be reviewed with applications aligned to the scope region. A local scope will require the source locally assigned to the region and appropriate control methods need to be determined for the local application not being transported across WAN infrastructure. Application containment within a scope can be used to limit bandwidth or local application dependency. Adding multiple local scopes may also increase the administrative overhead; the choice of local scope should be aligned to the outcome and to the benefits to the network infrastructure.

- Care should be taken to maintain the scopes to manageable limits permitted by the application.

- Multicast address group selection with an RP for each local scope should be considered.

3. **Downstream propagation** : Dynamic or static propagation:

The propagation method for an RP should be aligned to the multicast scope addresses. The static propagation method is to add a static RP address with an associated scope at every downstream router. This is a painstaking administrative task. Using a dynamic propagation method is preferred because the configuration for RP and ACL can be done only at the RP responsible for the scope.

[Table 5-4](#) explains the mapping of design features to the RP propagation scheme covered in [Chapter 4](#) :

RP Propagation	High-Availability Mode	Scoping Requirement	Downstream Propagation	Design Score
Auto-RP	Active/Standby	Yes	Dynamic	2/3
BSR	Active/Standby	No	Dynamic	1/3
Static	No HA	Yes	Static	1/3
Anycast	Active/Active	Yes	Static (for RFC 3446 or 3610)	2/3
Hybrid Design (Combined Propagation Methods)	Active/Active	Yes	Dynamic	3/3

Table 5-4 *Comparison of RP Distribution Methods*

As shown in [Table 5-4](#) , the actual choice for an enterprise RP design for ASM is not available by any of the known methods today. The best choice for an Architect would be to have an active/active implementation, which takes care of scoping and dynamic failover. (A score of 3/3 meets all the three requirements—Active/Active for high availability, supports scoping, and supports dynamic propagation.). This is possible by using the hybrid design. Yes, there is a hybrid design for an RP that leverages multiple protocols to achieve a desired effect for enterprise-scoped multicast design. [Table 5-5](#) outlines the mix of protocols to achieve this design state.

Design Features	Desired Design	Hybrid Design
High-Availability mode	Active/ Active	Anycast
Scoping support	Yes	Auto-RP
Downstream Propagation	Dynamic	Auto-RP

Table 5-5 *Hybrid Design Comparison*

This hybrid RP design is achieved by using Anycast to establish RP state information and Auto-RP for propagation of the RP information aligned to scope ranges to downstream routers. Please see [Figure 5-1](#) to understand the function of the hybrid design.

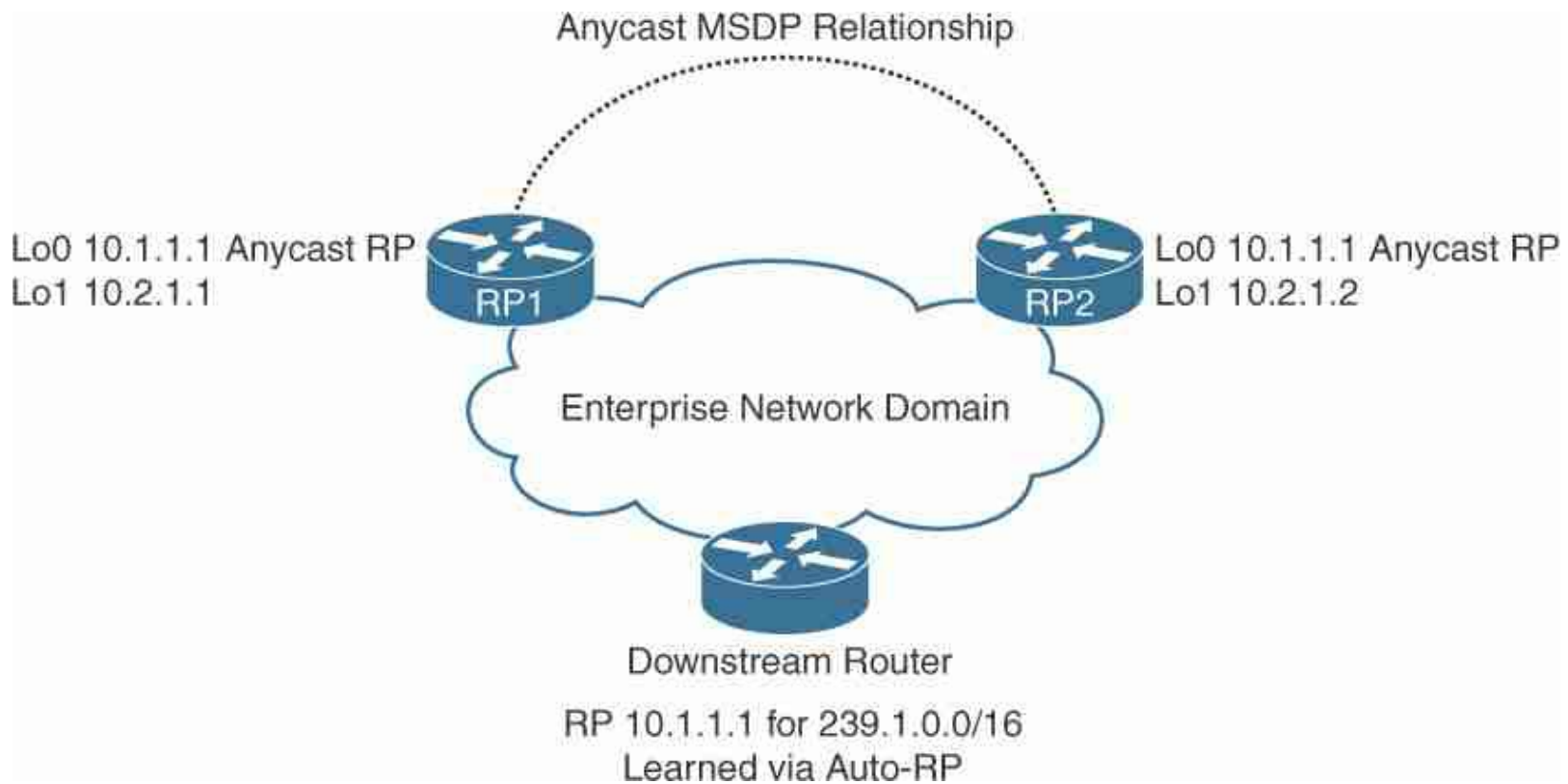


Figure 5-1 *Hybrid RP Design*

In the diagram, RP1 and RP2 act as the RP for the entire enterprise domain. The RP information is maintained by an Anycast MSDP relationship built between RP1 (10.2.1.1) and RP2 (10.2.1.2). The candidate RP (10.1.1.1) is used as the Auto-RP candidate. Auto-RP uses 10.1.1.1 as the elected candidate RP address, because RP1 and RP2 both advertise the same candidate RP address. Auto-RP ties the multicast scoping access-list at the RP; this ensures the downstream router receives the RP information with the ACL list attached to the RP scope range dynamically. [Example 5-1](#) provides a sample configuration.

Example 5-1 *Hybrid Design Configuration: Anycast RP with Auto-RP*

[Click here to view code image](#)

RP1

```
ip multicast-routing
ip cef
!
interface Loopback0
ip address 10.1.1.1 255.255.255.255
ip pim sparse-mode
!
interface Loopback1
ip address 10.2.1.1 255.255.255.255
ip pim sparse-mode
!
interface Ethernet0/0
ip address 192.168.1.1 255.255.255.0
ip pim sparse-mode
!
```

```
interface Ethernet1/0
 ip address 192.168.2.1 255.255.255.0
 ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
 eigrp router-id 10.2.1.1
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.2 connect-source Loopback1
ip msdp cache-sa-state
ip msdp default-peer 10.2.1.2
!
access-list 1 permit 239.1.0.0 0.0.255.255
    RP2
ip multicast-routing
ip cef
!
interface Loopback0
 ip address 10.1.1.1 255.255.255.255
 ip pim sparse-mode
!
interface Loopback1
 ip address 10.2.1.2 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet0/0
 ip address 192.168.1.2 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet1/0
 ip address 192.168.3.1 255.255.255.0
 ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
 eigrp router-id 10.2.1.2
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.1 connect-source Loopback1
ip msdp cache-sa-state
```

```
ip msdp default-peer 10.2.1.1
```

```
!
```

```
access-list 1 permit 239.1.0.0 0.0.255.255
```

[Example 5-2](#) shows the configuration for the downstream router.

Example 5-2 *Hybrid Design: Downstream Router Configuration*

[Click here to view code image](#)

```
ip multicast-routing
```

```
ip cef
```

```
!
```

```
!
```

```
interface Loopback0
```

```
ip address 10.2.1.3 255.255.255.255
```

```
!
```

```
interface Ethernet1/0
```

```
ip address 192.168.2.2 255.255.255.0
```

```
ip pim sparse-mode
```

```
!
```

```
interface Ethernet2/0
```

```
ip address 192.168.3.2 255.255.255.0
```

```
ip pim sparse-mode
```

```
!
```

```
router eigrp 1
```

```
network 0.0.0.0
```

```
!
```

```
ip pim autorp listener
```

[Example 5-3](#) shows the RP mapping command output at the downstream router.

Example 5-3 *Hybrid Design: Downstream RP Mapping*

[Click here to view code image](#)

```
R3# show ip pim rp mapping
```

```
PIM Group-to-RP Mappings
```

```
Group(s) 239.1.0.0/16 ← Scoped masked range configured using ACL 1 applied to  
the candidate RP configuration
```

```
RP 10.1.1.1 (?), v2v1
```

```
Info source: 10.1.1.1 (?), elected via Auto-RP <- Anycast RP 10.1.1.1 is  
propagated via Autp-RP
```

```
Uptime: 03:21:32, expires: 00:00:24
```

```
R3#
```

```
Anycast MSDP relationship at the RP
```

```
R2# show ip msdp summary
```

```
MSDP Peer Status Summary
```

Peer Address	AS	State	Uptime/	Reset SA	Peer Name
			Downtime	Count	Count

```
*10.2.1.1    ?    Connect 00:13:45 0    0    ?  
R2#
```

Multicast RP Design with MSDP Mesh Group

We previously discussed the concept of Anycast MSDP based on RFC 3618. The implementation included a default MSDP peer to create an MSDP Anycast relationship between two RPs to create an active/active solution. When you are faced with three regions and have to create an enterprise wide scope between them, using a default peer will not scale because you will only have two RPs in an active/active mode. For larger scale implementations, Anycast MSDP mesh groups are used.

Anycast MSDP mesh groups operate in the following way: When the RP for a domain receives an SA message from an MSDP peer, the RP verifies the receiver join requests for the group the SA message describes. If the (*,G) entry exists, the RP triggers an (S,G) join toward the source. After the (S,G) join reaches the source DR, a branch of the source tree is built from the source to the RP in the remote domain. If an MSDP peer receives the same SA message from a non-RPF peer toward the originating RP, it drops the message.

[Figure 5-2](#) explains the functionality for Anycast mesh groups.

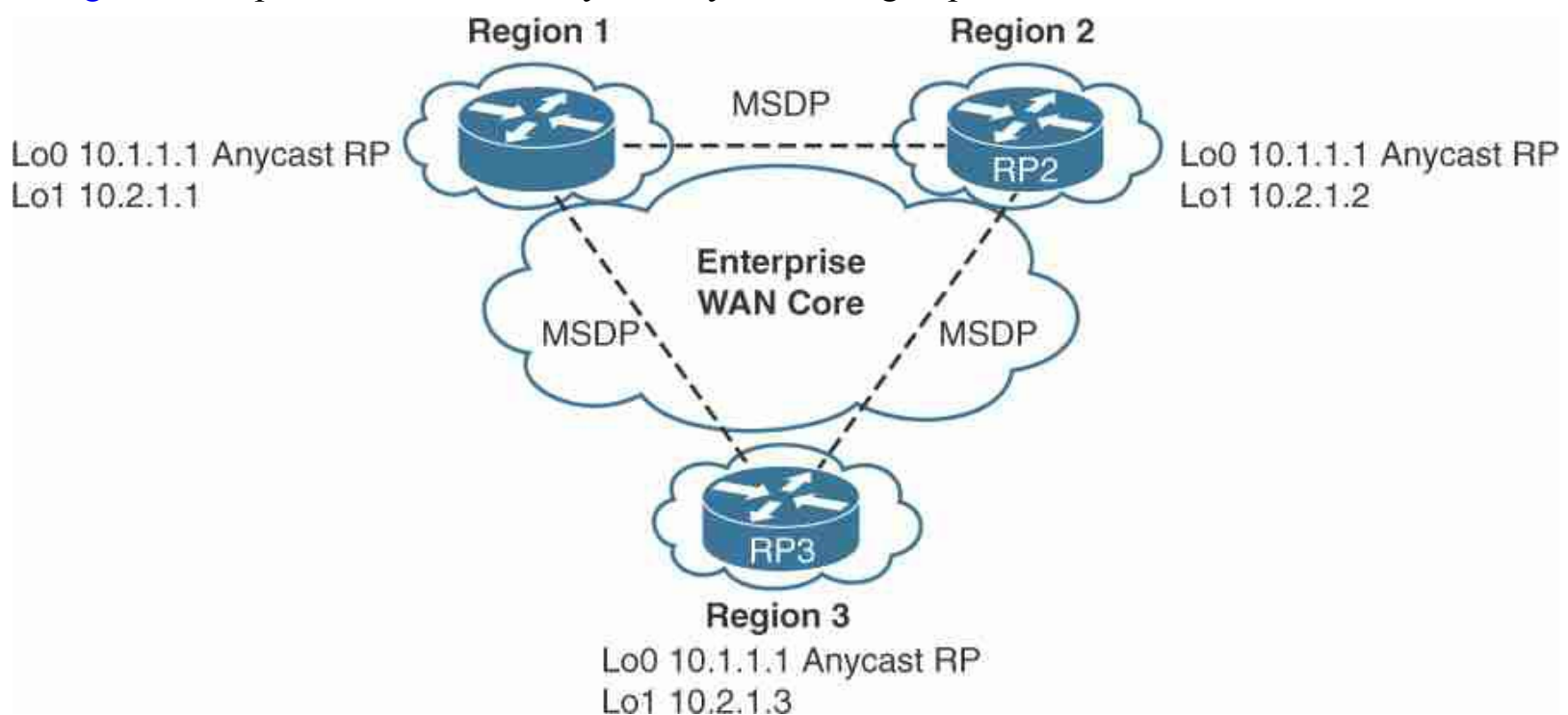


Figure 5-2 *Anycast Mesh Group*

Three regions are represented in [Figure 5-2](#). Each region has local sources that have global receivers and also receivers who participate in enterprise wide multicast streams. To create an active/active RP model localized to each region participating in the enterprise multicast domain, we leverage the same hybrid design concept, but with mesh groups. This provides active/active RP distribution for each region. The design allows localization of local multicast sources and receivers from state maintenance across the WAN. [Example 5-4](#) demonstrates the configuration.

Example 5-4 Anycast Mesh Group Configuration

[Click here to view code image](#)

RP1

interface Loopback0

```
ip address 10.1.1.1 255.255.255.255
ip pim sparse-mode
!
interface Loopback1
ip address 10.2.1.1 255.255.255.255
!
interface Ethernet0/0
ip address 192.168.1.1 255.255.255.0
ip pim sparse-mode
!
interface Ethernet1/0
ip address 192.168.2.1 255.255.255.0
ip pim sparse-mode
!
router eigrp 1
network 0.0.0.0
eigrp router-id 10.2.1.1
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.2 connect-source Loopback1
ip msdp peer 10.2.1.3 connect-source Loopback1
ip msdp cache-sa-state
ip msdp originator-id Loopback1
ip msdp mesh-group ENT 10.2.1.2
ip msdp mesh-group ENT 10.2.1.3
!
access-list 1 permit 239.1.0.0 0.0.255.255
    RP2
    ip multicast-routing
ip cef
!
interface Loopback0
ip address 10.1.1.1 255.255.255.255
ip pim sparse-mode
!
interface Loopback1
ip address 10.2.1.2 255.255.255.255
ip pim sparse-mode
!
interface Ethernet0/0
ip address 192.168.1.2 255.255.255.0
ip pim sparse-mode
```

```
!
interface Ethernet1/0
 ip address 192.168.3.1 255.255.255.0
 ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
 eigrp router-id 10.2.1.2
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.1 connect-source Loopback1
ip msdp peer 10.2.1.3 connect-source Loopback1
ip msdp cache-sa-state
ip msdp originator-id Loopback1
ip msdp mesh-group ENT 10.2.1.1
ip msdp mesh-group ENT 10.2.1.3
!
access-list 1 permit 239.1.0.0 0.0.255.255
    RP3
    ip multicast-routing
ip cef
!
!
interface Loopback0
 ip address 10.1.1.1 255.255.255.255
 ip pim sparse-mode
!
interface Loopback1
 ip address 10.2.1.3 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet1/0
 ip address 192.168.2.2 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet2/0
 ip address 192.168.3.2 255.255.255.0
 ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
!
```



```

ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.1 connect-source Loopback1
ip msdp peer 10.2.1.2 connect-source Loopback1
ip msdp cache-sa-state
ip msdp originator-id Loopback1
ip msdp mesh-group ENT 10.2.1.1
ip msdp mesh-group ENT 10.2.1.2
!
access-list 1 permit 239.1.0.0 0.0.255.255

```

[Example 5-5](#) demonstrates a functioning solution.

Example 5-5 *Anycast Mesh Group: RP Mapping and MSDP Summary*

[Click here to view code image](#)

```

r3#show ip pim rp mapping
PIM Group-to-RP Mappings
This system is an RP (Auto-RP)
This system is an RP-mapping agent (Loopback0)

  Group(s) 239.1.0.0/16
  RP 10.1.1.1 (?), v2v1
    Info source: 10.1.1.1 (?), elected via Auto-RP
      Uptime: 00:17:44, expires: 00:00:25

r3#show ip msdp summary
MSDP Peer Status Summary
Peer Address   AS   State  Uptime/  Reset SA  Peer Name
              Downtime Count Count
10.2.1.1       ?   Up     00:27:17 0    0    ?
10.2.1.2       ?   Up     00:27:26 0    0    ?

```

Multicast RP Hybrid Design with Scoped Multicast Domains

You learned about the importance of multicast scoping in [Chapter 3](#) and earlier in this chapter. It is very simple to overlay the hybrid RP design on the top of the multicast scoped design. Initially, you must review the local multicast groups that need to participate in the campus or branch domain. Then consider the requirements for enterprise-wide applications. Now, align these applications with the multicast IPv4 addressing scheme. When this is complete, use the hybrid RP design to address the active/active control plane. In [Figure 5-3](#), we review the enterprise-wide design and overlay the RP control-plane design.

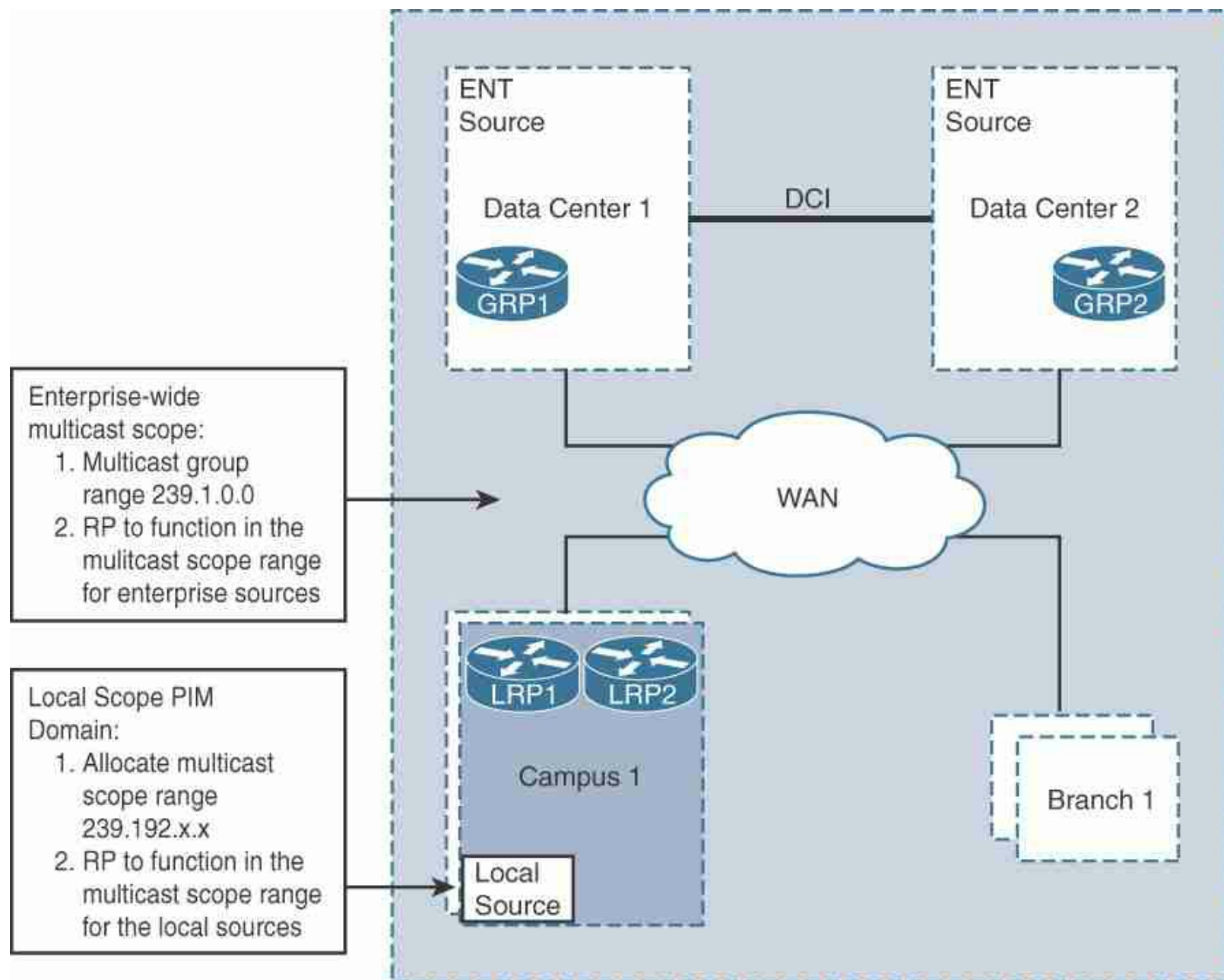


Figure 5-3 *Enterprise Multicast Scoped Domains*

In this example, the multicast application requirement is for enterprise-wide webcasting, local desktop imaging, and campus security camera multicast video. It is simple to categorize the multicast into two groups, enterprise-wide and campus. To optimize data transport and control plane for multicast, the campus source is scoped as a separate multicast domain. The multicast addressing scheme for the campus is planned accordingly. The RP selection has to be aligned to the multicast domain, as shown in [Figure 5-3](#). A global RP is selected with an enterprise-wide scope, and a local RP is selected with a scope for the local campus. In addition, the enterprise wide RP scope will cover the campus. The downstream routers at the campus location will learn about the two RPs, one for the enterprise-wide scope with the multicast address range of 239.1.0.0/16 and one for the local campus scope with the address of 239.192.0.0/16. Multicast best practices are covered in later sections of this chapter.

Using the same hybrid design methodology for the RP, [Example 5-6](#) shows the configuration for the global RP.

Example 5-6 *Enterprise Scoped Domain: Global RP Configuration*

[Click here to view code image](#)

G_RP1

```
ip multicast-routing
ip cef
!
interface Loopback0
 ip address 10.1.1.1 255.255.255.255
 ip pim sparse-mode
!
interface Loopback1
 ip address 10.2.1.1 255.255.255.255
 ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
 eigrp router-id 10.2.1.1
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.2 connect-source Loopback1
ip msdp cache-sa-state
ip msdp default-peer 10.2.1.2
!
access-list 1 permit 239.1.0.0 0.0.255.255
```

G_RP2

```
ip multicast-routing
ip cef
!
interface Loopback0
 ip address 10.1.1.1 255.255.255.255
 ip pim sparse-mode
!
interface Loopback1
 ip address 10.2.1.2 255.255.255.255
 ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
 eigrp router-id 10.2.1.2
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
```

```
ip msdp peer 10.2.1.1 connect-source Loopback1
ip msdp cache-sa-state
ip msdp default-peer 10.2.1.1
!
access-list 1 permit 239.1.0.0 0.0.255.255
```

[Example 5-7](#) shows the configuration for the local RP.

Example 5-7 *Enterprise Scoped Domain: Local RP Configuration*

[Click here to view code image](#)

```
L_RP1
ip multicast-routing
ip cef
!
interface Loopback0
! description- this loopback should be !unique to each campus !for multicast local !domain
ip address 10.1.1.10 255.255.255.255
ip pim sparse-mode
!
interface Loopback1
ip address 10.2.1.10 255.255.255.255
ip pim sparse-mode
!
router eigrp 1
network 0.0.0.0
eigrp router-id 10.2.1.10
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.20 connect-source Loopback1
ip msdp cache-sa-state
ip msdp default-peer 10.2.1.20
!
access-list 1 permit 239.192.0.0 0.0.255.255

L_RP2
ip multicast-routing
ip cef
!
interface Loopback0
! description- this loopback should be !unique to each campus !for multicast local !domain
ip address 10.1.1.10 255.255.255.255
ip pim sparse-mode
!
interface Loopback1
ip address 10.2.1.20 255.255.255.255
```

```

ip pim sparse-mode
!
router eigrp 1
 network 0.0.0.0
 eigrp router-id 10.2.1.20
!
ip pim autorp listener
ip pim send-rp-announce Loopback0 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback0 scope 20 interval 10
ip msdp peer 10.2.1.10 connect-source Loopback1
ip msdp cache-sa-state
ip msdp default-peer 10.2.1.10
!
access-list 1 permit 239.192.0.0 0.0.255.255

```

Using this configuration, we have achieved an active/active RP implementation in a scoped multicast environment that addresses the global and local scopes.

The downstream router in the campus will be part of two multicast RP domains, and the RP cache will appear as shown in [Example 5-8](#).

Example 5-8 *Enterprise Scoped Domain: Campus RP Mapping*

[Click here to view code image](#)

```

R2#show ip pim rp mapping
PIM Group-to-RP Mappings
Group(s) 239.1.0.0/16
RP 10.1.1.1 (?), v2v1
  Info source: 10.1.1.1 (?), elected via Auto-RP
    Uptime: 00:07:43, expires: 00:00:07
Group(s) 239.192.0.0/16
RP 10.1.1.10 (?), v2v1
  Info source: 10.1.1.10 (?), elected via Auto-RP
    Uptime: 00:07:43, expires: 00:00:07

```

The branch in this example only participates in the enterprise scope, as shown in [Example 5-9](#).

Example 5-9 *Enterprise Scope Domain: Branch RP Mapping*

[Click here to view code image](#)

```

R3#show ip pim rp mapping
PIM Group-to-RP Mappings
Group(s) 239.1.0.0/16
RP 10.1.1.1 (?), v2v1
  Info source: 10.1.1.1 (?), elected via Auto-RP
    Uptime: 00:07:43, expires: 00:00:07

```

RP Placement

RP placement is another key aspect in the multicast design. The details that need to be considered are as follows:

- The RP should be aligned to the multicast scope.

- Prefer placing the RP close to the source if possible. This is only applicable to a few sources that are of key importance to the business. Enterprise-wide deployments for multicast normally use RPs in the data center for the enterprise scope.

- Localization of the RP for local domains reduces the control plane state across the WAN. This is applicable when we have a MPLS-based service provider circuit in which the number of multicast states in the WAN is governed by a contractual agreement.

- If the number of states in the control plane is between 20 and 50, then another functional device such as a core switch or a WAN router can be used. Normally, it is not a mandate to have a separate RP; however, if the number of states is more than 100, a separate RP should be considered at least for the enterprise global scope.

Multicast Traffic Engineering and Forwarding

An ideal IP multicast network overlay matches the IP unicast underlay in a complimentary way. Where PIM is concerned, it is simply easier to make forwarding decisions if the underlying unicast forwarding paths are congruent and ultimately match the unicast forwarding paths. This type of implementation offers the benefits of low management and operational overhead. Consider [Figure 5-4](#).

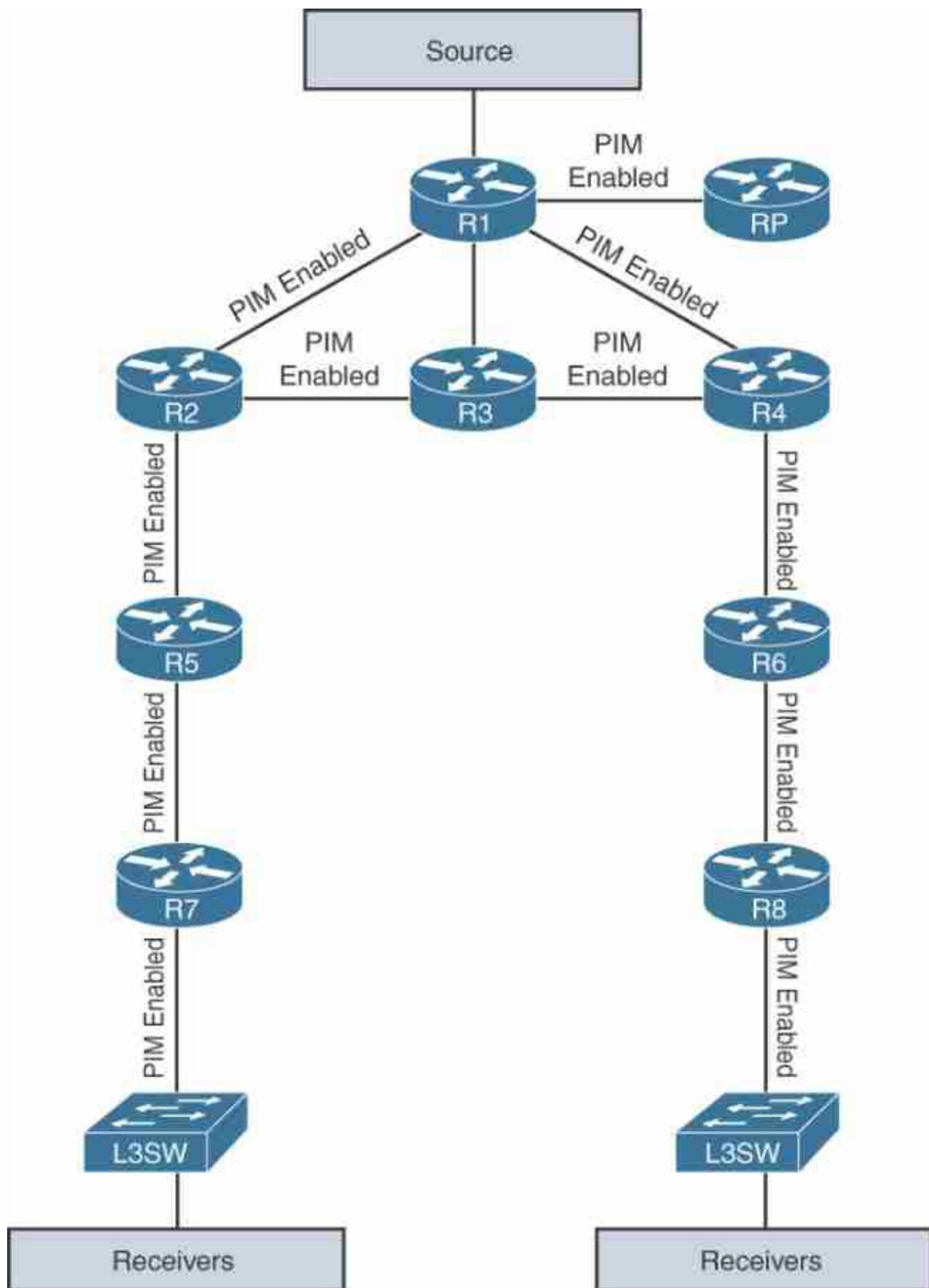


Figure 5-4 Simple PIM Domain

[Figure 5-4](#) shows a multicast domain in which the underlying IP paths are all unique and simple. If PIM is enabled on all links, as shown, multicast traffic can simply follow the network paths between source and receivers as dictated by the unicast routing table. It is clean and simple, and it is definitely a desirable design goal.

However, anyone who understands networking knows that this type of uniformity or simplicity is not always reality. Sometimes we have to make very specific changes to the multicast overlay to achieve certain desirable forwarding results. Typically, when we want IP traffic forwarding to conform to a method designed to improve some specific operational aspect of the network, we call this *traffic engineering*. Examples of traffic engineering include sending traffic over multiple load-balancing paths or specific paths with specific characteristics. Let's explore multicast traffic engineering a little further by first looking closer at the multicast state maintenance and forwarding mechanics.

More on mRIB, mFIB, and RPF Checks

A router (L3 device) interface is assigned an IP address from a subnet; this represents the final physical location of all hosts on a given segment. Reaching a host on a physical segment requires forwarding packets toward the destination router. IP routing protocols (such as static routing, OSPF, EIGRP, RIP, or BGP) either manually or dynamically learn the physical paths toward all networks. The L3 device uses the learned combined address and path information to create a forwarding table and decision tree. There is a table of all learned network addresses and associated physical paths with route ranking information, and a subsequent table that indicates which L3 interfaces the router has chosen to forward toward the destination.

Hierarchically speaking, we refer to these two separate tables as the *router information base (RIB)* and the *forwarding information base (FIB)*. The router populates the RIB by pulling routing information from tables built by the routing protocols. The IP routing table would be the RIP/ISIS/OSPF databases, the EIGRP topology table, or the BGP table. The router then derives the forwarding tree or FIB for each packet from the RIB.

There is a common-sense reason for the separation of these two table types. A router may run many protocols, and each protocol may record several paths toward an IP destination. The router first selects the best path(s) from the protocol tables and then ranks each protocol. The RIB consists of only the best route(s) from the most trusted protocol. This happens at the control-plane layer of the router. To forward packets, the router must make another recursive decision. The router must relate the appropriate interface(s) to the route. This is where the FIB comes into play. The FIB is used to make forwarding decisions or packet manipulation decisions. The use of application-specific integrated circuits (ASIC) may allow this function to be conducted in hardware, which will improve the throughput of the device. The FIB is a function of the forwarding or data plane of the router. [Figure 5-5](#) illustrates the process of building forwarding decision tables. Separating the control plane from the data plane makes the topology more robust and resilient, allowing the control plane to make on-the-fly changes or corrections without affecting packet-forwarding until changes are confirmed.

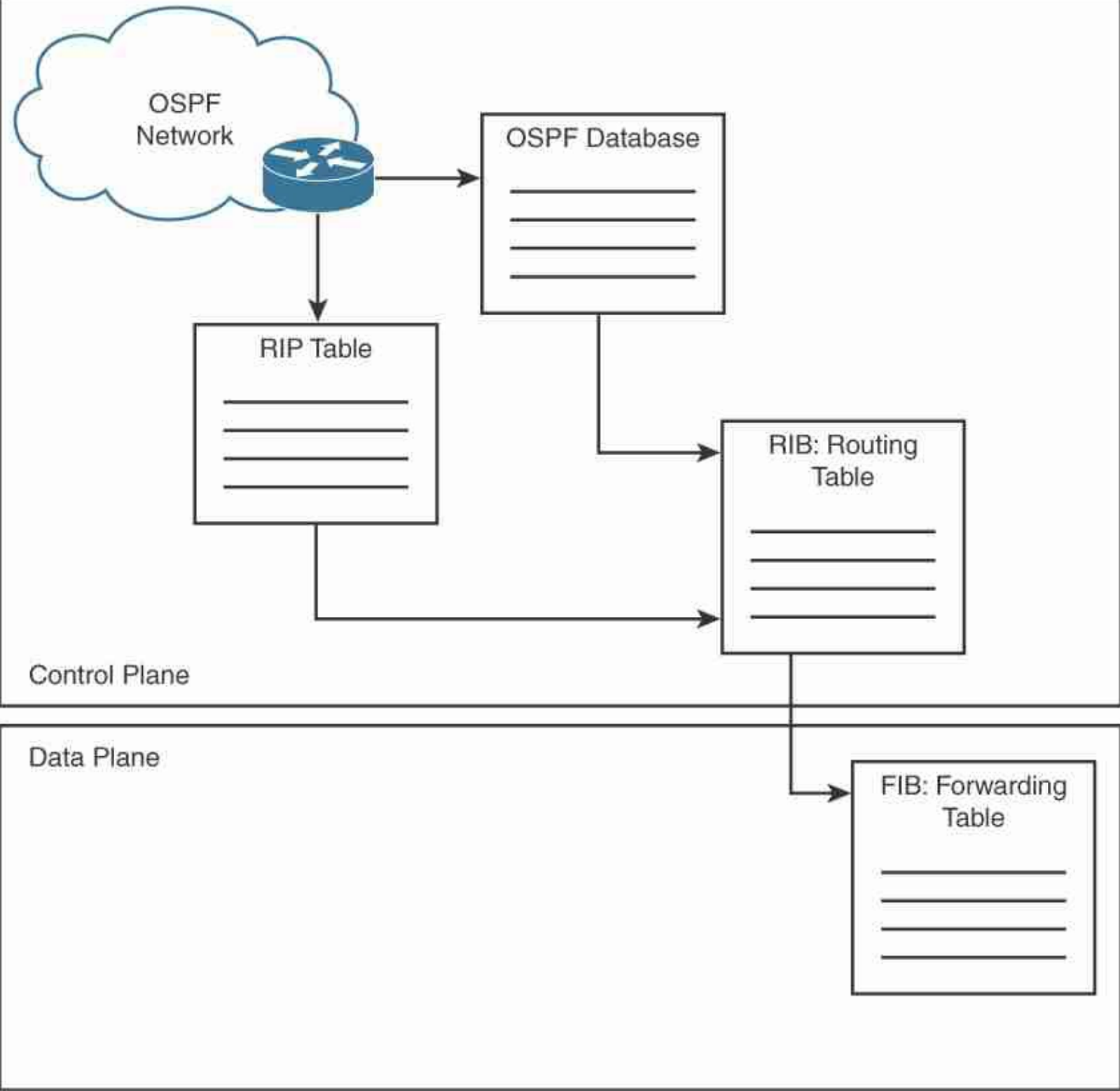


Figure 5-5 *RIB and FIB Population*

In the traditional Cisco routing environment, the **show ip route** command reveals the RIB. The output in [Example 5-10](#) shows the RIB for a small three-router network. You can see there are routes learned from OSPF, RIP, static, and from connected networks.

Example 5-10 *Basic IOS Unicast RIB: show ip route*

[Click here to view code image](#)

```
ASR1K-2#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```
R   10.0.0.0/8 [120/2] via 192.168.2.1, 00:00:15, GigabitEthernet0/0/0
    172.16.0.0/24 is subnetted, 1 subnets
S   172.16.1.0 is directly connected, GigabitEthernet0/0/0
    192.168.0.0/24 is variably subnetted, 3 subnets, 2 masks
R   192.168.0.0/24
    [120/2] via 192.168.2.1, 00:00:15, GigabitEthernet0/0/0
O   192.168.0.2/32
    [110/3] via 192.168.2.1, 00:15:51, GigabitEthernet0/0/0
C   192.168.0.3/32 is directly connected, Loopback0
O   192.168.1.0/24 [110/2] via 192.168.2.1, 00:16:01, GigabitEthernet0/0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L   192.168.2.2/32 is directly connected, GigabitEthernet0/0/0
O   192.168.3.0/24 [110/3] via 192.168.2.1, 00:15:51, GigabitEthernet0/0/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.4.0/24 is directly connected, GigabitEthernet0/0/3
L   192.168.4.1/32 is directly connected, GigabitEthernet0/0/3
```

Notice that the RIB table contains information from multiple protocols (RIP, static, and OSPF). The data plane does not need to know or understand the source of the routing information. It only needs to derive the best forwarding path for each known IP destination network or subnet. For those familiar with Cisco Express Forwarding (CEF), the **show ip cef** command displays the FIB at the data plane of most IOS routers as demonstrated in [Example 5-11](#) . This CEF table was derived from the above RIB.

Example 5-11 Basic IOS Unicast FIB: show ip cef

[Click here to view code image](#)

ASR1K-2#show ip cef

Prefix	Next Hop	Interface
10.0.0.0/8	192.168.2.1	GigabitEthernet0/0/0
127.0.0.0/8	drop	
172.16.1.0/24	attached	GigabitEthernet0/0/0
192.168.0.0/24	192.168.2.1	GigabitEthernet0/0/0
192.168.0.2/32	192.168.2.1	GigabitEthernet0/0/0
192.168.0.3/32	receive	Loopback0
192.168.1.0/24	192.168.2.1	GigabitEthernet0/0/0
192.168.2.0/24	attached	GigabitEthernet0/0/0
192.168.2.0/32	receive	GigabitEthernet0/0/0

192.168.2.1/32	attached	GigabitEthernet0/0/0
192.168.2.2/32	receive	GigabitEthernet0/0/0
192.168.2.255/32	receive	GigabitEthernet0/0/0
192.168.3.0/24	192.168.2.1	GigabitEthernet0/0/0
192.168.4.0/24	attached	GigabitEthernet0/0/3
192.168.4.0/32	receive	GigabitEthernet0/0/3
192.168.4.1/32	receive	GigabitEthernet0/0/3

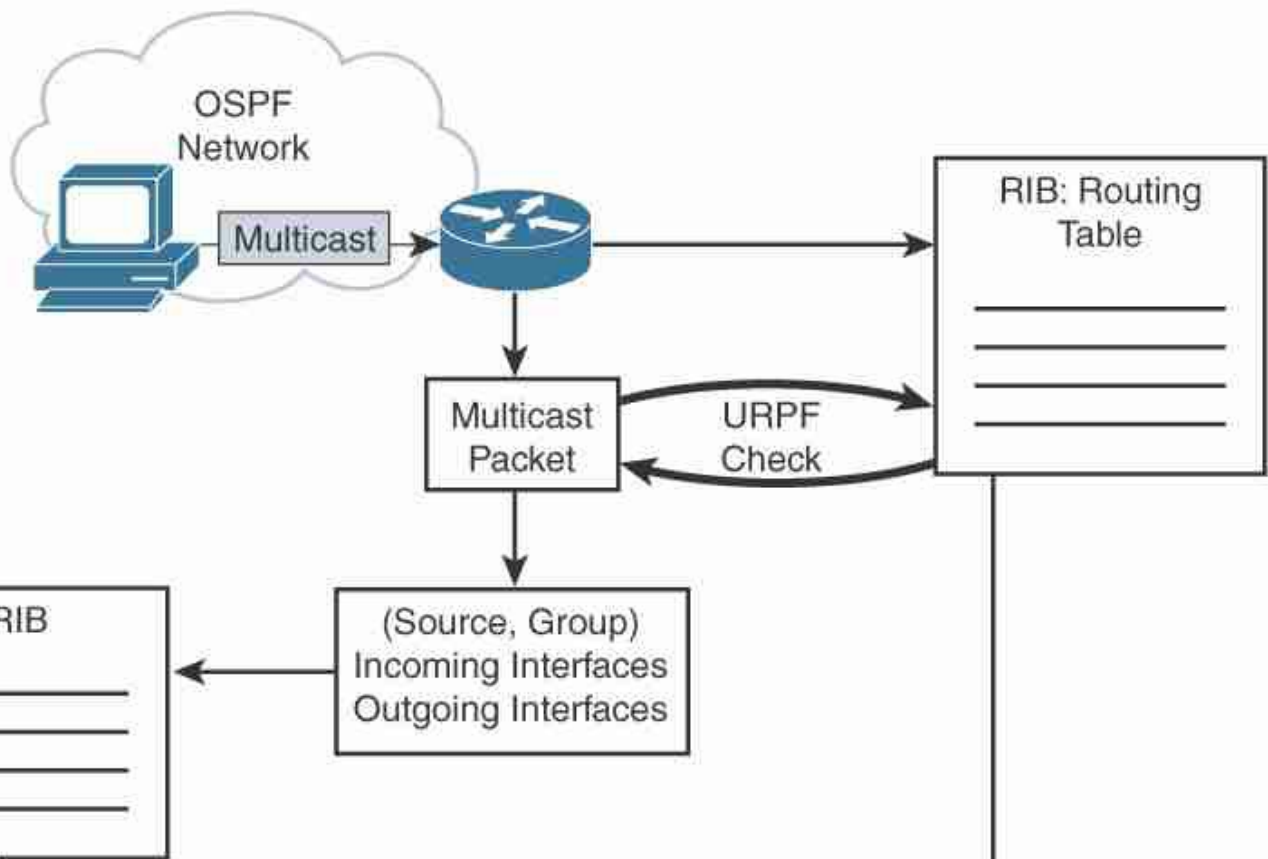
IP multicasting does not change basic unicast RIB information in any way. The process of unicast RIB derivation and population is the same for unicast, broadcast, and multicast routes. Routers also forward multicast packets from the source toward receivers based on the information learned from the RIB. An inherent danger exists in multicast-forwarding that does not exist in unicast and broadcast packet-forwarding.

When a network device receives a unicast or a broadcast packet, only a single copy of that packet exists because it transits Layer 3 interfaces. Broadcasts may have many intended recipients, but a Layer 3 interface does not make additional copies to send to host interfaces. That is the job of the Layer 2 switch. Layer 2 switches copy each broadcast frame and flood the copies out each interface in the associated Layer 2 domain; this is known as *packet replication*.

IP multicast packets come from a single source but are forwarded toward many Layer 3 destinations. It is very common to have both physical and logical redundancy in Layer 3 networks. Routers also do not have any inherent way of telling whether a packet is an original or a copy. Consequently, a Layer 3 router must make an important decision: It must choose which interfaces must forward a copy of the packet without creating a forwarding loop. The router must therefore have a way to determine which network paths multicast sources are sending from, where subscribed receivers are located, and which interfaces are in the path toward those receivers. This is further complicated by the fact that multicast receivers subscribe only to specific groups. Routers must also have a way to learn and share information about the groups that have current subscribers, the specific subscribers, and the sources generating multicast packets.

PIM is the most widely deployed multicast routing protocol; however, the term *multicast routing protocol* confuses many engineers. PIM does not learn and share routing information. PIM does not change, manipulate, or insert information into the unicast RIB of a router. The primary concern of the multicast routing protocol is to ensure loop-free forwarding over the existing IP network, acting as a control-plane overlay. This is why a router must maintain a separate multicast RIB and multicast FIB (mRIB and mFIB) specific to multicast packets. Routers must also populate multicast RIB and FIB tables using a combination of information from the unicast RIB and the learned source, group, and receiver information, using RPF checks to determine loop-free forwarding. Refer to the diagram in [Figure 5-6](#) for a visual illustration of this process.

Control Plane



Data Plane



Figure 5-6 *mRIB and mFIB Population*

The **show ip mroute** command in Cisco IOS reveals the multicast RIB. The **show ip mroute** output in [Example 5-12](#) shows the multicast RIB, the same router whose RIB and FIB were previously examined.

Example 5-12 *Basic IOS MRIB: show ip mroute*

[Click here to view code image](#)

```
ASR1K-2#show ip mroute
```

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
V - RD & Vector, v - Vector

Outgoing interface flags: H - Hardware switched, A - Assert winner

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:01:12/stopped, RP 192.168.0.1, flags: SJCL

Incoming interface: GigabitEthernet0/0/0, RPF nbr 192.168.2.1

Outgoing interface list:

Loopback0, Forward/Sparse-Dense, 00:01:12/00:02:12

(192.168.1.2, 239.1.1.1), 00:00:07/00:02:52, flags: LJT

Incoming interface: GigabitEthernet0/0/0, RPF nbr 192.168.2.1

Outgoing interface list:

Loopback0, Forward/Sparse-Dense, 00:00:07/00:02:52

(192.168.0.1, 239.1.1.1), 00:00:19/00:02:40, flags: LJT

Incoming interface: GigabitEthernet0/0/0, RPF nbr 192.168.2.1

Outgoing interface list:

Loopback0, Forward/Sparse-Dense, 00:00:19/00:02:40

(* , 224.0.1.40), 00:01:12/00:02:59, RP 192.168.0.1, flags: SJCL

Incoming interface: GigabitEthernet0/0/0, RPF nbr 192.168.2.1

Outgoing interface list:

Loopback0, Forward/Sparse-Dense, 00:01:12/00:02:10

GigabitEthernet0/0/3, Forward/Sparse-Dense, 00:01:12/00:02:59

If the device is using CEF, the multicast FIB is integrated into the CEF table and appears as follows:

ASR1K-2#show ip cef 239.1.1.1

224.0.0.0/4

multicast

This particular CEF output may not be very helpful. More advanced and modular operating systems like Cisco IOS-XR process the multicast RIB and FIB more independently. The output from the commands **show mrrib route** and **show mfrib route** executed on a router running IOS-XR, as demonstrated in [Example 5-13](#) , shows the distinction between RIB and FIB more acutely.

Example 5-13 IOS-XR MRIB and MFIB: show mrrib/mfrib route

[Click here to view code image](#)

RP/0/RSP0/CPU0:A9K#show mrrib route

IP Multicast Routing Information Base

Entry flags: L - Domain-Local Source, E - External Source to the Domain,

C - Directly-Connected Check, S - Signal, IA - Inherit Accept,

IF - Inherit From, D - Drop, ME - MDT Encap, EID - Encap ID,

MD - MDT Decap, MT - MDT Threshold Crossed, MH - MDT interface handle
CD - Conditional Decap, MPLS - MPLS Decap, MF - MPLS Encap, EX - Extranet
MoFE - MoFRR Enabled, MoFS - MoFRR State, MoFP - MoFRR Primary
MoFB - MoFRR Backup, RPFID - RPF ID Set

Interface flags: F - Forward, A - Accept, IC - Internal Copy,
NS - Negate Signal, DP - Don't Preserve, SP - Signal Present,
II - Internal Interest, ID - Internal Disinterest, LI - Local Interest,
LD - Local Disinterest, DI - Decapsulation Interface
EI - Encapsulation Interface, MI - MDT Interface, LVIF - MPLS Encap,
EX - Extranet, A2 - Secondary Accept, MT - MDT Threshold Crossed,
MA - Data MDT Assigned, LMI - mLDP MDT Interface, TMI - P2MP-TE MDT Interface
IRMI - IR MDT Interface

(*224.0.0.0/4) RPF nbr: 192.168.0.1 Flags: L C P

Up: 00:06:38

Outgoing Interface List

Decapstunnel0 Flags: NS DI, Up: 00:06:38

(*239.1.1.1) RPF nbr: 192.168.0.1 Flags: C

Up: 00:03:19

Incoming Interface List

Decapstunnel0 Flags: A, Up: 00:03:19

Outgoing Interface List

GigabitEthernet0/1/0/1 Flags: F NS LI, Up: 00:03:19

(192.168.0.1,239.1.1.1) RPF nbr: 192.168.0.1 Flags: L

Up: 00:01:05

Incoming Interface List

Loopback0 Flags: A, Up: 00:01:05

Outgoing Interface List

GigabitEthernet0/1/0/1 Flags: F NS, Up: 00:01:05

(192.168.1.2,239.1.1.1) RPF nbr: 192.168.1.2 Flags:

Up: 00:00:57

Incoming Interface List

GigabitEthernet0/1/0/0 Flags: A, Up: 00:00:57

Outgoing Interface List

GigabitEthernet0/1/0/1 Flags: F NS, Up: 00:00:57

(192.168.2.2,239.1.1.1) RPF nbr: 192.168.2.2 Flags:

Up: 00:02:29

Incoming Interface List

GigabitEthernet0/1/0/1 Flags: F A, Up: 00:01:58

Outgoing Interface List

GigabitEthernet0/1/0/1 Flags: F A, Up: 00:01:58

RP/0/RSP0/CPU0:A9K#show mfib route

IP Multicast Forwarding Information Base

Entry flags: C - Directly-Connected Check, S - Signal, D - Drop,

IA - Inherit Accept, IF - Inherit From, EID - Encap ID,

ME - MDT Encap, MD - MDT Decap, MT - MDT Threshold Crossed,
MH - MDT interface handle, CD - Conditional Decap,
DT - MDT Decap True, EX - Extranet, RPFID - RPF ID Set,
MoFE - MoFRR Enabled, MoFS - MoFRR State

Interface flags: F - Forward, A - Accept, IC - Internal Copy,
NS - Negate Signal, DP - Don't Preserve, SP - Signal Present,
EG - Egress, EI - Encapsulation Interface, MI - MDT Interface,
EX - Extranet, A2 - Secondary Accept

Forwarding/Replication Counts: Packets in/Packets out/Bytes out
Failure Counts: RPF / TTL / Empty Olist / Encap RL / Other

(*224.0.0.0/4), Flags: C

Up: 00:07:02
Last Used: never
SW Forwarding Counts: 0/0/0
SW Replication Counts: 0/0/0
SW Failure Counts: 0/0/0/0/0
Decapstunnel0 Flags: NS, Up:00:07:02

(*239.1.1.1), Flags: C

Up: 00:03:43
Last Used: 00:01:29
SW Forwarding Counts: 1/0/0
SW Replication Counts: 1/0/0
SW Failure Counts: 0/0/0/0/0
Decapstunnel0 Flags: A, Up:00:03:43
GigabitEthernet0/1/0/1 Flags: NS, Up:00:02:23

(192.168.0.1,239.1.1.1), Flags:

Up: 00:01:29
Last Used: 00:01:29
SW Forwarding Counts: 1/1/100
SW Replication Counts: 1/0/0
SW Failure Counts: 0/0/0/0/0
Loopback0 Flags: A, Up:00:01:29
GigabitEthernet0/1/0/1 Flags: NS, Up:00:01:29

(192.168.1.2,239.1.1.1), Flags:

Up: 00:01:21
Last Used: never
SW Forwarding Counts: 0/0/0
SW Replication Counts: 0/0/0
SW Failure Counts: 0/0/0/0/0
GigabitEthernet0/1/0/0 Flags: A, Up:00:01:21
GigabitEthernet0/1/0/1 Flags: NS, Up:00:01:21

(192.168.2.2,239.1.1.1), Flags:

Up: 00:02:53
Last Used: never
SW Forwarding Counts: 0/0/0
SW Replication Counts: 0/0/0
SW Failure Counts: 0/0/0/0/0
GigabitEthernet0/1/0/1 Flags: A, Up:00:02:23

As you can see from the output, the multicast RIB is a table of the sources and groups the router is currently receiving updates for, from multicast routing protocols like PIM and the host subscription protocol Internet Group Message Protocol (IGMP). As per the process, the list of sources and receivers is compared against the unicast routing table, checking for exit interfaces and ensuring loop-free packet delivery using the unicast reverse path.

Even though IP multicast is inherent in the IP stack, multicast overlays are analogous to any other overlay network, like MPLS, VPNs, GRE, and so on. Each of these protocols also creates and maintains additional forwarding information and requires an underlying IP network that is complete and converged. More specifically, the RIB and FIB of the underlying network are the foundation of any forwarding decision made by the Layer 3 device.

Finally, as previously mentioned in this and earlier chapters, unicast reverse path forwarding (RPF) checking is the way routers ensure loop-free multicast forwarding. Let us quickly review the RPF check process: When a multicast packet is received, the router looks up the unicast route toward the source in the packet header. If the multicast packet entered into the router through an interface that is in the preferred destination path (derived from the unicast RIB) for the source, the packet is deemed trustworthy and the router can add the (S,G) to the MRIB table. This is a reverse check, and the router records the list of incoming interface(s) of the source packet. That does not mean the router forwards the multicast packet or adds the entry to the MRIB. Additional information is still required. In order for the router to add the (S,G) entry to the table, an existing (*,G) entry must be in the MRIB, applicable to Any Source Multicast (ASM). In other words, a downstream source must have previously expressed interest in the group; otherwise, the router simply prunes the packets, preventing the multicast stream from flooding the network unnecessarily.

When a router receives an IGMP subscription packet or multicast route update (PIM Join) for a (*,G), the same RPF check process is followed. Remember, however, that the (*,G) represents a shared tree. The source is not included in the update, making the root of the tree the RP specified for the group in the router group to RP mapping. Thus, in the case of a (*,G) update from either PIM or IGMP, the router RPF checks the forwarding tree against the unicast path toward the RP.

The router builds a list of interfaces that are downstream from the router with interested hosts, the outgoing interface list (OIL). The OIL in the (*,G) entry represents all the interfaces to which a multicast packet for the group specified requires packet replication. Now that the router has both an (S,G) and a (*,G) entry for a particular group, it is ready to replicate and forward packets from the sources listed in the incoming list toward the receivers listed in the outgoing interface list. In most cases, routers forward multicast packets, obeying split-horizon rules. The packet must come from an incoming interface and will be replicated and forwarded down only those interfaces in the OIL. As you can see, RPF checking is used to govern entries in both the control plane and the forwarding plane of every multicast router. If any packet or any updated (S,G) or (*,G) fails the RPF check, the packet is not forwarded and the entry is removed from the mRIB and mFIB.

Traffic Engineering Using IP Multipath Feature

We have discussed at length the relationship between the unicast RIB, the mRIB, and RPF checks. But what happens when the unicast RIB has multiple equal-cost path entries for a source, the RP, or receivers for a given group? Consider the network in [Figure 5-7](#).

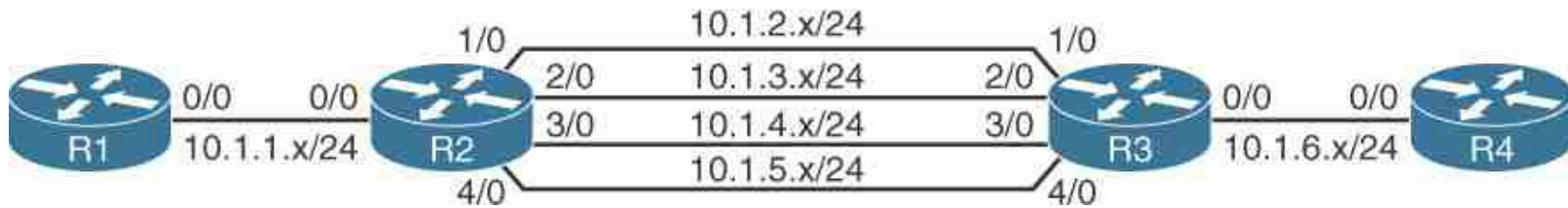


Figure 5-7 *Multiple IP Paths*

In this very simple network diagram, four equal-cost EIGRP paths lie between the two multicast routers. The network is purposefully designed to utilize all four paths to maximize efficiency. With unicast routing, this is referred to as equal-cost multi-path (ECMP). The default behavior of PIM states that we can only have one RPF neighbor interface in the multicast state table for (*,G) and (S,G) entries. By default, PIM uses the following rule to declare which interface is the appropriate RPF interface:

The RPF interface of a (S,G) entry is the interface with the lowest cost path (administrative distance, or if from the same protocol, the routing metric) to the IP address of the source. The RPF interface of a (,G) entry is the lowest cost path to the IP address of the RP. If multiple paths exist and have the same cost, the interface with the highest IP address is chosen as the tiebreaker.*

It is possible to change this default behavior to allow load-splitting between two or more paths, if required, and there may be many good reasons to do so. Configuring load-splitting with the **ip multicast multipath** command causes the system to load-split multicast traffic across multiple equal-cost paths based on source address using the S-hash algorithm. This feature load-splits the traffic and does not load-balance the traffic. Based on the S-hash algorithm, the multicast stream from a source uses only one path. The PIM joins will be distributed over the different ECMP links based on a hash of the source address. This enables streams to be divided across different network paths. The S-hash method can be used to achieve a diverse path for multicast data flow that is split between two multicast groups to achieve redundancy in transport of the real-time packets. The redundant flow for the same data stream is achieved using an intelligent application that can encapsulate the same data in two separate multicast streams. These applications are often seen in financial networks. This feature is leveraged from the network side to achieve redundancy. By using this feature, the network availability increases the overall resiliency because now a single failure in the network could potentially affect only 50% of the traffic streams. Furthermore, if you have an intelligent application that provides redundancy to the same stream by encapsulating in two multicast addresses, then delivery of the data across the network is guaranteed based on the IP multicast multipath feature. Things to consider while using this feature in a design are:

- Multicast traffic from different sources are load-split across the different equal-cost paths.
- Load-splitting does not occur across equal-cost paths for multicast traffic from the same source sent to different multicast groups.

Note

The multipath hashing algorithm is similar to other load-splitting algorithms in that it is unlikely that true 50-50 load-splitting will ever occur. It is possible that two unique flows (source to

receivers) could be hashed to forward down the same link, but a single stream from a single source will not be hashed over more than one link. [Table 5-6](#) delineates the different hash options.

[Table 5-6](#) provides the basic syntax for enabling the feature in IOS and IOS-XE systems.

Feature	Comments
ip multicast multipath	Enables ECMP multicast load-splitting on a router based on source address using the S-hash algorithm
ip multicast multipath s-g-hash basic	Enables ECMP multicast load-splitting on a router based on source and group address using the basic S-G-hash algorithm. This is, however, subjected to polarization because for a given source and group, the same hash is always picked irrespective of the router on which the hash is being calculated.
ip multicast multipath s-g-hash next-hop-based	Enables ECMP multicast load-splitting on a router based on source, group, and next-hop address using the next-hop-based S-G-hash algorithm. This is a more complex hash based on source, group, and next-hop address. The next-hop-based S-G-hash algorithm is predictable because no randomization is used in coming up with the hash value.

Table 5-6 *IOS Multipath Command Comparison*

The topology reviewed in [Figure 5-7](#) provides a use case for multicast multipath. Consider [Figure 5-8](#) , which adds a multicast PIM domain configuration.

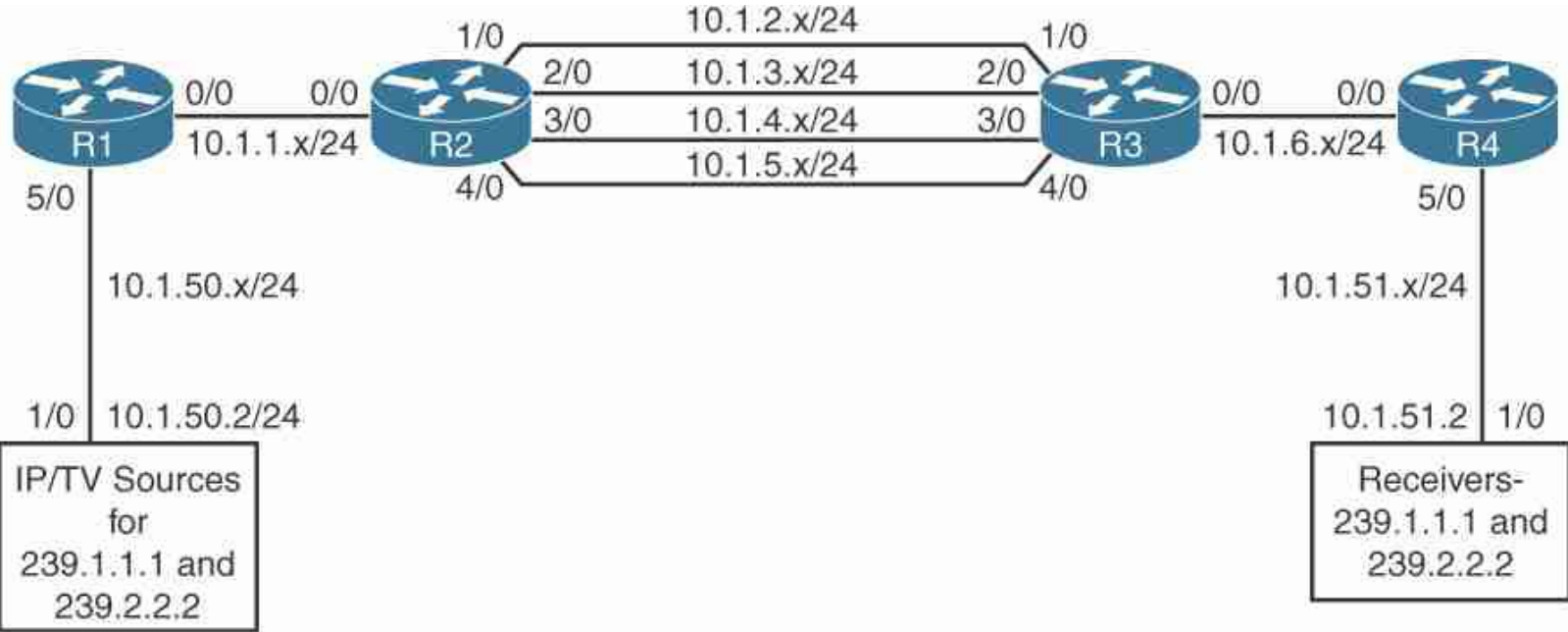


Figure 5-8 *Multiple IP Multicast Paths*

This configuration is shown in [Example 5-14](#) . R1, R2, R3, and R4 represent a multicast domain with multiple redundant links. There is a desire to split multicast traffic across the four different links between R2 and R3 to provide distribution of the multicast traffic. This can be accomplished using the **multicast multipath** command on both R2 and R3. The source generates traffic for two multicast groups, 239.1.1.1 and 239.2.2.2. The multicast domain configuration in this example is a simple PIM ASM with static RP.

Example 5-14 *IOS Multipath Configuration*

[Click here to view code image](#)

R3 Configuration

```
<..>
ip multicast-routing
ip multicast multipath s-g-hash next-hop-based
ip cef
!
!
interface Loopback0
 ip address 10.3.3.3 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet0/0
 ip address 10.1.6.1 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet1/0
 ip address 10.1.2.2 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet2/0
 ip address 10.1.3.2 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet3/0
 ip address 10.1.4.2 255.255.255.0
 ip pim sparse-mode
!
router eigrp 1
 network 10.0.0.0
!
 ip pim rp-address 10.3.3.3
```

Examine what happens to the RPF checks in the state entries for our multicast groups. Both the 239.1.1.1 and 239.2.2.2 sources send traffic to the respective receiver in the multicast topology.

[Example 5-15](#) shows the path taken by the two streams at R3.

Example 5-15 *IOS Multipath RPF*

[Click here to view code image](#)

```
R3#show ip rpf 10.1.50.2 239.1.1.1
RPF information for ? (10.1.50.2)
RPF interface: Ethernet2/0
RPF neighbor: ? (10.1.3.1)
RPF route/mask: 10.1.50.0/24
```

```

RPF type: unicast (eigrp 1)
Doing distance-preferred lookups across tables
Multicast Multipath enabled. algorithm: next-hop-based
Group: 239.1.1.1
RPF topology: ipv4 multicast base, originated from ipv4 unicast base
R3#show ip rpf 10.1.50.2 239.2.2.2
RPF information for ? (10.1.50.2)
RPF interface: Ethernet3/0
RPF neighbor: ? (10.1.4.1)
RPF route/mask: 10.1.50.0/24
RPF type: unicast (eigrp 1)
Doing distance-preferred lookups across tables
Multicast Multipath enabled. algorithm: next-hop-based
Group: 239.2.2.2
RPF topology: ipv4 multicast base, originated from ipv4 unicast base

```

Multicast Traffic Engineering: Deterministic Path Selection

In the previous multipath scenario, the network architects have chosen equal-cost paths for network-forwarding. This decision was likely made to maximize network-forwarding efficiency. If multicast application traffic is dense enough to consume significant bandwidth, it seems like a wise course of action to enable multicast multipath. However, in some networks, like financial networks, there may arise a need to separate multicast data transmission from unicast transmissions across WAN links.

This design choice could also achieve better optimization of bandwidth for multicast and unicast applications. Consider a network that is transporting an IP-TV multicast stream. The stream may be small enough to need only one pipe but large enough to cause resource constraints on unicast traffic, thereby justifying its own WAN link.

[Figure 5-9](#) illustrates just such a scenario. The administrators of the network have decided that a corporate IP-TV application consumes a great deal of bandwidth, consuming enough resources to put critical unicast traffic (Path 2) at risk. The network architect has asked that a redundant, non-unicast link (Path 1) be maintained for this purpose.

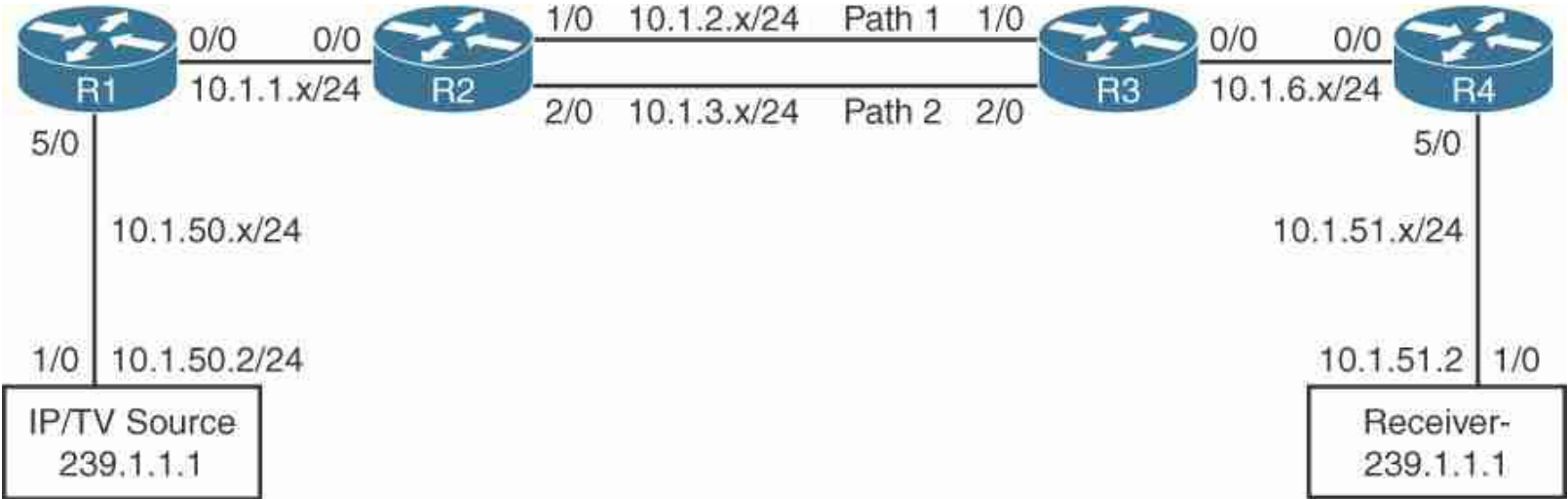


Figure 5-9 *Deterministic Multicast Paths*

Remember the rule of one RPF path that we just discussed? By default, in this topology, Path1 and

Path 2 are equal-cost links in the EIGRP topology table for 10.1.50.x and 10.1.51.x subnets (multicast source and receiver subnets). Based on RPF rules, the highest interface IP address is selected as the RPF interface for the outgoing interface list (OIL), which in this case is Path 2, as shown in [Example 5-16](#).

Example 5-16 *Unicast RIB with Multiple Paths*

[Click here to view code image](#)

R3# show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
D    10.1.1.0/24 [90/307200] via 10.1.3.1, 00:00:11, Ethernet2/0
      [90/307200] via 10.1.2.1, 00:00:11, Ethernet1/0
C    10.1.2.0/24 is directly connected, Ethernet1/0
L    10.1.2.2/32 is directly connected, Ethernet1/0
C    10.1.3.0/24 is directly connected, Ethernet2/0
L    10.1.3.2/32 is directly connected, Ethernet2/0
C    10.1.4.0/24 is directly connected, Ethernet3/0
L    10.1.4.2/32 is directly connected, Ethernet3/0
C    10.1.6.0/24 is directly connected, Ethernet0/0
L    10.1.6.1/32 is directly connected, Ethernet0/0
D    10.1.50.0/24 [90/332800] via 10.1.3.1, 00:00:11, Ethernet2/0
      [90/332800] via 10.1.2.1, 00:00:11, Ethernet1/0
D    10.1.51.0/24 [90/307200] via 10.1.6.2, 00:00:11, Ethernet0/0
C    10.3.3.3/32 is directly connected, Loopback0
```

R3# show ip rpf 10.1.50.2

RPF information for ? (10.1.50.2)

RPF interface: Ethernet2/0

RPF neighbor: ? (10.1.3.1)

RPF route/mask: 10.1.50.0/24

RPF type: unicast (eigrp 1)

Doing distance-preferred lookups across tables

RPF topology: ipv4 multicast base, originated from ipv4 unicast base

Because of the RPF rules, interface e2/0 will always be selected for the multicast flow. Further complicating the design, e2/0 might also be taken by unicast traffic, meaning ECMP load-sharing will occur between both Paths 1 and 2. To prevent critical unicast traffic from taking Path 1 (the multicast-

only path) EIGRP is configured to prefer Path 2 (10.1.2.x link) over Path 1 (10.1.3.x). To prevent multicast-forwarding over the unicast-only path, PIM is not configured on the Path 2 interfaces. If the network is configured in this manner, the PIM state for IP-TV application traffic has failed RPF checks and is incomplete.

How can we resolve this issue? What we need is a way to manually adjust the state table—in this case, the mroute table—to consider the PIM-enabled interface for Path 1 as a potential RPF interface. Great news! This is the exact purpose of static table entries (in this case, “mroutes”), and they are easy to understand and configure! [Figure 5-10](#) illustrates this configuration.

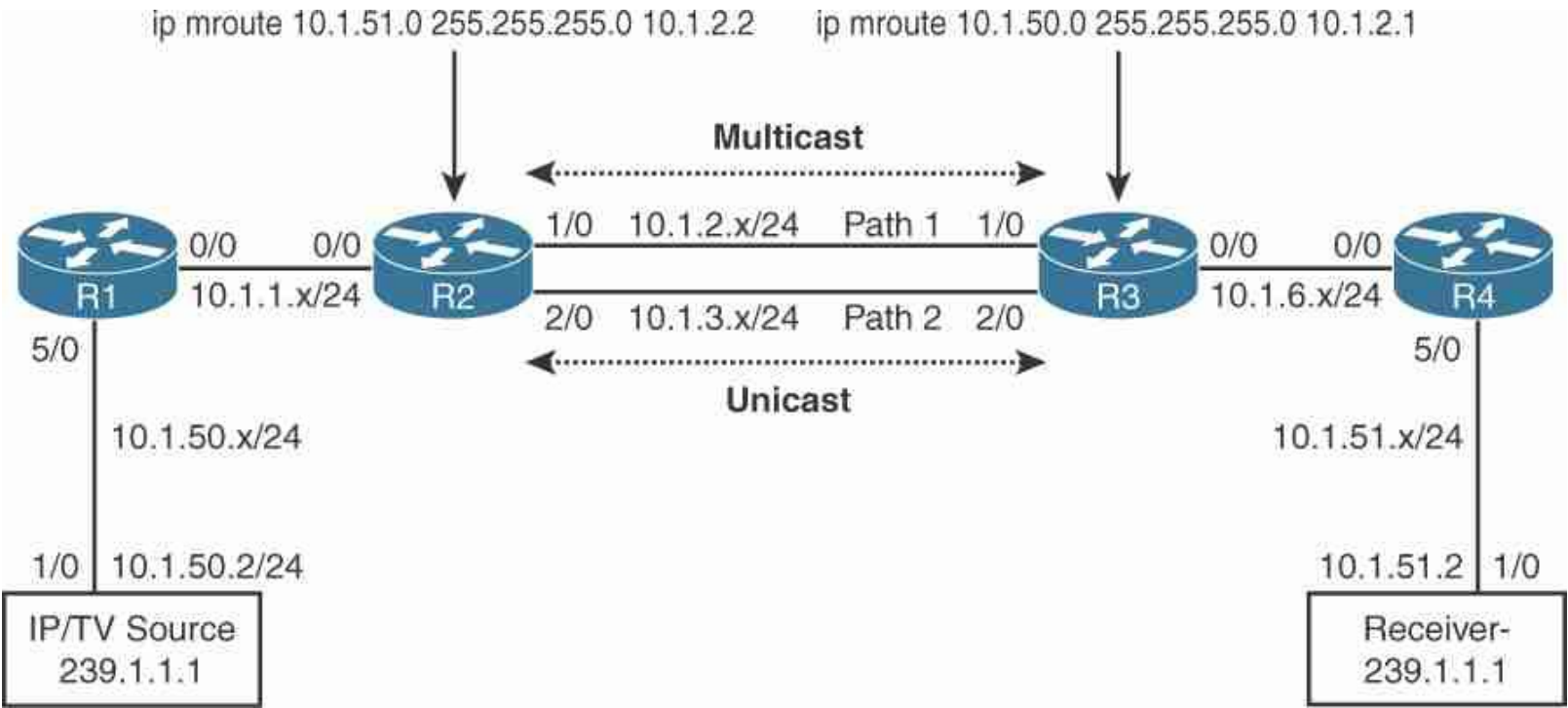


Figure 5-10 Static Multicast State Entries

To add static state entries, use the commands outlined in [Table 5-7](#) .

IOS/XE	<code>ip mroute [vrf vrf-name] source-address mask { fallback-lookup [global vrf vrf-name] [protocol] [rpf-address] interface-type interface-number } } [distance]</code>
IOS XR	<code>static-rpf prefix-address prefix-mask type path-id next-bop-address</code>
NX-OS	<code>ip mroute { ip-addr ip-mask ip-prefix } { { next-bop nb-prefix } if-type if-number } [pref] [vrf vrf-name]</code>

Table 5-7 Static mroute CLI Commands

Note

Notice the language used in the command syntax for IOS/IOS-XE. Do not let the word *mroute* confuse you. Remember, the mroute table is *not* a table of routes but rather a table of multicast state entries. Adding a static mroute does *not* add a static state entry in and of itself. What it does add is a static RPF “OK” when the PIM process checks RPF during state creation. You cannot add state to a non-PIM interface or when no source or subscribed receivers are present, where the potential for forwarding does not exist. Much like a unicast static route entry, the underlying physical and logical

infrastructure must match the configured entry to cause the state to occur within the table. Configuring a static unicast route where no underlying interface or address exists results in a failed route. The same is true for multicast state entries.

[Example 5-17](#) shows using a static mroute entry to adjust the behavior of PIM state creation to include Path 1 in the RPF calculation.

Example 5-17 *Static mroute Entry Output and Change in Forwarding Path*

[Click here to view code image](#)

```
R3# sh running-config | include ip mroute
ip mroute 10.1.50.0 255.255.255.0 10.1.2.1
r3# sh ip rpf 10.1.50.0
RPF information for ? (10.1.50.0)
RPF interface: Ethernet1/0
RPF neighbor: ? (10.1.2.1)
RPF route/mask: 10.1.50.0/24
RPF type: multicast (static)
Doing distance-preferred lookups across tables
RPF topology: ipv4 multicast base
```

Static state entries can be a useful way to perform multicast traffic engineering in a given simple network to separate multicast and unicast flows. This is especially valuable in scenarios like the above, or when asynchronous routing is desired in the network design.

When deploying multicast, it is wise to consider the underlying IP unicast network. The most preferred network design for IP multicast networks is one where the multicast and unicast forwarding paths fully align. It certainly simplifies troubleshooting and network management in a profound way. Network architects should deviate from this practice only when the requirements provide no alternatives. Static state entries are of course only one way to control deterministic forwarding in these scenarios.

In large networks with redundant links, to achieve the separation of the multicast traffic from the unicast, a dynamic way is more desirable. This is achieved using the BGP multicast address family. With BGP address families, the multicast network needs to be advertised and the next-hop prefix needs to be resolved via recursive lookup in the IGP to find the upstream RPF interface. In our example, the address 10.1.50.x (source) and 10.1.51.x (receiver) is advertised in the multicast BGP address family. [Figure 5-11](#) depicts using eBGP routing to achieve similar results to using static state entries for traffic engineering.

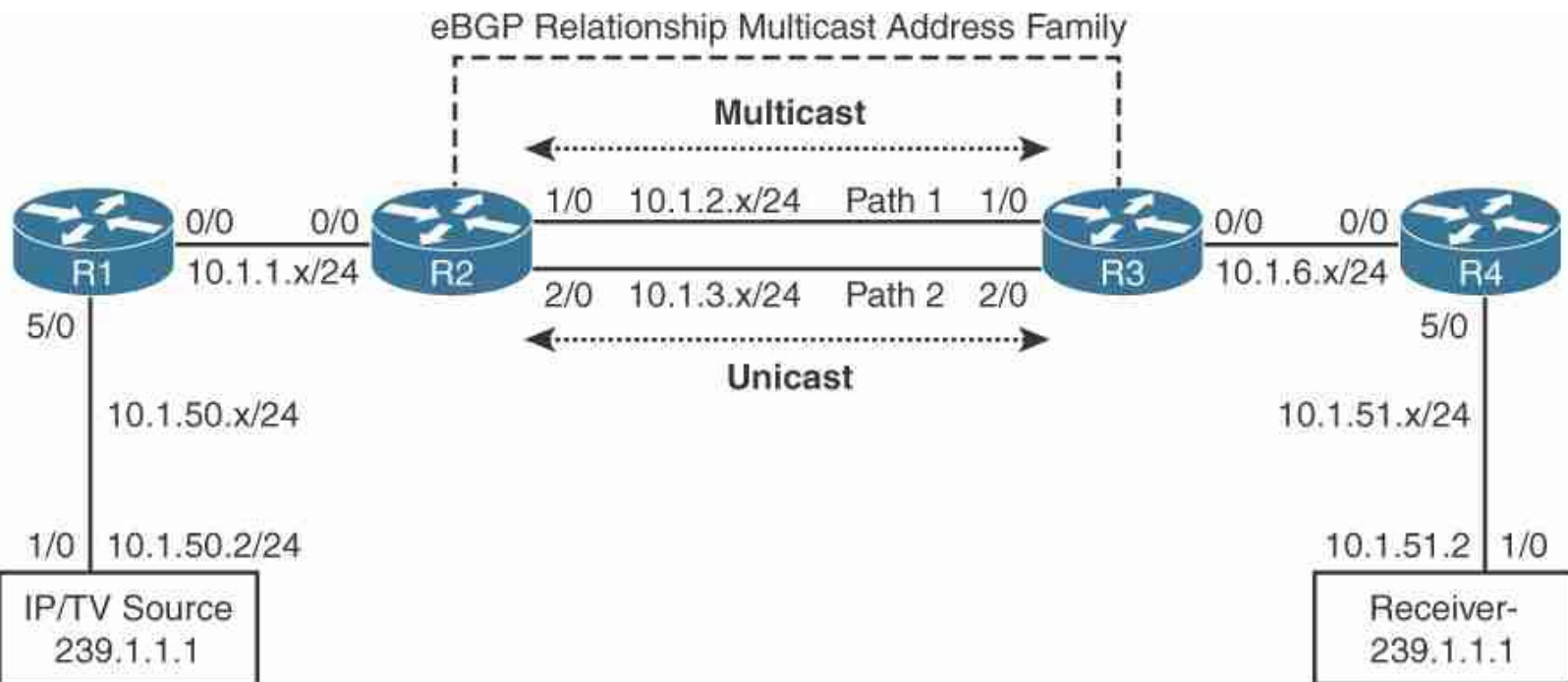


Figure 5-11 *Deterministic Multicast Pathing Using eBGP*

The network administrator would use the eBGP IOS configurations in [Example 5-18](#) on R2 and R3 to achieve path determinism.

Example 5-18 *Deterministic Multicast BGP Configuration*

[Click here to view code image](#)

R2

```
router bgp 65002
bgp log-neighbor-changes
neighbor 10.1.2.2 remote-as 65003
!
address-family ipv4
neighbor 10.1.2.2 activate
exit-address-family
!
address-family ipv4 multicast
network 10.1.50.0 mask 255.255.255.0
neighbor 10.1.2.2 activate
exit-address-family
```

R3

```
router bgp 65003
bgp log-neighbor-changes
neighbor 10.1.2.1 remote-as 65002
!
address-family ipv4
neighbor 10.1.2.1 activate
exit-address-family
!
address-family ipv4 multicast
```



```
network 10.1.51.0 mask 255.255.255.0
neighbor 10.1.2.1 activate
exit-address-family
```

Examine the changes in multicast path selection behavior using this configuration. The **show ip route** command on the same topology shows the IGP RIB with the respective multicast routes. The command captured at R3 displays the output in [Example 5-19](#) .

Example 5-19 IGP RIB

[Click here to view code image](#)

```
10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
D 10.1.1.0/24 [90/307200] via 10.1.3.1, 03:59:35, Ethernet2/0
    [90/307200] via 10.1.2.1, 03:59:35, Ethernet1/0
C 10.1.2.0/24 is directly connected, Ethernet1/0
L 10.1.2.2/32 is directly connected, Ethernet1/0
C 10.1.3.0/24 is directly connected, Ethernet2/0
L 10.1.3.2/32 is directly connected, Ethernet2/0
C 10.1.4.0/24 is directly connected, Ethernet3/0
L 10.1.4.2/32 is directly connected, Ethernet3/0
C 10.1.6.0/24 is directly connected, Ethernet0/0
L 10.1.6.1/32 is directly connected, Ethernet0/0
D 10.1.50.0/24 [90/332800] via 10.1.3.1, 03:59:35, Ethernet2/0
    [90/332800] via 10.1.2.1, 03:59:35, Ethernet1/0
D 10.1.51.0/24 [90/307200] via 10.1.6.2, 03:59:35, Ethernet0/0
C 10.3.3.3/32 is directly connected, Loopback0
r3#
```

However, the BGP multicast address family takes precedence on the MRIB selection, as demonstrated in [Example 5-20](#) .

Example 5-20 BGP RIB

[Click here to view code image](#)

```
R3# show ip bgp ipv4 multicast
BGP table version is 3, local router ID is 10.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i- internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i-- IGP, e-- EGP, ?-- incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop        Metric LocPrf Weight Path
*> 10.1.50.0/24    10.1.2.1         307200         0 65002 i
*> 10.1.51.0/24    10.1.6.2         307200        32768 i
R3#
```

The RPF for 10.1.50.1 from R3 shows Path 1 as preferred based on the BGP multicast topology table, as demonstrated in the output in [Example 5-21](#) .

Example 5-21 Deterministic Multicast BGP RPF Result

[Click here to view code image](#)

```
R3# show ip rpf 10.1.50.0
```

```
RPF information for ? (10.1.50.0)
```

```
RPF interface: Ethernet1/0
```

```
RPF neighbor: ? (10.1.2.1)
```

```
RPF route/mask: 10.1.50.0/24
```

```
RPF type: multicast (bgp 65003)
```

```
Doing distance-preferred lookups across tables
```

```
RPF topology: ipv4 multicast base
```

This method of creating RPF entries for the multicast state machine is a more dynamic than using static mroute entries, as shown in the previous examples.

In some enterprise networks, customers must transport multicast across provider or non-enterprise controlled segments. In order to route multicast across these provider links, the service provider network needs to support the enterprise implementation of PIM and to become a natural part of the PIM domain. Some providers may not support direct multicast interaction like this on certain types of links, such as, for example, MPLS WAN services, or they may not support the type of PIM mode deployed by the enterprise.

In the next example, we use the same deterministic routing scenario from above, but add a non-enterprise controlled network segment that does not support multicast. As shown in [Figure 5-12](#), Segment A (encompassing both Path 1 and Path 2) is the non-enterprise controlled segment that needs multicast support. In this example, the provider does not support multicast transport, leaving Segment A configured with PIM disabled. This would obviously cause RPF failures, spawning incomplete state entries in the mroute tables of all routers. [Figure 5-12](#) also shows that an easy solution exists for this type of problem: the use of a GRE tunnel to forward multicast packets.

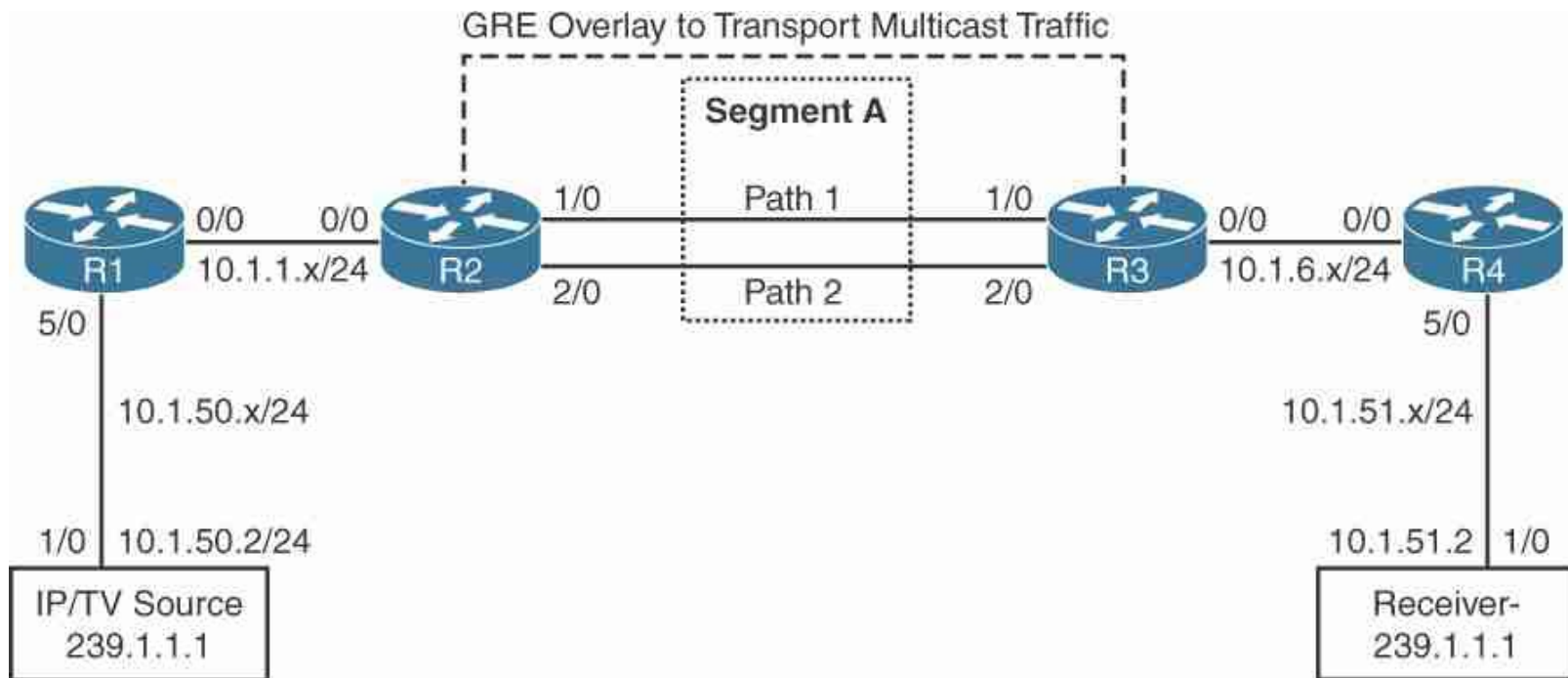


Figure 5-12 *Tunneling Multicast over PIM Disabled Paths*

Under such a scenario, the GRE tunnel must establish full IP connectivity between routers R2 and R3. The GRE tunnel interfaces should be configured for PIM, and a PIM neighborship should exist across the tunnel. It would not be prudent to configure unicast routing over the newly formed tunnel,

but the transport of MRIB and multicast RPF check still needs to be moved to the overlay segment. Without the existence of unicast routing, the tunnel interface will fail RPF checking.

In this situation, you have a choice to use static state entries or dynamic BGP multicast address families to enable multicast transport across Segment A. The principles of MRIB build-up will be the same and will follow the same rules. The GRE tunnel interface must become the RPF interface.

IP Multicast Best Practices and Security

Every implementation of multicast is unique, in part because every IP unicast underlay is unique. Multicast networks are often unique in and of themselves, which may place special constraints on network design. In spite of this uniqueness, certain elements should exist in every multicast network design.

Following current best practices for network architectures is paramount. Some of these items include hierarchical design, redundancy, resiliency, high-availability, limiting Layer 2 scope, security, and so on. Building a strong foundation from which to add services only enhances your ability to manage, operate, and troubleshoot your network infrastructure. When adding multicast to the network, the following sections are elements to strongly consider.

Before Enabling PIM

Many network engineers make the mistake of simply turning on multicast in complex network topologies with the expectation that it will instantly function in an ideal manner. Remember, if it was that easy, we would be out of work.

There are several items that must be considered before configuring a network for IP multicast:

- CEF/dCEF/MLS CEF considerations for those platforms that require it. Without CEF on these platforms, multicast packets are process-switched which can overwhelm the central processing unit (CPU) of the networking device (very bad).
- Unicast routing must be enabled and operational on the network. Remember that PIM is an overlay to a successful L3 unicast design. Carefully consider any redundant paths in the network, looking for possible asynchronous routing that could cause RPF check failures.
- Consider the multicast applications being placed on the network. Network architects should select the most ideal multicast features and configurations for these applications.
- Remember that groups in the 224.0.0.* range are reserved for routing control packets. A proper schema should be a design requirement. When creating your schema, do not forget to account for (and, if necessary, eliminate) MAC overlapping group ranges, as described in [Chapter 2](#) !
- The administrator should be familiar with IPv4 and IPv6 multicast routing configuration tasks and concepts.
- Administrators should be aware of the various multicast configurations and features for a given platform. Not all platforms support all features or modes. Make sure you do not select a PIM mode (such as, for example, dense-mode or PIM-Bidir) if it is not supported universally across the intended PIM domain. This chapter establishes the following protocol selection guidelines:
 - Dense-mode is not recommended except in legacy environments where it may already exist. It is likely that DM is not supported by your current platform.
 - In general, if the application is one-to-many or many-to-many in nature, then PIM-SM can be used successfully.

- For optimal one-to-many application performance, SSM is appropriate, but it requires IGMP version 3 client support.
- For optimal many-to-many application performance, bidirectional PIM is appropriate, but hardware support is limited to certain Cisco devices

[Table 5-8](#) provides an example of multicast applications and the relationships between sources and receivers.

One to Many	Many to One	Many to Many
Television	Data collection	Audio and video communication
Radio	Service discovery	Document sharing
Distance learning	Polling	Data distribution, caching, and synchronizing
Presentation sharing		Group chat applications
Computer imaging		Polling applications
Data distribution		Multiplayer games

Table 5-8 *Application Examples*

- You should have a proper PIM design for each desired protocol version, with an understanding of which protocol you will run and why, before moving to implementation.

In addition, each platform and each operating system has specific tasks and configuration parameters required to enable IP multicast functionality. For more detailed information, please refer to the individual configuration guides for each operating system found at Cisco.com. This book uses examples from the latest versions of each operating system at the time of writing. Remember to review current configuration guides for changes.

General Best Practices

Multicast can be your best friend or your worst enemy. As the manual for the dangerous equipment hidden away in your garage suggests, “be sure to read, understand, and follow all the safety procedures.”

Tuning the Network for Multicast

Most of the control-plane stress in a multicast network will be at the access edge, as well as at any RP routers. This occurs because receivers and sources are located on the edge of the network. A routed network may have only a few branches, but the edge devices must efficiently replicate packets for many potential interfaces, in addition to managing the IGMP subscription process. It is best to maximize efficiency at the edge for this reason, especially if the expected multicast usage is high with multiple types of many-to-many or one-to-many applications.

Architects should start by ensuring that the Layer 2 forwarding domain is fast, efficient, and loop-free. Multicast can substantially increase Layer 2 flooding. Ultimately you should strive for a design that limits flooding domain size by controlling VLAN sprawl and using Spanning Tree Protocol (STP) wisely. Limiting VLAN sprawl and excessive use of VLANs on access switches eliminates massive packet flooding across a number of switches. Unnecessary VLANs should be effectively pruned from

switch trunks. Manual configuration of interfaces and the use of virtual trunking protocol (VTP) transparent mode should be considered to enforce this policy. In addition, storm control can help alleviate potential configuration issues with multicast sources, protecting switches from inappropriate multicast and broadcast flooding.

IGMP snooping is also an excellent way to limit flooding behavior at the Layer 2 edge. Remember, without IGMP snooping, flooding of multicast packets will occur VLAN- or switch-wide, depending on configuration. If a VLAN spans many switches, the results of excessive flooding can take a toll on switch-forwarding resources. Remember that IGMP snooping limits replication and flooding to only those ports with subscribed hosts.

Note

If you have a network in which multicast is only local to a given Layer 2 domain (there is no multicast-enabled L3 gateway and no PIM), IGMP snooping is still your friend. However, remember that IGMP requires that an IGMP querier be elected for each VLAN with IGMP subscriptions. If there is no Layer 3 device, and no PIM configuration, switches by default assume there is no gateway and will not elect a querier as part of the natural process. To resolve this issue, a network administrator should either configure one device in the switch domain with PIM, or manually configure one switch as the IGMP querier.

Network architects should consider designs that make use of new technologies like the Cisco Virtual Switching System (VSS) to eliminate STP-blocked ports in the forwarding path. This helps optimize failure convergence times as well as packet flow through the edge, maximizing available bandwidth and predictability. If such a design is not possible, STP should be tuned to improve STP convergence. Rapid STP should be a minimum requirement for the multicast network edge.

Because multicast is an overlay on the unicast topology, multicast traffic will not work if there is an IP unicast network outage. If multicast communications are mission critical, or at the very least are important to the business, the same care and effort put into the unicast network design should be put into the multicast overlay design. It is also wise to tune and adjust the IP unicast network to maximize IP multicast traffic throughput and to minimize network disruption.

IP unicast interior gateway protocols (IGPs) used for routing (RIP, EIGRP, OSPF, and so on) should be both secured and optimized in the network. Exterior gateway protocols (BGP) should also be secured and optimized, especially if they are a part of the multicast domain control plane as described earlier. When possible, routing protocols should be locked down and adjacencies should be verified using ACLs and MD5 encryption when possible. This prevents intruders from injecting attack-based routing information into the network, possibly disrupting the flow of multicast traffic.

Protocol timers should be tuned in favor of fast convergence. For example, EIGRP can use lowered hello and dead timers to increase failure detection and recovery speeds. OSPF timer adjustments may also be warranted, including optimizing SPF, hello, dead-interval, and LSA timers. BGP perhaps allows for the most flexibility for adjusted and optimized timers. Be sure to understand fully the implications of any protocol tuning before you proceed with configuration, and ensure that any timer changes are implemented universally. Mismatched timers can cause major protocol adjacency flaps for some protocols. The bottom line: the faster the unicast network converges on changes and disruptions, the fewer interruptions there will be to IP multicast traffic.

Remember also, multicast convergence is not aligned to the convergence results you get with unicast. If you tweak unicast convergence to one second, then multicast convergence will be a factor

of five to six times unicast convergence. Tweaking of the multicast control plane via sub-second PIM and RPF back-off feature can reduce this gap in convergence between two and three times compared to the standard. If you choose to make multicast timing adjustments a requirement, then multicast tweaks should be assessed with the stability of the control plane as the main goal, keeping in mind the total number of states. When these timers are changed on any router in the network, *all* PIM routers in the network should be configured with matching timers.

Finally, network architects should ensure that the IP unicast network is robust, reliable, and simple. Architects should look for any situations that may affect multicast-forwarding. Look for tunnel interfaces, or asynchronous routing that may negatively affect RPF checks. Be aware of other network protocols like MPLS or VPNs (for example, GetVPN or DMVPN). Special considerations may be required for certain technologies or topologies when it comes to multicast. Always check with the latest design guide for those technologies implemented on the network.

Manually Selecting Designated Routers

On any segments with multiple PIM speakers, PIM software will select a PIM designated router (DR). Remember, the role of the DR is to forward multicast data for any groups attached to the segment. That means that it serves as the segment multicast-forwarder, as well as the control point for communication with any RPs for each group. The DR is essentially the PIM manager for that segment.

For an ASM network, this means the DR sends PIM join/prune messages to any RPs for any group subscriptions on the segment. The ASM DR will look up the corresponding RP mapping for each group and begin the control process. This also includes sending unicast-encapsulated multicast messages to the RP from any source on the segment, registering the source and completing the shared tree.

When the DR receives a direct IGMP membership report from a directly connected receiver, it is easy to make a complete shortest-path tree because the DR is obviously in the forwarding path. However, there is no rule that the PIM DR must be in the shortest-path tree. Examine the PIM network in [Figure 5-13](#) .

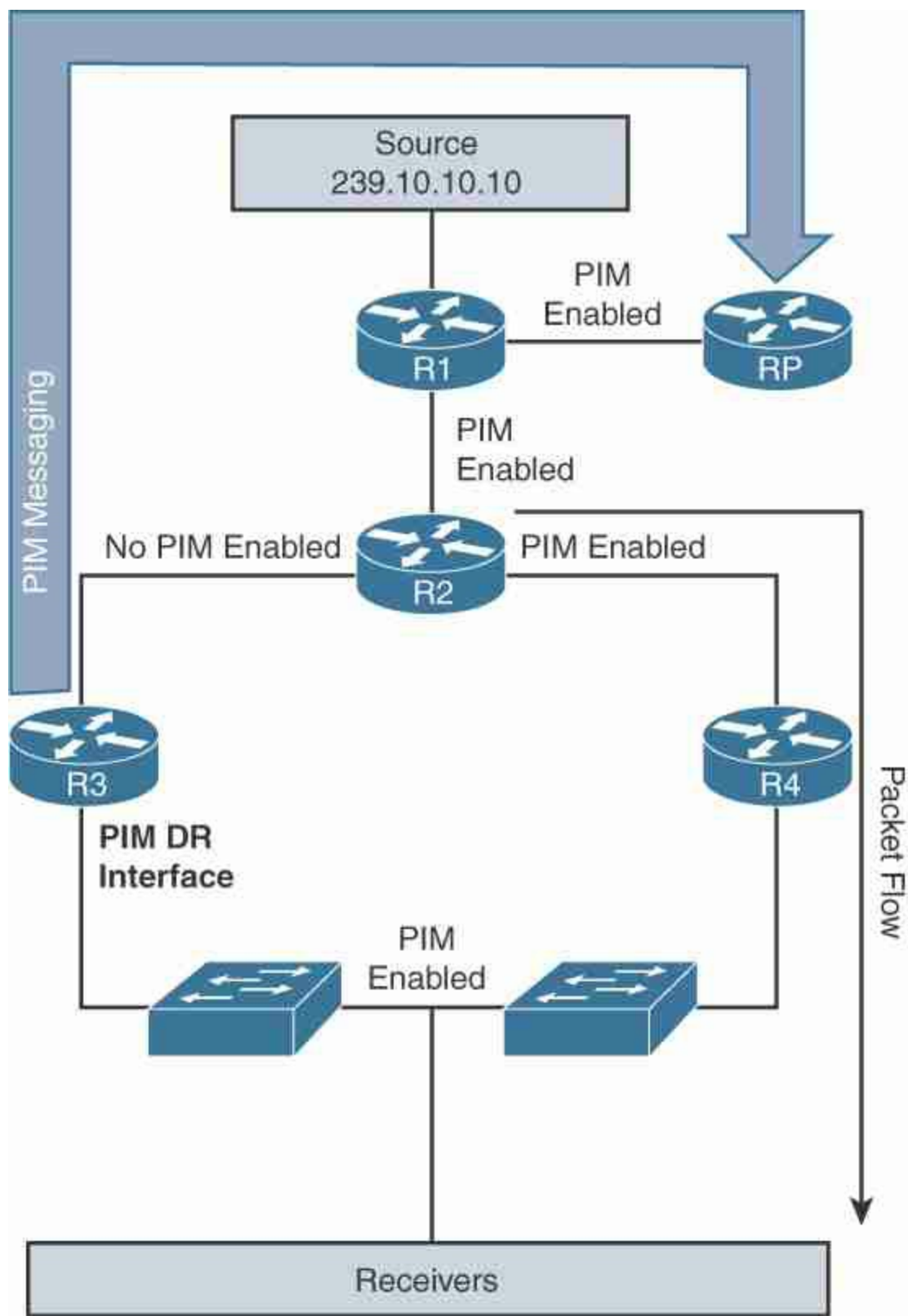


Figure 5-13 *Out-of-Path Designated Router*

In this network, routers R3 and R4 provide redundant paths for the unicast network. The source, 239.10.10.10, however, is reachable only via the primary unicast path running between R4 and R2. If the PIM process has elected R3 as the designated router for the LAN segment connecting R3 and R4, the DR for the segment would not be in the forwarding path. Although this design would still work, it would also be inefficient. Why not make R4, the in-path next-hop router, the PIM-DR? It would certainly improve efficiency, especially if there are a large number of hosts on the LAN segment and many groups to manage.

You should also consider the impact of making all redundant paths PIM-enabled in this example

network. Look at the adjustments made to the network in [Figure 5-14](#) . In this case, routers R3 and R4 are now redundant routers using a mechanism like Hot Standby Router Protocol (HSRP) to load-balance unicast traffic and all upstream paths are PIM-enabled. Like with HSRP, the administrator would also like to load-balance the PIM management between the two hosts. If there are many multicast-enabled VLAN segments terminating on these routers, you can achieve similar results by alternating the DR between R3 and R4 for each VLAN (even and odd), as shown. The router providing the DR should align with the active HSRP peer for that VLAN. This helps align traffic between unicast and multicast flows using the same gateway router, while also providing failover for multicast flows. This is configured using the DR **priority** interface command option, as explained in [Chapter 3](#) .

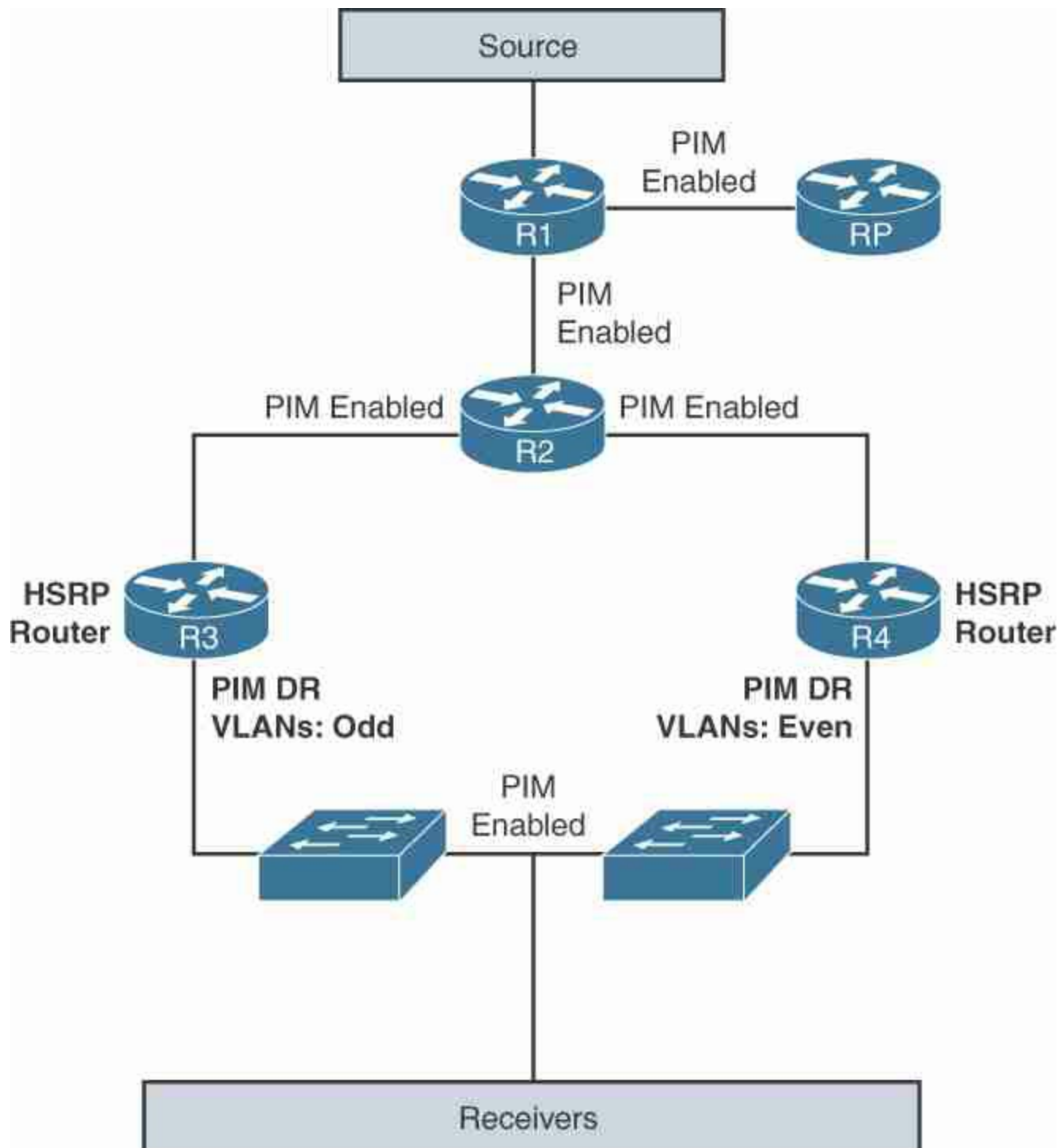


Figure 5-14 *Load Balancing with Designated Routers*

Note

In many modern Cisco networks, the concept of discrete paths and gateways is fading in part because of technologies like Cisco's Virtual Switching System (VSS) and virtual PortChannel (vPC) that allow a pair of L2/L3 switches to appear as a single switch and gateway. This eliminates the need to specify the DR function in designs like the one above. Consider that in a VSS design, for example, the two switches/routers in [Figure 5-14](#) , R3 and R4, could actually be a single pair of Layer 3 switches acting as a single gateway. This is the preferred way to design LAN access for multicast if the technology is available.

For PIM-SSM networks, inefficient DR placement in the path can be more problematic. The SSM-DR generates (S, G) PIM join messages that propagate through the path back toward the source. The path from the receiver to the source is determined hop by hop, and the source must be known and reachable by the receiver or the DR. If the DR is not in the direct path toward either the source or the receiver, unintended consequences can occur.

In either case, for large networks, manually forcing the outcome of DR elections to optimize network behavior is sometimes best. This is especially true at the network edge where sources and receivers are connected. Properly selecting the DR can improve both control plane and data-plane efficiency.

As discussed previously, PIM routers use information contained in the PIM hello message headers to determine the DR. Any PIM-speaking router on the segment can become the DR, assuming it meets the selection criteria. The rules of PIM-DR selection force the router with the highest priority to become the DR. If the DR priority is the same, the router with the highest IP address in the segment is elected DR. PIM-DR priority values range from 1 to 4,294,967,295, with the default being 1. If no priority is selected, the IP address of the PIM router is used, the highest address becoming the DR. Remember, the PIM address is derived from the interface that sent the hello message.

To change the PIM-DR priority on IOS or NX-OS interfaces, use the **ip pim dr-priority <0-4294967294 >** command. The same function in IOS-XR is done under the **router pim** submode using the command **dr-priority <0-4294967294 >**; this can be applied as an interface default or per interface under the submode.

You can display the configured DR priority and DR-elected router address for each interface by issuing the command **show ip pim interfaces** in IOS/XE or **show pim interface** in IOX-XR. For NX-OS, use the **show ip pim neighbors** command instead. The output in [Example 5-22](#) is from an IOS router, notice the **DR Prior** field and the **DR** address field in the output:

Example 5-22 show ip pim interface Command Output

[Click here to view code image](#)

CR1#show ip pim interface

Address	Interface	Ver/	Nbr	Query	DR	DR
	Mode	Count	Intvl	Prior		
192.168.63.3	Ethernet0/0	v2/D	1	30	1	192.168.63.6
192.168.43.3	Ethernet0/1	v2/S	1	30	1	192.168.43.4
192.168.31.3	Ethernet0/2	v2/S	1	30	1	192.168.31.3
192.168.8.1	Ethernet0/3	v2/S	0	30	1	192.168.8.1

Basic Multicast Security

Security is an important part of any network or application design. In years past, many engineers

considered security a set of point features that were an afterthought to the design of the network, much like an overlay. The technology industry in general has learned that this is both an ineffective and dangerous approach to securing networks and applications. Today's networks must be designed from the ground up with intrinsic security as a main essential objective.

Attacks to multicast networks can come in many forms. Because multicast is an overlay to an existing, functional unicast network, the same attack vectors and weaknesses that affect a unicast network impact multicast-forwarding. If the unicast network is not secure, the multicast network is equally vulnerable. In addition, IP multicast inherently increases the surface area of possible attack vectors.

Abstractly speaking, the key factors to security design are integrity, confidentiality, and availability. When it comes to IP multicast end points, any security mechanism that enables these factors to protect unicast applications should also be applied to multicast applications. For example, an enterprise security policy may require that encryption with authentication be enabled on any mission critical, secure applications. Because multicast packets are essentially just IP packets, this type of security policy should be applied equally to multicast applications. Another example is protecting multicast routers, switches, and other infrastructure from unauthorized access.

Generally, the objective of most network-concentrated security policies is to protect network availability for applications and users. Attack vectors and weaknesses used to induce availability security incidents are either focused on network resources (such as, for example, bandwidth or queue space), or the network devices themselves. Examples of these types of attacks and vulnerabilities include denial-of-service (DoS) attacks, unauthorized access of network devices, protocol spoofing, packet interception, and others. This section does not address the myriad ways in which a network or multicast domain can be compromised, but instead it attempts to deal mainly with protecting the availability of the multicast network.

Protecting Multicast Control-plane and Data-plane Resources

A network infrastructure or device that is overwhelmed with a specific task will not be able to adequately address the requests of other processes. As we require the network to perform additional functionality or run more processes, we begin to spread those resources (CPU, memory, TCAM, and so on) across multiple functions. Care must be taken to limit how those resources are utilized. Control-plane policing (CoPP) is beyond the scope of this book; however, it is prudent to understand how this technology can be utilized to protect network devices and ultimately the integrity of the entire network infrastructure.

One common way to protect multicast control-plane resources is to limit the number of state entries that a multicast router allows in the MRIB. This protects the router from misconfigured network consumption as well as potential denial-of-service attacks. It also protects the underlying IP unicast network control-plane resources by preventing the CPU and memory from being overwhelmed by multicast route churn. This is known as a *state maximum* or a *route-limit*. See [Table 5-9](#) for command details.

Operating System	Command
IOS/XE	<code>ip multicast vrf <i>vrf-name</i> route-limit <i>limit</i> [<i>threshold</i>]</code>
IOS XR	<code>global maximum register states route-interfaces routes <i>number</i> OR maximum routes <i>number</i></code>
NX-OS	<code>ip pim state-limit <i>max-states</i> [reserved <i>policy-name</i> <i>max-reserved</i>]</code>

Table 5-9 *Limiting Multicast State Entries*

These commands ultimately limit the total number of (*,G) and (S,G) entries that can be installed in a router. Valid limits are dependent on platform, but for a typical IOS router they are between 1 and 2,147,483,646. The default value is to allow the maximum. When the router reaches the configured route limit and a new PIM join is received with a new potential group, the router issues a warning message and fails to install state for that entry. No existing entries are removed.

You can also apply a similar limit to gateway routers running IGMP by limiting the number of IGMP-managed group entries in the state table. The **ip igmp limit** , or **maximum groups** command prevents any installation of additional group entries into the IGMP cache after the limit is reached. This also prevents the router from issuing any PIM messages for any groups exceeding the limit. [Table 5-10](#) shows the command usage.

Operating System	Command
IOS/XE	<code>ip igmp limit <i>number</i> [except <i>access-list</i>]</code>
IOS XR	<code>maximum groups <i>number</i> OR maximum groups-per-interface <i>number</i></code>
NX-OS	<code>ip igmp state-limit <i>max-states</i> [reserved <i>reserve-policy</i> <i>max-reserved</i>]</code>

Table 5-10 *Limiting IGMP Subscription Entries*

These commands can be applied either globally or at the interface level. IOS XR uses a different command for each command locality. In addition, the IOS version of this command allows administrators to create a list of exceptions using an ACL. An explicitly excepted group will not be limited, regardless of cache size. Nexus platforms can also make exceptions, but they do so by policy maps as opposed to ACLs.

Another way to allow IGMP to prevent unwanted state creation and PIM messaging is to create an explicit list of allowed and denied groups. This is a common security requirement in many networks. [Table 5-11](#) provides the commands necessary to limit group management by ACL.

Operating System	Command
IOS/XE	<code>ip igmp access-group <i>policy-name</i></code>
IOS XR	<code>access-group <i>access-list</i></code>
NX-OS	<code>ip igmp access-group <i>policy-name</i></code>

Table 5-11 *Use ACLs to Permit Multicast Subscriptions*

Finally, protecting data-plane resources is also an important part of basic multicast security. One way this can be achieved is by simply rate-limiting the number of allowed multicast packets on a given interface. In IOS, this is done using the **ip multicast rate-limit** command, as follows.

[Click here to view code image](#)

```
ip multicast rate-limit {in | out} [video | whiteboard] [group-list access-  
list] [source-list access-list] kbps
```

To achieve rate-limiting for multicast in XR and NX-OS, a simple rate-limit can be used under the normal modular quality-of-service (QoS) configuration CLI. IOS devices can also apply multicast traffic limits in the same manner. For more information on how to use modular QoS to rate-limit multicast traffic, see the individual QoS configuration guides for each platform. The multicast-rate-limit should be taken care by the QoS design and its parameters; it is not a must to have multicast rate-limiting unless there is a specific requirement for application usage.

Securing Multicast Domains with Boundaries and Borders

Just as with unicast networks, multicast boundaries should exist where the security policy and the needs of the application dictate. Remember the earlier discussion around scoping. Scoping addresses and bounding a domain within the construct of the IGP are important for securing the domain. You do not want your multicast packets leaking outside the logical domain scope. A well-planned addressing schema, wise RP placement, and a clear understanding of natural network boundaries make scoping a domain significantly easier.

In many cases, a boundary can occur between two domains overlaid on the same IGP. These would not necessarily be natural boundaries and should be enforced with policy. In other cases, like in large network-wide domains, it means that a natural boundary will occur at the IGP edge of the network. For example, the scope of an internal multicast application that is corporate-wide will end at any Internet-facing interfaces or at the edge of the autonomous system (AS).

The network administrator can make an AS boundary simply by not configuring PIM on any external interfaces. If PIM is not required outside the AS, there is no need to have those interfaces included in the PIM domain. This creates a natural PIM control-plane boundary between the routers within and without the PIM domain. [Figure 5-15](#) depicts this natural control-plane edge with the Internet, as well as a configured boundary for the global multicast domain.

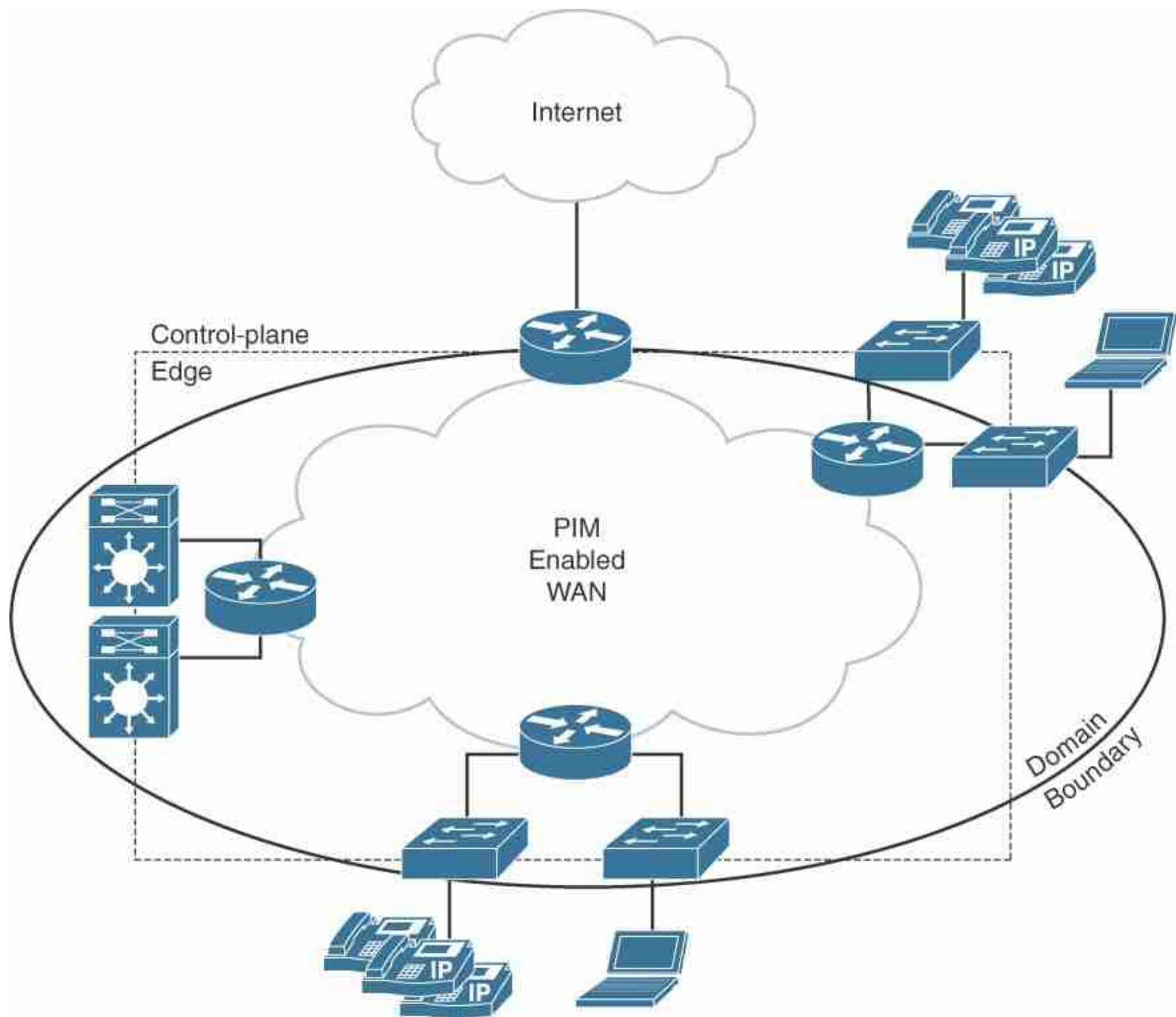


Figure 5-15 *Multicast Boundaries*

The definition of a domain that we have used thus far is fairly loose and, in the case of [Figure 5-15](#), follows the natural control-plane edge. Some applications and environments may need a more restrictive way to define boundaries. In addition, for locally scoped applications, those defined by administratively scoped group addresses, it may be necessary to create boundaries and scope outside those services offered by routers. Firewalls and router access lists (ACLs) can (and in most cases should) use rules to prevent the spread of multicast information beyond a specific boundary. In the same way, multicast hosts can also be configured for similar security measures if it is warranted.

Application developers can also be part of the bounding process. One way the application can be bound is by effectively using the time-to-live (TTL) counter in multicast source packets to limit the scope of transmission. Scope, in a network, is the number of times (usually by a router) a packet can be forwarded, known as hops. IP multicast packets have IP headers identical to those used in unicast, and the normal rules of transmission apply. Each router in the path inspects the TTL counter of the packet. If the router forwards the packet the TTL counter is decremented by one. If the application

sets the multicast IP TTL to 4, then the region is scoped to four hops. This is a good way to ensure that local data stays local.

Perhaps the best way to segment a domain or region is to configure hard network boundaries. There are different ways of achieving this. One easy way is to prevent the spread of dynamic multicast group mappings. All Cisco router operating systems provide a way to limit the scope of both the Auto-RP and BSR updates. If routers outside the configured scope do not have the RP mappings, they will not be able to participate in tree-building within that domain. There are two ways to limit the scope, depending on the protocol announcement method in use.

The first should be fairly obvious, using the **scope** option in the Auto-RP commands, as shown here:

[Click here to view code image](#)

```
ip pim send-rp-announce Loopback0 scope 3
ip pim send-rp-discovery Loopback0 scope 3
```

The **scope** option sets a TTL in the RP-Candidate and RP-Announce messages. If the domain has only four routers, then setting a scope to three should be sufficient to prevent the proliferation of the group mappings for that domain. Using the following IOS/XE commands, for example, accomplishes this boundary requirement. The ability to control the multicast RP control plane via TTL is one of the advantages of using Auto-RP in a scoped domain.

There are, of course, more granular methods of controlling Auto-RP announcements as well. But this method also applies to creating a secure border for any multicast group address or schema. One easy way to provide a boundary around a domain and Auto-RP announcements is to use the **boundary** interface-level configuration command, as shown in [Table 5-12](#) .

Operating System	Command
IOS/XE	ip multicast boundary <i>access-list</i> [filter-autorp]
IOS XR	boundary <i>access-list</i>
NX-OS	ip multicast boundary <i>access-list</i> [filter-autorp]

Table 5-12 *Configuring Multicast Boundaries*

To see how this works in practice, review the setup in diagram [Figure 5-16](#) . This simple multicast implementation uses an Auto-RP configuration and two multicast domains: global scope 239.1.1.0/24 and local scope 239.192.0.0/16. The diagram only shows two RPs each for global and local; the RP HA portion of the setup is not shown.

Multicast Boundary Configuration

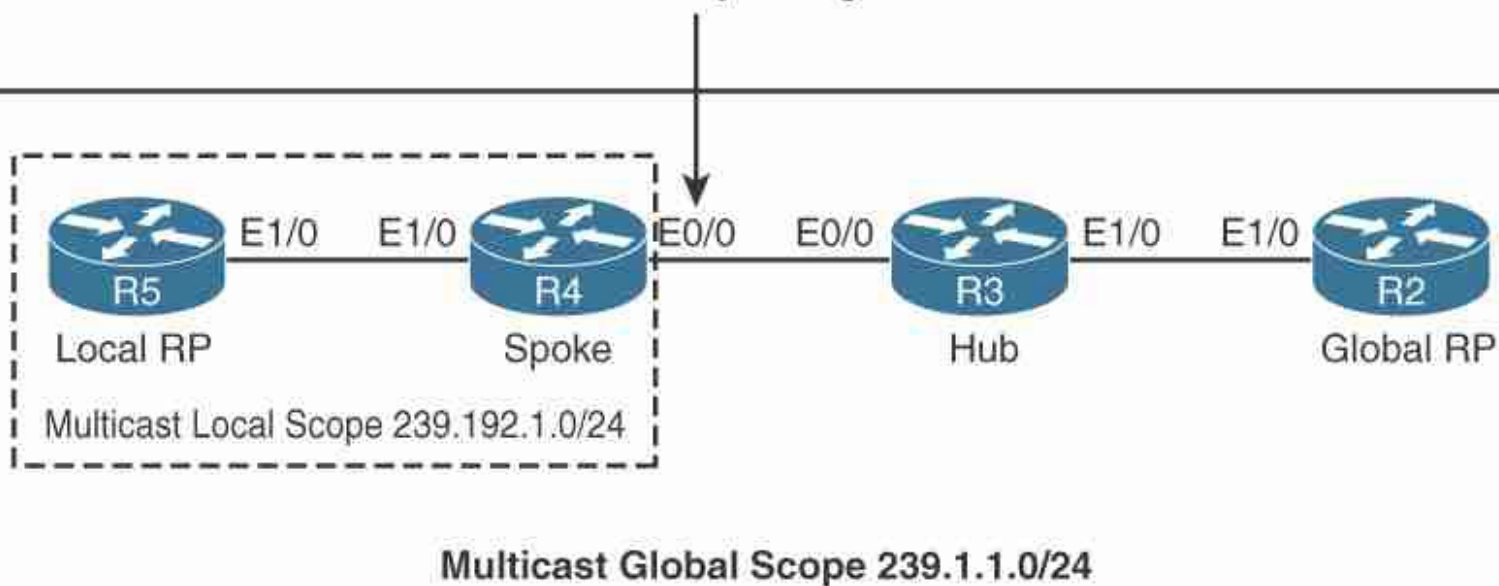


Figure 5-16 *Multicast Boundary Example*

We can now examine the configuration required for this setup. R4 router is the spoke WAN router; we will add a multicast boundary configuration on this router. [Example 5-23](#) shows the commands needed to enable the boundary configuration.

Example 5-23 *Configuring a Proper Multicast Boundary List*

[Click here to view code image](#)

```
ip access-list standard LSCOPE
```

```
deny 224.0.1.39
```

```
deny 224.0.1.40
```

```
deny 239.192.0.0 0.0.255.255
```

```
permit any
```

The ACL **LSCOPE** denies Auto-RP advertisement leakage outside the domain (which will end at interface Ethernet0/0) by denying groups 224.0.1.39 and .40. The data plane for the local groups in the 239.192.0.0/16 supernet is also contained within the local scope by using a **deny** statement. All other groups are allowed in the access-list with the **permit any** statement at the end.

The next step is to apply the access-list to the boundary interface (Ethernet0/0), using the **ip multicast boundary** command with the ACL name, as shown in the output in [Example 5-24](#).

Example 5-24 *Applying a Multicast Boundary List*

[Click here to view code image](#)

```
r4-spoke#show running-config interface ethernet 0/0
```

```
Building configuration...
```

```
Current configuration: 118 bytes
```

```
!
```

```
interface Ethernet0/0
```

```
ip address 10.1.3.2 255.255.255.0
```

```
ip pim sparse-mode
```

```
ip multicast boundary LSCOPE out
```

end

Note

Make special note of the **out** keyword used in [Example 5-24](#) . If the **out** keyword is not applied, the router assumes that it is an implicit deny and will not allow important group mappings to occur. In this example, it would not allow global group mappings to occur in the local scope.

Issuing the **show ip pim rp mapping** command on R3-WAN shows the global multicast group block 239.1.1.0/24 and global RP mappings, as demonstrated in [Example 5-25](#) .

Example 5-25 *Boundary Group Mapping*

[Click here to view code image](#)

```
r3-WAN# show ip pim rp mapping
PIM Group-to-RP Mappings
  Group(s) 239.1.1.0/24
  RP 192.168.2.2 (?), v2v1
  Info source: 192.168.2.2 (?), elected via Auto-RP
  Uptime: 07:39:13, expires: 00:00:21
```

r3-WAN#

The 'show ip pim rp mapping' at local domain node (R5) is:

```
r5-LRP# show ip pim rp map
PIM Group-to-RP Mappings
This system is an RP (Auto-RP)
This system is an RP-mapping agent (Loopback0)
  Group(s) 239.1.1.0/24
  RP 192.168.2.2 (?), v2v1
  Info source: 192.168.2.2 (?), elected via Auto-RP
  Uptime: 07:22:34, expires: 00:00:26
Group(s) 239.192.1.0/24
  RP 192.168.5.5 (?), v2v1
  Info source: 192.168.5.5 (?), elected via Auto-RP
  Uptime: 07:22:34, expires: 00:00:25
```

r5-LRP#

The local RP will have overlapping of two multicast domains and two RPs (global and local RP). BSR uses a less centralized but more sophisticated method of limiting the scope of BSR advertisements, as shown in [Table 5-13](#) .

Operating System	Command
IOS/XE	ip pim bsr border
IOS XR	bsr border
NX-OS	ip pim bsr border

Table 5-13 *PIM Border Configuration Commands*

The **bsr border** command is issued globally in NX-OS, on the interface in IOS/XE and under the

router **pim** interface subcommand mode in IOS-XR. The command creates a BSR border (boundary), preventing BSR advertisements from being forwarded on any participating interfaces. Consequently, if the scope of a region is still four routers, each router with an external-facing interface would need to be configured with this command. Although this requires additional configuration work, the border becomes more flexible and granular, addressing a major weakness in the Auto-RP **scope** option. (The scope should be equal to the network diameter—the number of hops—but the placement of the RP may not be ideal in relation to the required hop count.)

Note

NX-OS also provides the option of applying the **bsr border** command globally and then allowing specific interfaces to forward using the **ip pim bsr forward** interface configuration command.

The primary intent of this section is to illustrate that multicast domains are a little different and more fluid by definition than their unicast counterparts. Where and how you divide domains is a critical design decision that has a myriad of consequences. It affects the scope and topology of an application, the size of the MRIB and MFIB tables, the security of the applications, the need for specialized configuration, and the types of devices that will access the application.

There are other ways to control dynamic RP-to-group mapping. The **ip pim accept-rp** command is used in conjunction with the RP address as an optional ACL used to limit dynamic RP learning to specific group addresses. If the RP is also the first hop designated router (DR) for directly connected sources, PIM register packets will not be filtered using the **ip pim accept-register** command. [Table 5-14](#) shows the command syntax for the **accept-rp** configuration option.

Operating System	Command
IOS/XE	<code>ip pim [vrf vrf-name] accept-rp { rp-address auto-rp } [access-list]</code>
IOS XR	Not Available in IOS-XR
NX-OS	<code>ip pim bsr bsr-policy policy-name</code> or <code>ip pim auto-rp rp-candidate-policy policy-name</code>

Table 5-14 *Accept-RP Commands*

The configuration in [Example 5-26](#) accepts join or prune messages destined for the RP with an IP address of 10.3.3.3 destined for the multicast group of 239.1.1.1.

Example 5-26 *Accept-RP Configuration Example*

[Click here to view code image](#)

```
ip pim accept-rp 10.3.3.3 Group_Address
!
ip access-list standard Group_Address
permit 239.1.1.1
```

It is important to have a clear delineation between the edge of the multicast network control plane and other network connections, especially external connections, such as the Internet. The first and most important step to accomplish this is to disable PIM on any interfaces not in the PIM forwarding path or any externally facing interfaces. This should prevent a neighbor relationship from forming on

these interfaces. It is common practice to install a protective ACL on external interfaces as well (such as anti-spoofing or other basic filters). These ACLs should end with a **deny ip any any** and should not include a **permit** for any unwanted multicast traffic.

However, it occurs in some broadcast networks, like Metro-Ethernet, that PIM is needed on a broadcast media where both internal and external paths may lie. In other cases, PIM neighbor relationships may be undesirable for certain peers but desirable for others. Either way, a method is needed to filter out unwanted neighbors, permitting PIM hello messages only from explicit addresses. The **neighbor-filter** command, shown in [Table 5-15](#) , can assist with this configuration need.

Operating System	Command
IOS/XE	<code>ip pim neighbor-filter <i>access-list</i></code>
IOS XR	<code>neighbor-filter <i>access-list</i></code>
NX-OS	<code>ip pim neighbor-policy <i>policy-name</i></code>

Table 5-15 *Neighbor Filter Commands*

Finally, any other measure that a network administrator would use to create a boundary between networks is appropriate to apply to the multicast edge. Firewalls are an excellent device for edge-network separation, and many firewalls, including the Cisco ASA firewall appliance, can inspect multicast packets. A single-context firewall in routed mode will be able to participate in the multicast control and data plane. For multi-context firewall deployment, it is recommended to deploy firewalls in transparent mode for multicast support for security and state inspection. Also, network administrators should ensure that corporate security policies include IP multicast security measures at any level where it is appropriate in the infrastructure or application architecture.

Protecting Multicast RPs

The key to protecting multicast RPs is to protect the control-plane resources of the RP itself. Typically, RPs are placed near the heart of the network, away from the network edge. With the exception of PIM-BiDir, the RP does very little multicast forwarding, unless it is directly in the flow of traffic. Consequently, protecting the data plane should be relatively easy to do by removing the RP from the main multicast data path.

If the network is large, or if there are many multicast streams, the RP can be taxed. Remember that when a new source starts transmitting in a PIM sparse-mode network, the packets will be encapsulated and sent using unicast messages to the RP. By default, a RP is therefore vulnerable to accepting registrations from inappropriate DRs for inappropriate groups, as well as to accepting more registrations than it may be able to handle in memory.

It is a good idea, to lock the RP down so that it accepts registrations only for groups that are part of an explicit list. Using an ACL, an **accept-register** list does just that. This security feature allows control over which sources and groups can register with the RP from the FHR. [Table 5-16](#) lists the commands needed to lock out unwanted group registrations at the RP.

Operating System	Command
IOS/XE	<code>ip pim [vrf <i>vrf-name</i>] accept-register {list <i>access-list</i> route-map <i>map-name</i>}</code>
IOS XR	<code>accept-register <i>access-list-name</i></code>
NX-OS	<code>ip pim register-policy <i>policy-name</i></code>

Table 5-16 *accept-register Commands for RPs*

Aside from explicitly limiting the exact entries allowed to create state on the RP, you can also limit the rate at which state entries can be created. The multicast registration process can be taxing on the CPU of the designated router (DR) and the RP if the source is running at a high data rate or if there are many new sources starting at the same time. This scenario can potentially occur immediately after a network failover.

Limiting the registration rate protects the RP control plane from being overwhelmed during significant multicast state churn. Remember, that multicast state churn can be a consequence of something happening in both the multicast overlay or the underlying unicast network. [Table 5-17](#) displays the commands used to limit the rate of RP registrations.

Operating System	Command
IOS/XE	<code>ip pim [vrf <i>vrf-name</i>] register-rate-limit <i>packets-per-second</i></code>
IOS XR	<code>maximum register-states <i>number</i></code>
NX-OS	<code>ip pim register-rate-limit <i>rate</i></code>

Table 5-17 *RP Group Register Rate-Limiting Commands*

The number to which register packets should be limited depends largely on the number of potential sources registering at the same time and their data rates. Determining a baseline for multicast applications provides the administrator a general idea of what the communication characteristics look like. A typical setting in a PIM sparse-mode (PIM-SM) network is between 4 and 10 messages per second.

In an Auto-RP environment, you can use this feature in your design to filter certain RPs for certain multicast groups. This is needed if you have unique requirements to split the multicast groups aligned to each RP. This is achieved by using the command **ip pim rp-announce-filter rp-list *access-list* group-list *access-list*** . This is normally configured at the mapping agent for Auto-RP. [Table 5-18](#) displays the commands used to configure RP announce filters for mapping agents.

Operating System	Command
IOS/XE	<code>ip pim [vrf <i>vrf-name</i>] rp-announce-filter {group-list <i>access-list</i> rp-list <i>access-list</i> [group-list <i>access-list</i>] }</code>
IOS XR	Not available in IOS-XR
NX-OS	<code>ip pim auto-rp mapping-agent-policy <i>policy-name</i></code>

Table 5-18 *RP Announce Filter Commands*

Finally, the commands listed for IOS/XE and NX-OS in [Table 5-18](#) allow an RP to filter RP group mapping announcements. This allows an administrator to control domain learning from the Auto-RP mapping agent. IOS-XR does not support this command because it is now assumed that domains will have sufficient controls in place, prior to mapping agent dynamics.

Best Practice and Security Summary

The last several sections provide many solutions to protect the multicast control plane and ultimately the integrity of the entire network. Determining the appropriate solution depends on the landscape of the network infrastructure, the capabilities of the organization managing the solution, and the risk you are willing to take. [Table 5-19](#) provides an overview of the different solutions.

Feature	Recommendation
ip pim sparse-mode	Modern PIM networks should use PIM-SM or PIM-SSM in all cases. This is an interface configuration and should be enabled on every non-boundary-facing domain interface.
ip igmp snooping	This feature should not be disabled and is enabled by default in most Cisco switches.
ip pim dr-priority <i>value</i>	Configures the neighbor priority used for PIM designated router (DR) election. The router with the largest <i>value</i> on an interface becomes the PIM-DR. If gateway redundancy is used at the source/receiver edge of the PIM domain, DR priority should be manually configured to support optimal forwarding.
RP placement, protection, redundancy, and scoping	When assigning RPs, choose routers that can handle the additional load and are ideally located for shared-tree forwarding paths. (An edge router may not be the best place to place an RP.) Protect RP routers with ACLs and rate-limiting. Control-plane policing (CoPP) is desirable. If IP multicast services are mission-critical, be sure to include an RP redundancy scheme. For especially large installations, set the RP manually and use Anycast-RP to ensure fast convergence during a failure, as well as local redundancy during domain fracturing. Finally, make sure that dynamic RP propagation protocols are properly scoped and bounded to protect the domain from outside influence.

Multicast addressing schema	Create an addressing schema from the very first multicast application roll-out. Make sure your schema is specific and meaningful. This helps eliminate multicast group address sprawl and misconfigurations.
Multicast traffic engineering	Network architects and engineers should always use multicast multipath when redundancy over equal-cost multiple paths (ECMP) is available and desired. Outside of this, traffic engineering should be avoided unless there is a specific need, such as desired asynchronous routing. MBGP provides a dynamic way to engineer multicast-forwarding and is ideal for large networks with complex needs. Otherwise, static mroute state entries will suffice. Tunnels made specifically for multicast forwarding should be avoided at all costs, except when there is a need to bridge a gap in PIM peering within a forwarding domain, such as across service-provider links that do not support multicast networking.
ip pim register-rate-limit <i>value</i> and ip igmp limit <i>number</i> [except <i>access-list</i>]	Always rate-limit multicast and IGMP messaging on any source/receiver edge gateways. This prevents multicast storms from consuming resources, whether they are intentional or unintentional. Also, consider using ACLs to limit the groups to only those in your approved address schema. This protects the domain against attacks using rogue multicast group addresses.
ip multicast route-limit <i>limit threshold</i>	Make sure you are setting limits on the number of multicast state entries that each domain router should allow. This prevents both intentional and unintentional resource exhaustion, which could affect unicast routing as well as multicast routing.

Table 5-19 *Summary of Basic Multicast Deployment Recommendations*

Putting It All Together

Let's go back and take a look at our example company, Multicaster's Bank Corp. [Figure 5-17](#) illustrates the high-level topology for the Multicaster's Bank Corp. network. This scenario is a good example to bring all the concepts about configuring PIM together.

Multicaster's Bank Corp.

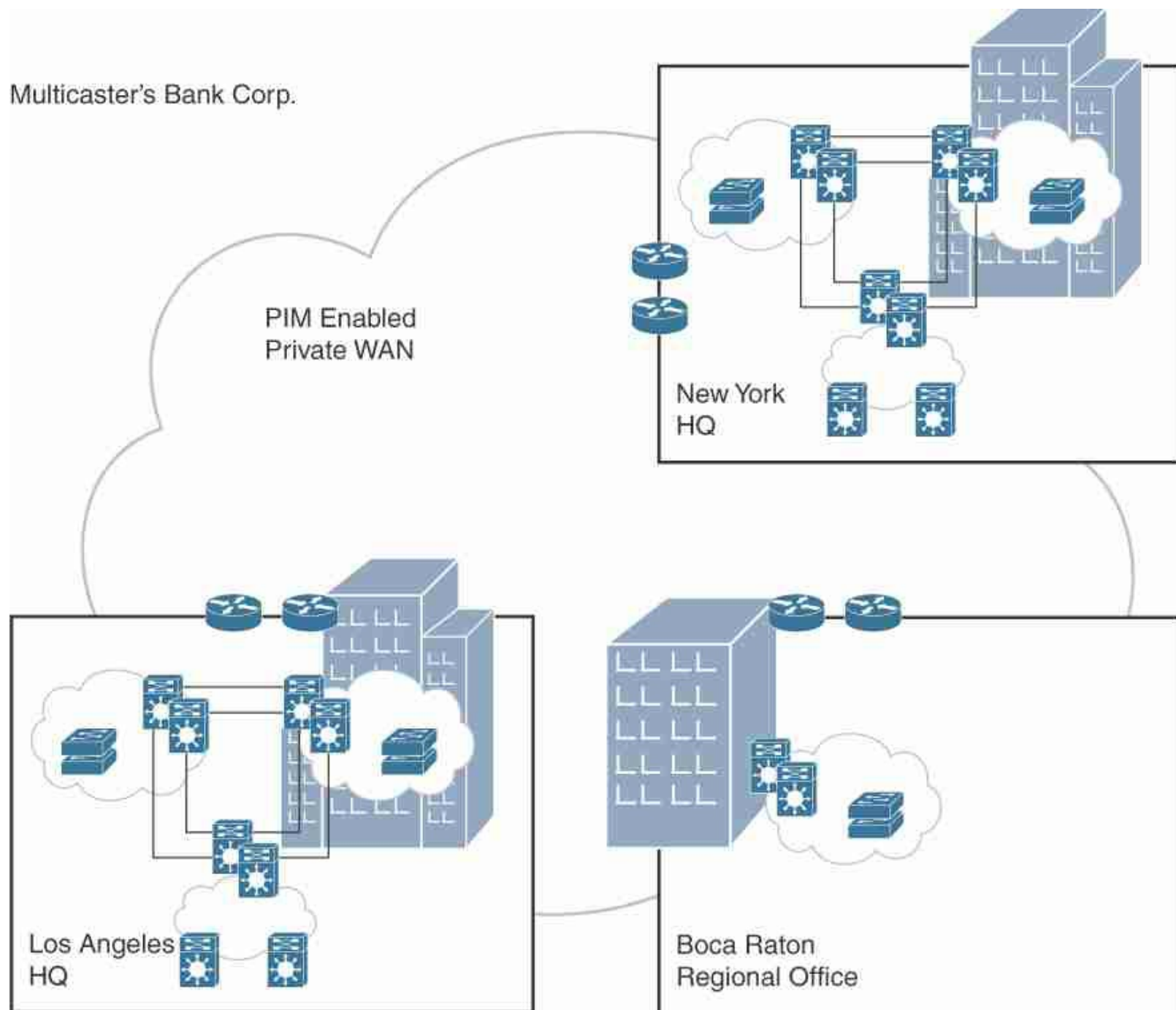


Figure 5-17 *Multicaster's Bank Corp. Network*

Scenario: Multicaster's Bank Corp. Media Services

Multicaster's Bank Corp.'s marketing department has decided to make some upgrades to the way they manage their brand locally at the Los Angeles HQ, New York City HQ, and Boca Raton regional offices. The HQ campuses share a very similar design, and the company wishes to deploy localized digital signage to each campus, giving media control to the local marketing teams. They purchased a digital signage media engine for each HQ office. They have also purchased a new webcasting TV system for internal management communications to employees (IPTV) that will be used corporate-wide. As new products are announced, the IPTV service can update employees with important messages about product changes.

In this scenario, we work for the director of infrastructure. The marketing department has given her some high-level requirements about the two systems. In addition, she is concerned that adding a lot of unicast media traffic to the campus LAN could impact daily operations at all three buildings. Both the digital signage and IPTV servers support multicast, and she has asked us to look over the

requirements and make the engineering and configuration recommendation to meet the requirements using multicast.

Requirements:

- There will be 20 unique feeds of IPTV media so individual departments can select between different services or advertisements from the service.
- Each of the HQ campuses should have the ability to support up to eight unique streams.
- The initial media systems will not be built with redundancy until Phase Two. However, this is a business-critical application for marketing; thus, the infrastructure supporting both systems should be as dynamic, reliable, and as redundant as possible.
- Because these systems are an integral part of the company's branding, they should be made as secure as possible network-wide.
- Each of the HQ digital signage systems should be separate from each other and no streams should appear at the Boca Raton office. There should be no chance that signage media can leak from one campus to another.
- All workstations across the campus should be able to tap into the IPTV feed. The IPTV server will be located at the Boca Raton office, where the principal IT is located. The location of the IPTV server is shown at a high-level in [Figure 5-18](#) . [Figure 5-19](#) shows the NYC HQ office network with the digital signage media server connected. The LA campus has an identical configuration to that of the NYC campus.

Note

Not all service providers offer the same services when it comes to multicast. One SP may not allow native multicast over the MPLS links, whereas another may. Some SPs may limit multicast traffic by default. It is important for enterprise customers to work with their service providers to determine which services are available, the fees that may apply (if any), and how services may affect a given design.

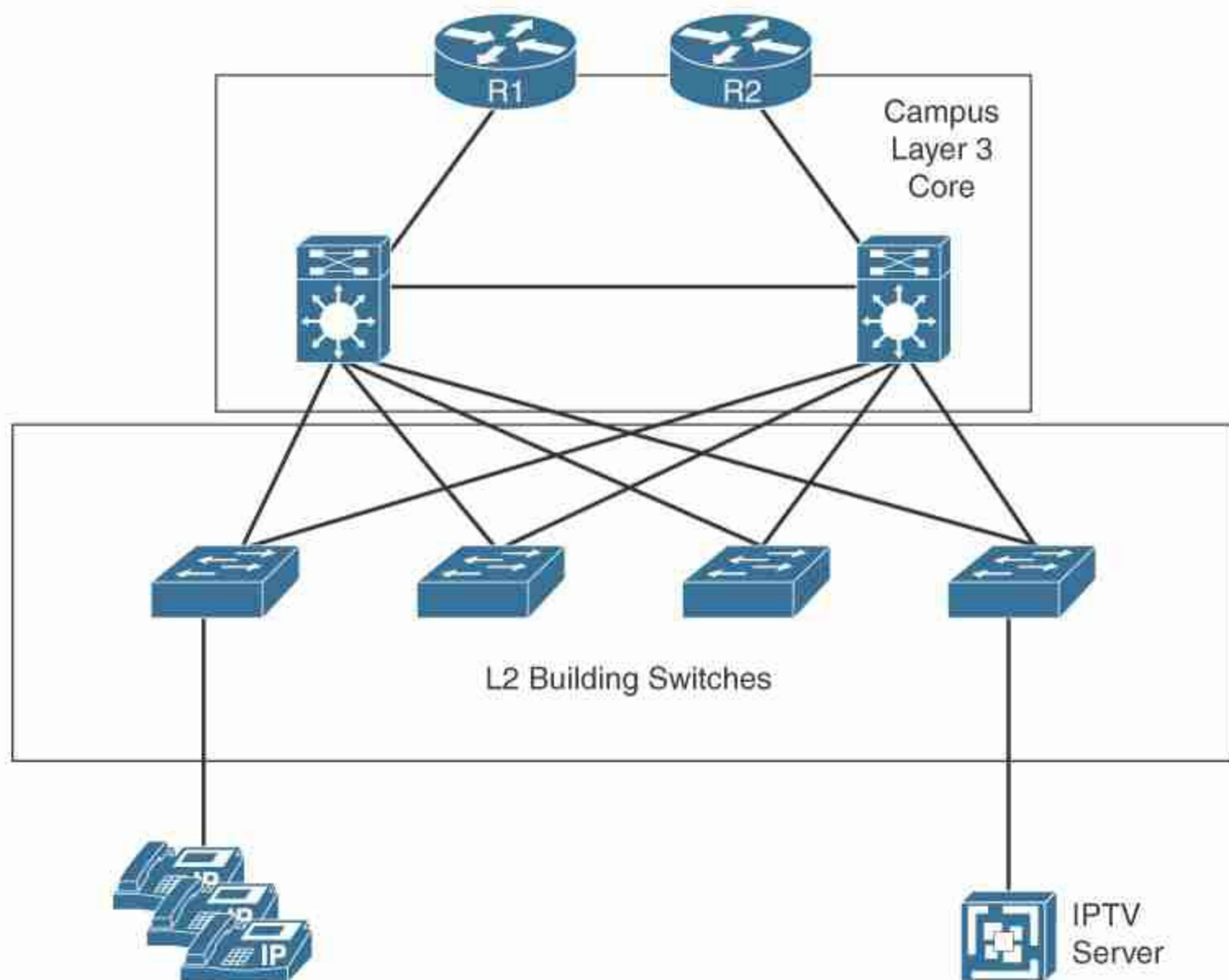


Figure 5-18 *Multicaster’s Boca Raton Office*

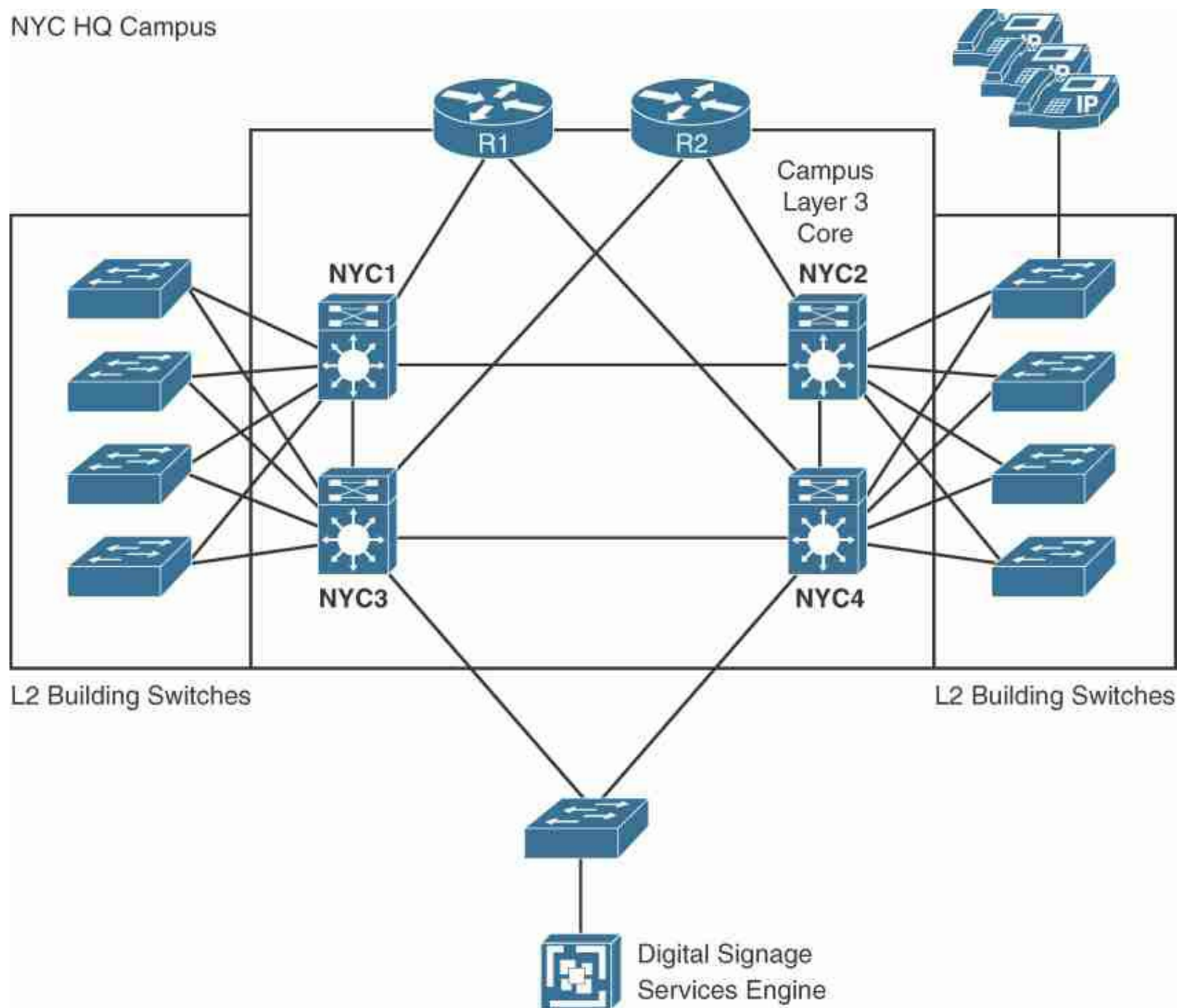


Figure 5-19 *NYC Office*

In addition to these requirements, our director has reminded us of a few technical details relevant to enabling multicast:

- A single MPLS cloud, obtained through a national service provider, connects the three corporate campuses. Each of the three campuses has two routers, each with one link to the provider, as shown in [Figure 5-20](#).

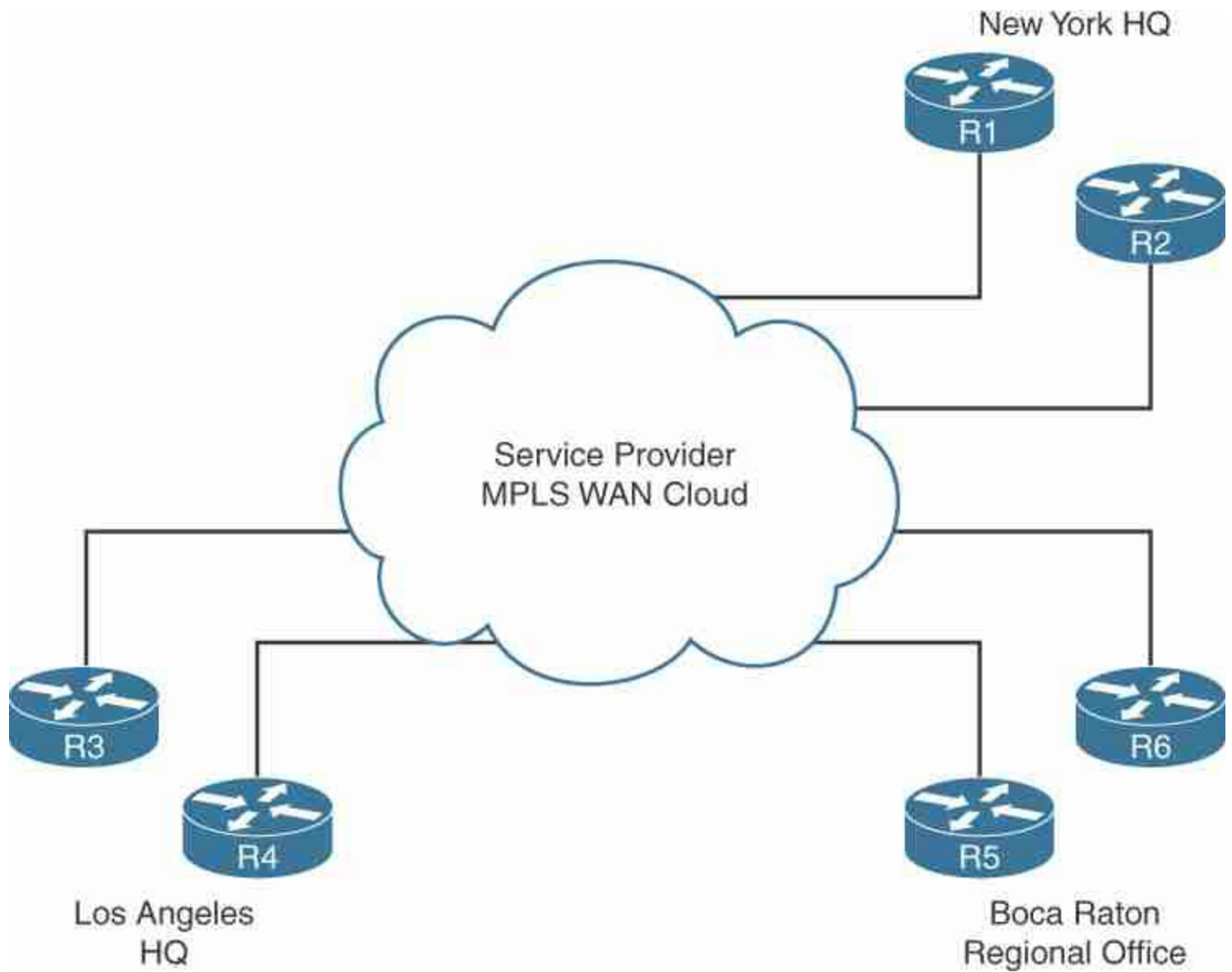


Figure 5-20 *Multicaster's MPLS WAN Cloud*

- The MPLS service provider supports full PIM sparse-mode multicast over the MPLS backbone, and the provider uses Cisco edge routers. The six Multicaster's routers each have an EIGRP neighbor relationship with the MPLS cloud.
- The service provider has an enforced multicast state limit of 24 entries per customer. (This depends on the service-level agreement that the enterprise has with the service provider.) Any entries forwarded beyond this state limit will not be installed on the provider routers.
- Because firewalls are not yet in the path of the media servers, network security has recommended that multicast filtering be used to ensure that group state can be formed where only appropriate and that IGMP be locked down on gateway routers.
- The IPTV server and the two digital signage media servers only support IGMPv2.
- Network configurations should always be identical (with the exception of IP addresses) between the two HQ campuses. This creates network operations consistency. To ease stream identification, however, each HQ campus should use different multicast group addresses between the eight streams in each campus.

Using this information, we should be able to derive an appropriate multicast design and configuration that will suit Multicaster's needs. It is clear from the requirements that multiple

overlapping domains will provide the separation needed between the local media engines and the IPTV service. There should be one corporate-wide domain that can encompass IPTV and reach all IP phones in the three campus networks. In addition, the digital signage streams will be best localized if a single overlapping domain is created for each HQ location (NYC and LA respectively). We will propose dividing the network into the three domains shown in [Figure 5-21](#).

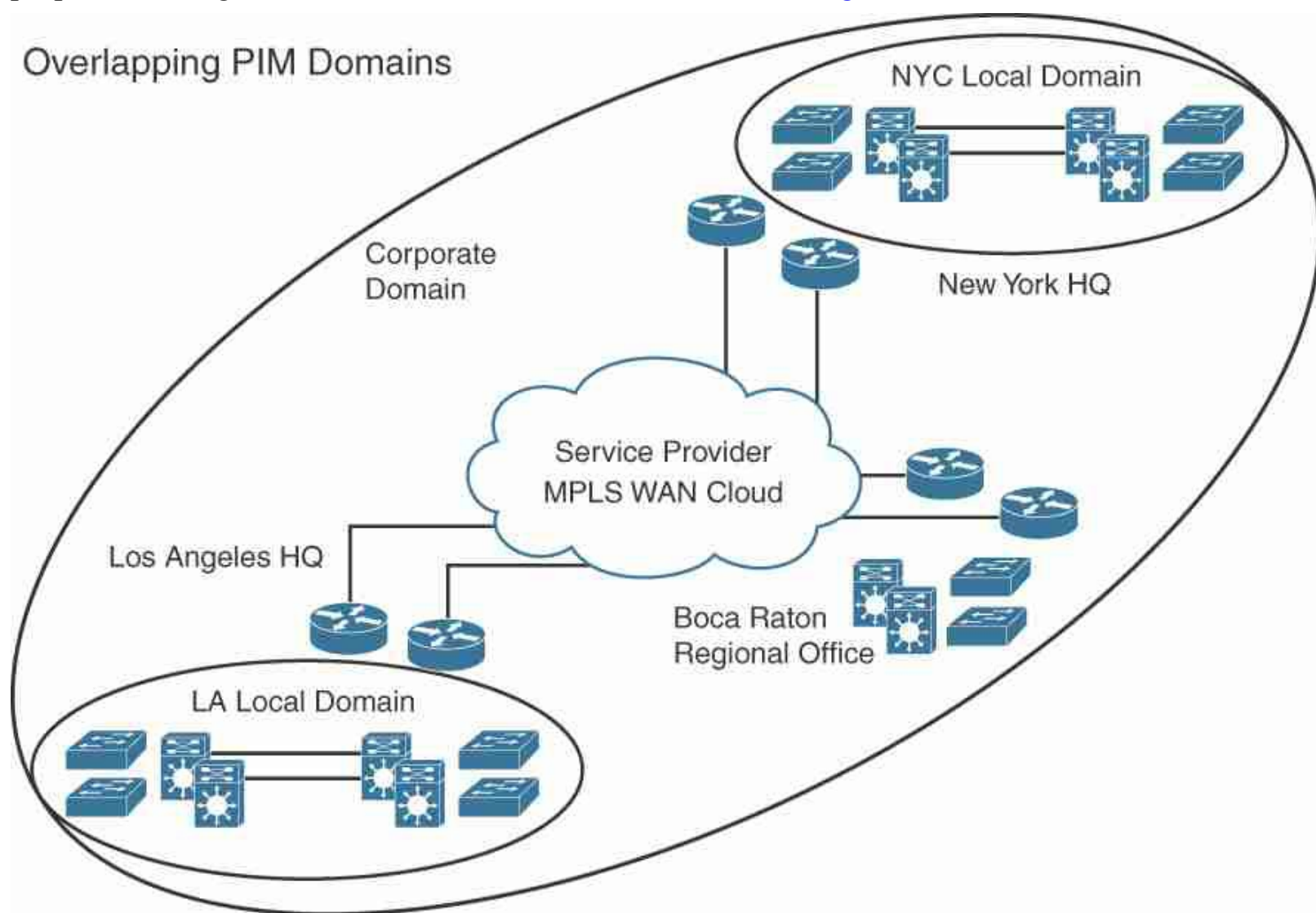


Figure 5-21 *Overlapping PIM Domains*

Domains should be segmented and secured using appropriate filters at the edge of each domain. Because the servers do not support IGMPv3, we will not be able to use SSM; consequently, sparse-mode PIM is our best option. To achieve dynamic ASM state consistency and reliability, we should use a dynamic RP feature in combination with Anycast RP redundancy within each domain. Auto-RP will be a good choice to provide our dynamic RP-mapping mechanism.

All routers and switches fall within the larger corporate domain. The WAN routers R1 and R3 connecting to the MPLS cloud will provide the Anycast RP and Auto-RP candidate and mapping agent functions for the corporate domain. [Figure 5-22](#) shows the high-level design for this configuration, with R1 and R3 acting as both Anycast peers and RP candidates. The primary loopback (Loopback0) interfaces on R1 and R3 will act as redundant mapping agents.

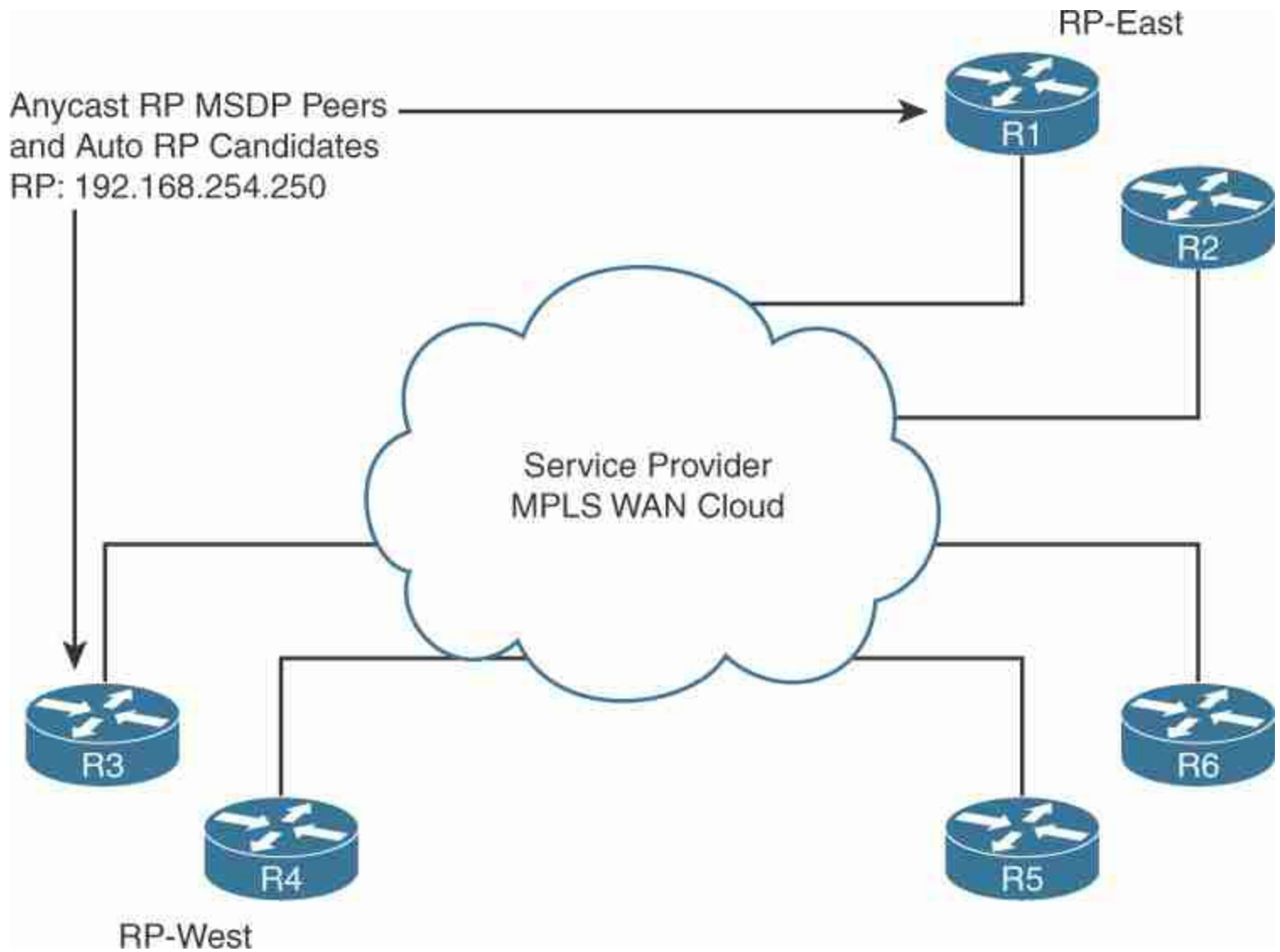


Figure 5-22 *Multicaster's Global Domain*

To support the local HQ domains, we will propose using Anycast RP and Auto-RP; ensuring network operations consistency. Because these domains are much smaller, we will consolidate RP-candidate and mapping agent functions onto the same devices. Campus Layer 3 core switches C3 and C4 will act as the RPs for these domains. The core switches should also be the boundary of the local domain. Because both the LA and NYC campuses have identical configurations, we can use the same design for both. [Figure 5-23](#) shows the proposed high-level design for the NYC campus domain.

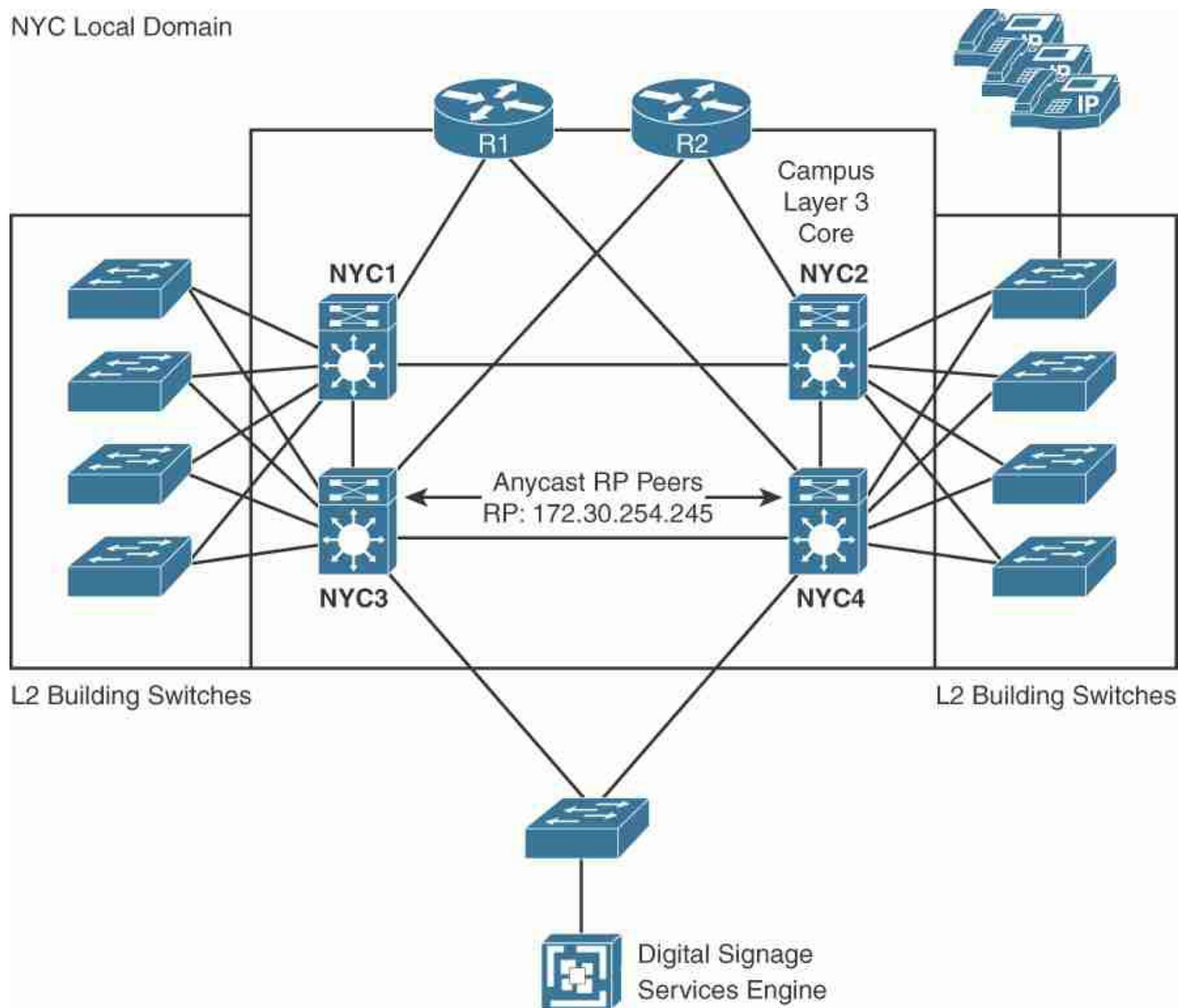


Figure 5-23 NYC Office RP Configuration

Let us examine the configuration steps on the network routers and switches to enable PIM across the network.

All routers and Layer 3 core switches:

Step 1. Enable multicast routing globally using the **ip multicast-routing** command.

Step 2. Enable PIM on *all* interfaces in the multicast distribution path, including interfaces facing either sources or receivers.

Step 3. Enable PIM on all relevant loopback interfaces (this step is critical for future configuration steps).

Step 4. Enable **ip pim auto-rp listener** on all multicast-enabled routers.

Step 5. Ensure that PIM is disabled on all other interfaces not in the forwarding path, in particular on any interfaces that face external entities, including VPN or other tunnel interfaces, and on Internet-

facing interfaces.

Step 6. If required by security policy, manually lock down PIM neighbor relationships on PIM interfaces using the **ip pim neighbor filter** command, especially on those interfaces connecting to the MPLS service provider.

Step 7. Use an ACL denying all PIM and all multicast for any of the interfaces identified in Step 5.

Step 8. Tune all unicast routing for fast convergence, if applicable to the infrastructure and application (not a requirement for multicast), and turn on multipath multicast when and where it is warranted.

All Layer 2 switches:

Step 1. Enable PIM on any Layer 3 interfaces facing sources or receivers, including switch virtual interface (SVI) interfaces.

Step 2. Ensure that IGMP snooping is enabled.

Step 3. Tune STP to ensure maximum convergence and forwarding efficiency throughout the Layer 2 domain.

When the network is up and PIM is communicating across the network, it is time to create the address schemas that will be used for each application and domain. Remember that the requirements ask for the same addresses to be used in both NYC and LA. Because both the IPTV and digital signage applications are private ASM, we will use the 239.0.0.0 group address block to assign the groups. Our addressing requirements are very small. We can allow the second octet in the block to indicate the scope of the domain (global or local), the third octet to indicate the application type, and the fourth octet to indicate the stream number. If we use .10 and .20 to represent global and local scopes respectively, while using .1 to represent the IPTV application and .2 the digital signage streams, the group schema would look like the one shown in [Figure 5-24](#).

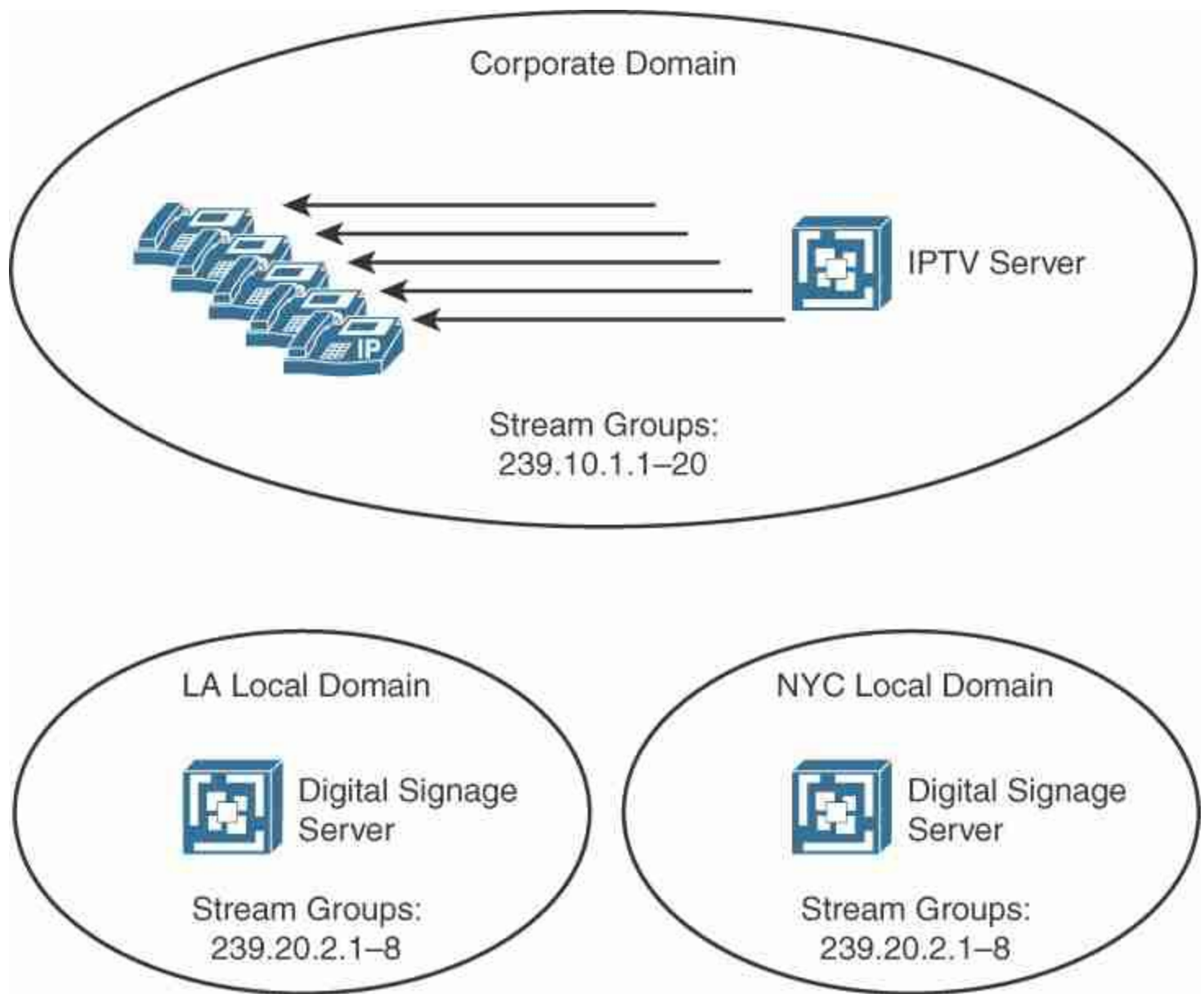


Figure 5-24 *Multicast Address Schema*

Next, we need to create the RPs, segment the domains, and secure the borders. We also need to set up Auto-RP boundaries for the network using a scope. Let us start with the global, IPTV domain. The following configuration steps should be taken on routers R1–R4 to create the Anycast RPs and Auto-RP mappings required for all L3 devices.

Routers R1 and R3 (global RPs):

Step 1. Create an Anycast RP loopback interface with an identical address on each router. (In this case, we can use Loopback 100 with IP address 192.168.254.250, making sure the interface is PIM sparse-mode-enabled.)

Step 2. Establish Anycast MSDP peering between routers R1 and R3 using the main loopback (Loopback0) interface as the peering sources. Loopback 0 should also be configured as the router-id.

Step 3. Configure the new interface as the Auto-RP candidate using a scope that covers the maximum breadth (in hop count) of the service provider network and Multicaster's routers.

Step 4. Create and apply an accept-register filter that limits multicast group registrations to only those in the global group address schema (deny the NYC and LA local groups).

Step 5. Ensure that the new loopback (Loopback 100) is distributed into the EIGRP routing unicast

routing domain.

All downstream routers:

Step 1. Configure Loopback0 on R2 and R4 as Auto-RP mapping agents using a scope that covers the maximum breadth of the service provider network and Multicaster's routers.

Step 2. Configure all edge gateway routers with an IGMP accept-list limited to only those groups included in the group schema.

Step 3. Create and apply a multicast boundary list on every Layer 3 device/interface that makes up the edge of the multicast network, such as Internet-facing interfaces (not shown in the diagrams).

Step 4. Use ACLs on the service provider facing interfaces to limit IP multicast packets to only those groups specified in the schema.

In the preceding configuration, interface Loopback 0 is the primary EIGRP and router management loopback on each router. We used Interface Loopback 100 on R1 and R3 with address 192.168.254.250/32 as the Anycast RP address. A **network** statement is added to EIGRP to propagate this address to the rest of the network. Remember that the network routers will build the shared tree to the closest unicast RP, meaning the routers will perform a L3 recursive look-up on the RP address, and whichever path has the shortest distance will be used as the RPF path. Auto-RP also uses this loopback address as the Candidate RP, giving us a hybrid RP design.

We need to make similar configuration changes for the local HQ domains (Local RPs). Here we show only the NYC domain configuration steps for brevity.

Step 1. On NYC3 and NYC4 L3 switches, create a new loopback interface (in this case, we will use Loopback200 on each), enabling PIM sparse-mode and assigning IP address 172.30.254.245.

Step 2. Establish Anycast MSDP peering between the primary loopback (Loopback0) interfaces on both routers.

Step 3. Configure the new loopback, Loopback 200, as both Auto-RP candidate and Auto-RP mapping agent with scope 2 on NYC3 and NYC4.

Step 4. Boundary router: It is recommended to filter Auto-RP on the boundary routers to prevent local scope leakage. The TTL (Auto-RP scope) value should be one more than the one configured on the candidate RP or mapping agent. This configuration prevents announce and discovery messages from leaking from the local site.

We would repeat these steps exactly in the LA local domain. However, in this scenario, we want the LA and NYC local domains to be identical, including the group addresses. Keep in mind that the larger Layer 3 unicast network will end up having multiple routes for *all* the Anycast peers. What would happen if we use the same loopback IP addresses for the Anycast RPs in both domains?

Note

All Layer 3 devices will use the unicast RIB entry for the loopback to build the RPF neighbor relationship. EIGRP will calculate a path to each Anycast loopback, as well as a feasible successor for each. EIGRP will select the path with the lowest metric to place in the RIB. Because it is a very large unicast domain, EIGRP would have a path to all four Anycast RPs with the same IP address (C3 and C4 in both the NYC and LA domains). To further ensure domain isolation, it is recommended to use different IP addresses for the Anycast loopbacks in the NYC and LA domains (the local admin scoped domains); one address for LA and one address for NYC. An alternative would be to use

distribute lists to control Layer 3 unicast updates, but many might consider this the more difficult option. To have complete control over the domain, use boundary best practices previously discussed.

When all of these configuration tasks are complete, the domains should be operational and isolated. Further security and domain configuration is likely warranted. We would use the preceding sections and other published configuration guides to derive the appropriate additional configurations for each PIM router in the network. Ultimately, we need to ensure the RPs are protected, and that the mapping process on all routers is limited to those groups explicitly configured. Other security measures should be based on an individual organization's policy and therefore is not included in this scenario.

Summary

This chapter discussed the importance of creating a functional schema for IP multicast addressing, how this schema provides better control over security implementations and builds a foundation to easily manage applications by establishing location and application identity into the address of the multicast messages.

Design elements were considered on the proper placement and implementation strategies using active/active and active/standby, and the different solutions using Auto-RP, BSR, static, Anycast, and MSDP mesh groups.

We generally desire the traffic flow in a unicast network and the multicast overlay to be congruent, but there are additional considerations for multicast that need to be addressed when equal-cost multipath unicast routes are involved. These include load-sharing using multipath selection, static entries, and BGP.

Security has been a growing concern over the years, and increasing the footprint of the infrastructure by adding multicast capability adds to the challenge. Fortunately, several mechanisms are available to protect the control plane and data plane for multicast. Some of these include control-plane protection, multicast filtering, and scoping.

Chapter 6. IPv6 Multicast Networks

Chapter 6. IPv6 Multicast Networks

This chapter begins by covering the fundamentals of IPv6 and why there is such a great need for it today. It explains IPv6 multicast group addressing, along with the concept of scoping and IPv6 multicast group address assignments. You also learn how IPv6 multicast addresses are mapped to MAC addresses. We discuss the Layer 2 and Layer 3 multicast environment, how subscriptions to multicast streams are controlled, how IPv6 multicast messages are routed across the network, and the different options for configuring a rendezvous point. Included in this chapter are many configuration examples to get you started on the road to implementing IPv6 multicast.

IPv6 Fundamentals: A Quick Overview

The rapid growth of the Internet has caused the depletion of IPv4 address space. Consequently, IPv6 is taking hold to provide for the expansion of the Internet and our ability to access any device on the planet.

IPv4 addresses use 32 bits for a numerical value to distinguish individual devices, whereas IPv6 uses 128 bits. This increase offers tremendous scalability. The implementation of IPv6 has another interesting characteristic in that there is no longer support for traditional network broadcasts. The two methods of communication available with IPv6 are unicast or multicast. Because multicast was considered during the creation of the protocol, some inherent capabilities improve that operation. Besides the greater amount of address space, the rendezvous point (RP) address and scoping information can be embedded in the message.

The biggest difference between IPv4 and IPv6 is the size of the IP address. An IPv6 address has 128 bits (16 bytes). These 128 bits are divided into 8 16-bit hexadecimal blocks separated by colons (:) in the format: X:X:X:X:X:X:X:X. Two examples of IPv6 addresses are as follows:

2002:8888:9999:AAAA:BBBB:CCCC:DDDD:1234

2002:8888:0:0:0:0:0:AAAA

An IPv6 address like the one above, 2002:8888:0:0:0:0:0:AAAA, can contain consecutive zeroes within the address. Two colons (::) within the address can be used to represent a group of consecutive zeroes. The two-colon marker can occur at the beginning, middle, or end of an IPv6 address, but it can only occur once within the entire address, representing the longest set of consecutive zeros. [Table 6-1](#) shows a list of compressed format IPv6 addresses.

IPv6 Address Type	Long Format	Compressed Format
Unicast	2002:10:0:0:0:1600:0:1234	2002:10::1600:0:1234
Multicast	FF05:0:0:0:0:0:0:1	FF05::1
Unicast	2002:0:0:0:0:10:20:1234	2002::10:20:1234
Multicast	FF06:10:0:0:0:0:0:100	FF06:10::100

Table 6-1 *Compressed IPv6 Address Formats*

Just as with IPv4, an IPv6 unicast address is an identifier for a single host interface on a single networked device. It is a logical address that represents a physical location. A packet that is sent to a

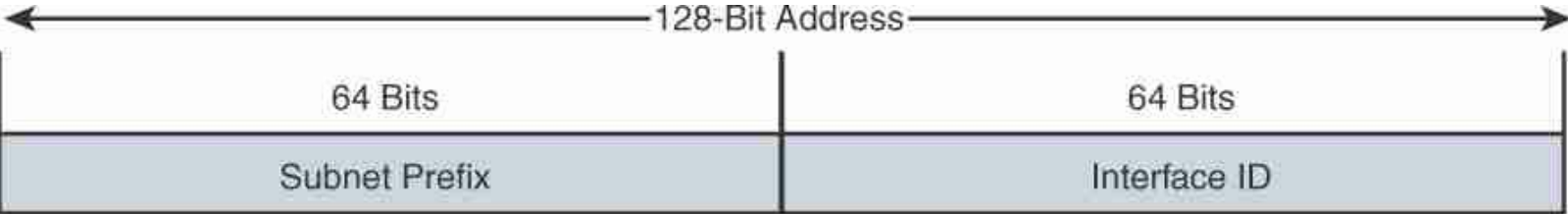
unicast destination address is routed and ultimately forwarded to the physical interface identified by that address. Unlike IPv4, IPv6 has different types of unicast addresses, and they serve different functions. The two most important to this text are *global* (often referred to as *aggregatable global*) addresses and *link-local* addresses.

A global unicast address is similar to a typical IPv4 address. Its structure enables address supernetting and subnetting and ultimately the strict aggregation of routed prefixes in the global (Internet) IPv6 table. Global addresses are assigned throughout a network and eventually are aggregated upward through organizations. Eventually, an aggregated block or blocks of address space are advertised by the organization to the Internet service providers (ISPs). These addresses are public and are assigned by IANA just like their IPv4 counterparts.

Global IPv6 addresses are defined by a global routing prefix, a subnet ID, and an interface ID. Except for addresses that start with binary 000, all global unicast addresses have a 64-bit interface ID. Thus, the global address has essentially two major sections, a 64-bit network prefix (equivalent to a major class network with the ability to subnet) and a 64-bit host identifier. This makes the assignment of addresses much easier for network administrators.

The assignable range of addresses for each 64-bit prefix is enormous, creating millions of potential networks with millions of potential hosts. This solves for the problem of address depletion in IPv4 networks. Additional subnetting can encroach on the 64-bit host ID of the address as well. This enables the address to convey not only the subnet but also any potential scoping included by the administrator. The final 48-bits in the address for many IPv6 hosts might simply be the MAC address of the host interface where the address is applied, connecting physical and logical in a single field.

The IPv6 global unicast address allocation uses the range of addresses that start with binary value 001 (2000::/3). [Figure 6-1](#) shows the structure of an aggregatable global address.



IPv6 Address Format



Global Unicast Address Format

Figure 6-1 IPv6 Address Format with Global Unicast Addressing

Note

We have only provided here a very brief overview of IPv6, but this is enough to get us moving forward on IPv6 multicast. If you need additional information on IPv6, we suggest you seek

additional Cisco publications specific to the subject.

The other type of addresses that we are concerned with are known as *link-local addresses* . A link-local address is an IPv6 unicast address that is typically automatically configured on any IPv6 interface. Consequently, if you configure a global address on a Cisco router interface, the router will by default, also assign a link-local IPv6 address. According to RFC 4291, link-local addresses must begin with the prefix FE80::/10 (1111 1110 1000 0000) and end with interface identifier in the modified EUI-64 format (normally automatically assigned using the physical address of the interface; Media Access Control [MAC] addresses in the case of Ethernet).

Link-local addresses are an absolute requirement for any IPv6 deployment. They are used in the IPv6 Neighbor Discovery Protocol (NDP), many infrastructure protocols, and the unique IPv6 stateless auto-configuration process. Nodes on a local link can use link-local addresses to communicate. This is very important to understand, IPv6 nodes do not need globally unique addresses to communicate.

Also of significance, is that IPv6 routers cannot forward packets that have link-local source or destination addresses to other links. The name link-local implies its scope. This scope is automatically enforced by any operating system that is conforming to IETF specifications for IPv6 address scoping (this should be universal). Link-local addresses are intended for local host-to-host communication only. Automatic assignment of these addresses can obscure implementations, especially inside the network infrastructure. Fortunately, the administrator can assign more meaning to the address by statically changing link-local addresses on Cisco routers and switches. [Figure 6-2](#) shows link-local address format.

10 Bits	54 Bits	64 Bits
1111111010 (FE80)	0	Interface ID

Figure 6-2 *Link-local Address Format*

We felt that a quick introduction to IPv6 addressing was important to exploring IPv6 multicast. Of course, anyone who has studied IPv6 can tell you from a protocol perspective, there is a lot more going on than just updating the size of IP addresses. Many parts of protocol intercommunication have also evolved with the addressing capabilities. Of course, multicast is no different.

IPv6 Layer 3 Multicast Group Addressing

As with IPv4 multicast addressing, IPv6 addressing is defined by the IETF. RFC 2373 and RFC 4291 specifically outline the foundations of IPv6 multicast group addressing, including how they are scoped. Scoping is an important part of IPv6 theory and consequently is an integral part of IPv6 addresses, which we will discuss.

The basic purpose of the address has not changed. The IPv6 multicast address still identifies a group of subscribing nodes, and a node may belong to any number of groups. Because of the differences in address size and scope, IPv6 multicast group addresses have a unique format. [Figure 6-3](#) illustrates the IPv6 group address format as defined by RFC 2373 and updated by RFC 4291.



Figure 6-3 *IPv6 Multicast Group Address Format*

The list that follows describes the fields shown in [Figure 6-3](#) :

- **11111111**: All group addresses must begin with this bit pattern.

- **Flags**: There are four one-bit fields equaling four flags that can be set, 0RPT. Flag values are:

- The first (highest order bit) flag is reserved and thus always set to 0.

- If the T bit is set to 0, this indicates a permanently assigned (“well-known”) multicast address, assigned by the Internet Assigned Numbers Authority (IANA).

- When the T bit is set to 1, this indicates a non-permanently assigned (“transient” or “dynamically” assigned) multicast address.

- The P flag is defined by RFC 3306:

- If the P-bit = 0, this indicates a multicast address that is not assigned based on the network prefix.

- If the P-bit = 1, this indicates a multicast address that is assigned based on the network prefix.

- When P = 1, T must also be set to 1, otherwise the setting of the T bit is defined in Section 2.7 of [ADDRARCH] in RFC 3306.

- The R flag is defined by RFC3956 and used in defining embedded RP addresses, an IPv6 inherent method for dynamically updating RP mappings on downstream routers without the use of a secondary distribution protocol like BSR or Auto-RP. Embedded RP is discussed in subsequent sections of this chapter.

- **Scope**: A four-bit field used to indicate the scope (breadth of forwarding) of the group. Scope values are:

- 0000 – 0: reserved

- 0001 – 1: node-local scope

- 0010 – 2: link-local scope

- 0011 – 3: (unassigned)

- 0100 – 4: (unassigned)

- 0101 – 5: site-local scope

- 0110 – 6: (unassigned)

- 0111 – 7: (unassigned)

- 1000 – 8: organization-local scope

- 1001 – 9: (unassigned)

- 1010 – A: (unassigned)

- 1011 – B: (unassigned)

- 1100 – C: (unassigned)

- 1101 – D: (unassigned)

- 1110 – E: global scope

- 1111 – F: reserved

- **Group ID** : Assigned by administrator and identifies the group as either permanent or transient within the given scope.

The scope value, however, does not indicate the meaning of the group address, only the scope in which the group should be found. Any non-permanently assigned group address is meaningful only in the scope that precedes it. The same address may appear in multiple physical locations in the network, but need not interfere with each other, or indicate any membership beyond the local network. For example, if a group of servers is assigned a permanent multicast address with a group ID of 111 (in hexadecimal), then:

FF01:0:0:0:0:0:0:111 is destined for all servers on the same node as the sender, and only that sender.

FF02:0:0:0:0:0:0:111 is destined for all servers on the same link as the sender, and only that sender.

FF05:0:0:0:0:0:0:111 is destined for all servers at the same site as the sender, and only that sender.

FF0E:0:0:0:0:0:0:111 is destined for all servers in the Internet.

Why is this relevant? Remember our discussion about overlapping domains in [Chapter 5](#) , especially the domain design and addressing schema for Multicaster's Bank Corp.? If IPv4 addressing had a similar scoping mechanism, the design and schema could be simplified. In addition, scopes can be nested within each other. [Figure 6-4](#) is a good way to visualize this relationship.

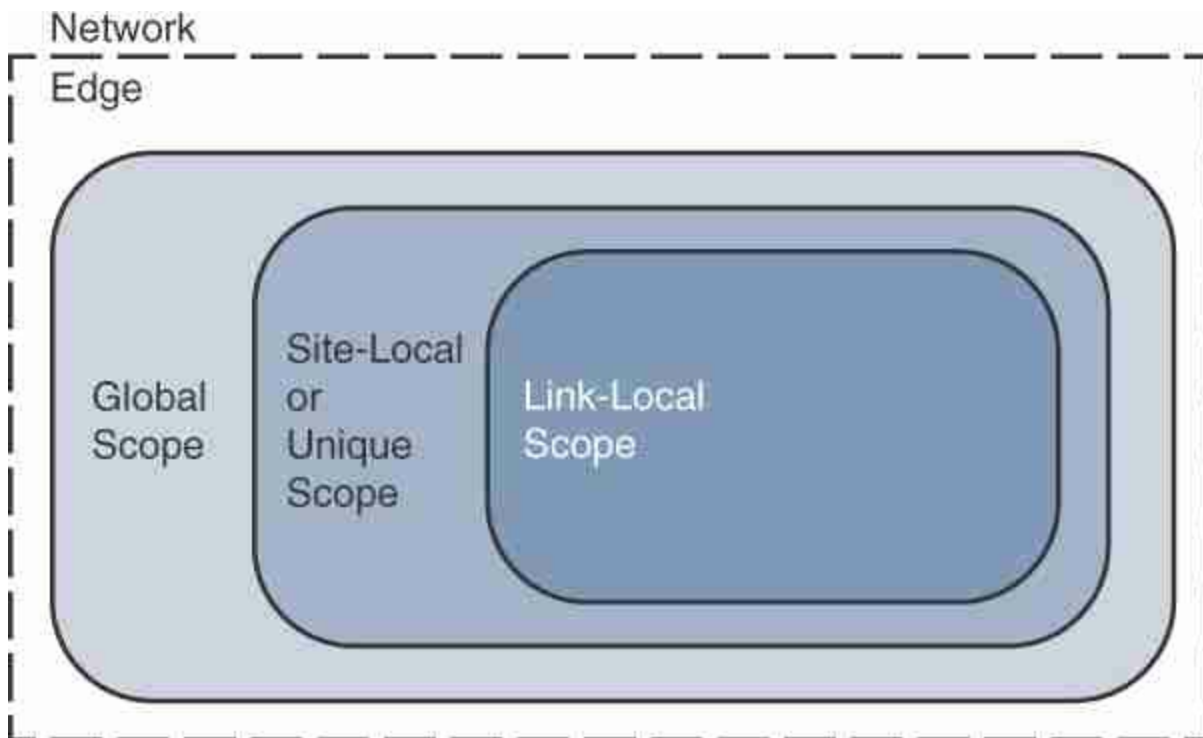


Figure 6-4 *Nested Group Scoping*

Furthermore, IANA breaks scope for group address assignments down into two types:

- **Variable-scope Group Addresses:** These addresses are similar across all scopes but represent distinct groups. Variable scope addresses are reserved for certain types of services or applications.

For example, Network Time Protocol (NTP), which has a multicast address of FF0X:0:0:0:0:0:101, where X masks the address scope.

■ **Fixed-scope Group Addresses:** These group addresses have a certain meaning only within the scope they are defined in. Fixed-scope addresses should not be changed. This type of address is important in the basic operation of IPv6. For this reason, a few common addresses are listed in [Table 6-2](#).

Multicast Address	Scope	Group Within the Scope
FF01:0:0:0:0:0:0:1	Node Local	All-nodes address
FF01:0:0:0:0:0:0:2	Node Local	All-routers address
FF02:0:0:0:0:0:0:1	Link Local	All-nodes address
FF02:0:0:0:0:0:0:2	Link Local	All-nodes address
FF02:0:0:0:0:0:0:5	Link Local	OSPFv3 IGP
FF02:0:0:0:0:0:0:6	Link Local	OSPFv3 designated routers
FF02:0:0:0:0:0:0:D	Link Local	All PIM routers
FF02:0:0:0:0:0:0:16	Link Local	All MLDv2-capable routers
FF02:0:0:0:0:0:1:2	Link Local	All DHCP agents
FF02:0:0:0:0:1:FFXX:XXXX	Link Local	Solicited-node address
FF05:0:0:0:0:0:0:2	Site Local	All-routers address
FF05:0:0:0:0:0:1:3	Site Local	All DHCP servers

Table 6-2 *Fixed-Scope IPv6 Multicast Addresses*

Note

According to RFC 2373, the source addresses in IPv6 packets cannot be multicast addresses, nor can a multicast address appear in any routing header.

IPv6 Multicast Group Address Assignments

IPv6 RFC 2373 also details a predefined and well-known set of multicast groups. These group addresses are reserved and may not be assigned to any public or private group, other than those expressed by the RFC.

The reserved multicast addresses are as follows:

- FF00:0:0:0:0:0:0:0
- FF01:0:0:0:0:0:0:0
- FF02:0:0:0:0:0:0:0
- FF03:0:0:0:0:0:0:0
- FF04:0:0:0:0:0:0:0
- FF05:0:0:0:0:0:0:0
- FF06:0:0:0:0:0:0:0

FF07:0:0:0:0:0:0:0

FF08:0:0:0:0:0:0:0

FF09:0:0:0:0:0:0:0

FF0A:0:0:0:0:0:0:0

FF0B:0:0:0:0:0:0:0

FF0C:0:0:0:0:0:0:0

FF0D:0:0:0:0:0:0:0

FF0E:0:0:0:0:0:0:0

FF0F:0:0:0:0:0:0:0

Other groups with the scopes listed above can be assigned to make a unique group. For example, the All-Nodes IPv6 multicast group addresses are as follows:

FF01:0:0:0:0:0:0:1

FF02:0:0:0:0:0:0:1

The All-Nodes group starting with FF01 represents an all nodes (all IPv6 addressed interfaces on this node, or node-local) group. The All-Nodes group starting with FF02 represents a link-local nodes group (the nodes on a particular subnet).

Another reserved group, the All-Routers group, could be any of the following group addresses:

FF01:0:0:0:0:0:0:2

FF02:0:0:0:0:0:0:2

FF05:0:0:0:0:0:0:2

There is a breakdown for these groups as well. Each group address above identifies a specific set of all IPv6 routers within scope 1 (node-local), 2 (link-local), or 5 (site-local).

There are many other reserved addresses, and there is an order to the assignment of group addresses for public and private use within the various scopes as defined earlier. One example to pay special attention to is the solicited-node group address of FF02:0:0:0:0:1:FFXX:XXXX.

A solicited-node multicast address is made from the low-order 24-bits of a reserved address (unicast or Anycast) in combination with appended prefix bits to the prefix FF02:0:0:0:0:1:FF00::/104. The resulting range of solicited-node group multicast addresses is FF02:0:0:0:0:1:FF00:0000–FF02:0:0:0:0:1:FFFF:FFFF. Therefore, the solicited-node multicast group address that corresponds to the IPv6 address 4037::01:800:200E:8C6C is FF02::1:FF0E:8C6C. An IPv6 node is required (due to the solicited nature) to join and process any associated solicited-node group addresses derived from the unicast or Anycast addresses assigned to its interfaces. This is a convenient way of targeting a specific group of nodes in a specific network or subnet; it's similar to a directed broadcast but with a different purpose.

For all other multicast address assignments, the IPv6 multicast group assignment uses the low-order 32-bits of the IPv6 address to create a MAC address. This is to avoid the MAC layer address overlap problem experienced in IPv4, which was discussed in the [Chapter 2](#).

New IPv6 multicast addresses should be assigned so that the group identifier is always in the low-order 32 bits, as shown in [Figure 6-5](#).



Figure 6-5 *IPv6 Multicast Address Format with MAC*

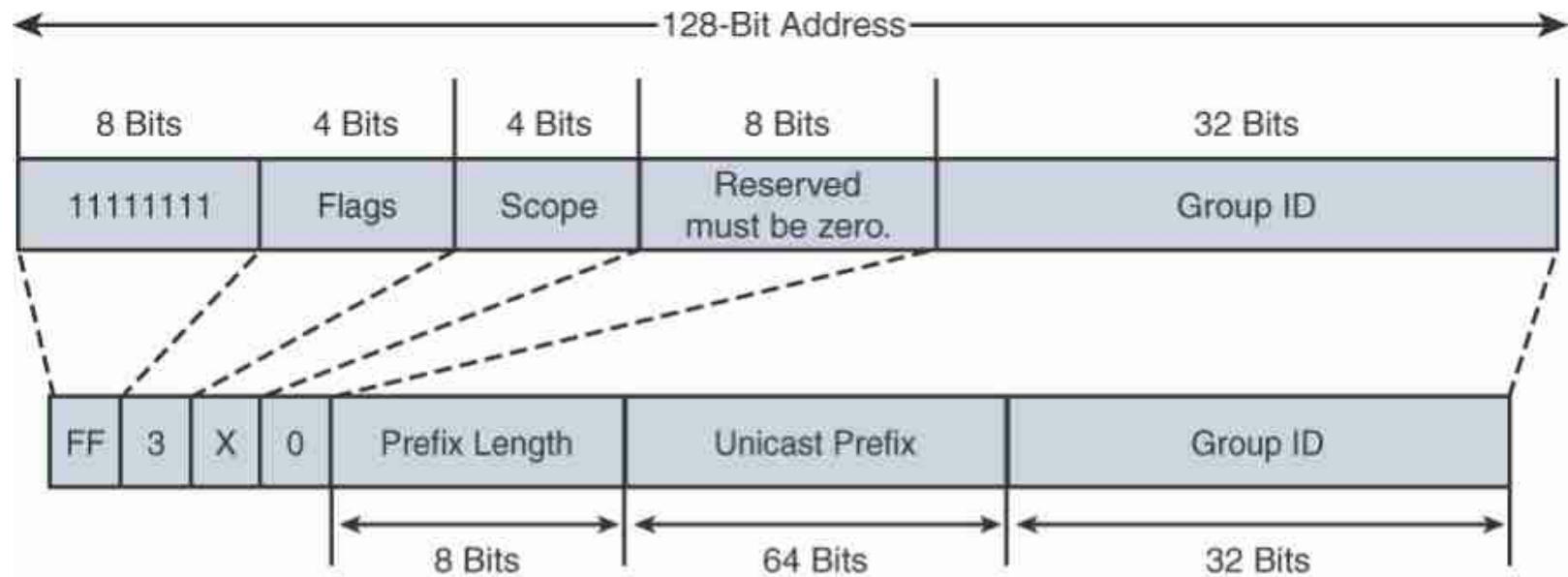
Any other restricted or public group assignments are managed by IANA.

IANA Unicast-Prefix–Based Multicast Address

IANA controls global IPv6 group address assignments for any organizations that apply. These are most often organizations that have Internet presence and are assigned individual autonomous system numbers (ASN). In IPv4, IANA uses GLOP addresses to make these types of public assignments, which includes the ASN’s assigned BGP AS number in the group. For IPv4 networks this is the type of group assignment that could be made dynamic, such that the IANA assigned prefix can be used for routing group packets across the Internet to the appropriate organization.

Because IPv6 does not have a GLOP equivalent, the IANA assigned unicast prefix for the organization can be used to make the global group assignment instead. The resulting address is a unique unicast-prefix–based multicast group address block, as defined by RFC 3306. The address provides a way to dynamically assign IPv6 group numbers and also allows organizations to assign private address space, including those who do not have global ASNs, or who receive address assignments from service providers.

In these addresses, the 64 bits provided for the Unicast-Prefix field are sufficient based on the structure of the IPv6 unicast addresses (64 bits are used by the interface ID). The format presented in [Figure 6-6](#) illustrates how any given IPv6 unicast prefix can become a multicast address, with 2^{32} possible groups per unicast prefix. The reserved bits must still always be set to 0, and the unicast prefix will follow. Thus, the address will always start with FF3X:. If an organization was assigned the IPv6 prefix 3F7E:FFFF:1/48, the resulting assignable group block would be F3X:0030:3F7E:FFFF:0001::/96. This gives each organization a unique, private block of group addresses with which to work. It is important to remember that the scope of the unicast-prefix–based multicast address should not exceed that of the embedded unicast prefix.



Group Address = FF3X::/96

Figure 6-6 Unicast-Prefix-Based Group Addressing

IPv6 Source-Specific Addressing

IPv6 source-specific multicast (SSM) group addresses make use of the penultimate four bits of the first hextet (the flag field) and the unicast-prefix-based address assignment rule. IPv6 SSM addresses use a 3 (decimal, or 0011 in binary) as a flag to indicate SSM compatibility. The group addresses used in an SSM deployment model are defined as a subset of the unicast-prefix-based multicast addresses, where the **Prefix Length** and the **Unicast-Prefix** fields are set to 0. [Figure 6-7](#) illustrates how SSM addresses are derived.



SSM Group Address = FF3X::[32-Bit Group ID]

Figure 6-7 IPv6 SSM Addressing

Solicited-Node Multicast Addresses

In addition to these group assignments, IPv6 multicast also incorporates the concept of *solicitation*. Address solicitation occurs when a unicast host sends out a request for a link-local IPv6 group address derived from the local host unicast network. The scope of the request is link-local and is typically used only for control plane functionality. The request derives and generates a multicast prefix for each unicast and IPv6 Anycast prefix assigned to the network.

The solicited-node group address format is FF02::1:FF00:0000/104, where the low-order 24 bits are the same as the unicast or Anycast address that generated it. The derived group address represents a deterministic way to identify the local-link multicast group(s) to which an IPv6 unicast address host

is listening. Thus, if all the requisite host information needed to establish unicast connectivity (the MAC address, and so on) is not available, the destination can still be reached via multicast sent to its solicited-node multicast address.

IPv6 Address Scoping and Schema Considerations

IPv6 addressing has clear advantages over its IPv4 counterpart when it comes to schema design. With multicast addressing, the biggest advantage is the size of the group portion of the address: 112 bits in length (more than four and a half times the size of the configurable portion of an IPv4 group address). In addition, masking on each hex character in the address is simple. This makes creating a unique schema much simpler.

Administrators should also create an organizational address schema for IPv6 multicast group assignments. Wildcard masks serve an identical function for IPv6 addressing. Any meaningful split in the group address section of the address is appropriate.

At press time, most of the Internet still uses IPv4. In fact, few global multicast applications currently use IPv6 multicast. However, with the exponential growth of mobile computing, this is likely to change quickly in the coming years. Mobile computing and the Internet of Everything will require significantly more addressing than IPv4 can provide, and multicast resource efficiency will become a de facto requirement for many applications.

This also means that dynamic and temporary group address assignments may become more commonplace. The IETF has developed an addressing methodology to meet this objective. This methodology is defined in RFC 3307: *Allocation Guidelines for IPv6 Multicast Addresses*. Architects and administrators who are creating an IPv6 group address schema should thoroughly understand RFC 3307. Architects should also consider IETF RFC 3956 (updated by RFC 7371), which describes embedding the rendezvous point (RP) address in an IPv6 multicast address, when creating an IPv6 group address schema. Implementing RFC 3956 can greatly simplify multicast domain boundaries.

Multicast-IPv6-Address-to-MAC-Address Mapping

Like IPv4 multicast addresses, IPv6 group addresses must be mapped into the MAC sublayer for Ethernet transport. Remember, the underlying Layer 2 technology has not changed. Switches will still need to create or segment forwarding domains, flooding domains, and broadcast domains. Obviously, with the expanded address size, this cannot happen with multicast the same way we did it for IPv4.

Does more address space mean more binary math? Fortunately not! Converting an IPv6 multicast address to a multicast MAC is much easier using IPv6, although there is far greater overlap of IPv6 addresses to MAC addresses. The addresses in the organizationally unique identifier (OUI) reserved block for IPv6 multicast MAC addresses are from 0X3333.0000.0000 to 0X3333.FFFF.FFFF.

To map the IPv6 multicast of FF02:0:0:0:0:0:E040:0707 to a MAC multicast, the low-order 32-bits of the IPv6 address are concatenated with the high-order bits of 0X3333, as shown in [Figure 6-8](#).

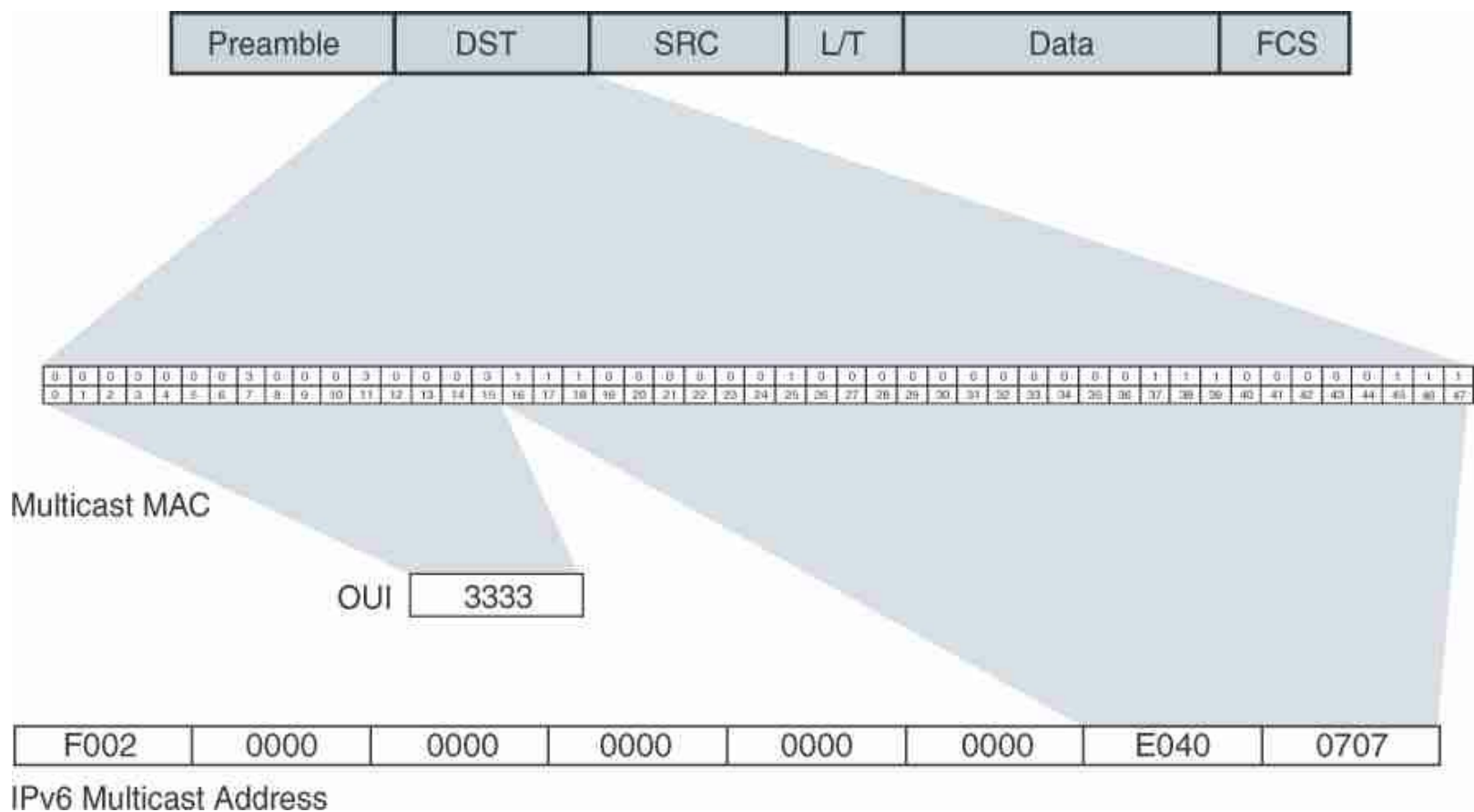


Figure 6-8 *IPv6-Multicast-Group-Address-to-MAC-Address Conversion*

As you can see, this is much easier than converting IPv4 multicast. It also makes for much cleaner multicast topologies. What IPv6 adds in address complexity, it makes up for with simplicity in the network design, or at least that is the intention.

IPv6 Layer 2 and Layer 3 Multicast

In order to establish appropriate and efficient IPv6 multicast communication, both the switching and routing environments must be configured appropriately. Layer 2 devices use Multicast Listener Discovery (MLD) protocol to configure the infrastructure so that it sends multicast messages to the appropriate devices. PIM6 is used at Layer 3, and the components you learned from IPv4 still apply.

Multicast Listener Discovery for IPv6

Through the use of Multicast Listener Discovery (MLD) protocol, IPv6 routers discover on a network the multicast devices that are requesting a multicast stream(s). In other words, the purpose of MLD is to serve as the group subscription method for local segments, in much the same way IGMP provides subscription services to IPv4 hosts and gateways. There are currently two versions of MLD, version 1 and version 2. MLDv1 is the IPv4 equivalent to IGMPv2 and MLDv2 is equivalent to IGMPv3. The IGMP message format was discussed earlier in [Chapter 3](#).

MLD is built on standard Internet Control Messaging Protocol (ICMP), the same protocol that enables traceroute, ping, and echo replies. MLD leverages ICMP messages to carry information about group subscription. This is very unlike IGMP, which has its own messaging format and process. In spite of this difference, both IGMP and MLD provide networks with a way to automatically control and limit the flow of multicast traffic. This control is achieved through the use of special multicast queriers. Queriers and hosts do not perform the same functions. Let us quickly review the role of each:

■ A *querier* is usually a network device, such as a gateway multicast router, that sends query messages to discover which network devices are members of a given multicast group. Queries can be specific to a group (“Are there any listeners for group G on the segment?”), or they can be general (as in, “Are there any listeners for any groups on this segment?”).

■ A *host* is simply a receiver, which can also be routers or other network devices. Hosts send group report messages to inform the querier of group subscriptions.

MLDv1

MLDv1 was originally defined by RFC 2710 and updated by RFCs 3590 and 3810. MLDv1 has functions similar to IGMPv2 and consequently has similar messages and message formats. [Figure 6-9](#) shows the packet header format for MLD messages.

Type	Code	Checksum
Maximum Response Delay		Reserved
Multicast Address		

Figure 6-9 *MLD Message Format*

Note

Because MLD is built on ICMP, the packet header is the same as ICMP and the Type field is used to distinguish the packet as an MLD message. The code field for an MLD message is always set to 0.

Outside of the Multicast Address field, the most important field in the header is the Type field. The type in the message header indicates what type of message is being issued:

■ **Multicast Listener Query:** As previously mentioned, there are two types of queries, both designed to locate multicast hosts on a segment. General queries are sent to the link-local, all-nodes multicast address of FF02::1. The MLD Multicast Address field is set to groups unspecified (::), which should solicit a group report (called a listener report) response from each receiver. A group-specific query is used to identify the listeners for a given group that is listed in the MLD Multicast Address field of the message and is sent to the queried multicast address.

■ MLD queries use a Maximum Response Delay field. This is the maximum allowed delay (in milliseconds) in which a node has to send a report if it has a listener. In all other MLD messages, this field is not used and is therefore set to 0. The Multicast Listener Query message uses a message header Type field equal to 130 (0X82).

■ **Multicast Listener Report:** Hosts use the report message type as a response to an MLD query. The destination group address for a report is the multicast address being reported about. In report and done messages, the Multicast Address field contains the multicast group to which a member listens or the group it is leaving. The Multicast Listener Report message uses a message header Type field equal to 131 (0X83).

■ **Multicast Listener Done:** This message is sent by a node to indicate that it stopped listening to a multicast address. It is the equivalent of an IPv4 IGMP leave group message. The destination group address for a done message is the link-local, all-routers multicast address (FF02::2). It is assumed that the local multicast querier will receive this message, which is also the likely Layer 3 gateway. The Multicast Listener Done message uses a message header Type field equal to 132 (0X84).

Note

All MLD packets are sent with a link-local address as the IPv6 source address. The packet time-to-live (TTL) is set to 1. This is done to ensure the packet is not routed outside of the segment. MLD reports must be sent with a valid IPv6 link-local source address. If the sending interface has not yet acquired a valid link-local address, the unspecified group (::) source address is used. Sending this type of report with an unspecified address is allowed to support the use of IPv6 multicast in the Neighbor Discovery Protocol.

MLDv2

MLD is defined by RFC 3810, and like IPv4 IGMPv3, is designed to improve upon the efficiency of group subscription messaging. It also supports source specific group reports and requests making it capable of supporting SSM, just like the IPv4 counterpart. Outside of these improvements, the message types and formats for MLDv2 are identical to MLDv1 with one notable exception: MLDv2 does not use Listener Done messages.

The biggest difference between the versions is the way MLDv2 improves the Listener Report message. Rather than sending a response for each group, it concatenates a set of records, each record containing information that relates to multicast group address subscriptions. This provides MLDv2 Report messages the capability of leaving a group, thereby eliminating the need for the MLDv1 Done message.

MLDv2 is backward compatible with MLDv1 and can coexist comfortably in the same LAN as MLDv1-only hosts. Keep in mind that when a host using MLD version 1 sends a Done message, the querier router needs to send query messages to reconfirm that this host was the last MLD version 1 host joined to the group before it can stop forwarding traffic, and this process takes about two seconds. This is known as *leave latency* and is also present in IPv4 IGMP version 2-enabled networks.

Configuring MLD and the MLD Message Process

We will use the same network in [Figure 6-10](#) to explain the mechanics of MLD and MLD configuration. In this case, the sender, Sender-1, sends multicast traffic to IPv6 multicast address FF37::7. The receiver, Receiver-1, requests the multicast stream.

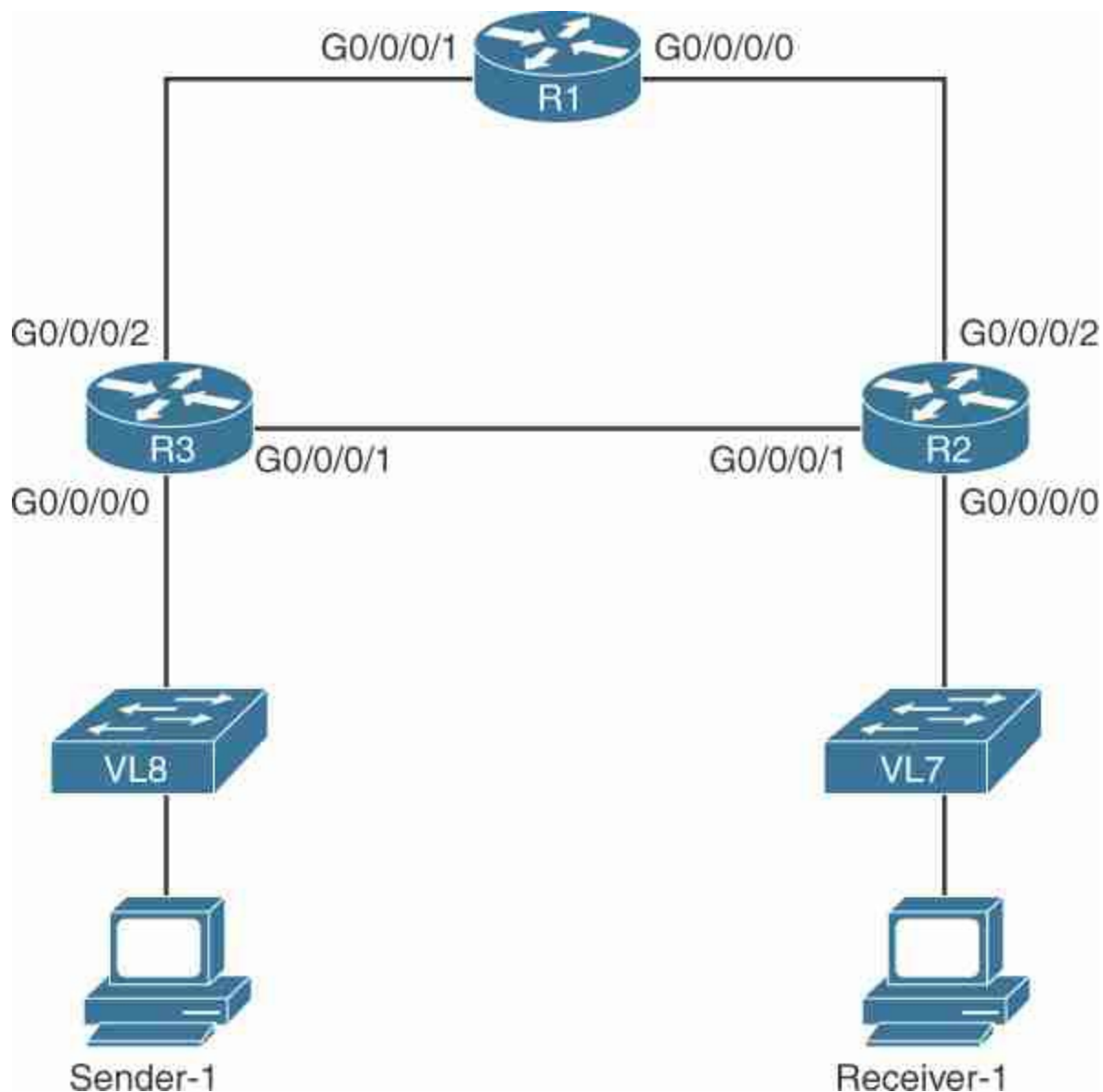


Figure 6-10 *MLD Example Network*

The prerequisite for configuring IPv6 multicast routing requires IPv6 unicast routing to be properly configured and running. Here, we provide the basic MLD configuration, but unicast routing is beyond the scope of this book.

Begin by configuring the router for IPv6 unicast routing in global configuration mode, as shown in the list that follows. In this example, we use IOS-XE. Refer to the command references for other operating systems.

Step 1. Enable IPv6 unicast routing:

[Click here to view code image](#)

```
R1(config)#ipv6 unicast-routing
```

Step 2. Assign IPv6 address to the appropriate interfaces in interface configuration mode, for example:

[Click here to view code image](#)

```
R1(config-if)#ipv6 address 2001:192:168:7::1/64
```

Step 3. Configure the IGP of your choice. You are on your own for this one!

```
R1(config)#router ?
```


Step 4. Enable IPv6 multicast routing in global configuration mode. By default, this enables PIM and MLD on any interfaces configured with an IPv6 address.

[Click here to view code image](#)

```
R1(config)#ipv6 multicast-routing
```

To ease the configuration burden for IPv6, it is assumed that MLD should be enabled on all IPv6-enabled interfaces when **ipv6 multicast-routing** is configured. In addition, many of the same configuration options that are available for IGMPv1-3 are also available for MLD, in particular those configurations that affect MLD on the segment. Examples include joining groups, limiting group addresses that are accepted in a response by ACL, and other query timers/parameters.

The configuration in [Example 6-1](#) shows some of the MLD commands as they would be applied in an IOS-XE router's running configuration file.

Example 6-1 *MLD Interface Commands*

[Click here to view code image](#)

```
!  
interface GigabitEthernet0/0  
ipv6 enable  
ipv6 mld join-group FF37::1 [include | exclude]  
ipv6 mld static-group FF37::1 [include | exclude]  
ipv6 mld access-group <access-list-name>  
ipv6 mld query-timeout <seconds>  
ipv6 mld query-interval <seconds>  
ipv6 mld query-max-response-time <seconds>  
!
```

Note

The include and exclude for the **join-group** and **static-group** commands are used to limit message processing by the gateway router (likely the querier). The **include** parameter allows the administrator to set specific groups for which messages will be received, excluding all others (whitelisting). The **exclude** parameter enforces the opposite logic, specifying addresses to exclude messages from, and accepting all others (blacklisting).

Multicast Listener Discovery Joining a Group and Forwarding Traffic

The list that follows outlines the high-level steps involved for the host to begin receiving the multicast stream.

1. The host (Receiver-1) sends a multicast join message to the group address FF37::7 in this example. Below is a packet capture of the message. There are a few items to note: The destination address for a MLDv2 report messages is FF02::16, and this corresponds to the MAC address of 33:33:00:00:00:16. The IPv6 address of Receiver-1 that sources the messages is a link-local address because it starts with FE80:

[Click here to view code image](#)

```
Ethernet II, Src: 80:ee:73:07:7b:61, Dst: 33:33:00:00:00:16  
Destination: 33:33:00:00:00:16  
Address: 33:33:00:00:00:16
```


.... 1. = LG bit: Locally administered address

(this is NOT the factory default)

.... 1 = IG bit: Group address

(multicast/broadcast)

Source: 80:ee:73:07:7b:61

Address: 80:ee:73:07:7b:61

.... 0. = LG bit: Globally unique address

(factory default)

.... 0 = IG bit: Individual address (unicast)

Type: IPv6 (0x86dd)

Internet Protocol Version 6, Src: fe80::82ee:73ff:fe7:7b61 Dst: ff02::16

0110 = Version: 6

.... 0000 0000 = Traffic class: 0x00000000

.... 0000 0000 0000 0000 0000 0000 = Flowlabel: 0x00000000

Payload length: 36

Next header: IPv6 hop-by-hop option (0)

Hop limit: 1

Source: fe80::82ee:73ff:fe07:7b61

[Source SA MAC: 80:ee:73:07:7b:61]

Destination: ff02::16

Hop-by-Hop Option

Next header: ICMPv6 (58)

Length: 0 (8 bytes)

IPv6 Option (Router Alert)

Type: Router Alert (5)

Length: 2

Router Alert: MLD (0)

IPv6 Option (PadN)

Type: PadN (1)

Length: 0

PadN: <MISSING>

Internet Control Message Protocol v6

Type: Multicast Listener Report Message v2 (143)

Code: 0

Checksum: 0xfec7 [correct]

Reserved: 0000

Number of Multicast Address Records: 1

Multicast Address Record Changed to exclude: ff37::7

Record Type: Changed to exclude (4)

Aux Data Len: 0

Number of Sources: 0

Multicast Address: ff37::7

2. The router receives the report from Receiver-1. The following command verifies the entry:

[Click here to view code image](#)

```
ASR1K-2#show ipv6 mld groups FF37::7 detail
```

```
Interface: GigabitEthernet0/0/0
Group: FF37::7
Uptime: 00:05:10
Router mode: EXCLUDE (Expires: 00:03:55)
Host mode: INCLUDE
Last reporter: FE80::82EE:73FF:FE07:7B61
Source list is empty
```

3. The (*,G) entry and (S,G) entries are added to the multicast routing table, which can be seen using the following command:

[Click here to view code image](#)

```
ASR1K-2#show ipv6 mroute
```

Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected, L - Local, I - Received Source Specific Host Report,
P - Pruned, R - RP-bit set, F - Register flag, T - SPT-bit set,
J - Join SPT

Timers: Uptime/Expires

Interface state: Interface, State

(*, FF05::1:3), 00:18:27/never, RP 2001:192:168::1, flags: SCLJ

Incoming interface: GigabitEthernet0/0/1

RPF nbr: FE80::D68C:B5FF:FE43:1E01

Immediate Outgoing interface list:

GigabitEthernet0/0/0, Forward, 00:18:27/never

(*, FF37::7), 00:04:04/never, RP 2001:192:168::1, flags: SCJ

Incoming interface: GigabitEthernet0/0/1

RPF nbr: FE80::D68C:B5FF:FE43:1E01

Immediate Outgoing interface list:

GigabitEthernet0/0/0, Forward, 00:04:04/never

(2001:192:168:8::10, FF37::7), 00:01:34/00:01:55, flags: SJT

Incoming interface: GigabitEthernet0/0/1

RPF nbr: FE80::D68C:B5FF:FE43:1E01

Inherited Outgoing interface list:

GigabitEthernet0/0/0, Forward, 00:04:04/never

4. The router performs an RPF check and updates the Multicast Forwarding Information Base (MFIB) table with the outgoing interface; this is where Receiver-1 is located and where the outbound interface list (OIF) is added.

The following output displays the IPv6 MFIB table for the multicast address of FF37::7. The Hardware (HW) forwarding entry indicates that traffic is flowing:

[Click here to view code image](#)

```
ASR1K-2#show ipv6 mfib FF37::7 verbose
```

Entry Flags: C - Directly Connected, S - Signal, IA - Inherit A flag,

ET - Data Rate Exceeds Threshold, K - Keepalive

DDE - Data Driven Event, HW - Hardware Installed

I/O Item Flags: IC - Internal Copy, NP - Not platform switched,

NS - Negate Signalling, SP - Signal Present,

A - Accept, F - Forward, RA - MRIB Accept, RF - MRIB Forward,

MA - MFIB Accept

Forwarding Counts: Pkt Count/Pkts per second/Avg Pkt Size/Kbits per second

Other counts: Total/RPF failed/Other drops

I/O Item Counts: FS Pkt Count/PS Pkt Count

Default

(*FF37::7) Flags: C K HW

0x9E OIF-IC count: 0, OIF-A count: 1

SW Forwarding: 0/0/0/0, Other: 0/0/0

HW Forwarding: 48/0/1390/0, Other: 0/0/0

GigabitEthernet0/0/1 Flags: RA A MA NS

GigabitEthernet0/0/0 Flags: RF F NS

CEF: Adjacency with MAC: 33330000000700225561250086DD

Pkts: 0/0

(2001:192:168:8::10,FF37::7) Flags: K HW DDE

0x9F OIF-IC count: 0, OIF-A count: 1

SW Forwarding: 0/0/0/0, Other: 0/0/0

HW Forwarding: 61559/278/1390/3018, Other: 0/0/0

GigabitEthernet0/0/1 Flags: RA A MA

GigabitEthernet0/0/0 Flags: RF F NS

CEF: Adjacency with MAC: 33330000000700225561250086DD

Pkts: 0/0

Leaving a MLD Group

When Receiver-1 is no longer interested in the multicast traffic, a MLD report message is sent to be removed from the group, as depicted in the packet capture in [Example 6-2](#) .

Example 6-2 MLD Leave Group Message Received

[Click here to view code image](#)

Ethernet II, Src: 80:ee:73:07:7b:61, Dst: 33:33:00:00:00:16

Destination: 33:33:00:00:00:16

Source: 80:ee:73:07:7b:61

Type: IPv6 (0x86dd)

Internet Protocol Version 6, Src: fe80::82ee:73ff:fe07:7b61, Dst: ff02::16

0110 = Version: 6

.... 0000 0000 = Traffic class: 0x00000000

.... 0000 0000 0000 0000 0000 = Flowlabel: 0x00000000

Payload length: 36

Next header: IPv6 hop-by-hop option (0)

Hop limit: 1

Source: fe80::82ee:73ff:fe07:7b61

Destination: ff02::16

Hop-by-Hop Option

Next header: ICMPv6 (58)

Length: 0 (8 bytes)

IPv6 Option (Router Alert)

IPv6 Option (PadN)

Internet Control Message Protocol v6

Type: Multicast Listener Report Message v2 (143)

Code: 0

Checksum: 0xffc7 [correct]

Reserved: 0000

Number of Multicast Address Records: 1

Multicast Address Record Changed to include: ff37::7

Record Type: Changed to include (3)

Aux Data Len: 0

Number of Sources: 0

Multicast Address: ff37::7

Multicast Listener Discovery (MLD) Snooping

MLD snooping for IPv6 works in much the same way as IGMP snooping for IPv4. The overall objective is to minimize the delivery of multicast traffic to devices that are not interested in receiving it on a common Layer 2 flooding domain. Remember, MLDv1 is the IPv4 equivalent to IGMPv2, and MLDv2 is equivalent to IGMPv3.

Note

MLDv1 and MLDv2 are product- and software-specific; please refer to product documentation to determine support.

Let us examine MLD snooping configuration and mechanics. As shown in [Figure 6-11](#), Host A and Host B are receiving an IPv6 multicast stream from FF02::E040:707.

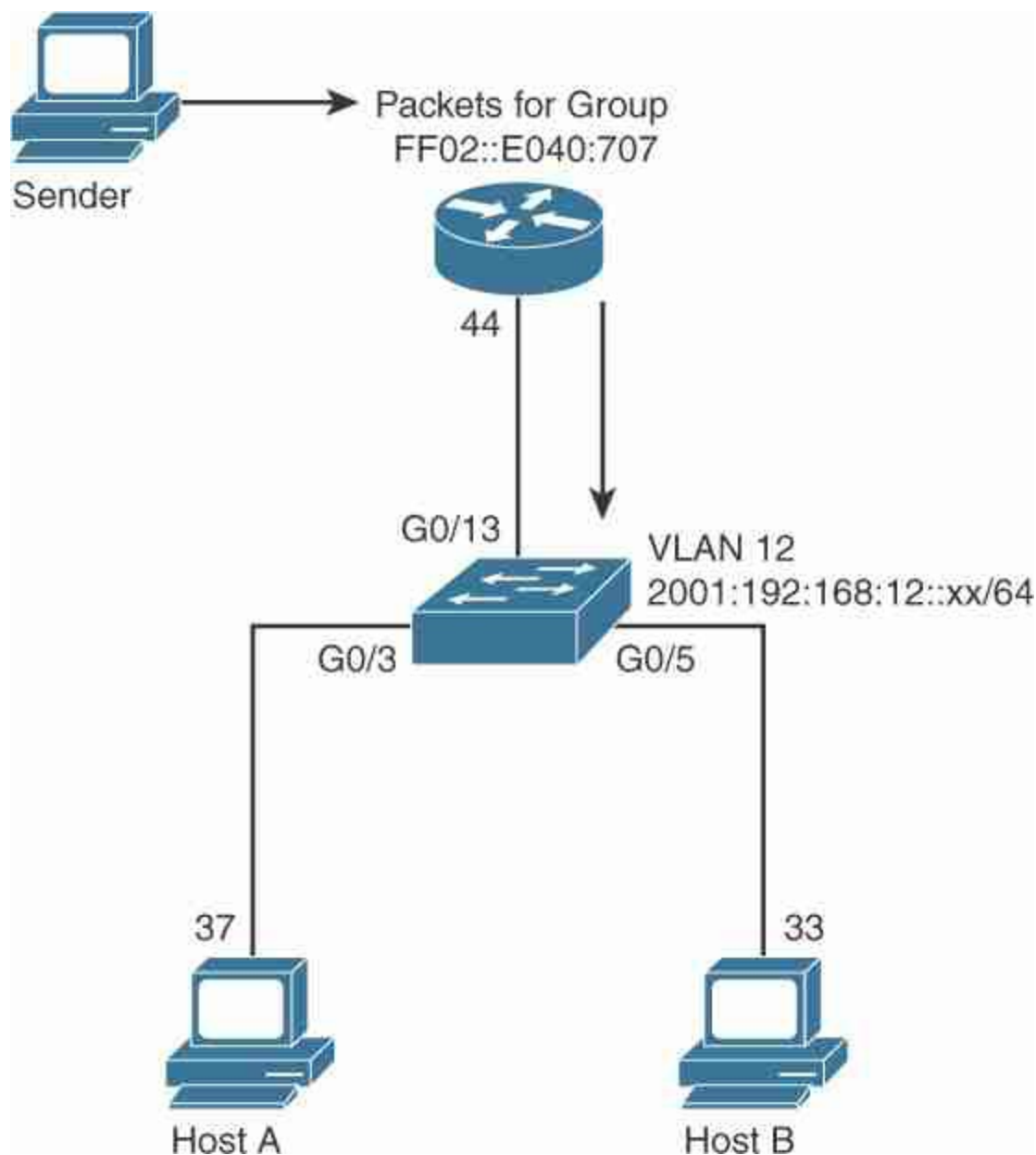


Figure 6-11 *MLD Snooping Example*

Configuring MLD Snooping

The first step in configuring MLD snooping is to make sure your switch supports IPv6 MLD. Some switches support MLD enablement out of the box. Other switches might need to have IPv6-switching capabilities added. For example, some older IOS switches use a switch database management (SDM) template to allocate switching resources to various protocols. If SDM templates are used, you can see the template assignment by issuing the **show sdm prefer** command, as demonstrated in [Example 6-3](#).

Example 6-3 Verifying IPv6 MLD Support

[Click here to view code image](#)

```
Switch#show sdm prefer
```

The current template is "desktop IPv4 and IPv6 routing" template.
The selected template optimizes the resources in the switch to support this level of features for 8 routed interfaces and 1024 VLANs.

number of unicast mac addresses:	1.5K
number of IPv4 IGMP groups + multicast routes:	1K

number of IPv4 unicast routes:	2.75K
number of directly-connected IPv4 hosts:	1.5K
number of indirect IPv4 routes:	1.25K
number of IPv6 multicast groups:	1K
number of directly-connected IPv6 addresses:	1.5K
number of indirect IPv6 unicast routes:	1.25K
number of IPv4 policy based routing aces:	0.25K
number of IPv4/MAC qos aces:	0.5K
number of IPv4/MAC security aces:	0.5K
number of IPv6 policy based routing aces:	0.25K
number of IPv6 qos aces:	0.5K
number of IPv6 security aces:	0.5K

The switch database management (SDM) templates allow you to change the behavior of the switch. In order to support IPv6 multicast, you must select a template that supports IPv6. If your switch does not support IPv6 multicast, you will need to change the template and reboot the switch, as demonstrated in [Example 6-4](#) .

Note

Always check the specifications, requirements, and limitations of your switching platform. Not all platforms support every IPv6 feature.

Example 6-4 *SDM Template for IPv6 and IPv4 Support*

[Click here to view code image](#)

```
C3560E(config)#sdm prefer dual-ipv4-and-ipv6 routing
```

Changes to the running SDM preferences have been stored, but cannot take effect until the next reload.

Use 'show sdm prefer' to see what SDM preference is currently active.

Now that IPv6 is enabled, you can complete the remainder of the configuration. Enable IPv6 MLD using the following command:

[Click here to view code image](#)

```
C3560E(config)#ipv6 mld snooping
```

Using the **show ipv6 mld snooping mrouter** command, you can verify the attached router has been detected as demonstrated in [Example 6-5](#) .

Example 6-5 *Show ipv6 mld snooping mrouter Command Output*

[Click here to view code image](#)

```
C3560E#show ipv6 mld snooping mrouter
```

```
Vlan  ports
```

```
----
```

```
12  Gi0/13(dynamic)
```

The output in [Example 6-6](#) shows Host A connected to Gi0/3 and Host B connected to Gi0/5, and both are accessing the FF02::E040:707 multicast stream.

Example 6-6 *Show MLD Snooping Per VLAN*

[Click here to view code image](#)


```
C3560E#show ipv6 mld snooping address vlan 12
```

Vlan	Group	Type	Version	Port List
12	FF02::E040:707	mld	v2	Gi0/3, Gi0/5, Gi0/13

The **debug ipv6 mld** provides a great deal of information. The following output shows the MLD report from Host B:

[Click here to view code image](#)

MLDSN: Processing v2 Report as a MLDv1 Report for group FF02::E040:707 on port Gi0/5 in vlan 12

When Host B leaves the multicast group, the following debug message is displayed:

[Click here to view code image](#)

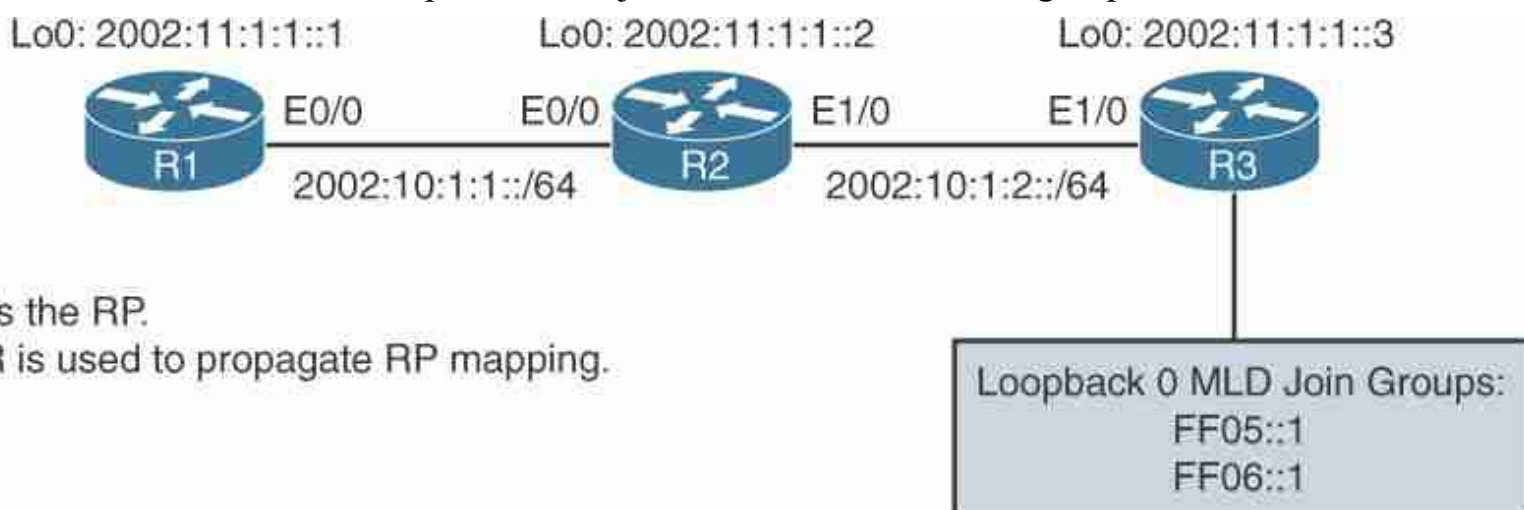
MLDSN: Processing v2 Report as MLDv1 Leave for group FF02::E040:707 on port Gi0/5 in vlan 12

IPv6 Layer 3 Multicast and Protocol Independent Multicast 6 (PIM6)

It's important to remember that although some of the protocols and rules have changed, and the addresses have changed between IP versions 4 and 6, for the most part it is all just Internet Protocol (IP). That means the PIM rules and features that apply to IPv4 multicast also apply to IPv6 multicast. We discussed how MLD replaces IGMP in the IPv6 network. For PIM, the other important multicast infrastructure protocol, the IETF drafted new PIMv2 rules to handle IPv6 addresses and packets but essentially left PIMv2 intact.

The important aspect to remember is that no new multicast routing protocol is required. When we use PIM in an IPv6 network, with IPv6 rules enabled, we sometimes call it PIM6. This is not a new protocol or protocol version, but again, indicates IPv6 support in PIMv2 implementations. This allows both IPv4 and IPv6 to reside simultaneously on the same infrastructure while still supporting multicast for each.

To illustrate this point, we take a look at some of the same output we examined in earlier chapters regarding how basic Layer 3 multicast operations work. In this case, we use a new IPv6 network without IPv4-enabled, running PIM sparse-mode (PIM-SM) on Cisco IOS-XE. We keep the network very simple, with only three routers (R1–R3), as shown in [Figure 6-12](#). R1 is the RP and is using BSR to advertise itself. R3 Loopback 0 has joined two IPv6 multicast groups, FF05::1 and FF06::1.



R1 is the RP.
BSR is used to propagate RP mapping.

Figure 6-12 Example IPv6 Multicast Network

Now, let us examine the multicast state table on R2, where the forwarding path and the reverse path converge. For IPv4 we would use the command **show ip mroute** to see the IPv4 multicast state table. To collect the equivalent IPv6 output, we need only replace **ip** with **ipv6**. The command is **show ipv6 mroute**, as demonstrated in [Example 6-7](#).

Example 6-7 Displaying the IPv6 Multicast State Table

[Click here to view code image](#)

```
R2#show ipv6 mroute
```

Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,

C - Connected, L - Local, I - Received Source Specific Host Report,

P - Pruned, R - RP-bit set, F - Register flag, T - SPT-bit set,

J - Join SPT

Timers: Uptime/Expires

Interface state: Interface, State

```
(2002:11:1:1::1, FF35::1), 00:36:40/00:02:47, flags: ST
```

```
Incoming interface: Ethernet0/0
```

```
RPF nbr: FE80::A8BB:CCFF:FE00:100
```

```
Immediate Outgoing interface list:
```

```
Ethernet1/0, Forward, 00:36:40/00:02:47
```

```
(2002:11:1:1::1, FF36::1), 00:46:23/00:02:47, flags: STI
```

```
Incoming interface: Ethernet0/0
```

```
RPF nbr: FE80::A8BB:CCFF:FE00:100
```

```
Immediate Outgoing interface list:
```

```
Ethernet1/0, Forward, 00:36:40/00:02:47
```

The command output is essentially identical to the IPv4 output. First we are given the table flags indicating the PIM type and other tree information. The protocol and state flags should be very familiar because they are the same as those in IPv4. Next, we are given the states of the two (S,G) entries for groups FF05::1 and FF06::1. Notice how the state information in the output includes the RPF neighbor information as well as the outgoing interface list (OIL).

What might strike some administrators as odd is the RPF neighbor information. Take a closer look at the output for the (2002:11:1:1::1, FF35::1) state entry, paying attention to the highlighted portions.

The OIL and the RPF neighbor in this case are IPv6. However, the process for determining this information is the same as the process for IPv4. The only difference is that the IPv6 routing table (RIB) and forwarding table (FIB) are used to derive the OIL and RPF check information and populate the v6 MRIB and MFIB respectively. The source, 2002:11:1:1::1 in the (S,G) for each entry, is the loopback for R1.

If we show the RPF information for the source address on R2, using the **show ipv6 rpf** command (again, same command, just replace **ip** with **ipv6**), we see that the unicast address 2002:11:1:1::1 is in fact located out router interface Ethernet0/0, but that the RPF neighbor address is not the configured interface address of R1 located out E0/0, as shown in [Example 6-8](#).

Example 6-8 RPF for IPv6 Multicast

[Click here to view code image](#)

```
R2#show ipv6 rpf 2002:11:1:1::1
```

```
RPF information for 2002:11:1:1::1
```

```
RPF interface: Ethernet0/0
```

```
RPF neighbor: FE80::A8BB:CCFF:FE00:100
```

```
RPF route/mask: 2002:11:1:1::1/128
```

```
RPF type: Unicast
```

```
RPF recursion count: 0
```

```
Metric preference: 120
```

```
Metric: 2
```

Instead, the RPF address is the link-local address configured for R1's E0/0 interface. The IPv6 unicast route table (RIB) and CEF table (FIB) show us where this RPF information was derived from, by using the **show ipv6 route** and **show ipv6 cef** commands, as shown in [Example 6-9](#).

Example 6-9 *IPv6 RIB and FIB*

[Click here to view code image](#)

```
R2#show ipv6 route
```

```
IPv6 Routing Table - default - 9 entries
```

```
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
```

```
      B - BGP, M - MIPv6, R - RIP, I1 - ISIS L1
```

```
      I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
```

```
      EX - EIGRP external, ND - Neighbor Discovery
```

```
      O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
```

```
      ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
```

```
R  2002:11:1:1::1/128 [120/2]
```

```
  via FE80::A8BB:CCFF:FE00:100, Ethernet0/0
```

```
R2#show ipv6 cef
```

```
2002:11:1:1::1/128
```

```
  nexthop FE80::A8BB:CCFF:FE00:100 Ethernet0/0
```

```
FF00::/8
```

```
  multicast
```

As you can see, the **nexthop** information for the RPF check uses the link-local address of the neighboring router (R1) from the RIB and FIB as the forwarding address. That makes the link-local address of R1 the RPF neighbor, even though the global routes are learned via the configured global interface addresses (the **RPF route** from the **show ipv6 rpf** output). This is a function of IPv6 addressing rules, and it may require that you spend some time double-checking local connectivity in an IPv6 PIM domain.

You can also see that many protocols, including PIM, use the link-local addresses to form neighbor adjacencies and exchange protocol information. Look at the **show ipv6 pim neighbor** command on R2 in [Example 6-10](#). Here we find the neighbor relationships are established between the link-local addresses of the routers.

Example 6-10 *IPv6 PIM Neighbors Table*

[Click here to view code image](#)

```
R2#show ipv6 pim neighbor
```

PIM Neighbor Table

Mode: B - Bidir Capable, G - GenID Capable

Neighbor Address	Interface	Uptime	Expires	Mode	DR	pri
FE80::A8BB:CCFF:FE00:100	Ethernet0/0	00:46:37	00:01:16	B G		1
FE80::A8BB:CCFF:FE00:301	Ethernet1/0	00:36:52	00:01:33	B G	DR	1

We can see that R1, located out Ethernet0/0 has a link-local address of FE80::A8BB:CCFF:FE00:100, the same as that found in the RPF information. To find the link-local address of a given interface—in this case, Ethernet0/0 on R1—simply use the **show ipv6 interface [type (port/slot)]** command. If we issue this command, **show ipv6 interface Ethernet0/0** on R1, we find consistency with the table outputs from R2, as demonstrated in [Example 6-11](#) .

Example 6-11 *Finding the Link-Local Address*

[Click here to view code image](#)

```
R1#show ipv6 interface e0/0
```

Ethernet0/0 is up, line protocol is up

IPv6 is enabled, link-local address is FE80::A8BB:CCFF:FE00:100

No Virtual link-local address(es):

Global unicast address(es):

2002:10:1:1::1, subnet is 2002:10:1:1::/64

Joined group address(es):

FF02::1

FF02::2

FF02::9

FF02::D

FF02::16

FF02::1:FF00:1

FF02::1:FF00:100

FF06::1

MTU is 1500 bytes

ICMP error messages limited to one every 100 milliseconds

ICMP redirects are enabled

ICMP unreachable are sent

Output features: MFIB Adjacency

ND DAD is enabled, number of DAD attempts: 1

ND reachable time is 30000 milliseconds (using 42613)

ND advertised reachable time is 0 (unspecified)

ND advertised retransmit interval is 0 (unspecified)

ND router advertisements are sent every 200 seconds

ND router advertisements live for 1800 seconds

ND advertised default router preference is Medium

Hosts use stateless autoconfig for addresses.

The important component to understand is that the IPv6 RIB and FIB are used for RPF loop prevention and state population in the MRIB and MFIB of the router. This is the same process as the

one used for IPv4, and it is consistent with the rules of PIMv2 as laid out by IETF RFCs. Almost any command that you can issue for IPv4 multicast you can issue for IPv6 multicast. Most of the commands simply require a change in the syntax from **ip** to **ipv6** . This is true even of configuration commands in IOS.

Let us look at the basic PIM and interface configuration of R1 in [Example 6-12](#) to illustrate this point more fully.

Example 6-12 *R1 PIM and Interface Configuration*

[Click here to view code image](#)

```
hostname R1
!
!
ip cef
ipv6 unicast-routing
ipv6 cef
ipv6 multicast-routing
!
!
interface Loopback0
no ip address
ipv6 address 2002:11:1:1::1/128
ipv6 enable
ipv6 router isis TEST
!
interface Loopback100
no ip address
ipv6 address 2002:100:1:1::1/128
ipv6 enable
ipv6 router isis TEST
!
interface Ethernet0/0
no ip address
ipv6 address 2002:10:1:1::1/64
ipv6 enable
ipv6 router isis TEST
!
!
router isis TEST
net 49.0001.0000.0001.00
!
!
ipv6 pim bsr candidate bsr 2002:11:1:1::1
ipv6 pim bsr candidate rp 2002:11:1:1::1 group-list ACL priority 190
!
!
```

```
!
ipv6 access-list ACL
permit ipv6 any FF05::/64
```

In this configuration, we enabled PIM on the relevant interfaces (Ethernet0/0, Loopback0, and Loopback100) by simply enabling **ipv6 multicast routing**, as previously mentioned. The **show running-config all** command (see [Example 6-13](#)), shows us that this automatically configures PIM-SM for IPv6 using the command **ipv6 pim sparse-mode**, along with many other IPv6 multicast commands, as shown here for interface Ethernet0/0. (Remember that sparse-mode with or without SSM support is the only option for PIM6, and consequently the command to enable it per interface is simply **ipv6 pim**.)

Example 6-13 *Finding Automatic PIM Configuration for IPv6*

[Click here to view code image](#)

```
R1#show running-config all
!
interface Ethernet0/0
no ip address
ipv6 address 2002:10:1:1::1/64
ipv6 enable
ipv6 nd reachable-time 0
ipv6 nd ns-interval 0
ipv6 nd dad attempts 1
ipv6 nd prefix framed-ipv6-prefix
ipv6 nd nud igp
ipv6 nd ra lifetime 1800
ipv6 nd ra interval 200
ipv6 redirects
ipv6 unreachable
ipv6 mfib forwarding input
ipv6 mfib forwarding output
ipv6 mfib cef input
ipv6 mfib cef output
ipv6 mld query-max-response-time 10
ipv6 mld query-timeout 255
ipv6 mld query-interval 125
ipv6 mld router
ipv6 pim hello-interval 30
ipv6 pim dr-priority 1
ipv6 pim join-prune-interval 60
ipv6 pim
ipv6 router isis TEST
!
```

Note

Once again, enabling IPv6 multicast in IOS/XE is rudimentary. Unlike IPv4 multicast configuration,

when the **ipv6 multicast-routing** command is issued, all IPv6 configured interfaces on the router are automatically enabled for PIM6; no other configuration is necessary. The **ipv6 pim** command is enabled on the interface and does not appear in the interface configuration. To disable PIM on a particular interface, use the command **no ipv6 pim** . For routers and switches running NX-OS, you must configure PIM6 on each interface, but only after the PIM6 and PIM features are enabled using the commands **feature pim** and **feature pim6** . In IOS-XR PIM6 can be enabled globally or on an individual interface basis using the **router pim** configuration mode, the same as with IPv4, but using the **address-family ipv6** command option. For all operating systems, a PIM6-enabled interface always operates in sparse-mode (dense-mode operation is not supported). IOS-XR PIM6 features are all configured in a similar manner, under the **router pim** and **address-family** configuration sub-modes.

In addition, we have enabled PIM BSR for IPv6. This was accomplished using the same command as BSR for IPv4 (**ip pim bsr candidate bsr address**), except we replace the **ip** syntax with **ipv6** . This makes configuring IPv6 multicast a cinch, and it enables dual stack networking (running IPv4 and IPv6 natively over the same infrastructure) with ease. IPv4 multicast configuration complexities and interface mode management confusion have been eliminated in PIM6.

PIM6 Static mroute Entries

Traffic engineering is also a feature of PIM6. IPv6 static state entries (mroutes) behave in a similar manner to IPv4 static mroutes. IPv6 static mroutes share the same database as IPv6 static routes and are implemented by extending static route support. This is very different from the way static mroutes are configured. Static mroute entries support equal-cost multipath routing for multicast forwarding, as well as unicast forwarding.

Static mroutes for IPv6 multicast are configured as an extension of IPv6 static routes. An IPv6 static route can be configured for unicast routing only, static multicast route for multicast RPF selection only, or to use a static route for both unicast routing and multicast RPF selection. This is accomplished using the **command ipv6 route address/mask interface-type slot/port [multicast | unicast]**. If no keyword (multicast or unicast) is specified in the IPv6 static route, it is used for both unicast routing and multicast RPF selection.

The following is an example configuration for adding a static entry on R2 for RPF of source 2001:DDBB:200::10/128.

[Click here to view code image](#)

!

```
ipv6 route 2001:DDBB:200::10/128 Ethernet0/1 [multicast | unicast ]
```

!

Both IPv6 multipath-multicast and tunneling are supported on all Cisco platforms. These can come in very handy for v6 installs, for the same reasons they do for IPv4 networks. Tunneling can also be especially useful for crossing a non-PIM6 or non-IPV6-enabled boundary. This is common in some IPv6 networks, where IPv6 is deployed at the edge but tunneled over an IPv4 core network. In these situations, there are ways to propagate the multicast information from IPv6 across the IPv4 infrastructure, but many may find it easier to simply configure IPv6-enabled tunnels, where appropriate.

In addition to static entries for tunneling and multipath routing, IPv6 MP-BGP address-family

information (AFI) can be used for PIM6 traffic engineering. Multicast BGP with the IPv6 multicast address family provides multicast BGP extensions for IPv6 and supports the same features and functionality as IPv4 BGP. IPv6 enhancements to multicast BGP include support for an IPv6 multicast address family, network layer reachability information (NLRI), and next-hop (the next router in the path to the destination) attributes that use IPv6 addresses. A subsequent address family identifier (SAFI) provides information about the type of network layer reachability information that is carried in the attribute. Multiprotocol BGP unicast uses SAFI 1 messages, and multiprotocol BGP multicast uses SAFI 2 messages.

The IPv6 multicast address family can carry routes used for RPF lookup by the IPv6 PIM protocol, and multicast BGP IPv6. Users must use IPv6 multicast BGP AFI when using IPv6 multicast with BGP because the IPv6 unicast BGP learned routes will not be used for IPv6 multicast RPF. A separate BGP routing table is maintained to configure incongruent policies and topologies (for example, IPv6 unicast and multicast) by using IPv6 multicast RPF lookup. Multicast RPF lookup is very similar to the IP unicast route lookup.

PIM6 Group Modes

In Cisco IOS Software, each IP multicast group operates in exactly one mode of PIM:

- **General any-source multicast (ASM):** The group ranges specified in Generic Multicast Group Addresses Definition and Unicast-Prefix–Based Addresses will by default always be handled as sparse-mode groups. If Cisco IOS Software is not aware of an RP for a group, then a default group mode is used (sparse-mode for IPv6, for example).
- **Source-specific multicast (SSM):** The SSM group range (SSM address range) is hard-coded and will always be handled in the SSM mode (no wildcard joins processed, no RP definition for the group range required).
- **Embedded RP groups:** Embedded group ranges (embedded RP addresses) cannot be predefined (the RP address is not preliminarily known) in the SSM sense, but the router interprets the embedded RP group address only when the first PIM joins for that group appear or the first data appears from the directly connected source; prior to that, it is not possible to determine any RP for the group. Embedded RP is explained in the next section, but it is introduced here to complete the group mode discussion.

The group mode can be displayed using the **show ipv6 pim range-list** command (PIM range for an output example). By default, when multicast routing is enabled, the group modes are listed in [Table 6-3](#).

ffX[0-2]::/16	Non-Routable
ff3X::/32	Unicast-prefix–based groups and SSM
ff00::/8	None

Table 6-3 *IPv6 Group Modes*

PIM6 RP Options

One element where there is less similarity between IPv4 PIM and PIM6 are the options that exist for configuring RPs in ASM mode. As shown in the previous network, PIM6 supports BSR. It also supports static RP configurations as well as the use of Anycast RP. The configurations for these are

nearly identical to the configurations for IPv4 (as we showed with BSR RP).

However, the IETF developed a new, open standard method of embedding RP information within multicast packets as an easier alternative for performing dynamic group mappings. This is known as *embedded RP* and is defined by IETF RFC 3956. Cisco did not extend proprietary Auto-RP support to PIM6. Consequently, there is no support for Auto-RP in an IPv6 multicast network. [Table 6-4](#) provides you a complete overview of the available methods to configure RP group mappings, with a comparison between IPv4 and IPv6 PIM.

	Static RP	BSR	Auto-RP	Embedded RP
Configuration required on every router?	Yes	No (only candidate-BSRs and candidate-RPs)	No (only candidate-RPs and mapping agents)	No (except RP routers)
IPv4 support?	Yes	Yes	Yes	No
IPv6 support?	Yes	Yes	No	Yes
RP redundancy?	No (unless Anycast RP is used)	Yes	Yes	No

Table 6-4 *PIM RP Options Compared*

As shown in [Table 6-4](#) , if the domain is not using embedded RP, BSR is the only other dynamic RP propagation and mapping option. Again, there is no Auto-RP support for IPv6. PIM6 devices in a domain must still be able to map each multicast group to the correct RP address regardless of IP version. With the PIM6 BSR feature, multicast routing devices will detect an unreachable RP. Any mapping tables will be modified after the failure so that the unreachable RP is no longer used. If failover is configured within the BSR domain using priority, new tables will be rapidly distributed throughout the domain when the primary RP failure has fully converged.

As shown earlier, the configuration for BSR is nearly identical to IPv4, changing only the key syntax word **ip** to **ipv6** . The domain still needs at least one of each, a candidate RP and a candidate BSR, configured in the domain. The PIM BSR messages are propagated through the network in the same way, except using the link-local IPv6 PIM connection instead of the IPv4 PIM network.

Just as with IPv4 PIM, PIM6 sparse-mode multicast groups need to associate with the IP or IPv6 address of an RP. If the domain is running dual-stack (simultaneous configuration of IPv4 and IPv6) an IPv4 RP can work as the RP for PIM6 managed groups. This is one of the big advantages of the IETF using the same PIM version (version 2) for both IPv4 and IPv6. When a new IPv6 multicast source starts sending multicast packets, its local gateway router DR encapsulates these data packets in a PIM register message and sends them to the RP for that multicast group. When a new multicast receiver joins a group, the local gateway DR sends a PIM join message to the RP for that multicast group. This functionality does not change, and the role of the RP is the same as it is for IPv4.

Don't forget the best practice and security recommendations for PIM domains that we discussed in the previous chapter. These all still apply for PIM6 domains. It is especially important to limit the

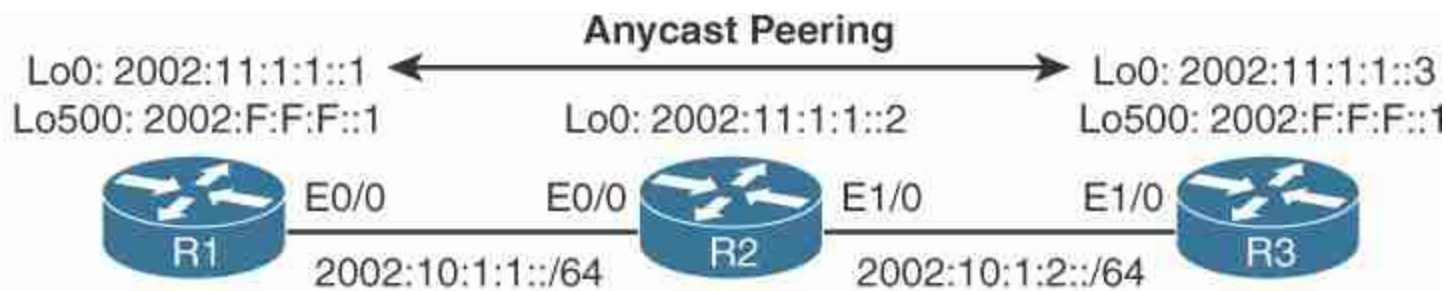
scope of the PIM6 domain. When using BSR, for example, make sure you have a well-defined BSR borders using the **ipv6 pim bsr border** command and well-defined domain boundaries using the **ipv6 multicast boundary** command.

PIM6 Anycast RP

IPv6 PIM supports Anycast services for PIM-SM RP in the same manner that IPv4 PIM does with one major difference: There is no MSDP or equivalent protocol for IPv6, and therefore it cannot be used in PIM6 Anycast RP. This also means that Anycast RP can only be used inside a domain that runs PIM only and cannot be propagated beyond a domain border or boundary. Outside of MSDP, the other principles of Anycast RP from IPv4 PIM still apply. It allows receivers and sources to rendezvous to the closest RP. Packets from a source must be replicated by the receiving RP to all other RPs to find joined receivers and reliably create both the shared tree and source tree. The unicast RP address can still be configured statically on each router or distributed using a dynamic protocol to all PIM6 devices throughout the domain. The mechanics of Anycast RP for PIM6 are the same as those introduced for IPv4 in [Chapter 4](#) based on RFC 4610.

Remember that Anycast RP offers the PIM domain robust RP services along with fast convergence when a PIM RP device fails. This is considered ideal for any IPv6 multicast services that are mission critical or of high importance to the organization where PIM6 is deployed. Let us examine the IOS-XE configuration for PIM6 Anycast RP. It is of course very similar to the non-MSDP configuration option for IPv4 and works using the same principles.

In this configuration example, we use the same three router network we used for the previous BSR configuration example. R1 still acts as an RP, but R3 will also be added to the RP set using Anycast RP global-scope address 2002:F:F:F::1/128 (Loopback 500 on each RP router). All routers (including the RP) will be configured with a static RP address command. Examine [Figure 6-13](#) to see how this configuration is arranged.



R1 and R3 are anycast RP peers.
The anycast RP address is configured statically on **ALL** domain routers.

Figure 6-13 PIM6 Anycast RP Example

Note

Normally, in a network of this small size, Anycast RP would not be recommended. We are using a three-node network to simplify the configurations for demonstration. Anycast RP is ideal for large multicast domains in which reliability and fast convergence are a must.

[Example 6-14](#) shows the router configurations used to support this Anycast RP deployment.

Example 6-14 Configuration to Support Anycast RP Deployment

[Click here to view code image](#)

RP: R1

```
hostname R1
!
ipv6 unicast-routing
ipv6 cef
ipv6 multicast-routing
!
interface Loopback0
no ip address
ipv6 address 2002:11:1:1::1/128
ipv6 enable
ipv6 router isis TEST
!
interface Loopback500
no ip address
ipv6 address 2002:F:F:F::1/128
ipv6 enable
ipv6 router isis TEST
!
interface Ethernet0/0
no ip address
ipv6 address 2002:10:1:1::1/64
ipv6 enable
ipv6 router isis TEST
!
router isis TEST
net 49.0001.0000.0001.00
!
ipv6 pim anycast-rp 2002:F:F:F::1 2002:11:1:1::3
ipv6 pim rp-address 2002:F:F:F::1 group-list ACL
!
ipv6 access-list ACL
permit ipv6 any FF05::/64
```

RP: R2

```
hostname R2
!
ipv6 unicast-routing
ipv6 cef
ipv6 multicast-routing
!
interface Loopback0
no ip address
ipv6 address 2002:11:1:1::2/128
```

```
ipv6 enable
ipv6 router isis TEST
!
interface Ethernet0/0
no ip address
ipv6 address 2002:10:1:1::2/64
ipv6 enable
ipv6 router isis TEST
!
interface Ethernet1/0
no ip address
ipv6 address 2002:10:1:2::1/64
ipv6 enable
ipv6 router isis TEST
!
router isis TEST
net 49.0002.0000.0002.00
!
ipv6 pim rp-address 2002:F:F:F::1 group-list ACL
!
ipv6 access-list ACL
permit ipv6 any FF05::/64

R3
hostname R3
!
ipv6 unicast-routing
ipv6 cef
ipv6 multicast-routing
!
interface Loopback0
no ip address
ipv6 address 2002:11:1:1::3/128
ipv6 enable
ipv6 router isis TEST
!
interface Loopback500
no ip address
ipv6 address 2002:F:F:F::1/128
ipv6 enable
ipv6 router isis TEST
!
interface Ethernet1/0
no ip address
ipv6 address 2002:10:1:2::2/64
ipv6 enable
```



```

ipv6 router isis TEST
!
router isis TEST
 net 49.0002.0000.0003.00
!
ipv6 pim anycast-rp 2002:F:F:F::1 2002:11:1:1::1
ipv6 pim rp-address 2002:F:F:F::1 group-list ACL
!
ipv6 access-list ACL
 permit ipv6 any FF05::/64
!

```

Embedded RP

Embedded RP defines an IPv6-unique address allocation policy in which the address of the RP is encoded in an IPv6 unicast-prefix-based group address. This deployment greatly simplifies intra-domain multicast designs, configurations, and operations. The rules for embedded RP are really quite simple.

RFC3956 specifies rendezvous point (RP) address encoding and is accomplished by adding a new field for the RP address and by defining a new flag in the Flgs field. The group address should start with FF7X::/12, where the flag value of 7 means embedded RP. [Figure 6-14](#) illustrates the format of the group.

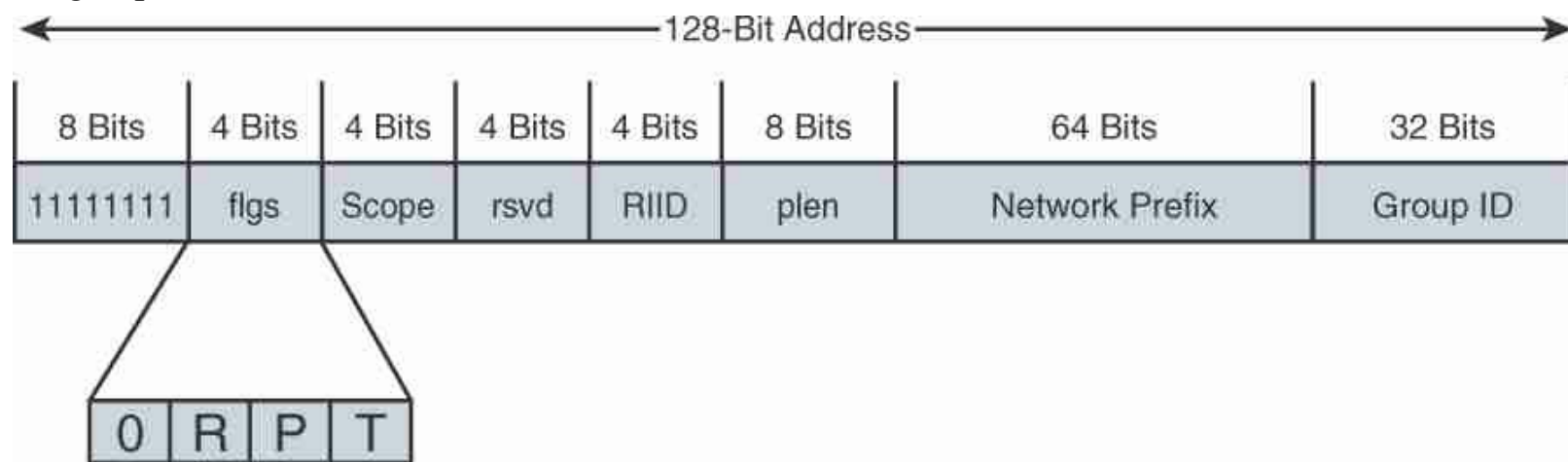


Figure 6-14 PIM6 Embedded RP Group Encoding Format

Elements in [Figure 6-14](#) related to PIM6 Embedded RP:

- **Binary Bits:** 11111111 at the start of the address identifies the address as being a multicast address.

- **Flgs:** The RFC redefines the four-bit “flag” field with the following option for using embedded RP:

R = 1 indicates this is an embedded RP multicast address and contains the address of the PIM-SM RP. When R = 1, P *must be* set to 1, and consequently T *must* also be set to 1, as specified in RFC3306 (Generic Multicast Group Addresses Definition). This address range is therefore represented as FF7X::/12 in the prefix format.

- **RIID:** The RP address of the PIM-SM RP for this unicast-prefix-based multicast address.

- **Plen:** The number of significant bits in the Network Prefix field.

■ **Network Prefix:** Contains the assigned unicast network prefix. The maximum length is 64 bits. Any unused prefix bits are not required to be zeroed anymore—a strict requirement from RFC 3306. RFC 3956 relaxes this requirement. The “plen” is used to construct the RP address, but the group address can contain a longer part of the unicast prefix.

■ **Group ID:** The address owner assigned 32-bit multicast group ID.

That means embedding an RP address in the multicast group is a simple operation. We need only examine the Flgs, RIID, and Network Prefix fields for address derivation or loading. [Figure 6-15](#) illustrates this process.

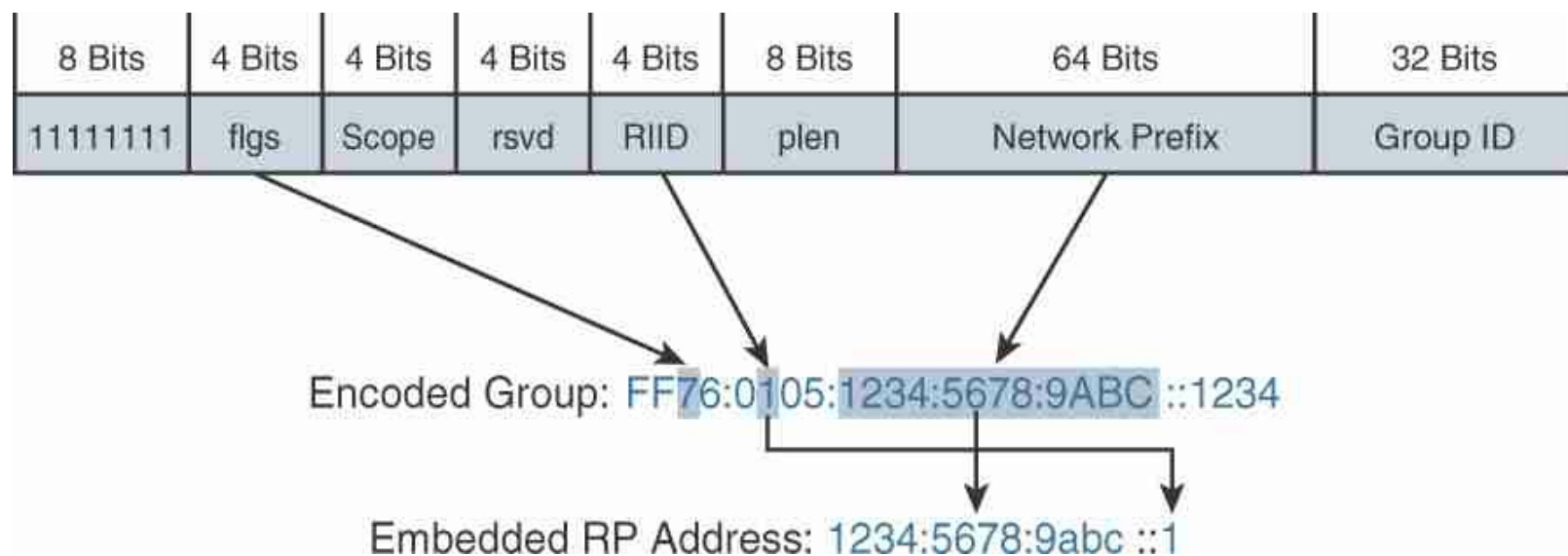


Figure 6-15 *Deriving the RP Address from the Multicast Group Address*

When using embedded RP, there is no need to configure routers with RP information or to use a dynamic protocol to perform RP mapping and propagation. Instead, multicast-enabled devices can extract and use the RP information from the group address, as shown above. That means a large number of RPs can be deployed anywhere, either local to the PIM domain or on the Internet for inter-domain multicast routing (multicast routing that typically occurs between two or more organizations or the Internet). No additional protocol or PIM6 changes are required to support embedded RP, it is built-in. Cisco router operating systems are programmed to automatically derive the RP information, making additional configuration unnecessary.

Any routers acting as a physical RP for an embedded group must have a statically configured RP address using an address assigned to one interface on the RP router. Routers automatically search for embedded RP group addresses in MLD reports or PIM messages and data packets coming from the source. When an embedded RP address is discovered, the router performs the group-to-RP mapping using the newly learned RP address. If no physical RP exists that matches the learned address, or if the learned RP address is not in the unicast RIB, then the router will not be able to complete the multicast group state. Embedded RP does not allow administrators to escape the need for a physical RP.

Also, keep in mind that a router can learn just one RP address for a multicast group using embedded RP. That means that you cannot have redundancy with embedded RP, because you cannot embed more than one RP address in the group. You can, however, use an Anycast RP set as the physical RP component, so long as the derived embedded RP address always points to the Anycast RP address configured on each router in the RP set. Also, as of this writing, embedded RP is not compatible with bidirectional PIM.

Let us adjust our BSR RP example (shown earlier in [Figure 6-12](#)) to include an embedded RP, removing the BSR configuration and replacing it with a static RP configuration. Below are the example configurations for each router. We statically configure our RP on R1 with a new Loopback400 interface that is assigned the IP address 1234:5678:9ABC::1/128 from our example group FF76:0150:1234:5678:9ABC::1234 (illustrated by [Figure 6-15](#)). On routers R2 and R3, we can now remove any irrelevant RP information. In fact, outside of the command **ipv6 multicast-routing** , no other multicast configuration is required on R2 and R3. We will, however, add an MLD join on R3's Loopback0 interface to simulate a host using the embedded RP group address from the example above, group FF76:0150:1234:5678:9ABC::1234. [Figure 6-16](#) depicts the updated topology.

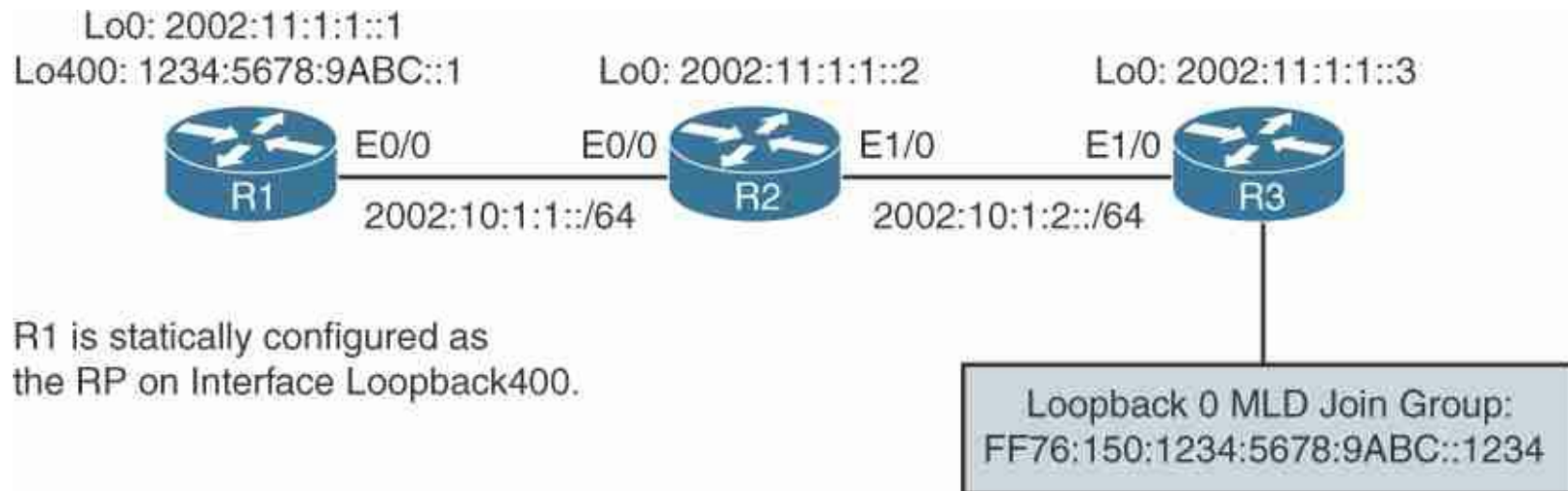


Figure 6-16 *Embedded RP Example Configuration Topology*

[Example 6-15](#) shows the router configurations for embedded RP support.

Example 6-15 *Configuration to Support Embedded RP*

[Click here to view code image](#)

RP: R1

```
hostname R1
!
ipv6 unicast-routing
ipv6 cef
ipv6 multicast-routing
!
interface Loopback0
no ip address
ipv6 address 2002:11:1:1::1/128
ipv6 enable
ipv6 router isis TEST
!
interface Loopback400
no ip address
ipv6 address 1234:5678:9ABC::1
ipv6 enable
ipv6 router isis TEST
```

```
!  
interface Ethernet0/0  
no ip address  
ipv6 address 2002:10:1:1::1/64  
ipv6 enable  
ipv6 router isis TEST  
!  
router isis TEST  
net 49.0001.0000.0001.00  
!  
!
```

```
RP: R2  
hostname R2  
!  
ipv6 unicast-routing  
ipv6 cef  
ipv6 multicast-routing  
!  
interface Loopback0  
no ip address  
ipv6 address 2002:11:1:1::2/128  
ipv6 enable  
ipv6 router isis TEST  
!  
interface Ethernet0/0  
no ip address  
ipv6 address 2002:10:1:1::2/64  
ipv6 enable  
ipv6 router isis TEST  
!  
interface Ethernet1/0  
no ip address  
ipv6 address 2002:10:1:2::1/64  
ipv6 enable  
ipv6 router isis TEST  
!  
router isis TEST  
net 49.0002.0000.0002.00  
!
```

```
R3  
hostname R3  
!  
ipv6 unicast-routing  
ipv6 cef
```

```

ipv6 multicast-routing
!
interface Loopback0
 no ip address
 ipv6 address 2002:11:1:1::3/128
 ipv6 enable
 ipv6 router isis TEST
 ipv6 mld join-group FF76:0150:1234:5678:9ABC::1234
!
interface Loopback500
 no ip address
 ipv6 address 2002:F:F:F::1/128
 ipv6 enable
 ipv6 router isis TEST
!
interface Ethernet0/1
 no ip address
 ipv6 address 2002:10:1:2::2/64
 ipv6 enable
 ipv6 router isis TEST
!
router isis TEST
 net 49.0002.0000.0003.00
!

```

Look at what happens on R3 first. R3 needs to have an RP-to-group mapping for the joined group of FF76:0150:1234:5678:9ABC::1234. Issuing a **show ipv6 pim group-map** on R3 reveals whether this mapping has occurred, as in [Example 6-16](#) .

Example 6-16 *Verifying RP-to-Group Mapping*

[Click here to view code image](#)

```

R3#show ipv6 pim group-map
IP PIM Group Mapping Table
(* indicates group mappings being used)
FF76:150:1234:5678:9ABC::/112*
SM, RP: 1234:5678:9ABC::1
RPF: ,::
Info source: Embedded
Uptime: 00:12:32, Groups: 1

```

Great! The group is in fact learned by R3 through the MLD join and the router has derived the RP 1234:5678:9ABC::1 from the group. That's good news. If the R1 RP is properly configured, we should be able to reach R3. Because no packets have been sourced to the group address yet, R1 should not yet have a group mapping for FF76:0150:1234:5678:9ABC::1234, as shown by executing the same **show ipv6 pim group-map** command on R1 in [Example 6-17](#) .

Example 6-17 *PIM6 Group Mappings*

[Click here to view code image](#)

```
R1#show ipv6 pim group-map
```

```
IP PIM Group Mapping Table
```

```
(* indicates group mappings being used)
```

```
FF05::/64*
```

```
SM, RP: 2002:F:F:F::1
```

```
RPF: Tu2,2002:F:F:F::1 (us)
```

```
Info source: Static
```

```
Uptime: 00:37:08, Groups: 0
```

```
FF70::/15*
```

```
NO
```

```
Info source: Default
```

```
Uptime: 00:37:51, Groups: 0
```

If we now send an IPv6 ping packet sourced from interface Ethernet0/0 on router R1, the group mapping should be derived from the embedded RP in the packet, and R1 should forward the packet toward R3 with success, as in [Example 6-18](#).

Example 6-18 *IPv6 Multicast Group Ping*

[Click here to view code image](#)

```
R1#ping ipv6 FF76:0150:1234:5678:9ABC::1234
```

```
Output Interface: ethernet0/0
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to FF76:150:1234:5678:9ABC::1234, timeout is 2 seconds:
```

```
Packet sent with a source address of 2002:10:1:1::1
```

```
Reply to request 0 received from 2002:11:1:1::3, 39 ms
```

```
Reply to request 1 received from 2002:11:1:1::3, 1 ms
```

```
Reply to request 2 received from 2002:11:1:1::3, 1 ms
```

```
Reply to request 3 received from 2002:11:1:1::3, 1 ms
```

```
Reply to request 4 received from 2002:11:1:1::3, 1 ms
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/39 ms
```

```
5 multicast replies and 0 errors.
```

Now, if we reissue the **show ipv6 pim group-map** command, we should find that R1, the RP, has learned the group mapping and constructed the appropriate RPF neighbor, as demonstrated in [Example 6-19](#).

Example 6-19 *Source Added to PIM6 Group Mapping*

[Click here to view code image](#)

```
R1#show ipv6 pim group-map
```

```
IP PIM Group Mapping Table
```

```
(* indicates group mappings being used)
```

```
FF76:150:1234:5678:9ABC::/112*
```

```
SM, RP: 1234:5678:9ABC::1
```

```
RPF: Et0/0, FE80::A8BB:CCFF:FE00:200
```


Info source: Embedded

Uptime: 00:00:26, Groups: 1

R2, the router between the source (R1) and the receiver (R3), needs to accept the incoming IPv6 multicast packet, derive the embedded RP information, create the appropriate group mappings and state entries, and, finally, forward the packet down the OIL to R3. If we issue the **show ipv6 pim group-map** and **show ipv6 mroute** commands on R2, we should see that this process is in fact complete, as demonstrated in [Example 6-20](#).

Example 6-20 *Final Group State in the MRIB*

[Click here to view code image](#)

```
R2#show ipv6 mroute
```

Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,

C - Connected, L - Local, I - Received Source Specific Host Report,

P - Pruned, R - RP-bit set, F - Register flag, T - SPT-bit set,

J - Join SPT, Y - Joined MDT-data group,

y - Sending to MDT-data group

g - BGP signal originated, G - BGP Signal received,

N - BGP Shared-Tree Prune received, n - BGP C-Mroute suppressed,

q - BGP Src-Active originated, Q - BGP Src-Active received

E - Extranet

Timers: Uptime/Expires

Interface state: Interface, State

```
(*, FF76:150:1234:5678:9ABC::1234), 00:07:22/00:03:07, RP 1234:5678:9ABC::1,  
flags: S
```

```
Incoming interface: Ethernet0/0
```

```
RPF nbr: FE80::A8BB:CCFF:FE00:100
```

```
Immediate Outgoing interface list:
```

```
Ethernet1/0, Forward, 00:07:22/00:03:07
```

```
R2#show ipv6 pim group-map
```

IP PIM Group Mapping Table

(* indicates group mappings being used)

```
FF76:150:1234:5678:9ABC::/112*
```

```
SM, RP: 1234:5678:9ABC::1
```

```
RPF: Et0/0,FE80::A8BB:CCFF:FE00:100
```

Info source: Embedded

Uptime: 00:12:34, Groups: 1

As this example shows, configuring embedded RP for PIM6 is very easy. It is enabled by default. R2 and R3 have virtually no PIM configuration requirements. Simply enable IPv6 multicast-routing and you are off to the races.

Which brings us to one final consideration:

Any multicast host can embed an RP address, meaning host applications on multicast sources. That means that if the embedded RP information is incorrectly configured by that application, the

infrastructure could have erroneous mappings for multicast groups. Embedded RP functionality is enabled by default on all Cisco routers when IPv6 multicast routing is enabled. That can pose an issue if the group has a large organizational scope. It also means that there is a chance that a low-end router could end up becoming the RP for many of high data rate sources. If these are concerns in the multicast domain, it may be best to use BSR and/or Anycast RP and to disable embedded RP capability altogether.

Use the following IOS-XE command to disable embedded RP functionality:

[Click here to view code image](#)

```
no ipv6 pim [vrf vrf-name] rp embedded
```

Summary

Understanding the nuances of IPv6 and how to implement multicast is paramount given the depletion of IPv4 address space and the number of devices that are being connected to the Internet.

One of the interesting aspects of IPv6 is the notion of a link-local address. These are intended for local device-to-device communication only, and it must be well understood because the link-local address is used for Neighbor Discovery Protocol (NDP), device communication, and so on.

Multicast Listener Discovery (MLD) protocol is used to discover devices requesting access to multicast streams at Layer 2. There are currently two implementations of MLD, version 1 and version 2. Layer 2 switches use MLD to direct traffic to the appropriate destinations.

Previous knowledge of protocol independent multicast (PIM) for IPv4 networks makes it easy to understand how PIM6 is implemented. Just as in IPv4, an additional routing protocol is not required to implement PIM6. The PIM6 modes of operation include general any-source multicast (ASM), source-specific multicast (SSM), and embedded RP groups.

Chapter 7. Operating and Troubleshooting IP Multicast Networks

Chapter 7. Operating and Troubleshooting IP Multicast Networks

This chapter covers the concepts of troubleshooting a multicast environment. We begin with an overview of the logical steps in troubleshooting a multicast environment. Next, we use an example network to explain the steps and commands necessary for troubleshooting in order to help you understand multicast packet flows. This chapter also examines troubleshooting case scenarios that are mapped to the multicast troubleshooting logic and the fundamental usage of that logic while troubleshooting a multicast problem.

Multicast Troubleshooting Logic

There is a wide range of techniques to determine the root cause of a problem. In this chapter, we take a systematic approach to troubleshooting multicast issues by using the following methodology:

1. Receiver check
2. Source check
3. State verification

Multicast Troubleshooting Methodology

The topology presented is a simple multicast network, as shown in [Figure 7-1](#). R3 and R4 are configured as RP (with Anycast with Auto-RP for downstream propagation). OSPF is the routing protocol configured for the unicast domain. R2 is the first-hop router that is connected to the source and R5 is the last-hop router connected to the receiver.

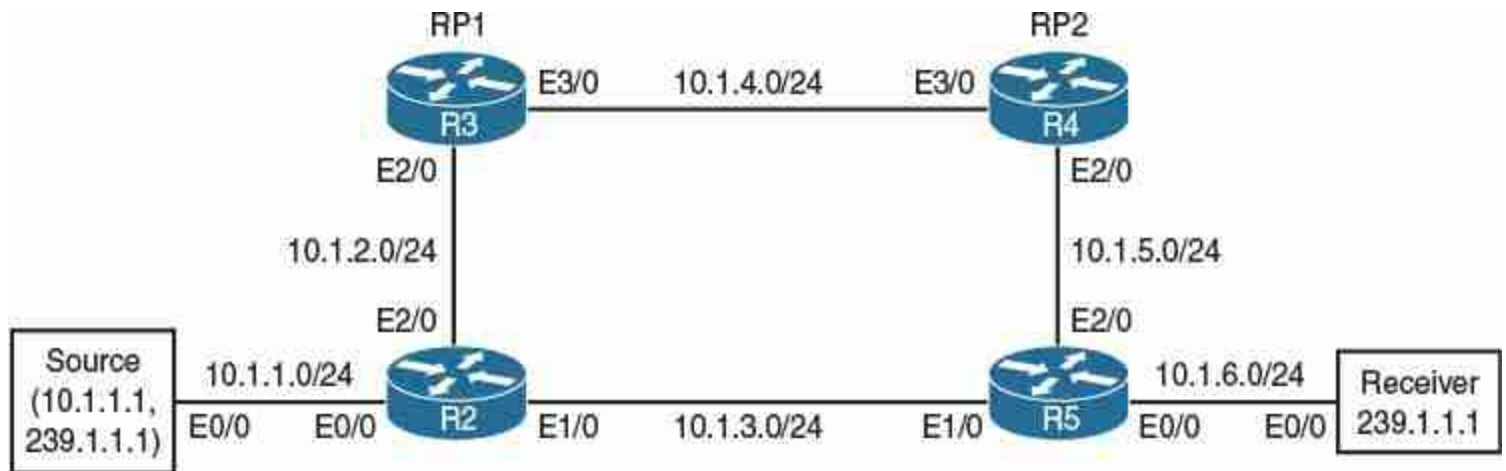


Figure 7-1 *Multicast Troubleshooting Sample Topology*

[Example 7-1](#) provides the configurations for the routers, and you should review these before proceeding with the chapter.

Example 7-1 Show R3 and R4 Multicast Configurations for Sample Topology

[Click here to view code image](#)

R3

```
R3#show running-config
```

```
..
```

```
hostname R3
```

```
!
```

```
no ip domain lookup
ip multicast-routing
ip cef
!
interface Loopback0
 ip address 192.168.3.3 255.255.255.255
 ip pim sparse-mode
!
interface Loopback100
 ip address 192.168.100.100 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet2/0
 ip address 10.1.2.2 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet3/0
 ip address 10.1.4.1 255.255.255.0
 ip pim sparse-mode
!
router ospf 1
 router-id 192.168.3.3
 network 0.0.0.0 255.255.255.255 area 0
!
ip forward-protocol nd
!
ip pim autorp listener
ip pim send-rp-announce Loopback100 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback100 scope 20 interval 10
ip msdp peer 192.168.4.4 connect-source Loopback0
ip msdp cache-sa-state
ip msdp default-peer 192.168.4.4
!
access-list 1 permit 239.1.0.0 0.0.255.255
```

R4

```
R4#show running-config
Building configuration...
```

..

```
hostname R4
```

!

```
no ip domain lookup
ip multicast-routing
ip cef
!
interface Loopback0
```

```

ip address 192.168.4.4 255.255.255.255
ip pim sparse-mode
!
interface Loopback100
ip address 192.168.100.100 255.255.255.255
ip pim sparse-mode
!
interface Ethernet2/0
ip address 10.1.5.1 255.255.255.0
ip pim sparse-mode
!
!
interface Ethernet3/0
ip address 10.1.4.2 255.255.255.0
ip pim sparse-mode
!
router ospf 1
router-id 192.168.4.4
network 0.0.0.0 255.255.255.255 area 0
!
ip pim autorp listener
ip pim send-rp-announce Loopback100 scope 20 group-list 1 interval 10
ip pim send-rp-discovery Loopback100 scope 20 interval 10
ip msdp peer 192.168.3.3 connect-source Loopback0
ip msdp cache-sa-state
ip msdp default-peer 192.168.3.3
!
access-list 1 permit 239.1.0.0 0.0.255.255
...

```

[Example 7-2](#) provides the downstream router configuration.

Example 7-2 *Downstream Router Configurations*

[Click here to view code image](#)

R2

```

R2#show running-config

```

...

```

hostname R2

```

```

!

```

```

ip multicast-routing

```

```

ip cef

```

```

!

```

```

interface Loopback0

```

```

ip address 192.168.2.2 255.255.255.255

```

```

ip pim sparse-mode

```

```

!

```

```
interface Ethernet0/0
 ip address 10.1.1.2 255.255.255.0
 ip pim sparse-mode
 load-interval 30
!
interface Ethernet1/0
 ip address 10.1.3.1 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet2/0
 ip address 10.1.2.1 255.255.255.0
 ip pim sparse-mode
!
router ospf 1
 network 0.0.0.0 255.255.255.255 area 0
!
ip pim autorp listener
....

R5
R5#show running-config
...
hostname R5
!
no ip domain lookup
ip multicast-routing
ip cef
interface Loopback0
 ip address 192.168.5.5 255.255.255.255
 ip pim sparse-mode
!
interface Ethernet0/0
 ip address 10.1.6.1 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet1/0
 ip address 10.1.3.2 255.255.255.0
 ip pim sparse-mode
!
interface Ethernet2/0
 ip address 10.1.5.2 255.255.255.0
 ip pim sparse-mode
!
router ospf 1
 network 0.0.0.0 255.255.255.255 area 0
```


! ip pim autorp listener

...

Baseline Check: Source and Receiver Verification

A baseline check is used to determine unicast connectivity between specific elements in the network. Remember, multicast routing uses the unicast routing infrastructure. If unicast routing is not working, neither will multicast!

Before starting the multicast baseline check, verify that the source and receiver have reachability. This can be accomplished with a ping test.

To perform the baseline check and gather the baseline information, use the following steps:

Step 1. Receiver check: Make sure a receiver is subscribed via IGMP and that (*,G) to the RP exists before trying to troubleshoot:

- a. Check the group state on the last-hop router.
- b. Check IGMP membership on the last-hop PIM-DR.
- c. Verify the (*,G) state at the LHR and check the RP for the (*,G) entry and RPF.

Step 2. Source check: Make sure you have an active source before trying to troubleshoot:

- a. Verify that the source is sending the multicast traffic to the first-hop router.
- b. Confirm that the FHR has registered the group with the RP.
- c. Determine that the RP is receiving the registry messages.
- d. Confirm that the multicast state is built on the FHR.

Let's perform each of the steps above on our sample network to ensure that the FHR and LHR are operating as expected and that state for the flow is established between them. To do this, use the following steps:

Step 1. Make sure a receiver is subscribed via IGMP before trying to debug:

- a. Check the group state on the last-hop router.

Verify that the last-hop router (LHR) has the appropriate (*,G) entry:

[Click here to view code image](#)

```
R5#show ip mroute 239.1.1.1
```

```
(*, 239.1.1.1), 00:02:35/00:02:50, RP 192.168.100.100, flags: SJCF
```

```
Incoming interface: Ethernet2/0, RPF nbr 10.1.5.1
```

```
Outgoing interface list:
```

```
Ethernet0/0, Forward/Sparse, 00:02:35/00:02:50
```

Notice that the incoming interface (IIF) is Ethernet2/0, and the outgoing interface is Ethernet0/0. Referring to [Figure 7-1](#), we can see that this flow has been established using the proper interfaces.

- b. Check IGMP membership on the last-hop PIM-DR:

[Click here to view code image](#)

```
R5#show ip igmp groups 239.1.1.1
```

```
IGMP Connected Group Membership
```

Group Address	Interface	Uptime	Expires	Last Reporter	Group Accounted
239.1.1.1	Ethernet0/0	01:14:42	00:02:16	10.1.6.2	

R5#

Ensure that the router has the RP information aligned to the scope range of the multicast group (using the **show ip pim rp** mapping command) and document the outgoing interface to reach the RP (using the **show ip rpf RP_address** command):

[Click here to view code image](#)

```
R5# show ip pim rp mapping
PIM Group-to-RP Mappings
Group(s) 239.1.0.0/16
RP 192.168.100.100 (?), v2v1
Info source: 192.168.100.100 (?), elected via Auto-RP
Uptime: 01:13:19, expires: 00:00:20
```

```
R5#show ip rpf 192.168.100.100
RPF information for ? (192.168.100.100)
RPF interface: Ethernet2/0
RPF neighbor: ? (10.1.5.1)
RPF route/mask: 192.168.100.100/32
RPF type: unicast (ospf 1)
Doing distance-preferred lookups across tables
RPF topology: ipv4 multicast base, originated from ipv4 unicast base
```

c. Verify the (*,G) state at the LHR and check the RP for the (*,G) entry and RPF.

The objective is to verify the registration of the receiver to the RP, consequently seeing the (*,G) entry.

Next, confirm that R5 is connected to the receiver, using the **show ip mroute** command:

[Click here to view code image](#)

```
R5#show ip mroute 239.1.1.1
IP Multicast Routing Table
Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,
C - Connected,
L - Local, P - Pruned, R - RP-bit set, F - Register flag,
T - SPT-bit set, J - Join SPT, M - MSDP created entry,
E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP
Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
```

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 1d22h/00:03:02, RP 192.168.100.100, flags: SJC

Incoming interface: Ethernet2/0, RPF nbr 10.1.5.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 1d22h/00:03:02

The incoming interface is Ethernet2/0 for the shared tree (*,G) connected to the RP (192.168.100.100), and the outgoing interface shows the receiver connection.

Verify the multicast state at the RP (R4):

[Click here to view code image](#)

R4#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,

C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP

Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,

N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,

Q - Received BGP S-A Route, q - Sent BGP S-A Route,

V - RD & Vector, v - Vector, p - PIM Joins on route,

x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,

p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 1d19h/00:02:43, RP 192.168.100.100, flags: S

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list:

Ethernet2/0, Forward/Sparse, 1d19h/00:02:43

The output for the RP shows the outgoing interface (Ethernet2/0) connected to the LHR; this matches the steady state topology.

The next step is to confirm that the RP address (*,G) entry is installed on the LHR and to check RPF.

The **show ip rpf RP** command points to the next hop in the (*,G) tree. Use this command in

conjunction with the **show ip mroute** command to verify the appropriate interface:

[Click here to view code image](#)

```
R5#show ip rpf 192.168.100.100
```

```
RPF information for ? (192.168.100.100)
```

```
RPF interface: Ethernet2/0
```

```
RPF neighbor: ? (10.1.5.1)
```

```
RPF route/mask: 192.168.100.100/32
```

```
RPF type: unicast (eigrp 1)
```

```
Doing distance-preferred lookups across tables
```

```
RPF topology: ipv4 multicast base, originated from ipv4 unicast base
```

```
R5#
```

```
R5#show ip mroute 239.1.1.1
```

```
(*, 239.1.1.1), 00:02:35/00:02:50, RP 192.168.100.100, flags: SJCF
```

```
Incoming interface: Ethernet2/0, RPF nbr 10.1.5.1
```

```
Outgoing interface list:
```

```
Ethernet0/0, Forward/Sparse, 00:02:35/00:02:50
```

```
(10.1.1.1, 239.1.1.1), 00:02:21/00:00:38, flags: T
```

```
Incoming interface: Ethernet1/0, RPF nbr 10.1.3.1
```

```
Outgoing interface list:
```

```
Ethernet0/0, Forward/Sparse, 00:02:21/00:03:28
```

The highlighted text indicates symmetry between the mroute state and RPF state.

Step 2. Make sure you have an active source before trying to debug:

a. Verify multicast traffic on the incoming interface (Ethernet0/0) of the first-hop router (FHR) R2:

[Click here to view code image](#)

```
R2#show interface Ethernet0/0 | include multicast
```

```
Received 1182 broadcasts (33 IP multicasts)
```

```
R2#show interface Ethernet0/0 | include multicast
```

```
Received 1182 broadcasts (35 IP multicasts)
```

```
R2#show interface Ethernet0/0 | include multicast
```

```
Received 1183 broadcasts (36 IP multicasts)
```

```
R2#show interface Ethernet0/0 | include multicast
```

```
Received 1184 broadcasts (37 IP multicasts)
```

The output shows the increase in the IP multicast packets received on the interface that connect to the source.

b. Confirm that the FHR has registered the group with the RP:

Make sure the FHR is aware of the RP using the **show ip pim rp mapping** command:

[Click here to view code image](#)

```
R2#show ip pim rp mapping
```

```
PIM Group-to-RP Mappings
```

```
Group(s) 239.1.0.0/16
```

```
RP 192.168.100.100 (?), v2v1
```

Info source: 192.168.100.100 (?), elected via Auto-RP

Uptime: 01:25:28, expires: 00:00:24

R2#

Verify that the FHR is sending packets to the RP (unicast register packets):

[Click here to view code image](#)

R2# show interface tunnel 0 | include output

Last input never, output 00:00:19, output hang never

Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0

5 minute output rate 0 bits/sec, 0 packets/sec

15 packets output, 1920 bytes, 0 underruns

0 output errors, 0 collisions, 0 interface resets

0 output buffer failures, 0 output buffers swapped out

R2#clear ip mroute 239.1.1.1

R2# show interface tunnel 0 | include output

Last input never, output 00:00:04, output hang never

Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0

5 minute output rate 0 bits/sec, 0 packets/sec

16 packets output, 2048 bytes, 0 underruns

0 output errors, 0 collisions, 0 interface resets

0 output buffer failures, 0 output buffers swapped out

R2#

You may be wondering about where the tunnel interface came from. The tunnel 0 interface is automatically created by enabling multicast on the downstream router. This tunnel is used to encapsulate the unicast register packets to the RP.

c. Determine that the RP is receiving the registry messages.

Use the **debug ip pim** *<mcast_group_address>* command.

R3 shows the registry message received and register stop sent. The register stop will be sent only if the multicast state has active receivers in the shared tree:

[Click here to view code image](#)

Mar 15 20:11:13.030: PIM(0): Received v2 Register on Ethernet2/0 from 10.1.2.1

*Mar 15 20:11:13.030: for 10.1.1.1, group 239.1.1.1

*Mar 15 20:11:13.030: PIM(0): Send v2 Register-Stop to 10.1.2.1 for 10.1.1.1, group 239.1.1.1

d. Confirm that the multicast state is built on the FHR (R2):

[Click here to view code image](#)

R2#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,

C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,
E - Extranet,
X - Proxy Join Timer Running, A - Candidate for MSDP
Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,
p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:00:36/stopped, RP 192.168.100.100, flags: SPF

Incoming interface: Ethernet2/0, RPF nbr 10.1.2.2

Outgoing interface list: Null

(10.1.1.1, 239.1.1.1), 00:00:36/00:02:23, flags: FT

Incoming interface: Ethernet0/0, RPF nbr 10.1.1.1

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:00:17/00:03:12

The multicast state indicates the (S,G) is built with FT flags. The incoming interface is connected to the source and the outgoing interface shows that the packet-forwarding takes the best path between the source and receiver based on the unicast routing protocol.

The results shown in this section are steady state. It is important to understand the commands and steady state condition. During troubleshooting, use the steady state information with the output collected during troubleshooting to compare and assess the problems.

State Verification

There are two primary components to state verification; these include RP control-plane check and hop-by-hop state validation.

RP Control-Plane Check

[Figure 7-2](#) is used to provide an example of state behavior in which the SPT uses the following path, Source -> R2 -> R3 -> R4 -> R5 -> Receiver. In this example, the interface between R2 and R5 is shut down, forcing R2 to choose the path to R3 and R4. This is done to verify the switchover of the multicast flow from (*,G) to (S,G) following the best path selected by the unicast routing protocol.

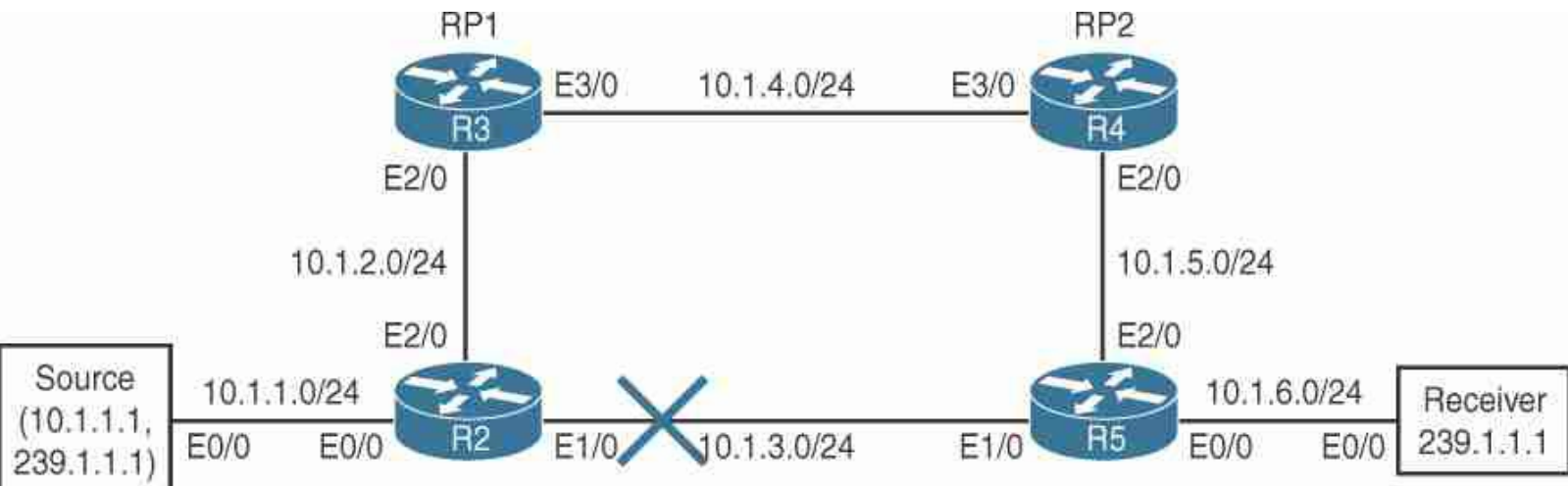


Figure 7-2 Multicast State Flow via the RP

This would essentially be step 3 in the baselining process, as shown here:

Step 3. Verify RP and SPT state entries across the path:

- a. Check the MSDP summary to verify peering is operational.
- b. Verify the group state at each active RP.
- c. Verify SPT changes.

This configuration, covered in [Chapter 4](#) , uses the hybrid RP design and Auto-RP for downstream propagation of the RP.

Step 3a. Check the MSDP summary to verify peering is operational:

[Click here to view code image](#)

```
R3# show ip msdp summary
MSDP Peer Status Summary
Peer Address    AS   State  Uptime/  Reset SA  Peer Name
                Downtime Count Count
*192.168.4.4    ?   Up     00:02:16 1    0    ?
```

The show **ip msdp summary** command verifies that the MSDP relationship is operational between the RPs. This relationship is not impacted by the multicast data flow state in the mroute table.

Step 3b. Verify the group state at each active RP (R3 and R4):

[Click here to view code image](#)

```
R3#show ip mroute 239.1.1.1
(*, 239.1.1.1), 00:08:39/stopped, RP 192.168.100.100, flags: SP
Incoming interface: Null, RPF nbr 0.0.0.0
Outgoing interface list: Null
(10.1.1.1, 239.1.1.1), 00:00:16/00:02:43, flags: TA
Incoming interface: Ethernet2/0, RPF nbr 10.1.2.1
Outgoing interface list:
Ethernet3/0, Forward/Sparse, 00:00:16/00:03:13

R4# show ip mroute 239.1.1.1
(*, 239.1.1.1), 00:08:45/00:03:02, RP 192.168.100.100, flags: S
Incoming interface: Null, RPF nbr 0.0.0.0
```

Outgoing interface list:

Ethernet2/0, Forward/Sparse, 00:01:08/00:02:38

(10.1.1.1, 239.1.1.1), 00:00:27/00:02:32, flags: MT

Incoming interface: Ethernet3/0, RPF nbr 10.1.4.1

Outgoing interface list:

Ethernet2/0, Forward/Sparse, 00:00:27/00:03:02

The RP (R4) is closest to the receiver and will consequently be the RP that receives the join request from the LHR (R5). Comparing the output above, notice the outgoing interface list is NULL on R3 and R4 is using Ethernet2/0. This shows that R4 is the active RP. The “T” flags on both R3 and R4 indicate the SPT has been established. The flags “A” and “M” on routers R3 and R4, respectively, indicate that R3 is the candidate RP and that the entry was created by MSDP.

Step 3c. Verify SPT changes.

Review the multicast state at the RP after the connection between R2 and R5 is restored; refer to [Figure 7-3](#).

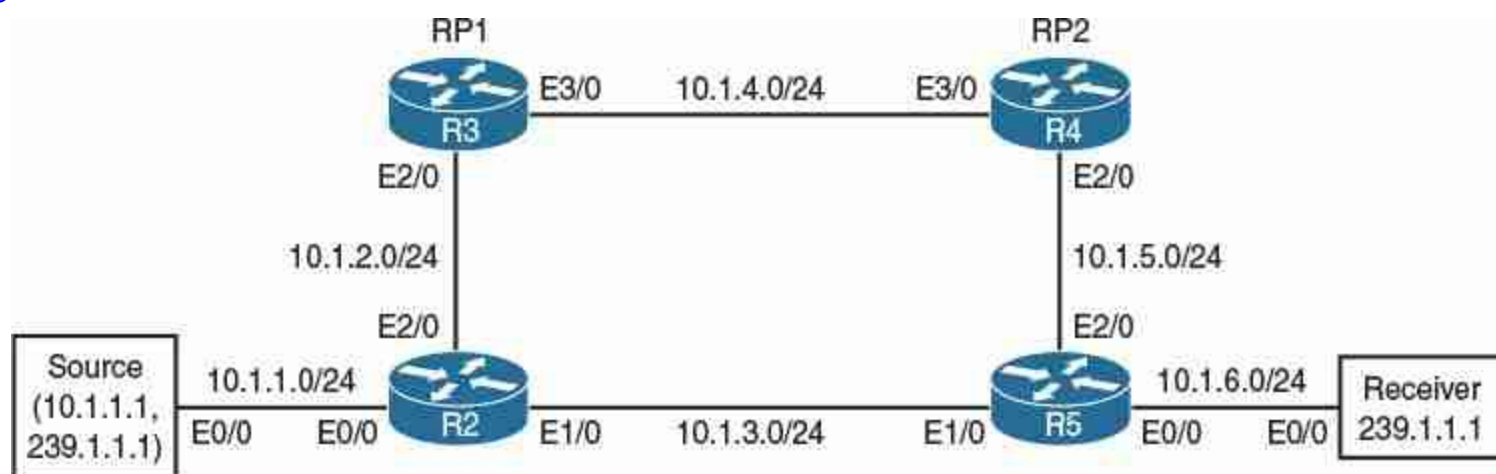


Figure 7-3 *Multicast Steady State Topology*

With the R2 to R5 connection restored, the SPT flow no longer traverses the R2, R3, R4, R5 path but rather the R2, R5 path. Notice the P flag set in (S,G) entry. This shows the multicast stream has been pruned. The (*,G) state still will have receiver information via Ethernet2/0 as the outgoing interface:

[Click here to view code image](#)

R4#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group,

C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry,

E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP

Advertisement,

U - URD, I - Received Source Specific Host Report,

Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,

G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner,
p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 1d21h/00:03:08, RP 192.168.100.100, flags: S

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list:

Ethernet2/0, Forward/Sparse, 1d21h/00:03:08

(10.1.1.1, 239.1.1.1), 00:07:57/00:02:19, flags: PMT

Incoming interface: Ethernet2/0, RPF nbr 10.1.5.2

Outgoing interface list: Null

You should be very familiar with multicast flows traversing through the RP and those that do not. You will undoubtedly encounter both situations as you implement and troubleshoot multicast networks.

It would be terribly inefficient for multicast messages to always traverse the RP. We saw that multicast traffic moved to the SPT in Step 3c. How does this happen? Let's review the SPT process.

R5 is the LHR and also the location where there is a better (lower-cost) unicast route to the source.

Use the **show ip mroute** command to monitor the incoming interface for the (S,G) and (*,G) shift, as demonstrated in [Example 7-3](#).

Example 7-3 show ip mroute *Command Output*

[Click here to view code image](#)

R5#show ip mroute 239.1.1.1

..

(* , 239.1.1.1), 00:02:31/00:02:55, RP 192.168.100.100, flags: SJC

Incoming interface: Ethernet2/0, RPF nbr 10.1.5.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:02:31/00:02:55

(10.1.1.1, 239.1.1.1), 00:02:31/00:00:28, flags: T

Incoming interface: Ethernet1/0, RPF nbr 10.1.3.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:02:31/00:03:14

In this example, R5 shows Ethernet1/0 will be selected as the RPF towards the source address 10.1.1.1. This is shown using the **show ip rpf** command, as demonstrated in [Example 7-4](#).

Example 7-4 show ip rpf *Command Output*

[Click here to view code image](#)

R5#show ip rpf 10.1.1.1

RPF information for ? (10.1.1.1)
RPF interface: Ethernet1/0
RPF neighbor: ? (10.1.3.1)
RPF route/mask: 10.1.1.0/24
RPF type: unicast (ospf 1)
Doing distance-preferred lookups across tables
RPF topology: ipv4 multicast base, originated from ipv4 unicast base
R5#

We can also see additional details using the **debug ip pim** command, as shown in the output in [Example 7-5](#) . Pay particular attention to the highlighted section.

Example 7-5 debug ip pim Output

[Click here to view code image](#)

! Start the debug and wait for the state to change

```
R5#
R5#debug ip pim 239.1.1.1
PIM debugging is on
R5#
*Mar 26 18:44:59.351: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.2 on Ethernet1/0 from
LOADING to FULL, Loading Done
R5#
*Mar 26 18:45:08.556: PIM(0): Received v2 Join/Prune on Ethernet0/0 from 10.1.6.2,
to us
*Mar 26 18:45:08.557: PIM(0): Join-list: (10.1.1.1/32, 239.1.1.1), S-bit set
*Mar 26 18:45:08.557: PIM(0): Update Ethernet0/0/10.1.6.2 to (10.1.1.1, 239.1.1.1),
Forward state, by PIM SG Join
R5#
*Mar 26 18:45:10.261: PIM(0): Insert (10.1.1.1,239.1.1.1) join in nbr 10.1.3.1's queue
*Mar 26 18:45:10.261: PIM(0): Insert (10.1.1.1,239.1.1.1) prune in nbr 10.1.5.1's
queue
*Mar 26 18:45:10.261: PIM(0): Building Join/Prune packet for nbr 10.1.5.1
*Mar 26 18:45:10.261: PIM(0): Adding v2 (10.1.1.1/32, 239.1.1.1), S-bit Prune
*Mar 26 18:45:10.261: PIM(0): Send v2 join/prune to 10.1.5.1 (Ethernet2/0)
*Mar 26 18:45:10.261: PIM(0): Building Join/Prune packet for nbr 10.1.3.1
*Mar 26 18:45:10.261: PIM(0): Adding v2 (10.1.1.1/32, 239.1.1.1), S-bit Join
*Mar 26 18:45:10.261: PIM(0): Send v2 join/prune to 10.1.3.1 (Ethernet1/0)
R5#
*Mar 26 18:45:10.324: PIM(0): Building Periodic (*,G) Join / (S,G,RP-bit) Prune mes-
sage for 239.1.1.1
*Mar 26 18:45:10.324: PIM(0): Insert (*,239.1.1.1) join in nbr 10.1.5.1's queue
*Mar 26 18:45:10.324: PIM(0): Insert (10.1.1.1,239.1.1.1) sgr prune in nbr
10.1.5.1's queue
*Mar 26 18:45:10.324: PIM(0): Building Join/Prune packet for nbr 10.1.5.1
*Mar 26 18:45:10.324: PIM(0): Adding v2 (192.168.100.100/32, 239.1.1.1), WC-bit,
RPT-bit, S-bit Join
```

```

*Mar 26 18:45:10.324: PIM(0): Adding v2 (10.1.1.1/32, 239.1.1.1), RPT-bit, S-bit
Prune
*Mar 26 18:45:10.324: PIM(0): Send v2 join/prune to 10.1.5.1 (Ethernet2/0)
R5#
*Mar 26 18:45:24.124: PIM(0): Insert (10.1.1.1,239.1.1.1) join in nbr 10.1.3.1's
queue
*Mar 26 18:45:24.125: PIM(0): Building Join/Prune packet for nbr 10.1.3.1
*Mar 26 18:45:24.125: PIM(0): Adding v2 (10.1.1.1/32, 239.1.1.1), S-bit Join
*Mar 26 18:45:24.125: PIM(0): Send v2 join/prune to 10.1.3.1 (Ethernet1/0)
*Mar 26 18:45:24.256: PIM(0): Received v2 Join/Prune on Ethernet0/0 from 10.1.6.2,
to us
*Mar 26 18:45:24.256: PIM(0): Join-list: (10.1.1.1/32, 239.1.1.1), S-bit set
*Mar 26 18:45:24.256: PIM(0): Update Ethernet0/0/10.1.6.2 to (10.1.1.1, 239.1.1.1),
Forward state, by PIM SG Join
R5#sh ip mroute 239.1.1.1
*Mar 26 18:45:41.349: PIM(0): Received v2 Join/Prune on Ethernet0/0 from 10.1.6.2,
to us
*Mar 26 18:45:41.349: PIM(0): Join-list: (*, 239.1.1.1), RPT-bit set, WC-bit set,
S-bit set
*Mar 26 18:45:41.349: PIM(0): Update Ethernet0/0/10.1.6.2 to (*, 239.1.1.1), Forward
state, by PIM *G Join
*Mar 26 18:45:41.349: PIM(0): Update Ethernet0/0/10.1.6.2 to (10.1.1.1, 239.1.1.1),
Forward state, by PIM *G Join

```

From the previous debug, please note the change from 10.1.5.1 to 10.1.3.1 (PIM join state for S,G) and note also the “S-Bit Join” message that indicates the transition to the SPT. During the triggered (*,G) state, the DR creates a join/prune message with the WC-bit and RPT-bit set to 1. The WC-bit set to 1 indicates that any source may match this entry, and the flow will follow the shared tree. When the RPT-bit is set to 1, it shows that the join is associated with the shared tree and the join/prune message is sent along the shared tree towards the RP.

Hop-by-Hop State Validation

Up to this point in the troubleshooting process, we have covered the fundamentals—source and receiver verification and RP control-plane confirmation. If we have not solved the multicast problem, the troubleshooting methodology will be done on a hop-by-hop basis from the LHR to the FHR. For this to be accomplished correctly, you must have a thorough understanding of the entire topology.

Note

When you are under pressure to solve a problem, this is not the time to begin documenting the network. There should already be a comprehensive network diagram available.

The network diagram in [Figure 7-4](#) aids in identifying the path between the source and the receiver. This begins the next phase of the troubleshooting process—understanding the flow.

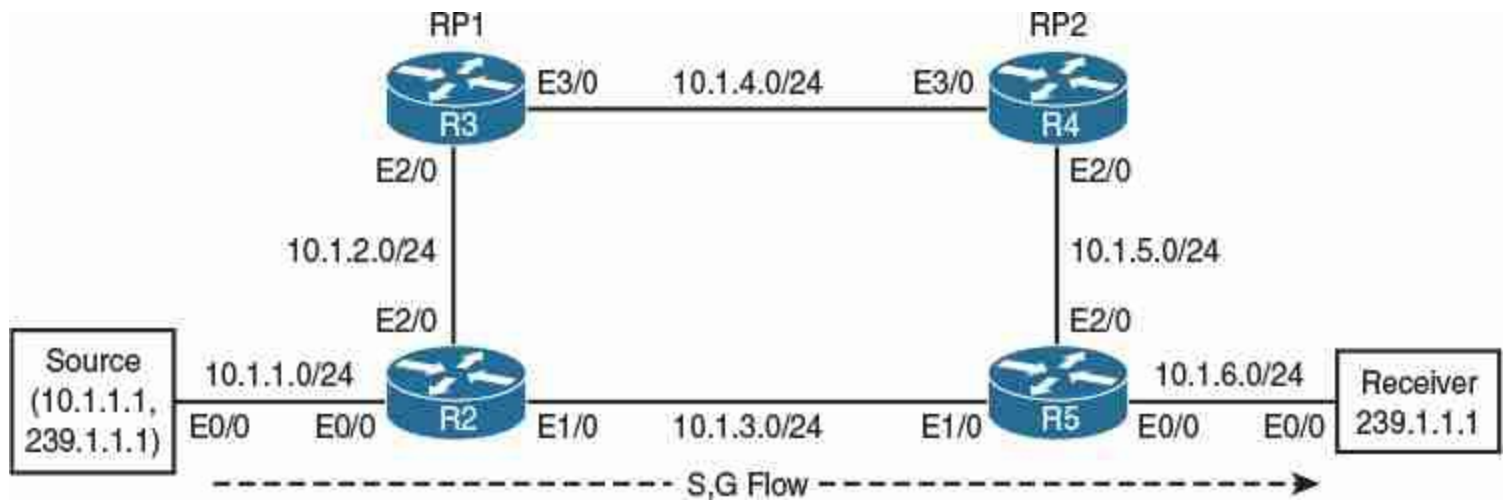


Figure 7-4 *Multicast (S,G) Path Flow*

[Figure 7-4](#) is a simple reference topology. Whether you are troubleshooting something very simple or very complex, the process is still the same in assessing the health of the multicast network.

We begin by establishing the appropriate (S,G) path. Because this is a network we are familiar with from previous examples, we know that the (S,G) path uses R2 and R5. We begin our troubleshooting effort at the LHR and then work towards the FHR.

High-level steps:

Step 4. Verify the mroute state information for the following elements:

- Validated that the IIF is correct?
- Verify that the OIF is correct?
- Are the “flags” for (*, G) and (S, G) entries correct?
- Verify that the RP information is correct?

If there are anomalies from the previous steps, verify that each entry contains RFP information with the **show ip rpf ip-address** command and move up the shortest-path toward the source. The questions to ask yourself based on the output are:

- Does this align with the information in the mroute entry?
- Is this what you would expect when looking at the unicast routing table?

Now let’s review each element shown in Step 4. R5 is the LHR where we begin the process. Using the **show ip mroute** command, we can verify that the state information for 239.1.1.1 is maintained correctly by examining the elements, as shown in [Example 7-6](#) :

Example 7-6 *Multicast State at R5(LHR)*

[Click here to view code image](#)

R5#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,
 L - Local, P - Pruned, R - RP-bit set, F - Register flag,
 T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,
 X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
 U - URD, I - Received Source Specific Host Report,
 Z - Multicast Tunnel, z - MDT-data group sender,

Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 2d01h/00:03:22, RP 192.168.100.100, flags: SJC

Incoming interface: Ethernet2/0, RPF nbr 10.1.5.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 2d01h/00:03:22

(10.1.1.1, 239.1.1.1), 00:04:53/00:02:31, flags: T

Incoming interface: Ethernet1/0, RPF nbr 10.1.3.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse, 00:04:53/00:03:22

From the output in [Example 7-6](#) , we can answer the following questions and verify the appropriate operation:

Step 4a. Validated that the IIF is correct?

■ (*,G) Ethernet2/0 points to the RP.

■ (S,G) Ethernet 1/0 based on RPF to the source.

Step 4b. Verify that the OIF is correct?

■ (*,G) Ethernet 0/0 points towards the receiver.

■ (S,G) Ethernet 0/0 points towards the receiver.

Step 4c. Are the “flags” for (*, G) and (S, G) entries correct?

■ (*,G) state: SJC.

■ (S,G) state: T.

Step 4d. Verify that the RP information is correct?

■ 192.168.100.100 (The Anycast Auto-RP learned from R3 and R4.)

Taking a systematic approach to determining the root cause of the problem, we continue examining each device in the path toward the FHR. For brevity, here we jump to the FHR and use the **show ip mroute** command to inspect the multicast state, as demonstrated in [Example 7-7](#) .

Example 7-7 Multicast State at R2 (FHR)

[Click here to view code image](#)

R2#show ip mroute 239.1.1.1

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, B - Bidir Group, s - SSM Group, C - Connected,

L - Local, P - Pruned, R - RP-bit set, F - Register flag,

T - SPT-bit set, J - Join SPT, M - MSDP created entry, E - Extranet,

X - Proxy Join Timer Running, A - Candidate for MSDP Advertisement,
U - URD, I - Received Source Specific Host Report,
Z - Multicast Tunnel, z - MDT-data group sender,
Y - Joined MDT-data group, y - Sending to MDT-data group,
G - Received BGP C-Mroute, g - Sent BGP C-Mroute,
N - Received BGP Shared-Tree Prune, n - BGP C-Mroute suppressed,
Q - Received BGP S-A Route, q - Sent BGP S-A Route,
V - RD & Vector, v - Vector, p - PIM Joins on route,
x - VxLAN group

Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:08:48/stopped, RP 192.168.100.100, flags: SPF

Incoming interface: Ethernet2/0, RPF nbr 10.1.2.2

Outgoing interface list: Null

(10.1.1.1, 239.1.1.1), 00:08:48/00:02:41, flags: FT

Incoming interface: Ethernet0/0, RPF nbr 10.1.1.1

Outgoing interface list:

Ethernet1/0, Forward/Sparse, 00:08:48/00:02:36

The “T” flag indicates that multicast traffic is flowing using the (S, G) state entry.

We can use the output of [Example 7-7](#) to validate the correct operation and answer the following troubleshooting questions.

Step 4a. Validated that the IIF is correct?

- (*,G) Ethernet2/0 points to the RP.
- (S,G) Ethernet0/0 based on RPF to the source.

Step 4b. Verify that the OIF is correct?

- (*,G) NULL; no receiver state on shared tree.
- (S,G) Ethernet1/0 points towards the receiver.

Step 4c. Are the “flags” for (*, G) and (S, G) entries correct?

- (*,G) state: SPF.
- (S,G) state: FT.

Step 4d. Verify that the RP information is correct?

- 192.168.100.100.

The (*,G) is in a pruned state after registration. This is based on the switch from the shared to the source tree following the unicast best path between the source and receiver. The (S,G) shows the “FT” flags. The packet is flowing via the shortest path tree, and it is the first-hop router for registry.

The fundamental troubleshooting considerations previously covered are applicable for IPv4 and IPv6 multicast environments. In the following section, we review some of the common tools used in IOS for multicast troubleshooting.

Overview of Common Tools for Multicast Troubleshooting

In many situations during troubleshooting, we may need to generate synthetic traffic using a traditional tool such as **ping**, or we may be required to determine the source of intermittent problems or network performance challenges that would require the use of a more sophisticated tool such as an IP service level agreement (SLA). We may also be required to move beyond the traditional **show** commands to gain additional insight into the operation or the lack of correct operation in the network.

Ping Test

The multicast ping test is a traditional troubleshooting tool used by network engineers to verify the control and data planes. The test does not remove application centric problems, but it can be used to verify the network infrastructure.

As shown in [Figure 7-5](#), we begin by assigning a router interface to a join-group.

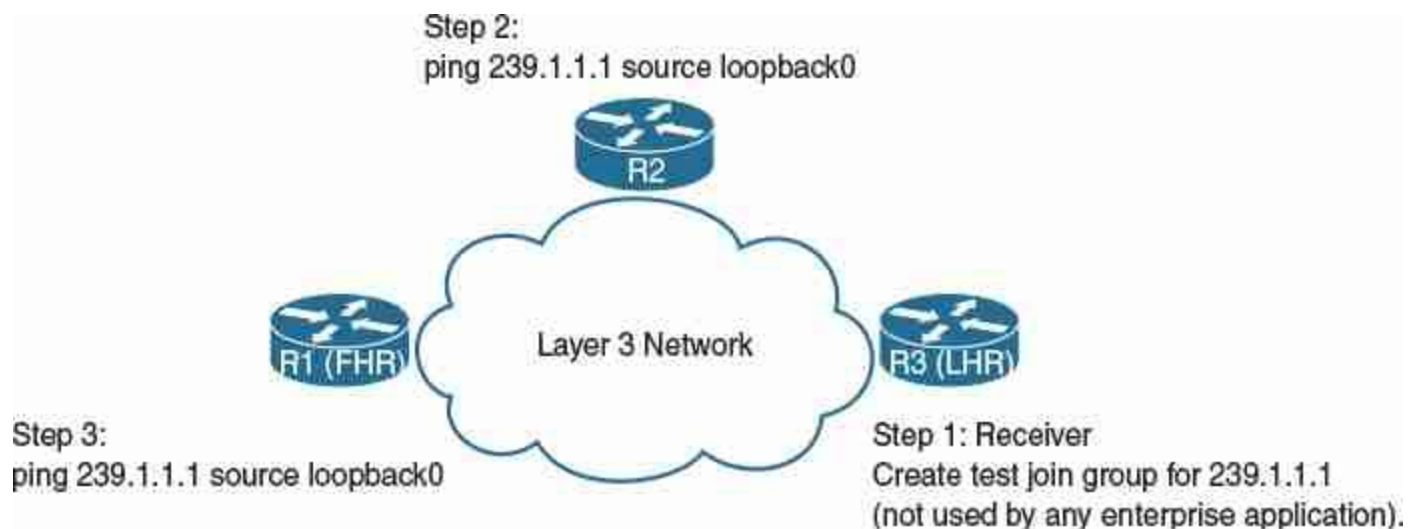


Figure 7-5 Multicast Ping Test Procedure

Step 1. Add an IGMP Join group to simulate the receiver at the LHR:

[Click here to view code image](#)

```
R3#show run interface Ethernet0/0
```

Building configuration...

Current configuration : 114 bytes

!

```
interface Ethernet0/0
```

```
ip address 10.1.6.2 255.255.255.0
```

```
ip pim sparse-mode
```

```
ip igmp join-group 239.1.1.1
```

```
end
```

Step 2. Use multicast ping to send packets from the FHR:

[Click here to view code image](#)

```
R1#ping 239.1.1.1
```

Type escape sequence to abort.

Sending 1, 100-byte ICMP Echos to 239.1.1.1, timeout is 2 seconds:

```
*Mar 23 21:27:15.209: ICMP: echo reply rcvd, src 10.1.6.2, dst  
10.1.1.1, topology BASE, dscp 0 topoid 0
```

```
Reply to request 0 from 10.1.6.2, 34 ms
```

R1#

You can now verify the multicast state in the mroute table. This is a very simple method to troubleshoot a multicast network. It also serves as a method to verify multicast functionality prior to implementing a multicast service.

Note

Use caution when using the **ip igmp join-group** command because it causes the device to process switch all packets to the configured multicast address. This can lead to high CPU utilization. This command should be removed immediately after troubleshooting, and it should not be used to test live multicast application traffic.

SLA Test

The IP service level agreement (SLA) feature provides the ability to analyze applications and services through the generation of synthetic traffic. With the IP SLA feature, you can measure one-way latency, jitter, and packet loss.

This is accomplished using two elements, a sender that generates UDP packets at a given interval to a particular destination or destinations, and a receiver, or multiple receivers, that collect and analyze the synthetic traffic and send a response back to the sender.

At the time of this writing, the following code releases support SLA multicast:

- 15.2(4)M

- 15.3(1)S

- Cisco IOS XE Release 3.8S

- 15.1(2)SG

- Cisco IOS XE Release 3.4SG

[Example 7-8](#) and [Example 7-9](#) show the sender and receiver configurations.

Example 7-8 *Sender Configuration*

[Click here to view code image](#)

```
ip sla endpoint-list type ip RCVRS
ip-address 10.1.6.2 port 5000
ip sla 10
udp-jitter 239.1.1.101 5000 endpoint-list RCVRS source-ip 10.1.1.1 source-port 5000
 num-packets 50 interval 25
request-data-size 128
tos 160
verify-data
tree-init 1
control timeout 4
control retry 2
dscp 10
threshold 1000
timeout 10000
frequency 10
history hours-of-statistics-kept 4
```

history distributions-of-statistics-kept 5
history statistics-distribution-interval 10
history enhanced interval 60 buckets 100
ip sla schedule 10 life forever start-time now

Example 7-9 Receiver Configuration

[Click here to view code image](#)

ip sla responder
ip sla responder udp-echo ipaddress 10.1.1.1 port 5000

Use the **show** commands in [Example 7-10](#) to verify the condition of the multicast probes.

Example 7-10 Verifying Multicast Probe Condition

[Click here to view code image](#)

R1#show ip sla configuration
IP SLAs Infrastructure Engine-III
Entry number: 10
Owner:
Tag:
Operation timeout (milliseconds): 10000
Type of operation to perform: mcast
Endpoint list: RCVRS
Target address/Source address: 239.1.1.100/10.1.1.1
Target port/Source port: 5000/5001
Type Of Service parameter: 0xA0
Request size (ARR data portion): 128
Packet Interval (milliseconds)/Number of packets: 25/50
Verify data: Yes
Vrf Name:
DSCP: 10
Number of packets to set multicast tree: 1
SSM: disabled
Control message timeout(seconds): 4
Control message retry count: 2
Schedule:
Operation frequency (seconds): 10 (not considered if randomly scheduled)
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Randomly Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 1000
Distribution Statistics:
Number of statistic hours kept: 4

Number of statistic distribution buckets kept: 5
Statistic distribution interval (milliseconds): 10

Enhanced History:

Aggregation Interval:60 Buckets:100

Entry number: 2132304746

Type of operation: mcast

Multicast operation id: 10

Target address/Source address: 10.1.6.2/10.1.1.1

Target port/Source port: 5000/5001

Multicast address: 239.1.1.100

R6#show ip sla endpoint-list type ip

Endpoint-list Name: RCVRS

Description:

List of IPV4 Endpoints:

ip-address 10.1.6.2 port 5000

Common Multicast Debug Commands

It is advisable to enable the debug commands during a maintenance window with Cisco TAC assistance. The following debug commands help provide additional insight during troubleshooting.

debug ip mpacket Command

The **debug ip mpacket** command provides packet details for the multicast packets traversing the L3 device, as shown in [Example 7-11](#) .

Example 7-11 Verifying Multicast Probe Condition

[Click here to view code image](#)

*Sep 04 14:48:01.651: IP: s=10.1.1.1 (Ethernet1/0) d=239.1.1.1 (Serial0/0) ld

*Sep 04 14:48:02.651: IP: s=10.1.1.1 (Ethernet1/0) d=239.1.1.1 (Serial0/0) ld

*Sep 04 14:48:03.651: IP: s=10.1.1.1 (Ethernet1/0) d=239.1.1.1 (Serial0/0) ld

debug ip pim Command

The **debug ip pim** command shows the all PIM packets flowing through the device and is useful for understanding PIM state.

The debug is taken from a RP configured in hybrid ASM mode with Auto-RP. The messages show a group join using 239.1.1.1 and group 239.1.1.2 that has a receiver with no active sources. The output also indicates the RP election of 192.168.100.100 as a candidate. [Example 7-12](#) shows the output of **debug ip pim** .

Example 7-12 debug ip pim Output

[Click here to view code image](#)

R3#show logging

*Mar 17 21:41:37.637: PIM(0): check pim_rp_announce 1

*Mar 17 21:41:37.637: PIM(0): send rp announce

*Mar 17 21:41:44.657: PIM(0): Received v2 Join/Prune on Ethernet2/0 from 10.1.2.1,
to us


```

*Mar 17 21:41:44.658: PIM(0): Join-list: (*, 239.1.1.2), RPT-bit set, WC-bit set,
S-bit set
*Mar 17 21:41:44.658: PIM(0): Check RP 192.168.100.100 into the (*, 239.1.1.2) entry
*Mar 17 21:41:44.658: PIM(0): Adding register decap tunnel (Tunnel1) as accepting
interface of (*, 239.1.1.2).
*Mar 17 21:41:44.658: PIM(0): Add Ethernet2/0/10.1.2.1 to (*, 239.1.1.2), Forward
state, by PIM *G Join
*Mar 17 21:41:45.658: PIM(0): Received v2 Register on Ethernet2/0 from 10.1.2.1
*Mar 17 21:41:45.658: for 10.1.1.1, group 239.1.1.1
*Mar 17 21:41:45.658: PIM(0): Check RP 192.168.100.100 into the (*, 239.1.1.1) entry
*Mar 17 21:41:45.658: PIM(0): Adding register decap tunnel (Tunnel1) as accepting
interface of (*, 239.1.1.1).
*Mar 17 21:41:45.658: PIM(0): Adding register decap tunnel (Tunnel1) as accepting
interface of (10.1.1.1, 239.1.1.1).
*Mar 17 21:41:45.659: PIM(0): Send v2 Register-Stop to 10.1.2.1 for 10.1.1.1, group
239.1.1.1
*Mar 17 21:41:47.633: PIM(0): check pim_rp_announce 1
*Mar 17 21:41:47.633: PIM(0): send rp announce
*Mar 17 21:41:47.634: PIM(0): Received v2 Assert on Ethernet3/0 from 10.1.4.2
*Mar 17 21:41:47.634: PIM(0): Assert metric to source 192.168.100.100 is [0/0]
*Mar 17 21:41:47.634: PIM(0): We lose, our metric [0/0]
*Mar 17 21:41:47.634: PIM(0): Insert (192.168.100.100,224.0.1.39) prune in nbr
10.1.4.2's queue
*Mar 17 21:41:47.634: PIM(0): Send (192.168.100.100, 224.0.1.39) PIM-DM prune to oif
Ethernet3/0 in Prune state
*Mar 17 21:41:47.634: PIM(0): (192.168.100.100/32, 224.0.1.39) oif Ethernet3/0 in Prune state
*Mar 17 21:41:47.634: PIM(0): Building Join/Prune packet for nbr 10.1.4.2
*Mar 17 21:41:47.634: PIM(0): Adding v2 (192.168.100.100/32, 224.0.1.39) Prune
*Mar 17 21:41:47.634: PIM(0): Send v2 join/prune to 10.1.4.2 (Ethernet3/0)
*Mar 17 21:41:51.022: PIM(0): Received v2 Assert on Ethernet3/0 from 10.1.4.2
*Mar 17 21:41:51.022: PIM(0): Assert metric to source 192.168.100.100 is [0/0]

```

debug ip igmp Command

The output of **debug ip igmp** command at LHR (R5) provides details regarding the IGMP packets. As shown in [Example 7-13](#), the router receives a join from 239.1.1.1 and then adds a multicast (*,G) entry to add the receiver state in the multicast entry.

Example 7-13 debug ip igmp Output

[Click here to view code image](#)

```

*Mar 17 21:50:45.703: IGMP(0): Received v2 Report on Ethernet0/0 from 10.1.6.2 for
239.1.1.1
*Mar 17 21:50:45.703: IGMP(0): Received Group record for group 239.1.1.1, mode 2
from 10.1.6.2 for 0 sources
*Mar 17 21:50:45.703: IGMP(0): Updating EXCLUDE group timer for 239.1.1.1
*Mar 17 21:50:45.703: IGMP(0): MRT Add/Update Ethernet0/0 for (*,239.1.1.1) by 0

```

*Mar 17 21:50:45.704: IGMP(0): Received v2 Report on Ethernet0/0 from 10.1.6.2 for 239.1.1.1
*Mar 17 21:50:45.704: IGMP(0): Received Group record for group 239.1.1.1, mode 2 from 10.1.6.2 for 0 sources
*Mar 17 21:50:45.704: IGMP(0): Updating EXCLUDE group timer for 239.1.1.1
Mar 17 21:50:45.704: IGMP(0): MRT Add/Update Ethernet0/0 for (,239.1.1.1) by 0

Multicast Troubleshooting

[Figure 7-6](#) shows the three basic blocks that you need to understand to support our multicast troubleshooting efforts.



Figure 7-6 *Multicast Troubleshooting Blocks*

Before you begin troubleshooting, you need to understand each domain for your multicast environment and then tie it to the appropriate troubleshooting steps to the respective domain.

I. Application Scope Review (the scope of the application that is not working):

- Are the receivers local to the source?
- Are the receivers across the WAN?
- Verify the bandwidth of the multicast group for the application and validate latency-specific parameters.
- Verify the addressing scheme used in the enterprise.

II. Unicast Domain Scope Review:

- What is the Unicast routing protocol design?
- What is the path between the source and destination for the multicast flow?
- What are the WAN technologies such as IPSEC, MPLS-VPN? Is this offering from a service provider or self-managed, and so on?
- What is the QoS design and does it align to the class that is configured for that multicast service?

III. Multicast Domain:

- What is the PIM configuration? (ASM, SSM, or BiDir)
- Verify the number of multicast states in the network.
- Verify the best practices have been followed for PIM control-plane configuration, explained in [Chapter 5](#).
- Verify the configuration of the downstream routers.

These are some high-level steps used to understand the multicast environment. Earlier in the chapter we reviewed a through step-by-step approach to identify a multicast issue. Let us review a

case study to understand the troubleshooting process.

Multicast Troubleshooting Case Study

Problem: Multicast is not working for few multicast groups in the network.

Troubleshooting:

I. Application Scope Review: Items to consider while you review the application:

- The scope of the application.

1.1 Are any multicast applications in the network currently functioning?

Answer: Multicast groups across in other regions are working.

1.2 Are all the groups that are not working a part of the same multicast RP group?

Answer: Yes.

1.3 Are there any working groups with the scope range?

Answer: No.

1.4 Was it working before?

Answer: Yes.

1.5 What changes were made in the environment?

Answer: I don't know.

- The bandwidth of the stream for the single multicast group.

Comment: At this point, when you have a nonworking group, the bandwidth does not play a significant role.

- What are the latency specific parameters for multicast stream distribution?

Comment: With a nonfunctional group, latency information does not play an important part.

- What type of multicast address scheme is used by the source?

Answer: Scoped using the range of 239.x.x.x space.

II. Unicast Domain Scope Review: Things to consider in this domain:

- Unicast routing protocol architecture.

- What is the underlying unicast protocol?

Answer: OSPF and BGP.

- Is there any unicast routing reachability issues?

Answer: No.

- What is the path between the source and destination and identify the source and destination (single out a multicast group)?

Answer: Draw the path between the, source, RP, and receiver. It is beneficial to understand this path while troubleshooting the multicast control plane.

- Is there a WAN technology overlay, such as IPsec, MPLS-VPN?

Answer: Dedicated point-to-point connections with no IPsec or MPLS-VPNs.

Note

IPsec does not natively support multicast transport. An overlay technology should be considered, such as GRE, DMVPN or GETVPN, to transport multicast with IPSEC encryption.

- Does the QoS architecture and class align to the multicast transport?

Comment: When there is no latency or sporadic user experience issue, then a review of QoS really is not generally warranted.

III. PIM Domain: Things to consider in this domain:

- What is the PIM configuration? (ASM, SSM, or BiDir)

Answer: ASM.

- What is the multicast state in the network?

Answer: Not all the best practices are deployed; only ip pim register limit, boundary, and scoping of the hybrid RP with Auto-RP has been deployed.

- Are the best practices for PIM control plane deployed?

Answer: Only **ip pim register-limit** is configured.

- Identify the RP location in the network for the application group (ASM).

Answer: The RP is in the data path, and the RP also functions as the campus core.

- What is the configuration of the downstream router?

Answer: Only **ip pim register-limit** has been deployed.

[Figure 7-7](#) will be used to explain the troubleshooting methodology.

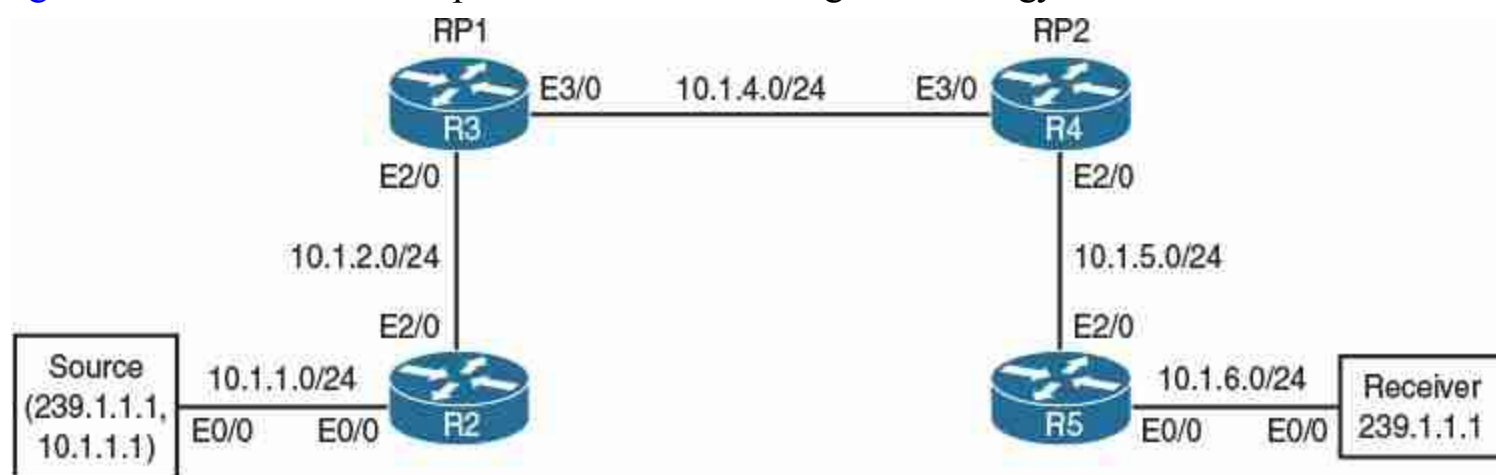


Figure 7-7 Case Study: Multicast Troubleshooting Topology

Baseline Check: Source and Receiver Verification

Before you start the multicast baseline check, verify that the source and receiver have ping reachability. [Example 7-14](#) shows a unicast ping test from the receiver to the source IP (10.1.1.1).

Example 7-14 Verification of the Source Reachability from the Receiver

[Click here to view code image](#)

```
Receiver#ping 10.1.1.1
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:

!!!!

Step 1: Receiver Check

[Table 7-1](#) Outlines the receiver verification steps.

Step 1: Receiver			
Check	Verify	Command	Check
Step 1a and 1b	Did the LHR Receive the IGMP join message and does it reflect in the mroute table?	<pre> R5# show ip mroute 239.1.1.1 IP Multicast Routing Table ... (*, 239.1.1.1), 5d20h/00:02:56, RP 192.168.100.100, flags: SJC Incoming interface: Ethernet2/0, RPF nbr 10.1.5.1 Outgoing interface list: Ethernet0/0, Forward/Sparse, 5d20h/00:03:01 R5#show ip igmp group 239.1.1.1 IGMP Connected Group Membership Group Address Interface Uptime Expires Last Reporter Group Accounted 239.1.1.1 Ethernet0/0 3d18h 00:02:32 10.1.6.2 R5# </pre>	Yes
Step 1c	Check RP information, multicast state information for RP and RPF	<pre> R5# show ip pim rp mapping PIM Group-to-RP Mappings Group(s) 239.1.0.0/16 RP 192.168.100.100 (?), v2vl Info source: 192.168.100.100 (?), elected via Auto-RP </pre>	Yes

R5# show ip mroute 239.1.1.1

IP Multicast Routing Table

--

(*, 239.1.1.1), 5d20h/00:02:56, RP
192.168.100.100, flags: SJC

Incoming interface: Ethernet2/0,
RPF nbr 10.1.5.1

Outgoing interface list:

Ethernet0/0, Forward/Sparse,
5d20h/00:03:01

R5#show ip rpf 192.168.100.100

RPF information for ? (192.168.100.100)

RPF interface: Ethernet2/0

RPF neighbor: ? (10.1.5.1)

RPF route/mask: 192.168.100.100/32

RPF type: unicast (ospf 1)

Doing distance-preferred lookups
across tables

RPF topology: ipv4 multicast base,
originated from ipv4 unicast base

Table 7-1 Receiver Verification Steps

Step 2: Source Check

[Table 7-2](#) outlines the source verification steps before starting the multicast baseline check.

Step 2:	Verify	Commands	Check
Step 2a	Is the source sending the multicast entry to the first-hop router?	R2# show interface e0/0 include multicast Received 105855 broadcasts (4477 IP multicasts) R2# show interface e0/0 include multicast Received 105855 broadcasts (4478 IP multicasts) R2# show interface e0/0 include multicast Received 105855 broadcasts (4479 IP multicasts) R2# show interface e0/0 include multicast Received 105856 broadcasts (4480 IP multicasts)	Yes
Step 2b	Is the first-hop router registering the multicast group at the RP?	R2# show ip pim rp mapping Group(s) 239.1.0.0/16 RP 192.168.100.100 (?), v2v1 Info source: 192.168.100.100 (?), elected via Auto-RP Uptime: 5d17h, expires: 00:00:26 R2# show interface tunnel 0 in rate Queueing strategy: fifo 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec	No
Step 2c	Is the multicast state being built at the FHR?	R2# show ip mroute 239.1.1.1 ... (*, 239.1.1.1), 00:12:30/stopped, RP 192.168.100.100, flags: SPF Incoming interface: Ethernet2/0, RPF nbr 10.1.2.2 Outgoing interface list: Null (10.1.1.1, 239.1.1.1), 00:12:30/00:01:55, flags: PFT Incoming interface: Ethernet0/0, RPF nbr 10.1.1.1 Outgoing interface list: Null	No

Table 7-2 *Source Verification Steps*

Make sure you have an active source before trying to troubleshoot.

The Outgoing interface list points to NULL in Step 2c. At steady state, the outgoing interface should point towards the RP in initial state. This is a problem in the state build that needs to be reviewed further. Rather than reviewing Step 2d (debug message to verify state build-up at FHR), let's move to next step, state verification.

Step 3: State Verification

[Table 7-3](#) outlines the baseline RP and control-plane verification steps.

Step 3	Verify	Command	Check
Step 3a	Is the MSDP state "Up"?	<pre>R3#show ip msdp summary MSDP Peer Status Summary Peer Address AS State Uptime/ Reset SA Peer Name Downtime Count Count *192.168.4.4 ? Up 5d18h 1 0 ? R3#</pre>	Yes.
Step 3b	Does the state entry at R3 and R4 relate to the baseline working state?	<pre>R4 State R4#show ip mroute 239.1.1.1 ... (*, 239.1.1.1), 5d19h/stopped, RP 192.168.100.100, flags: S Incoming interface: Null, RPF nbr 0.0.0.0 Outgoing interface list: Ethernet2/0, Forward/Sparse, 5d18h/00:02:38 (10.1.1.1, 239.1.1.1), 00:00:27/00:02:32, flags: M Incoming interface: Null, RPF nbr 0.0.0.0 Outgoing interface list: Ethernet2/0, Forward/Sparse, 00:00:27/00:03:02</pre>	No. Comment: The state at R3 does not look correct. due to the "P" flag set for (S,G) and the outgoing interface is NULL.

R3 State

R3#show ip mroute 239.1.1.1

...

(*, 239.1.1.1), 00:00:20/stopped, RP
192.168.100.100, flags: SP

Incoming interface: Null, RPF nbr
0.0.0.0

Outgoing interface list: Null
(10.1.1.1, 239.1.1.1),
00:00:20/00:02:39, flags: PA

Incoming interface: Ethernet2/0,
RPF nbr 10.1.2.1

Outgoing interface list: Null

Step 3c	Does the RP control plane align to the routing plane?	R5#show ip rpf 192.168.100.100	Yes.
---------	---	--------------------------------	------

RPF information for ? (192.168.100.100)

RPF interface: Ethernet2/0

RPF neighbor: ? (10.1.5.1)

RPF route/mask: 192.168.100.100/32

RPF type: unicast (ospf 1)

Doing distance-preferred lookups
across tables

RPF topology: ipv4 multicast base,
originated from ipv4 unicast base

R5#show ip route 192.168.100.100

Routing entry for 192.168.100.100/32

Known via "ospf 1", distance 110,
metric 11, type intra area

Last update from 10.1.5.1 on
Ethernet2/0, 1w6d ago

Routing Descriptor Blocks:

* 10.1.5.1, from 192.168.4.4, 1w6d
ago, via Ethernet2/0

Route metric is 11, traffic share
count is 1

Table 7-3 *RP and Control-Plane Check*

As you observed earlier in Step 3, the RP control-plane check had inconsistencies. There is no point in proceeding with Step 4, hop-by-hop state validation, until the inconsistencies are resolved. This means it could be a control-plane problem. To further eliminate suspicion of application-based issues, perform a multicast ping from FHR to LHR in the topology, as outlined in [Figure 7-8](#) . The multicast ping will simulate the application, becoming the new multicast source. This illuminates how the state is built from Steps 1, 2, and 3 with a network source.

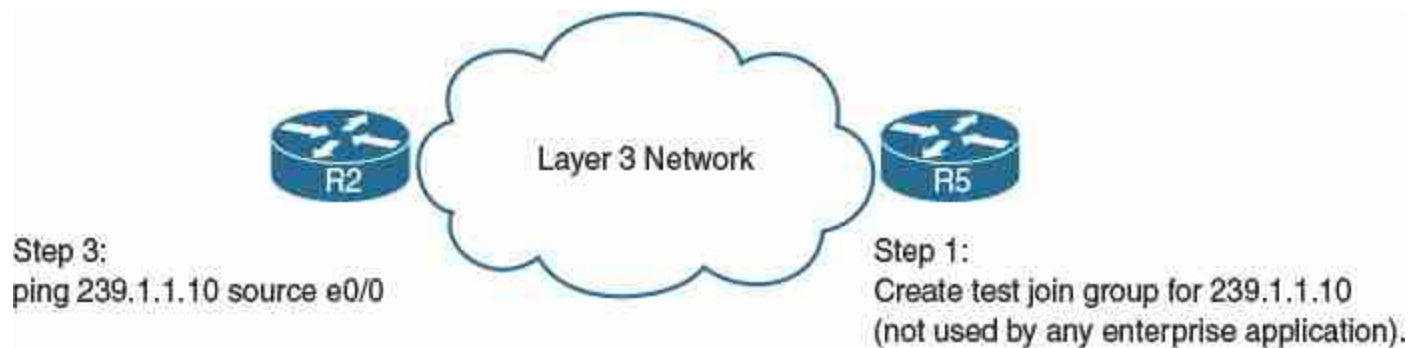


Figure 7-8 *Multicast Ping Test Procedure*

To create a receiver, configure an IGMP join group at R5(LHR) closest to the receiver. Use the **ip igmp join-group** command. Use a multicast group that is not used in any production traffic and falls in the multicast RP scope range. [Example 7-15](#) shows the configuration.

Example 7-15 *Create IGMP Join Group*

[Click here to view code image](#)

```
R5#show run interface loopback 0
Building configuration...

Current configuration : 118 bytes
!
interface Loopback0
 ip address 192.168.5.5 255.255.255.255
 ip pim sparse-mode
 ip igmp join-group 239.1.1.10
end

R5#
```

Verify connectivity to the unicast and multicast control plane from R2 (closest to the receiver), as demonstrated in [Example 7-16](#) .

Example 7-16 *Multicast and Unicast Connectivity Check*

[Click here to view code image](#)

```
R2#ping 192.168.5.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.5.5, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R2#ping 239.1.1.10
Type escape sequence to abort.
Sending 1, 100-byte ICMP Echos to 239.1.1.100, timeout is 2 seconds:
```

R2#

The unicast ping test is successful at R2, but the multicast ping is not successful. This test is done merely to remove application anomalies. The output clearly shows that this test is not successful and confirms that this is a control-plane issue in the network.

Summary of the failed tests:

- Steps 2b and 2c: FHR registration to the RP failed, and the RP state flags for multicast were incorrect.

- Step 3c: Failure of the RP state exchanges; the state at R3 did not provide correct results because the (S,G) group was in a pruned state.

[Figure 7-9](#) shows the fundamental overview of the three troubleshooting steps:

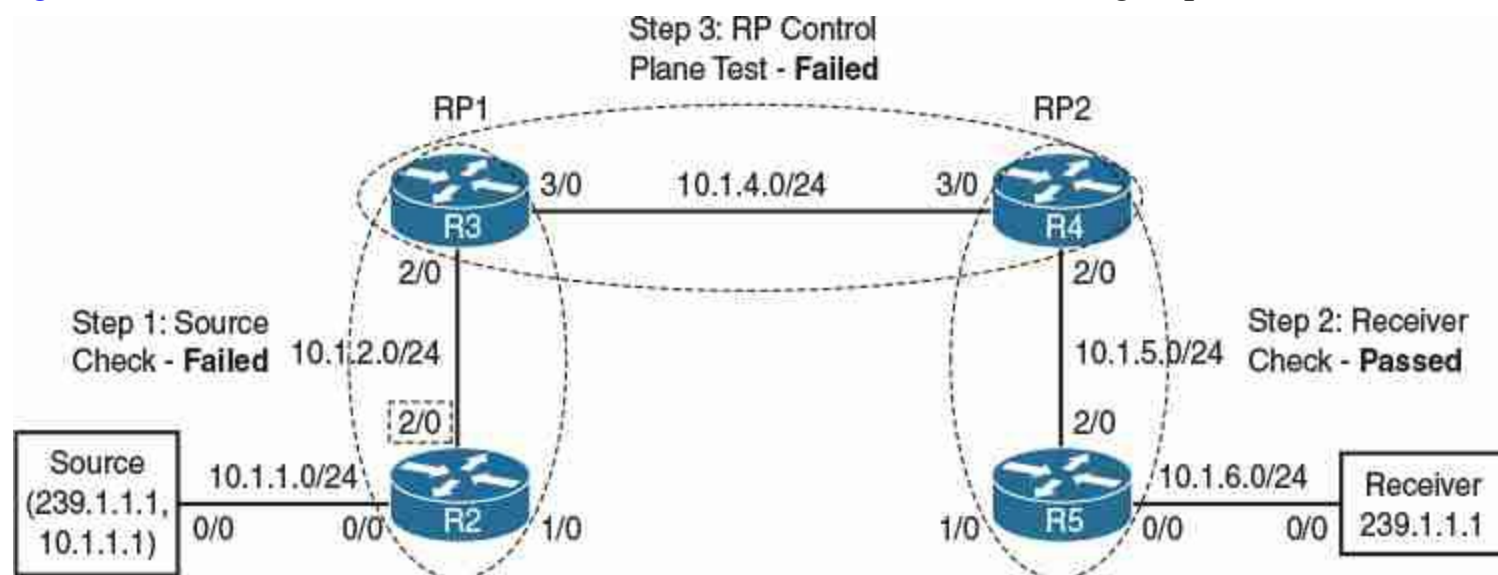


Figure 7-9 *Troubleshooting Case Study: Focus Area*

The multicast control plane is broken at Step 1 or Step 3. The areas of concern include R2, R3, and R4. It is recommended to work with Cisco TAC during troubleshooting because we will be using debug commands to understand the PIM control-plane flow. It is also recommended to plan a change control window prior to executing debug commands, because it might impact other network functions.

To verify the registry process, use the **debug ip pim 239.1.1.1** command at R3, the output for which is shown in [Example 7-17](#).

Example 7-17 *Control-plane Debug Capture at R3*

[Click here to view code image](#)

```
*Mar 21 20:48:06.866: PIM(0): Received v2 Register on Ethernet2/0 from 10.1.2.1
*Mar 21 20:48:06.866:   for 10.1.1.1, group 239.1.1.1
*Mar 21 20:48:06.866: PIM(0): Check RP 192.168.100.100 into the (*, 239.1.1.1) entry
*Mar 21 20:48:06.866: PIM(0): Adding register decap tunnel (Tunnel1) as accepting
interface of (*, 239.1.1.1).
*Mar 21 20:48:06.866: PIM(0): Adding register decap tunnel (Tunnel1) as accepting
interface of (10.1.1.1, 239.1.1.1).
*Mar 21 20:48:06.866: PIM(0): Send v2 Register-Stop to 10.1.2.1 for 10.1.1.1, group
239.1.1.1
```


PIM register packets are seen being sent and received. The question arises, why does the state at R3 not show the receiver and instead shows the pruned state for the (S,G) entry? We need to do some further investigation to determine the root cause; this moves our attention to Step 3.

We begin troubleshooting this by using the **show ip mroute command** on R3, as demonstrated in [Example 7-18](#).

Example 7-18 show ip mroute Command Output

[Click here to view code image](#)

```
R3#show ip mroute 239.1.1.1
```

```
...
```

```
(*, 239.1.1.1), 00:04:21/stopped, RP 192.168.100.100, flags: SP
```

```
Incoming interface: Null, RPF nbr 0.0.0.0
```

```
Outgoing interface list: Null
```

```
(10.1.1.1, 239.1.1.1), 00:04:21/00:02:38, flags: PA
```

```
Incoming interface: Ethernet2/0, RPF nbr 10.1.2.1
```

```
Outgoing interface list: Null→ Problem Area
```

This state flag relationship for the (S,G) entry does not look correct; the message is in the pruned state; and the outgoing interface is set to NULL. The (*,G) also does not show any receiver information.

The peer MSDP relationship between R3 and R4 shows the information of the source learned at R4. This can be seen using the **show ip msdp sa-cache** command, as demonstrated in [Example 7-19](#).

Example 7-19 show ip msdp sa-cache Command Output

[Click here to view code image](#)

```
R4#show ip msdp sa-cache
```

```
MSDP Source-Active Cache - 1 entries
```

```
(10.1.1.1, 239.1.1.1), RP 192.168.100.100, AS ?,00:08:46/00:05:22, Peer 192.168.3.3
```

```
R4#
```

R4 has the receiver information in its multicast state table, shown using the **ip mroute** command, as demonstrated in [Example 7-20](#).

Example 7-20 Multicast State Table at R4

[Click here to view code image](#)

```
(*, 239.1.1.1), 5d22h/stopped, RP 192.168.100.100, flags: S
```

```
Incoming interface: Null, RPF nbr 0.0.0.0
```

```
Outgoing interface list:
```

```
Ethernet2/0, Forward/Sparse, 5d22h/00:03:04
```

```
(10.1.1.1, 239.1.1.1), 00:00:18/00:02:41, flags: M
```

```
Incoming interface: Null, RPF nbr 0.0.0.0
```

```
Outgoing interface list:
```

```
Ethernet2/0, Forward/Sparse, 00:00:18/00:02:41
```

The control plane from the RPs (R3 and R4) looks fine; however, the state entry at R3 does not show the multicast stream flowing. It cannot be an application issue because the multicast ping test

did not succeed. The multicast state does not show R3 forwarding the (S,G) state to R4, which has the receiver entry in the (*,G) state table, and consequently the group is in a pruned state for the (S,G) at R3 (refer to the “problem area” identified in the **show ip mroute** at R3). The R3 and R4 multicast state table exchange between the RP is based on MSDP feature.

Verify the data plane interface with the multicast control plane for congruency.

At R3, check the Layer 3 routing relationship with the PIM relationship. This is accomplished using the **show ip ospf neighbor** command to determine the state of the routing adjacency and the **show ip pim neighbor** command to show the PIM neighbor relationship. [Example 7-21](#) shows output from the **show ip ospf neighbor** command.

Example 7-21 show ip ospf neighbor and show ip pim neighbor Command Output for R3

[Click here to view code image](#)

R3#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.4.4	1	FULL/BDR	00:00:38	10.1.4.2	Ethernet3/0
192.168.2.2	1	FULL/DR	00:00:37	10.1.2.1	Ethernet2/0

R3# show ip pim neighbor

PIM Neighbor Table

Mode: B - Bidir Capable, DR - Designated Router, N - Default DR Priority,
P - Proxy Capable, S - State Refresh Capable, G - GenID Capable

Neighbor Address	Interface	Uptime/Expires	Ver	DR Prio/Mode
10.1.2.1	Ethernet2/0	6d02h/00:01:34	v2	1 / S P G

R3#

Notice that Ethernet3/0 does not have a PIM relationship. This interface connects R3 to R4 and could be the reason why the receiver PIM join did not make it to R3 from R4. To verify the inconsistency, execute the same command at R4 and check the Layer 3 routing relationship with the PIM relationship, as demonstrated in [Example 7-22](#) .

Example 7-22 show ip ospf neighbor and show ip pim neighbor Command Output for R4

[Click here to view code image](#)

R4#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.3.3	1	FULL/DR	00:00:32	10.1.4.1	Ethernet3/0
192.168.5.5	1	FULL/DR	00:00:37	10.1.5.2	Ethernet2/0

R4#show ip pim neighbor

PIM Neighbor Table

Mode: B - Bidir Capable, DR - Designated Router, N - Default DR Priority,
P - Proxy Capable, S - State Refresh Capable, G - GenID Capable

Neighbor Address	Interface	Uptime/Expires	Ver	DR Prio/Mode
10.1.5.2	Ethernet2/0	6d02h/00:01:24	v2	1 / DR S P G

R4#

The previous command verifies the failure of the PIM relationship between R3 and R4.

Check the configuration on R4, Ethernet3/0 and verify the existence of PIM routing using the command **show ip pim interface** , as demonstrated in [Example 7-23](#) .

Example 7-23 show ip pim interface Command Output for R4

[Click here to view code image](#)

R4#show ip pim interface

Address	Interface	Ver/ Mode	Nbr Count	Query Intvl	DR	Prior	DR
192.168.4.4	Loopback0	v2/S	0	30	1		192.168.4.4
192.168.100.100	Loopback100	v2/S	0	30	1		192.168.100.100
10.1.5.1	Ethernet2/0	v2/S	1	30	1		10.1.5.2
10.1.4.2	Ethernet3/0	v2/S	1	30	1		10.1.4.2

R4#

From the output we see that PIM sparse-mode is enabled on the interface.

Now we check the configuration on R3 Ethernet3/0 and verify the existence of PIM routing using the **show ip pim interface** command on R3, as demonstrated in [Example 7-24](#) .

Example 7-24 show ip pim interface Command Output for R3

[Click here to view code image](#)

R3#show ip pim interface

Address	Interface	Ver/ Mode	Nbr Count	Query Intvl	DR	Prior	DR
10.1.2.2	Ethernet2/0	v2/S	1	30	1		10.1.2.2
192.168.3.3	Loopback0	v2/S	0	30	1		192.168.3.3
192.168.100.100	Loopback100	v2/S	0	30	1		192.168.100.100

R3#

We can see from the output of this command that PIM has not been enabled on the Ethernet3/0 interface. This clearly indicates the problem.

To rectify the problem, we enable PIM sparse-mode on interface Ethernet 3/0 at R3 using the commands shown in [Example 7-25](#) .

Example 7-25 Enabling PIM Sparse-Mode on R3's Ethernet 3/0 Interface

[Click here to view code image](#)

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#interface ethernet 3/0

R3(config-if)#ip pim sparse-mode

Now we need to verify that the configuration change resolved the problem using the **show ip pim neighbor** command, as demonstrated in [Example 7-26](#) .

Example 7-26 PIM Neighbor Overview

[Click here to view code image](#)

```
R3#show ip pim neighbor
```

PIM Neighbor Table

Mode: B - Bidir Capable, DR - Designated Router, N - Default DR Priority,

P - Proxy Capable, S - State Refresh Capable, G - GenID Capable

Neighbor Address	Interface	Uptime/Expires	Ver	DR Prio/Mode
10.1.2.1	Ethernet2/0	6d03h/00:01:34	v2	1 / S P G
10.1.4.2	Ethernet3/0	00:00:55/00:01:18	v2	1 / DR S P G

```
R3#
```

```
R3#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.4.4	1	FULL/BDR	00:00:35	10.1.4.2	Ethernet3/0
192.168.2.2	1	FULL/DR	00:00:33	10.1.2.1	Ethernet2/0

This output shows that the neighbor adjacency is established.

Now we can test to verify if the source and receiver for 239.1.1.1 are able to communicate. To verify the source is sending multicast traffic, we execute **show ip mroute** to check the multicast state flags on R3, as demonstrated in [Example 7-27](#) .

Example 7-27 Multicast State Table at R3

[Click here to view code image](#)

```
R3#show ip mroute 239.1.1.1
```

IP Multicast Routing Table

..

Outgoing interface flags: H - Hardware switched, A - Assert winner, p - PIM Join

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode

(* , 239.1.1.1), 00:04:59/stopped, RP 192.168.100.100, flags: SP

Incoming interface: Null, RPF nbr 0.0.0.0

Outgoing interface list: Null

(10.1.1.1, 239.1.1.1), 00:04:59/00:02:17, flags: TA

Incoming interface: Ethernet2/0, RPF nbr 10.1.2.1

Outgoing interface list:

Ethernet3/0, Forward/Sparse, 00:03:49/00:03:25

The flag “T” indicates that multicast flow is via R3 router. Now verify with multicast ping that it is working properly, as demonstrated in [Example 7-28](#) .

Example 7-28 Verifying the Multicast Ping

[Click here to view code image](#)

```
R2# ping 239.1.1.10 source 10 repeat 100
```

Type escape sequence to abort.

Sending 100, 100-byte ICMP Echos to 239.1.1.10, timeout is 2 seconds:

Packet sent with a source address of 192.168.2.2

Reply to request 0 from 192.168.5.5, 19 ms

Reply to request 0 from 192.168.5.5, 19 ms
Reply to request 1 from 192.168.5.5, 1 ms
Reply to request 1 from 192.168.5.5, 1 ms
Reply to request 2 from 192.168.5.5, 1 ms
Reply to request 2 from 192.168.5.5, 1 ms

The test shows the ping is successful. The multicast of state of the group 239.1.1.1 on R3 shows the appropriate information, as shown in [Example 7-29](#) .

Example 7-29 *Multicast State Table at R3*

[Click here to view code image](#)

```
R3#sh ip mroute 239.1.1.10
IP Multicast Routing Table
..
(*, 239.1.1.10), 00:00:24/stopped, RP 192.168.100.100, flags: SP
Incoming interface: Null, RPF nbr 0.0.0.0
Outgoing interface list: Null
(192.168.2.2, 239.1.1.10), 00:00:24/00:02:35, flags: TA
Incoming interface: Ethernet2/0, RPF nbr 10.1.2.1
Outgoing interface list:
Ethernet3/0, Forward/Sparse, 00:00:24/00:03:05
```

The problem was caused due to PIM not being enabled on one of the links. Another command you can verify to see the flow of multicast is the **show ip mfib count** , as shown in [Example 7-30](#) .

Example 7-30 *Multicast FIB Table to Check the Data Plane*

[Click here to view code image](#)

```
R3#show ip mfib count | begin 239.1.1.1
Group: 239.1.1.1
RP-tree,
SW Forwarding: 0/0/0/0, Other: 0/0/0
Source: 10.1.1.1,
SW Forwarding: 9/0/100/0, Other: 0/0/0
Totals - Source count: 1, Packet count: 9
Group: 239.1.1.2
RP-tree,
SW Forwarding: 0/0/0/0, Other: 0/0/0
Group: 239.1.1.10
RP-tree,
SW Forwarding: 0/0/0/0, Other: 0/0/0
Source: 192.168.2.2,
SW Forwarding: 36/0/100/0, Other: 0/0/0
Totals - Source count: 1, Packet count: 36
```

This command changes based on the network platform you are using; however, it is an excellent command to know if you want to verify data plane traffic.

Important Multicast show Commands

This section explains several IOS, NX-OS, and IOS-XR commands useful in collecting pertinent information on multicast. We recommend that you review Cisco.com command documentation to understand other multicast commands that can be leveraged during troubleshooting.

show ip igmp group Command

This command is used to verify if the receiver has sent an IGMP request to join group 239.1.2.3. The command syntax is the same for IOS, Nexus, and IOS-XR:

```
show ip igmp group group
```

[Example 7-31](#) demonstrates sample output from this command.

Example 7-31 show ip igmp group Command Output

[Click here to view code image](#)

```
R1#sh ip igmp group 239.1.2.3
```

IGMP Connected Group Membership

Group Address	Interface	Uptime	Expires	Last Reporter
239.1.2.3	Ethernet1/0	00:01:07	never	172.16.8.6

show ip igmp interface/show igmp interface Commands

This command is used to verify the IGMP version, query interval/timeout, IGMP activity, query router, and the multicast group that has been joined.

The command syntax for IOS and Nexus is:

[Click here to view code image](#)

```
show ip igmp interface interface-type
```

The command syntax for IOS-XR is:

[Click here to view code image](#)

```
show igmp interface interface-type
```

[Example 7-32](#) demonstrates sample output from the **show ip igmp interface** command for IOS. (The NX-OS output will be similar, but it is not shown here.)

Example 7-32 show ip igmp interface Command Output for IOS

[Click here to view code image](#)

```
R6#sh ip igmp interface Ethernet1/0
```

Ethernet1/0 is up, line protocol is up

Internet address is 172.16.8.6/24

IGMP is enabled on interface

Current IGMP version is 2

CGMP is disabled on interface

IGMP query interval is 60 seconds

IGMP querier timeout is 120 seconds

IGMP max query response time is 10 seconds

Last member query response interval is 1000 ms

Inbound IGMP access group is not set

IGMP activity: 1 joins, 0 leaves

Multicast routing is enabled on interface

Multicast TTL threshold is 0
Multicast designated router (DR) is 172.16.8.6 (this system)
IGMP querying router is 172.16.8.6 (this system)
Multicast groups joined (number of users):

239.1.2.3(1)

[Example 7-33](#) demonstrates sample output from the **show igmp interface** command for IOS-XR.

Example 7-33 show igmp interface *Command Output for IOS-XR*

[Click here to view code image](#)

```
RP/0/0/CPU0:R1#sh igmp interface loopback 1
```

Wed Mar 23 07:52:13.139 UTC

```
Loopback1 is up, line protocol is up
Internet address is 10.100.1.1/32
IGMP is enabled on interface
Current IGMP version is 3
IGMP query interval is 60 seconds
IGMP querier timeout is 125 seconds
IGMP max query response time is 10 seconds
Last member query response interval is 1 seconds
IGMP activity: 4 joins, 0 leaves
IGMP querying router is 10.100.1.1 (this system)
Time elapsed since last query sent 00:00:48
Time elapsed since IGMP router enabled 00:02:07
Time elapsed since last report received 00:00:45
```

show ip mroute/show mrib route **Command**

This is a very useful command to understand the multicast flow. The information available from this command helps understand the flags, RPF neighbor, RP information, OIL, and IIF. Note the difference in the flag nomenclature between IOS and NX-OS. The information relayed is the same.

The command syntax for IOS and Nexus is:

```
show ip mroute
```

The command syntax for IOS-XR is:

```
show mrib route
```

[Example 7-34](#) demonstrates sample output from the **show ip mroute** command for IOS.

Example 7-34 show ip mroute *Command Output for IOS*

[Click here to view code image](#)

```
R6> show ip mroute
```

IP Multicast Routing Table

Flags: D - Dense, S - Sparse, C - Connected, L - Local, P - Pruned

R - RP-bit set, F - Register flag, T - SPT-bit set, J - Join SPT

M - MSDP created entry, X - Proxy Join Timer Running

A - Advertised via MSDP

Timers: Uptime/Expires

Interface state: Interface, Next-Hop or VCD, State/Mode
(*, 239.1.1.1), 00:13:28/00:02:59, RP 10.1.5.1, flags: SCJ
Incoming interface: Ethernet0, RPF nbr 10.1.2.1,
Outgoing interface list:
Ethernet1, Forward/Sparse, 00:13:28/00:02:32
Serial0, Forward/Sparse, 00:4:52/00:02:08
(171.68.37.121/32, 239.1.1.1), 00:01:43/00:02:59, flags: CJT
Incoming interface: Serial0, RPF nbr 192.10.2.1
Outgoing interface list:
Ethernet1, Forward/Sparse, 00:01:43/00:02:11
Ethernet0, forward/Sparse, 00:01:43/00:02:11

[Example 7-35](#) demonstrates sample output from the **show ip mroute** command for Nexus.

Example 7-35 show ip mroute *Command Output for Nexus*

[Click here to view code image](#)

```
nexusr1# show ip mroute
IP Multicast Routing Table for VRF "default"
(*, 232.0.0.0/8), uptime: 00:08:09, pim ip
Incoming interface: Null, RPF nbr: 0.0.0.0
Outgoing interface list: (count: 0)
(*, 239.1.1.1/32), uptime: 00:07:22, pim ip
Incoming interface: loopback1, RPF nbr: 10.100.1.1
Outgoing interface list: (count: 1)
Ethernet2/2, uptime: 00:07:22, pim
(*, 239.1.1.100/32), uptime: 00:07:27, igmp ip pim
Incoming interface: loopback1, RPF nbr: 10.100.1.1
Outgoing interface list: (count: 1)
loopback0, uptime: 00:07:27, igmp
```

[Example 7-36](#) demonstrates sample output from the **show mrrib route** command for IOS-XR.

Example 7-36 show mrrib route *Command Output for IOS-XR*

[Click here to view code image](#)

```
RP/0/0/CPU0:R1#show mrrib route
Wed Mar 23 07:53:27.604 UTC

IP Multicast Routing Information Base
Entry flags: L - Domain-Local Source, E - External Source to the Domain,
C - Directly-Connected Check, S - Signal, IA - Inherit Accept,
IF - Inherit From, D - Drop, ME - MDT Encap, EID - Encap ID,
MD - MDT Decap, MT - MDT Threshold Crossed, MH - MDT interface handle
CD - Conditional Decap, MPLS - MPLS Decap, MF - MPLS Encap, EX - Extranet
MoFE - MoFRR Enabled, MoFS - MoFRR State, MoFP - MoFRR Primary
MoFB - MoFRR Backup, RPFID - RPF ID Set, X - VXLAN
Interface flags: F - Forward, A - Accept, IC - Internal Copy,
NS - Negate Signal, DP - Don't Preserve, SP - Signal Present,
```

II - Internal Interest, ID - Internal Disinterest, LI - Local Interest,
LD - Local Disinterest, DI - Decapsulation Interface
EI - Encapsulation Interface, MI - MDT Interface, LVIF - MPLS Encap,
EX - Extranet, A2 - Secondary Accept, MT - MDT Threshold Crossed,
MA - Data MDT Assigned, LMI - mLDP MDT Interface, TMI - P2MP-TE MDT Interface
IRMI - IR MDT Interface

(*239.0.0.0/8) RPF nbr: 10.100.1.1 Flags: L C RPF P

Up: 00:03:23

Outgoing Interface List

Decapstunnel0 Flags: NS DI, Up: 00:03:17

(*239.1.1.100) RPF nbr: 10.100.1.1 Flags: C RPF

Up: 00:02:15

Incoming Interface List

Decapstunnel0 Flags: A NS, Up: 00:02:15

Outgoing Interface List

Loopback1 Flags: F IC NS II LI, Up: 00:02:15

RP/0/0/CPU0:R1#

show ip pim interface/show pim interface Commands

This command is useful for verifying the interfaces that are configured with PIM. The designated router (DR) for segment is also shown using this command.

The command syntax for IOS and Nexus is:

show ip pim interface

The command syntax for IOS-XR is:

show pim interface

[Example 7-37](#) demonstrates sample output from the **show ip pim interface** command for IOS. (The NX-OS output is similar, but it is not shown here.)

Example 7-37 show ip pim interface Command Output for IOS

[Click here to view code image](#)

R6#show ip pim interface

Address	Interface	Version/Mode	Nbr	Query	DR
		Count Intvl			
172.16.10.6	Serial0/0	v2/Sparse	1	30	0.0.0.0
172.16.7.6	Ethernet0/1	v2/Sparse	1	30	172.16.7.6
172.16.8.6	Ethernet1/0	v2/Sparse	0	30	172.16.8.6

[Example 7-38](#) demonstrates sample output from the **show pim interface** command for IOS-XR.

Example 7-38 show pim interface Command Output for IOS-XR

[Click here to view code image](#)

RP/0/0/CPU0:R1#show pim interface

Wed Mar 23 07:55:37.675 UTC

PIM interfaces in VRF default

Address	Interface	PIM Nbr	Hello	DR	DR
---------	-----------	---------	-------	----	----

```

192.168.0.1    Loopback0          on 1 30 1 this system
10.100.1.1    Loopback1          on 1 30 1 this system
192.168.12.1  GigabitEthernet0/0/0/0  on 2 30 1 192.168.12.2
192.168.23.1  GigabitEthernet0/0/0/1  on 2 30 1 192.168.23.2
RP/0/0/CPU0:R1#

```

show ip pim neighbor/show pim neighbor Commands

After checking the PIM interface, this command helps verify the PIM adjacency between Layer 3 neighbors.

The command syntax for IOS and Nexus is:

```
show ip pim neighbor
```

The command syntax for IOS-XR is:

```
show pim neighbor
```

[Example 7-39](#) demonstrates sample output from the **show ip pim neighbor** command for IOS. (The NX-OS output is similar, but it is not shown here.)

Example 7-39 show ip pim neighbor Command Output for IOS

[Click here to view code image](#)

```
R6#show ip pim neighbor
```

PIM Neighbor Table

Neighbor Address	Interface	Uptime	Expires	Ver	Mode
172.16.10.3	Serial0/0	7w0d	00:01:26	v2	
172.16.7.5	Ethernet0/1	7w0d	00:01:30	v2	

[Example 7-40](#) demonstrates sample output from the **show pim neighbor** command for IOS-XR.

Example 7-40 show pim neighbor Command Output for IOS-XR

[Click here to view code image](#)

```
RP/0/0/CPU0:R1#show pim neighbor
```

Wed Mar 23 07:56:42.811 UTC

PIM neighbors in VRF default

Flag: B - Bidir capable, P - Proxy capable, DR - Designated Router,
E - ECMP Redirect capable

* indicates the neighbor created for this router

Neighbor Address	Interface	Uptime	Expires	DR pri	Flags
192.168.12.1*	GigabitEthernet0/0/0/0	00:06:36	00:01:36	1	B P E
192.168.12.2	GigabitEthernet0/0/0/0	00:06:32	00:01:35	1 (DR)	P
192.168.23.1*	GigabitEthernet0/0/0/1	00:06:36	00:01:37	1	B P E
192.168.23.2	GigabitEthernet0/0/0/1	00:06:34	00:01:35	1 (DR)	B P
192.168.0.1*	Loopback0	00:06:37	00:01:20	1 (DR)	B P E
10.100.1.1*	Loopback1	00:06:37	00:01:19	1 (DR)	B P E

```
RP/0/0/CPU0:R1#
```

show ip pim rp Command

This command checks the RP information on a router. The Nexus command shows scope range and RP propagation similar to the IOS **show ip pim rp mapping** command.

The command syntax for IOS and Nexus is:

```
show ip pim rp
```

[Example 7-41](#) demonstrates sample output from the **show ip pim rp** command for IOS.

Example 7-41 show ip pim rp Command Output for IOS

[Click here to view code image](#)

```
R6#sh ip pim rp 239.1.2.3
```

```
Group: 239.1.2.3, RP: 111.1.1.1, v2, uptime 00:23:36, expires never
```

[Example 7-42](#) demonstrates sample output from the **show ip pim rp** command for NX-OS.

Example 7-42 show ip pim rp Command Output for NX-OS

[Click here to view code image](#)

```
nexusr1# show ip pim rp
```

```
PIM RP Status Information for VRF "default"
```

```
BSR disabled
```

```
Auto-RP disabled
```

```
BSR RP Candidate policy: None
```

```
BSR RP policy: None
```

```
Auto-RP Announce policy: None
```

```
Auto-RP Discovery policy: None
```

```
Anycast-RP 192.168.0.1 members:
```

```
10.100.1.1*
```

```
Anycast-RP 192.168.0.2 members:
```

```
10.100.1.1*
```

```
RP: 10.100.1.1*, (0), uptime: 00:04:42, expires: never,
```

```
priority: 0, RP-source: (local), group ranges:
```

```
239.0.0.0/8
```

```
nexusr1#
```

show ip pim rp mapping/show pim rp mapping Commands

This command checks the RP information on a router. The groups scoped for the RP and the RP propagation method. NX-OS displays RP information details using the **show ip pim rp** command.

The command syntax for IOS is:

```
show ip pim rp mapping
```

The command syntax for IOS-XR is:

```
show pim rp mapping
```

[Example 7-43](#) demonstrates sample output from the **show ip pim rp mapping** command for IOS.

Example 7-43 show ip pim rp mapping Command Output for IOS

[Click here to view code image](#)

```
R5# show ip pim rp mapping
```

PIM Group-to-RP Mappings

Group(s) 239.1.0.0/16

RP 192.168.100.100 (?), v2v1

Info source: 192.168.100.100 (?), elected via Auto-RP

Uptime: 1w0d, expires: 00:00:24

[Example 7-44](#) demonstrates sample output from the **show pim rp mapping** command for IOS-XR.

Example 7-44 show pim rp mapping *Command Output for IOS-XR*

[Click here to view code image](#)

```
RP/0/0/CPU0:R1#show pim rp mapping
```

```
Wed Mar 23 07:57:43.757 UTC
```

```
PIM Group-to-RP Mappings
```

```
Group(s) 239.0.0.0/8
```

```
RP 10.100.1.1 (?), v2
```

```
Info source: 0.0.0.0 (?), elected via config
```

```
Uptime: 00:07:39, expires: never
```

```
RP/0/0/CPU0:R1#
```

Summary

Troubleshooting is a learned skill that requires time and effort. To become proficient at solving multicast problems quickly, you need to understand the infrastructure, establish a baseline, and follow the fundamental troubleshooting methodology of application, unicast domain, and multicast domain. The more time you spend reviewing the output of the show commands explained in this chapter, the better understanding you will have of the operation of multicast. Remember, practice, practice, practice!

Code Snippets

Code Snippets

Ethernet Packet: 60 bytes

Dest Addr: 0100.5E00.0001, Source Addr: 0022.5561.2501

Protocol: 0x0800

IP Version: 0x4, HdrLen: 0x6, TOS: 0xC0 (Prec=Internet Contrl)

Length: 32, ID: 0x03E6, Flags-Offset: 0x0000

TTL: 1, Protocol: 2 (IGMP), Checksum: 0x7387 (OK)

Source: 192.168.12.1, Dest: 224.0.0.1

Options: Length = 4

Router Alert Option: 94 0000

IGMP VersionType: 0x11, Max Resp: 0x64, Checksum: 0xEE9B (OK)

Version 2 Membership Query

Group Address: 0.0.0.0

Router#show ip igmp interface e1/0

Loopback0 is up, line protocol is up

Internet address is 192.168.2.2/32

IGMP is enabled on interface

Current IGMP host version is 2

Current IGMP router version is 2

IGMP query interval is 60 seconds

IGMP configured query interval is 60 seconds

IGMP querier timeout is 120 seconds

IGMP configured querier timeout is 120 seconds

IGMP max query response time is 10 seconds

Last member query count is 2

Last member query response interval is 1000 ms

Inbound IGMP access group is not set

IGMP activity: 3 joins, 0 leaves

Multicast routing is enabled on interface

Multicast TTL threshold is 0

Multicast designated router (DR) is 192.168.2.2 (this system)

IGMP querying router is 192.168.2.2 (this system)

Multicast groups joined by this system (number of users):

224.0.1.40(1) 224.0.1.39(1) 239.1.1.1(1)

Ethernet II, Src: (80:ee:73:07:7b:61), Dst: (01:00:5e:00:00:16)

Type: IP (0x0800)

Internet Protocol Version 4, Src: 192.168.7.14, Dst: 224.0.0.22

Version: 4

Header length: 24 bytes

Differentiated Services Field: 0xc0 (DSCP 0x30: Class Selector 6; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))

Total Length: 52

Identification: 0x0000 (0)

Flags: 0x02 (Don't Fragment)

Fragment offset: 0

Time to live: 1

Protocol: IGMP (2)

Header checksum: 0x3c37 [validation disabled]

Source: 192.168.7.14

Destination: 224.0.0.22

Options: (4 bytes), Router Alert

Internet Group Management Protocol

[IGMP Version: 3]

Type: Membership Report (0x22)

Header checksum: 0x4a06 [correct]

Num Group Records: 2

Group Record : 224.64.7.7 Mode Is Include

Record Type: Mode Is Include (1)

Aux Data Len: 0

Num Src: 1

Multicast Address: 224.64.7.7

Source Address: 192.168.8.10

Group Record : 224.0.0.251 Mode Is Exclude

Record Type: Mode Is Exclude (2)

Aux Data Len: 0

Num Src: 0

Multicast Address: 224.0.0.251

```

Ethernet Packet: 60 bytes
  Dest Addr: 0100.5E00.0001,   Source Addr: 0C85.2545.9541
  Protocol: 0x0800

IP   Version: 0x4,  HdrLen: 0x6,  TOS: 0xC0 (Prec=Internet Contrl)
     Length: 32,    ID: 0x1FC5,   Flags-Offset: 0x0000
     TTL: 1,      Protocol: 2 (IGMP),  Checksum: 0x57A8 (OK)
     Source: 192.168.12.1,      Dest: 224.0.0.1

     Options: Length = 4
     Router Alert Option: 94 0000

IGMP VersionType: 0x11,  Max Resp: 0x64,  Checksum: 0xEE9B (OK)

Version 2 Membership Query
  Group Address: 0.0.0.0

```

```
Switch#show ip igmp snooping mrouter
```

```

Vlan    ports
----    -
 12     Gi0/12(dynamic)

```

```
Switch#debug ip igmp snooping router
```

```

router debugging is on
01:49:07: IGMP SN: router: Received non igmp pak on Vlan 12, port Gi0/12
01:49:07: IGMP SN: router: PIMV2 Hello packet received in 12
01:49:07: IGMP SN: router: Is not a router port on Vlan 12, port Gi0/12
01:49:07: IGMP SN: router: Created router port on Vlan 12, port Gi0/12
01:49:07: IGMP SN: router: Learning port: Gi0/12 as rport on Vlan 12

```

```
Switch#debug ip igmp snooping group
```

```

router debugging is on
01:58:47: IGMP SN: Received IGMPv2 Report for group 224.64.7.7 received on Vlan 12,
port Gi0/2
01:58:47: IGMP SN: group: Adding client ip 192.168.12.20, port_id Gi0/2, on vlan 12

```

```
Switch#show ip igmp snooping groups
```

Vlan	Group	Type	Version	Port List
12	224.0.1.40	igmp	v2	Gi0/12
12	224.64.7.7	igmp	v2	Gi0/2, Gi0/12

```

Ethernet Packet: 60 bytes
  Dest Addr: 0100.5E40.0707,   Source Addr: 000F.F7B1.67E0
  Protocol: 0x0800

IP   Version: 0x4,  HdrLen: 0x5,  TOS: 0xC0 (Prec=Internet Contrl)
     Length: 28,    ID: 0x0000,   Flags-Offset: 0x0000
     TTL: 1,      Protocol: 2 (IGMP),  Checksum: 0x051D (OK)
     Source: 192.168.12.20,      Dest: 224.64.7.7

IGMP VersionType: 0x16,  Max Resp: 0x00,  Checksum: 0x02B8 (OK)

Version 2 Membership Report
  Group Address: 224.64.7.7

```

```
Switch#show ip igmp snooping groups
```

Vlan	Group	Type	Version	Port List
12	224.0.1.40	igmp	v2	Gi0/15
12	224.64.7.7	igmp	v2	Gi0/1, Gi0/2, Gi0/4, Gi0/15

```

Ethernet Packet: 60 bytes
  Dest Addr: 0100.5E00.0002,   Source Addr: 0013.19C6.A60F
  Protocol: 0x0800

IP   Version: 0x4,  HdrLen: 0x6,  TOS: 0xC0 (Prec=Internet Contrl)
     Length: 32,    ID: 0x0000,   Flags-Offset: 0x0000
     TTL: 1,      Protocol: 2 (IGMP),  Checksum: 0x7745 (OK)
     Source: 192.168.12.40,      Dest: 224.0.0.2

     Options: Length = 4
     Router Alert Option: 94 0000

IGMP VersionType: 0x17,  Max Resp: 0x00,  Checksum: 0x01B8 (OK)

Version 2 Leave Group
  Group Address: 224.64.7.7

```

```
Switch#show interface Gi0/12
```

```

GigabitEthernet0/12 is up, line protocol is up
  Hardware is Gigabit Ethernet, address is 0013.19C6.A60F (bia 0013.19C6.A60F)
  MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,

```

```
C2970(config)#ip igmp snooping
```

```
Switch#show ip igmp snooping
```

```
Global IGMP Snooping configuration:
```

```
-----  
IGMP snooping : Enabled  
IGMPv3 snooping (minimal) : Enabled  
Report suppression : Enabled  
TCN solicit query : Disabled  
TCN flood query count : 2  
Robustness variable : 2  
Last member query count : 2  
Last member query interval : 1000
```

```
storm-control broadcast level <0.00 - 100.00> / bps / pps
```

```
storm-control multicast level <0.00 - 100.00> / bps / pps
```

```
storm-control unicast level <0.00 - 100.00> / bps / pps
```

```
storm-control action trap
```

```
storm-control action shutdown
```

```
Switch(config)#interface gigabitEthernet 0/2
```

```
Switch(config-if)#storm-control broadcast level 50
```

```
Switch(config-if)#storm-control action trap
```

```
%STORM_CONTROL-3-FILTERED: A Broadcast storm detected on Gi0/2. A packet filter  
action has been applied on the interface.
```

```
Switch(config-if)#storm-control action shutdown
```

```
%PM-4-ERR_DISABLE: storm-control error detected on Gi0/2, putting Gi0/2 in  
err-disable state
```

```
%STORM_CONTROL-3-SHUTDOWN: A packet storm was detected on Gi0/2. The interface has  
been disabled.
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state  
to down
```

```
%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down
```

```
Switch(config)#errdisable recovery cause storm-control
```

```
Switch(config)#errdisable recovery interval 30
```

```
2d07h: %PM-4-ERR_RECOVER: Attempting to recover from storm-control err-disable  
state on Gi0/2
```

```
2d07h: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
```

```
2d07h: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed  
state to up
```